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P802.1AS-2020-Rev/D1.1

April 25, 2025

(Revision of IEEE Std 802.1AS™-2020)

Draft Standard for Local and metropolitan area networks—

Timing and Synchronization for Time-Sensitive Applications

Unapproved draft, prepared by the
Maintenance Task Group of IEEE 802.1

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4 Information on participation in this project, and in the IEEE 802.1 Working Group can be found [here](#).

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34 A vote of "Approve" on this draft is also an affirmation by the balloter that the PAR is still valid.

1 **Project Authorization Request, Scope, Purpose, and Criteria for Standards** 2 **Development (CSD)**

3 The complete PAR, as approved by IEEE NesCom 5 June 2023, can be found at:

4 <https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/10637>

5 The 'Scope of the Proposed changes' and the 'Need for the Project' specify the changes to be made by this
6 amendment (see below).

7 **Scope of the Proposed changes:**

8 This standard specifies protocols, procedures, and managed objects used to ensure that the synchronization
9 requirements are met for time-sensitive applications, such as audio, video, and time-sensitive control, across
10 networks, for example, IEEE 802 and similar media. This includes the maintenance of synchronized time
11 during normal operation and following addition, removal, or failure of network components and network
12 reconfiguration. It specifies the use of IEEE 1588(TM) specifications where applicable in the context of IEEE
13 Std 802.1Q(TM). Synchronization to an externally provided timing signal [e.g., a recognized timing standard
14 such as Coordinated Universal Time (UTC) or International Atomic Time (TAI)] is not part of this standard but
15 is not precluded.

16 **Need for the Project:**

17 This revision project is needed in order to incorporate approved amendments and corrigenda, to incorporate
18 technical and editorial corrections to existing functionality, and to ensure that consistency is maintained in the
19 consolidated text.

20 **Criteria for Standards Development:**

21 **There is no CSD statement since this maintenance project is not intended to provide any new**
22 **functionality**

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5 **Draft Standard for** 6 **Local and Metropolitan Area Networks—** 7 **Timing and Synchronization for** 8 **Time-Sensitive Applications**

9 Unapproved draft, prepared by the
10 **Maintenance Task Group of IEEE 802.1**

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1 **Abstract:** This is a Maintenance roll-up of IEEE Std 802.1AS™-2020 with the corrigendum IEEE
2 Std 802.1AS-2020/Cor1 and amendments IEEE Std 802.1ASdr, IEEE Std 802.1ASdm, and IEEE
3 Std 802.1ASdn.

4 **Keywords:** best timeTransmitter, frequency offset, Grandmaster Clock, Grandmaster PTP
5 Instance, IEEE 802.1AS™, IEEE 802.1ASdr™, phase offset, PTP End Instance, PTP Relay
6 Instance, synchronization, syntonization, time-aware system

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10

1 Introduction

This introduction is not part of IEEE Std 802.1AS-2020-Rev, IEEE Standard for Local and Metropolitan Area Networks—Timing and Synchronization for Time-Sensitive Applications.

2 The first edition of IEEE Std 802.1AS was published in 2011. A first corrigendum, IEEE Std
3 802.1AS-2011/Cor1-2013, provided technical and editorial corrections. A second corrigendum, IEEE Std
4 802.1AS-2011/Cor2-2015 provided additional technical and editorial corrections.

5 The second edition, IEEE Std 802.1AS-2020, added support for multiple gPTP domains, Common Mean
6 Link Delay Service, external port configuration, and Fine Timing Measurement for IEEE 802.11 transport.
7 Backward compatibility with IEEE Std 802.1AS-2011 was maintained. A corrigendum, IEEE Std
8 802.1AS-2020/Cor1-2021, provides technical and editorial corrections.

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10 IEEE Std 802.1AS-2020/Cor1 and amendments IEEE Std 802.1ASdr, IEEE Std 802.1ASdm, and IEEE Std
11 802.1ASdn. Backward compatibility with IEEE Std 802.1AS-2011 and IEEE Std 802.1AS-2020 is
12 maintained.

13

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2 **Draft Standard for** 3 **Local and Metropolitan Area Networks—** 4 **Timing and Synchronization for** 5 **Time-Sensitive Applications**

6 **1. Overview**

7 **1.1 Scope**

8 This standard specifies protocols, procedures, and managed objects used to ensure that the synchronization
9 requirements are met for time-sensitive applications, such as audio, video, and time-sensitive control, across
10 networks, for example, IEEE 802 and similar media. This includes the maintenance of synchronized time
11 during normal operation and following addition, removal, or failure of network components and network
12 reconfiguration. It specifies the use of IEEE 1588™ specifications where applicable in the context of
13 IEEE Std 802.1Q™.¹ Synchronization to an externally provided timing signal [e.g., a recognized timing
14 standard such as Coordinated Universal Time (UTC) or International Atomic Time (TAI)] is not part of this
15 standard but is not precluded.

16 **1.2 Purpose**

17 This standard enables systems to meet the respective jitter, wander, and time-synchronization requirements
18 for time-sensitive applications, including those that involve multiple streams delivered to multiple end
19 stations. To facilitate the widespread use of packet networks for these applications, synchronization
20 information is one of the components needed at each network element where time-sensitive application data
21 are mapped or demapped or a time-sensitive function is performed. This standard leverages the work of the
22 IEEE 1588 Working Group by developing the additional specifications needed to address these
23 requirements.

24

¹ Information on references can be found in Clause 2.

12. Normative references

The following referenced documents are indispensable for the application of this standard (i.e., they must be understood and used; therefore, each referenced document is cited in text, and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 754™-2008, IEEE Standard for Floating-Point Arithmetic.^{2, 3}

IEEE Std 802®-2024, IEEE Standard for Local and Metropolitan Area Networks—Overview and Architecture.

IEEE Std 802.1AC™-2016, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Service Definition.

IEEE Std 802.1AX™-2020, IEEE Standard for Local and metropolitan area networks—Link Aggregation.

IEEE Std 802.1Q™-2022, IEEE Standard for Local and Metropolitan Area Networks—Bridges and Bridged Networks.

IEEE Std 802.3™-2022, IEEE Standard for Ethernet.

IEEE Std 802.11™-2016, IEEE Standard for Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements, Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications.

IEEE Std 1588™-2019, IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems.

IEEE Std 1588e™-2024, IEEE Standard for Precision Clock Synchronization Protocol for Networked Measurement and Control Systems—Amendment 5: MIB and YANG Modules.

IERS Bulletin C (see <https://www.iers.org/IERS/EN/Publications/Bulletins/bulletins.html>).

IETF RFC 2863 (June 2000), The Interfaces Group MIB, K. McCloghrie and F. Kastenholz.⁴

IETF RFC 3410 (Dec. 2002), Introduction and Applicability Statements for Internet Standard Management Framework, J. Case, R. Mundy, D. Partain, and B. Stewart.

IETF RFC 3418 (Dec. 2002), Management Information Base (MIB) for the Simple Network Management Protocol (SNMP), R. Presuhn, ed.

IETF RFC 7950 (Aug. 2016), The YANG 1.1 Data Modeling Language.⁵

ISO 80000-3:2006, Quantities and units — Part 3: Space and time.⁶

² IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org>).

³ The IEEE standards or products referenced in this clause are trademarks owned by The Institute of Electrical and Electronics Engineers, Incorporated.

⁴ IETF Requests for Comments (RFCs) are available from the Internet Engineering Task Force (<https://www.rfc-editor.org>).

⁵ IETF RFCs are available from the Internet Engineering Task Force (<https://www.ietf.org/>).

⁶ ISO publications are available from the International Organization for Standardization (<https://www.iso.org>) and the American National Standards Institute (<https://www.ansi.org>).

1 ITU-T Recommendation G.984.3 (Nov. 2009), Amendment 2, Gigabit-capable Passive Optical Networks
2 (G-PON): Transmission convergence layer specification—Time-of-day distribution and maintenance
3 updates and clarifications.⁷

4 ITU-T Recommendation G.9960 (Nov. 2018), Unified high-speed wire-line based home networking
5 transceivers—System architecture and physical layer specification [with ITU-T G.9961, commonly referred
6 to as “ITU-T G.hn”].

7 ITU-T Recommendation G.9961 (Nov. 2018), Data link layer (DLL) for unified high-speed wire-line based
8 home networking transceivers [with ITU-T G.9960, commonly referred to as “ITU-T G.hn”].

9 MoCA[®] MAC/PHY Specification v2.0, MoCA-M/P-SPEC-V2.0-20100507, Multimedia over Coax
10 Alliance (MoCA).⁸

11

⁷ ITU-T publications are available from the International Telecommunications Union (<https://www.itu.int>).

⁸ MoCA specifications are available from the Multimedia over Coax Alliance (<http://www.mocalliance.org/specs>).

1 3. Definitions

2 For the purposes of this standard, the following terms and definitions apply. The *IEEE Standards Dictionary*
3 *Online* should be consulted for terms not defined in this clause.⁹

4 **3.1 accuracy:** The mean of the time or frequency error between the clock under test and a reference clock
5 over an ensemble of measurements (see IEEE Std 1588-2019).

6 **3.2 Bridge:** Either a MAC Bridge or a VLAN-aware Bridge, as specified in Clause 5 of IEEE Std 802.1Q-
7 2022.

8 **3.3 clock:** A physical device that is capable of providing a measurement of the passage of time since a
9 defined epoch.

10 **3.4 default:** The value of an implementation parameter or capability selection, in the absence of an
11 overriding profile standard requirement or explicit configuration.

12 **3.5 device:** An entity implementing some functionality, e.g., a clock, a time-aware system, a port.

13 **3.6 direct communication:** A communication of IEEE 802.1AS information between two PTP Instances
14 with no intervening PTP Instance.

15 **3.7 end station:** A device attached to a local area network (LAN) or metropolitan area network (MAN) that
16 acts as a source of, and/or destination for, traffic carried on the LAN or MAN.

17 **3.8 epoch:** The origin of a timescale.

18 **3.9 event message:** A message that is timestamped on egress from a PTP Instance and ingress to a PTP
19 Instance.

20 NOTE—See 8.4.3.¹⁰

21 **3.10 fractional frequency offset:** The fractional offset, y , between a measured clock and a reference clock
22 as defined by the following:

23
$$y = \frac{f_m - f_r}{f_r}$$

24 where f_m is the frequency of the measured clock and f_r is the frequency of the reference clock. The
25 measurement units of f_m and f_r are the same.

26 **3.11 frequency offset:** The offset between a measured frequency and a reference frequency as defined by
27 $f_m - f_r$, where f_m is the frequency of the measured clock and f_r is the frequency of the reference clock. The
28 measurement units of f_m and f_r are the same.

29 **3.12 general message:** A message that is not timestamped.

30 **3.13 gPTP communication path:** A segment of a generalized precision time protocol (gPTP) domain that
31 enables direct communication between two PTP Instances.

32 NOTE—See 8.1.

⁹ *IEEE Standards Dictionary Online* is available at <https://dictionary.ieee.org/>.

¹⁰ Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement the standard.

1 **3.14 grandmaster-capable PTP Instance:** A PTP Instance that is capable of being a Grandmaster
2 PTP Instance.

3 **3.15 Grandmaster Clock:** In the context of a single gPTP domain, the synchronized time of a PTP Instance
4 that is the source of time to which all other PTP Instances in the domain are synchronized.

5 **3.16 Grandmaster PTP Instance:** A PTP Instance containing the Grandmaster Clock.

6 NOTE—None of the PTP Ports of a Grandmaster PTP Instance is in the TimeReceiverPort state.

7 **3.17 local area network (LAN):** A network of devices, whether indoors or outdoors, covering a limited
8 geographic area, e.g., a building or campus.

9 **3.18 local clock:** A free-running clock, embedded in a respective entity (e.g., PTP Instance, CSN node), that
10 provides a common time to that entity relative to an arbitrary epoch.

11 **3.19 message timestamp point:** A point within an event message serving as a reference point for when a
12 timestamp is taken.

13 **3.20 message type:** The message type of a message is the name of the respective message, e.g., Sync,
14 Announce, Timing Measurement Frame.

15 **3.21 precision:** A measure of the deviation from the mean of the time or frequency error between the clock
16 under test and a reference clock (see IEEE Std 1588-2019).

17 **3.22 primary reference:** A source of time and/or frequency that is traceable to international standards.
18 *See also: traceability.*

19 **3.23 profile standard:** A standard specifying the capabilities, options, and parameter values of one or more
20 base standards for use in a specific target environment or application.

21 **3.24 PTP End Instance:** A PTP Instance that has exactly one PTP Port.

22 **3.25 PTP Instance:** An instance of the IEEE 802.1AS protocol, operating in a single time-aware system
23 within exactly one domain. A PTP Instance implements the portions of IEEE Std 802.1AS indicated as
24 applicable to either a PTP Relay Instance or a PTP End Instance.

25 NOTE—As used in IEEE Std 802.1AS, the term *PTP Instance* refers to an IEEE 1588 PTP Instance that conforms to the
26 requirements of IEEE Std 802.1AS.

27 **3.26 PTP Link:** Within a domain, a network segment between two PTP Ports using the peer-to-peer delay
28 mechanism of IEEE Std 802.1AS. The peer-to-peer delay mechanism is designed to measure the
29 propagation time over such a link.

30 NOTE—A PTP Link between PTP Ports of PTP Instances is also a gPTP Communication Path (see 3.13).

31 **3.27 PTP Relay Instance:** A PTP Instance that is capable of communicating synchronized time received on
32 one PTP Port to other PTP Ports, using the IEEE 802.1AS protocol.

33 NOTE—A PTP Relay Instance could, for example, be contained in a bridge, a router, or a multi-port end station.

34 **3.28 recognized timing standard:** A recognized standard time source that is a source external to IEEE 1588
35 precision time protocol (PTP) and provides time that is traceable to the international standards laboratories
36 maintaining clocks that form the basis for the International Atomic Time (TAI) and Coordinated Universal
37 Time (UTC) timescales. Examples of these sources are National Institute of Standards and Technology
38 (NIST) timeservers and global navigation satellite systems (GNSSs).

13.29 reference plane: The boundary between a PTP Port of a PTP Instance and the network physical medium. Timestamp events occur as frames cross this interface.

3.30 residence time: The duration of the time interval between the receipt of a time-synchronization event message by a PTP Instance and the sending of the next subsequent time-synchronization event message on another PTP Port of that PTP Instance. Residence time can be different for different PTP Ports. The term *residence time* applies only to the case where `syncLocked` is `TRUE`.

NOTE 1—See 10.2.5.15 for the definition of the variable `syncLocked`.

NOTE 2—If a PTP Port of a PTP Instance sends a time-synchronization event message without having received a time-synchronization event message, e.g., if `syncLocked` is `FALSE` or if `sync receipt timeout` occurs (see 10.7.3.1), the duration of the interval between the most recently received time-synchronization event message and the sent time-synchronization event message is mathematically equivalent to residence time; however, this interval is not normally called a *residence time*.

3.31 sdoid: An attribute that is the primary mechanism for providing isolation of PTP Instances operating under a PTP profile specified by one Qualified Standards Development Organization (QSDO) from PTP Instances operating under a PTP profile specified by a different QSDO (see 7.1.3 and 7.1.4 of IEEE Std 1588-2019).

3.32 stability (of a clock or clock signal): A measure of the variations over time of the frequency error (of the clock or clock signal). The frequency error typically varies with time due to aging and various environmental effects, e.g., temperature.

3.33 synchronized time: The time of an event relative to the Grandmaster Clock.

NOTE—If there is a change in the Grandmaster PTP Instance or its time base, the *synchronized time* can experience a phase and/or frequency step.

3.34 synchronized clocks: Absent relativistic effects, two clocks are synchronized to a specified uncertainty if they have the same epoch and their measurements of the time of a single event at an arbitrary time differ by no more than that uncertainty.

NOTE—See 8.2.2.

3.35 syntonized clocks: Absent relativistic effects, two clocks are syntonized to a specified uncertainty if the duration of a second is the same on both, which means the time as measured by each advances at the same rate within the specified uncertainty. The two clocks might or might not share the same epoch.

3.36 time-aware system: A device that contains one or more PTP Instances and/or PTP services (e.g., Common Mean Link Delay Service).

NOTE 1—See 11.2.17 for a description of the Common Mean Link Delay Service.

NOTE 2—A time-aware system can contain more than one PTP Instance in the same domain and/or different domains.

3.37 timescale: A measure of elapsed time since an epoch.

3.38 timestamp measurement plane: The plane at which timestamps are captured. If the timestamp measurement plane is different from the reference plane, the timestamp is corrected for `ingressLatency` and/or `egressLatency`. See: **reference plane**.

NOTE—For timestamping on egress and ingress, see 8.4.3.

3.39 traceability: See 3.1.81 in IEEE Std 1588-2019.

40

14. Acronyms and abbreviations

2 Ack	acknowledgment
3 ADEV	Allan deviation
4 ARB	arbitrary
5 BC	Boundary Clock
6 BTC	best timeTransmitter clock
7 BTCA	best timeTransmitter clock algorithm
8 CID	Company identification (allocated by the IEEE)
9 CMLDS	Common Mean Link Delay Service
10 CSN	coordinated shared network
11 CTC	channel time clock
12 EPON	IEEE 802.3™ Ethernet Passive Optical Network, as specified in IEEE Std 802.3-2022
13 FTM	Fine Timing Measurement
14 G.hn	ITU-T G.9960 and ITU-T G.9961
15 GM	Grandmaster
16 GMT	Greenwich mean time
17 GNSS	global navigation satellite system
18 GPS	global positioning (satellite) system
19 gPTP	generalized precision time protocol (IEEE Std 802.1AS)
20 IERS	International Earth Rotation and Reference Systems Service
21 IP	Internet Protocol
22 ISS	Internal Sublayer Service
23 LAN	local area network
24 LCI	location configuration information
25 LLC	logical link control
26 MAC	media access control
27 MACsec	media access control security

1 MIB	Management Information Base
2 MLME	IEEE 802.11™ MAC layer management entity
3 MPCP	IEEE 802.3 multipoint control protocol
4 MPCPDU	IEEE 802.3 MPCP data unit
5 MD	media-dependent
6 NETCONF	Network Configuration Protocol
7 NMS	Network Management System
8 NTP	network time protocol ¹¹
9 OLT	IEEE 802.3 optical line terminal
10 ONU	IEEE 802.3 optical network unit
11 OSSP	organization-specific slow protocol
12 OUI	Organizationally Unique Identifier
13 P2P	peer-to-peer
14 PICS	Protocol Implementation Conformance Statement
15 POSIX®	portable operating system interface (see ISO/IEC 9945:2003 [B17] ¹²)
16 PTP	IEEE 1588 precision time protocol
17 PTPDEV	PTP deviation
18 QSDO	Qualified Standards Development Organization
19 RTT	round-trip time
20 SI	international system of units
21 SMI	Structure of Management Information
22 SMIv2	Structure of Management Information version 2
23 SNMP	Simple Network Management Protocol
24 STA	station
25 TAI	International Atomic Time
26 TC	Transparent Clock

¹¹ Information available at <https://www.ietf.org/rfc/rfc1305.txt>.

¹² The numbers in brackets correspond to the numbers in the bibliography in Annex H.

1 TDEV	time deviation
2 TDM	time division multiplexing
3 TDMA	time division multiple access
4 TLV	type, length, value
5 TM	Timing Measurement
6 UCT	unconditional transfer
7 UML [®]	Unified Modeling Language [™] ¹³
8 UTC	Coordinated Universal Time
9 VLAN	virtual local area network
10 WLAN	wireless local area network

¹³ UML[®] is a registered trademark of the Object Management Group, Inc., and Unified Modeling Language[™] is a trademark of the Object Management Group, Inc.

1 5. Conformance

2 This clause specifies the mandatory and optional capabilities provided by conformant implementations of
3 this standard.

4 5.1 Requirements terminology

5 For consistency with existing IEEE and IEEE 802.1 standards, requirements placed upon conformant
6 implementations of this standard are expressed using the following terminology:

- 7 a) **shall** is used for mandatory requirements.
- 8 b) **may** is used to describe implementation or administrative choices (“may” means “is permitted to,”
9 and hence, “may” and “may not” mean precisely the same thing).
- 10 c) **should** is used for recommended choices (the behaviors described by “should” and “should not” are
11 both permissible but not equally desirable choices).

12 The Protocol Implementation Conformance Statement (PICS) proforma (see Annex A) reflects the
13 occurrences of the words shall, may, and should within the standard. The words shall, may, and should, as
14 used in Annex A itself, reflect the use of the PICs and not conformance to the standard.

15 The standard avoids needless repetition and apparent duplication of its formal requirements by using **is**, **is**
16 **not**, **are**, and **are not** for definitions and the logical consequences of conformant behavior. Behavior that is
17 permitted but is neither always required nor directly controlled by an implementer or administrator, or
18 whose conformance requirement is detailed elsewhere, is described by **can**. Behavior that never occurs in a
19 conformant implementation or system of conformant implementations is described by **cannot**. The word
20 **allow** is used as a replacement for the phrase “support the ability for,” and the word **capability** means “can
21 be configured to.”

22 5.2 Protocol Implementation Conformance Statement (PICS)

23 The supplier of an implementation that is claimed to conform to this standard shall complete a copy of the
24 PICS proforma provided in Annex A and shall provide the information necessary to identify both the
25 supplier and the implementation.

26 5.3 Time-aware system requirements

27 An implementation of a time-aware system shall support at least one IEEE 1588 precision time protocol
28 (PTP) Instance.

29 5.4 PTP Instance requirements and options

30 5.4.1 PTP Instance requirements

31 An implementation of a PTP Instance shall:

- 32 a) Implement the generalized precision time protocol (gPTP) requirements specified in Clause 8.
- 33 b) Support the requirements for time-synchronization state machines (10.1.2, 10.2.1, 10.2.2, 10.2.3,
34 10.2.4, 10.2.5, and 10.2.6).
- 35 c) Support at least one PTP Port.
- 36 d) On each supported PTP Port, implement the PortSyncSyncReceive state machine (10.2.8).

- 1 e) Implement the ClockTimeReceiverSync state machine (10.2.13).
- 2 f) Implement the SiteSyncSync state machine (10.2.7).
- 3 g) Implement the state machines related to signaling gPTP capability (10.4).
- 4 NOTE 1—5.4.1 g) does not include the GtpCapableIntervalSetting state machine (10.4.4), despite its name
- 5 containing “GtpCapable”.
- 6 NOTE 2—The GtpCapableTransmit and GtpCapableReceive state machines are required; however, they can
- 7 be disabled by setting the managed object gtpCapableStateMachinesEnabled (see 14.10.55) to FALSE.
- 8 h) For receipt of all messages and for transmission of all messages except Announce (see 10.6.3),
- 9 support the message requirements as specified in 10.5, 10.6, and 10.7.
- 10 NOTE 3—The support of Signaling messages is mandatory [5.4.1 h)] because the state machines related to the
- 11 signaling of gPTP capability are mandatory [5.4.1 g)].
- 12 i) Support the performance requirements in B.1 and B.2.4.

13 5.4.2 PTP Instance options

14 An implementation of a PTP Instance should:

- 15 a) Support the performance requirements in B.2.2 and B.2.3.

16 An implementation of a PTP Instance may:

- 17 b) Support the following media-independent timeTransmitter capability on at least one PTP Port:
 - 18 1) Implement the PortSyncSyncSend state machine (10.2.12).
 - 19 2) Implement the PortAnnounceTransmit state machine (10.3.16).
 - 20 3) Implement the AnnounceIntervalSetting state machine (10.3.17).
 - 21 4) For transmit of the Announce message, support the message requirements as specified in 10.5,
 - 22 10.6, and 10.7.
- 23 c) Support the following for Grandmaster PTP Instance capability:
 - 24 1) Support the media-independent timeTransmitter capability specified in item b) of 5.4.2.
 - 25 2) Support the requirements for a grandmaster-capable PTP Instance (10.1.3).
 - 26 3) Implement the ClockTimeTransmitterSyncSend state machine (10.2.9).
 - 27 4) Implement the ClockTimeTransmitterSyncOffset state machine (10.2.10).
 - 28 5) Implement the ClockTimeTransmitterReceive state machine (10.2.11).
- 29 d) Support more than one PTP Port as a PTP Relay Instance (5.4.4).
- 30 e) Support more than one PTP Instance; such support allows for more than one domain (7.3.2).
- 31 f) Support the following external port configuration capability on at least one port:
 - 32 1) Implement specifications for externalPortConfigurationEnabled value of true (10.3.1).
 - 33 2) Implement the PortAnnounceInformationExt state machine (10.3.14).
 - 34 3) Implement the PortStateSettingExt state machine (10.3.15).
- 35 g) Implement the SyncIntervalSetting state machine (10.3.18).
- 36 h) Implement the GtpCapableIntervalSetting state machine.
- 37 i) Implement one or more of the application interfaces specified in Clause 9; A PTP Instance that
- 38 claims to support application interfaces shall state which application interfaces are supported.
- 39 j) Support timing and synchronization management as specified in Clause 14.

- 1 k) Support the use of a remote management protocol. A PTP Instance that claims to support remote
2 management shall:
 - 3 1) State which remote management protocol standard(s) or specification(s) are supported
4 (see A.19).
 - 5 2) State which standard(s) or specification(s) for managed object definitions and encodings are
6 supported for use by the remote management protocol (see A.19).
 - 7 3) If the Simple Network Management Protocol (SNMP) is supported as a remote management
8 protocol, support the managed object definitions specified as Structure of Management
9 Information version 2 (SMIv2) Management Information Base (MIB) modules in Clause 15.
 - 10 4) If YANG is supported with a remote management protocol, support the YANG data model
11 ieee802-dot1as-gtp in Clause 17.
 - 12 5) If YANG is supported with a remote management protocol, and if hot standby is supported,
13 support the YANG data model ieee802-dot1as-hs in Clause 17.
- 14 l) Implement both BTCA and external port configuration on domains other than domain 0; if both
15 possibilities are implemented on domains other than domain 0, the default value of
16 externalPortConfigurationEnabled shall be FALSE.
- 17 m) Implement hot standby as specified in Clause 18, 14.8, 9.3.3.4, 9.4.3.4, 9.5.3.4, and 9.6.2.6.
- 18 n) Implement the Drift_Tracking TLV as specified in 10.2.4.26, 10.2.4.27, 10.2.4.28, 11.2.14, 11.2.15,
19 and 11.4.4.

20 5.4.3 PTP Instance defaults and recommendations

21 This standard identifies specific capabilities and parameter values as defaults. If these defaults are not
22 further qualified by reference to the use of other capabilities or by profile standards, the defaults are intended
23 to facilitate deployment and to provide interoperability in target environments and applications with
24 time-aware system implementations conformant to this standard (including its 2011 and 2020 editions).
25 Mandatory requirements that this standard does not identify as defaults, or call out in some other way, are
26 not subsetted or modified by profile standards.

27 Changes to capability support or parameter defaults arising from conformance to a profile standard should
28 be identified in the PCS for that profile standard. Changes to defaults by pre-configuration of parameter
29 values by the supplier of a protocol implementation should be identified by Additional Information
30 (see A.3.2) in a completed PICS for this standard.

31 For domain 0, an implementation of a PTP Instance shall, by default, support the following best
32 timeTransmitter clock algorithm (BTCA) requirements:

- 33 a) Have the BTCA as the default mode of operation, with externalPortConfigurationEnabled FALSE.
- 34 b) Implement the BTCA (10.3.1.1, 10.3.1.2, 10.3.2, 10.3.3, 10.3.4, 10.3.5, 10.3.6, 10.3.8, and 10.3.10).
- 35 c) Implement specifications for an externalPortConfigurationEnabled value of FALSE (10.3.1).
- 36 d) Implement the PortAnnounceReceive state machine (10.3.11).
- 37 e) Implement the PortAnnounceInformation state machine (10.3.12).
- 38 f) Implement the PortStateSelection state machine (10.3.13).

39 For domains other than domain 0, an implementation of a PTP Instance shall, by default, support the
40 following best timeTransmitter clock algorithm (BTCA) requirements:

- 41 g) Implement at least one of the possibilities for externalPortConfigurationEnabled (i.e., FALSE,
42 meaning the BTCA is used, and TRUE, meaning external port configuration is used) on domains
43 other than domain 0.

- 1 h) If BTCA is used then:
- 2 1) Implement the BTCA (10.3.1.1, 10.3.1.2, 10.3.2, 10.3.3, 10.3.4, 10.3.5, 10.3.6, 10.3.8, and
- 3 10.3.10).
- 4 2) Implement specifications for an externalPortConfigurationEnabled value of FALSE (10.3.1).
- 5 3) Implement the PortAnnounceReceive state machine (10.3.11).
- 6 4) Implement the PortAnnounceInformation state machine (10.3.12).
- 7 5) Implement the PortStateSelection state machine (10.3.13).

8 5.4.4 PTP Relay Instance requirements

9 An implementation of a PTP Relay Instance shall:

- 10 a) Support more than one PTP Port.
- 11 b) Support the media-independent timeTransmitter capability specified in item b) of 5.4.2.

12 5.5 MAC-specific timing and synchronization methods for full-duplex IEEE 802.3 13 links

14 An implementation of a time-aware system with IEEE 802.3 media access control (MAC) services to
15 physical ports shall:

- 16 a) Support full-duplex operation, as specified in 4.2 and Annex 4A of IEEE Std 802.3-2022.
- 17 b) Support the requirements as specified in Clause 11 [with the exception of requirements more
- 18 specifically addressed in this subclause (5.5)].
- 19 c) Support two-step capability on receive as specified in 11.2.14.
- 20 d) Support two-step capability on transmit as specified in 11.2.15
- 21 e) Provide the CMLDS if the time-aware system implements more than one domain, as specified in
- 22 11.2.17 and 11.2.18.

23 An implementation of a PTP Instance with IEEE 802.3 MAC services to physical ports may:

- 24 f) Support asymmetry measurement mode as specified in 10.3.12, 10.3.13, 10.3.16, 11.2.14, 11.2.15,
- 25 11.2.19, and 14.10.45.
- 26 g) Support one-step capability on receive as specified in 11.2.14.
- 27 h) Support one-step capability on transmit as specified in 11.2.15.
- 28 i) Support the OneStepTxOperSetting state machine specified in 11.2.16.
- 29 j) Support propagation delay averaging, as specified in 11.2.19.3.4.
- 30 k) Provide the CMLDS if the time-aware system implements only one domain, as specified in 11.2.17
- 31 and 11.2.18.
- 32 l) Implement the LinkDelayIntervalSetting state machine (11.2.21).

33 5.6 MAC-specific timing and synchronization methods for IEEE Std 802.11-2024

34 An implementation of a time-aware system with IEEE 802.11 MAC services to physical ports shall:

- 35 a) Support the requirements as specified in Clause 12.
- 36 b) Support at least one of
- 37 1) the media-dependent timeTransmitter state machines (12.5.1), or

1 2) the media-dependent timeReceiver state machine (12.5.2).

2 An implementation of a PTP End Instance with IEEE 802.11 MAC services to physical ports shall:

3 c) Support at least one of TIMINGMSMT as specified in IEEE Std 802.11-2024 or
4 FINETIMINGMSMT as specified in IEEE Std 802.11-2024.

5 An implementation of a PTP Relay Instance with IEEE 802.11 MAC services to physical ports shall:

6 d) Support the requirements of TIMINGMSMT as specified in IEEE Std 802.11-2024.

7 An implementation of a PTP Relay Instance with IEEE 802.11 MAC services to physical ports should:

8 e) Support FINETIMINGMSMT as specified in IEEE Std 802.11-2024.

9 NOTE—In order to maintain backward compatibility with existing TM-based PTP End Instances that support only
10 Timing Measurement (TM), the PTP Relay Instance is required to support TM. PTP End Instances are allowed to
11 support TM or Fine Timing Measurement (FTM), or both to permit PTP End Instances compliant with this standard to
12 implement only the FTM standard, which requires a PTP Relay Instance that supports FTM.

13 **5.7 MAC-specific timing and synchronization methods for IEEE 802.3 EPON**

14 An implementation of a time-aware system with IEEE 802.3 Ethernet Passive Optical Network (EPON)
15 MAC services to physical ports shall:

16 a) Support the requirements as specified in IEEE Std 802.3-2022 for multipoint MAC Control (64.2
17 and 64.3) and multipoint physical coding sublayer (PCS) and physical medium attachment (PMA)
18 extensions (Clause 65).

19 b) Support the requirements as specified in Clause 13.

20 **5.8 MAC-specific timing and synchronization methods for coordinated shared** 21 **network (CSN)**

22 An implementation of a time-aware system with CSN MAC services to physical ports shall:

23 a) Support the requirements as specified in Clause 16.

24 b) Support at least one MoCA port (16.6.2) or ITU-T G.hn port (16.6.3).

1 6. Conventions

2 6.1 General

3 This clause defines various conventions and notation used in the standard, i.e., naming conventions, service
4 specification method and notation, and data type definitions.

5 6.2 Service specification method and notation

6 The method and notation for specifying service interfaces is described in Clause 7 of IEEE Std 802.1AC-
7 2016.

8 6.3 Lexical form syntax

9 A lexical form refers to the following:

- 10 — A name
- 11 — A data type

12 The conventions illustrated in the following list regarding lexical forms are used in this standard:

- 13 a) Type names: e.g., ClockQuality (no word separation, initial letter of each word capitalized).
- 14 b) Enumeration members and global constants: e.g., ATOMIC_CLOCK (word separation underscored,
15 all letters capitalized).
- 16 c) Fields within PTP messages, instances of structures, and variables: e.g., secondsField, clockQuality,
17 clockIdentity (two-word field names at a minimum, no word separation, initial word not capitalized,
18 initial letter capitalization on subsequent words).
- 19 d) Members of a structure: e.g., clockQuality.clockClass (no word separation, structure name followed
20 by a period followed by the member name).
- 21 e) Data set names: e.g., defaultDS, parentDS, portDS, currentDS, timePropertiesDS (no word
22 separation, initial word not capitalized, initial letter capitalization on subsequent words, end marked
23 by the letters DS).
- 24 f) Data set members: e.g., defaultDS.clockQuality.clockClass (data set name followed by a period
25 followed by a member name followed by a period followed by a member name).
- 26 g) PTP message names: e.g., Sync, Pdelay_Req (word separation underscored, initial letter of each
27 word capitalized).

28 When a lexical form appears in text, as opposed to in a type, or a format definition, the form is to be
29 interpreted as singular, plural, or possessive as appropriate to the context of the text.

30 6.4 Data types and on-the-wire formats

31 6.4.1 General

32 The data types specified for the various variables and message fields define logical properties that are
33 necessary for correct operation of the protocol or interpretation of IEEE 1588 precision time protocol (PTP)
34 or IEEE 802.11 message content.

35 NOTE—Implementations are free to use any internal representation of data types if the internal representation does not
36 change the semantics of any quantity visible via communications using the IEEE 802.1AS protocol or in the specified
37 operations of the protocol.

1 6.4.2 Primitive data types specifications

2 All non-primitive data types are derived from the primitive types in Table 6-1. Signed integers are
3 represented in twos complement form.

Table 6-1—Primitive data types

Data type	Definition
Boolean	TRUE or FALSE
EnumerationN	N-bit enumerated value
UIntegerN	N-bit unsigned integer
IntegerN	N-bit signed integer
Nibble	4-bit field not interpreted as a number
Octet	8-bit field not interpreted as a number
OctetN	N-octet field not interpreted as a number, with $N > 1$
Float64 (see NOTE)	IEEE 754™ binary64 (64-bit double-precision floating-point format)
NOTE—The Float64 data type was called Double in the 2011 edition of this standard. The semantics of the data type has not changed.	

4 6.4.3 Derived data type specifications

5 6.4.3.1 ScaledNs

6 The ScaledNs type represents signed values of time and time interval in units of 2^{-16} ns.

7 typedef Integer96 ScaledNs;

8 For example: -2.5 ns is expressed as:

9 0xFFFF FFFF FFFF FFFF FFFD 8000

10 Positive or negative values of time or time interval outside the maximum range of this data type are encoded
11 as the largest positive or negative value of the data type, respectively.

12 6.4.3.2 UScaledNs

13 The UScaledNs type represents unsigned values of time and time interval in units of 2^{-16} ns.

14 typedef UInteger96 UScaledNs;

15 For example: 2.5 ns is expressed as:

16 0x0000 0000 0000 0000 0002 8000

17 Values of time or time interval greater than the maximum value of this data type are encoded as the largest
18 positive value of the data type.

1 6.4.3.3 TimeInterval

2 The TimeInterval type represents time intervals, in units of 2^{-16} ns.

3 typedef Integer64 TimeInterval;

4 For example: 2.5 ns is expressed as:

5 0x0000 0000 0002 8000

6 Positive or negative time intervals outside the maximum range of this data type are encoded as the largest
7 positive and negative values of the data type, respectively.

8 6.4.3.4 Timestamp

9 The Timestamp type represents a positive time with respect to the epoch.

```
10 struct Timestamp
11 {
12     UInteger48 seconds;
13     UInteger32 nanoseconds;
14 };
```

15 The seconds member is the integer portion of the timestamp in units of seconds.

16 The nanoseconds member is the fractional portion of the timestamp in units of nanoseconds.

17 The nanoseconds member is always less than 10^9 .

18 For example:

19 +2.000000001 seconds is represented by seconds = 0x0000 0000 0002 and nanoseconds= 0x0000 0001

20 6.4.3.5 ExtendedTimestamp

21 The ExtendedTimestamp type represents a positive time with respect to the epoch.

```
22 struct ExtendedTimestamp
23 {
24     UInteger48 seconds;
25     UInteger48 fractionalNanoseconds;
26 };
```

27 The seconds member is the integer portion of the timestamp in units of seconds.

28 The fractionalNanoseconds member is the fractional portion of the timestamp in units of 2^{-16} ns.

29 The fractionalNanoseconds member is always less than $(2^{16})(10^9)$.

30 For example:

31 +2.000000001 seconds is represented by seconds = 0x0000 0000 0002 and fractionalNanoseconds =
32 0x0000 0001 0000

1 6.4.3.6 ClockIdentity

2 The ClockIdentity type identifies a PTP Instance.

3 typedef Octet8 ClockIdentity;

4 6.4.3.7 PortIdentity

5 The PortIdentity type identifies a port of a PTP Instance.

```
6 struct PortIdentity
7 {
8     ClockIdentity clockIdentity;
9     UInteger16 portNumber;
10 };
```

11 6.4.3.8 ClockQuality

12 The ClockQuality represents the quality of a clock.

```
13 struct ClockQuality
14 {
15     UInteger8 clockClass;
16     Enumeration8 clockAccuracy;
17     UInteger16 offsetScaledLogVariance;
18 };
```

19 6.4.4 Protocol data unit (PDU) formats

20 6.4.4.1 General

21 The data types defined in 6.4.2 and 6.4.3 shall be mapped onto the wire according to the mapping rules for
22 the respective medium, e.g., IEEE Std 802.3-2022 and IEEE Std 802.11-2024, and the terms of 6.4.4.

23 IEEE 802.1AS PDUs consist of the messages defined or referenced in Clause 10, Clause 11, Clause 12, and
24 Clause 13, based on the data types defined in 6.4.2 and 6.4.3. The internal ordering of the fields of the
25 IEEE 802.1AS PDUs is specified in 6.4.4.3 to 6.4.4.5.

26 6.4.4.2 Numbering of bits within an octet

27 Bits are numbered with the most significant bit being 7 and the least significant bit being 0.

28 NOTE—The numbering and ordering of bits within an octet of a PDU, described here, is independent of and unrelated to
29 the order of transmission of the bits on the underlying physical layer.

30 6.4.4.3 Primitive data types

31 Numeric primitive data types defined in 6.4.2 shall be formatted with the most significant octet nearest to the
32 beginning of the PDU followed in order by octets of decreasing significance.

33 The Boolean data type TRUE shall be formatted as a single bit equal to 1 and FALSE as a single bit equal
34 to 0.

1 Enumerations of whatever length shall be formatted as though the assigned values are unsigned integers of
2 the same length, e.g., Enumeration16 shall be formatted as though the value had type UInteger16.

3 **6.4.4.4 Arrays of primitive types**

4 All arrays shall be formatted with the member having the lowest numerical index nearest to the beginning of
5 the PDU followed by successively higher numbered members, without any padding. In octet arrays, the
6 octet with the lowest numerical index is termed the most significant octet.

7 When a field containing more than one octet is used to represent a numeric value, the most significant octet
8 shall be nearest to the beginning of the PDU, followed by successively less significant octets.

9 When a single octet contains multiple fields of primitive data types, the bit positions within the octet of each
10 of the primitive types as defined in the message field specification shall be preserved. For example, the first
11 field of the header of PTP messages is a single octet composed of two fields, one of type Nibble bit 4
12 through bit 7, and one of type Enumeration4 bit 0 through bit 3 (see 11.4.2 and 10.6.2).

13 **6.4.4.5 Derived data types**

14 Derived data types defined as structs shall be formatted with the first member of the struct nearest to the
15 beginning of the PDU followed by each succeeding member, without any padding. Each member shall be
16 formatted according to its data type.

17 Derived data types defined as typedefs shall be formatted according to its referenced data type.

17. Time-synchronization model for a packet network

7.1 General

This clause provides a model for understanding the operation of the generalized precision time protocol (gPTP), which specifies the operation of time-aware systems on a packet network. Although this standard is based on the precision time protocol (PTP) described in IEEE Std 1588-2019 (and, indeed, is a proper profile of IEEE Std 1588-2019 in particular configurations), there are differences, which are summarized in 7.6.

Although this standard has been written as a stand-alone document, it is useful to understand the IEEE 1588 architecture as described in Clause 6 of IEEE Std 1588-2019.

7.2 Architecture of a time-aware network

7.2.1 General

A time-aware network consists of a number of interconnected time-aware systems that support the gPTP defined within this standard. These time-aware systems can be any networking device, including, for example, bridges, routers, and end stations. A set of time-aware systems that are interconnected by gPTP-capable network elements is called a *gPTP network*. Each instance of gPTP that the time-aware systems support is in one *gPTP domain*, and the instances of gPTP are said to be part of that gPTP domain. A time-aware system can support, and therefore be part of, more than one gPTP domain. The entity of a single time-aware system that executes gPTP in one gPTP domain is called a *PTP Instance*. A time-aware system can contain multiple PTP Instances, which are each associated with a different gPTP domain. There are two types of PTP Instances:

- a) PTP End Instance, which, if not a Grandmaster PTP Instance, is a recipient of time information, and
- b) PTP Relay Instance, which, if not a Grandmaster PTP Instance, receives time information from the Grandmaster PTP Instance (perhaps indirectly through other PTP Relay Instances), applies corrections to compensate for delays in the local area network (LAN) and the PTP Relay Instance itself, and retransmits the corrected information.

This standard specifies gPTP in terms of media-independent and media-dependent elements for different media. The supported media are as follows:

- c) IEEE 802.3 Ethernet using full-duplex point-to-point links as specified in Clause 11.
- d) IEEE 802.3 EPON links as specified in Clause 13.
- e) IEEE 802.11 wireless as specified in Clause 12.
- f) Generic coordinated shared networks (CSNs, e.g., MoCA and ITU-T G.hn) as specified in Clause 16.

This standard specifies time distribution mechanisms that are tolerant to some faults if the network paths are redundant. One of these time distribution mechanisms updates the distribution path in reaction to changes (see 10.3.1.2), whereas the other one relies on redundant PTP instances on top of the redundant network paths (see Clause 18).

This standard specifies multiple mechanisms for providing fault tolerant synchronized time:

- a) The best timeTransmitter clock algorithm (BTCA) that allows dynamic reconfiguration of the synchronization tree of a gPTP domain in case of failure, see 10.3.1.2.

- 1 b) Hot standby (see Clause 18) that provides fault tolerance for GM and path failures.
- 2 c) The use of multiple gptp domains that allows the use of multiple timescales. It also allows
- 3 independent instances of the BTCA providing an independent synchronization tree for each domain.

4 If a time-aware network implements hot standby, the BTCA algorithm is disabled for the domains providing
5 hot standby, and the synchronization tree for these domains is statically configured by network management.

6 7.2.4 gPTP domains

7 gPTP requires that all systems in the gPTP domain be time-aware systems, i.e., the protocol does not transfer
8 timing over systems that are not time-aware. A time-aware system uses media-dependent mechanisms to
9 distinguish between time-aware and non-time-aware systems, as specified in the media-dependent clauses of
10 this standard.

11 NOTE—A synchronization tree is calculated by the BTCA or statically configured to achieve the best synchronization
12 result in a network. A forwarding spanning tree such as IEEE 802.1Q RSTP can differ because it serves different
13 purposes.

14 7.3 Examples of time-aware networks

15 7.3.1 Time-aware network consisting of a single gPTP domain

16 Figure 7-1 illustrates an example time-aware network consisting of a single gPTP domain, using all the
17 above network technologies [i.e., item c) through item f) of 7.2.1], where end stations on several local

1 networks are connected to a Grandmaster PTP Instance on a backbone network via an EPON access
2 network.

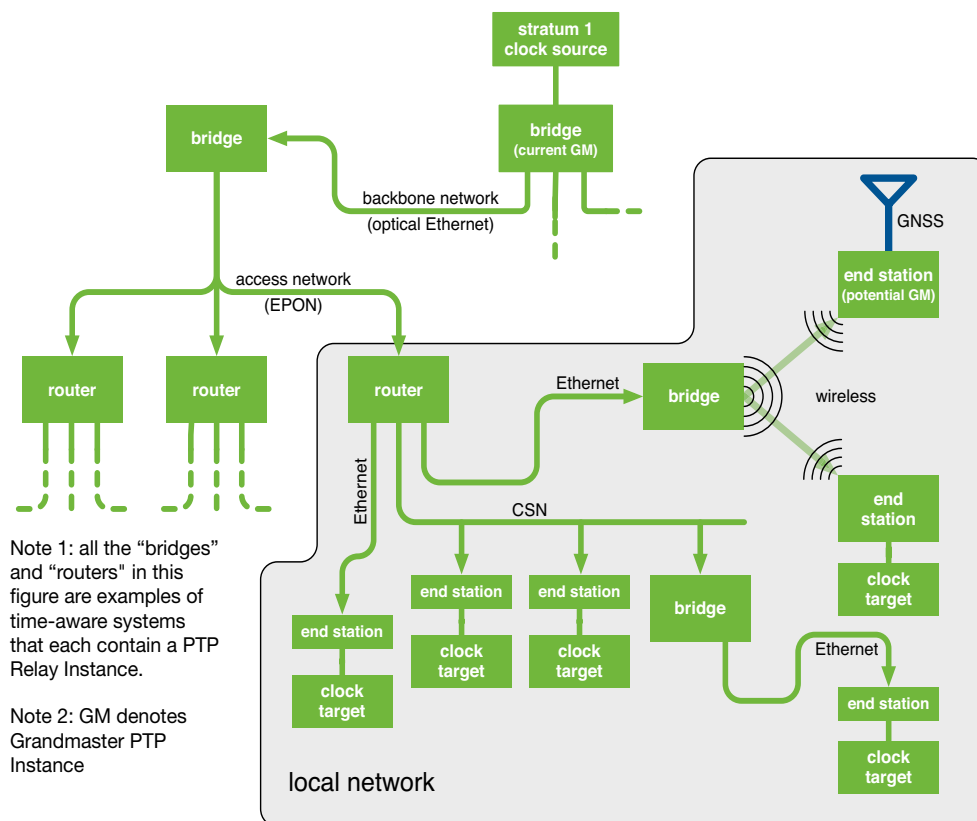


Figure 7-1—Time-aware network example

3 Any PTP Instance with clock sourcing capabilities can be a potential Grandmaster PTP Instance, and a
4 selection method (the *best timeTransmitter clock algorithm*, or BTCA) ensures that all of the PTP Instances
5 in the gPTP domain use the same Grandmaster PTP Instance.¹⁴ The BTCA is largely identical to that used in
6 IEEE Std 1588-2019, but somewhat simplified. In Figure 7-1 the BTCA process has resulted in the
7 Grandmaster PTP Instance being on the network backbone. If, however, the access network fails, the
8 systems on a local network automatically switch over to one of the potential Grandmaster PTP Instances on
9 the local network that is as least as “good” as any other. For example, in Figure 7-2, the access network link
10 has failed, and a potential Grandmaster PTP Instance that has a GNSS reference source has become the
11 active Grandmaster PTP Instance. As a result, now two gPTP domains exist where there used to be one.

12

¹⁴ There are, however, short periods during network reconfiguration when more than one Grandmaster PTP Instance might be active while the BTCA process is taking place.

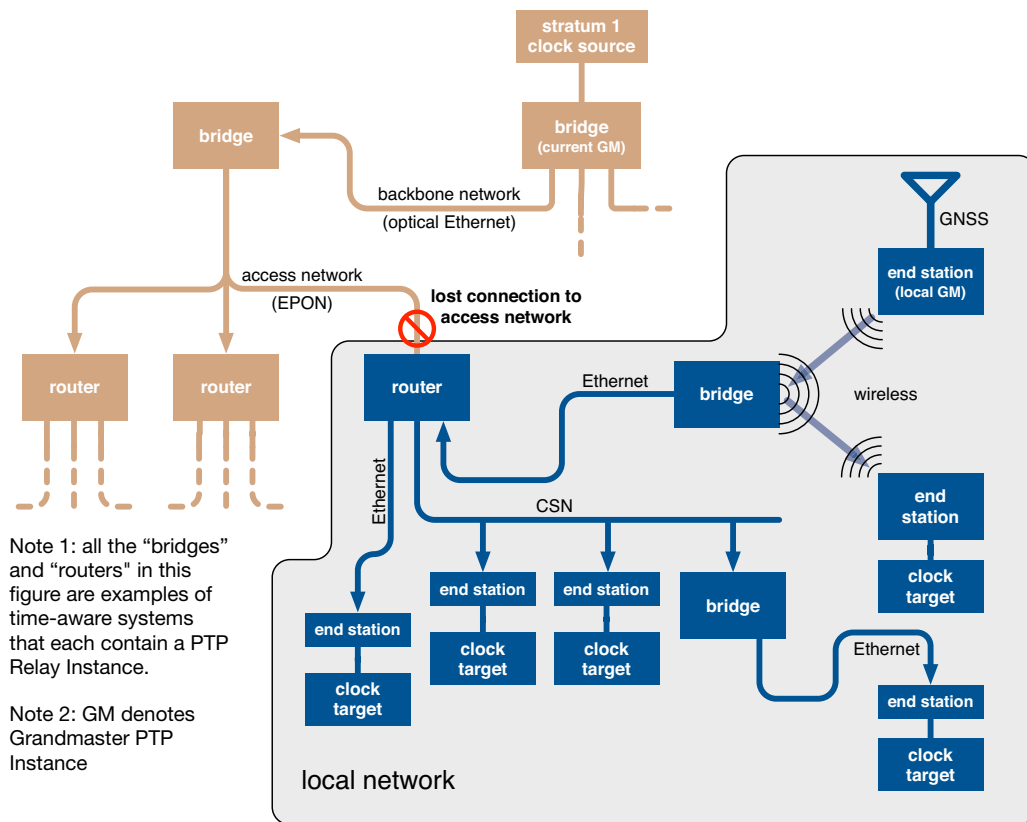
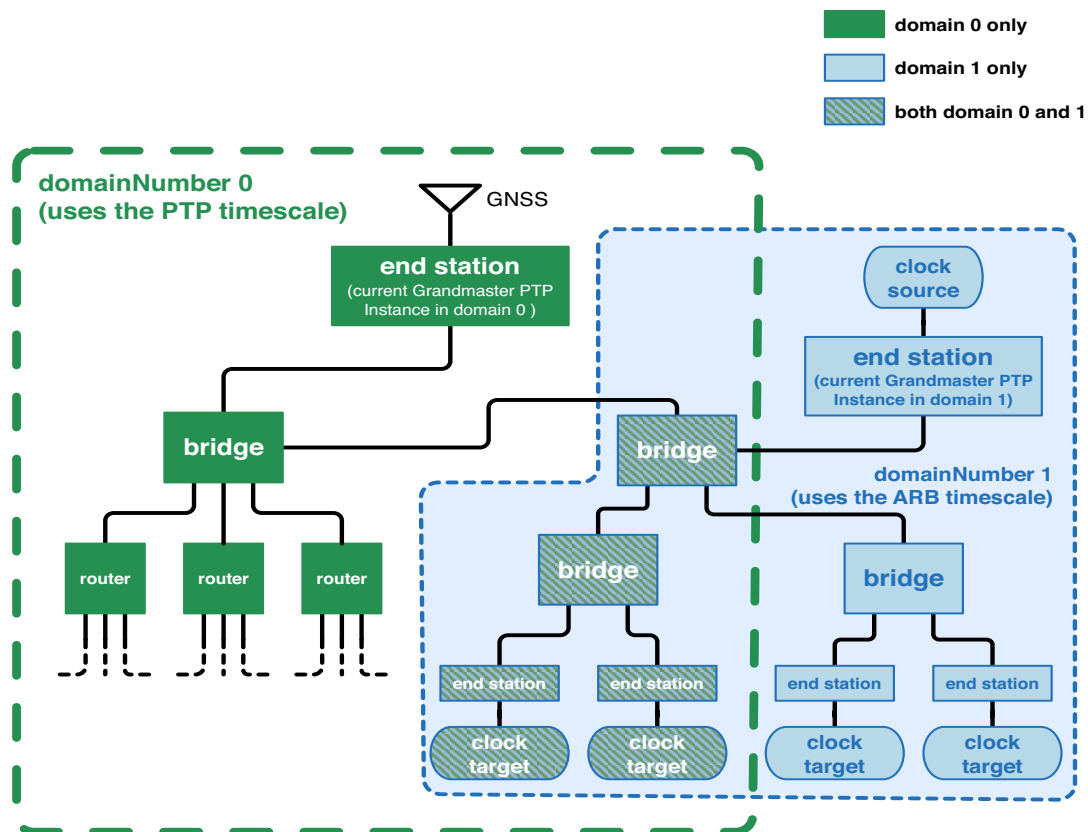


Figure 7-2—Time-aware network of Figure 7-1 after an access network link failure

Figure 7-3 illustrates an example time-aware network consisting of multiple gPTP domains that could be used in an industrial application. Specifically, in this example the network has two timescales/domains, where domain 0 uses the PTP timescale and domain 1 uses the arbitrary (ARB) timescale (see 8.2). Notice that not all PTP Instances in domain 1 (within the blue shorter-dashed area) have domain 0 active in this example, even though every time-aware system supports domain 0 for backward compatibility with the 2011 edition of this standard. In addition, it is required that all PTP Instances belonging to the same domain have direct connections among them in their physical topology (e.g., time cannot be transported from one PTP Instance in domain 0 to another PTP Instance in domain 0 via a time-aware system that does not have domain 0 active). In addition, the time-aware systems for which both domains are active are depicted by slanted internal hatching, representing two independent, active PTP Instances.

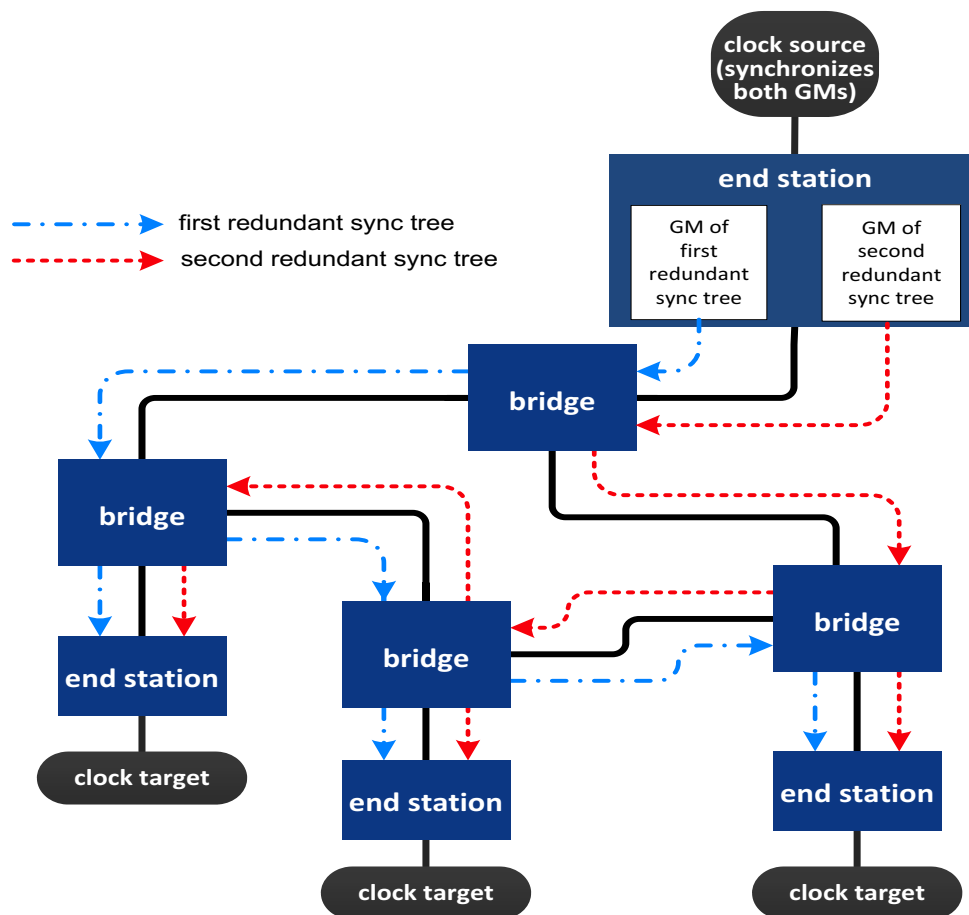


NOTE —All the “bridges” and “routers” in this figure are examples of time-aware systems that contain at least one PTP Relay Instance, and the end stations are time-aware systems that contain at least one PTP End Instance.

Figure 7-3—Time-aware network example for multiple gPTP domains

17.3.3 Time-aware networks using BTCA

2 Figure 7-4 shows an example network realizing two redundant synchronization trees from a single GM, each
3 with its own GM. The two GMs receive timing from the single clock source.



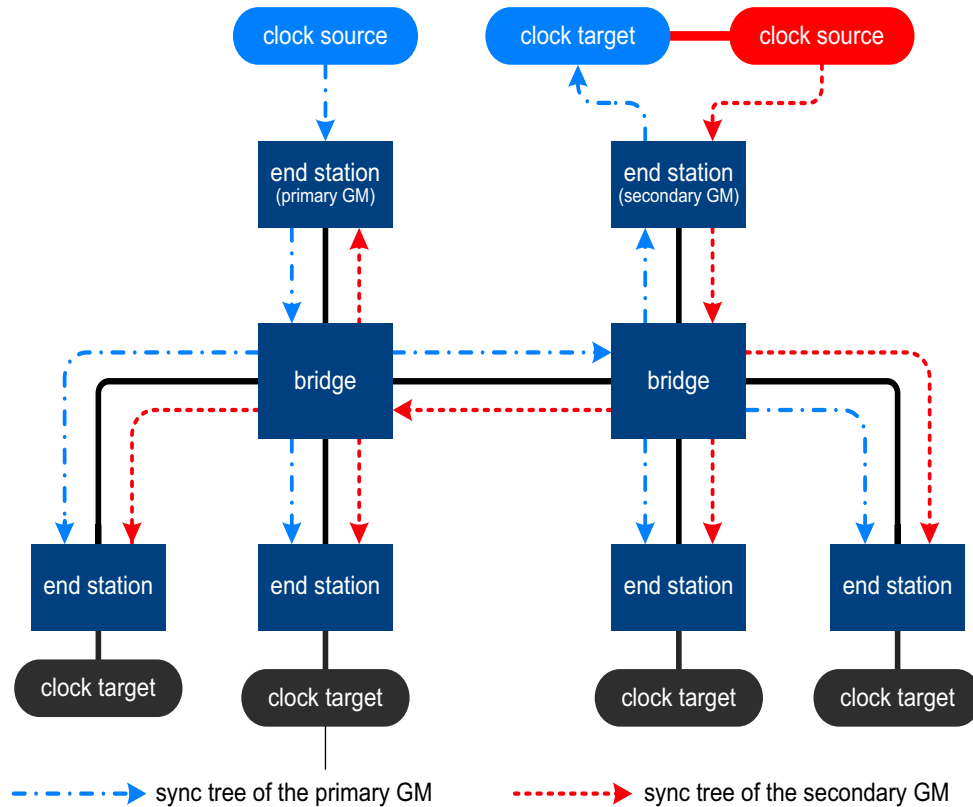
NOTE 1—The methods used for merging the redundant Sync messages received at each end station are not specified in this standard.

NOTE 2—All the “bridges” in this figure are examples of time-aware systems that contain PTP Relay Instances, and the end stations are examples of time-aware systems that contain PTP End Instances.

NOTE 3—GM denotes Grandmaster PTP Instance

Figure 7-4—Time-aware network example for synchronization path redundancy, with one clock source providing time to two domains

Figure 7-5 shows an example network with two redundant GMs, one as primary GM and the other as secondary GM, where each GM has one of the two redundant synchronization trees originating from it. This example supports hot-standby operating mode. In this mode, the secondary GM has to be synchronized to the primary GM, because it is part of the synchronization tree of the primary GM as shown in the figure. The secondary GM does not start sending Sync messages until it is synchronized to the primary GM.



NOTE 1—The methods used for merging the redundant Sync messages received at each end station are not specified in this standard.
NOTE 2—All the “bridges” in this figure are examples of time-aware systems that contain PTP Relay Instances, and the end stations are examples of time-aware systems that contain PTP End Instances.
NOTE 3—GM denotes Grandmaster PTP Instance.

Figure 7-5—Time-aware network example for GM redundancy with one primary GM and one hot-standby GM, which are separated in two gPTP domains

6 7.3.4 Time-aware network with hot standby

Figure 7-6 shows an example of a time-aware network with hot standby (see Clause 18). The network has 8 GM redundancy and path redundancy to all of the Bridges and end stations 1, 4, and 6 (path redundancy for 9 end stations 2, 3, and 5 is possible, but is not illustrated to avoid clutter in the figure). In addition, for 10 simplicity the optional split functionality (see 18.6.4) is assumed to be disabled.

End station 1 is the primary GM (see Clause 18), and its synchronization tree is illustrated by the arrows of 12 the primary grandmaster synchronization tree. End station 4 is the secondary GM, and its synchronization 13 tree is illustrated by the arrows of the secondary grandmaster synchronization tree. Except for the links to 14 end stations 2, 3, and 5, the network can tolerate the failure of any single link and the Bridges and end

1 stations remain synchronized by either the primary or the secondary GM. For example, if the link between
 2 bridge 3 and end station 6 fails, end station 6 is still synchronized by the primary domain via bridge 2. If the
 3 link between bridge 5 and bridge 6 fails, bridge 6 is still synchronized by the primary domain and bridge 5 is
 4 synchronized by the secondary domain. If the secondary GM fails, all the Bridges and end stations are still
 5 synchronized by the primary GM.

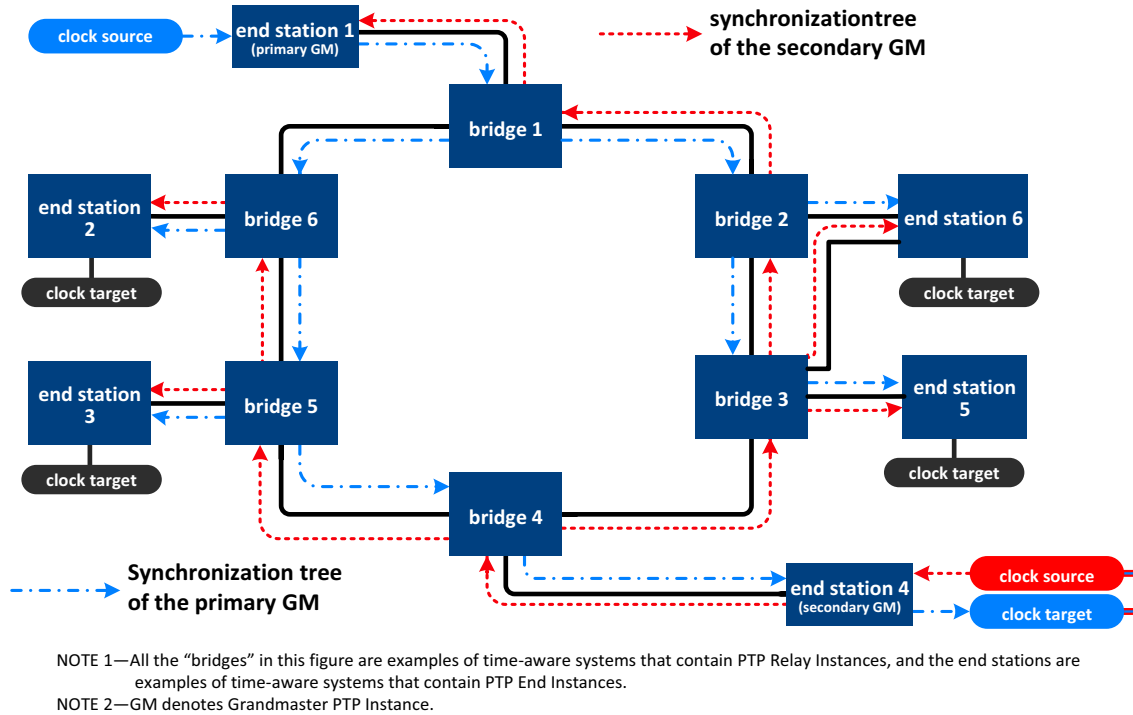


Figure 7-6—Time-aware network example for hot standby with both GM and partial path redundancy

6 If the primary GM fails, all the Bridges and end stations are still synchronized by the secondary GM. In this
 7 case, if syncLocked is TRUE for any node in the path from end station 1 to end station 4, the primary PTP
 8 Instance of end station 4 stops receiving Sync messages and isSynced changes to FALSE; the
 9 hotStandbySystemState of end station 4 is then changed to NOT_REDUNDANT (see Clause 18). If
 10 syncLocked is FALSE, and assuming the Bridges and end stations other than the GMs are not grandmaster
 11 capable, the time difference between the primary and secondary domain eventually exceeds the
 12 primarySecondaryOffsetThresh (see Clause 18) and the hotStandbySystemState of end station 4 changes to
 13 NOT_REDUNDANT. Irrespective of the syncLocked state, the secondary GM no longer takes timing from
 14 the primary domain.

15 If the link between bridge 4 and bridge 5 fails, all the PTP Instances except for bridge 4 and end station 4 are
 16 then synchronized by the primary domain. Bridge 4 and end station 4 are synchronized by the secondary
 17 domain; in addition, bridges 1, 2, and 3, and end stations 1, 4, 5, and 6 receive timing on the secondary
 18 domain. Since end station 4 does not receive timing on the primary domain, its hotStandbySystemState
 19 changes to NOT_REDUNDANT, and the secondary domain GM does not take timing from the primary
 20 domain.

17.4 Time synchronization

2 7.4.1 General

3 Time synchronization in gPTP is done the same way (in the abstract) as is done in IEEE Std 1588-2019: a
4 Grandmaster PTP Instance sends information including the current synchronized time to all directly attached
5 PTP Instances. Each of these PTP Instances must correct the received synchronized time by adding the
6 propagation time needed for the information to transit the gPTP communication path from the Grandmaster
7 PTP Instance. If the PTP Instance is a PTP Relay Instance, then it must forward the corrected time
8 information (including additional corrections for delays in the forwarding process) to all the other attached
9 PTP Instances.

10 To make this all work, there are two time intervals that must be precisely known: the forwarding delay
11 (called the *residence time*), and the time taken for the synchronized time information to transit the gPTP
12 communication path between two PTP Instances.

13 7.4.2 Delay measurement

14 Each type of LAN or gPTP communication path has different methods for measuring propagation time, but
15 they are all based on the same principle: measuring the time that a well-known part of a message is
16 transmitted from one device and the time that the same part of the same message is received by the other
17 device, then sending another message in the opposite direction and doing the same measurement as shown in
18 Figure 7-7.

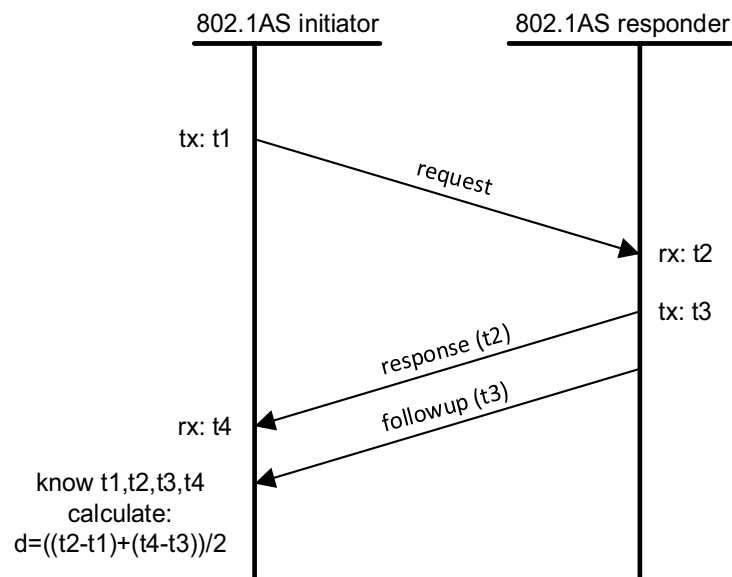


Figure 7-7—Conceptual medium delay measurement

19 This basic mechanism is used as the foundation for media-dependent measurement mechanisms specified in
20 the media dependent clauses of this standard (see 7.2.1).

21 7.4.3 Logical syntonization

22 The time-synchronization correction previously described is dependent on the accuracy of the delay and
23 residence time measurements. If the clock used for this purpose is frequency locked (syntonized) to the
24 Grandmaster Clock, then all the time interval measurements use the same time base. Since actually adjusting

1 the frequency of an oscillator (e.g., using a phase-lock loop) is slow and prone to gain peaking effects, PTP
2 Relay Instances can correct time interval measurements using the Grandmaster Clock frequency ratio.

3 Each PTP Instance measures, at each PTP Port, the ratio of the frequency of the PTP Instance at the other
4 end of the link attached to that PTP Port to the frequency of its own clock. The cumulative ratio of the
5 Grandmaster Clock frequency to the local clock frequency is accumulated in a standard organizational type,
6 length, value (TLV) attached to the Follow_Up message (or the Sync message if the optional one-step
7 processing is enabled). The frequency ratio of the Grandmaster Clock relative to the local clock is used in
8 computing synchronized time, and the frequency ratio of the neighbor relative to the local clock is used
9 in correcting the propagation time measurement (NRR is not shown in figure 7-7).

10 The Grandmaster Clock frequency ratio is measured by accumulating neighbor frequency ratios for two
11 main reasons. First, if there is a network reconfiguration and a new Grandmaster PTP Instance is elected, the
12 nearest neighbor frequency ratios do not have to be newly measured as they are constantly measured using
13 the Pdelay messages. This results in the frequency offset relative to the new Grandmaster Clock being
14 known when the first Follow_Up message (or first Sync message if the optional one-step processing is
15 enabled) is received, which reduces the duration of any transient error in synchronized time during the
16 reconfiguration. This is beneficial to many high-end audio applications. Second, there are no gain peaking
17 effects because an error in frequency offset at one PTP Relay Instance, and resulting residence time error,
18 does not directly affect the frequency offset at a downstream PTP Relay Instance.

19 **7.4.4 Energy efficiency**

20 Sending PTP messages at relatively high rates when there is otherwise little or no traffic conflicts with the
21 goal of reducing energy consumption. This standard specifies a way to request that a neighbor PTP Port
22 reduce the rate of sending Sync (and Follow_Up if optional one-step processing is not enabled), peer delay,
23 and Announce messages and also to inform the neighbor not to compute neighbor rate ratio and/or
24 propagation delay on this link. A time-aware system could do this when it enters low-power mode, but this
25 standard does not specify the conditions under which this is done; it specifies only the actions a time-aware
26 system takes.

17.5 PTP Instance architecture

2 The model of a PTP Instance and its interfaces to higher-layer applications are shown in Figure 7-8.

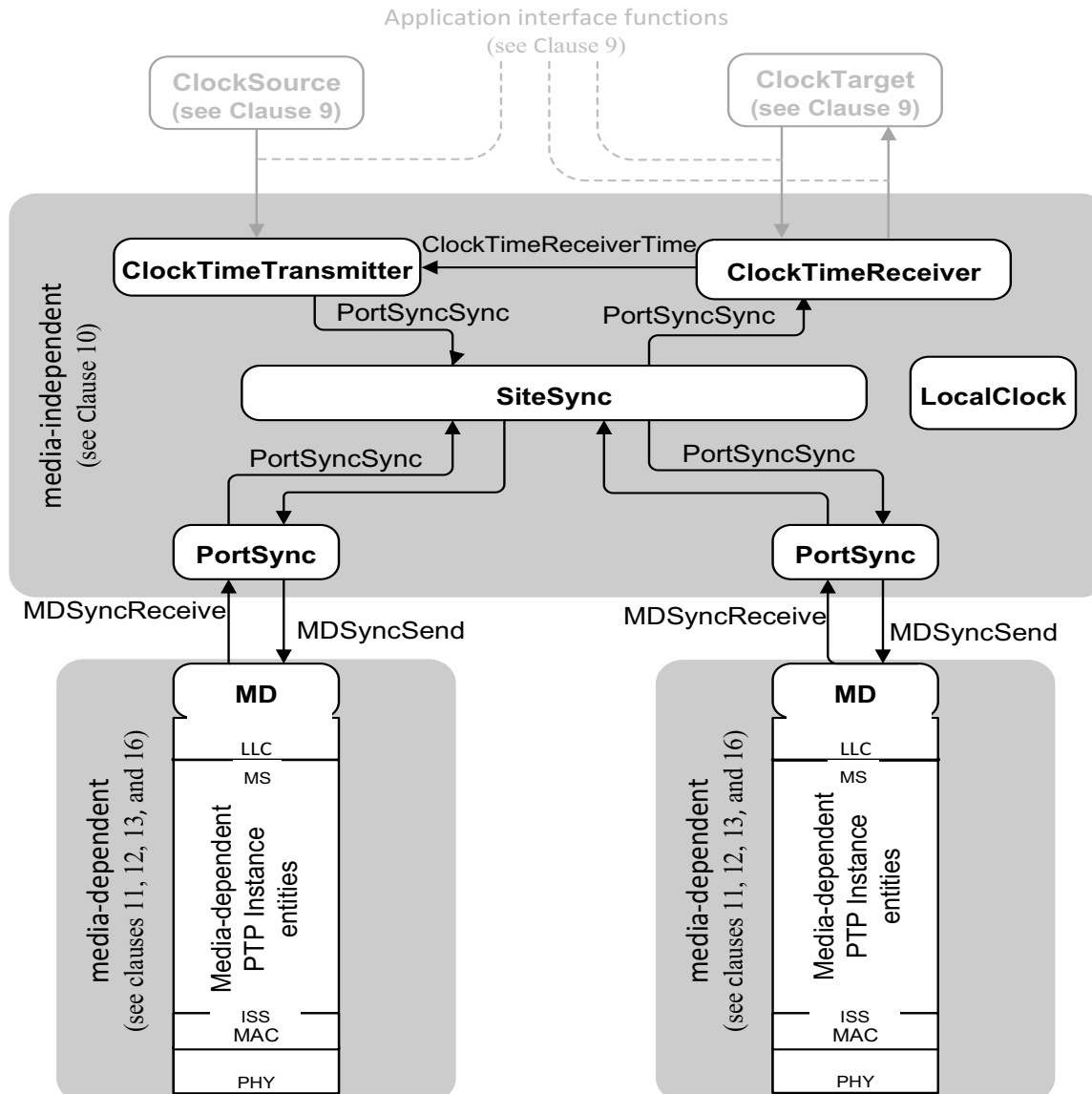


Figure 7-8—Model for a PTP Instance and its interfaces to higher-layer applications

3 A PTP Instance consists of the following major parts:

- 4 a) If the PTP Instance includes application(s) that either use or source time information, then they
- 5 interface with the gPTP information using the application interfaces specified in Clause 9.
- 6 b) A single media-independent part that consists of ClockTimeTransmitter, ClockTimeReceiver, and
- 7 SiteSync logical entities, one or more PortSync entities, and a LocalClock entity. The BTCA and
- 8 forwarding of time information between logical ports and the ClockTimeReceiver and
- 9 ClockTimeTransmitter is done by the SiteSync entity, while the computation of PTP Port-specific
- 10 delays needed for time-synchronization correction is done by the PortSync entities. The media-
- 11 independent layer can be extended by hot standby (see Clause 18).

- 1 c) Media-dependent ports, which translate the abstract “MDSyncSend” and “MDSyncReceive”
2 structures received from or sent to the media-independent layer and corresponding methods used for
3 the particular LAN attached to the port.

4 **7.6 Differences between gPTP (IEEE Std 802.1AS) and PTP (IEEE Std 1588-2019)**

- 5 a) gPTP assumes all communication between PTP Instances is done only using IEEE 802 MAC PDUs
6 and addressing, while IEEE Std 1588-2019 supports various layer 2 and layer 3-4 communication
7 methods.
- 8 b) gPTP specifies a media-independent sublayer that simplifies the integration within a single timing
9 domain of multiple different networking technologies with radically different media access
10 protocols. gPTP specifies a media-dependent sublayer for each medium. The information exchanged
11 between PTP Instances has been generalized to support different packet formats and management
12 schemes appropriate to the particular networking technology. IEEE Std 1588-2019, on the other
13 hand, has introduced a new architecture based on media-independent and media-dependent
14 sublayers (see 6.5.2, Figure 5, and Figure 6 of IEEE Std 1588-2019); however, this architecture is
15 optional. The architecture of IEEE Std 1588-2008 [B10], which is not based on media-independent
16 and media-dependent layers, has been retained for Internet Protocol (IP) version 4, IP version 6,
17 Ethernet LANs, and several industrial automation control protocols. The intent in IEEE Std 1588-
18 2019 is that the new architecture, based on media-independent and media-dependent layers, will be
19 used for IEEE 802.11 networks, IEEE 802.3 EPON, and CSN using the specifications of gPTP, and
20 that the architecture must be used for transports that define native timing mechanisms if those native
21 timing mechanisms are used.
- 22 c) In gPTP there are only two types of PTP Instances: PTP End Instances and PTP Relay Instances,
23 while IEEE Std 1588-2019 has Ordinary Clocks, Boundary Clocks, end-to-end Transparent Clocks,
24 and P2P Transparent Clocks. A PTP End Instance corresponds to an IEEE 1588 Ordinary Clock, and
25 a PTP Relay Instance is a type of IEEE 1588 Boundary Clock where its operation is very tightly
26 defined, so much so that a PTP Relay Instance with Ethernet ports can be shown to be
27 mathematically equivalent to a P2P Transparent Clock in terms of how synchronization is
28 performed, as shown in 11.1.3. In addition, a PTP Relay Instance can operate in a mode (i.e., the
29 mode where the variable syncLocked is TRUE; see 10.2.5.15) where the PTP Relay Instance is
30 equivalent to a P2P Transparent Clock in terms of when time-synchronization messages are sent. A
31 time-aware system measures link delay and residence time and communicates these in a correction
32 field. In summary, a PTP Relay Instance conforms to the specifications for a Boundary Clock in
33 IEEE Std 1588-2019, but a PTP Relay Instance does not conform to the complete specifications for
34 a P2P Transparent Clock in IEEE Std 1588-2019 because:
35 1) When syncLocked is FALSE, the PTP Relay Instance sends Sync according to the
36 specifications for a Boundary Clock, and
37 2) The PTP Relay Instance invokes the BTCA and has PTP Port states.
- 38 d) PTP Instances communicate gPTP information only directly with other PTP Instances. That is, a
39 gPTP domain consists ONLY of PTP Instances. Non-PTP Relay Instances cannot be used to relay
40 gPTP information. In IEEE Std 1588-2019, it is possible to use non-IEEE-1588-aware relays in an
41 IEEE 1588 domain, although this slows timing convergence and introduce extra jitter and wander
42 that must be filtered by any IEEE 1588 clock.
- 43 e) For full-duplex Ethernet links, gPTP requires the use of the peer-to-peer delay mechanism, while
44 IEEE Std 1588-2019 also allows the use of end-to-end delay measurement.
- 45 f) For full-duplex Ethernet links, gPTP requires the use of two-step processing (use of Follow_Up and
46 Pdelay_Resp_Follow_Up messages to communicate timestamps), with an optional one-step
47 processing mode that embeds timestamps in the Sync “on the fly” as they are being transmitted
48 (gPTP does not specify one-step processing for peer delay messages). IEEE Std 1588-2019 allows

- 1 either two-step or one-step processing to be required (for both Sync and peer delay messages)
2 depending on a specific profile.
- 3 g) All PTP Instances in a gPTP domain are logically syntonized; in other words, they all measure time
4 intervals using the same frequency. This is done by the process described in 7.4.3 and is mandatory.
5 Syntonization in IEEE Std 1588-2019 is optional. The syntonization method used by gPTP is
6 supported as an option in IEEE Std 1588-2019, but uses a TLV standardized as part of
7 IEEE Std 1588-2019 (this feature is new for IEEE Std 1588-2019), while gPTP uses the
8 ORGANIZATION_EXTENSION TLV specified in 11.4.4.3.
- 9 h) Finally, this standard includes formal interface definitions, including primitives, for the time-aware
10 applications (see Clause 9). IEEE Std 1588-2019 describes external interfaces without describing
11 specific interface primitives.

12

1 8. IEEE 802.1AS concepts

2 8.1 gPTP domain

3 A gPTP domain consists of one or more PTP Instances and links that meet the requirements of this standard
4 and communicate with each other as defined by the IEEE 802.1AS protocol. A gPTP domain defines the
5 scope of gPTP message communication, state, operations, data sets, and timescale.

6 A gPTP domain is identified by two attributes: domainNumber and sdoId. The sdoId of a gPTP domain is a
7 12-bit unsigned integer. The sdoId is structured as a two-part attribute as follows:

- 8 — The most significant 4 bits are named the majorSdoId, and
- 9 — The least significant 8 bits are named the minorSdoId.

10 A time-aware system shall support one or more gPTP domains, each with a distinct domainNumber in the
11 range 0 through 127. Unless otherwise specified in this standard, the operation of gPTP and the timescale in
12 any given gPTP domain is independent of operation in any other domain.

13 The value of majorSdoId for a gPTP domain shall be 0x1. The value of minorSdoId for a gPTP domain shall
14 be 0x00 (see 16.5 of IEEE Std 1588-2019).

15 Since the IEEE 802.1 Working Group has only one single unique sdoId value, the PTP profile specified in
16 the present standard is isolated from other PTP profiles (see 16.5.2 of IEEE Std 1588-2019).

17 Both the domainNumber and the sdoId are carried in the common header of all PTP messages (see 10.6.2.2).

18 NOTE 1—The above requirements for majorSdoId and minorSdoId are for gPTP domains. The requirements for the
19 CMLDS are given in 11.2.17.

20 NOTE 2—In the 2011 edition of this standard, the attribute majorSdoId was named transportSpecific, and its value was
21 specified as 0x1 in 10.5.2.2.1 of Corrigendum 1. The attribute minorSdoId did not exist in the 2011 edition, but its
22 location in the common header was a reserved field, which was specified to be transmitted as 0 and ignored on receipt.

23 Unless otherwise stated, information in the remainder of this document is per gPTP domain.

24 NOTE 3—In steady state, all PTP Instances in a gPTP domain are traceable to a single Grandmaster PTP Instance.

25 8.2 Timescale

26 8.2.1 Introduction

27 The timescale for a gPTP domain is established by the Grandmaster Clock. There are two types of
28 timescales supported by gPTP:

- 29 — The timescale PTP: The epoch is the PTP epoch (see 8.2.2), and the timescale is continuous. The
30 unit of measure of time is the second defined by International Atomic Time (TAI) (for the definition
31 of TAI, see Service de la Rotation Terrestre [B28], with further amplification in IAU [B27]; and for
32 more information on TAI, see Jekeli [B22], international system of units (SI) brochure [B14], and
33 Petit and Luzum [B26]). The timescale of domain 0 shall be PTP. See IEEE Std 1588-2019 for
34 more details.
- 35 — The timescale ARB (arbitrary): The epoch is the domain startup time and can be set by an
36 administrative procedure. Between invocations of the administrative procedure, the timescale is
37 continuous. Additional invocations of the administrative procedure can introduce discontinuities in

1 the overall timescale. The unit of measure of time is determined by the Grandmaster Clock. The
2 second used in the operation of the protocol can differ from the SI second.

3 8.2.2 Epoch

4 The epoch is the origin of the timescale of a gPTP domain.

5 The PTP epoch (epoch of the timescale PTP) is 1 January 1970 00:00:00 TAI.

6 NOTE—The common portable operating system interface (POSIX) algorithms can be used for converting elapsed
7 seconds since the PTP epoch to the ISO 8601:2004 [B15] printed representation of time of day on the TAI timescale (see
8 also ISO/IEC 9945:2003 [B17]).

9 See Annex C for information on converting between common timescales.

10 8.2.3 UTC offset

11 When the timescale is PTP, it is possible to calculate Coordinated Universal Time (UTC) time using the
12 value of `currentUtcOffset`. The value of `currentUtcOffset` is given by

13 $\text{currentUtcOffset} = \text{TAI} - \text{UTC}$

14 where the difference $\text{TAI} - \text{UTC}$ is derived from $\text{UTC} - \text{TAI}$ (the negative of `currentUtcOffset`) specified in
15 IERS Bulletin C.

16 The value of `currentUtcOffset` for the current Grandmaster PTP Instance is maintained in the
17 `currentUtcOffset` member of the time properties data set (see 14.5.2).

18 NOTE 1—As of 0 hours 1 January 2017 UTC, UTC was behind TAI by 37 s, i.e., $\text{TAI} - \text{UTC} = +37$ s. At that moment,
19 the IEEE-802.1AS-defined value of `currentUtcOffset` was +37 s, as designated in the applicable IERS Bulletin C
20 (see Clause 2; see also Service de la Rotation Terrestre [B28] and U.S. Naval Observatory [B29]).

21 NOTE 2—Leap second events and the value of $\text{UTC} - \text{TAI}$ are posted well in advance in IERS Bulletin C. A list of all
22 leap second events is maintained by the U.S. Naval Observatory [B29], which also offers an extensive discussion of
23 timescales, leap seconds, and related time issues.

24 NOTE 3—The value of `currentUtcOffset` represents the difference between TAI and UTC. Since 1972, only integral
25 values are permitted for this difference. Due to leap seconds, UTC cannot be correctly represented as a single integer but
26 can be expressed in the ISO 8601:2004 [B15] print form. See also C.2 for an example of the use of the POSIX algorithm
27 to compute the correct print form.

28 When the timescale is PTP, it is possible to calculate local time from the time provided by a gPTP domain
29 using the `currentUtcOffset`, `leap59`, and `leap61` member values of the time properties data set and knowledge
30 of the local time zone and whether and when daylight savings time is observed.

31 When the timescale is ARB, the values of `currentUtcOffset`, `leap59`, and `leap61` cannot be used to compute
32 UTC.

33 The mechanism for computing UTC or any other time does not change the synchronized time (see 3.33), i.e.,
34 the PTP Instance time, of a PTP Instance.

35 8.2.4 Measurement of time within a gPTP domain

36 Time in a gPTP domain shall be measured as elapsed time since the epoch of the timescale of that domain.

18.3 Link asymmetry

This standard requires the measurement of the mean propagation time (also known as the *mean propagation delay*) between the time-aware systems that comprise the two endpoints of a link. The measurement is performed when one of the time-aware systems (the initiator time-aware system) sends a message to the other time-aware system (the responder time-aware system). The responder then sends a message back to the initiator at a later time. The departure of the message sent by the initiator time-aware system is timestamped, and the timestamp value is retained by that system. The arrival of this message at the responder time-aware system is timestamped, and the timestamp value is conveyed to the initiator time-aware system in a subsequent message. The departure of the response message sent by the responder time-aware system (in response to the message it receives from the initiator time-aware system) is timestamped, and the timestamp value is conveyed to the initiator time-aware system in a subsequent message. The arrival of this response message at the initiator time-aware system is timestamped, and the timestamp value is retained by that system. The mean propagation time is computed by the initiator time-aware system after receiving the response message, from the four timestamp values it has at this point.

Typically, the propagation time is not exactly the same in both directions, and the degree to which it differs in the two directions is characterized by the delay asymmetry. The relation between the individual propagation times in the two directions, the mean propagation time, and the delay asymmetry is as follows. Let t_{ir} be the propagation time from the initiator to the responder, t_{ri} be the propagation time from the responder to the initiator, meanLinkDelay be the mean propagation time (see 10.2.5.8), and delayAsymmetry be the delay asymmetry. The propagation times in the two directions are illustrated in Figure 8-1.

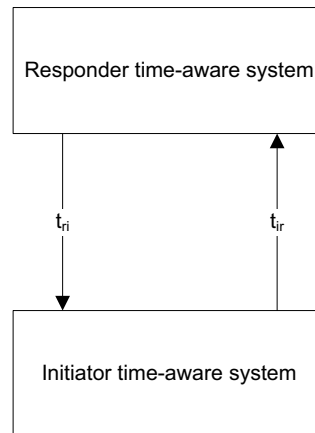


Figure 8-1—Propagation asymmetry

The meanLinkDelay is the mean value of t_{ir} and t_{ri} , i.e., $\text{meanLinkDelay} = (t_{ir} + t_{ri}) / 2$. The delayAsymmetry is defined as:

$$t_{ir} = \text{meanLinkDelay} - \text{delayAsymmetry}$$

$$t_{ri} = \text{meanLinkDelay} + \text{delayAsymmetry}$$

In other words, delayAsymmetry is defined to be positive when the responder to initiator propagation time is longer than the initiator to responder propagation time.

This standard does not explicitly require the measurement of delayAsymmetry ; however, if delayAsymmetry is modeled, it shall be modeled as specified in this clause.

1 NOTE 1—A time-aware system can change the value of delayAsymmetry during operation (see 14.10.11, 14.18.9, and
2 Annex G).

3 NOTE 2—A time-aware system PTP Port cannot measure the value of delayAsymmetry during live operation of the
4 system (i.e., asymmetryMeasurementMode is FALSE; see 10.2.5.2); therefore, the value of delayAsymmetry must be
5 defined separately using information from the supplier or additional testing before running the live system. The methods
6 for measuring asymmetry are not specified in this standard. These values can be added to the system configuration to
7 improve the accuracy of time synchronization. The inaccuracy caused by asymmetry is half the value of the difference
8 between t_{ri} and t_{ir} , and these inaccuracies can either accumulate over successive hops or, if the successive asymmetries
9 have different signs, cancel each other over the successive hops.

10 8.4 Messages

11 8.4.1 General

12 All communications occur via PTP messages and/or media-dependent messages.

13 8.4.2 Message attributes

14 8.4.2.1 General

15 All messages used in this standard have the following attributes:

- 16 a) Message class
- 17 b) Message type

18 The message class attribute is defined in this clause. The message type attribute is defined in 3.20. Some
19 messages have additional attributes; these are defined in the subclauses where the respective messages are
20 defined.

21 8.4.2.2 Message class

22 There are two message classes, the event message class and the general message class. Event messages are
23 timestamped on egress from a PTP Instance and ingress to a PTP Instance. General messages are not
24 timestamped. Every message is either an event message or a general message.

25 8.4.3 Generation of event message timestamps

26 All event messages are timestamped on egress and ingress. The timestamp shall be the time, relative to the
27 LocalClock entity (see 10.1) at which the message timestamp point passes the reference plane marking the
28 boundary between the PTP Instance and the network media.

29 The definition of the timestamp measurement plane (see 3.38), along with the corrections defined as
30 follows, allows transmission delays to be measured in such a way (at such a low layer) that they appear fixed
31 and symmetrical to gPTP even though the MAC client might otherwise observe substantial asymmetry and
32 transmission variation. For example, the timestamp measurement plane is located below any retransmission
33 and queuing performed by the MAC.

34 NOTE 1—If an implementation generates event message timestamps using a point other than the message timestamp
35 point, then the generated timestamps should be appropriately corrected by the time interval (fixed or otherwise) between
36 the actual time of detection and the time the message timestamp point passed the reference plane. Failure to make these
37 corrections results in a time offset between PTP Instances.

38 NOTE 2—In general, the timestamps can be generated at a timestamp measurement plane that is removed from the
39 reference plane. Furthermore, the timestamp measurement plane, and therefore the time offset of this plane from the

1 reference plane, is likely to be different for inbound and outbound event messages. To meet the requirement of this
2 clause, the generated timestamps should be corrected for these offsets. Figure 8-2 illustrates these offsets. Based on this
3 model the appropriate corrections are as follows:

$$4 \quad \text{egressTimestamp} = \text{egressMeasuredTimestamp} + \text{egressLatency}$$

$$5 \quad \text{ingressTimestamp} = \text{ingressMeasuredTimestamp} - \text{ingressLatency}$$

6 where the timestamps relative to the reference plane, egressTimestamp and ingressTimestamp, are computed from the
7 timestamps relative to the timestamp measurement plane, egressMeasuredTimestamp and ingressMeasuredTimestamp,
8 respectively, using their respective latencies, egressLatency and ingressLatency. Failure to make these corrections results
9 in a time offset between the timeReceiver and timeTransmitter clocks.

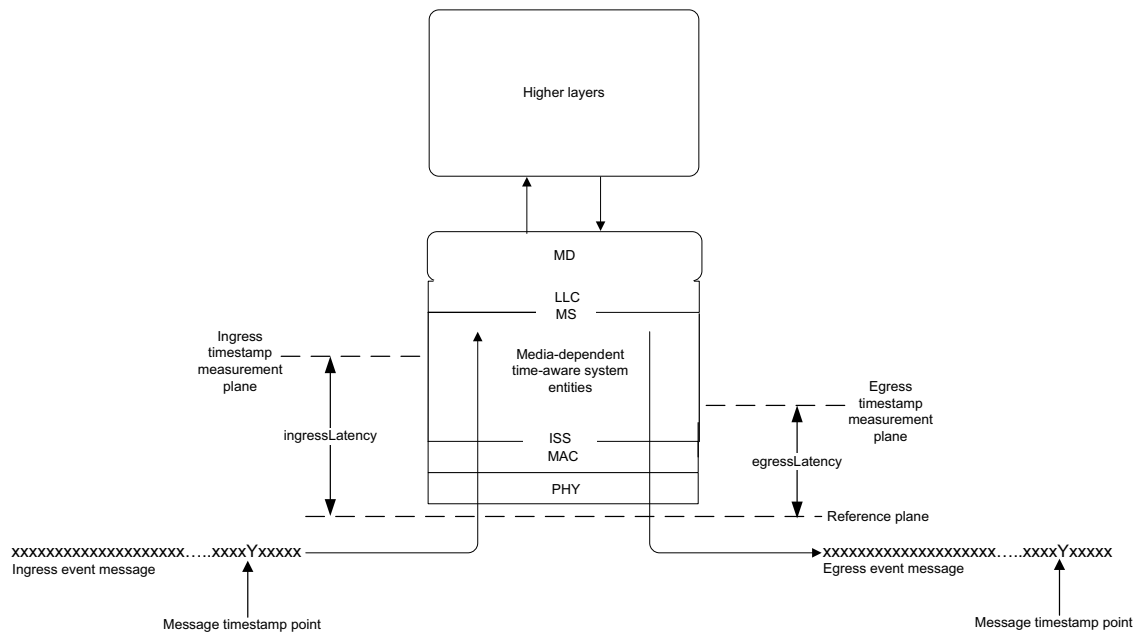


Figure 8-2—Definition of message timestamp point, reference plane, timestamp measurement plane, and latency constants

10 It is advised to use the ingress and egress timestamping measurement planes recommended by the specification for the
11 medium used. For example, for 802.3 networks, the recommendation is to follow Clause 90 of IEEE Std 802.3-2022, as
12 amended by IEEE Std 802.3cx-2023.

13 8.4.4 Priorities

14 IEEE Std 802.1AS messages shall be transmitted in expedited manner compared to other traffic (e.g., best
15 effort).

16 NOTE 1—If IEEE Std 802.1AS messages are not expedited in the internal queueing, long bursts of other traffic can
17 cause loss of synchronization due to timeouts.

18 NOTE 2—For example, two outbound queues can be supported: one outbound queue is used to transmit nonexpedited
19 traffic (e.g., best-effort), and another outbound queue is used to transmit frames in an expedited manner, including
20 IEEE 802.1AS messages. Outbound queues are often implemented in hardware, but software implementation is
21 possible.

22 NOTE 3—When IEEE Std 802.1Q-2022 is supported, the outbound queue used for transmitting IEEE 802.1AS
23 messages uses a traffic class (see 3.273 of IEEE Std 802.1Q-2022) greater than zero.

1 NOTE 4—Frames carrying IEEE 802.1AS messages are neither VLAN-tagged nor priority-tagged, i.e., they are
2 untagged (see 11.3.3).

3 8.5 Ports

4 8.5.1 General

5 The PTP Instances in a gPTP domain interface with the network media via physical ports. gPTP defines a
6 logical port, i.e., a PTP Port, in such a way that communication between PTP Instances is point-to-point
7 even over physical ports that are attached to shared media. One logical port, consisting of one PortSync
8 entity and one media-dependent (MD) entity, is instantiated for each PTP Instance with which the
9 PTP Instance communicates. For shared media, multiple logical ports can be associated with a single
10 physical port.

11 Unless otherwise qualified, each instance of the term *port* refers to a *logical port*.

12 8.5.2 Port identity

13 8.5.2.1 General

14 A PTP Port is identified by a port identity of type PortIdentity (see 6.4.3.7). The value is maintained in
15 portDS.portIdentity (see 14.10.2). A port identity consists of the following two attributes:

- 16 a) portIdentity.clockIdentity
- 17 b) portIdentity.portNumber

18 8.5.2.2 clockIdentity

19 The clockIdentity attribute shall be as specified in 7.5.2.2 of IEEE Std 1588-2019.

20 8.5.2.3 Port number

21 The portNumber values for the PTP Ports on a time-aware system shall be distinct in the range 0x0001
22 through 0xFFFFE.

23 The portNumber value 0 is assigned to the interface between the ClockTimeTransmitter and ClockSource
24 entities (see 10.1 and Figure 10-1). The assignment of portNumber 0 to this interface helps to simplify the
25 SiteSync and PortStateSelection state machines.

26 The portNumber value 0xFFFF is reserved.

27 8.5.2.4 Ordering of clockIdentity and portIdentity values

28 Two clockIdentity values X and Y are compared as follows. Let x be the unsigned integer formed by
29 concatenating octets 0 through 7 of X such that octet $j+1$ follows octet j (i.e., is less significant than octet j)
30 in x ($j = 0, 1, \dots, 6$). Let y be the unsigned integer formed by concatenating octets 0 through 7 of Y such that
31 octet $j+1$ follows octet j (i.e., is less significant than octet j) in y ($j = 0, 1, \dots, 6$). Then:

- 32 a) $X = Y$ if and only if $x = y$,
- 33 b) $X > Y$ if and only if $x > y$, and
- 34 c) $X < Y$ if and only if $x < y$.

Two portIdentity values A and B with members clockIdentity and portNumber are compared as follows. Let a be the unsigned integer formed by concatenating octets 0 through 7 of A.clockIdentity, such that octet $j+1$ follows octet j (i.e., is less significant than octet j) in a ($j = 0, 1, \dots, 6$), followed by octet 0 of A.portNumber, followed by octet 1 of A.portNumber. Let b be the unsigned integer formed by concatenating octets 0 through 7 of B.clockIdentity, such that octet $j+1$ follows octet j (i.e., is less significant than octet j) in b ($j = 0, 1, \dots, 6$), followed by octet 0 of B.portNumber, followed by octet 1 of B.portNumber. Then:

- d) $A = B$ if and only if $a = b$,
- e) $A > B$ if and only if $a > b$, and
- f) $A < B$ if and only if $a < b$.

A portIdentity A with members clockIdentity and portNumber and a clockIdentity B are compared as follows. The unsigned integer a is formed from portIdentity A as described above. The unsigned integer b is formed by first forming a portIdentity B' whose clockIdentity is B and portNumber is 0. b is then formed from B' as described above. A and B are then compared as described in items d) through f) in this subclause.

8.6 PTP Instance characterization

8.6.1 PTP Instance type

There are two types of PTP Instances used in a gPTP domain, as follows:

- a) PTP End Instance
- b) PTP Relay Instance

All PTP Instances are identified by clockIdentity.

In addition, PTP Instances are characterized by the following attributes:

- c) priority1
- d) clockClass
- e) clockAccuracy
- f) offsetScaledLogVariance
- g) priority2
- h) clockIdentity
- i) timeSource
- j) numberPorts

NOTE—Attributes c) through i) can be considered to be associated with the ClockTimeTransmitter entity of the PTP Instance.

8.6.2 PTP Instance attributes

8.6.2.1 priority1

priority1 is used in the execution of the BTCA (see 10.3). The value of priority1 is an integer selected from the range 0 through 255. The ordering of priority1 in the operation of the BTCA (see 10.3.4 and 10.3.5) is specified as follows. A ClockTimeTransmitter A shall be deemed better than a ClockTimeTransmitter B if the value of priority1 of A is numerically less than that of B.

1 The value of priority1 shall be 255 for a PTP Instance that is not grandmaster-capable. The value of priority1
2 shall be less than 255 for a PTP Instance that is grandmaster-capable. The value 0 shall be reserved for
3 management use, i.e., the value of priority1 shall be set to 0 only via management action. The default value
4 shall be set to one of the values listed in Table 8-1, and the choice of value from Table 8-1 is left to the
5 implementer of the PTP Instance.

Table 8-1—Default values for priority1, for the respective media

PTP Instance type	Default value for priority1
PTP Instances that are a central/critical part of the network and are not expected to ever be turned off during normal network operation (e.g., Bridges, wireless access points, and grandmaster-capable end nodes that are intended by the manufacturer to be Grandmaster PTP Instances). (End node Grandmaster PTP Instances are not a ‘central’ part of the network, but they are “critical” to this gPTP operation.)	246
PTP Instances that (A) can be turned off at any time and therefore cannot be considered a central/critical part of the network; or (B) are a central/critical part of the network, are not expected to ever be turned off during normal network operation, and are grandmaster-capable, but are not intended by the manufacturer to be Grandmaster PTP Instances. PTP Instances that can be turned off and are turned off affect only the function(s) they support and do not affect any other functions of the network (e.g., desktop computers, fixed (heavy or otherwise) end nodes, speakers, receivers, amplifiers, and televisions).	248
PTP Instances that can go away (physically or otherwise) at any time (e.g., PTP Instances that are designed to be transient to the network such as laptop computers, cell phones, and battery-powered speakers).	250
PTP Instance that is not grandmaster-capable.	255

6 NOTE 1—Care must be applied to multi-function device design, specifically for an end station that also contains a
7 Bridge. The Bridge function inside multi-function devices must not be powered down when the end station function is
8 powered down, or else the data and controls (like gPTP) passing through the Bridge function to other network devices
9 will cease. Example devices of this are a television and/or amplifier/receiver, each of which contains a Bridge.

10 NOTE 2—The BTCA (see 10.3) considers priority1 before other attributes; the priority1 attribute can therefore be used
11 to force a desired ordering of PTP Instances for best timeTransmitter selection.

12 NOTE 3—The settings for priority1 in Table 8-1 guarantee that a PTP Instance that is grandmaster-capable is always
13 preferred by the BTCA over a PTP Instance that is not grandmaster-capable.

14 NOTE 4—These values are assigned so that PTP Instances with priority1 value of 246 are selected as Grandmaster PTP
15 Instances over PTP Instances with priority1 values of 248 or 250. (PTP Instances with priority1 value of 255 are never
16 selected as Grandmaster PTP Instances.)

17 NOTE 5—These default values are suitable for applications in which the availability of the Grandmaster PTP Instance is
18 the most important criterion for Grandmaster PTP Instance selection. A PTP Instance built for a specific application for
19 which this is not the case can be capable of having priority1 changed via management.

20 8.6.2.2 clockClass

21 The clockClass attribute denotes the traceability of the synchronized time distributed by a
22 ClockTimeTransmitter when it is the Grandmaster PTP Instance.

23 The value shall be selected as follows:

- 24 a) If defaultDS.gmCapable is TRUE (see 14.2.7), then

- 1) clockClass is set to the value that reflects the combination of the LocalClock and ClockSource entities; else
- 2) If the value that reflects the LocalClock and ClockSource entities is not specified or not known, clockClass is set to 248;
- b) If the defaultDS.gmCapable is FALSE, clockClass is set to 255 (see 8.6.2.1).

The ordering of clockClass in the operation of the best timeTransmitter clock algorithm (see 10.3.4 and 10.3.5) is specified as follows. When comparing clockClass values, PTP Instance A shall be deemed better than PTP Instance B if the value of the clockClass of A is lower than that of B.

See 7.6.2.5 of IEEE Std 1588-2019 for a more detailed description of clockClass.

NOTE—The PTP Instance has a LocalClock entity, which can be the free-running quartz crystal that just meets the IEEE 802.3 requirements, but could also be better. There can be a ClockSource entity, e.g., timing taken from a GNSS, available in the local system that provides timing to the ClockSource entity. The time provided by the PTP Instance, if it is the Grandmaster PTP Instance, is reflected by the combination of these two entities, and the clockClass reflects this combination as specified in 7.6.2.5 of IEEE Std 1588-2019. For example, when the LocalClock entity uses a quartz oscillator that meets the requirements of IEEE Std 802.3-2022 and B.1 of this standard, clockClass is set to 248. But, if a GNSS receiver is present and synchronizes the PTP Instance, then the clockClass is set to the value 6, indicating traceability to a primary reference time source (see 7.6.2.5 of IEEE Std 1588-2019).

8.6.2.3 clockAccuracy

The clockAccuracy attribute indicates the expected time accuracy of a ClockTimeTransmitter.

The value shall be selected as follows:

- a) clockAccuracy is set to the value that reflects the combination of the LocalClock and ClockSource entities; else
- b) If the value that reflects the LocalClock and ClockSource entities is not specified or unknown, clockAccuracy is set to 254 (0xFE).

The ordering of clockAccuracy in the operation of the best timeTransmitter clock algorithm (see 10.3.4 and 10.3.5) is specified as follows. When comparing clockAccuracy values, PTP Instance A shall be deemed better than PTP Instance B if the value of the clockAccuracy of A is lower than that of B.

See 7.6.2.6 of IEEE Std 1588-2019 for more detailed description of clockAccuracy.

8.6.2.4 offsetScaledLogVariance

The offsetScaledLogVariance is a scaled, offset representation of an estimate of the PTP variance. The PTP variance characterizes the precision and frequency stability of the ClockTimeTransmitter. The PTP variance is the square of PTP Deviation (PTPDEV) (see B.1.3.2).

The value shall be selected as follows:

- a) offsetScaledLogVariance is set to the value that reflects the combination of the LocalClock and ClockSource entities; else
- b) If the value that reflects these entities is not specified or not known, offsetScaledLogVariance is set to 17258 (0x436A). This value corresponds to the value of PTPDEV for observation interval equal to the default Sync message transmission interval (i.e., observation interval of 0.125 s; see 11.5.2.3 and B.1.3.2).

1 The ordering of offsetScaledLogVariance in the operation of the best timeTransmitter clock algorithm (see
2 10.3.4 and 10.3.5) is specified as follows. When comparing offsetScaledLogVariance values, PTP Instance
3 A shall be deemed better than PTP Instance B if the value of the offsetScaledLogVariance of A is lower than
4 that of B.

5 See 7.6.3 of IEEE Std 1588-2019 for more detailed description of PTP variance and
6 offsetScaledLogVariance. (Subclause 7.6.3.3 of IEEE Std 1588-2019 provides a detailed description of the
7 computation of offsetScaledLogVariance from PTP variance, along with an example.)

8 8.6.2.5 priority2

9 priority2 is used in the execution of the BTCA (see 10.3). The value of priority2 shall be an integer selected
10 from the range 0 through 255. The ordering of priority2 in the operation of the BTCA is the same as the
11 ordering of priority1 (see 8.6.2.1).

12 The default value of priority2 shall be 247 or 248. The default value for a PTP Relay Instance should be 247.
13 The default value for a PTP End Instance should be 248. See 7.6.2.4 of IEEE Std 1588-2019 for a more
14 detailed description of priority2.

15 NOTE—IEEE 802.1AS performance is improved when the number of hops between the Grandmaster PTP Instance and
16 a timeReceiver PTP End Instance is reduced. When BTCA attributes are equal in a network, the preceding
17 recommendations for priority2 select a PTP Relay Instance in order to reduce the number of hops (rather than use
18 clockIdentity alone).

19 8.6.2.6 clockIdentity

20 The clockIdentity value for a PTP Instance shall be as specified in 8.5.2.2.

21 8.6.2.7 timeSource

22 The timeSource is an information only attribute indicating the type of source of time used by a
23 ClockTimeTransmitter. The value is not used in the selection of the Grandmaster PTP Instance. The data
24 type of timeSource shall be TimeSource, which is an Enumeration8. The values of TimeSource are specified
25 in Table 8-2. These represent categories. For example, the global positioning system (GPS) entry includes
26 not only the GPS system of the U.S. Department of Defense but the European Galileo system and other
27 present and future GNSSs.

Table 8-2—TimeSource enumeration

Value	Time source	Description
0x10	ATOMIC_CLOCK	Any PTP Instance, or PTP Instance directly connected to such a device, that is based on atomic resonance for frequency and that has been calibrated against international standards for frequency and time.
0x20	GPS (see NOTE 1)	Any PTP Instance synchronized to any of the satellite systems that distribute time and frequency tied to international standards.
0x30	TERRESTRIAL_RADIO	Any PTP Instance synchronized via any of the radio distribution systems that distribute time and frequency tied to international standards.
0x40	PTP	Any PTP Instance synchronized to an IEEE 1588 PTP-based source of time external to the gPTP domain (see NOTE 2).

Table 8-2—TimeSource enumeration (continued)

Value	Time source	Description
0x50	NTP	Any PTP Instance synchronized via the network time protocol (NTP) to servers that distribute time and frequency tied to international standards.
0x60	HAND_SET	Used in all cases for any PTP Instance whose time has been set by means of a human interface based on observation of an international standards source of time to within the claimed clock accuracy.
0x90	OTHER	Any source of time and/or frequency not covered by other values, or for which the source is not known.
0xA0	INTERNAL_OSCILLATOR	Any PTP Instance whose frequency is not based on atomic resonance nor calibrated against international standards for frequency, and whose time is based on a free-running oscillator with epoch determined in an arbitrary or unknown manner.
NOTE 1—In this standard, this value refers to any GNSS or Regional Navigation Satellite System (i.e., not only GPS).		
NOTE 2—For example, a clock that implements both a gPTP domain and a separate IEEE 1588 (i.e., PTP) domain, and is synchronized by the separate IEEE 1588 domain, would have time source of PTP in the gPTP domain.		

1 All unused values are reserved.

2 See 7.6.2.8 of IEEE Std 1588-2019 for a more detailed description of timeSource.

3 The initialization value is selected as follows:

- 4 a) If the timeSource (8.6.2.7 and Table 8-2) is known at the time of initialization, the value is derived
- 5 from the table, else
- 6 b) The value is set to 0xA0 (INTERNAL_OSCILLATOR).

7 8.6.2.8 numberPorts

8 The numberPorts indicates the number of PTP Ports of the PTP Instance.

1 9. Application interfaces

2 9.1 Overview of the interfaces

3 Unless otherwise stated, information in this clause is per domain.

4 The following subclauses define one application interface between the ClockSource entity and
5 ClockTimeTransmitter entity (see 10.1.2) and four application interfaces between the ClockTarget entity and
6 ClockTimeReceiver entity (see 10.1.2). The ClockSource is an entity that can be used as an external timing
7 source for the gPTP domain. The ClockSource entity either contains or has access to a clock (see 3.3). The
8 ClockTarget entity represents any application that uses information provided by the ClockTimeReceiver
9 entity via any of the application interfaces.

10 NOTE 1—The manner in which the ClockSource entity obtains time from a clock is outside the scope of this standard.
11 The manner in which the ClockTarget uses the information provided by application interfaces is outside the scope of this
12 standard.

13 The five interfaces are illustrated in Figure 9-1:

- 14 a) ClockSourceTime interface, which provides external timing to a PTP Instance,
- 15 b) ClockTargetEventCapture interface, which returns the synchronized time of an event signaled by a
16 ClockTarget entity,
- 17 c) ClockTargetTriggerGenerate interface, which causes an event to be signaled at a synchronized time
18 specified by a ClockTarget entity,
- 19 d) ClockTargetClockGenerator interface, which causes a periodic sequence of results to be generated,
20 with a phase and rate specified by a ClockTarget entity, and
- 21 e) ClockTargetPhaseDiscontinuity interface, which supplies information that an application can use to
22 determine if a discontinuity in Grandmaster Clock phase or frequency has occurred.

23 NOTE 2—The application interfaces described in this clause are models for behavior and not application program
24 interfaces. Other application interfaces besides items a) through e) in this subclause are possible, but are not described
25 here. In addition, there can be multiple instances of a particular interface.

1

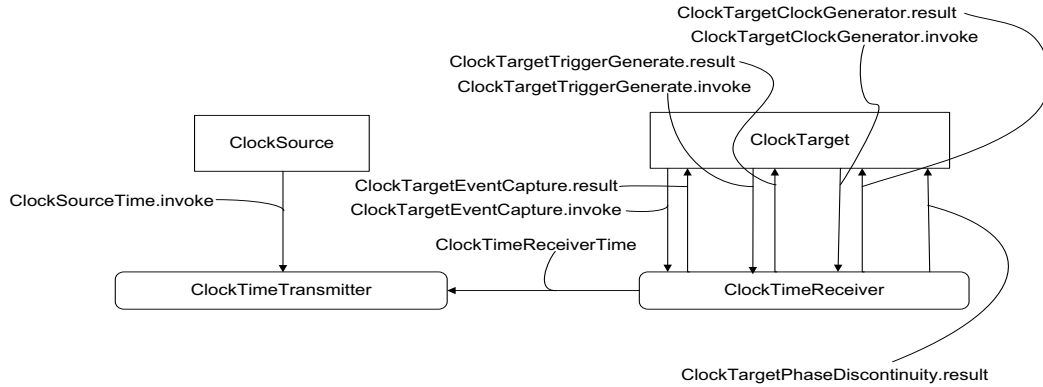


Figure 9-1—Application interfaces

2 9.2 ClockSourceTime interface

3 9.2.1 General

4 This interface is used by the ClockSource entity to provide time to the ClockTimeTransmitter entity of a
5 PTP Instance. The ClockSource entity invokes the ClockSourceTime.invoke function. The function
6 provides the time, relative to the ClockSource, at which the function was invoked.

7 9.2.2 ClockSourceTime.invoke function parameters

```

8 ClockSourceTime.invoke {
9     domainNumber,
10    sourceTime,
11    timeBaseIndicator,
12    lastGmPhaseChange,
13    lastGmFreqChange
14 }
```

15 9.2.2.1 domainNumber (UInteger8)

16 This parameter is the domainNumber of the gPTP domain to which this ClockSource entity is providing
17 time.

18 9.2.2.2 sourceTime (ExtendedTimestamp)

19 The value of sourceTime is the time this function is invoked by the ClockSource entity.

20 9.2.2.3 timeBaseIndicator (UInteger16)

21 The timeBaseIndicator is a value that is set by the ClockSource entity. The ClockSource entity changes the
22 value whenever its time base changes. The ClockSource entity shall change the value of timeBaseIndicator
23 if and only if there is a phase or frequency change.

1 NOTE—While the clock that supplies time to the ClockSource entity can be lost, the ClockSource entity itself is not
2 lost, the ClockSource entity ensures that timeBaseIndicator changes if the source of time is lost.

3 9.2.2.4 lastGmPhaseChange (ScaledNs)

4 The value of lastGmPhaseChange is the phase change (i.e., change in sourceTime) that occurred on the most
5 recent change in timeBaseIndicator. The value is initialized to 0.

6 9.2.2.5 lastGmFreqChange (Float64)

7 The value of lastGmFreqChange is the fractional frequency change (i.e., frequency change expressed as a
8 pure fraction) that occurred on the most recent change in timeBaseIndicator. The value is initialized to 0.

9 9.3 ClockTargetEventCapture interface

10 9.3.1 General

11 This interface is used by the ClockTarget entity to request the synchronized time of an event that it signals to
12 the ClockTimeReceiver entity of a PTP Instance. The ClockTarget entity invokes the
13 ClockTargetEventCapture.invoke function to signal an event to the ClockTimeReceiver entity. The
14 ClockTimeReceiver entity invokes the ClockTargetEventCapture.result function to return the time of the
15 event relative to the current Grandmaster Clock or, if no PTP Instance is grandmaster-capable, the
16 LocalClock. The ClockTargetEventCapture.result function also returns gmPresent, to indicate to the
17 ClockTarget whether a Grandmaster PTP Instance is present.

18 9.3.2 ClockTargetEventCapture.invoke parameters

```
19 ClockTargetEventCapture.invoke {  
20     domainNumber  
21 }
```

22 9.3.2.1 domainNumber (UInteger8)

23 This parameter is the domainNumber of the ClockTimeReceiver entity that is requested to provide the
24 synchronized time of the signaled event.

25 9.3.3 ClockTargetEventCapture.result parameters

```
26 ClockTargetEventCapture.result {  
27     domainNumber,  
28     timeReceiverTimeCallback,  
29     gmPresent  
30     isSynced,  
31     grandmasterIdentity  
32 }
```

33 9.3.3.1 domainNumber (UInteger8)

34 This parameter is the domainNumber of the ClockTimeReceiver entity that is providing the synchronized
35 time of the signaled event.

1 9.3.3.2 timeReceiverTimeCallback (ExtendedTimestamp)

2 The value of timeReceiverTimeCallback is the time, relative to the Grandmaster Clock, that the
3 corresponding ClockTargetEventCapture.invoke function is invoked.

4 NOTE—The invocation of the ClockTargetEventCapture.invoke function and the detection of this invocation by the
5 ClockTimeReceiver entity are simultaneous in this abstract interface.

6 9.3.3.3 gmPresent (Boolean)

7 The value of gmPresent is set equal to the value of the global variable gmPresent (see 10.2.4.13). This
8 parameter indicates to the ClockTarget whether a Grandmaster PTP Instance is present.

9 9.3.3.4 isSynced (Boolean)

10 The value of the variable isSynced (see 18.4.1.1). This parameter shall be present if the optional hot standby
11 feature is implemented (see Clause 18).

12 9.3.3.5 grandmasterIdentity (ClockIdentity)

13 If driftTrackingTlvSupport is TRUE and the value of the global variable syncGrandmasterIdentity is not
14 0xFFFFFFFFFFFFFFFF, grandmasterIdentity is the value of the global variable syncGrandmasterIdentity.
15 Else, the value of grandmasterIdentity is the clockIdentity of the Grandmaster PTP Instance.

16 NOTE—If the transfer of valid grandmasterIdentity information in the Drift_Tracking TLV provides a value that
17 identifies the grandmaster upon which the API call is triggered or the time information conveyed, this value is used. In
18 all other cases, the grandmasterIdentity as defined in 10.6.3.2.5 is used.

19 9.4 ClockTargetTriggerGenerate interface

20 9.4.1 General

21 This interface is used by the ClockTarget entity to request that the ClockTimeReceiver entity send a result at
22 a specified time relative to the Grandmaster Clock. The ClockTarget entity invokes the
23 ClockTargetTriggerGenerate.invoke function to indicate the synchronized time of the event. The
24 ClockTimeReceiver entity invokes the ClockTargetTriggerGenerate.result function to either signal the event
25 at the requested synchronized time or indicate an error condition.

26 9.4.2 ClockTargetTriggerGenerate.invoke parameters

```
27 ClockTargetTriggerGenerate.invoke {  
28     domainNumber,  
29     timeReceiverTimeCallback  
30 }
```

31 9.4.2.1 domainNumber (UInteger8)

32 This parameter is the domainNumber of the ClockTimeReceiver entity that is requested to signal an event at
33 the specified time.

34 9.4.2.2 timeReceiverTimeCallback (ExtendedTimestamp)

35 If timeReceiverTimeCallback is nonzero, its value is the synchronized time the corresponding
36 ClockTargetTriggerGenerate.result function, i.e., the trigger, is to be invoked. If timeReceiverTimeCallback

1 is zero, any previous `ClockTargetTriggerGenerate.invoke` function for which a
2 `ClockTargetTriggerGenerate.result` function has not yet been issued is canceled.

3 9.4.3 `ClockTargetTriggerGenerate.result` parameters

```
4 ClockTargetTriggerGenerate.result {  
5     domainNumber,  
6     errorCondition,  
7     gmPresent  
8     isSynced,  
9     grandmasterIdentity  
10 }
```

11 9.4.3.1 `domainNumber` (`UInteger8`)

12 This parameter is the `domainNumber` of the `ClockTimeReceiver` entity that is triggering an event at the
13 specified time.

14 9.4.3.2 `errorCondition` (`Boolean`)

15 A value of `FALSE` indicates that the `ClockTargetTriggerGenerate.result` function was invoked at the time,
16 relative to the Grandmaster Clock, contained in the corresponding `ClockTargetTriggerGenerate.invoke`
17 function. A value of `TRUE` indicates that the `ClockTargetTriggerGenerate.result` function could not be
18 invoked at the synchronized time contained in the corresponding `ClockTargetTriggerGenerate.invoke`
19 function.

20 NOTE—For example, the `ClockTargetTriggerGenerate.result` function is invoked with `errorCondition = TRUE` if the
21 requested `timeReceiverTimeCallback` is a time prior to the synchronized time when the corresponding
22 `ClockTargetTriggerGenerate.invoke` function is invoked. As another example, the `ClockTargetTriggerGenerate.result`
23 function is invoked with `errorCondition = TRUE` if a discontinuity in the synchronized time causes the requested
24 `timeReceiverTimeCallback` to be skipped over.

25 9.4.3.3 `gmPresent` (`Boolean`)

26 As specified in 9.3.3.3.

27 9.4.3.4 `isSynced` (`Boolean`)

28 As specified in 9.3.3.4.

29 9.4.3.5 `grandmasterIdentity` (`ClockIdentity`)

30 As specified in 9.3.3.5.

31 9.4.4 `ClockTargetTriggerGenerate` interface definition

32 The invocation of the `ClockTargetTriggerGenerate.invoke` function causes the `ClockTimeReceiver` entity to
33 store the value of the `timeReceiverTimeCallback` parameter in an internal variable (replacing any previous
34 value of that variable) until the synchronized time, or `LocalClock` time if `gmPresent` is `FALSE`, equals the
35 value of that variable, at which time the `ClockTargetTriggerGenerate.result` function is invoked with
36 `errorCondition = FALSE`. If it is not possible to invoke the `ClockTargetTriggerGenerate.result` function at
37 `timeReceiverTimeCallback`, e.g., if `timeReceiverTimeCallback` is earlier than the synchronized time (or
38 `LocalClock` time if `gmPresent` is `FALSE`) when the `ClockTargetTriggerGenerate.invoke` function is invoked,
39 the `ClockTargetTriggerGenerate.result` function is invoked with `errorCondition = TRUE`. Invocation of the

1 ClockTargetTriggerGenerate.invoke function with timeReceiverTimeCallback = 0 (which is earlier than any
2 synchronized time) is used to cancel a pending request.

3 9.5 ClockTargetClockGenerator interface

4 9.5.1 General

5 This interface is used by the ClockTarget entity to request that the ClockTimeReceiver entity deliver a
6 periodic clock signal of specified period and phase. The ClockTarget entity invokes the
7 ClockTargetClockGenerator.invoke function to request that the ClockTimeReceiver entity generate the
8 periodic clock signal. The ClockTimeReceiver entity invokes the ClockTargetClockGenerator.result
9 function at significant instants of the desired clock signal.

10 9.5.2 ClockTargetClockGenerator.invoke parameters

```
11 ClockTargetClockGenerator.invoke {
12     domainNumber,
13     clockPeriod,
14     timeReceiverTimeCallbackPhase
15 }
```

16 9.5.2.1 domainNumber (UInteger8)

17 This parameter is the domainNumber of the ClockTimeReceiver entity that is requested to deliver a periodic
18 clock signal.

19 9.5.2.2 clockPeriod (TimeInterval)

20 The value of clockPeriod is the period between successive invocations of the
21 ClockTargetClockGenerator.result function. A value that is zero or negative causes any existing periodic
22 clock signal generated via this application interface to be terminated.

23 9.5.2.3 timeReceiverTimeCallbackPhase (ExtendedTimestamp)

24 The value of timeReceiverTimeCallbackPhase describes phase of the generated clock signal by specifying a
25 point on the timescale in use such that ClockTargetClockGenerator.result invocations occur at synchronized
26 times that differ from timeReceiverTimeCallbackPhase by $n \times \text{clockPeriod}$, where n is an integer.

27 NOTE—The value of timeReceiverTimeCallbackPhase can be earlier or later than the synchronized time the
28 ClockTargetClockGenerator.invoke function is invoked; use of a timeReceiverTimeCallbackPhase value in the future
29 does not imply that the initiation of the periodic clock signal is suppressed until that synchronized time.

30 9.5.3 ClockTargetClockGenerator.result parameters

```
31 ClockTargetClockGenerator.result {
32     domainNumber,
33     gmPresent,
34     timeReceiverTimeCallback,
35     isSynced,
36     grandmasterIdentity
37 }
```

1 9.5.3.1 domainNumber (UInteger8)

2 This parameter is the domainNumber of the ClockTimeReceiver entity that is delivering a periodic clock
3 signal.

4 9.5.3.2 gmPresent

5 As specified in 9.3.3.3.

6 9.5.3.3 timeReceiverTimeCallback (ExtendedTimestamp)

7 The value of timeReceiverTimeCallback is the synchronized time of this event.

8 9.5.3.4 isSynced (Boolean)

9 As specified in 9.3.3.4.

10 9.5.3.5 grandmasterIdentity (ClockIdentity)

11 As specified in 9.3.3.5.

12 9.6 ClockTargetPhaseDiscontinuity interface

13 9.6.1 General

14 This interface provides discontinuity information, sent from the Grandmaster PTP Instance, to an
15 application within an end station. It is used by the ClockTimeReceiver entity to supply sufficient
16 information to the ClockTarget entity to enable the ClockTarget entity to determine whether a phase or
17 frequency discontinuity has occurred. The ClockTimeReceiver invokes the
18 ClockTargetPhaseDiscontinuity.result function in the SEND_SYNC_INDICATION block of the
19 ClockTimeReceiverSync state machine (see 10.2.13 and Figure 10-9). The invocation occurs when a
20 PortSyncSync structure is received, after the needed information has been computed by the
21 ClockTimeReceiverSync state machine.

22 9.6.2 ClockTargetPhaseDiscontinuity.result parameters

```
23 ClockTargetPhaseDiscontinuity.result {  
24     domainNumber,  
25     grandmasterIdentity,  
26     gmTimeBaseIndicator,  
27     lastGmPhaseChange,  
28     lastGmFreqChange  
29     isSynced,  
30 }
```

31 9.6.2.1 domainNumber (UInteger8)

32 This parameter is the domainNumber of the ClockTimeReceiver entity that is providing discontinuity
33 information.

1 **9.6.2.2 grandmasterIdentity (ClockIdentity)**

2 If gmPresent (see 10.2.4.13) is TRUE, the value of grandmasterIdentity is the ClockIdentity of the current
3 Grandmaster PTP Instance. If gmPresent is FALSE, the value of grandmasterIdentity is 0x0.

4 **9.6.2.3 gmTimeBaseIndicator (UInteger16)**

5 The value of gmTimeBaseIndicator is the timeBaseIndicator of the current Grandmaster PTP Instance.

6 **9.6.2.4 lastGmPhaseChange (ScaledNs)**

7 The value of the global lastGmPhaseChange parameter (see 10.2.4.16) received from the Grandmaster PTP
8 Instance.

9 **9.6.2.5 lastGmFreqChange (Float64)**

10 The value of lastGmFreqChange parameter (see 10.2.4.17) received from the Grandmaster PTP Instance.

11 **9.6.2.6 isSynced (Boolean)**

12 As specified in 9.3.3.4.

13

I

1 10. Media-independent layer specification

2 10.1 Overview

3 10.1.1 General

4 Unless otherwise stated, information in this clause is per domain.

5 10.1.2 Model of operation

6 A PTP Instance contains a best timeTransmitter selection function and a synchronization function. These
7 functions include PTP Port-specific aspects and aspects associated with the PTP Instance as a whole. The
8 functions are distributed among a number of entities, which together describe the behavior of a compliant
9 implementation. The functions are specified by a number of state machines.

10 The model for the media-independent layer of a PTP Instance is shown in Figure 10-1. It includes a single
11 SiteSync entity, ClockTimeTransmitter entity, and ClockTimeReceiver entity for the PTP Instance as a
12 whole, plus one PortSync for each PTP Port. The PTP Instance also includes one MD entity for each PTP
13 Port, which is part of the media-dependent layer. The media-dependent functions performed by the MD
14 entity are described in the clauses for the respective media. In addition to the entities, Figure 10-1 shows the
15 information that flows between the entities via the PortSyncSync, MDSyncSend, and MDSyncReceive
16 structures (see 10.2.2.3, 10.2.2.1, and 10.2.2.2, respectively).

17 The SiteSync, ClockTimeTransmitter, ClockTimeReceiver, and PortSync entities each contain a number of
18 cooperating state machines, which are described later in this clause (the MD entity state machines are
19 described in the respective media-dependent clauses). The ClockTimeTransmitter entity receives
20 information from an external time source, known as a *ClockSource entity* (see 9.2), via an application
21 interface, and provides the information to the SiteSync entity. The ClockTimeReceiver entity receives
22 Grandmaster Clock time-synchronization and current Grandmaster PTP Instance information from the
23 SiteSync entity, and makes the information available to an external application, known as a *clockTarget*
24 *entity* (see 9.3 through 9.6), via one or more application service interfaces. If the PTP instance uses the
25 BTCA, then the SiteSync entity executes the portion of best timeTransmitter clock selection associated with
26 the PTP Instance as a whole, i.e., it uses the best timeTransmitter information received on each PTP Port to
27 determine which PTP Port has received the best information, and updates the states of all the ports (see
28 10.3.1.1 for a discussion of PTP Port states). If the PTP instance uses external port configuration, port roles
29 are defined by engineering and the best timeTransmitter information received is not used for this purpose. It
30 also distributes synchronization information received on the TimeReceiverPort to all the ports whose state is
31 TimeTransmitterPort (see 10.3.1.1). The PortSync entity for a TimeReceiverPort receives best
32 timeTransmitter selection information from the PTP Instance at the other end of the associated link,
33 compares this to the current best timeTransmitter information that it has, and forwards the result of the
34 comparison to the Site Sync entity. The PortSync entity for a TimeReceiverPort also receives time-
35 synchronization information from the MD entity associated with the PTP Port, and forwards it to the
36 SiteSync entity. The PortSync entity for a TimeTransmitterPort sends best timeTransmitter selection and
37 time-synchronization information to the MD entity for the PTP Port, which in turn sends the respective
38 messages.

39 NOTE—This clause does not require a one-to-one correspondence between the PortSync entities of PTP Instances
40 attached to the same gPTP communication path (see 3.13), i.e., more than two PTP Instances can be attached to a gPTP
41 communication path that uses a shared medium and meet the requirements of this clause. However, it is possible for a
42 media-dependent clause to have additional requirements that limit the gPTP communication paths to point-to-point links
43 for that medium; in this case, each link has exactly two PortSync entities, which can be considered to be in one-to-one
44 correspondence. One example of this is the full-duplex point-to-point media-dependent layer specified in Clause 11. In
45 addition, one or more gPTP communication paths can be logically point-to-point but traverse the same shared medium.

1

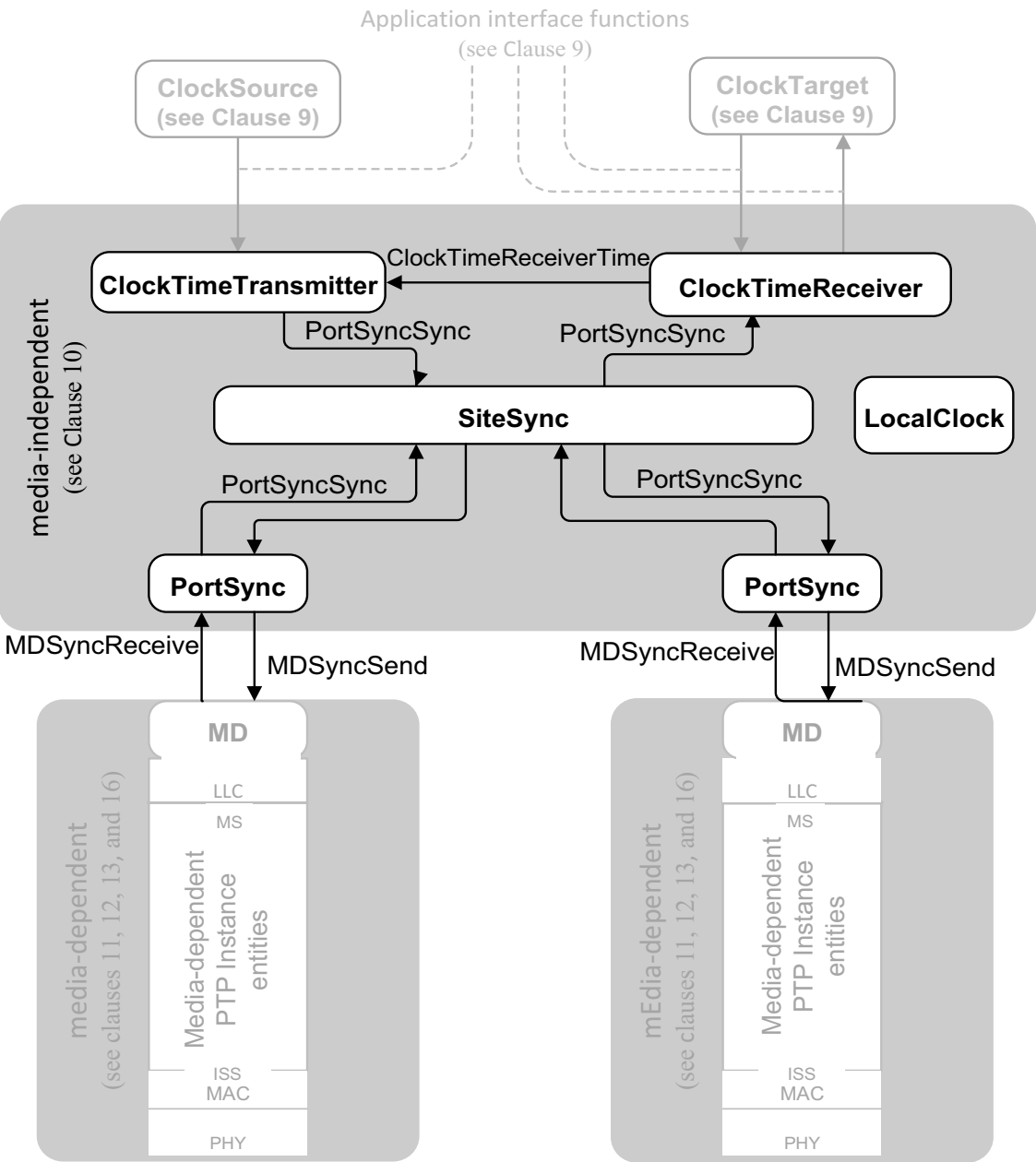


Figure 10-1—Model for media-independent layer of PTP Instance

2 The time-synchronization state machines are described in 10.2. The best timeTransmitter clock selection
3 state machines are described in 10.3. The attributes and format of the Announce message are described in
4 10.5 and 10.6. The timing characterization of the protocol is described in 10.7.

1 10.1.2.1 LocalClock entity

2 The LocalClock entity is a free-running local clock (see 3.18) that provides a common time to the PTP
3 Instance, relative to an arbitrary epoch. A PTP Instance contains a LocalClock entity. The requirements for
4 the LocalClock entity are specified in B.1. All timestamps are taken relative to the LocalClock entity (see
5 8.4.3). The LocalClock entity also provides the value of currentTime (see 10.2.4.12), which is used in the
6 state machines to specify the various timers.

7 NOTE—The epoch for the LocalClock entity can be the time that the PTP Instance is powered on.

8 10.1.3 Grandmaster-capable PTP Instance

9 A PTP Instance may be grandmaster-capable. An implementation may provide the ability to configure a PTP
10 Instance as grandmaster-capable via a management interface.

11 NOTE 1—The managed object gmCapable is read only (see Table 14-1). gmCapable is configured by setting the value
12 of the managed object priority1, which is read/write (see Table 14-1). If the value of priority1 is 255, then

- 13 a) gmCapable is set to FALSE (see 8.6.2.1), and
- 14 b) The value of the managed object clockClass, which is read only, is set to 255 (see 8.6.2.2).

15 NOTE 2—While a PTP Instance that is not grandmaster-capable can never be the Grandmaster PTP Instance of the
16 gPTP domain, such a PTP Instance contains a best timeTransmitter selection function, invokes the best timeTransmitter
17 selection algorithm, and conveys synchronization information received from the current Grandmaster PTP Instance.

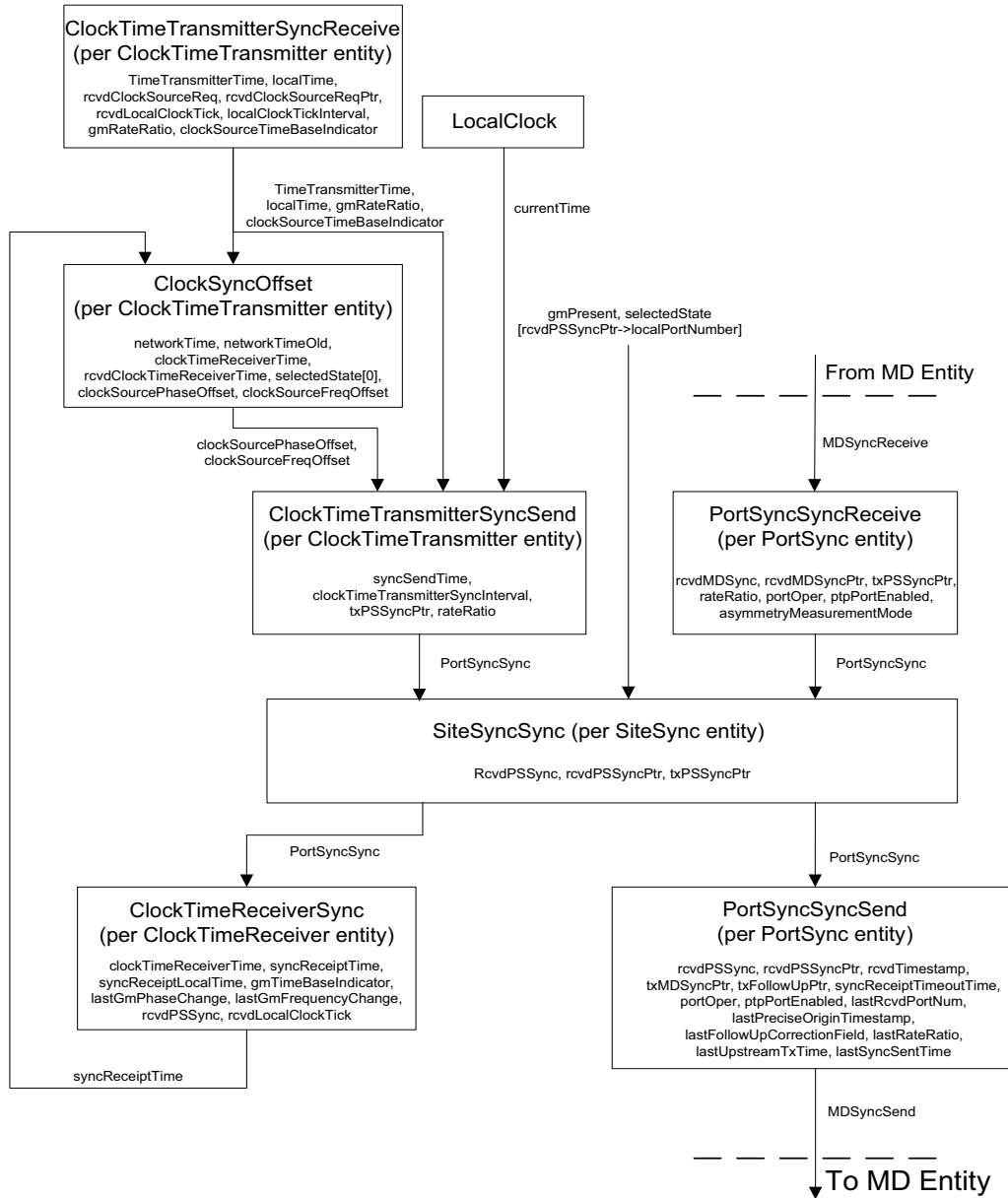
18 10.2 Time-synchronization state machines

19 10.2.1 Overview

20 The time-synchronization function in a PTP Instance is specified by a number of cooperating state
21 machines. Figure 10-2 illustrates these state machines, their local variables, their interrelationships, and the
22 global variables and structures used to communicate between them. The figure indicates the interaction
23 between the state machines and the media-dependent layer and LocalClock entity.

24 The ClockTimeTransmitterSyncReceive, ClockTimeTransmitterSyncOffset, and
25 ClockTimeTransmitterSyncSend state machines are optional for PTP Instances that are not grandmaster-
26 capable (see 8.6.2.1 and 10.1.3). These state machines may be present in a PTP Instance that is not
27 grandmaster-capable; however, any information supplied by them, via the ClockTimeTransmitterSyncSend
28 state machine, to the SiteSyncSync state machine is not used by the SiteSyncSync state machine if the PTP
29 Instance is not grandmaster-capable.

1



Notes:

- a) selectedState for each port and gmPresent are set by Port State Selection state machine (see 10.3.12)
- b) currentTime is a global variable that is always equal to the current time relative to the local oscillator
- c) application interfaces to higher layers are not shown
- d) the ClockTimeTransmitterSyncReceive, ClockTimeTransmitterSyncSend, and ClockTimeTransmitterSyncOffset state machines are optional for PTP Instances that are not grandmaster-capable.

Figure 10-2—Time-synchronization state machines—overview and interrelationships

1 The media-independent layer state machines in Figure 10-2 are as follows:

- 2 a) `ClockTimeTransmitterSyncReceive` (one instance per PTP Instance): receives
3 `ClockSourceTime.invoke` functions from the `ClockSource` entity and notifications of `LocalClock`
4 entity ticks (see 10.2.4.18), updates `timeTransmitterTime`, and provides `timeTransmitterTime` to
5 `ClockTimeTransmitterSyncOffset` and `ClockTimeTransmitterSyncSend` state machines.
- 6 b) `ClockTimeTransmitterSyncOffset` (one instance per PTP Instance): receives `syncReceiptTime` from
7 the `ClockTimeReceiver` entity and `timeTransmitterTime` from the
8 `ClockTimeTransmitterSyncReceive` state machine, computes phase offset and frequency offset
9 between `timeTransmitterTime` and `syncReceiptTime` if the PTP Instance is not the Grandmaster PTP
10 Instance, and provides the frequency and phase offsets to the `ClockTimeTransmitterSyncSend` state
11 machine.
- 12 c) `ClockTimeTransmitterSyncSend` (one instance per PTP Instance): receives `timeTransmitterTime`
13 from the `ClockTimeTransmitterSyncReceive` state machine, receives phase and frequency offset
14 between `timeTransmitterTime` and `syncReceiptTime` from the `ClockTimeTransmitterSyncOffset`
15 state machine, and provides `timeTransmitterTime` (i.e., synchronized time) and the phase and
16 frequency offset to the `SiteSync` entity using a `PortSyncSync` structure.
- 17 d) `PortSyncSyncReceive` (one instance per PTP Instance, per PTP Port): receives time-synchronization
18 information from the MD entity of the corresponding PTP Port, computes accumulated `rateRatio`,
19 computes `syncReceiptTimeoutTime`, and sends the information to the `SiteSync` entity.
- 20 e) `SiteSyncSync` (one instance per PTP Instance): receives time-synchronization information,
21 accumulated `rateRatio`, and `syncReceiptTimeoutTime` from the `PortSync` entity of the current
22 timeReceiver port or from the `ClockTimeTransmitter` entity; and sends the information to the
23 `PortSync` entities of all the ports and to the `ClockTimeReceiver` entity.
- 24 f) `PortSyncSyncSend` (one instance per PTP Instance, per PTP Port): receives time-synchronization
25 information from the `SiteSync` entity, requests that the MD entity of the corresponding PTP Port
26 send a time-synchronization event message, receives the `syncEventEgressTimestamp` for this event
27 message from the MD entity, uses the most recent time-synchronization information received from
28 the `SiteSync` entity and the timestamp to compute time-synchronization information that will be sent
29 by the MD entity in a general message (e.g., for full-duplex IEEE 802.3 media) or a subsequent
30 event message (e.g., for IEEE 802.11 media), and sends this latter information to the MD entity.
- 31 g) `ClockTimeReceiverSync` (one instance per PTP Instance): receives time-synchronization
32 information from the `SiteSync` entity; computes `clockTimeReceiverTime` and `syncReceiptTime`; sets
33 `syncReceiptLocalTime`, `GmTimeBaseIndicator`, `lastGmPhaseChange`, and `lastGmFreqChange`;
34 sends `clockTimeReceiverTime` to the `ClockTimeTransmitter` entity; and provides information to the
35 `ClockTarget` entity (via the `ClockTargetPhaseDiscontinuity` interface; see 9.6) to enable that entity to
36 determine if a phase or frequency discontinuity has occurred.

37 10.2.2 Data structures communicated between state machines

38 The following subclauses describe the data structures communicated between the time-synchronization state
39 machines.

40 10.2.2.1 MDSyncSend

41 10.2.2.1.1 General

42 This structure contains information that is sent by the `PortSync` entity of a PTP Port to the MD entity of that
43 PTP Port when requesting that the MD entity cause time-synchronization information to be sent. The
44 structure contains information that reflects the most recent time-synchronization information received by
45 this PTP Instance and is used to determine the contents of the time-synchronization event message and
46 possibly separate general message that will be sent by this PTP Port.

```

1 MDSyncSend {
2     domainNumber,
3     followUpCorrectionField,
4     sourcePortIdentity,
5     logMessageInterval,
6     preciseOriginTimestamp,
7     upstreamTxTime,
8     rateRatio,
9     gmTimeBaseIndicator,
10    lastGmPhaseChange,
11    lastGmFreqChange
12 }

```

13 The members of the structure are defined in the following subclauses.

14 **10.2.2.1.2 domainNumber (UInteger8)**

15 This parameter is the domainNumber of the gPTP domain in which this structure is sent.

16 NOTE—The domainNumber member is not essential because the state machines that send and receive this structure
17 are per domain, and each state machine implicitly knows the number of the domain in which it operates.

18 **10.2.2.1.3 followUpCorrectionField (ScaledNs)**

19 The followUpCorrectionField contains the accumulated time since the preciseOriginTimestamp was
20 captured by the Grandmaster PTP Instance. This is equal to the elapsed time, relative to the Grandmaster
21 Clock, between the time the Grandmaster PTP Instance sent the received time-synchronization event
22 message, truncated to the nearest nanosecond, and the time at which that event message was sent by the
23 upstream PTP Instance. The followUpCorrectionField is equal to the value of the followUpCorrectionField
24 member of the most recently received PortSyncSync structure from the PortSync entity of this PTP Port (see
25 10.2.2.3.5).

26 **10.2.2.1.4 sourcePortIdentity (PortIdentity)**

27 The sourcePortIdentity is the portIdentity of this PTP Port (see 8.5.2).

28 **10.2.2.1.5 logMessageInterval (Integer8)**

29 The logMessageInterval is the value of currentLogSyncInterval for this PTP Port (see 10.7.2.3).

30 **10.2.2.1.6 preciseOriginTimestamp (Timestamp)**

31 The preciseOriginTimestamp is the sourceTime of the ClockTimeTransmitter entity of the Grandmaster PTP
32 Instance, with any fractional nanoseconds truncated, when the received time-synchronization information
33 was sent by the Grandmaster PTP Instance. The preciseOriginTimestamp is the value of the
34 preciseOriginTimestamp member of the most recently received PortSyncSync structure from the PortSync
35 entity of this PTP Port (see 10.2.2.3.8).

36 **10.2.2.1.7 upstreamTxTime (UScaledNs)**

37 The upstreamTxTime is given by the following equation:

$$38 \quad \text{upstreamTxTime} = \text{syncEventIngressTimestamp} - \frac{\text{meanLinkDelay}}{\text{neighborRateRatio}}$$

1 where

2	syncEventIngressTimestamp	corresponds to the receipt of the time-synchronization information at the
3		timeReceiver port of this PTP Instance
4	meanLinkDelay	is defined in 10.2.5.8
5	neighborRateRatio	as defined in 10.2.5.7
6	upstreamTxTime	is the value of the upstreamTxTime member of the most recently
7		received PortSyncSync structure from the PortSync entity of this PTP
8		Port (see 10.2.2.3.9)

9 NOTE 1—When using Clause 11 MD entity, nrrPdelay is used instead of neighborRateRatio (see 11.2.13.14).

10 NOTE 2—Depending on the medium, delayAsymmetry also needs to be considered.

11 10.2.2.1.8 rateRatio (Float64)

12 The rateRatio is the value of the rateRatio member of the most recently received PortSyncSync structure
13 from the PortSync entity of this PTP Port (see 10.2.2.3.10). It is equal to the ratio of the frequency of the
14 Grandmaster Clock to the frequency of the LocalClock entity of this PTP Instance (see 10.2.8.1.4).

15 10.2.2.1.9 gmTimeBaseIndicator (UInteger16)

16 The gmTimeBaseIndicator is the timeBaseIndicator of the ClockSource entity of the current Grandmaster
17 PTP Instance. It is set equal to the gmTimeBaseIndicator of the received time-synchronization information.
18 The gmTimeBaseIndicator is the value of the gmTimeBaseIndicator member of the most recently received
19 PortSyncSync structure from the PortSync entity of this PTP Port (see 10.2.2.3.11).

20 10.2.2.1.10 lastGmPhaseChange (ScaledNs)

21 The lastGmPhaseChange is the time of the current Grandmaster Clock minus the time of the previous
22 Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the Grandmaster PTP
23 Instance, or the step change in the time of the current Grandmaster Clock at the time of the most recent
24 gmTimeBaseIndicator change. It is set equal to the lastGmPhaseChange of the received time-
25 synchronization information. The lastGmPhaseChange is the value of the lastGmPhaseChange member of
26 the most recently received PortSyncSync structure from the PortSync entity of this PTP Port
27 (see 10.2.2.3.12).

28 10.2.2.1.11 lastGmFreqChange (Float64)

29 The lastGmFreqChange is the fractional frequency offset of the current Grandmaster Clock relative to the
30 previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the
31 Grandmaster PTP Instance, or relative to itself prior to the last change in gmTimeBaseIndicator. It is set
32 equal to the lastGmFreqChange of the received time-synchronization information. The lastGmFreqChange
33 is the value of the lastGmFreqChange member of the most recently received PortSyncSync structure from
34 the PortSync entity of this PTP Port (see 10.2.2.3.13).

35 10.2.2.2 MDSyncReceive

36 10.2.2.2.1 General

37 This structure contains information that is sent by the MD entity of a PTP Port to the PortSync entity of that
38 PTP Port. It provides the PortSync entity with timeTransmitter clock timing information and timestamp of
39 receipt of a time-synchronization event message compensated for propagation time on the upstream link.
40 The information is sent to the PortSync entity upon receipt of time-synchronization information by the MD
41 entity of the PTP Port. The information is in turn provided by the PortSync entity to the SiteSync entity. The

1 information is used by the PortSyncSyncReceive state machine of the PortSync entity to compute the rate
 2 ratio of the Grandmaster Clock relative to the local clock and is communicated to the SiteSync entity, and
 3 then the SiteSync entity communicates it to the other PortSync entities for use in computing timeTransmitter
 4 clock timing information.

```

5 MDSyncReceive {
6     domainNumber,
7     followUpCorrectionField,
8     sourcePortIdentity,
9     logMessageInterval,
10    preciseOriginTimestamp,
11    upstreamTxTime,
12    rateRatio,
13    gmTimeBaseIndicator,
14    lastGmPhaseChange,
15    lastGmFreqChange
16 }
```

17 The members of the structure are defined in the following subclauses.

18 **10.2.2.2.2 domainNumber (UInteger8)**

19 This parameter is the domainNumber of the gPTP domain in which this structure is sent.

20 NOTE—The domainNumber member is not essential because the state machines that send and receive this structure
 21 are per domain, and each state machine implicitly knows the number of the domain in which it operates.

22 **10.2.2.2.3 followUpCorrectionField (ScaledNs)**

23 The followUpCorrectionField contains the elapsed time, relative to the Grandmaster Clock, between the
 24 time the Grandmaster PTP Instance sent the received time-synchronization information, truncated to the
 25 nearest nanosecond, and the time at which this information was sent by the upstream PTP Instance.

26 NOTE 1—The sum of followUpCorrectionField and preciseOriginTimestamp is the synchronized time that corresponds
 27 to the time the most recently received time-synchronization event message was sent by the upstream PTP Instance.

28 NOTE 2—For a medium that uses separate event and general messages (for example, full-duplex point-to-point media
 29 described in 11), the event message corresponding to the most recently received network synchronization information is
 30 the event message that corresponds to the most recently received general message. For a medium that places
 31 synchronization information based on the event message timestamp in the next event message (for example,
 32 IEEE 802.11 media described in Clause 12), the event message corresponding to the most recently received network
 33 synchronization information is the previous event message; in this case, the time-synchronization information in the
 34 current event message refers to the previous event message.

35 **10.2.2.2.4 sourcePortIdentity (PortIdentity)**

36 The sourcePortIdentity is the value of the sourcePortIdentity of the time-synchronization event message
 37 received by this PTP Port. It is the portIdentity of the upstream TimeTransmitterPort that sent the event
 38 message.

39 **10.2.2.2.5 logMessageInterval (Integer8)**

40 The logMessageInterval is the value of the logMessageInterval of the time-synchronization event message
 41 received by this PTP Port. It is the currentLogSyncInterval (see 10.7.2.3) of the upstream
 42 TimeTransmitterPort that sent the event message.

1 10.2.2.2.6 preciseOriginTimestamp (Timestamp)

2 The preciseOriginTimestamp is the sourceTime of the ClockTimeTransmitter entity of the Grandmaster PTP
3 Instance, with any fractional nanoseconds truncated, when the time-synchronization event message was sent
4 by the Grandmaster PTP Instance.

5 10.2.2.2.7 upstreamTxTime (UScaledNs)

6 The upstreamTxTime is given by the following equation:

$$7 \quad \text{upstreamTxTime} = \text{syncEventIngressTimestamp} - \frac{\text{meanLinkDelay}}{\text{neighborRateRatio}}$$

8 where

9 syncEventIngressTimestamp	corresponds to the receipt of the time-synchronization information at the
10	timeReceiver port of this PTP Instance (i.e., at this PTP Port)
11 meanLinkDelay	is defined in 10.2.5.8
12 neighborRateRatio	as defined in 10.2.5.7

13 NOTE—When using Clause 11 MD entity, nrrPdelay is used instead of neighborRateRatio (see 11.2.13.14).

14 10.2.2.2.8 rateRatio (Float64)

15 The rateRatio is the value of rateRatio of the received time-synchronization information. It is equal to the
16 ratio of the frequency of the Grandmaster Clock to the frequency of the LocalClock entity of the PTP
17 Instance at the other end of the link attached to this PTP Port, i.e., the PTP Instance that sent the most
18 recently received time-synchronization event message (see 10.2.8.1.4).

19 10.2.2.2.9 gmTimeBaseIndicator (UInteger16)

20 The gmTimeBaseIndicator is the timeBaseIndicator of the ClockSource entity of the current Grandmaster
21 PTP Instance. It is set equal to the gmTimeBaseIndicator of the received time-synchronization information.

22 10.2.2.2.10 lastGmPhaseChange (ScaledNs)

23 The lastGmPhaseChange is the time of the current Grandmaster Clock minus the time of the previous
24 Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the Grandmaster PTP
25 Instance, or the step change in the time of the current Grandmaster Clock at the time of the most recent
26 gmTimeBaseIndicator change. It is set equal to the lastGmPhaseChange of the received time-
27 synchronization information.

28 10.2.2.2.11 lastGmFreqChange (Float64)

29 The lastGmFreqChange is the fractional frequency offset of the current Grandmaster Clock relative to the
30 previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the
31 Grandmaster PTP Instance, or relative to itself prior to the last change in gmTimeBaseIndicator. It is set
32 equal to the lastGmFreqChange of the received time-synchronization information.

1 10.2.2.3 PortSyncSync

2 10.2.2.3.1 General

3 This structure is sent by the PortSync and ClockTimeTransmitter entities to the SiteSync entity and also
4 from the SiteSync entity to the PortSync and ClockTimeReceiver entities.

5 When sent from the PortSync or ClockTimeTransmitter entity, it provides the SiteSync entity with
6 timeTransmitter clock timing information, timestamp of receipt of a time-synchronization event message
7 compensated for propagation time on the upstream link, and the time at which sync receipt timeout occurs if
8 a subsequent Sync message is not received by then. The information is used by the SiteSync entity to
9 compute the rate ratio of the Grandmaster Clock relative to the local clock and is communicated to the other
10 PortSync entities for use in computing timeTransmitter clock timing information.

11 When sent from the SiteSync entity to the PortSync or ClockTimeReceiver entity, the structure contains
12 information needed to compute the synchronization information that will be included in respective fields of
13 the time-synchronization event and general messages that will be sent and also to compute the synchronized
14 time that the ClockTimeReceiver entity will supply to the ClockTarget entity.

```
15 PortSyncSync {
16     domainNumber,
17     localPortNumber,
18     syncReceiptTimeoutTime,
19     followUpCorrectionField,
20     sourcePortIdentity,
21     logMessageInterval,
22     preciseOriginTimestamp,
23     upstreamTxTime,
24     rateRatio,
25     gmTimeBaseIndicator,
26     lastGmPhaseChange,
27     lastGmFreqChange
28 }
```

29 The parameters of the PortSyncSync structure are defined in the following subclauses for when the structure
30 is sent from the PortSync or ClockTimeTransmitter entity to the SiteSync entity. If the structure is sent from
31 the SiteSync entity to the PortSync or ClockTimeReceiver entity, the member values are copied from the
32 most recently received PortSyncSync structure where the PTP Port that received this structure has PTP Port
33 state of TimeReceiverPort.

34 10.2.2.3.2 domainNumber (UInteger8)

35 This parameter is the domainNumber of the gPTP domain in which this structure is sent.

36 NOTE—The domainNumber member is not essential because the state machines that send and receive this structure
37 are per domain, and each state machine implicitly knows the number of the domain in which it operates.

38 10.2.2.3.3 localPortNumber (UInteger16)

39 If the structure is sent by a PortSync entity, the localPortNumber is the port number of the PTP Port whose
40 PortSync entity sent this structure. If the structure is sent by a ClockTimeTransmitter entity, the
41 localPortNumber is zero.

1 10.2.2.3.4 syncReceiptTimeoutTime (UScaledNs)

2 If the structure is sent by a PortSync entity, the syncReceiptTimeoutTime is the value of the local time
3 (i.e., the free-running, local clock time) at which sync receipt timeout occurs if a subsequent
4 time-synchronization event message is not received by that time. If the structure is sent by a
5 ClockTimeTransmitter entity, the syncReceiptTimeoutTime is 0xFFFFFFFFFFFFFFFF [see item h) in
6 10.2.9.2.1].

7 10.2.2.3.5 followUpCorrectionField (ScaledNs)

8 If the structure is sent by a PortSync entity, the followUpCorrectionField is the value of the
9 followUpCorrectionField member of the MDSyncReceive structure whose receipt caused the sending of this
10 structure (see 10.2.2.2.2). If the structure is sent by a ClockTimeTransmitter entity, the
11 followUpCorrectionField is the sub-nanosecond portion of the ClockTimeTransmitter time.

12 10.2.2.3.6 sourcePortIdentity (PortIdentity)

13 If the structure is sent by a PortSync entity, the sourcePortIdentity is the value of the sourcePortIdentity
14 member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.4).
15 If the structure is sent by a ClockTimeTransmitter entity, the clockIdentity member of the sourcePortIdentity
16 is the clockIdentity of this PTP Instance, and the portNumber member of the sourcePortIdentity is 0.

17 10.2.2.3.7 logMessageInterval (Integer8)

18 If the structure is sent by a PortSync entity, the logMessageInterval is the value of the logMessageInterval
19 member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.5).
20 If the structure is sent by a ClockTimeTransmitter entity, the logMessageInterval is the value of
21 clockTimeTransmitterLogSyncInterval (see 10.7.2.4).

22 10.2.2.3.8 preciseOriginTimestamp (Timestamp)

23 If the structure is sent by a PortSync entity, the preciseOriginTimestamp is the value of the
24 preciseOriginTimestamp member of the MDSyncReceive structure whose receipt caused the sending of this
25 structure (see 10.2.2.2.6). If the structure is sent by a ClockTimeTransmitter entity, the
26 preciseOriginTimestamp is the ClockTimeTransmitter time truncated to the next lower nanosecond.

27 10.2.2.3.9 upstreamTxTime (UScaledNs)

28 If the structure is sent by a PortSync entity, the upstreamTxTime is the value of the upstreamTxTime
29 member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.7).
30 If the structure is sent by a ClockTimeTransmitter entity, the upstreamTxTime is the local clock time
31 corresponding to the ClockTimeTransmitter time.

32 10.2.2.3.10 rateRatio (Float64)

33 If the structure is sent by a PortSync entity, the rateRatio is the value of the rateRatio member of the
34 MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.8). It is equal to
35 the ratio of the frequency of the Grandmaster Clock to the frequency of the LocalClock entity of the PTP
36 Instance at the other end of the link attached to this PTP Port, i.e., the PTP Instance that sent the most
37 recently-received time-synchronization event message (see 10.2.8.1.4). If the structure is sent by a
38 ClockTimeTransmitter entity, the rateRatio is equal to gmRateRatio (see 10.2.4.14).

1 10.2.2.3.11 gmTimeBaseIndicator (UInteger16)

2 If the structure is sent by a PortSync entity, the gmTimeBaseIndicator is the value of the
3 gmTimeBaseIndicator member of the MDSyncReceive structure whose receipt caused the sending of this
4 structure (see 10.2.2.2.9). If the structure is sent by a ClockTimeTransmitter entity, the
5 gmTimeBaseIndicator is equal to clockSourceTimeBaseIndicator (see 10.2.4.8).

6 10.2.2.3.12 lastGmPhaseChange (ScaledNs)

7 If the structure is sent by a PortSync entity, the lastGmPhaseChange is the value of the lastGmPhaseChange
8 member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.9).
9 If the structure is sent by a ClockTimeTransmitter entity, the lastGmPhaseChange is equal to
10 clockSourcePhaseOffset (see 10.2.4.7).

11 10.2.2.3.13 lastGmFreqChange (Float64)

12 If the structure is sent by a PortSync entity, the lastGmFreqChange is the value of the lastGmFreqChange
13 member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.9).
14 If the structure is sent by a ClockTimeTransmitter entity, the lastGmFreqChange is equal to
15 clockSourceFreqOffset (see 10.2.4.6).

16 10.2.3 Overview of global variables used by time synchronization state machines

17 Subclauses 10.2.4 and 10.2.5 define global variables used by time synchronization state machines whose
18 scopes are as follows:

- 19 — Per PTP Instance (i.e., per domain)
- 20 — Per PTP Instance, per PTP Port
- 21 — Instance used by the CMLDS (see 11.2.17) (i.e., variable is common across all Link Ports)
- 22 — Instance used by CMLDS, per Link Port

1 Table 10-1 summarizes the scope of each global variable of 10.2.4 and 10.2.5.

Table 10-1—Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5)

Variable name	Subclause of definition	Data Type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
BEGIN	10.2.4.1	Boolean	Per PTP Instance	Yes	No
clockTimeTransmitterSyncInterval	10.2.4.2	UScaledNs	Per PTP Instance	No	No
clockTimeReceiverTime	10.2.4.3	ExtendedTimestamp	Per PTP Instance	No	No
syncReceiptTime	10.2.4.4	ExtendedTimestamp	Per PTP Instance	No	No
syncReceiptLocalTime	10.2.4.5	UScaledNs	Per PTP Instance	No	No
clockSourceFreqOffset	10.2.4.6	Float64	Per PTP Instance	No	No
clockSourcePhaseOffset	10.2.4.7	ScaledNs	Per PTP Instance	No	No
clockSourceTimeBaseIndicator	10.2.4.8	UInteger16	Per PTP Instance	No	No
clockSourceTimeBaseIndicatorOld	10.2.4.9	UInteger16	Per PTP Instance	No	No
clockSourceLastGmPhaseChange	10.2.4.10	ScaledNs	Per PTP Instance	No	No
clockSourceLastGmFreqChange	10.2.4.11	Float64	Per PTP Instance	No	No
currentTime	10.2.4.12	UScaledNs	Per PTP Instance	No	No
gmPresent	10.2.4.13	Boolean	Per PTP Instance	No	No
gmRateRatio	10.2.4.14	Float64	Per PTP Instance	No	No
gmTimeBaseIndicator	10.2.4.15	UInteger16	Per PTP Instance	No	No
lastGmPhaseChange	10.2.4.16	ScaledNs	Per PTP Instance	No	No
lastGmFreqChange	10.2.4.17	Float64	Per PTP Instance	No	No
localClockTickInterval	10.2.4.18	TimeInterval	Per PTP Instance	No	No
localTime	10.2.4.19	UScaledNs	Per PTP Instance	No	No

Table 10-1—Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5) (continued)

Variable name	Subclause of definition	Data Type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
selectedState	10.2.4.20	Enumeration2 array	Per PTP Instance	No	No
timeTransmitterTime	10.2.4.21	ExtendedTimestamp	Per PTP Instance	No	No
thisClock	10.2.4.22	ClockIdentity	Per PTP Instance	Yes	No
parentLogSyncInterval	10.2.4.23	Integer8	Per PTP Instance	No	No
instanceEnable	10.2.4.24	Boolean	Per PTP Instance	No	No
syncReceiptTimeoutTime	10.2.4.25	UScaledNs	Per PTP Instance	No	No
asCapable	10.2.5.1	Boolean	Per PTP Port	No	No
asymmetryMeasurementMode ^a	10.2.5.2	Boolean	Per PTP Port	No	Yes
syncReceiptTimeoutTimeInterval	10.2.5.3	UScaledNs	Per PTP Port	No	No
currentLogSyncInterval	10.2.5.4	Integer8	Per PTP Port	No	No
initialLogSyncInterval	10.2.5.5	Integer8	Per PTP Port	No	No
syncInterval	10.2.5.6	UScaledNs	Per PTP Port	No	No
neighborRateRatio	10.2.5.7	Float64	Per PTP Port	No	No
meanLinkDelay	10.2.5.8	ScaledNs	Per PTP Port	No	Yes
delayAsymmetry	10.2.5.9	ScaledNs	Per PTP Port	No	Yes
computeNeighborRateRatio	10.2.5.10	Boolean	Per PTP Port	No	Yes
computeMeanLinkDelay	10.2.5.11	Boolean	Per PTP Port	No	Yes
portOper ^a	10.2.5.12	Boolean	Per PTP Port	No	Yes
ptpPortEnabled	10.2.5.13	Boolean	Per PTP Port	No	No
thisPort	10.2.5.14	UInteger16	Per PTP Port	No	Yes
syncLocked	10.2.5.15	Boolean	Per PTP Port	No	No
neighborGptpCapable	10.2.5.16	Boolean	Per PTP Port	No	No
syncSlowdown	10.2.5.17	Boolean	Per PTP Port	No	No
oldSyncInterval	10.2.5.18	UScaledNs	Per PTP Port	No	No
gPtpCapableMessageSlowdown	10.2.5.19	Boolean	Per PTP Port	No	No
gPtpCapableMessageInterval	10.2.5.20	UScaledNs	Per PTP Port	No	No
oldGptpCapableMessageInterval	10.2.5.21	UScaledNs	Per PTP Port	No	No
currentLogGptpCapableMessageInterval	10.2.5.22	Integer8	Per PTP Port	No	No

Table 10-1—Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5) (continued)

Variable name	Subclause of definition	Data Type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
initialLogGtpCapableMessageInterval	10.2.5.23	Integer8	Per PTP Port	No	No
syncGrandmasterIdentity	10.2.4.26	ClockIdentity	Per PTP Instance	No	No
syncStepsRemoved	10.2.4.27	UInteger16	Per PTP Instance	No	No
driftTrackingTlvSupport	10.2.4.28	Boolean	Per PTP Instance	No	No
rcvdPSSyncCSS	10.2.4.29	Boolean	Per PTP Instance	No	No
rcvdLocalClockTickCSS	10.2.4.30	Boolean	Per PTP Instance	No	No
rateRatioDrift	10.2.4.31	Integer32	Per PTP Instance	No	No

^a There is one instance of this variable per physical port, which is accessible by all PTP Ports and Link Ports associated with the physical port.

1 10.2.4 Per PTP Instance global variables

2 **10.2.4.1 BEGIN:** A Boolean controlled by the system initialization. If BEGIN is true, all state machines,
3 including per-PTP Port state machines, continuously execute their initial state. See Annex E of
4 IEEE Std 802.1Q-2022.

5 **10.2.4.2 clockTimeTransmitterSyncInterval:** A variable containing the mean time interval between
6 successive messages providing time-synchronization information by the ClockTimeTransmitter entity to the
7 SiteSync entity. This value is given by $1000000000 \times 2^{\text{clockTimeTransmitterLogSyncInterval}}$ ns, where
8 clockTimeTransmitterLogSyncInterval is the logarithm to base 2 of the mean time between the successive
9 providing of time-synchronization information by the ClockTimeTransmitter entity (see 10.7.2.4).

10 **10.2.4.3 ClockTimeReceiverTime:** The synchronized time maintained, at the timeReceiver, at the
11 granularity of the LocalClock entity [i.e., a new value is computed every localClockTickInterval (see
12 10.2.4.18) by the ClockTimeReceiver entity].

13 **10.2.4.4 syncReceiptTime:** The synchronized time computed by the ClockTimeReceiver entity at the
14 instant time-synchronization information, contained in a PortSyncSync structure, is received.

15 **10.2.4.5 syncReceiptLocalTime:** The value of currentTime (i.e., the time relative to the LocalClock entity)
16 corresponding to syncReceiptTime.

17 **10.2.4.6 clockSourceFreqOffset:** The fractional frequency offset of the ClockSource entity frequency
18 relative to the current Grandmaster Clock frequency.

1 **10.2.4.7 clockSourcePhaseOffset:** The time provided by the ClockSource entity, minus the synchronized
2 time.

3 **10.2.4.8 clockSourceTimeBaseIndicator:** A global variable that is set equal to the timeBaseIndicator
4 parameter of the ClockSourceTime.invoke application interface function (see 9.2.2.3), by the
5 ClockTimeTransmitter entity. The parameter timeBaseIndicator of ClockSourceTime.invoke is set by the
6 ClockSource entity and is changed by that entity whenever the time base changes.

7 **10.2.4.9 clockSourceTimeBaseIndicatorOld:** A global variable that is set equal to the previous value of
8 clockSourceTimeBaseIndicator.

9 **10.2.4.10 clockSourceLastGmPhaseChange:** A global variable that is set equal to the lastGmPhaseChange
10 parameter of the ClockSourceTime.invoke application interface function (see 9.2.2.4). That parameter is set
11 by the ClockSource entity and is changed by that entity whenever the time base changes.

12 **10.2.4.11 clockSourceLastGmFreqChange:** A global variable that is set equal to the lastGmFreqChange
13 parameter of the ClockSourceTime.invoke application interface function (see 9.2.2.5). That parameter is set
14 by the ClockSource entity and is changed by that entity whenever the time base changes.

15 **10.2.4.12 currentTime:** The current value of time relative to the LocalClock entity clock.

16 **10.2.4.13 gmPresent:** A Boolean that indicates whether a grandmaster-capable PTP Instance is present in
17 the domain. If TRUE, a grandmaster-capable PTP Instance is present; if FALSE, a grandmaster-capable PTP
18 Instance is not present.

19 **10.2.4.14 gmRateRatio:** The measured ratio of the frequency of the ClockSource entity to the frequency of
20 the LocalClock entity.

21 **10.2.4.15 gmTimeBaseIndicator:** The most recent value of gmTimeBaseIndicator provided to the
22 ClockTimeReceiverSync state machine via a PortSyncSync structure. The data type for
23 gmTimeBaseIndicator is UInteger16.

24 **10.2.4.16 lastGmPhaseChange:** The most recent value of lastGmPhaseChange provided to the
25 ClockTimeReceiverSync state machine via a PortSyncSync structure.

26 **10.2.4.17 lastGmFreqChange:** The most recent value of lastGmFreqChange provided to the
27 ClockTimeReceiverSync state machine via a PortSyncSync structure.

28 **10.2.4.18 localClockTickInterval:** The time interval between two successive significant instants (i.e.,
29 “ticks”) of the LocalClock entity.

30 **10.2.4.19 localTime:** The value of currentTime when the most recent ClockSourceTime.invoke function
31 (see 9.2) was received from the ClockSource entity, or when the LocalClock entity most recently updated its
32 time.

33 **10.2.4.20 selectedState:** An Enumeration2 array of length numberPorts+1 (see 8.6.2.8). selectedState[j] is
34 set equal to the PTP Port State (see Table 10-2) of the PTP Port whose portList index is j.

35 **10.2.4.21 timeTransmitterTime:** The time maintained by the ClockTimeTransmitter entity, based on
36 information received from the ClockSource and LocalClock entities.

37 **10.2.4.22 thisClock:** The clockIdentity of the current PTP Instance.

1 **10.2.4.23 parentLogSyncInterval:** The most recent logMessageInterval value received on the
 2 timeReceiver port. If this PTP Instance is the Grandmaster PTP Instance, then this is the
 3 clockTimeTransmitterLogSyncInterval (see 10.7.2.4).

4 **10.2.4.24 instanceEnable:** A per-domain Boolean used to enable gPTP on all ports that are enabled for that
 5 domain (i.e., ports for which portOper and ptpPortEnabled are both TRUE). Setting instanceEnable to
 6 FALSE causes all per-domain state machines to go to the initial state.

7 NOTE—instanceEnable has no effect on the operation of the MDPdelayReq (see 11.2.19) and MDPdelayResp (see
 8 11.2.20) state machines because those state machines are not per domain (i.e., there is a single instance of each of those
 9 state machines, per link, for all domains).

10 **10.2.4.25 syncReceiptTimeoutTime:** The value of the syncReceiptTimeoutTime member of the most
 11 recently received PortSyncSync structure.

12 **10.2.4.26 syncGrandmasterIdentity:** The clockIdentity carried in the syncGrandmasterIdentity field of
 13 the Drift_Tracking TLV carried by the most recently received Sync message (twoStep flag FALSE) or
 14 Follow_Up message (twoStep flag TRUE). If the received Sync or Follow_Up message does not carry a
 15 Drift_Tracking TLV, syncGrandmasterIdentity is set to the null value 0xFFFF FFFF FFFF FFFF (see the
 16 section “Unassigned and NULL EUI values” of the IEEE Registration Authority tutorial “Guidelines for
 17 Use of Extended Unique Identifier (EUI), Organizationally Unique Identifier (OUI), and Company ID
 18 (CID)” [B5a]).

19 **10.2.4.27 syncStepsRemoved:** The value of the syncStepsRemoved field of the Drift_Tracking TLV carried
 20 by the most recently received Sync message (twoStep flag FALSE) or Follow_Up message (twoStep flag
 21 TRUE). If the received Sync or Follow_Up message does not carry a Drift_Tracking TLV,
 22 syncStepsRemoved is set to 0xFFFF.

23 **10.2.4.28 driftTrackingTlvSupport:** An indicator of whether the PTP Instance supports the Drift_Tracking
 24 TLV and the feature is enabled. The value is TRUE if the Drift_Tracking TLV is supported and the managed
 25 object driftTrackingTlvSupportEnabled (see 14.9) is TRUE. The value is FALSE if the Drift_Tracking TLV
 26 is not supported, or if the TLV is supported and the managed object driftTrackingTlvSupport is FALSE.

27 NOTE—The Drift_Tracking TLV is transported only on full-duplex, point-to-point links as specified in Clause 11. Even
 28 if driftTrackingTlvSupport is TRUE, the Drift Tracking TLV is not transported on links other than full-duplex,
 29 point-to-point links, and is not received by the PTP Ports at the other end of these links.

30 **10.2.4.29 rcvdPSSyncCSS:** A Boolean variable that is set to TRUE when a PortSyncSync structure is
 31 received from the SiteSyncSync state machine of the SiteSync entity by the ClockTimeReceiverSync state
 32 machine. This variable is reset by the ClockTimeReceiverSync state machine.

33 **10.2.4.30 rcvdLocalClockTickCSS:** A Boolean variable that is set to TRUE when the LocalClock entity
 34 updates its time. This variable is reset by the ClockTimeReceiverSync state machine.

35 **10.2.4.31 rateRatioDrift:** The value of the rateRatioDrift field of the Drift_Tracking TLV carried by the
 36 most recently received Sync message (twoStep flag FALSE) or Follow_Up message (twoStep flag TRUE).
 37 If the received Sync or Follow_Up message does not carry a Drift_Tracking TLV, rateRatioDrift is set to
 38 0xFFFFFFFF.

39 10.2.5 Per-port global variables

40 **10.2.5.1 asCapable:** A Boolean that is TRUE if and only if it is determined that this PTP Instance and the
 41 PTP Instance at the other end of the link attached to this PTP Port can interoperate with each other via the
 42 IEEE 802.1AS protocol. As a result,

- 43 a) This PTP Instance is capable of executing the IEEE 802.1AS protocol,

- 1 b) The PTP Instance at the other end of the link is capable of executing the IEEE 802.1AS protocol,
- 2 and
- 3 c) There are no non-IEEE-802.1AS systems in between this PTP Instance and the PTP Instance at the
- 4 other end of the link that introduce sufficient impairments that the end-to-end time-synchronization
- 5 performance of B.3 cannot be met.

6 The determination of asCapable is different for each medium and is described in the respective media-
7 dependent clauses.

8 NOTE—The per-port global variable asCapableAcrossDomains (see 11.2.13.13) is common across, and accessible by,
9 all the domains. It is computed by the MDPdelayReq state machine (see 11.2.19). For full-duplex point-to-point links
10 (see 11), asCapableAcrossDomains is used when setting the instance of asCapable for each domain (for the link in
11 question).

12 **10.2.5.2 asymmetryMeasurementMode:** A Boolean that contains the value of the managed object
13 asymmetryMeasurementMode (see 14.10.45). For full-duplex IEEE 802.3 media, the value is TRUE if an
14 asymmetry measurement is being performed for the link attached to this port and FALSE otherwise. For all
15 other media, the value is FALSE. There is one instance of this variable for all the domains, i.e., all the PTP
16 Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The
17 variable is accessible by all the domains.

18 **10.2.5.3 syncReceiptTimeoutTimeInterval:** The time interval after which sync receipt timeout occurs if
19 time-synchronization information has not been received during the interval. The value of
20 syncReceiptTimeoutTimeInterval is equal to syncReceiptTimeout (see 10.7.3.1) multiplied by the
21 syncInterval (see 10.2.5.6) for the PTP Port at the other end of the link to which this PTP Port is attached.
22 The value of syncInterval for the PTP Port at the other end of the link is computed from logMessageInterval
23 of the received Sync message (see 10.6.2.2.14).

24 **10.2.5.4 currentLogSyncInterval:** The current value of the logarithm to base 2 of the mean time interval, in
25 seconds, between the sending of successive time-synchronization event messages (see 10.7.2.3). This value
26 is set in the SyncIntervalSetting state machine (see 10.3.18).

27 **10.2.5.5 initialLogSyncInterval:** The initial value of the logarithm to base 2 of the mean time interval, in
28 seconds, between the sending of successive time-synchronization event messages (see 10.7.2.3).

29 **10.2.5.6 syncInterval:** A variable containing the mean time-synchronization event message transmission
30 interval for the PTP Port. This value is set in the SyncIntervalSetting state machine (see 10.3.18).

31 **10.2.5.7 neighborRateRatio:** The measured ratio of the frequency of the LocalClock entity of the time-
32 aware system at the other end of the link attached to this port, to the frequency of the LocalClock entity of
33 this time-aware system. There is one instance of this variable for each domain, i.e., each PTP Instance (per
34 port).

35 **10.2.5.8 meanLinkDelay:** The measured mean propagation delay (see 8.3) on the link attached to this port,
36 relative to the LocalClock entity of the time-aware system at the other end of the link (i.e., expressed in the
37 time base of the time-aware system at the other end of the link). There is one instance of this variable for all
38 the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if
39 domain 0 is implemented. The variable is accessible by all the domains.

40 NOTE—The variable meanLinkDelay was named neighborPropDelay in the 2011 edition of this standard.

10.2.5.9 delayAsymmetry: The asymmetry in the propagation delay on the link attached to this port. If propagation delay asymmetry is not modeled, then delayAsymmetry is zero. There is one instance of this variable for CMLDS (see 11.2.17), and there is also one instance of this variable for each domain. The instance of this variable for CMLDS is relative to the local clock. The instance of this variable for each domain is relative to the grandmaster time base for that domain. The instance of delayAsymmetry for CMLDS is used where needed in all computations done by the CMLDS, and also by computations using the syncEgressTimestamp field of the Drift_Tracking TLV (see 11.4.4.2.3) if CMLDS is present. The instance of delayAsymmetry for a domain is used where needed in all computations done for that domain, and also by computations using the Drift_Tracking TLV if CMLDS is not present.

10.2.5.10 computeNeighborRateRatio: A Boolean, set by the LinkDelayIntervalSetting state machine (see 11.2.21), that indicates whether nrrPdelay is computed for this port. There is one instance of this variable for each domain, i.e., each PTP Instance (per port), and one instance of this variable for CMLDS. If the instance of this variable for a domain is TRUE, nrrPdelay for that domain is computed.

10.2.5.11 computeMeanLinkDelay: A Boolean, set by the LinkDelayIntervalSetting state machine (see 11.2.21), that indicates whether meanLinkDelay is to be computed by this port. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

10.2.5.12 portOper: A Boolean that is TRUE if and only if the port is up and able to send and receive messages. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port). The variable is accessible by all the domains. The term *port* in this definition is a physical port.

NOTE 1—portOper is an indicator, and not a control, and reflects the operational status of the underlying medium. It is not administratively set by gPTP.

NOTE 2—The variable portOper corresponds to the variable portEnabled in the 2011 edition of this standard. The change is reflected in many state machines.

NOTE 3—portOper is the same as MAC_Operational (see IEEE Std 802.1AC-2016).

NOTE 4—When portOper is referenced for a logical port, the reference is to portOper for the physical port that corresponds to the logical port.

10.2.5.13 ptpPortEnabled: A Boolean that is administratively set to TRUE if time-synchronization is to be enabled on this PTP Port.

NOTE 1—It is expected that the value of ptpPortEnabled is set via the management interface (see 14.10.4). A physical port can be enabled for data transport but not for synchronization transport.

NOTE 2—The variable ptpPortEnabled was named pttPortEnabled in the 2011 edition of this standard. Only the name of this variable has changed; the definition and function of this variable are the same as in the 2011 edition. The name change is reflected in many state machines.

10.2.5.14 thisPort: The portNumber of the current port.

10.2.5.15 syncLocked: A Boolean, set by the PortSyncSyncSend state machine (see 10.2.12.3), that indicates that this PTP Port, when operating as a timeTransmitter port, shall transmit a Sync as soon as possible after the timeReceiver port received a Sync (ignoring syncInterval). If FALSE, the PTP Port shall use the timing set by syncInterval.

10.2.5.16 neighborGptpCapable: A Boolean, set by the GptpCapableReceive state machine (see 10.4.3), that indicates that the neighbor of this PTP Port (i.e., the PTP Port at the other end of the link attached to this PTP Port) is capable of invoking gPTP.

10.2.5.17 syncSlowdown: A Boolean that is set to TRUE if the SyncIntervalSetting state machine (see Figure 10-20 in 10.3.18.3) receives a TLV that requests a larger sync interval (see 10.7.2.3) and FALSE otherwise. When syncSlowdown is set to TRUE, the PortSyncSyncSend state machine (see Figure 10-8) continues to send time synchronization event messages (see 11.4.3, 12.1, 12.2, and 13.3.1) at the old (i.e., faster) rate until the number of time synchronization event messages equal to syncReceiptTimeout (see 10.7.3.1) have been sent, but with the respective time synchronization event message transmission interval field (see 11.4.2.9, 12.7, Figure 12-8, and 13.3.1.2.10) of the time synchronization event message set equal to the new sync interval (i.e., corresponding to the slower rate). After syncReceiptTimeout Sync messages have been sent, subsequent time synchronization event messages are sent at the new (i.e., slower) rate and with the respective time synchronization event message transmission interval field of the time synchronization event message set to the new sync interval. When syncSlowdown is set to FALSE, the PortSyncSyncSend state machine immediately sends time synchronization event messages at the new (i.e., faster or the same) rate.

NOTE—If a receiver of time synchronization event messages (see 11.4.3, 12.1, 12.2, and 13.3.1) requests a slower rate, the receiver will continue to use the upstream sync interval value, which it obtains from the respective time synchronization event message transmission interval field (see 11.4.2.9, 12.7, Figure 12-8, and 13.3.1.2.10) of the received time synchronization event message, until it receives a time synchronization event message where that value has changed. If, immediately after requesting a slower time synchronization event message rate, up to syncReceiptTimeout consecutive time synchronization event messages sent to the receiver are lost, sync receipt timeout could occur if the sender had changed to the slower rate immediately. Delaying the slowing down of the sending rate of time synchronization event messages for syncReceiptTimeout messages prevents this timeout from happening.

10.2.5.18 oldSyncInterval: The saved value of the previous sync interval, when a new time-synchronization event message transmission interval is requested via a Signaling message that contains a message interval request TLV.

10.2.5.19 gPtpCapableMessageSlowdown: A Boolean that is set to TRUE if the GtpCapableIntervalSetting state machine (see Figure 10-19 in 10.3.17.3) receives a TLV that requests a larger gPTP-capable message interval (see 10.7.2.5) and FALSE otherwise. When gPtpCapableMessageSlowdown is set to TRUE, the GtpCapableTransmit state machine (see Figure 10-21 in 10.4.2.3) continues to send Signaling messages containing the gPTP-capable TLV at the old (i.e., faster) rate until a number of Signaling messages containing the gPTP-capable TLV, equal to gPtpCapableReceiptTimeout (see 10.7.3.3), have been sent, but with the logGtpCapableMessageInterval field of the gPTP-capable TLV (see 10.6.4.5.6) set equal to the new gPTP-capable message interval (i.e., corresponding to the slower rate). After gPtpCapableReceiptTimeout Signaling messages containing the gPTP-capable TLV have been sent, subsequent such Signaling messages are sent at the new (i.e., slower) rate and with the logGtpCapableMessageInterval field of the gPTP-capable TLV set to the new gPTP-capable message interval. When gPtpCapableSlowdown is set to FALSE, the GtpCapableTransmit state machine immediately sends Signaling messages containing the gPTP-capable TLV at the new (i.e., faster or the same) rate.

NOTE—If a receiver of Signaling messages containing the gPTP-capable TLV requests a slower rate, the receiver will continue to use the old gPTP-capable message interval value in determining, via the GtpCapableReceive state machine (see 10.4.3), if its neighbor is no longer capable of invoking gPTP, until it has received gPtpCapableReceiptTimeout such Signaling messages. If, immediately after requesting a slower rate, up to gPtpCapableReceiptTimeout consecutive Signaling messages, containing the gPTP-capable TLV, sent to the receiver are lost, a declaration that the sender is no longer capable of invoking gPTP could occur if the sender had changed to the slower rate immediately. Delaying the slowing down of the sending rate of Signaling messages containing the gPTP-capable TLV for gPtpCapableReceiptTimeout messages prevents this timeout from happening.

10.2.5.20 gPtpCapableMessageInterval: A variable whose value is the mean time, in seconds, between the sending of successive Signaling messages that carry the gPTP-capable TLV (see 10.7.2.5 and 10.6.4.4).

10.2.5.21 oldGtpCapableMessageInterval: The saved value of the previous interval between the sending of successive Signaling messages that carry the gPTP-capable TLV (see 10.2.5.20), when a new such interval is requested via a Signaling message that contains a gPTP-capable message interval request TLV.

1 **10.2.5.22 currentLogGtpCapableMessageInterval:** The current value of the logarithm to base 2 of the
2 mean time interval, in seconds, between the sending of successive Signaling messages that carry the gPTP-
3 capable TLV (see 10.6.4.4). This value is set in the GtpCapableIntervalSetting state machine (see 10.4.4).

4 **10.2.5.23 initialLogGtpCapableMessageInterval:** The initial value of the logarithm to base 2 of the
5 mean time interval, in seconds, between the sending of successive Signaling messages that carry the gPTP-
6 capable TLV (see 10.6.4.4).

7 **10.2.6 Function used by multiple state machines**

8 **10.2.6.1 random():** Returns a uniformly-distributed pseudo-random number whose data type is UInteger16
9 (i.e., the function returns a uniformly distributed, pseudo-random integer in the range $[0, 2^{16} - 1]$).

10 **10.2.7 SiteSyncSync state machine**

11 **10.2.7.1 State machine variables**

12 The following variables are used in the state diagram in Figure 10-3 (in 10.2.7.3):

13 **10.2.7.1.1 revdPSSyncSSS:** A Boolean variable that notifies the current state machine when a
14 PortSyncSync structure (see 10.2.2.3) is received from the PortSyncSyncReceive state machine of a
15 PortSync entity or from the ClockTimeTransmitterSyncSend state machine of the ClockTimeTransmitter
16 entity. This variable is reset by this state machine.

17 **10.2.7.1.2 revdPSSyncPtrSSS:** A pointer to the received PortSyncSync structure indicated by
18 revdPSSyncSSS.

19 **10.2.7.1.3 txPSSyncPtrSSS:** A pointer to the PortSyncSync structure transmitted by the state machine.

20 **10.2.7.2 State machine functions**

21 **10.2.7.2.1 setPSSyncSend (revdPSSyncPtrSSS):** Creates a PortSyncSync structure to be transmitted, and
22 returns a pointer to this structure. The members are copied from the received PortSyncSync structure
23 pointed to by revdPSSyncPtrSSS.

24 **10.2.7.2.2 txPSSync (txPSSyncPtrSSS):** Transmits a copy of the PortSyncSync structure pointed to by
25 txPSSyncPtrSSS to the PortSyncSyncSend state machine of each PortSync entity and the
26 ClockTimeReceiverSync state machine of the ClockTimeReceiver entity of this PTP Instance.

27 **10.2.7.3 State diagram**

28 The SiteSyncSync state machine shall implement the function specified by the state diagram in Figure 10-3,
29 the local variables specified in 10.2.7.1, the functions specified in 10.2.7.2, the structure specified in
30 10.2.2.3, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state
31 machine receives time-synchronization information, accumulated rateRatio, and syncReceiptTimeoutTime
32 from the PortSync entity (PortSyncSyncReceive state machine) of the current timeReceiver port or from the
33 ClockTimeTransmitter entity (ClockTimeTransmitterSyncSend state machine). If the information was sent
34 by a PortSync entity, the state machine also receives the portIdentity of the PTP Port on the upstream PTP
35 Instance that sent the information to this PTP Instance (if the information was sent by the
36 ClockTimeTransmitter entity, the portIdentity is that of the ClockTimeTransmitter entity, i.e., it has
37 clockIdentity equal to the clockIdentity of this PTP Instance and portNumber 0). The state machine sends a
38 PortSyncSync structure to the PortSync entities of all the ports and to the ClockTimeReceiver entity.

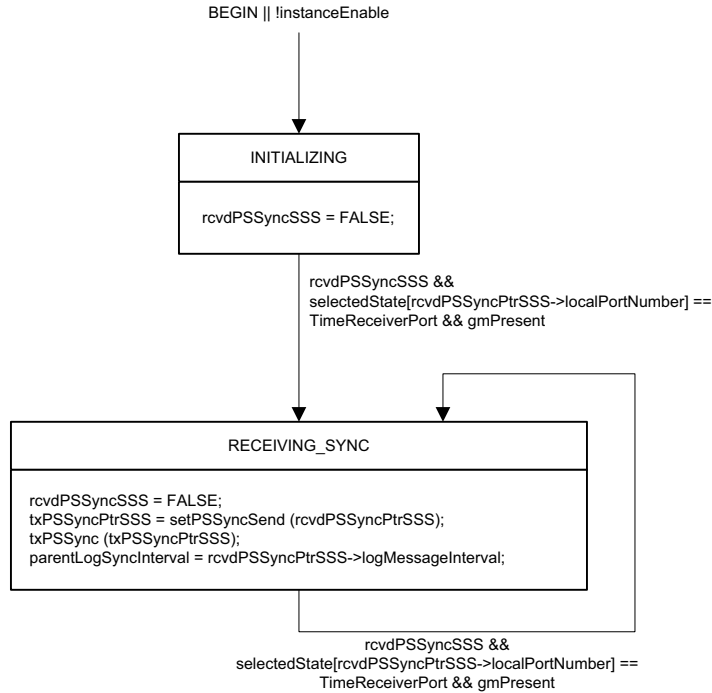


Figure 10-3—SiteSyncSync state machine

1 10.2.8 PortSyncSyncReceive state machine

2 10.2.8.1 State machine variables

3 The following variables are used in the state diagram in Figure 10-4 (in 10.2.8.3):

4 **10.2.8.1.1 rcvdMDSyncPSSR:** A Boolean variable that notifies the current state machine when an
5 MDSyncReceive structure is received from an MD entity of the same PTP Port (see 10.2.2.1). This variable
6 is reset by this state machine.

7 **10.2.8.1.2 rcvdMDSyncPtrPSSR:** A pointer to the received MDSyncReceive structure indicated by
8 rcvdMDSyncPSSR.

9 **10.2.8.1.3 txPSSyncPtrPSSR:** A pointer to the PortSyncSync structure transmitted by the state machine.

10 **10.2.8.1.4 rateRatio:** A Float64 variable that holds the ratio of the frequency of the Grandmaster Clock to
11 the frequency of the LocalClock entity. This frequency ratio is computed by:

- 12 a) Measuring the ratio of the Grandmaster Clock frequency to the LocalClock frequency at the
13 Grandmaster PTP Instance and initializing rateRatio to this value in the
14 ClockTimeTransmitterSyncSend state machine of the Grandmaster PTP Instance and
- 15 b) Accumulating, in the PortSyncSyncReceive state machine of each PTP Instance, the frequency
16 offset of the LocalClock entity of the PTP Instance at the remote end of the link attached to that PTP
17 Port to the frequency of the LocalClock entity of this PTP Instance.

1 **10.2.8.1.5 TEMP:** A temporary variable used to reduce clutter in the state diagram (see Figure 10-4). The
2 data type for TEMP is Integer16.

3 **10.2.8.2 State machine functions**

4 **10.2.8.2.1 setPSSyncPSSR (rcvdMDSyncPtrPSSR syncReceiptTimeoutTimeInterval, rateRatio):**
5 Creates a PortSyncSync structure to be transmitted, and returns a pointer to this structure. The members are
6 set as follows:

- 7 a) localPortNumber is set equal to thisPort.
- 8 b) domainNumber, followUpCorrectionField, sourcePortIdentity, logMessageInterval,
9 preciseOriginTimestamp, and upstreamTxTime are copied from the received MDSyncReceive
10 structure pointed to by rcvdMDSyncPtrPSSR.
- 11 c) syncReceiptTimeoutTime is set equal to currentTime plus syncReceiptTimeoutTimeInterval (see
12 10.2.5.3).
- 13 d) The function argument rateRatio is set equal to the local variable rateRatio (computed just prior to
14 invoking setPSSyncPSSR (see Figure 10-4)). The rateRatio member of the PortSyncSync structure
15 is then set equal to the function argument rateRatio.

16 **10.2.8.2.2 txPSSyncPSSR (txPSSyncPtrPSSR):** Transmits a copy of the PortSyncSync structure pointed to
17 by txPSSyncPtrPSSR to the SiteSyncSync state machine of this PTP Instance.

1 10.2.8.3 State diagram

2 The PortSyncSyncReceive state machine shall implement the function specified by the state diagram in
3 Figure 10-4, the local variables specified in 10.2.8.1, the functions specified in 10.2.8.2, the structures
4 specified in 10.2.2.1 and 10.2.2.3, and the relevant global variables and functions specified in 10.2.4 through
5 10.2.6. The state machine receives time-synchronization information, accumulated rateRatio, and
6 syncReceiptTimeoutTime from the MD entity of the same PTP Port. The state machine adds, to rateRatio,
7 the fractional frequency offset of the LocalClock entity relative to the LocalClock entity of the upstream
8 PTP Instance at the remote end of the link attached to this PTP Port. The state machine computes
9 syncReceiptTimeoutTime. The state machine sends this information to the SiteSync entity (SiteSyncSync
10 state machine).

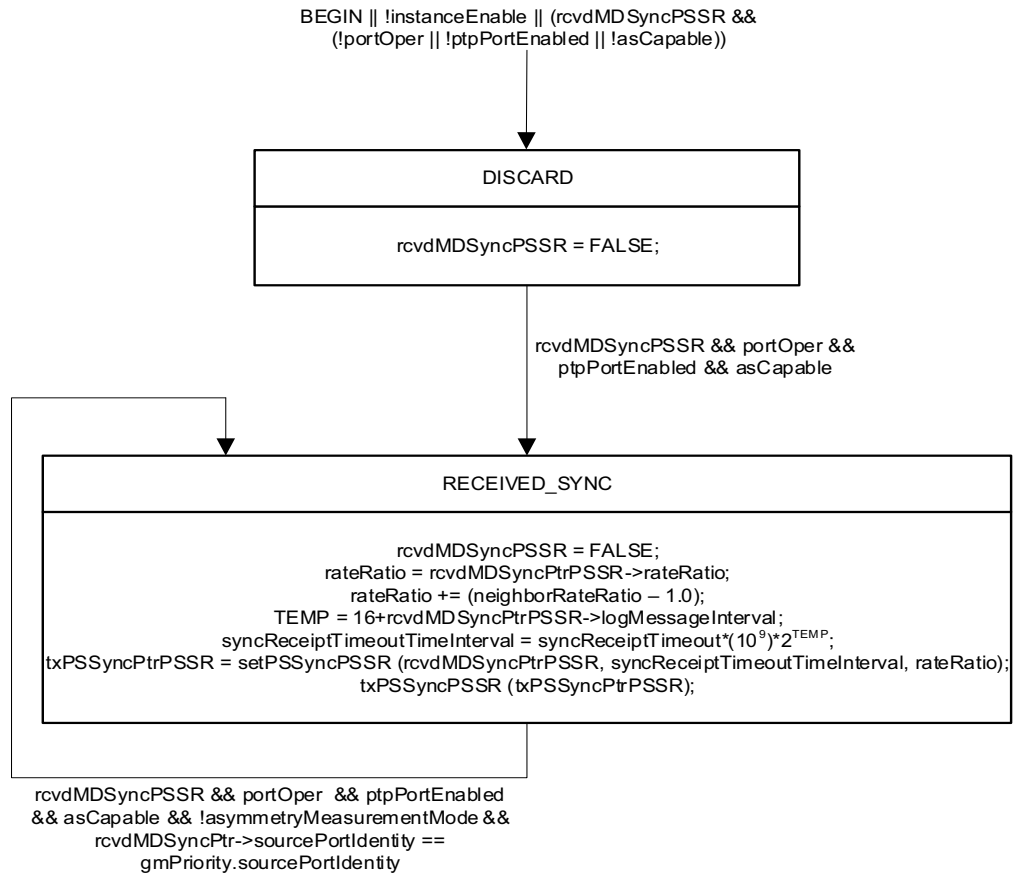


Figure 10-4—PortSyncSyncReceive state machine

1 10.2.9 ClockTimeTransmitterSyncSend state machine

2 10.2.9.1 State machine variables

3 The following variables are used in the state diagram in Figure 10-5 (in 10.2.9.3):

4 **10.2.9.1.1 syncSendTime:** The time in seconds, relative to the LocalClock entity, when synchronization
5 information will next be sent to the SiteSync entity, via a PortSyncSync structure. The data type for
6 syncSendTime is UScaledNs.

7 **10.2.9.1.2 txPSSyncPtrCMSS:** A pointer to the PortSyncSync structure transmitted by the state machine.

8 10.2.9.2 State machine functions

9 **10.2.9.2.1 setPSSyncCMSS (gmRateRatio):** Creates a PortSyncSync structure to be transmitted, and
10 returns a pointer to this structure. The members are set as follows:

- 11 a) localPortNumber is set to 0.
- 12 b) preciseOriginTimestamp is set equal to the timeTransmitterTime, with any fractional nanoseconds
13 truncated.
- 14 c) followUpCorrectionField is set equal to the sum of
 - 15 1) The fractional nanoseconds portion of timeTransmitterTime.fractionalNanoseconds and
 - 16 2) The quantity gmRateRatio × (currentTime – localTime).

17 NOTE 1—Both localTime (10.2.4.19) and timeTransmitterTime (10.2.4.21) are updated by the
18 ClockTimeTransmitterSyncReceive state machine (10.2.11). The updates occur both when sourceTime (9.2.2.2) is
19 received from the ClockSource (9.2) - indicated by the variable rcvdClockSourceReq (10.2.11.1.1) being TRUE - and
20 when the LocalClock entity (10.1.2.1) itself updates by one tick - indicated by the variable rcvdLocalClockTickCMSR
21 (10.2.11.1.3) being TRUE. timeTransmitterTime is updated by updateTimeTransmitterTime() (see 10.2.11.2.2).
22 localTime is updated by setting it equal to currentTime (Figure 10-7). The result is that localTime is equal to the value of
23 currentTime when timeTransmitterTime was most recently updated. localTime, currentTime, and timeTransmitterTime
24 are per PTP Instance global variables. When they are updated, their values are known to all state machines.

25 NOTE 2—It is possible that currentTime (10.2.4.12) and localTime (10.2.4.19) are not equal, despite the statement
26 localTime = currentTime in the RECEIVE_SOURCE_TIME state of the ClockTimeTransmitterSyncReceive state
27 machine (10.2.11). For example, if the ClockTimeTransmitterSyncReceive state machine is implemented as a different
28 module than the ClockTimeTransmitterSyncSend state machine (10.2.9), with timeTransmitterTime (10.2.4.21) and
29 localTime being sent from the ClockTimeTransmitterSyncReceive state machine to the ClockTimeTransmitterSyncSend
30 state machine, then currentTime and localTime will not be equal. In this case, currentTime in c) 2) is the value of the
31 LocalClock entity (10.1.2.1) when the ClockTimeTransmitterSyncSend state machine receives timeTransmitterTime and
32 localTime from the ClockTimeTransmitterSyncReceive state machine.

- 33 d) The clockIdentity member of sourcePortIdentity is set equal to the clockIdentity of this PTP
34 Instance.
- 35 e) The portNumber member of the sourcePortIdentity is set to 0.

36 NOTE 3—This quantity and localPortNumber are redundant; both are retained so that the SiteSync entity can process
37 PortSyncSync structures received from a PortSync entity or the ClockTimeTransmitter entity in the same manner.

- 38 f) logMessageInterval is set to clockTimeTransmitterLogSyncInterval.
- 39 g) upstreamTxTime is set equal to localTime.
- 40 h) syncReceiptTimeoutTime is set equal to 0xFFFFFFFFFFFFFFFF, which indicates that there is no
41 sync receipt timeout.

42 NOTE 4—A ClockTimeTransmitter entity does not receive Sync messages, and there is no notion of sync receipt
43 timeout.

- 1 i) rateRatio is set equal to gmRateRatio.
- 2 j) gmTimeBaseIndicator is set equal to clockSourceTimeBaseIndicator.
- 3 k) lastGmPhaseChange is set equal to clockSourcePhaseOffset.
- 4 l) lastGmFreqChange is set equal to clockSourceFreqOffset.
- 5 m) domainNumber is set equal to the domainNumber of this gPTP domain.

6 **10.2.9.2.2 txPSSyncCMSS (txPSSyncPtrCMSS):** Transmits a copy of the PortSyncSync structure pointed
7 to by txPSSyncPtrCMSS to the SiteSync state machine.

8 **10.2.9.2.3 computeClockTimeTransmitterSyncInterval():** Computes the value of
9 clockTimeTransmitterSyncInterval (see 10.2.4.2) as $1000000000 \times 2^{\text{clockTimeTransmitterLogSyncInterval}}$ ns,
10 where clockTimeTransmitterLogSyncInterval is the minimum currentLogSyncInterval value, taken over all
11 the PTP Ports of the PTP Instance (see 10.7.2.4).

12 **10.2.9.3 State diagram**

13 The ClockTimeTransmitterSyncSend state machine shall implement the function specified by the state
14 diagram in Figure 10-5, the local variables specified in 10.2.9.1, the functions specified in 10.2.9.2, the
15 structure specified in 10.2.2.3, and the relevant global variables and functions specified in 10.2.4 through
16 10.2.6. The state machine receives timeTransmitterTime and clockSourceTimeBaseIndicator from the
17 ClockTimeTransmitterSyncReceive state machine, and phase and frequency offset between
18 timeTransmitterTime and syncReceiptTime from the ClockTimeTransmitterSyncOffset state machine. It
19 provides timeTransmitterTime (i.e., synchronized time) and the phase and frequency offset to the SiteSync
20 entity via a PortSyncSync structure.

21 The ClockTimeTransmitterSyncSend state machine is optional for PTP Instances that are not grandmaster-
22 capable (see 8.6.2.1, 10.1.3, and 10.2.1). This state machine may be present in a PTP Instance that is not
23 grandmaster-capable; however, any information supplied by it to the SiteSyncSync state machine is not used
24 by the SiteSyncSync state machine if the PTP Instance is not grandmaster-capable.

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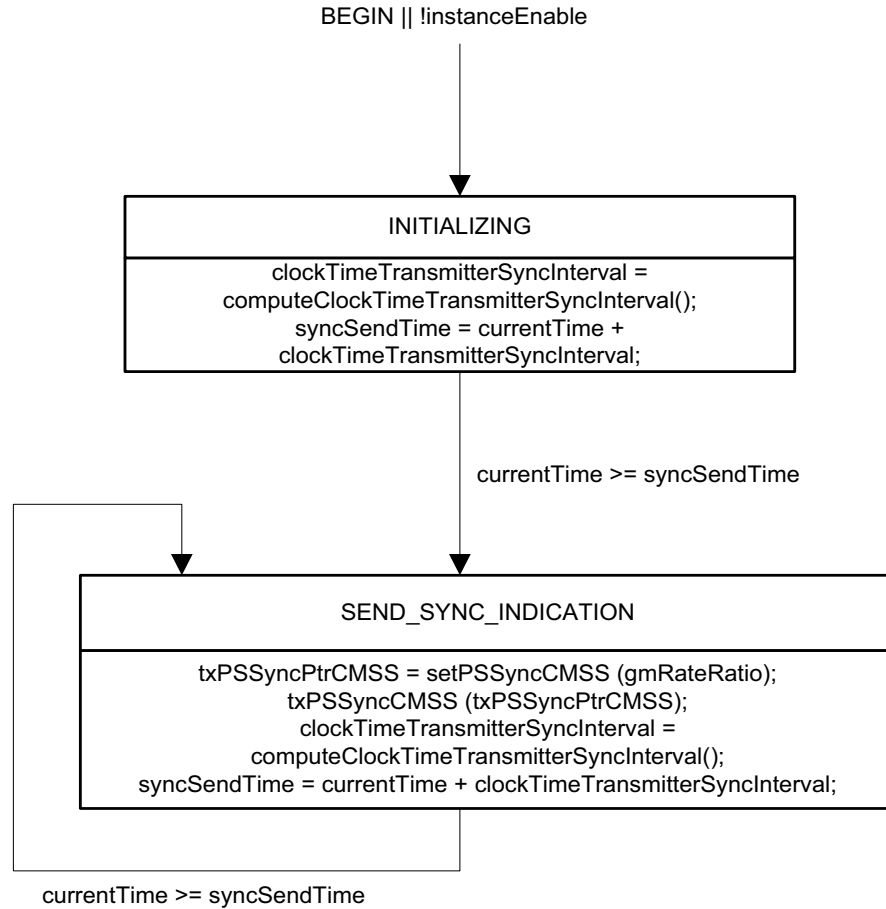


Figure 10-5—ClockTimeTransmitterSyncSend state machine

2 10.2.10 ClockTimeTransmitterSyncOffset state machine

3 10.2.10.1 State machine variables

4 The following variable is used in the state diagram in Figure 10-6 (in 10.2.10.3):

5 **10.2.10.1.1 rcvdSyncReceiptTime:** A Boolean variable that notifies the current state machine that
6 syncReceiptTime has been updated by the ClockTimeReceiver entity. This variable is reset by this state
7 machine.

8 10.2.10.2 State machine functions

9 **10.2.10.2.1 computeClockSourceFreqOffset():** Computes and returns clockSourceFreqOffset (see
10 10.2.4.6), using successive values of timeTransmitterTime computed by the
11 ClockTimeTransmitterSyncReceive state machine (see 10.2.11) and successive values of syncReceiptTime
12 computed by the ClockTimeReceiverSync state machine (see 10.2.13). The data type for the returned value

1 is Float64. Any scheme that uses this information to compute clockSourceFreqOffset is acceptable as long
2 as the performance requirements specified in B.2.4 are met.

3 NOTE—As one example, clockSourceFreqOffset can be estimated as the ratio of the duration of a time interval
4 measured by the ClockSource entity to the duration of the same time interval computed from ClockTimeReceiverTime
5 values, minus 1.

6 10.2.10.3 State diagram

7 The ClockTimeTransmitterSyncOffset state machine shall implement the function specified by the state
8 diagram in Figure 10-6, the local variable specified in 10.2.10.1, the function specified in 10.2.10.2, and the
9 relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives
10 syncReceiptTime from the ClockTimeReceiverSync state machine and timeTransmitterTime from the
11 ClockTimeTransmitterSyncReceive state machine. It computes clockSourcePhaseOffset and
12 clockSourceFrequency offset if this PTP Instance is not currently the Grandmaster PTP Instance, i.e., if
13 selectedState[0] is equal to PassivePort.

14 The ClockTimeTransmitterSyncOffset state machine is optional for PTP Instances that are not grandmaster-
15 capable (see 8.6.2.1, 10.1.3, and 10.2.1). This state machine may be present in a PTP Instance that is not
16 grandmaster-capable; however, any information supplied by it, via the ClockTimeTransmitterSyncSend
17 state machine, to the SiteSyncSync state machine is not used by the SiteSyncSync state machine if the PTP
18 Instance is not grandmaster-capable.

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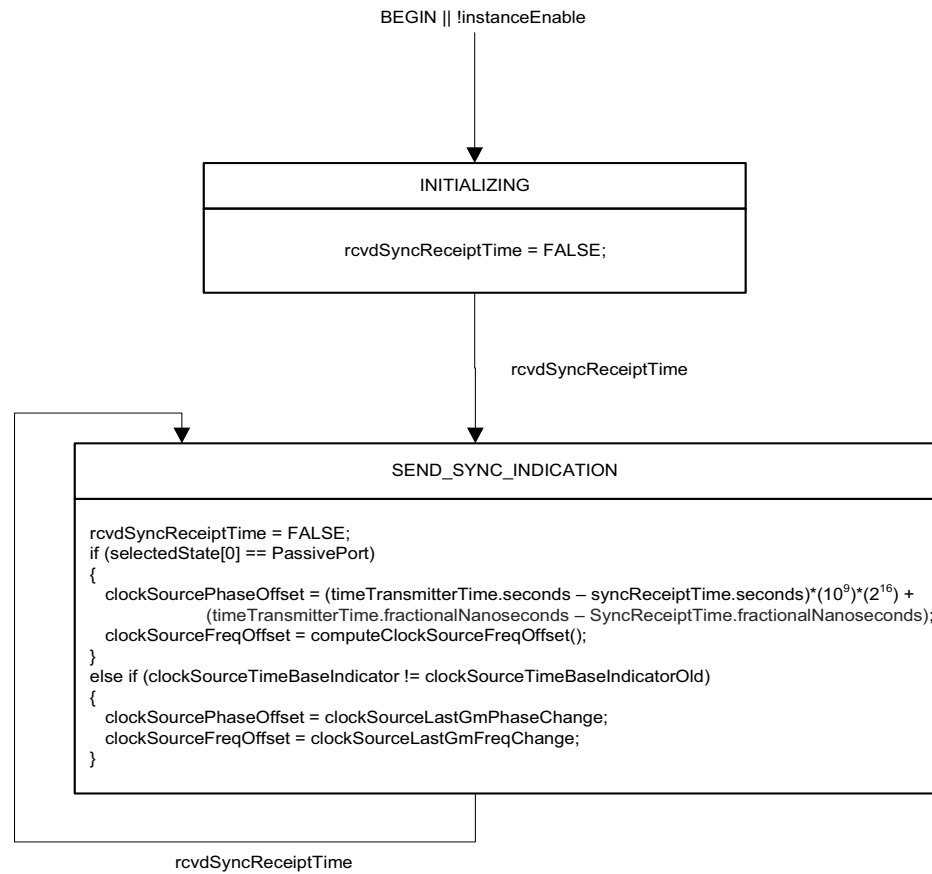


Figure 10-6—ClockTimeTransmitterSyncOffset state machine

2 10.2.11 ClockTimeTransmitterSyncReceive state machine

3 10.2.11.1 State machine variables

4 The following variables are used in the state diagram in Figure 10-7 (in 10.2.11.3):

5 **10.2.11.1.1 rcvdClockSourceReq:** A Boolean variable that notifies the current state machine when
6 sourceTime is received from the ClockSource entity, due to the ClockSourceTime.invoke primitive having
7 been invoked at that entity. This variable is reset by this state machine.

8 **10.2.11.1.2 rcvdClockSourceReqPtr:** A pointer to the received ClockSourceTime.invoke function
9 parameters.

10 **10.2.11.1.3 rcvdLocalClockTickCMSR:** A Boolean variable that notifies the current state machine when
11 the LocalClock entity updates its time. This variable is reset by this state machine.

1 10.2.11.2 State machine functions

2 **10.2.11.2.1 computeGmRateRatio():** Computes gmRateRatio (see 10.2.4.14), using values of sourceTime
3 conveyed by successive ClockSourceTime.invoke functions (see 9.2.2.1), and corresponding values of
4 localTime (see 10.2.4.19). Any scheme that uses this information, along with any other information
5 conveyed by the successive ClockSourceTime.invoke functions and corresponding values of localTime, to
6 compute gmRateRatio is acceptable as long as the performance requirements specified in B.2.4 are met.

7 NOTE—As one example, gmRateRatio can be estimated as the ratio of the elapsed time of the ClockSource entity that
8 supplies time to this PTP Instance, to the elapsed time of the LocalClock entity of this PTP Instance. This ratio can be
9 computed for the time interval between a received ClockSourceTime.invoke function and a second received
10 ClockSourceTime.invoke function some number of ClockSourceTime.invoke functions later, i.e.,

$$11 \quad \frac{\text{ClockSource.invoke.sourceTime}_N - \text{ClockSource.invoke.sourceTime}_0}{\text{localTime}_N - \text{localTime}_0}$$

12 where the successive received ClockSourceTime.invoke functions are indexed from 0 to N , with the first such function
13 indexed as 0, and localTime_j is the value of localTime when the ClockSourceTime.invoke function whose index is j is
14 received. Another example is to use a sliding window for the calculation of gmRateRatio.

15 **10.2.11.2.2 updateTimeTransmitterTime():** Updates the global variable timeTransmitterTime (see
16 10.2.4.21), based on information received from the ClockSource and LocalClock entities. It is the
17 responsibility of the application to filter timeTransmitter times appropriately. As one example,
18 timeTransmitterTime can be set equal to the sourceTime member of the ClockSourceTime.invoke function
19 when this function is invoked at the ClockSource entity and can be incremented by localClockTickInterval
20 (see 10.2.4.18) multiplied by gmRateRatio (see 10.2.4.14) when rcvdLocalClockTickCMSR is TRUE.

21 10.2.11.3 State diagram

22 The ClockTimeTransmitterSyncReceive state machine shall implement the function specified by the state
23 diagram in Figure 10-7, the local variables specified in 10.2.11.1, the functions specified in 10.2.11.2, and
24 the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine updates the
25 global variable timeTransmitterTime with information received from the ClockSource entity via the
26 ClockSourceTime.invoke function and information received from the LocalClock entity. It also computes
27 gmRateRatio, i.e., the ratio of the ClockSource entity frequency and the LocalClock entity frequency.

28 The ClockTimeTransmitterSyncReceive state machine is optional for PTP Instances that are not
29 grandmaster-capable (see 8.6.2.1, 10.1.3, and 10.2.1). This state machine may be present in a PTP Instance
30 that is not grandmaster-capable; however, any information supplied by it, via the
31 ClockTimeTransmitterSyncSend state machine, to the SiteSyncSync state machine is not used by the
32 SiteSyncSync state machine if the PTP Instance is not grandmaster-capable.

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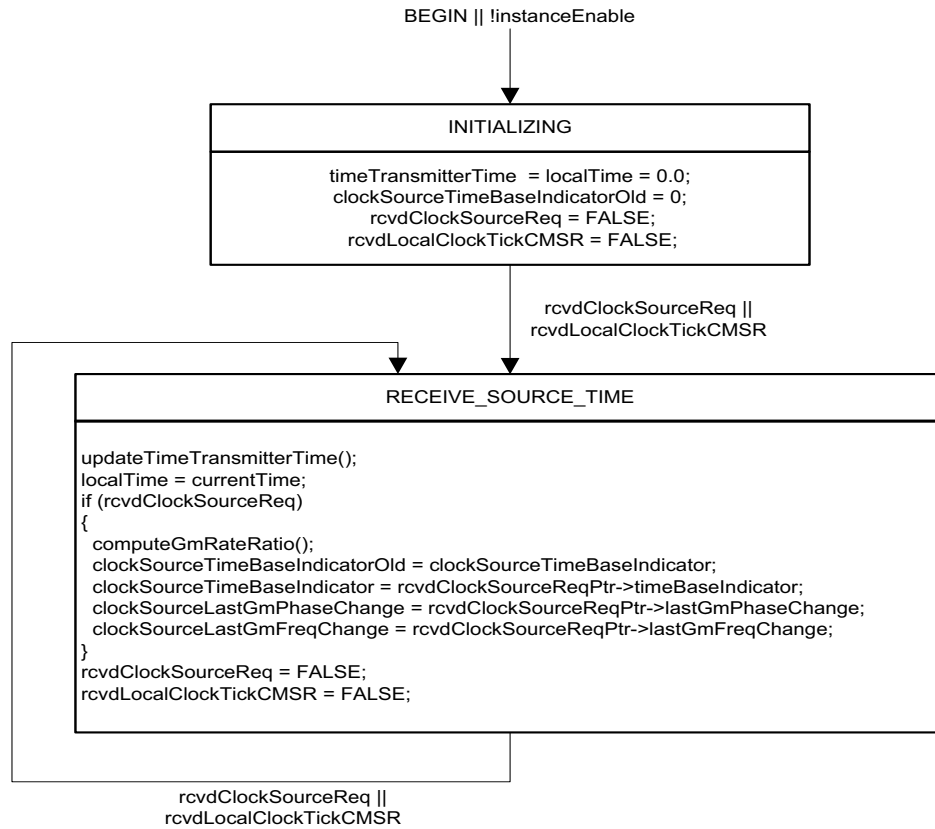


Figure 10-7—ClockTimeTransmitterSyncReceive state machine

2 10.2.12 PortSyncSyncSend state machine

3 10.2.12.1 State machine variables

4 The following variables are used in the state diagram in Figure 10-8 (in 10.2.12.3):

5 **10.2.12.1.1 rcvdPSSyncPSSS:** A Boolean variable that notifies the current state machine when a
6 PortSyncSync structure is received from the SiteSyncSync state machine of the SiteSync entity of the PTP
7 Instance (see 10.2.2.3). This variable is reset by this state machine.

8 **10.2.12.1.2 rcvdPSSyncPtrPSSS:** A pointer to the received PortSyncSync structure indicated by
9 rcvdPSSyncPSSS.

10 **10.2.12.1.3 lastPreciseOriginTimestamp:** The preciseOriginTimestamp member of the most recently
11 received PortSyncSync structure. The data type for lastPreciseOriginTimestamp is Timestamp.

12 **10.2.12.1.4 lastFollowUpCorrectionField:** The followUpCorrectionField member of the most recently
13 received PortSyncSync structure. The data type for lastFollowUpCorrectionField is ScaledNs.

1 **10.2.12.1.5 lastRateRatio:** The rateRatio member of the most recently received PortSyncSync structure.
 2 The data type for lastRateRatio is Float64.

3 **10.2.12.1.6 lastUpstreamTxTime:** The upstreamTxTime member of the most recently received
 4 PortSyncSync structure. The data type for lastUpstreamTxTime is UScaledNs.

5 **10.2.12.1.7 lastSyncSentTime:** The value of currentTime (i.e., the time relative to the LocalClock entity)
 6 when the most recent MDSyncSend structure was sent. The data type for lastSyncSentTime is UScaledNs.

7 NOTE—lastSyncSentTime is the time the abstract MDSyncSend structure was sent, NOT the time the corresponding
 8 Sync message (or equivalent) was sent on a physical link.

9 **10.2.12.1.8 lastRcvdPortNum:** The portNumber of the PTP Port on which time-synchronization
 10 information was most recently received. The data type for lastRcvdPortNum is UInteger16.

11 **10.2.12.1.9 lastGmTimeBaseIndicator:** The gmTimeBaseIndicator of the most recently received
 12 PortSyncSync structure. The data type for lastGmTimeBaseIndicator is UInteger16.

13 **10.2.12.1.10 lastGmPhaseChangePSSS:** The lastGmPhaseChange of the most recently received
 14 PortSyncSync structure. The data type for lastGmPhaseChange is ScaledNs.

15 **10.2.12.1.11 lastGmFreqChangePSSS:** The lastGmFreqChange of the most recently received
 16 PortSyncSync structure. The data type for lastGmPhaseChange is Float64.

17 **10.2.12.1.12 txMDSyncPtr:** A pointer to the MDSyncSend structure sent to the MD entity of this PTP Port.

18 **10.2.12.1.13 numberSyncTransmissions:** A count of the number of consecutive Sync message
 19 transmissions after the SyncIntervalSetting state machine (see Figure 10-20) has set syncSlowdown
 20 (see 10.2.5.17) to TRUE. The data type for numberSyncTransmissions is UInteger8.

21 **10.2.12.1.14 interval1:** A local variable that holds either syncInterval or oldSyncInterval. The data type for
 22 interval1 is UScaledNs.

23 **10.2.12.2 State machine functions**

24 **10.2.12.2.1 setMDSync():** Creates an MDSyncSend structure, and returns a pointer to this structure. The
 25 members are set as follows:

- 26 a) sourcePortIdentity is set to the portIdentity of this PTP Port (see 8.5.2).
- 27 b) logMessageInterval is set equal to the value of currentLogSyncInterval for this PTP Port
 28 (see 10.7.2.3).
- 29 c) preciseOriginTimestamp is set equal to lastPreciseOriginTimestamp (see 10.2.12.1.3).
- 30 d) rateRatio is set equal to lastRateRatio (see 10.2.12.1.5).
- 31 e) followUpCorrectionField is set equal to lastFollowUpCorrectionField (see 10.2.12.1.4).
- 32 f) upstreamTxTime is set equal to lastUpstreamTxTime (see 10.2.12.1.6).
- 33 g) gmTimeBaseIndicator is set to lastGmTimeBaseIndicator (see 10.2.12.1.9).
- 34 h) lastGmPhaseChange (structure member) is set to lastGmPhaseChangePSSS (see 10.2.12.1.10).
- 35 i) lastGmFreqChange (structure member) is set to lastGmFreqChangePSSS (see 10.2.12.1.11).
- 36 j) domainNumber is set equal to the domainNumber of this gPTP domain (see 8.1).

37 **10.2.12.2.2 txMDSync(txMDSyncPtr):** Transmits the MDSyncSend structure pointed to by txMDSyncPtr,
 38 to the MD entity of this PTP Port.

10.2.12.3 State diagram

The PortSyncSyncSend state machine shall implement the function specified by the state diagram in Figure 10-8, the local variables specified in 10.2.12.1, the functions specified in 10.2.12.2, the structures specified in 10.2.2.1 through 10.2.2.3, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives time-synchronization information from the SiteSyncSync state machine, corresponding to the receipt of the most recent synchronization information on either the timeReceiver port, if this PTP Instance is not the Grandmaster PTP Instance, or from the ClockTimeTransmitterSyncSend state machine, if this PTP Instance is the Grandmaster PTP Instance. The state machine causes time-synchronization information to be sent to the MD entity if this PTP Port is a TimeTransmitterPort.

1

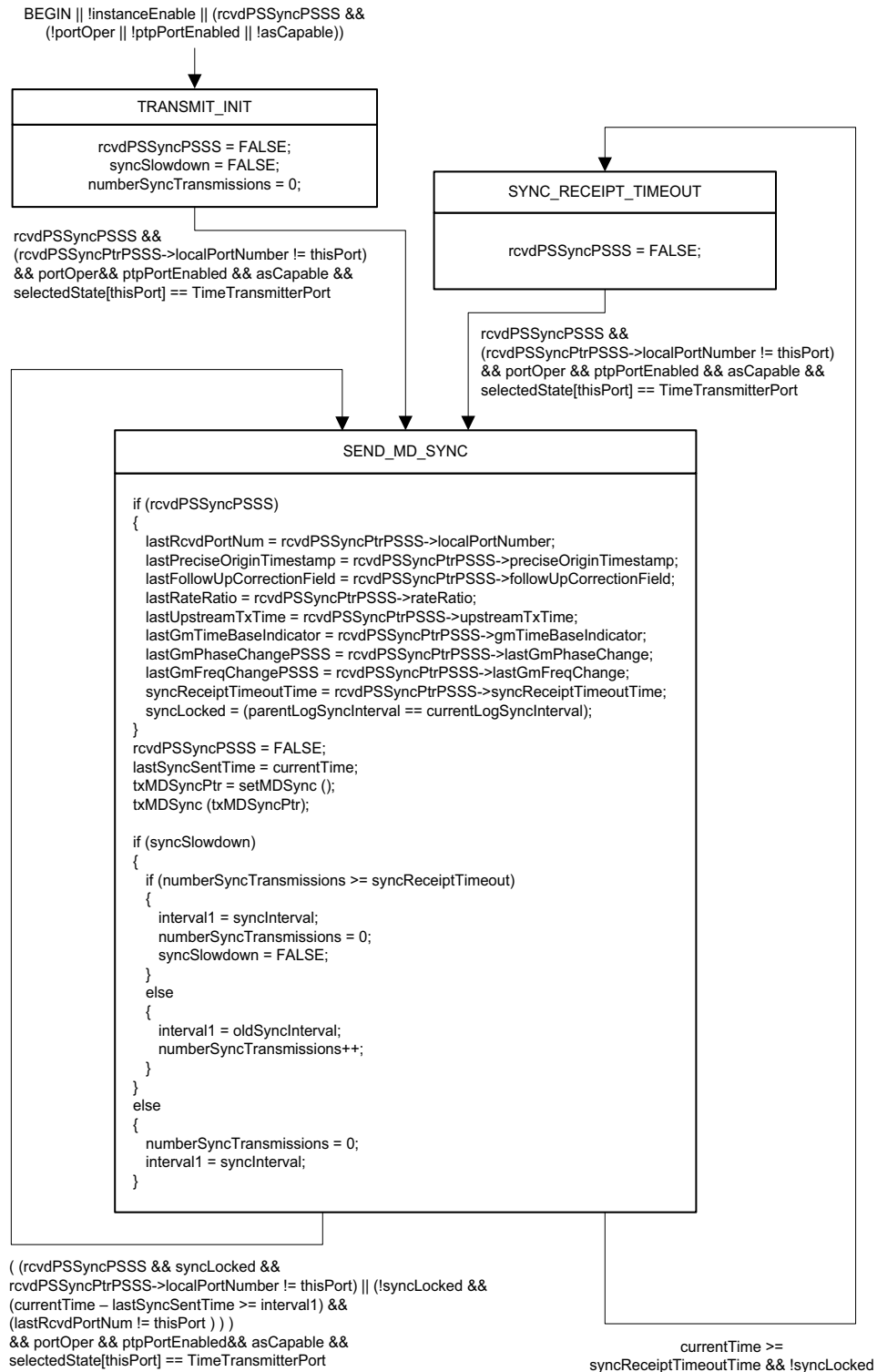


Figure 10-8—PortSyncSyncSend state machine

1 10.2.13 ClockTimeReceiverSync state machine

2 10.2.13.1 State machine variable

3 The following variable is used in the state diagram in Figure 10-9 (in 10.2.13.3):

4 **10.2.13.1.1 rcvdPSSyncPtrCSS:** A pointer to the received PortSyncSync structure.

5 10.2.13.2 State machine functions

6 **10.2.13.2.1 updateTimeReceiverTime():** Updates the global variable clockTimeReceiverTime (see
7 10.2.4.3), based on information received from the SiteSync and LocalClock entities. It is the responsibility
8 of the application to filter timeReceiver times appropriately (see B.3 and B.4 for examples). As one
9 example, clockTimeReceiverTime can be:

- 10 a) Set to syncReceiptTime at every LocalClock update immediately after a PortSyncSync structure is
11 received, and
- 12 b) Incremented by localClockTickInterval (see 10.2.4.18) multiplied by the rateRatio member of the
13 previously received PortSyncSync structure during all other LocalClock updates.

14 If no PTP Instance is grandmaster-capable, i.e., gmPresent is FALSE, then clockTimeReceiverTime is set to
15 the time provided by the LocalClock. This function is invoked when rcvdLocalClockTickCSS is TRUE.

16 **10.2.13.2.2 invokeApplicationInterfaceFunction (functionName):** Invokes the application interface
17 function whose name is functionName. For the ClockTimeReceiverSync state machine, functionName is
18 clockTargetPhaseDiscontinuity.result (see 9.6.2).

19 10.2.13.3 State diagram

20 The ClockTimeReceiverSync state machine shall implement the function specified by the state diagram in
21 Figure 10-9, the local variables specified in 10.2.13.1, the functions specified in 10.2.13.2, and the relevant
22 global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives a
23 PortSyncSync structure from the SiteSyncSync state machine. It computes syncReceiptTime and
24 clockTimeReceiverTime, and sets syncReceiptLocalTime (i.e., the time relative to the LocalClock entity
25 corresponding to syncReceiptTime), GmTimeBaseIndicator, lastGmPhaseChange, and lastGmFreqChange.
26 It provides clockTimeReceiverTime to the ClockTimeTransmitterSyncOffset state machine, and provides
27 information to the ClockTarget entity (via the ClockTargetPhaseDiscontinuity interface; see 9.6) to enable
28 that entity to determine if a phase or frequency discontinuity has occurred.

29 The per-PTP Port global variables used in the ClockTimeReceiverSync state machine are determined based
30 on rcvdPSSyncPtrCSS->localPortNumber, as follows:

- 31 a) If rcvdPSSyncPtrCSS->localPortNumber > 0, the per-PTP Port global variables of PTP Port number
32 rcvdPSSyncPtrCSS->localPortNumber are used.
- 33 b) If rcvdPSSyncPtrCSS->localPortNumber == 0, the values of the used per-PTP Port global variables
34 are fixed as follows:
 - 35 1) meanLinkDelay = 0
 - 36 2) delayAsymmetry = 0
 - 37 3) neighborRateRatio = 1.0

38 For Clause 11 nrrPdelay is used instead neighborRateRatio.

1

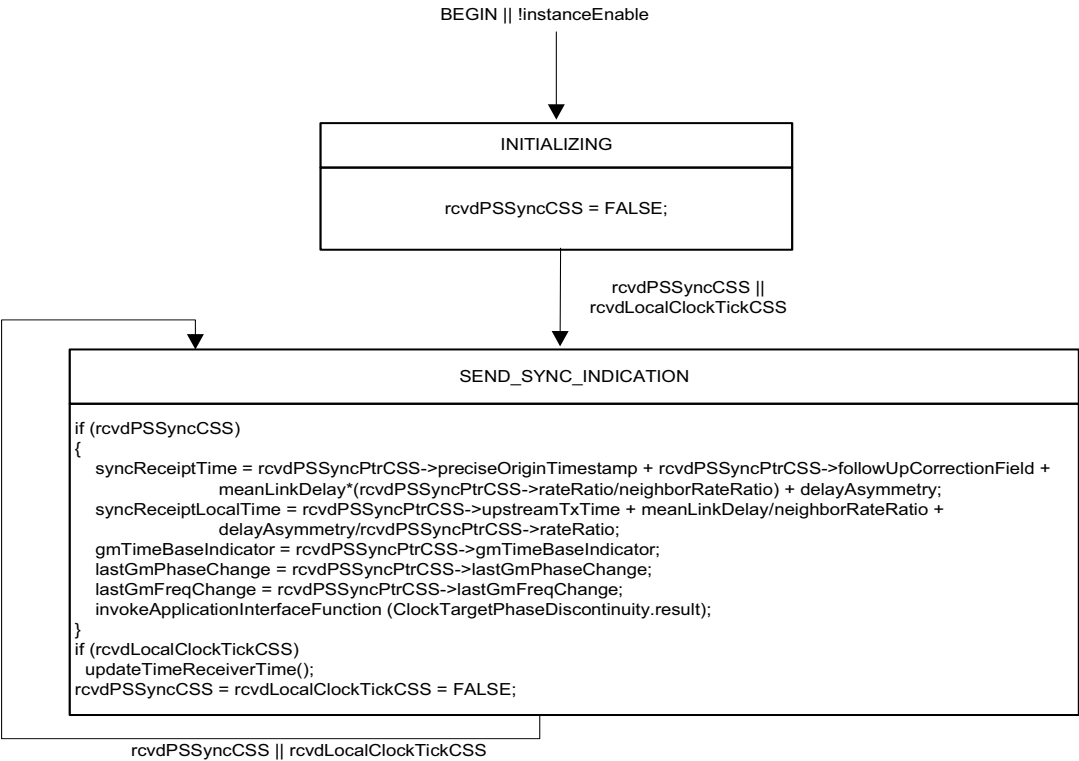


Figure 10-9—ClockTimeReceiverSync state machine

2 10.3 Best timeTransmitter clock selection, external port configuration, and
3 announce interval setting state machines

4 10.3.1 Best timeTransmitter clock selection and external port configuration overview

5 10.3.1.1 General

6 There are two methods for setting the Grandmaster PTP Instance and synchronization tree for a gPTP
7 domain:

- 8 a) The BTCA is used to determine the Grandmaster PTP Instance for a gPTP domain and construct the
9 synchronization tree with that Grandmaster PTP Instance as the root. In this case, the network is
10 configured automatically, i.e., the PTP Port states are set, using the results of the BTCA.
- 11 b) The PTP Port states are configured to force a desired Grandmaster PTP Instance and to construct a
12 desired synchronization tree with the Grandmaster PTP Instance as the root.

13 The PTP Port state definitions are given in Table 10-2.

14 The per PTP Instance global variable externalPortConfigurationEnabled indicates whether method a) or b)
15 is used; a value of TRUE indicates method b), and a value of FALSE indicates method a) (see 10.3.9.24).

16 For domain 0, the following statements apply:

- 1 c) Method a), i.e. BTCA is implemented.
- 2 d) Both methods a) and b) can be implemented
- 3 e) The default value of externalPortConfigurationEnabled is FALSE, unless Hot standby is activated.

4 For domains other than domain 0, the following statements apply:

- 5 f) At least one of the possibilities [method a) or b)] is implemented.
- 6 g) Both methods a) and b) can be implemented.
- 7 h) If both methods a) and b) are implemented, the default value of externalPortConfigurationEnabled is
- 8 FALSE, unless Hot standby is activated.

9 Once a GM is indicated in an Announce message transmitted by a PTP Port, subsequent timing information
10 (see 7.5) transmitted by that PTP Port is derived from the Grandmaster PTP Instance indicated in that
11 Announce message.

Table 10-2—PTP Port state definitions

PTP Port state	Description
TimeTransmitter Port	Any PTP Port, X, of the PTP Instance that is closer to the root than any other PTP Port of the gPTP communication path connected to X.
TimeReceiverPort	The one PTP Port of the PTP Instance that is closest to the root PTP Instance. If the root is grandmaster-capable, the TimeReceiverPort is also closest to the Grandmaster PTP Instance. The PTP Instance does not transmit Sync or Announce messages on the TimeReceiverPort.
PassivePort	Any PTP Port of the PTP Instance whose PTP Port state is not TimeTransmitterPort, TimeReceiverPort, or DisabledPort.
DisabledPort	Any PTP Port of the PTP Instance for which the variables portOper, ptpPortEnabled, and asCapable are not all TRUE.
NOTE—PTP Port states are per PTP Port and per domain (i.e., per PTP Instance; see 8.1).	

12 NOTE—Information contained in Sync and associated Follow Up messages received on PTP Ports whose PTP Port
13 state is PassivePort is discarded; the SiteSyncSync state machine (see 10.2.7) uses only information received from a PTP
14 Port whose PTP Port state is TimeReceiverPort.

15 An example timeTransmitter/timeReceiver hierarchy of PTP Instances is shown in Figure 10-10. The
16 Grandmaster PTP Instance ports all have PTP Port state of TimeTransmitterPort. All the other PTP Instances
17 have exactly one timeReceiver port. The synchronization tree is composed of the PTP Instances and the
18 links that do not have an endpoint PTP Port whose PTP Port state is PassivePort.

1

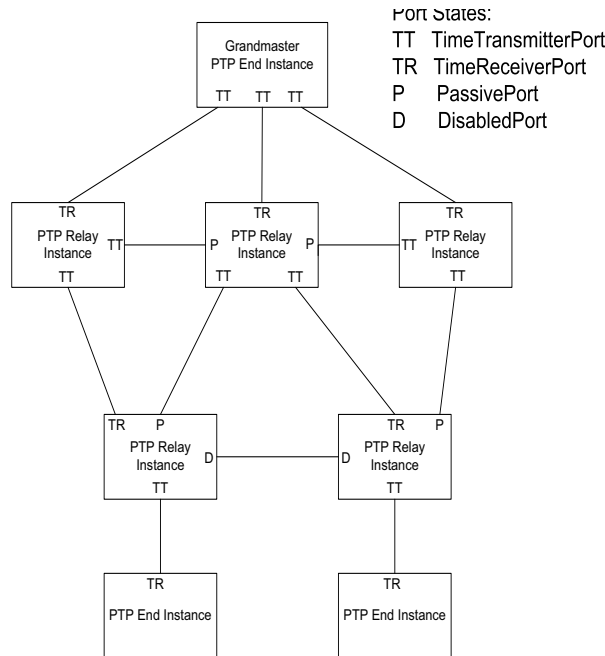


Figure 10-10—Example timeTransmitter/timeReceiver hierarchy of PTP Instances

2 10.3.1.2 Best timeTransmitter clock algorithm overview

3 In the BTCA (i.e., method a) of 10.3.1.1), best timeTransmitter selection information is exchanged between
4 PTP Instances of time-aware systems via Announce messages (see 10.5 and 10.6). Each Announce message
5 contains synchronization tree vector information that identifies one PTP Instance as the root of the
6 synchronization tree and, if the PTP Instance is grandmaster-capable, the Grandmaster PTP Instance. Each
7 PTP Instance in turn uses the information contained in the Announce messages it receives, along with its
8 knowledge of itself, to compute which of the PTP Instances that it has knowledge of ought to be the root of
9 the synchronization tree and, if grandmaster-capable, the Grandmaster PTP Instance. As part of constructing
10 the synchronization tree, each PTP Port of each PTP Instance is assigned a PTP Port state from Table 10-2
11 by state machines associated with the ports and with the PTP Instance as a whole.

12 NOTE—The BTCA described in this standard is the default BTCA according to the specifications of 9.3
13 of IEEE Std 1588-2019. It is also equivalent to a subset of the Rapid Spanning Tree Protocol (RSTP) described in
14 IEEE Std 802.1Q-2022 (though the full RSTP described in IEEE Std 802.1Q-2022 is not equivalent to the full BTCA
15 described in IEEE Std 1588-2019). The BTCA description here uses the formalism of the RSTP description
16 in IEEE Std 802.1Q-2022.

17 10.3.1.3 External port configuration overview

18 In external port configuration (i.e., method b) of 10.3.1.1), an external entity determines the synchronization
19 tree and sets the PTP Port states accordingly. The method used by the external entity to determine the
20 synchronization tree is outside the scope of this standard. However, as with the BTCA, Announce messages
21 are used to transport information on the synchronization tree and Grandmaster PTP Instance time properties
22 information from one PTP Instance to the next in the synchronization tree. The external entity sets the state
23 of a PTP Port by setting the value of externalPortConfigurationPortDS.desiredState to the desired state.

1 In the case of external port configuration, the synchronization tree and the desired PTP Port states are
 2 controlled by an external entity, but the Grandmaster PTP Instance might change. If the timeReceiver port of
 3 a PTP Instance that is gmCapable (priority1 < 255, see 8.6.2.1) is no longer asCapable, the state of this PTP
 4 Port changes from TimeReceiverPort to DisabledPort (see 10.3.6.2) and the PTP Instance becomes
 5 grandmaster for the synchronization (sub-)tree where it is the root (10.3.15.2.2 f) 3)). The original
 6 synchronization tree can be split into disjunct subtrees with different, mutually unsynchronized,
 7 Grandmaster PTP Instances. If the port becomes asCapable again, its PTP Port state is again set to the
 8 desired state. If the timeReceiver port of a PTP Instance that is not gmCapable is no longer asCapable, the
 9 state of this PTP Port changes from TimeReceiverPort to DisabledPort (see 10.3.6.2). However, in this case
 10 gmPresent is set to FALSE (see 10.3.15.2.2 f) 3)), and the PTP Instance does not send Sync messages on any
 11 of its ports.

12 In external port configuration, there is no supervision of announce receipt timeout (see 10.7.3.2).

13 10.3.2 systemIdentity

14 The systemIdentity attribute of a PTP Instance is a UInteger112 (i.e., a 14-byte, unsigned integer) formed by
 15 concatenating the following attributes, in the following order, from most significant to least significant octet:

- 16 a) priority1 (1 octet; see 8.6.2.1)
- 17 b) clockClass (1 octet; see 8.6.2.2 and 6.4.3.8)
- 18 c) clockAccuracy (1 octet; see 8.6.2.3 and 6.4.3.8)
- 19 d) offsetScaledLogVariance (2 octets; see 8.6.2.4 and 6.4.3.8)
- 20 e) priority2 (1 octet; see 8.6.2.5)
- 21 f) clockIdentity (8 octets; see 8.5.2.2 and 6.4.3.6)

22 The systemIdentity attribute is defined for convenience when comparing two PTP Instances to determine,
 23 when using the BTCA (i.e., method a) of 10.3.1.1), which is a better candidate for root and if the PTP
 24 Instance is grandmaster-capable (i.e., the value of priority1 is less than 255; see 8.6.2.1). Two PTP Instances
 25 are compared as follows. Let the systemIdentity of PTP Instance A be S_A and the systemIdentity of PTP
 26 Instance B be S_B . Let the clockIdentity of A be C_A and the clockIdentity of B be C_B . Then, if $C_A \neq C_B$, i.e.,
 27 A and B represent different PTP Instances,

- 28 g) A is better than B if and only if $S_A < S_B$, and
- 29 h) B is better than A if and only if $S_B < S_A$.

30 If $C_A = C_B$, i.e., A and B represent the same PTP Instance,

- 31 i) $S_A < S_B$ means that A represents an upgrading of the PTP Instance compared to B or, equivalently, B
 32 represents a downgrading of the PTP Instance compared to A,
- 33 j) $S_B < S_A$ means that B represents an upgrading of the PTP Instance compared to A or, equivalently,
 34 A represents a downgrading of the PTP Instance compared to B, and
- 35 k) $S_A = S_B$ means that A and B represent the same PTP Instance that has not changed.

36 Comparisons g) and h) in this subclause imply that, with the ordering of attributes in the systemIdentity, the
 37 clockIdentity is a tie-breaker when two different PTP Instances that have identical attributes a) through e)
 38 are compared.

39 Comparisons g) and h) also imply that a PTP Instance that is grandmaster-capable is always better than
 40 another PTP Instance that is not grandmaster-capable because the priority1 is less than 255 if the PTP
 41 Instance is grandmaster-capable and is equal to 255 if it is not grandmaster-capable (see 8.6.2.1).

1 The cases where A and B represent different PTP Instances and represent the same PTP Instance are handled
 2 separately in the BTCA. When comparing two different PTP Instances, the better PTP Instance is selected as
 3 the Grandmaster PTP Instance candidate. However, if A and B represent the same PTP Instance with
 4 attributes that have changed, the PTP Instance is considered as having the most recent attributes when doing
 5 subsequent comparisons with other PTP Instances.

6 10.3.3 stepsRemoved

7 Every PTP Instance has a stepsRemoved associated with it. For the root PTP Instance, and therefore the
 8 Grandmaster PTP Instance when the root is grandmaster-capable, it is zero. For all other PTP Instances, it is
 9 the number of gPTP communication paths in the path from the root to the respective PTP Instance.

10 NOTE—For example, stepsRemoved for a timeReceiver port on the same gPTP communication path as the
 11 Grandmaster PTP Instance will have a value of 1, indicating that a single path was traversed.

12 The stepsRemoved attributes of different ports of a PTP Instance are compared after comparisons of other
 13 attributes that take precedence (i.e., priority1, clockClass, clockAccuracy, offsetScaledLogVariance,
 14 priority2) do not result in one PTP Port being declared better than the other. Among the ports whose
 15 stepsRemoved attributes are compared, the PTP Port on the PTP Instance with the lowest stepsRemoved is
 16 assigned the state of TimeReceiverPort for that PTP Instance (the root PTP Instance does not have a
 17 TimeReceiverPort). This lowest stepsRemoved is also considered the stepsRemoved for the PTP Instance.
 18 If a PTP Instance has two or more ports with the same stepsRemoved, then the PTP Port with the smallest
 19 sourcePortIdentity is selected as the TimeReceiverPort. If a PTP Instance has two or more ports with the same
 20 stepsRemoved and the same sourcePortIdentity, then the PTP Port with the smallest portNumber is selected
 21 as the TimeReceiverPort (see 10.3.4).

22 10.3.4 Synchronization tree priority vectors

23 PTP Instances send best timeTransmitter selection information to each other in Announce messages. The
 24 information is structured in a synchronization tree priority vector. Synchronization tree priority vectors
 25 provide the basis for a concise specification of the BTCA's determination of the synchronization tree and
 26 Grandmaster PTP Instance. A priority vector is formed by concatenating the following attributes, in the
 27 following order, from most significant to least significant octet:

- 28 a) rootSystemIdentity (14 octets; see 10.3.2)
- 29 b) stepsRemoved (2 octets; see 10.3.3)
- 30 c) sourcePortIdentity (i.e., portIdentity of the transmitting PTP Instance; 10 octets; see 8.5.2 and
- 31 10.6.2)
- 32 d) portNumber of the receiving PTP Port (2 octets; see 8.5.2.3)

33 The first two components of a priority vector are significant throughout the gPTP domain; they are
 34 propagated via Announce messages and updated through invocation of BTCA state machines. The next
 35 component is assigned hop-by-hop for each gPTP communication path or PTP Instance and thus is of local
 36 significance only. It is used as a tie-breaker in decisions between synchronization tree priority vectors that
 37 are otherwise equal. The fourth component is not conveyed in Announce messages, but is used as a tie-
 38 breaker within a PTP Instance.

39 The set of all synchronization tree priority vectors is totally ordered. For all components, a lesser numerical
 40 value is better, and earlier components in the preceding list are more significant. In addition, as mentioned
 41 earlier, a priority vector that reflects a root PTP Instance that is grandmaster-capable is always better than a
 42 priority vector that reflects a root PTP Instance that is not grandmaster-capable. As each PTP Port receives a
 43 priority vector, via an Announce message, from ports closer to the root, additions are made to one or more
 44 components to yield a worse priority vector. This process of receiving information, adding to it, and passing
 45 it on, can be described in terms of the message priority vector received and a set of priority vectors used to

1 facilitate the computation of a priority vector for each PTP Port, to be transmitted in further Announce
2 Messages to PTP Instances further from the root.

3 10.3.5 Priority vector calculations

4 The portPriorityVector is the synchronization tree priority vector held for the PTP Port when the reception of
5 Announce messages and any pending update of information has been completed:

6
$$\text{portPriorityVector} = \{\text{rootSystemIdentity} : \text{stepsRemoved} : \text{sourcePortIdentity} : \text{portNumber}\}$$

7 A messagePriorityVector is the synchronization tree priority vector conveyed in a received Announce
8 Message. For a PTP Instance S receiving an Announce Message on PTP Port P_S with portNumber PN_S,
9 from a TimeTransmitterPort with portIdentity P_M on PTP Instance M claiming a rootSystemIdentity of R_M
10 and a stepsRemoved of SR_M:

11
$$\text{messagePriorityVector} = \{R_M : SR_M : P_M : PN_S\}$$

12 This messagePriorityVector is superior to the portPriorityVector and will replace it if, and only if, the
13 messagePriorityVector is better than the portPriorityVector, or the Announce message has been transmitted
14 from the same timeTransmitter PTP Instance and TimeTransmitterPort as the portPriorityVector, i.e., if the
15 following is true:

16
$$((R_M < \text{rootSystemIdentity})) \parallel$$

17
$$((R_M == \text{rootSystemIdentity}) \ \&\& \ (SR_M < \text{stepsRemoved})) \parallel$$

18
$$((R_M == \text{rootSystemIdentity}) \ \&\& \ (SR_M == \text{stepsRemoved}) \ \&\& \ (P_M < \text{sourcePortIdentity (of current}$$

19
$$\text{timeTransmitter PTP Instance)})) \parallel$$

20
$$((R_M == \text{rootSystemIdentity}) \ \&\& \ (SR_M == \text{stepsRemoved}))$$

21
$$\ \&\& \ (P_M == \text{sourcePortIdentity (of current timeTransmitter PTP Instance)}) \ \&\& \ (PN_S < \text{portNumber})) \parallel$$

22
$$((P_M.\text{clockIdentity} == \text{sourcePortIdentity.clockIdentity (of current timeTransmitter PTP Instance)}) \ \&\& \$$

23
$$(P_M.\text{portNumber} == \text{sourcePortIdentity.PortNumber (of the current timeTransmitter PTP Instance)}))$$

24 A gmPathPriorityVector can be calculated from a received portPriorityVector by adding one to the
25 stepsRemoved component:

26
$$\text{gmPathPriorityVector} = \{R_M : SR_M + 1 : P_M : PN_S\}$$

27 The systemPriorityVector for a PTP Instance S with systemIdentity S_S and clockIdentity C_S is the priority
28 vector that would, with the portIdentity of the TimeReceiverPort set equal to the portIdentity of the
29 transmitting PTP Port, be used as the message priority vector in Announce Messages transmitted on S's
30 ports whose state is TimeTransmitterPort if S was selected as the root:

31
$$\text{systemPriorityVector} = \{S_S : 0 : \{C_S : 0\} : 0\}$$

1 The gmPriorityVector for S is the best of the set comprising the systemPriorityVector vector plus every
 2 gmPathPriorityVector for which the clockIdentity of the timeTransmitter PTP Instance portIdentity is not the
 3 clockIdentity of S. If the systemPriorityVector is best, S has been selected as the root. When the best
 4 gmPathPriorityVector is that of PTP Port PN_S above, then:

5 gmPriorityVector = {S_S : 0 : {C_S : 0} : 0} if S is better than R_M, or

6 gmPriorityVector = {R_M : SR_M + 1 : P_M : PN_S} if S is worse than R_M.

7 The timeTransmitterPriorityVector for a PTP Port Q on PTP Instance S is the gmPriorityVector with S's
 8 clockIdentity C_S substituted for the clockIdentity of the timeTransmitter portIdentity, and Q's portNumber
 9 PN_Q substituted for the portNumber of the timeTransmitter portIdentity and for the portNumber of the
 10 receiving PTP Port:

11 timeTransmitterPriorityVector = {S_S : 0 : {C_S : PN_Q} : PN_Q} if S is better than R_M, or

12 timeTransmitterPriorityVector = {R_M : SR_M + 1 : {C_S : PN_Q} : PN_Q} if S is worse than R_M.

13 If the timeTransmitterPriorityVector is better than the portPriorityVector, the PTP Port will be the
 14 TimeTransmitterPort for the attached gPTP communication path and the portPriorityVector will be updated.
 15 The messagePriorityVector information in Announce messages transmitted by a PTP Port always includes
 16 the first three components of the timeTransmitterPriorityVector of the PTP Port.

17 NOTE—The consistent use of lower numerical values to indicate better information is deliberate as the
 18 TimeTransmitterPort that is closest to the root, i.e., has a numerically lowest path cost component, is selected from
 19 amongst potential alternatives for any given gPTP communication path. Adopting the conventions that lower numerical
 20 values indicate better information, that where possible more significant priority components are encoded earlier in the
 21 octet sequence of an Announce message, and that earlier octets in the encoding of individual components are more
 22 significant allows concatenated octets that compose a priority vector to be compared as if they were a multiple octet
 23 encoding of a single number, without regard to the boundaries between the encoded components. To reduce the
 24 confusion that naturally arises from having the lesser of two numerical values represent the better of the two, i.e., the one
 25 to be chosen all other factors being equal, this clause uses the following consistent terminology. Relative numeric values
 26 are described as “least,” “lesser,” “equal,” and “greater,” and their comparisons as “less than,” “equal to,” or “greater
 27 than,” while relative synchronization tree priorities are described as “best,” “better,” “the same,” “different,” and
 28 “worse” and their comparisons as “better than,” “the same as,” “different from,” and “worse than.” The operators “<”
 29 and “=” represent less than and equal to, respectively. The terms “superior” and “inferior” are used for comparisons
 30 that are not simply based on priority, but can include the fact that the priority vector of a TimeTransmitterPort can
 31 replace an earlier vector transmitted in an Announce message by the same PTP Port.

32 10.3.6 PTP Port state assignments

33 10.3.6.1 PTP Port state assignments when the BTCA is used

34 The BTCA assigns one of the following PTP Port states to each PTP Port: TimeTransmitterPort,
 35 TimeReceiverPort, PassivePort, or DisabledPort.

36 The DisabledPort state is assigned if portOper is FALSE (see 10.2.5.12), ptpPortEnabled is FALSE
 37 (see 10.2.5.13), or asCapable is FALSE (see 10.2.5.1).

38 A PTP Port for which portOper, ptpPortEnabled, and asCapable are all TRUE has its PTP Port state assigned
 39 according to the source and relative priority of the synchronization tree portPriorityVector (see 10.3.4 and
 40 10.3.5) as follows:

41 a) If the PTP Instance is not the root, the source of the gmPriorityVector is the TimeReceiverPort.

- 1 b) Each PTP Port whose portPriorityVector is its timeTransmitterPriorityVector is a
2 TimeTransmitterPort.
- 3 c) Each PTP Port, other than the TimeReceiverPort, whose portPriorityVector has been received from
4 another PTP Instance or another PTP Port on this PTP Instance is a PassivePort.

5 10.3.6.2 PTP Port state assignments when external port configuration is used

6 If external port configuration is used, one of the states TimeTransmitterPort, TimeReceiverPort, PassivePort,
7 or DisabledPort is assigned to each PTP Port by an external entity, as described in this subclause.

8 The DisabledPort state is assigned if portOper is FALSE (see 10.2.5.12), ptpPortEnabled is FALSE (see
9 10.2.5.13), or asCapable is FALSE (see 10.2.5.1).

10 The member externalPortConfigurationPortDS.desiredState (see 14.14.2) is used by an external entity to set
11 the state of the respective PTP Port to TimeTransmitterPort, TimeReceiverPort, or PassivePort. When this
12 member is set, its value is copied to the per PTP Port local variable portStateInd (see 10.3.15.1.5). If
13 portOper, ptpPortEnabled, and asCapable are all TRUE for this PTP Port, the PTP Port state is set equal to
14 the value of externalPortConfigurationPortDS.desiredState by copying the value of this member to the
15 element of the selectedState array (see 10.2.4.20) for this PTP Port.

16 10.3.7 Overview of best timeTransmitter clock selection, external port configuration, and 17 announce interval setting state machines

18 10.3.7.1 Best timeTransmitter clock selection state machines overview

19 The best timeTransmitter clock selection function in a PTP Instance is specified by a number of cooperating
20 state machines. Figure 10-11 is not itself a state machine, but illustrates the machines, their local variables,
21 their interrelationships, their performance parameters, and the global variables and structures used to
22 communicate between them.

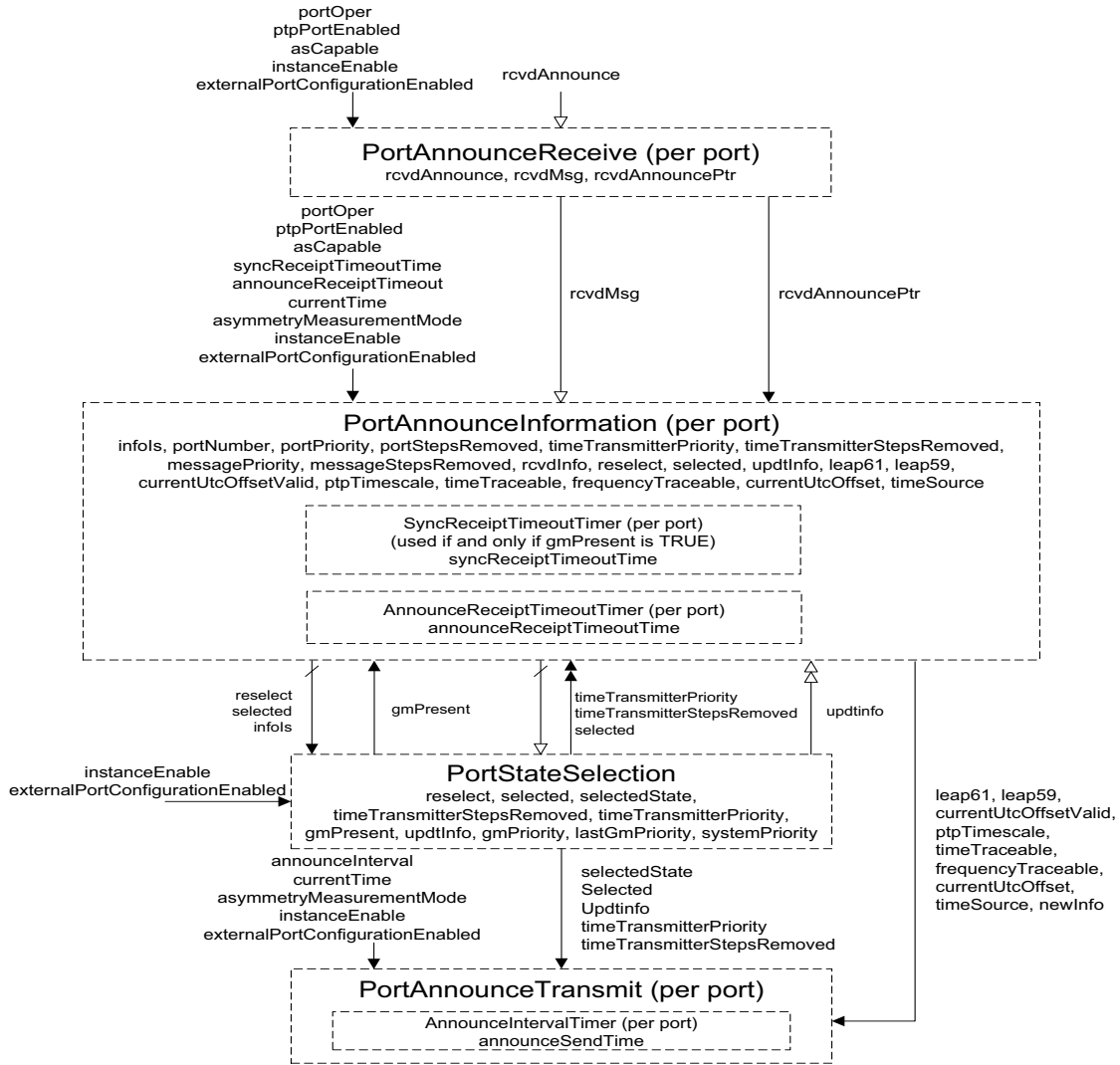
23 NOTE—The BTCA state machines are all invoked by the media-independent layer, i.e., by the SiteSync and PortSync
24 entities. The media-dependent layer, i.e., the MD entity, simply takes an Announce message received from the PortSync
25 entity of the same PTP Port and gives it to the next lower layer (e.g., IEEE 802.3, IEEE 802.11). It is the PortSync entity
26 that generates and consumes Announce messages.

27 The following media-independent layer state machines are in Figure 10-11:

- 28 a) PortAnnounceReceive (one instance per PTP Instance, per PTP Port): receives Announce
29 information from the MD entity of the same PTP Port, determines if the Announce message is
30 qualified and, if so, sets the rcvdMsg variable. This state machine is invoked by the PortSync entity
31 of the PTP Port.
- 32 b) PortAnnounceInformation (one instance per PTP Instance, per PTP Port): receives new qualified
33 Announce information from the PortAnnounceReceive state machine, determines if the Announce
34 information is better than the current best timeTransmitter information it knows about, and updates
35 the current best timeTransmitter information when it receives updated PTP Port state information
36 from the PortStateSelection state machine and when announce receipt timeout or, when gmPresent is
37 TRUE, sync receipt timeout occurs. This state machine is invoked by the PortSync entity of the PTP
38 Port.
- 39 c) PortStateSelection (one instance per PTP Instance): updates the gmPathPriority vector for each PTP
40 Port of the PTP Instance, the gmPriorityVector for the PTP Instance, and the
41 timeTransmitterPriorityVector for each PTP Port of the PTP Instance; determines the PTP Port state
42 for each PTP Port; and updates gmPresent. This state machine is invoked by the SiteSync entity of
43 the PTP Instance.

- 1 d) PortAnnounceTransmit (one instance per PTP Instance, per PTP Port): If the PTP Port state is
2 TimeTransmitterPort, transmits Announce information to the MD entity when an announce interval
3 has elapsed; or PTP Port states have been updated; or portPriority and portStepsRemoved
4 information has been updated with newly determined timeTransmitterPriority and
5 timeTransmitterStepsRemoved information. This state machine is invoked by the PortSync entity of
6 the PTP Port and is also used when external port configuration is used.

1



Notation:

Variables are shown both within the machine where they are principally used and between machines where they are used to communicate information. In the latter case a variety of arrow styles, running from one machine to another, show how each is typically used:

- Not changed by the target machine. Where the machines are both per port, this variable communicates between instances for the same port
- ⇨ Set (or cleared) by the originating machine, cleared (or set) by the target machine. Where the machines are both per port, this communicates between instances for the same port.
- ⇨⇨ As above, except that the originating per-port machine instance communicates with multiple port machine instances (by setting or clearing variables owned by those ports).
- ⇨⇨⇨ As above, except that multiple per-port instances communicate with (an)other instance(s) (by setting or clearing variables owned by the originating ports).

Figure 10-11—Best timeTransmitter clock selection state machines—overview and interrelationships

1 10.3.7.2 External port configuration state machines overview

2 The external port configuration function in a PTP Instance is specified by a number of cooperating state
 3 machines. Figure 10-12 is not itself a state machine, but illustrates the machines, their local variables, their
 4 interrelationships, their performance parameters, and the global variables and structures used to
 5 communicate between them.

6 NOTE—The external port configuration state machines are all invoked by the media-independent layer and are per PTP
 7 Port, i.e., they are invoked by the PortSync entity for the respective PTP Port. The media-dependent layer, i.e., the MD
 8 entity, simply takes an Announce message received from the PortSync entity of the same PTP Port and gives it to the
 9 next lower layer (e.g., IEEE 802.3, IEEE 802.11). It is the PortSync entity that generates and consumes Announce
 10 messages.

11 The following media-independent layer state machines are in Figure 10-12:

- 12 a) PortAnnounceInformationExt (one instance per PTP Instance, per PTP Port): Receives and stores
 13 new Announce information received in Announce messages.
- 14 b) PortStateSettingExt (one instance per PTP Instance): Copies the desired PTP Port state for the PTP
 15 Port to the respective selectedState array element, updates gmPresent, computes
 16 timeTransmitterStepsRemoved, stores the time properties information in the respective global
 17 variables, and computes the gmPriorityVector and timeTransmitterPriorityVector.
- 18 c) PortAnnounceTransmit (one instance per PTP Instance, per PTP Port): If the PTP Port state is
 19 TimeTransmitterPort, transmits Announce information to the MD entity when an announce interval
 20 has elapsed, or PTP Port states have been updated, or portPriority and portStepsRemoved
 21 information has been updated with newly determined TimeTransmitterPriority and
 22 timeTransmitterStepsRemoved information. This state machine is invoked by the PortSync entity of
 23 the PTP Port and is also used when the BTCA is used.

1

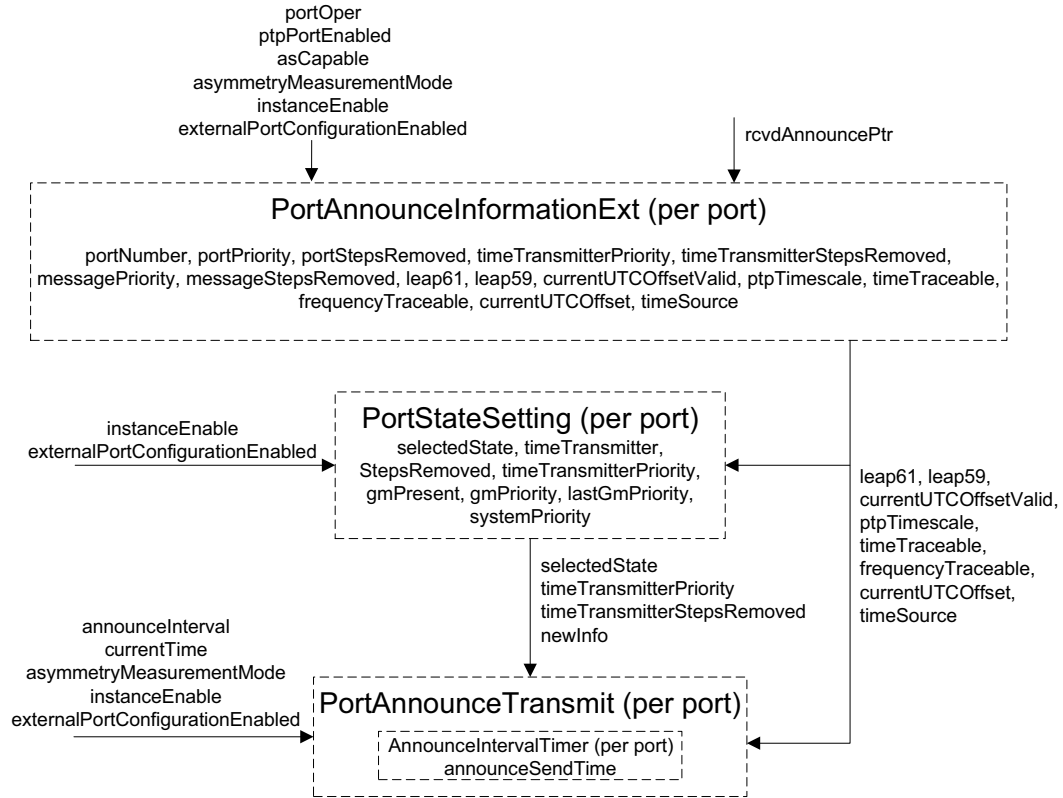


Figure 10-12—External port configuration state machines—overview and interrelationships

2 10.3.7.3 AnnounceIntervalSetting state machine overview

3 An additional state machine, the AnnounceIntervalSetting state machine, receives a Signaling message that
4 contains a Message Interval Request TLV (see 10.6.4.3) and sets the global variables that give the duration
5 of the mean interval between successive Announce messages.

6 10.3.8 Overview of global variables used by best timeTransmitter clock selection, external 7 port configuration, and announce interval setting state machines

8 10.3.9 and 10.3.10 define global variables used by best timeTransmitter clock selection, external port
9 configuration, and announce interval setting state machines whose scopes are as follows:

- 10 — Per PTP Instance (i.e., per domain)
- 11 — Per PTP Instance, per PTP Port
- 12 — Instance used by CMLDS (see 11.2.17) (i.e., variable is common across all Link Ports)
- 13 — Instance used by CMLDS, per Link Port

1 Table 10-3 summarizes the scope of each global variable of 10.3.9 and 10.3.10.

Table 10-3—Summary of scope of global variables used by best timeTransmitter clock selection, external port configuration, and announce interval setting state machines (see 10.3.9 and 10.3.10)

Variable name	Subclause of definition	Data Type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
reselect	10.3.9.1	Boolean array	Per PTP Instance	No	No
selected	10.3.9.2	Boolean array	Per PTP Instance	No	No
timeTransmitterStepsRemoved	10.3.9.3	UInteger16	Per PTP Instance	No	No
leap61	10.3.9.4	Boolean	Per PTP Instance	No	No
leap59	10.3.9.5	Boolean	Per PTP Instance	No	No
currentUtcOffsetValid	10.3.9.6	Boolean	Per PTP Instance	No	No
ptpTimescale	10.3.9.7	Boolean	Per PTP Instance	No	No
timeTraceable	10.3.9.8	Boolean	Per PTP Instance	No	No
frequencyTraceable	10.3.9.9	Boolean	Per PTP Instance	No	No
currentUtcOffset	10.3.9.10	UInteger16	Per PTP Instance	No	No
timeSource	10.3.9.11	TimeSource	Per PTP Instance	No	No
sysLeap61	10.3.9.12	Boolean	Per PTP Instance	No	No
sysLeap59	10.3.9.13	Boolean	Per PTP Instance	No	No
sysCurrentUtcOffsetValid	10.3.9.14	Boolean	Per PTP Instance	No	No
sysPtpTimescale	10.3.9.15	Boolean	Per PTP Instance	No	No
sysTimeTraceable	10.3.9.16	Boolean	Per PTP Instance	No	No
sysFrequencyTraceable	10.3.9.17	Boolean	Per PTP Instance	No	No
sysCurrentUtcOffset	10.3.9.18	Integer16	Per PTP Instance	No	No

Table 10-3—Summary of scope of global variables used by best timeTransmitter clock selection, external port configuration, and announce interval setting state machines (see 10.3.9 and 10.3.10) (continued)

Variable name	Subclause of definition	Data Type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
sysTimeSource	10.3.9.19	TimeSource	Per PTP Instance	No	No
systemPriority	10.3.9.20	UInteger224	Per PTP Instance	No	No
gmPriority	10.3.9.21	UInteger224	Per PTP Instance	No	No
lastGmPriority	10.3.9.22	UInteger224	Per PTP Instance	No	No
pathTrace	10.3.9.23	ClockIdentity[N]	Per PTP Instance	No	No
externalPortConfigurationEnabled	10.3.9.24	Boolean	Per PTP Instance	No	No
lastAnnouncePort	10.3.9.25	UInteger16	Per PTP Instance	No	No
announceReceiptTimeoutTimeInterval	10.3.10.1	UScaledNs	Per PTP Port	No	No
announceSlowdown	10.3.10.2	Boolean	Per PTP Port	No	No
oldAnnounceInterval	10.3.10.3	UScaledNs	Per PTP Port	No	No
infoIs	10.3.10.4	Enumeration2	Per PTP Port	No	No
timeTransmitterPriority	10.3.10.5	UInteger224	Per PTP Port	No	No
currentLogAnnounceInterval	10.3.10.6	Integer8	Per PTP Port	No	No
initialLogAnnounceInterval	10.3.10.7	Integer8	Per PTP Port	No	No
announceInterval	10.3.10.8	UScaledNs	Per PTP Port	No	No
messageStepsRemoved	10.3.10.9	UInteger16	Per PTP Port	No	No
newInfo	10.3.10.10	Boolean	Per PTP Port	No	No
portPriority	10.3.10.11	UInteger224	Per PTP Port	No	No
portStepsRemoved	10.3.10.12	UInteger16	Per PTP Port	No	No

Table 10-3—Summary of scope of global variables used by best timeTransmitter clock selection, external port configuration, and announce interval setting state machines (see 10.3.9 and 10.3.10) (continued)

Variable name	Subclause of definition	Data Type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
rcvdAnnouncePtr	10.3.10.13	Pointer to a structure	Per PTP Port	No	No
rcvdMsg	10.3.10.14	Boolean	Per PTP Port	No	No
updtInfo	10.3.10.15	Boolean	Per PTP Port	No	No
annLeap61	10.3.10.16	Boolean	Per PTP Port	No	No
annLeap59	10.3.10.17	Boolean	Per PTP Port	No	No
annCurrentUtcOffsetValid	10.3.10.18	Boolean	Per PTP Port	No	No
annPtpTimescale	10.3.10.19	Boolean	Per PTP Port	No	No
annTimeTraceable	10.3.10.20	Boolean	Per PTP Port	No	No
annFrequencyTraceable	10.3.10.21	Boolean	Per PTP Port	No	No
annCurrentUtcOffset	10.3.10.22	Integer16	Per PTP Port	No	No
annTimeSource	10.3.10.23	TimeSource	Per PTP Port	No	No
receivedPathTrace	10.3.10.24	clockIdentity[N]	Per PTP Port	No	No

1 10.3.9 Per PTP Instance global variables

2 **10.3.9.1 reselect:** A Boolean array of length numberPorts+1 (see 8.6.2.8). Setting reselect[j], where $0 \leq j \leq$
3 numberPorts, to TRUE causes the STATE_SELECTION block of the PortStateSelection state machine (see
4 10.3.13) to be re-entered, which in turn causes the PTP Port state of each PTP Port of the PTP Instance to be
5 updated (via the function updtStatesTree()); see 10.3.13.2.4). This variable is used only by the BTCA, i.e.,
6 not by the external port configuration option.

7 **10.3.9.2 selected:** A Boolean array of length numberPorts+1 (see 8.6.2.8). selected[j], where $0 \leq j \leq$
8 numberPorts, is set to TRUE immediately after the PTP Port states of all the ports are updated. This value
9 indicates to the PortAnnounceInformation state machine (see 10.3.12) that it can update the
10 portPriorityVector and other variables for each PTP Port. This variable is used by both the BTCA and the
11 external port configuration option; however, its value does not impact the external port configuration option
12 (see the NOTE in 10.3.16.3).

1 NOTE—Array elements 0 of the reselect and selected arrays are not used, except that the function clearReselectTree()
 2 sets reselect[0] to FALSE when it sets the entire array to zero and the function setSelectedTree() sets selected[0] to
 3 TRUE when it sets the entire array to TRUE. This action is taken only for convenience, so that array element j can
 4 correspond to PTP Port j. Note also that, in contrast, selectedState[0] is used (see 10.2.4.20).

5 **10.3.9.3 timeTransmitterStepsRemoved:** The value of stepsRemoved for the PTP Instance, after the PTP
 6 Port states of all the ports have been updated (see 10.3.13.2.4 for details on the computation of
 7 timeTransmitterStepsRemoved). This variable is used by both the BTCA and the external port configuration
 8 option.

9 **10.3.9.4 leap61:** A Boolean variable whose value is TRUE if the last minute of the current UTC day, relative
 10 to the current Grandmaster Clock, contains 61 s and FALSE if the last minute of the current UTC day does
 11 not contain 61 s. This variable is used by both the BTCA and the external port configuration option.

12 **10.3.9.5 leap59:** A Boolean variable whose value is TRUE if the last minute of the current UTC day, relative
 13 to the current Grandmaster Clock, contains 59 s and FALSE if the last minute of the current UTC day does
 14 not contain 59 s. This variable is used by both the BTCA and the external port configuration option.

15 **10.3.9.6 currentUtcOffsetValid:** A Boolean variable whose value is TRUE if currentUtcOffset (see
 16 10.3.9.10), relative to the current Grandmaster Clock, is known to be correct and FALSE if currentUtcOffset
 17 is not known to be correct. This variable is used by both the BTCA and the external port configuration
 18 option.

19 **10.3.9.7 ptpTimescale:** A Boolean variable whose value is TRUE if the timescale of the current
 20 Grandmaster Clock is PTP (see 8.2.1) and FALSE if the timescale is ARB. This variable is used by both the
 21 BTCA and the external port configuration option.

22 **10.3.9.8 timeTraceable:** A Boolean variable whose value is TRUE if both ClockTimeReceiverTime [i.e.,
 23 the synchronized time maintained at the timeReceiver (see 10.2.4.3)] and currentUtcOffset (see 10.3.9.10),
 24 relative to the current Grandmaster Clock, are traceable to a primary reference and FALSE if one or both are
 25 not traceable to a primary reference. This variable is used by both the BTCA and the external port
 26 configuration option.

27 **10.3.9.9 frequencyTraceable:** A Boolean variable whose value is TRUE if the frequency that determines
 28 clockTimeReceiverTime, i.e., the frequency of the LocalClockEntity multiplied by the most recently
 29 computed rateRatio by the PortSyncSyncReceive state machine (see 10.2.8.1.4), is traceable to a primary
 30 reference and FALSE if this frequency is not traceable to a primary reference. This variable is used by both
 31 the BTCA and the external port configuration option.

32 **10.3.9.10 currentUtcOffset:** The difference between TAI time and UTC time, i.e., TAI time minus UTC
 33 time, in seconds, and relative to the current Grandmaster Clock, when known. Otherwise, the value has no
 34 meaning (see 10.3.9.6). This variable is used by both the BTCA and the external port configuration option.

35 NOTE—For example, 2006-01-01 00:00:00 UTC and 2006-01-01 00:00:33 TAI represent the same instant of time. At
 36 this time, currentUtcOffset was equal to 33 s.¹⁵

37 **10.3.9.11 timeSource:** The value of the timeSource attribute of the current Grandmaster PTP Instance. The
 38 data type for timeSource is TimeSource (see 8.6.2.7). This variable is used by both the BTCA and the
 39 external port configuration option.

40 **10.3.9.12 sysLeap61:** A Boolean variable whose value is TRUE if the last minute of the current UTC day,
 41 relative to the ClockTimeTransmitter entity of this PTP Instance, contains 61 s and FALSE if the last minute

¹⁵ Note also that a leap second was not added at the end of the last UTC minute of 2005-12-31.

1 of the current UTC day does not contain 61 s. This variable is used by both the BTCA and the external port
2 configuration option.

3 **10.3.9.13 sysLeap59:** A Boolean variable whose value is TRUE if the last minute of the current UTC day,
4 relative to the ClockTimeTransmitter entity of this PTP Instance, contains 59 s and FALSE if the last minute
5 of the current UTC day does not contain 59 s. This variable is used by both the BTCA and the external port
6 configuration option.

7 **10.3.9.14 sysCurrentUtcOffsetValid:** A Boolean variable whose value is TRUE if currentUtcOffset (see
8 10.3.9.10), relative to the ClockTimeTransmitter entity of this PTP Instance, is known to be correct and
9 FALSE if currentUtcOffset is not known to be correct. This variable is used by both the BTCA and the
10 external port configuration option.

11 **10.3.9.15 sysPtpTimescale:** A Boolean variable whose value is TRUE if the timescale of the
12 ClockTimeTransmitter entity of this PTP Instance is PTP (see 8.2.1) and FALSE if the timescale of the
13 ClockTimeTransmitter entity of this PTP Instance is ARB. This variable is used by both the BTCA and the
14 external port configuration option.

15 **10.3.9.16 sysTimeTraceable:** A Boolean variable whose value is TRUE if both timeTransmitterTime [i.e.,
16 the time maintained by the ClockTimeTransmitter entity of this PTP Instance (see 10.2.4.21)] and
17 currentUtcOffset (see 10.3.9.10), relative to the ClockTimeTransmitter entity of this PTP Instance, are
18 traceable to a primary reference and FALSE if one or both are not traceable to a primary reference. This
19 variable is used by both the BTCA and the external port configuration option.

20 **10.3.9.17 sysFrequencyTraceable:** A Boolean variable whose value is TRUE if the frequency that
21 determines timeTransmitterTime of the ClockTimeTransmitter entity of this PTP Instance, i.e., the
22 frequency of the LocalClockEntity multiplied by the most recently computed gmRateRatio by the
23 ClockTimeTransmitterSyncReceive state machine (see 10.2.4.14 and 10.2.11), is traceable to a primary
24 reference and FALSE if this frequency is not traceable to a primary reference. This variable is used by both
25 the BTCA and the external port configuration option.

26 **10.3.9.18 sysCurrentUtcOffset:** The difference between TAI time and UTC time, i.e., TAI time minus UTC
27 time, in seconds, and relative to the ClockTimeTransmitter entity of this PTP Instance, when known.
28 Otherwise, the value has no meaning (see 10.3.9.14). This variable is used by both the BTCA and the
29 external port configuration option.

30 NOTE—See the NOTE in 10.3.9.10 for more detail on the sign convention.

31 **10.3.9.19 sysTimeSource:** The value of the timeSource attribute of the ClockTimeTransmitter entity of this
32 PTP Instance (see 8.6.2.7).

33 **10.3.9.20 systemPriority:** The systemPriority vector for this PTP Instance. The data type for systemPriority
34 is UInteger224 (see 10.3.5).

35 **10.3.9.21 gmPriority:** The current gmPriorityVector for the PTP Instance.

36 **10.3.9.22 lastGmPriority:** The previous gmPriorityVector for the PTP Instance, prior to the most recent
37 invocation of the PortStateSelection state machine. lastGmPriority is used only by the BTCA, i.e., not by the
38 external port configuration option.

39 **10.3.9.23 pathTrace:** An array that contains the clockIdentities of the successive PTP Instances that receive,
40 process, and send Announce messages. The data type for pathTrace is ClockIdentity[N], where N is the
41 number of PTP Instances, including the Grandmaster PTP Instance, that the Announce information has
42 traversed. This variable is used by both the BTCA and the external port configuration option.

1 NOTE 1—N is equal to stepsRemoved+1 (see 10.6.3.2.6). The size of the pathTrace array can change after each
 2 reception of an Announce message, up to the maximum size for the respective medium. For example, the maximum
 3 value of N for a full-duplex IEEE 802.3 medium is 179. This is obtained from the fact that the number of PTP octets in
 4 an Announce message is $68 + 8N$, where N is the number of entries in the pathTrace array (see 10.6.3.1 and Table 10-
 5 11), and the maximum payload size for full-duplex IEEE 802.3 media is 1500 octets. Setting $68 + 8N = 1500$, and
 6 solving for N gives $N = 179$.

7 NOTE 2—The current behavior for the path trace feature is documented in 10.3.11.2.1 and 10.3.16.2.1 and is as follows:

- 8 — Item c) of 10.3.11.2.1, the description of the qualifyAnnounce() function of the PortAnnounceReceive state
 9 machine, indicates that if a path trace TLV is present and one of the elements of the pathSequence array field is
 10 equal to the clockIdentity of the clock where the TLV is being processed, the Announce message is not
 11 qualified.
- 12 — Item d) of 10.3.11.2.1 (qualifyAnnounce() function) indicates that if the Announce message is qualified and a
 13 path trace TLV is present, the pathSequence array of the TLV is copied to the pathTrace array (described in this
 14 subclause) and the clockIdentity of the PTP Instance that processes the Announce message is appended to the
 15 array. However, if a path trace TLV is not present, the path trace array is empty.
- 16 — Item f) of 10.3.16.2.1, the description of the txAnnounce() function of the PortAnnounceTransmit state
 17 machine, indicates that a path trace TLV is constructed and appended to an Announce message just before the
 18 Announce message is transmitted only if the pathTrace array is not empty and appending the TLV does not
 19 cause the media-dependent layer frame to exceed any respective maximum size. If appending the TLV does
 20 cause a respective maximum frame size to be exceeded or if the pathTrace array is empty, the TLV is not
 21 appended.
- 22 — As a result of the behaviors of the qualifyAnnounce() and txAnnounce() functions described in this note, the
 23 path trace feature is no longer used, i.e., a path trace TLV is not appended to an Announce message and the
 24 pathTrace array is empty, when appending a clockIdentity to the TLV would cause the frame carrying the
 25 Announce message to exceed its maximum size.

26 NOTE 3—Once the value of stepsRemoved of an Announce message reaches 255, the Announce message is not
 27 qualified [see item b) of 10.3.11.2.1].

28 **10.3.9.24 externalPortConfigurationEnabled:** A variable whose value indicates whether PTP Port states
 29 are externally configured or determined by the BTCA. The value TRUE indicates that the PTP Port states
 30 are externally configured; the value FALSE indicates that the PTP Port states are determined by the BTCA.
 31 This variable is used by both the BTCA and the external port configuration option.

32 **10.3.9.25 lastAnnouncePort:** The PTP Port number of the PTP Port on which the most recent Announce
 33 message was received. This variable is used by the PortAnnounceInformationExt and PortStateSettingExt
 34 state machines for the external port configuration option. This variable is not used by the BTCA.

35 10.3.10 Per-port global variables

36 **10.3.10.1 announceReceiptTimeoutTimeInterval:** The time interval after which announce receipt timeout
 37 occurs if an Announce message has not been received during the interval. The value of
 38 announceReceiptTimeoutTimeInterval is equal to announceReceiptTimeout (see 10.7.3.2) multiplied by the
 39 announceInterval (see 10.3.10.8) for the PTP Port at the other end of the link to which this PTP Port is
 40 attached. The value of announceInterval for the PTP Port at the other end of the link is computed from
 41 logMessageInterval of the received Announce message (see 10.6.2.2.14). This variable is used only by the
 42 BTCA, i.e., not by the external port configuration option.

43 **10.3.10.2 announceSlowdown:** A Boolean that is set to TRUE if the AnnounceIntervalSetting state
 44 machine (see Figure 10-19 in item 10.3.17.3) receives a TLV that requests a larger Announce message
 45 transmission interval (see 10.7.2.2) and FALSE otherwise. When announceSlowdown is set to TRUE, the
 46 PortAnnounceTransmit state machine (see Figure 10-18) continues to send Announce messages at the old
 47 (i.e., faster) rate until a number of Announce messages equal to announceReceiptTimeout (see 10.7.3.2)
 48 have been sent, but with the logMessageInterval field of the PTP common header set equal to the new
 49 announce interval (i.e., corresponding to the slower rate). After announceReceiptTimeout Announce

1 messages have been sent, subsequent Announce messages are sent at the new (i.e., slower) rate and with the
 2 logMessageInterval field of the PTP common header set to the new announce interval. This variable is used
 3 by both the BTCA and the external port configuration option. When announceSlowdown is set to FALSE,
 4 the PortAnnounceTransmit state machine immediately sends Announce messages at the new (i.e., faster or
 5 the same) rate.

6 NOTE—If a receiver of Announce messages requests a slower rate, the receiver continues to use the upstream
 7 announceInterval value, which it obtains from the logMessageInterval field of received Announce messages, until it
 8 receives an Announce message where that value has changed. If, immediately after requesting a slower Announce
 9 message rate, up to announceReceiptTimeout minus one consecutive Announce messages sent to the receiver are lost,
 10 announce receipt timeout could occur if the sender had changed to the slower rate immediately. Delaying the slowing
 11 down of the sending rate of Announce messages for announceReceiptTimeout messages prevents announce receipt
 12 timeout from occurring until at least announceReceiptTimeout Announce messages have been lost. Note that networks
 13 with high packet loss can still experience announce receipt timeout under high-packet-loss conditions; however,
 14 the announce receipt timeout condition occurs only after at least announceReceiptTimeout Announce messages have
 15 been lost.

16 **10.3.10.3 oldAnnounceInterval:** The saved value of the previous announce interval, when a new announce
 17 interval is requested via a Signaling message that contains a message interval request TLV. This variable is
 18 used by both the BTCA and the external port configuration option.

19 **10.3.10.4 infoIs:** An Enumeration2 that takes the values Received, Mine, Aged, or Disabled to indicate the
 20 origin and state of the PTP Port's synchronization tree information:

- 21 a) If infoIs is Received, the PTP Port has received current information (i.e., announce receipt timeout
 22 has not occurred and, if gmPresent is TRUE, sync receipt timeout also has not occurred) from the
 23 timeTransmitter PTP Instance for the attached gPTP communication path.
- 24 b) If infoIs is Mine, information for the PTP Port has been derived from the TimeReceiverPort for the
 25 PTP Instance (with the addition of TimeReceiverPort stepsRemoved). This includes the possibility
 26 that the TimeReceiverPort is the PTP Port whose portNumber is 0, i.e., the PTP Instance is the root
 27 of the gPTP domain.
- 28 c) If infoIs is Aged, announce receipt timeout or, when gmPresent is TRUE, sync receipt timeout has
 29 occurred.
- 30 d) If portOper, ptpPortEnabled, and asCapable are not all TRUE, infoIs is Disabled.

31 The variable infoIs is used only by the BTCA, i.e., not by the external port configuration option.

32 **10.3.10.5 timeTransmitterPriority:** The timeTransmitterPriorityVector for the PTP Port. This variable is
 33 used by both the BTCA and the external port configuration option.

34 **10.3.10.6 currentLogAnnounceInterval:** The current value of the logarithm to base 2 of the mean time
 35 interval, in seconds, between the sending of successive Announce messages (see 10.7.2.2). This value is set
 36 in the AnnounceIntervalSetting state machine (see 10.3.17). This variable is used by both the BTCA and the
 37 external port configuration option.

38 **10.3.10.7 initialLogAnnounceInterval:** The initial value of the logarithm to base 2 of the mean time
 39 interval, in seconds, between the sending of successive Announce messages (see 10.7.2.2). This variable is
 40 used by both the BTCA and the external port configuration option.

41 **10.3.10.8 announceInterval:** A variable containing the mean Announce message transmission interval for
 42 the PTP Port. This value is set in the AnnounceIntervalSetting state machine (see 10.3.17). This variable is
 43 used by both the BTCA and the external port configuration option.

44 **10.3.10.9 messageStepsRemoved:** The value of stepsRemoved contained in the received Announce
 45 information. This variable is used by both the BTCA and the external port configuration option.

1 **10.3.10.10 newInfo:** A Boolean variable that is set to cause a PTP Port to transmit Announce information;
 2 specifically, it is set when an announce interval has elapsed (see Figure 10-18), or PTP Port states have been
 3 updated, or portPriority and portStepsRemoved information has been updated with newly determined
 4 timeTransmitterPriority and timeTransmitterStepsRemoved information. This variable is used by both the
 5 BTCA and the external port configuration option.

6 **10.3.10.11 portPriority:** The portPriorityVector for the PTP Port. This variable is used only by the BTCA,
 7 i.e., not by the external port configuration option.

8 **10.3.10.12 portStepsRemoved:** The value of stepsRemoved for the PTP Port. portStepsRemoved is set
 9 equal to timeTransmitterStepsRemoved (see 10.3.9.3) after timeTransmitterStepsRemoved is updated. This
 10 variable is used by both the BTCA and the external port configuration option.

11 **10.3.10.13 rcvdAnnouncePtr:** A pointer to a structure that contains the fields of a received Announce
 12 message. This variable is used by both the BTCA and the external port configuration option.

13 **10.3.10.14 rcvdMsg:** A Boolean variable that is TRUE if a received Announce message is qualified and
 14 FALSE if it is not qualified. This variable is used only by the BTCA, i.e., not by the external port
 15 configuration option.

16 **10.3.10.15 updtInfo:** A Boolean variable that is set to TRUE to indicate that the PortAnnounceInformation
 17 state machine (see 10.3.12) should copy the newly determined timeTransmitterPriority and
 18 timeTransmitterStepsRemoved to portPriority and portStepsRemoved, respectively. This variable is used by
 19 both the BTCA and the external port configuration option; however, its value does not impact the external
 20 port configuration option (see the NOTE in 10.3.16.3).

21 **10.3.10.16 annLeap61:** A global variable in which the leap61 flag (see 10.6.2.2.8) of a received Announce
 22 message is saved. This variable is used by both the BTCA and the external port configuration option.

23 **10.3.10.17 annLeap59:** A global variable in which the leap59 flag (see 10.6.2.2.8) of a received Announce
 24 message is saved. This variable is used by both the BTCA and the external port configuration option.

25 **10.3.10.18 annCurrentUtcOffsetValid:** A global variable in which the currentUtcOffsetValid flag (see
 26 10.6.2.2.8) of a received Announce message is saved. This variable is used by both the BTCA and the
 27 external port configuration option.

28 **10.3.10.19 annPtpTimescale:** A global variable in which the ptpTimescale flag (see 10.6.2.2.8) of a
 29 received Announce message is saved. This variable is used by both the BTCA and the external port
 30 configuration option.

31 **10.3.10.20 annTimeTraceable:** A global variable in which the timeTraceable flag (see 10.6.2.2.8) of a
 32 received Announce message is saved. This variable is used by both the BTCA and the external port
 33 configuration option.

34 **10.3.10.21 annFrequencyTraceable:** A global variable in which the frequencyTraceable flag (see
 35 10.6.2.2.8) of a received Announce message is saved. This variable is used by both the BTCA and the
 36 external port configuration option.

37 **10.3.10.22 annCurrentUtcOffset:** A global variable in which the currentUtcOffset field (see 10.6.3.2.1) of
 38 a received Announce message is saved. This variable is used by both the BTCA and the external port
 39 configuration option.

1 **10.3.10.23 annTimeSource:** A global variable in which the timeSource field (see 10.6.3.2.1) of a received
 2 Announce message is saved. This variable is used by both the BTCA and the external port configuration
 3 option.

4 **10.3.10.24 receivedPathTrace:** An array in which the pathSequence array field of the path trace TLV of the
 5 most recently received Announce message is saved. The data type for receivedPathTrace is
 6 clockIdentity[N], where N is the number of entries in the pathSequence array field.

7 **10.3.11 PortAnnounceReceive state machine**

8 **10.3.11.1 State machine variables**

9 The following variable is used in the state diagram in Figure 10-13 (in 10.3.11.3):

10 **10.3.11.1.1 rcvdAnnouncePAR:** A Boolean variable that notifies the current state machine when Announce
 11 message information is received from the MD entity of the same PTP Port. This variable is reset by this state
 12 machine.

13 **10.3.11.2 State machine functions**

14 **10.3.11.2.1 qualifyAnnounce (rcvdAnnouncePtr):** Qualifies the received Announce message pointed to
 15 by rcvdAnnouncePtr as follows:

- 16 a) If the Announce message was sent by the current PTP Instance, i.e., if
 17 sourcePortIdentity.clockIdentity (see 10.6.2.2.11 and 8.5.2) is equal to thisClock (see 10.2.4.22), the
 18 Announce message is not qualified, and FALSE is returned;
- 19 b) If the stepsRemoved field is greater than or equal to 255, the Announce message is not qualified, and
 20 FALSE is returned;
- 21 c) If a path trace TLV is present and one of the elements of the pathSequence array field of the path
 22 trace TLV is equal to thisClock (i.e., the clockIdentity of the current PTP Instance; see 10.2.4.22),
 23 the Announce message is not qualified, and FALSE is returned;
- 24 d) Otherwise, the Announce message is qualified, and TRUE is returned. If a path trace TLV is present,
 25 it is saved in the per port global variable receivedPathTrace. If a path trace TLV is not present, the
 26 per port global variable receivedPathTrace is set to the empty array.

27 **10.3.11.3 State diagram**

28 The PortAnnounceReceive state machine shall implement the function specified by the state diagram in
 29 Figure 10-13, the local variable specified in 10.3.11.1, the function specified in 10.3.11.2, and the relevant
 30 global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. The state machine is not used if
 31 externalPortConfigurationEnabled is TRUE. The machine receives Announce information from the MD
 32 entity of the same PTP Port, determines if the Announce message is qualified, and if so, sets the rcvdMsg
 33 variable.

34 **10.3.12 PortAnnounceInformation state machine**

35 **10.3.12.1 State machine variables**

36 The following variables are used in the state diagram in Figure 10-14 (in 10.3.12.3):

37 **10.3.12.1.1 announceReceiptTimeoutTime:** A variable used to save the time at which announce receipt
 38 timeout occurs. The data type for announceReceiptTimeoutTime is UScaledNs.

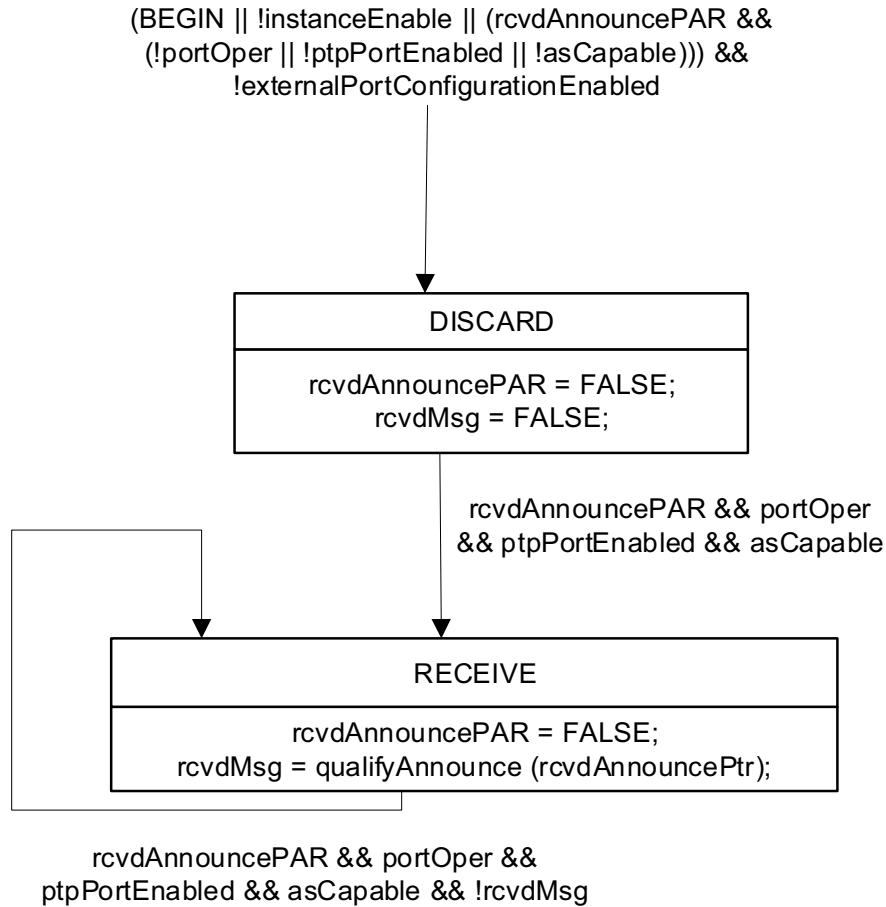


Figure 10-13—PortAnnounceReceive state machine

1 **10.3.12.1.2 messagePriorityPAI:** The messagePriorityVector corresponding to the received Announce
2 information. The data type for messagePriorityPAI is UInteger224 (see 10.3.4).

3 **10.3.12.1.3 rcvdInfo:** An Enumeration2 that holds the value returned by rcvInfo() (see 10.3.12.2.1).

4 **10.3.12.1.4 TEMP:** A temporary variable used to reduce clutter in the state diagram (see Figure 10-14). The
5 data type for TEMP is Integer16.

6 **10.3.12.2 State machine functions**

7 **10.3.12.2.1 rcvInfo (rcvdAnnouncePtr):** Decodes the messagePriorityVector (see 10.3.4 and 10.3.5) and
8 stepsRemoved 10.6.3.2.6) field from the Announce information pointed to by rcvdAnnouncePtr
9 (see 10.3.10.13), and then:

- 10 a) Stores the messagePriorityVector and stepsRemoved field value in messagePriorityPAI and
11 messageStepsRemoved, respectively, and then:
 - 12 1) If the received message conveys the PTP Port state TimeTransmitterPort and the
13 messagePriorityVector is the same as the portPriorityVector of the PTP Port, returns
14 RepeatedTimeTransmitterInfo; else

- 1 2) If the received message conveys the PTP Port state `TimeTransmitterPort` and the
2 `messagePriorityVector` is superior to the `portPriorityVector` of the PTP Port, returns
3 `SuperiorTimeTransmitterInfo`; else
- 4 3) If the received message conveys the PTP Port state `TimeTransmitterPort`, and the
5 `messagePriorityVector` is worse than the `portPriorityVector` of the PTP Port, returns
6 `InferiorTimeTransmitterInfo`; else
- 7 4) Returns `OtherInfo`.

8 NOTE—In accordance with 10.3.5, the `messagePriorityVector` is superior to the `portPriorityVector` of the PTP Port if,
9 and only if, the `messagePriorityVector` is better than the `portPriorityVector`, or the Announce message has been
10 transmitted from the same timeTransmitter PTP Instance and `TimeTransmitterPort` as the `portPriorityVector`. In
11 steps a) 1) to a) 4) in this subclause, `revInfo()` first checks whether the `messagePriorityVector` and `portPriorityVector` are
12 the same (and the received message conveys the PTP Port state `TimeTransmitterPort`), before checking whether the
13 `messagePriorityVector` is superior to the `portPriorityVector`. The reason for this sequence is that
14 `RepeatedTimeTransmitterInfo` needs to be returned if the `messagePriorityVector` and `portPriorityVector` are the same,
15 while `SuperiorTimeTransmitterInfo` needs to be returned in other instances where the Announce message has been
16 transmitted from the same timeTransmitter PTP Instance and `TimeTransmitterPort` as the `portPriorityVector` (if the test
17 for `SuperiorTimeTransmitterInfo` were done before the test for `RepeatedTimeTransmitterInfo`,
18 `SuperiorTimeTransmitterInfo` would be returned when `RepeatedTimeTransmitterInfo` is desired).

19 **10.3.12.2.2 `recordOtherAnnounceInfo()`:** Saves the flags `leap61`, `leap59`, `currentUtcOffsetValid`,
20 `ptpTimescale`, `timeTraceable`, and `frequencyTraceable`, and the fields `currentUtcOffset` and `timeSource`, of
21 the received Announce message for this PTP Port. The values are saved in the per-PTP Port global variables
22 `annLeap61`, `annLeap59`, `annCurrentUtcOffsetValid`, `annPtpTimescale`, `annTimeTraceable`,
23 `annFrequencyTraceable`, `annCurrentUtcOffset`, and `annTimeSource` (see 10.3.10.16 through 10.3.10.23).

24 **10.3.12.3 State diagram**

25 The `PortAnnounceInformation` state machine shall implement the function specified by the state diagram in
26 Figure 10-14, the local variables specified in 10.3.12.1, the functions specified in 10.3.12.2, and the relevant
27 global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. This state machine is used only if
28 `externalPortConfigurationEnabled` is `FALSE` (if this variable is `TRUE`, the `PortAnnounceInformationExt`
29 state machine of 10.3.14.3 is used instead). The state machine receives new qualified Announce information
30 from the `PortAnnounceReceive` state machine (see 10.3.11) of the same PTP Port and determines if the
31 Announce information is better than the current best timeTransmitter information it knows. The state
32 machine also updates the current best timeTransmitter information when it receives updated PTP Port state
33 information from the `PortStateSelection` state machine (see 10.3.13) and when announce receipt timeout or,
34 when `gmPresent` is `TRUE`, sync receipt timeout occurs.

1

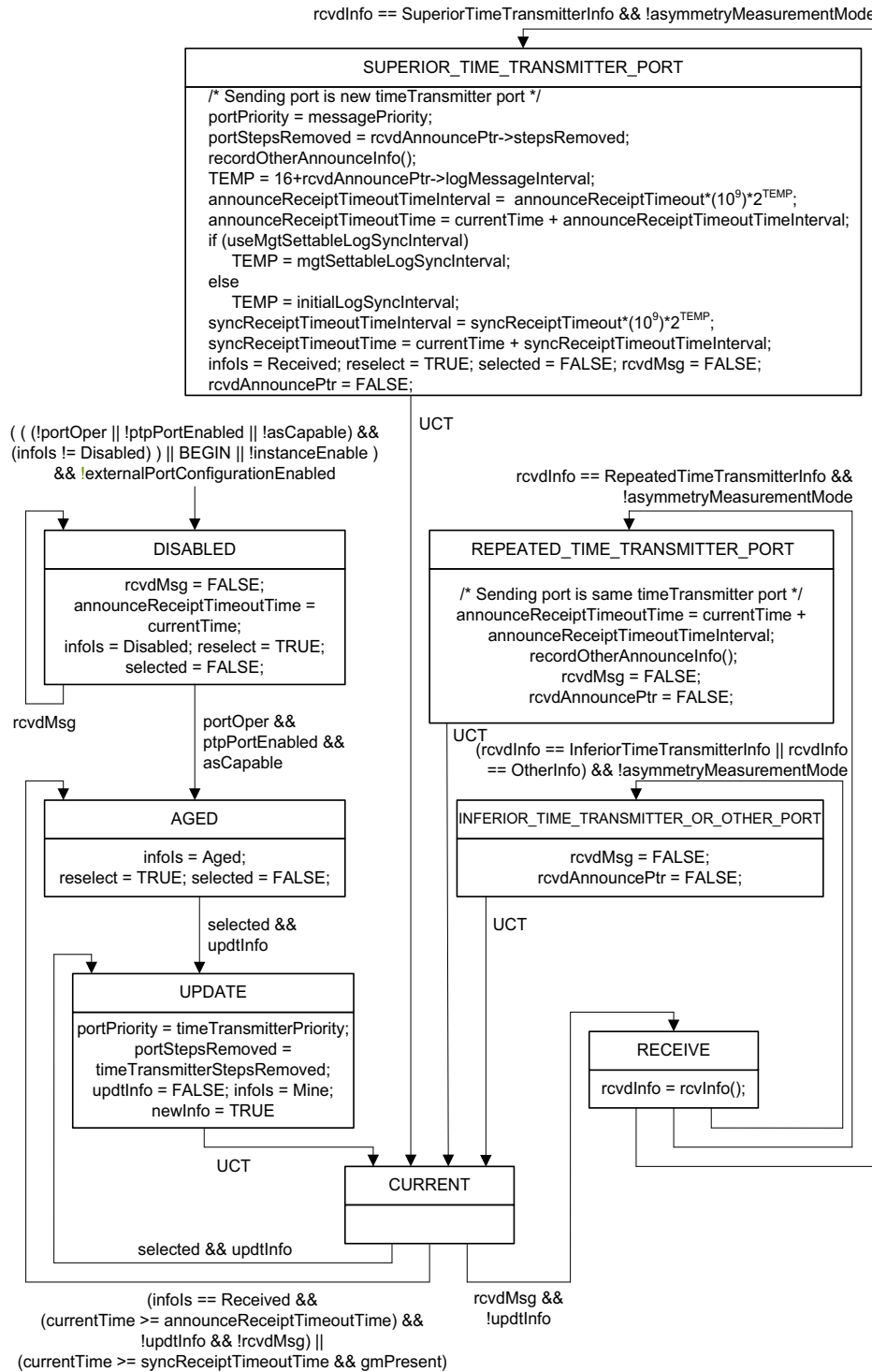


Figure 10-14—PortAnnounceInformation state machine

1 10.3.13 PortStateSelection state machine

2 10.3.13.1 State machine variables

3 The following variables are used in the state diagram in Figure 10-15 (in 10.3.13.3):

4 **10.3.13.1.1 systemIdentityChange:** A Boolean variable that notifies the current state machine when the
5 systemIdentity (see 10.3.2) of this PTP Instance has changed. This variable is reset by this state machine.

6 The systemIdentity changes when at least one of the attributes priority1, clockClass, clockAccuracy,
7 offsetScaledLogVariance, and priority2 changes (e.g., due to management action, degradation or loss of the
8 ClockSource). The systemIdentity also includes the attribute clockIdentity, but this attribute does not
9 change.

10 **10.3.13.1.2 asymmetryMeasurementModeChange:** A Boolean variable that notifies the current state
11 machine when the per-port variable asymmetryMeasurementMode (see 10.2.5.2) changes on any port. This
12 variable is reset by this state machine. There is one instance of asymmetryMeasurementModeChange for all
13 the domains (per port). The variable is accessible by all the domains.

14 10.3.13.2 State machine functions

15 **10.3.13.2.1 updtStateDisabledTree():** Sets all the elements of the selectedState array (see 10.2.4.20) to
16 DisabledPort. Sets lastGmPriority to all ones. Sets the pathTrace array (see 10.3.9.23) to contain the single
17 element thisClock (see 10.2.4.22).

18 **10.3.13.2.2 clearReselectTree():** Sets all the elements of the reselect array (see 10.3.9.1) to FALSE.

19 **10.3.13.2.3 setSelectedTree():** Sets all the elements of the selected array (see 10.3.9.2) to TRUE.

20 **10.3.13.2.4 updtStatesTree():** Performs the following operations (see 10.3.4 and 10.3.5 for details on the
21 priority vectors):

- 22 a) Computes the gmPathPriorityVector for each PTP Port that has a portPriorityVector and for which
23 neither announce receipt timeout nor, if gmPresent is TRUE, sync receipt timeout have occurred,
- 24 b) Saves gmPriority (see 10.3.9.21) in lastGmPriority (see 10.3.9.22), computes the gmPriorityVector
25 for the PTP Instance and saves it in gmPriority, chosen as the best of the set consisting of the
26 systemPriorityVector (for this PTP Instance) and the gmPathPriorityVector for each PTP Port for
27 which the clockIdentity of the timeTransmitter port is not equal to thisClock (see 10.2.4.22),
- 28 c) Sets the per PTP Instance global variables leap61, leap59, currentUtcOffsetValid, ptpTimescale,
29 timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource as follows:
 - 30 1) If the gmPriorityVector was set to the gmPathPriorityVector of one of the ports, then leap61,
31 leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable,
32 currentUtcOffset, and timeSource are set to annLeap61, annLeap59,
33 annCurrentUtcOffsetValid, annPtpTimescale, annTimeTraceable, annFrequencyTraceable,
34 annCurrentUtcOffset, and annTimeSource, respectively, for that PTP Port.
 - 35 2) If the gmPriorityVector was set to the systemPriorityVector, then leap61, leap59,
36 currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset,
37 and timeSource are set to sysLeap61, sysLeap59, sysCurrentUtcOffsetValid, sysPtpTimescale,
38 sysTimeTraceable, sysFrequencyTraceable, sysCurrentUtcOffset, and sysTimeSource,
39 respectively.
- 40 d) Updates the timePropertiesDS members leap61, leap59, currentUtcOffsetValid, ptpTimescale,
41 timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource with the values of the
42 corresponding global variables computed in item c) above.

- 1 e) Updates the parentDS members grandmasterIdentity, grandmasterClockQuality.clockClass,
2 grandmasterClockQuality.clockAccuracy, grandmasterClockQuality.offsetScaledLogVariance,
3 grandmasterPriority1, grandmasterPriority2 with the clockIdentity, clockClass, clockAccuracy,
4 offsetScaledLogVariance, priority1, and priority2 attributes, respectively, of the systemIdentity
5 component of the gmPriority computed in item b) above.
- 6 f) Updates the parentDS member parentPortIdentity with the sourcePortIdentity component of the
7 gmPriority computed in item b) above.
- 8 g) Computes the timeTransmitterPriorityVector for each PTP Port.
- 9 h) Computes timeTransmitterStepsRemoved, which is equal to one of the following:
 - 10 1) messageStepsRemoved (see 10.3.10.9) for the PTP Port associated with the gmPriorityVector,
11 incremented by 1, if the gmPriorityVector is not the systemPriorityVector, or
 - 12 2) 0 if the gmPriorityVector is the systemPriorityVector.
- 13 i) Sets currentDS.stepsRemoved equal to timeTransmitterStepsRemoved.
- 14 j) Assigns the PTP Port state for PTP Port j, and sets selectedState[j] equal to this PTP Port state, as
15 follows, for j = 1, 2, ..., numberPorts:
 - 16 1) If the PTP Port is disabled (infoIs == Disabled), then selectedState[j] is set to DisabledPort.
 - 17 2) If asymmetryMeasurementMode is TRUE, then selectedState[j] is set to PassivePort, and
18 updtInfo is set to FALSE.
 - 19 3) If announce receipt timeout, or sync receipt timeout with gmPresent set to TRUE, has occurred
20 (infoIs = Aged), then selectedState[j] is set to TimeTransmitterPort, and updtInfo is set to
21 TRUE.
 - 22 4) If the portPriorityVector was derived from another PTP Port on the PTP Instance or from the
23 PTP Instance itself as the root (infoIs == Mine), then selectedState[j] is set to
24 TimeTransmitterPort. In addition, updtInfo is set to TRUE if the portPriorityVector differs from
25 the timeTransmitterPriorityVector or portStepsRemoved differs from
26 timeTransmitterStepsRemoved.
 - 27 5) If the portPriorityVector was received in an Announce message, neither announce receipt
28 timeout nor sync receipt timeout with gmPresent TRUE has not occurred (infoIs == Received),
29 and the gmPriorityVector is now derived from the portPriorityVector, then selectedState[j] is set
30 to TimeReceiverPort, and updtInfo is set to FALSE. The per port global variable
31 receivedPathTrace, for this port, is copied to the per PTP Instance global array pathTrace, and,
32 if it is not empty, thisClock is appended to pathTrace.
 - 33 6) If the portPriorityVector was received in an Announce message, neither announce receipt
34 timeout nor sync receipt timeout with gmPresent TRUE has not occurred (infoIs == Received),
35 the gmPriorityVector is not now derived from the portPriorityVector, the
36 timeTransmitterPriorityVector is not better than the portPriorityVector, and the
37 sourcePortIdentity component of the portPriorityVector *does not* reflect another PTP Port on
38 the PTP Instance, then selectedState[j] is set to PassivePort, and updtInfo is set to FALSE.
 - 39 7) If the portPriorityVector was received in an Announce message, neither announce receipt
40 timeout nor sync receipt timeout with gmPresent TRUE has not occurred (infoIs == Received),
41 the gmPriorityVector is not now derived from the portPriorityVector, the
42 timeTransmitterPriorityVector is not better than the portPriorityVector, and the
43 sourcePortIdentity component of the portPriorityVector *does* reflect another PTP Port on the
44 PTP Instance, then selectedState[j] is set to PassivePort, and updtInfo is set to FALSE.
 - 45 8) If the portPriorityVector was received in an Announce message, neither announce receipt
46 timeout nor sync receipt timeout with gmPresent TRUE has not occurred (infoIs == Received),
47 the gmPriorityVector is not now derived from the portPriorityVector, and the
48 timeTransmitterPriorityVector is better than the portPriorityVector, then selectedState[j] is set to
49 TimeTransmitterPort, and updtInfo is set to TRUE.
- 50 k) Updates gmPresent as follows:

- 1) gmPresent is set to TRUE if the priority1 field of the rootSystemIdentity of the gmPriorityVector is less than 255.
- 2) gmPresent is set to FALSE if the priority1 field of the rootSystemIdentity of the gmPriorityVector is equal to 255.
- l) Assigns the PTP Port state for PTP Port 0 (see 8.5.2.3), and sets selectedState[0] as follows:
 - 1) if selectedState[j] is set to TimeReceiverPort for any PTP Port with portNumber j, j = 1, 2, ..., numberPorts, selectedState[0] is set to PassivePort.
 - 2) if selectedState[j] is *not* set to TimeReceiverPort for any PTP Port with portNumber j, j = 1, 2, ..., numberPorts, selectedState[0] is set to TimeReceiverPort.
- m) If the clockIdentity member of the systemIdentity (see 10.3.2) member of gmPriority (see 10.3.9.21) is equal to thisClock (see 10.2.4.22), i.e., if the current PTP Instance is the Grandmaster PTP Instance, the pathTrace array is set to contain the single element thisClock (see 10.2.4.22).

10.3.13.3 State diagram

The PortStateSelection state machine shall implement the function specified by the state diagram in Figure 10-15, the functions specified in 10.3.13.2, and the relevant global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. This state machine is used only if externalPortConfigurationEnabled is FALSE (if this variable is TRUE, the PortStateSettingExt state machine is used instead). The state machine updates the gmPathPriority vector for each PTP Port of the PTP Instance, the gmPriorityVector for the PTP Instance, and the timeTransmitterPriorityVector for each PTP Port of the PTP Instance. The state machine determines the PTP Port state for each PTP Port and updates gmPresent.

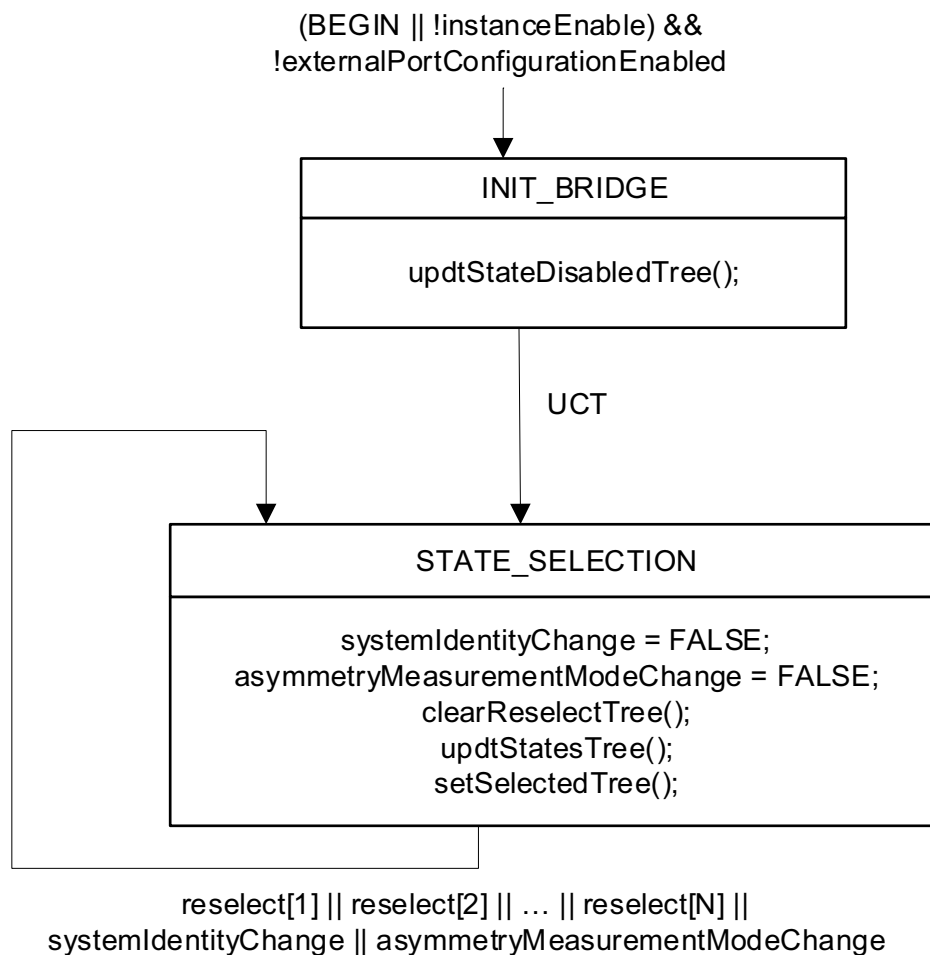


Figure 10-15—PortStateSelection state machine

1 10.3.14 PortAnnounceInformationExt state machine

2 10.3.14.1 State machine variables

3 The following variables are used in the state diagram in Figure 10-16 (in 10.3.14.3):

4 **10.3.14.1.1 rcvdAnnouncePAIE:** A Boolean variable that notifies the current state machine when
5 Announce message information is received from the MD entity of the same PTP Port. This variable is reset
6 by this state machine.

7 **10.3.14.1.2 messagePriorityPAIE:** The messagePriorityVector corresponding to the received Announce
8 information. The data type for messagePriorityPAIE is UInteger224 (see 10.3.4).

9 10.3.14.2 State machine functions

10 **10.3.14.2.1 rcvInfoExt (rcvdAnnouncePtr):** Decodes the messagePriorityVector (see 10.3.4 and 10.3.5)
11 and stepsRemoved (see 10.6.3.2.6) field from the Announce information pointed to by rcvdAnnouncePtr
12 (see 10.3.10.13), and then stores the messagePriorityVector and stepsRemoved field value in
13 messagePriorityPAIE and messageStepsRemoved, respectively. If a path trace TLV is present in the
14 Announce message and the portState of the PTP Port is TimeReceiverPort, the pathSequence array field of
15 the TLV is copied to the global array pathTrace, and thisClock is appended to pathTrace (i.e., is added to the
16 end of the array).

17 **10.3.14.2.2 recordOtherAnnounceInfo():** Saves the flags leap61, leap59, currentUtcOffsetValid,
18 ptpTimescale, timeTraceable, and frequencyTraceable, and the fields currentUtcOffset and timeSource, of
19 the received Announce message for this PTP Port. The values are saved in the per-PTP Port global variables
20 annLeap61, annLeap59, annCurrentUtcOffsetValid, annPtpTimescale, annTimeTraceable,
21 annFrequencyTraceable, annCurrentUtcOffset, and annTimeSource (see 10.3.10.16 through 10.3.10.23).

22 10.3.14.3 State diagram

23 The PortAnnounceInformationExt state machine shall implement the function specified by the state diagram
24 in Figure 10-16, the local variables specified in 10.3.14.1, the functions specified in 10.3.14.2, and the
25 relevant global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. This state machine is used only if
26 externalPortConfigurationEnabled is TRUE (if this variable is FALSE, the PortAnnounceInformation state
27 machine of 10.3.12.3 is used instead). The state machine receives Announce information from the MD entity
28 of the same PTP Port and saves the information.

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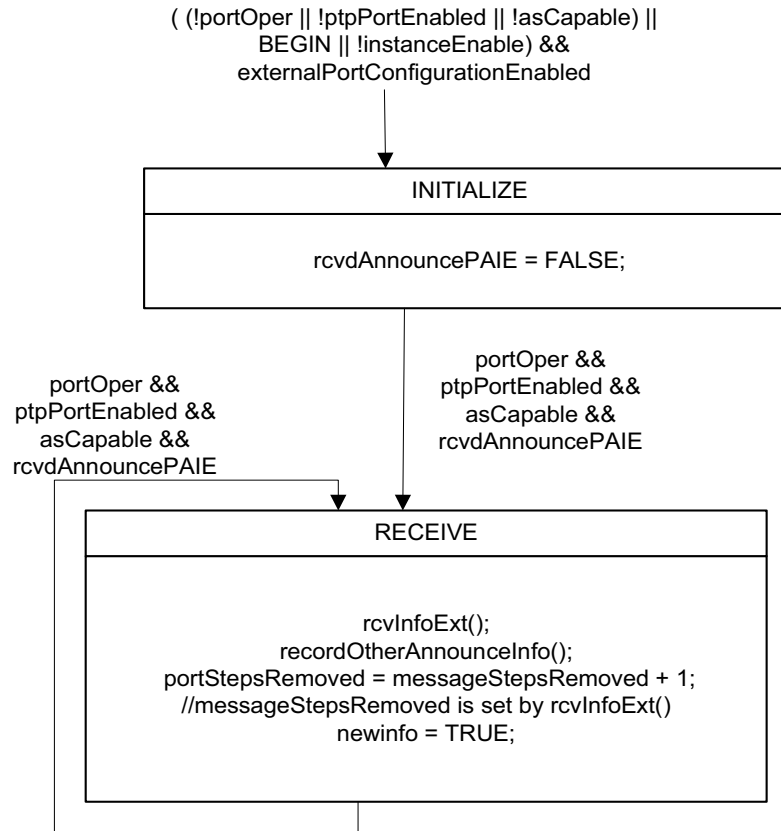


Figure 10-16—PortAnnounceInformationExt state machine

1 10.3.15 PortStateSettingExt state machine

2 10.3.15.1 State machine variables

3 The following variables are used in the state diagram in Figure 10-17 (in 10.3.15.3):

4 **10.3.15.1.1 disabledExt:** A Boolean variable that notifies the current state machine (i.e., when it is set to
5 TRUE) when at least one of the variables portOper, ptpPortEnabled, or asCapable, for this PTP Port, has
6 changed from TRUE to FALSE. This variable is reset by this state machine.

7 **10.3.15.1.2 reenabledExt:** A Boolean variable that notifies the current state machine (i.e., when it is set to
8 TRUE) when all of the variables portOper, ptpPortEnabled, and asCapable, for this PTP Port, that are
9 FALSE have changed to TRUE. This variable is reset by this state machine.

10 **10.3.15.1.3 asymmetryMeasurementModeChangeThisPort:** A Boolean variable that notifies the current
11 state machine when the per-port variable asymmetryMeasurementMode (see 10.2.5.2) changes on this port.
12 This variable is reset by this state machine. There is one instance of
13 asymmetryMeasurementModeChangeThisPort for all the domains (per port). The variable is accessible by
14 all the domains.

1 **10.3.15.1.4 rcvdPortStateInd:** A Boolean variable that notifies the current state machine (i.e., when it is set
2 to TRUE) when the PTP Port state of this PTP Port has been externally set. This variable is reset by this state
3 machine.

4 **10.3.15.1.5 portStateInd:** An Enumeration2 that indicates the PTP Port state that has been set. The values
5 are TimeTransmitterPort, TimeReceiverPort, and PassivePort.

6 NOTE—The PTP Port state can be externally set to DisabledPort by setting portOper or ptpPortEnabled to FALSE. The
7 PTP Port state is set to DisabledPort when asCapable becomes FALSE.

8 **10.3.15.2 State machine functions**

9 **10.3.15.2.1 resetStateTree(j):** Sets selectedState[j] (see 10.2.4.20) to
10 externalPortConfigurationPortDS.desiredState. Sets the pathTrace array (see 10.3.9.23) to contain the single
11 element thisClock (see 10.2.4.22) if no PTP Port of the PTP Instance has the PTP Port state
12 TimeReceiverPort.

13 **10.3.15.2.2 updtPortState(j):** Performs the following operations for PTP Port j (see 10.3.4 and 10.3.5 for
14 details on the priority vectors):

- 15 a) Sets the per PTP Instance global variables leap61, leap59, currentUtcOffsetValid, ptpTimescale,
16 timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource as follows:
 - 17 1) If the PTP Port state of any PTP Port of this PTP Instance (see 8.5.2.3) is TimeReceiverPort,
18 then leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable,
19 currentUtcOffset, and timeSource are set to annLeap61, annLeap59,
20 annCurrentUtcOffsetValid, annPtpTimescale, annTimeTraceable, annFrequencyTraceable,
21 annCurrentUtcOffset, and annTimeSource, respectively, for that PTP Port.
 - 22 2) If no PTP Port of this PTP Instance (see 8.5.2.3) has the PTP Port state TimeReceiverPort, then
23 leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable,
24 currentUtcOffset, and timeSource are set to sysLeap61, sysLeap59, sysCurrentUtcOffsetValid,
25 sysPtpTimescale, sysTimeTraceable, sysFrequencyTraceable, sysCurrentUtcOffset, and
26 sysTimeSource, respectively.
- 27 b) Update the timePropertiesDS members leap61, leap59, currentUtcOffsetValid, ptpTimescale,
28 timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource with the values of the
29 corresponding global variables computed in item a) above.
- 30 c) Computes timeTransmitterStepsRemoved as follows:
 - 31 1) If the PTP Port state of any PTP Port of this PTP Instance (see 8.5.2.3) is TimeReceiverPort,
32 then timeTransmitterStepsRemoved is set equal to portStepsRemoved for that PTP Port.
 - 33 2) If no PTP Port of this PTP Instance (see 8.5.2.3) has the PTP Port state TimeReceiverPort, then
34 timeTransmitterStepsRemoved is set equal to 0.
- 35 d) Sets currentDS.stepsRemoved equal to timeTransmitterStepsRemoved.
- 36 e) Assigns the PTP Port state for PTP Port j, and sets selectedState[j] equal to this PTP Port state, as
37 follows:
 - 38 1) If disabledExt is TRUE, selectedState[j] is set to DisabledPort, else
 - 39 2) If asymmetryMeasurementMode is TRUE, selectedState[j] is set to PassivePort, else
 - 40 3) selectedState[j] is set to portStateInd. If portStateInd is equal to TimeTransmitterPort, newInfo
41 is set to TRUE.
- 42 f) Updates gmPresent as follows:
 - 43 1) If the PTP Port state of any PTP Port of this PTP Instance (see 8.5.2.3) is TimeReceiverPort
44 and the priority1 field of the rootSystemIdentity of the messagePriorityPAIE of the
45 timeReceiver port is less than 255, gmPresent is set to TRUE, else

- 1 2) If the PTP Port state of any PTP Port of this PTP Instance (see 8.5.2.3) is TimeReceiverPort
2 and the priority1 field of the rootSystemIdentity of the messagePriorityPAIE of the
3 timeReceiver PTP Port is equal to 255, gmPresent is set to FALSE, else
- 4 3) If no PTP Port of this PTP Instance (see 8.5.2.3) has the PTP Port state TimeReceiverPort,
5 gmPresent is set to TRUE if priority1 for this PTP Instance is less than 255 and FALSE if
6 priority1 for this PTP Instance is equal to 255.
- 7 g) Assigns the PTP Port state for PTP Port 0, and sets selectedState[0] as follows:
8 1) If selectedState[j] is set to TimeReceiverPort, selectedState[0] is set to PassivePort.
9 2) If selectedState[j] is *not* set to TimeReceiverPort and selectedState[k] is not equal to
10 TimeReceiverPort for every k not equal to 0 or j, selectedState[0] is set to TimeReceiverPort.
- 11 h) Computes the gmPriorityVector as follows:
12 1) If selectedState[j] is set to TimeReceiverPort, the gmPriorityVector is set equal to
13 messagePriorityPAIE for PTP Port j.
14 2) If selectedState[j] is *not* set to TimeReceiverPort and selectedState[k] is not equal to
15 TimeReceiverPort for every k not equal to 0 or j, the gmPriorityVector is set equal to the
16 systemPriorityVector.
- 17 i) Update the parentDS members grandmasterIdentity, grandmasterClockQuality.clockClass,
18 grandmasterClockQuality.clockAccuracy, grandmasterClockQuality.offsetScaledLogVariance,
19 grandmasterPriority1, grandmasterPriority2 with the clockIdentity, clockClass, clockAccuracy,
20 offsetScaledLogVariance, priority1, and priority2 attributes, respectively, of the systemIdentity
21 component of the gmPriorityVector computed in item h) above.
- 22 j) Update the parentDS member parentPortIdentity with the sourcePortIdentity component of the
23 gmPriorityVector computed in item h) above.
- 24 k) Computes the timeTransmitterPriorityVector for PTP Port j.
- 25 l) If no PTP Port of this PTP Instance has the PTP Port state TimeReceiverPort, the pathTrace array is
26 set to contain the single element thisClock (see 10.2.4.22).

27 10.3.15.3 State diagram

28 The PortStateSettingExt state machine shall implement the function specified by the state diagram
29 in Figure 10-17, the local variables specified in 10.3.15.1, the functions specified in 10.3.15.2, and the
30 relevant global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. This state machine is used only if
31 externalPortConfigurationEnabled is TRUE (if this variable is FALSE, the PortStateSelection state machine
32 of 10.3.13.3 is used instead). A separate instance of this state machine runs on each PTP Port (unlike the
33 PortStateSelection state machine, for which a single instance runs in the PTP Instance and performs
34 operations on all the ports).

35 The state machine updates the gmPriorityVector for the PTP Instance and the timeTransmitterPriorityVector
36 for each PTP Port of the PTP Instance. The state machine determines the PTP Port state for each PTP Port
37 and updates gmPresent.

38

1 NOTE—It is possible to use the external port configuration mechanism to misconfigure the network, e.g., to produce a
2 configuration where one or more PTP Instances have more than one timeReceiver port. Detecting and correcting
3 misconfigurations is outside the scope of this standard.

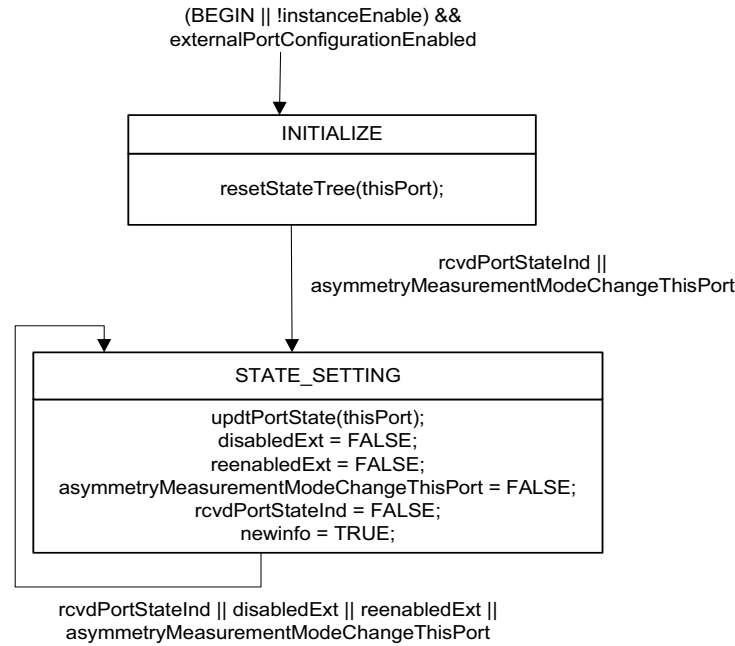


Figure 10-17—PortStateSettingExt state machine

4 10.3.16 PortAnnounceTransmit state machine

5 10.3.16.1 State machine variables

6 The following variables are used in the state diagram in Figure 10-18 (in 10.3.16.3):

7 **10.3.16.1.1 announceSendTime:** The time, relative to the LocalClock, at which the next transmission of
8 Announce information is to occur. The data type for announceSendTime is UScaledNs.

9 **10.3.16.1.2 numberAnnounceTransmissions:** A count of the number of consecutive Announce message
10 transmissions after the AnnounceIntervalSetting state machine (see Figure 10-19 in 10.3.17.3) has set
11 announceSlowdown (see 10.3.10.2) to TRUE. The data type for numberAnnounceTransmissions is
12 UInteger8.

13 **10.3.16.1.3 interval2:** A local variable that holds either announceInterval or oldAnnounceInterval. The data
14 type for interval2 is UScaledNs.

15 10.3.16.2 State machine functions

16 **10.3.16.2.1 txAnnounce():** Transmits Announce information to the MD entity of this PTP Port. The
17 Announce information is set as follows:

- 18 a) The components of the messagePriorityVector are set to the values of the respective components of
19 the timeTransmitterPriorityVector of this PTP Port.

- 1 b) The `grandmasterIdentity`, `grandmasterClockQuality`, `grandmasterPriority1`, and
2 `grandmasterPriority2` fields of the Announce message are set equal to the corresponding
3 components of the `messagePriorityVector`.
- 4 c) The value of the `stepsRemoved` field of the Announce message is set equal to
5 `timeTransmitterStepsRemoved`.
- 6 d) The Announce message flags `leap61`, `leap59`, `currentUtcOffsetValid`, `ptpTimescale`, `timeTraceable`,
7 and `frequencyTraceable`, and the Announce message fields `currentUtcOffset` and `timeSource`, are set
8 equal to the values of the global variables `leap61`, `leap59`, `currentUtcOffsetValid`, `ptpTimescale`,
9 `timeTraceable`, `frequencyTraceable`, `currentUtcOffset`, and `timeSource`, respectively (see 10.3.9.4
10 through 10.3.9.11).
- 11 e) The `sequenceId` field of the Announce message is set in accordance with 10.5.7.
- 12 f) A path trace TLV (see 10.6.3.3) is constructed, with its `pathSequence` field (see 10.6.3.3.4) set equal
13 to the `pathTrace` array (see 10.3.9.23). If appending the path trace TLV to the Announce message
14 does not cause the media-dependent layer frame to exceed any respective maximum size, the path
15 trace TLV is appended to the Announce message; otherwise, it is not appended. If the `pathTrace`
16 array is empty, the path trace TLV is not appended. See 10.3.9.23 for a description of the path trace
17 feature.

18 10.3.16.3 State diagram

19 The `PortAnnounceTransmit` state machine shall implement the function specified by the state diagram in
20 Figure 10-18, the local variables specified in 10.3.16.1, the functions specified in 10.3.16.2, and the relevant
21 global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. The state machine transmits Announce
22 information to the MD entity when an announce interval has elapsed, PTP Port states have been updated,
23 and `portPriority` and `portStepsRemoved` information has been updated with newly determined
24 `timeTransmitterPriority` and `timeTransmitterStepsRemoved` information.

25 NOTE—When the external port configuration option is used (i.e., `externalPortConfigurationEnabled` is `TRUE`; see
26 10.3.9.24) the values of the variables `updtInfo` and `selected` do not affect the operation of the `PortAnnounceTransmit`
27 state machine because the term of the conditions in which they appear, i.e., (`selected && !updtInfo`) ||
28 `externalPortConfigurationEnabled`, evaluates to `TRUE` when `externalPortConfigurationEnabled` is `TRUE`.

1

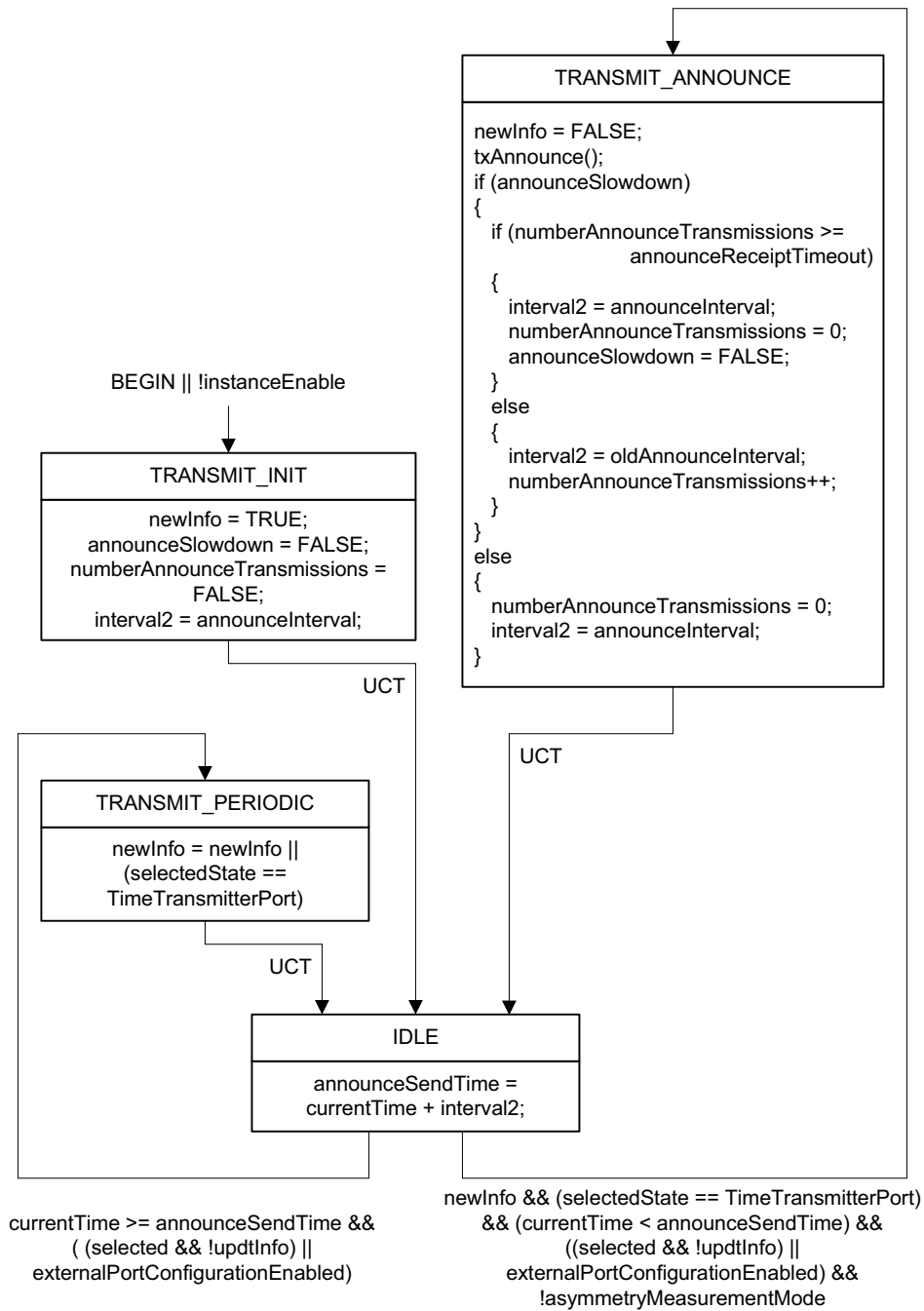


Figure 10-18—PortAnnounceTransmit state machine

2 10.3.17 AnnounceIntervalSetting state machine

1 10.3.17.1 State machine variables

2 The following variables are used in the state diagram in Figure 10-19 (in 10.3.17.3):

3 **10.3.17.1.1 rcvdSignalingMsg2:** A Boolean variable that notifies the current state machine when a
4 Signaling message that contains a Message Interval Request TLV (see 10.6.4.3) is received. This variable is
5 reset by the current state machine.

6 **10.3.17.1.2 rcvdSignalingPtrAIS:** A pointer to a structure whose members contain the values of the fields
7 of the received Signaling message that contains a Message Interval Request TLV (see 10.6.4.3).

8 **10.3.17.1.3 logSupportedAnnounceIntervalMax:** The maximum supported logarithm to base 2 of the
9 announce interval. The data type for logSupportedAnnounceIntervalMax is Integer8.

10 **10.3.17.1.4 logSupportedClosestLongerAnnounceInterval:** The logarithm to base 2 of the announce
11 interval, such that $\logSupportedClosestLongerAnnounceInterval > \logRequestedAnnounceInterval$, that is
12 numerically closest to $\logRequestedAnnounceInterval$, where $\logRequestedAnnounceInterval$ is the
13 argument of the function `computeLogAnnounceInterval()` (see 10.3.17.2.2). The data type for
14 `logSupportedClosestLongerAnnounceInterval` is Integer8.

15 **10.3.17.1.5 computedLogAnnounceInterval:** A variable used to hold the result of the function
16 `computeLogAnnounceInterval()`. The data type for `computedLogAnnounceInterval` is Integer8.

17 **10.3.17.1.6 TEMP:** A temporary variable used to reduce clutter in the state diagram (see Figure 10-19). The
18 data type for TEMP is Integer16.

19 10.3.17.2 State machine functions

20 **10.3.17.2.1 isSupportedLogAnnounceInterval (logAnnounceInterval):** A Boolean function that returns
21 TRUE if the announce interval given by the argument `logAnnounceInterval` is supported by the PTP Port
22 and FALSE if the announce interval is not supported by the PTP Port. The argument `logAnnounceInterval`
23 has the same data type and format as the field `logAnnounceInterval` of the message interval request TLV (see
24 10.6.4.3.8).

1 10.3.17.2.2 computeLogAnnounceInterval (logRequestedAnnounceInterval): An Integer8 function that
 2 computes and returns the logAnnounceInterval, based on the logRequestedAnnounceInterval. This function
 3 is defined as indicated below. It is defined here so that the detailed code that it invokes does not need to be
 4 placed into the state machine diagram.

```

5 Integer8 computeLogAnnounceInterval (logRequestedAnnounceInterval)
6 Integer8 logRequestedAnnounceInterval;
7 {
8     Integer8 logSupportedAnnounceIntervalMax,
9         logSupportedClosestLongerAnnounceInterval;
10    if (isSupportedLogAnnounceInterval (logRequestedAnnounceInterval))
11        // The requested Announce Interval is supported and returned
12        return (logRequestedAnnounceInterval)
13    else
14    {
15        if (logRequestedAnnounceInterval > logSupportedAnnounceIntervalMax)
16            // Return the largest supported logAnnounceInterval, even if smaller than
17 the requested interval
18            return (logSupportedAnnounceIntervalMax);
19        else
20            // Return the smallest supported logAnnounceInterval that is still larger
21 than
22            // the requested interval.
23            return (logSupportedClosestLongerAnnounceInterval);
24    }
25 }
```

26 10.3.17.3 State diagram

27 The AnnounceIntervalSetting state machine shall implement the function specified by the state diagram in
 28 Figure 10-19, the local variables specified in 10.3.17.1, the functions specified in 10.3.17.2, the messages
 29 specified in 10.6, the relevant global variables specified in 10.2.5 and 10.3.10, the relevant managed objects
 30 specified in 14.10, and the relevant timing attributes specified in 10.7. This state machine is responsible for
 31 setting the global variables that give the duration of the mean interval between successive Announce
 32 messages, both at initialization and in response to the receipt of a Signaling message that contains a Message
 33 Interval Request TLV (see 10.6.4.3).

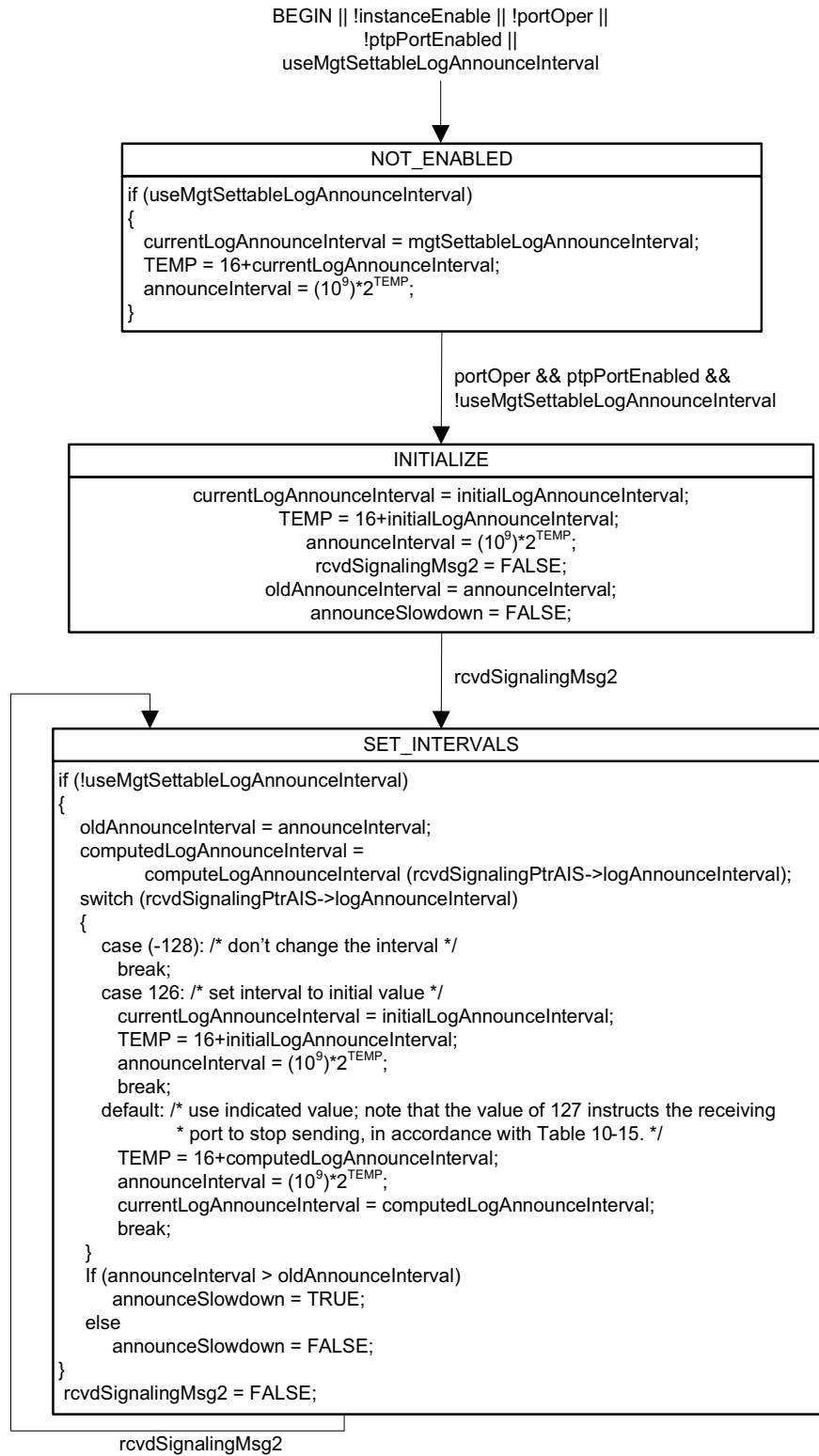


Figure 10-19—AnnounceIntervalSetting state machine

1 10.3.18 SyncIntervalSetting state machine

2 10.3.18.1 State machine variables

3 The following variables are used in the state diagram in Figure 10-20 (in 10.3.18.3):

4 **10.3.18.1.1 rcvdSignalingMsg3:** A Boolean variable that notifies the current state machine when a
5 Signaling message that contains a Message Interval Request TLV (see 10.6.4.3) is received. This variable is
6 reset by the current state machine.

7 **10.3.18.1.2 rcvdSignalingPtrSIS:** A pointer to a structure whose members contain the values of the fields
8 of the received Signaling message that contains a Message Interval Request TLV (see 10.6.4.3).

9 **10.3.18.1.3 logSupportedSyncIntervalMax:** The maximum supported logarithm to base 2 of the sync
10 interval. The data type for logSupportedSyncIntervalMax is Integer8.

11 **10.3.18.1.4 logSupportedClosestLongerSyncInterval:** The logarithm to base 2 of the sync interval, such
12 that $\text{logSupportedClosestLongerSyncInterval} > \text{logRequestedSyncInterval}$, that is numerically closest to
13 $\text{logRequestedSyncInterval}$, where $\text{logRequestedSyncInterval}$ is the argument of the function
14 $\text{computeLogSyncInterval}()$ (see 10.3.18.2.2). The data type for logSupportedClosestLongerSyncInterval is
15 Integer8.

16 **10.3.18.1.5 computedLogSyncInterval:** A variable used to hold the result of the function
17 $\text{computeLogSyncInterval}()$. The data type for computedLogSyncInterval is Integer8.

18 **10.3.18.1.6 TEMP:** A temporary variable used to reduce clutter in the state diagram (see Figure 10-20). The
19 data type for TEMP is Integer16.

20 10.3.18.2 State machine functions

21 **10.3.18.2.1 isSupportedLogSyncInterval (logSyncInterval):** A Boolean function that returns TRUE if the
22 sync interval given by the argument logSyncInterval is supported by the PTP Port and FALSE if the sync
23 interval is not supported by the PTP Port. The argument logSyncInterval has the same data type and format
24 as the field syncInterval of the message interval request TLV (see 10.6.4.3.7).

1 **10.3.18.2.2 computeLogSyncInterval (logRequestedSyncInterval):** An Integer8 function that computes
 2 and returns the logSyncInterval, based on the logRequestedSyncInterval. This function is defined as
 3 indicated below. It is defined here so that the detailed code that it invokes does not need to be placed into the
 4 state machine diagram.

```

5 Integer8 computeLogSyncInterval (logRequestedSyncInterval)
6 Integer8 logRequestedSyncInterval;
7 {
8     Integer8 logSupportedSyncIntervalMax, logSupportedClosestLongerSyncInterval;
9     if (isSupportedLogSyncInterval (logRequestedSyncInterval))
10         // The requested Sync Interval is supported and returned
11         return (logRequestedSyncInterval)
12     else
13     {
14         if (logRequestedSyncInterval > logSupportedSyncIntervalMax)
15             // Return the largest supported logSyncInterval, even if smaller than the
16 requested interval
17             return (logSupportedSyncIntervalMax);
18         else
19             // Return the smallest supported logSyncInterval that is still larger than
20             // the requested interval.
21             return (logSupportedClosestLongerSyncInterval);
22     }
23 }
```

24 **10.3.18.3 State diagram**

25 The SyncIntervalSetting state machine shall implement the function specified by the state diagram in
 26 Figure 10-20, the local variables specified in 10.3.18.1, the functions specified in 10.3.18.2, the messages
 27 specified in 10.6, the relevant global variables specified in 10.2.5, the relevant managed objects specified in
 28 14.10, and the relevant timing attributes specified in 10.7. This state machine is responsible for setting the
 29 global variables that give the duration of the mean intervals between successive Sync messages, both at
 30 initialization and in response to the receipt of a Signaling message that contains a Message Interval Request
 31 TLV (see 10.6.4.3).

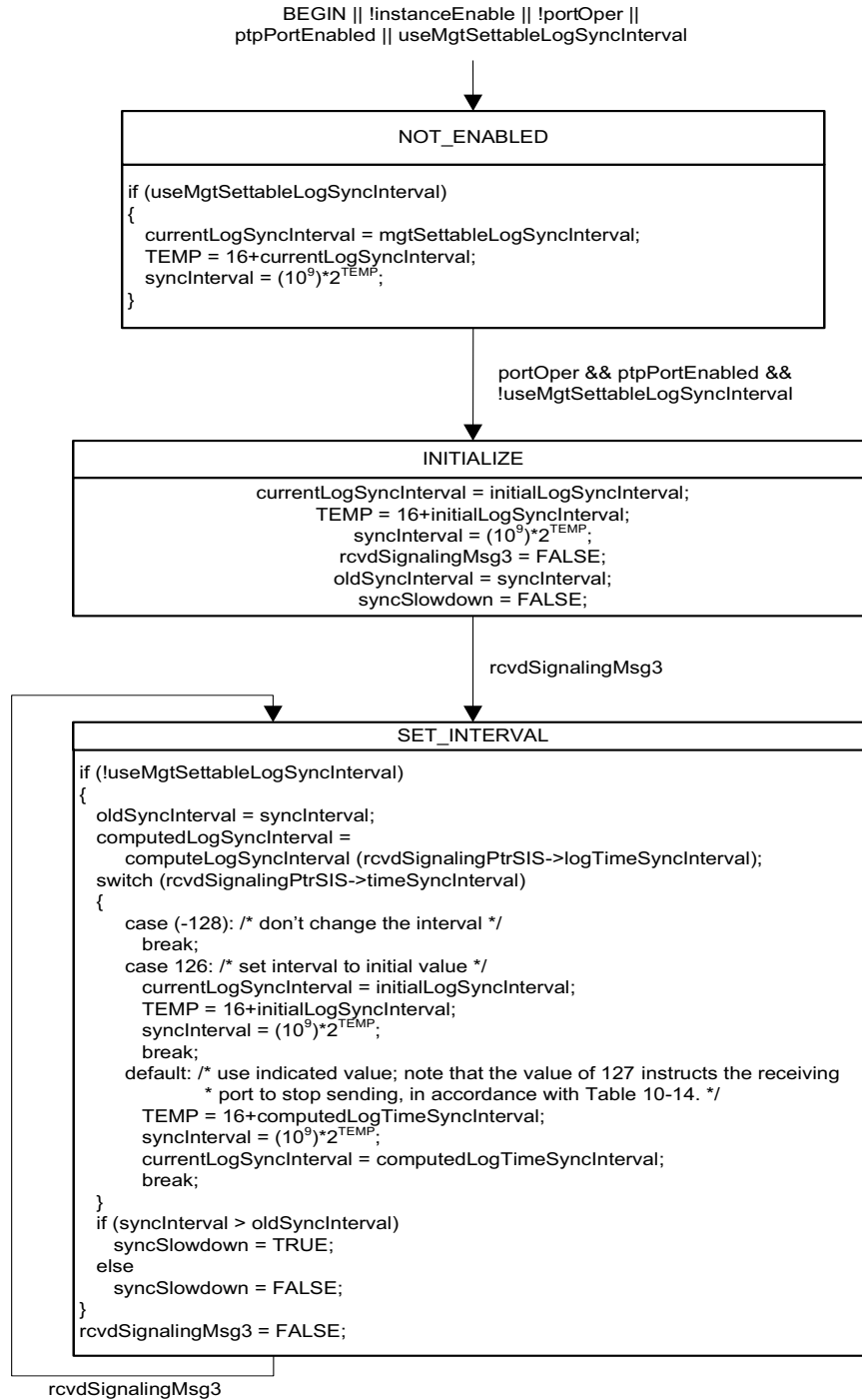


Figure 10-20—SyncIntervalSetting state machine

1 10.4 State machines related to signaling gPTP capability

2 10.4.1 Enabling and disabling the state machines

3 If the managed object `gtpCapableStateMachinesEnabled` (see 14.10.55) is `TRUE`, the
4 `GtpCapableTransmit` and `GtpCapableReceive` state machines shall be enabled and shall function as
5 specified in 10.4.2 and 10.4.3, respectively. The `GtpCapableIntervalSetting` state machine is enabled if it is
6 implemented.

7 If the managed object `gtpCapableStateMachinesEnabled` is `FALSE`, the `GtpCapableTransmit` and
8 `GtpCapableReceive` state machines shall be disabled. The `GtpCapableIntervalSetting` state machine shall
9 be disabled if it is implemented.

10 If the managed object `gtpCapableStateMachinesEnabled` is `FALSE`, the global variable
11 `neighborGtpCapable` for the port (see 10.2.5.16) shall be set to `TRUE`.

12 NOTE 1—The global variable `neighborGtpCapable` indicates to a PTP Port whether the PTP Port at the other end of the
13 attached PTP Link is capable of invoking gPTP, and is used in the determination of `asCapable` (see 10.2.5.1, 11.2.2, 12.4,
14 and 13.4). If `gtpCapableStateMachinesEnabled` is `TRUE`, the variable is set to `TRUE` or `FALSE` by the
15 `GtpCapableReceive` state machine. If `gtpCapableStateMachinesEnabled` is `FALSE`, the variable is automatically set to
16 `TRUE`, that is, it is assumed that the network is engineered such that the PTP Port at the other end of the attached link is
17 capable of invoking gPTP. In this case, it is essential that the network be engineered to fulfill this condition; if
18 `neighborGtpCapable` is `TRUE` and the PTP Port at the other end of the link is not capable of invoking gPTP,
19 undesirable behavior can occur.

20 NOTE 2—If the `GtpCapableTransmit` and `GtpCapableReceive` state machines are disabled, the exchange of peer
21 delay messages can occur immediately, that is, it is not necessary to wait until gPTP capability is determined by the
22 `GtpCapableTransmit` and `GtpCapableReceive` state machines.

23 10.4.2 GtpCapableTransmit state machine

24 10.4.2.1 State machine variables

25 The following variables are used in the state diagram in Figure 10-21 (in 10.4.2.3):

26 **10.4.2.1.1 gPtpCapableSendTime:** The time, relative to the `LocalClock` entity, when a Signaling message
27 containing a gPTP-capable TLV is next sent. The data type for `gPtpCapableSendTime` is `UScaledNs`.

28 **10.4.2.1.2 txSignalingMsgPtr:** A pointer to a structure whose members contain the values of the fields of a
29 Signaling message to be transmitted, which contains a gPTP-capable TLV (see 10.6.4.4).

30 **10.4.2.1.3 interval3:** A local variable that holds either `gPtpCapableMessageInterval` or
31 `oldGtpCapableMessageInterval`. The data type for `interval3` is `UScaledNs`.

32 **10.4.2.1.4 numberGtpCapableMessageTransmissions:** A count of the number of consecutive
33 transmissions of Signaling messages that contain a gPTP-capable TLV, after the `GtpCapableIntervalSetting`
34 state machine (see Figure 10-23 in 10.4.4.3) has set `gPtpCapableMessageSlowdown` (see 10.2.5.19) to
35 `TRUE`. The data type for `numberGtpCapableMessageTransmissions` is `UInteger8`.

36 10.4.2.2 State machine functions

37 **10.4.2.2.1 setGtpCapableTlv():** Creates a structure containing the parameters of a Signaling message that
38 contains a gPTP-capable TLV, to be transmitted (see 10.6.4), and returns a pointer to this structure. The
39 parameters are set as follows:

- 1 a) `logGtpCapableMessageInterval` is set to the value of the managed object
- 2 `currentLogGtpCapableMessageInterval` (see 14.10.28).
- 3 b) The remaining parameters are set as specified in 10.6.4 and 10.6.4.4.

4 **10.4.2.3 State diagram**

5 The `GtpCapableTransmit` state machine shall implement the function specified by the state diagram in
6 Figure 10-21, the local variables specified in 10.4.2.1, the functions specified in 10.4.2.2, the relevant
7 parameters specified in 10.6.4 and 10.6.4.4, and the relevant timing attributes specified in 10.7. This state
8 machine is responsible for setting the parameters of each Signaling message that contains the gPTP-capable
9 TLV, and causing these Signaling messages to be transmitted at a regular rate.

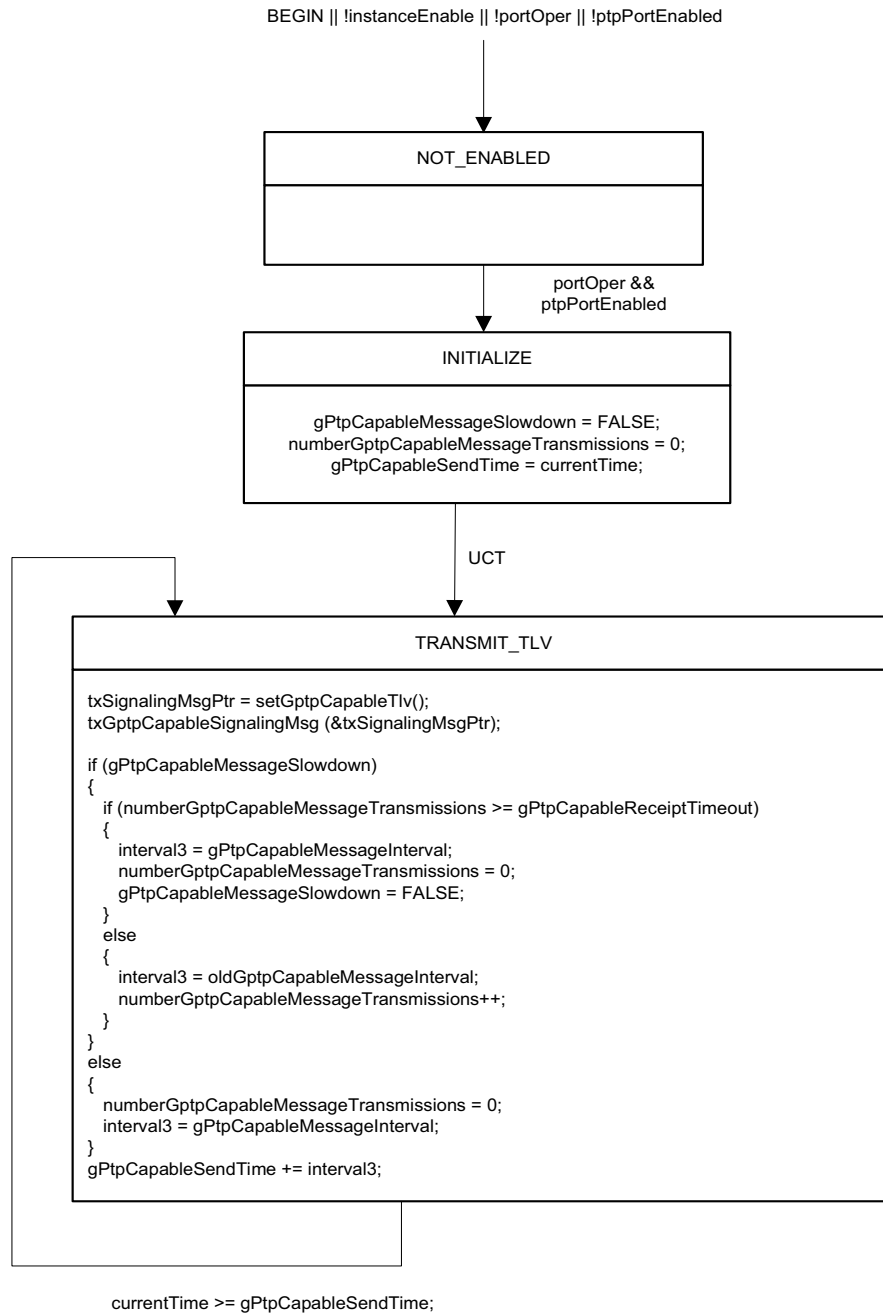


Figure 10-21—GtpCapableTransmit state machine

1 10.4.3 GtpCapableReceive state machine

2 10.4.3.1 State machine variables

3 The following variables are used in the state diagram in Figure 10-22 (in 10.4.3.2):

4 **10.4.3.1.1 rcvdGtpCapableTlv:** A Boolean variable that notifies the current state machine when a
5 Signaling message containing a gPTP-capable TLV is received. This variable is reset by the current state
6 machine.

7 **10.4.3.1.2 rcvdSignalingMsgPtr:** A pointer to a structure whose members contain the values of the fields of
8 a Signaling message whose receipt is indicated by rcvdGtpCapableTlv (see 10.4.3.1.1).

9 **10.4.3.1.3 gPtpCapableReceiptTimeoutTimeInterval:** The time interval after which, if a Signaling
10 message containing a gPTP-capable TLV is not received, the neighbor of this PTP Port is considered to no
11 longer be invoking gPTP. The data type for gPtpCapableReceiptTimeoutTimeInterval is UScaledNs.

12 **10.4.3.1.4 timeoutTime:** A variable used to save the time at which the neighbor of this PTP Port is
13 considered to no longer be invoking gPTP if a Signaling message containing a gPTP-capable TLV is not
14 received. The data type for timeoutTime is UScaledNs.

15 **10.4.3.1.5 TEMP:** A temporary variable used to reduce clutter in the state diagram (see Figure 10-22). The
16 data type for TEMP is Integer16.

17 10.4.3.2 State diagram

18 The GtpCapableReceive state machine shall implement the function specified by the state diagram in
19 Figure 10-22, the local variables specified in 10.4.3.1, the relevant parameters specified in 10.6.4 and
20 10.6.4.4, and the relevant timing attributes specified in 10.7. This state machine is responsible for setting
21 neighborGtpCapable to TRUE on receipt of a Signaling message containing the gPTP-capable TLV, and
22 setting the timeout time after which neighborGtpCapable is set to FALSE.

23 10.4.4 GtpCapableIntervalSetting state machine

24 10.4.4.1 State machine variables

25 The following variables are used in the state diagram in Figure 10-23 (in 10.4.4.3):

26 **10.4.4.1.1 rcvdSignalingMsg5:** A Boolean variable that notifies the current state machine when a Signaling
27 message that contains a gPTP-capable Message Interval Request TLV (see 10.6.4.5) is received. This
28 variable is reset by the current state machine.

29 **10.4.4.1.2 rcvdSignalingPtrGIS:** A pointer to a structure whose members contain the values of the fields of
30 the received Signaling message that contains a gPTP-capable Message Interval Request TLV (see 10.6.4.5).

31 **10.4.4.1.3 logSupportedGtpCapableMessageIntervalMax:** The maximum supported logarithm to base 2
32 of the gPTP-capable message interval. The data type for logSupportedGtpCapableMessageIntervalMax is
33 Integer8.

34 **10.4.4.1.4 logSupportedClosestLongerGtpCapableMessageInterval:** The logarithm to base 2 of the
35 gPTP-capable message interval, such that $\text{logSupportedClosestLongerGtpCapableMessageInterval} >$
36 $\text{logRequestedGtpCapableMessageInterval}$, that is numerically closest to
37 $\text{logRequestedGtpCapableMessageInterval}$, where $\text{logRequestedGtpCapableMessageInterval}$ is the

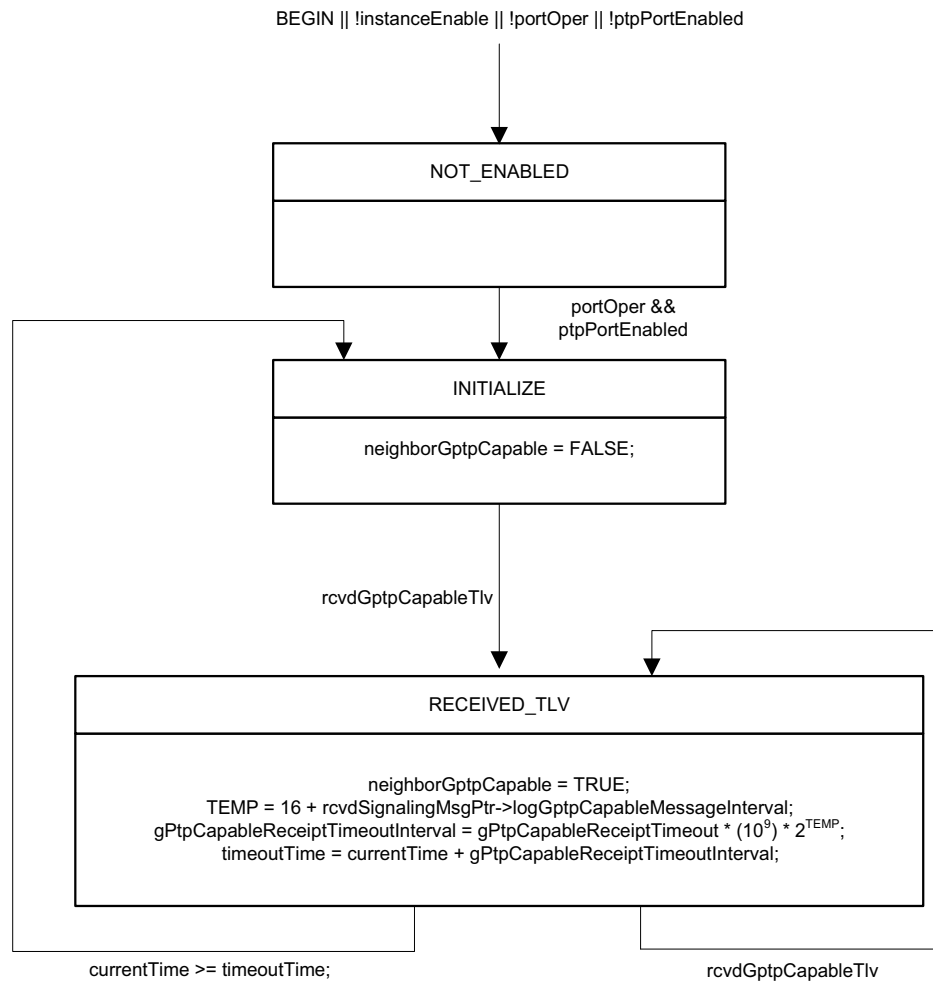


Figure 10-22—GtpCapableReceive state machine

1 argument of the function `computeLogGtpCapableMessageInterval()` (see 10.4.4.2.2). The data type for
2 `logSupportedClosestLongerGtpCapableMessageInterval` is `Integer8`.

3 **10.4.4.1.5 `computedLogGtpCapableMessageInterval`:** A variable used to hold the result of the function
4 `computeLogGtpCapableMessageInterval()`. The data type for `computedLogGtpCapableMessageInterval`
5 is `Integer8`.

6 **10.4.4.1.6 `TEMP`:** A temporary variable used to reduce clutter in the state diagram (see Figure 10-23). The
7 data type for `TEMP` is `Integer16`.

8 10.4.4.2 State machine functions

9 **10.4.4.2.1 `isSupportedLogGtpCapableMessageInterval (logGtpCapableMessageInterval)`:** A
10 Boolean function that returns `TRUE` if the gPTP-capable message interval given by the argument
11 `logGtpCapableMessageInterval` is supported by the PTP Port and `FALSE` if the gPTP-capable message
12 interval is not supported by the PTP Port. The argument `logGtpCapableMessageInterval` has the same data

1 type and format as the field logGtpCapableMessageInterval of the gPTP-capable message interval request
2 TLV (see 10.6.4.5.6).

3 10.4.4.2.2 computeLogGtpCapableMessageInterval (logRequestedGtpCapableMessageInterval):

4 An Integer8 function that computes and returns the logGtpCapableMessageInterval, based on the
5 logRequestedGtpCapableMessageInterval. This function is defined as indicated below. It is defined here so
6 that the detailed code that it invokes does not need to be placed into the state machine diagram.

```

7 Integer8 computeLogGtpCapableMessageInterval (logRequestedGtpCapableMessageInterval)
8 Integer8 logRequestedGtpCapableMessageInterval;
9 {
10     Integer8 logSupportedGtpCapableMessageIntervalMax,
11         logSupportedClosestLongerGtpCapableMessageInterval;
12     if (isSupportedLogGtpCapableMessageInterval
13         (logRequestedGtpCapableMessageInterval))
14         // The requested gPTP-capable Message Interval is supported and returned
15         return (logRequestedGtpCapableMessageInterval)
16     else
17     {
18         if (logRequestedGtpCapableMessageInterval >
19             logSupportedGtpCapableMessageIntervalMax)
20             // Return the largest supported logGtpCapableMessageInterval, even if
21             smaller than the requested interval
22             return (logSupportedGtpCapableMessageIntervalMax);
23         else
24             // Return the smallest supported logGtpCapableMessageInterval that is
25             still larger than
26             // the requested interval.
27             return (logSupportedClosestLongerGtpCapableMessageInterval);
28     }
29 }
```

30 10.4.4.3 State diagram

31 The GtpCapableIntervalSetting state machine shall implement the function specified by the state diagram
32 in Figure 10-23, the local variables specified in 10.4.4.1, the functions specified in 10.4.4.2, the messages
33 specified in 10.6, the relevant global variables specified in 10.2.5, the relevant managed objects specified in
34 14.10, and the relevant timing attributes specified in 10.7. This state machine is responsible for setting the
35 global variables that give the duration of the mean intervals between successive Signaling messages
36 containing the gPTP-capable TLV, both at initialization and in response to the receipt of a Signaling message
37 that contains a gPTP-capable Message Interval Request TLV (see 10.6.4.5).

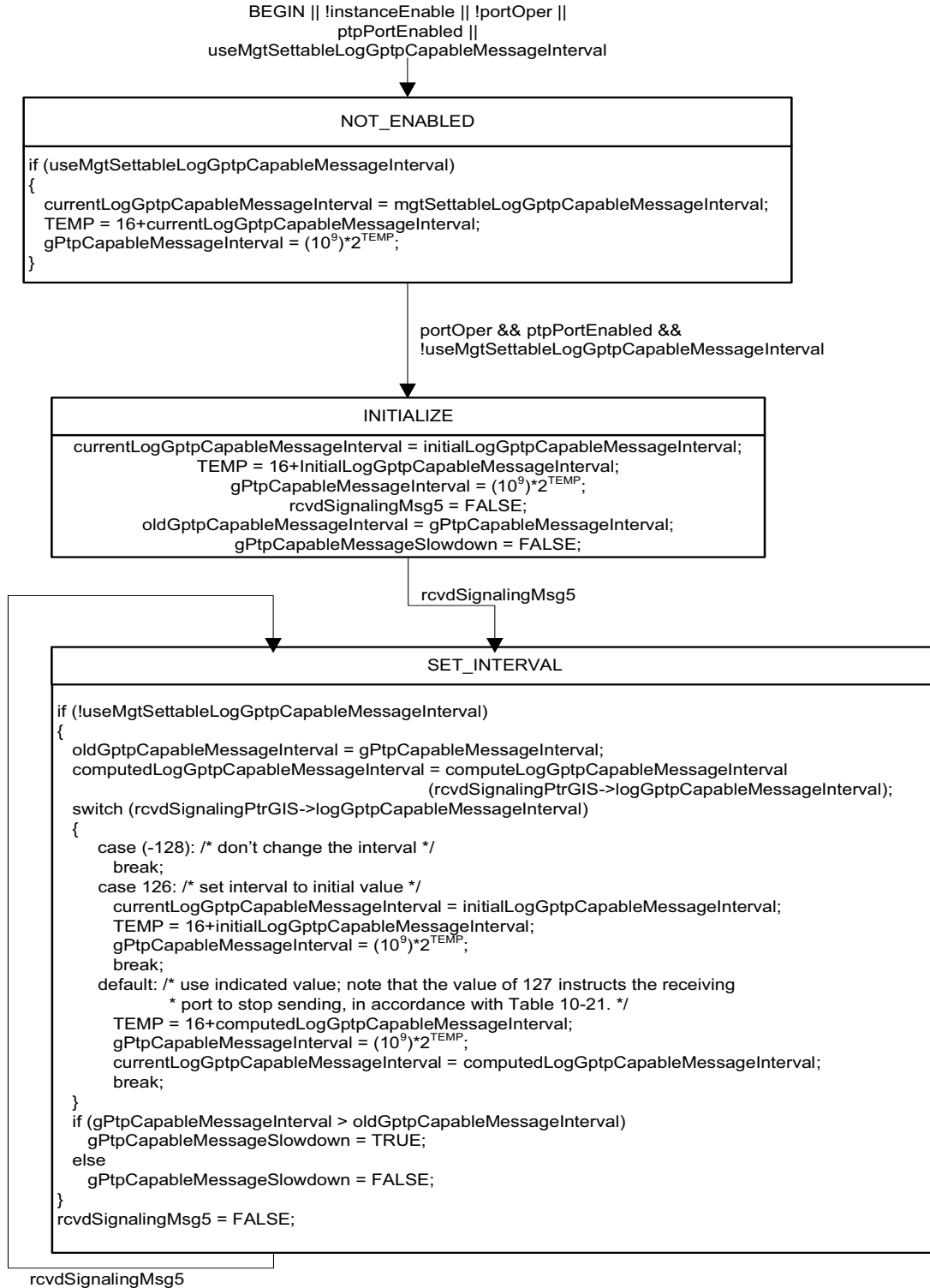


Figure 10-23—GtpCapableIntervalSetting state machine

10.5 Message attributes

10.5.1 General

This subclause describes media-independent attributes of the Announce message and the Signaling message that are not described in 8.4.2 and whose descriptions are not generic to all messages used in this standard. This subclause also describes media-independent attributes of all time-synchronization event messages.

10.5.2 Message class

The Announce message is a general message, i.e., it is not timestamped. An Announce message provides status and characterization information of the PTP Instance that transmitted the message and the Grandmaster PTP Instance. This information is used by the receiving PTP Instance when executing the BTCA.

The Signaling message is a general message, i.e., it is not timestamped. A Signaling message carries information, requests, and/or commands between PTP Instances, via one or more TLVs.

NOTE—In this standard, the Signaling message is used by a port of a PTP Instance to request that the port at the other end of the link send time-synchronization event messages, link delay measurement messages, or Announce messages at desired intervals; to indicate whether the port at the other end of the link should compute neighborRateRatio and/or meanLinkDelay; and to indicate whether a PTP Port can receive and correctly process one-step Syncs. The message interval request TLV is defined to carry this information (see 10.6.4.3). One usage of this functionality is to allow a time-aware system in power-saving mode to remain connected to a gPTP domain via the port on which the Signaling message is sent.

10.5.3 Addresses

The destination address of the Announce and Signaling messages shall be the reserved multicast address given in Table 10-4 unless otherwise specified in a media-dependent clause (see 12.2 and 16.2).

Table 10-4—Destination address for Announce and Signaling messages

Destination address
01-80-C2-00-00-0E
NOTE—This address is taken from Table 8-1, Table 8-2, and Table 8-3 of IEEE Std 802.1Q-2022.

NOTE—Frames whose destination address is the address of Table 10-4 are never forwarded, according to IEEE 802.1Q protocol. Use of this address is shared by IEEE 802.1AS and other IEEE 802.1 protocols.

If the transport is full-duplex IEEE 802.3, all Announce and Signaling messages shall use the MAC address of the respective egress physical port as the source address.

10.5.4 EtherType

The EtherType of the Announce and Signaling messages shall be the EtherType given in Table 10-5.

NOTE—This EtherType is used for all PTP messages.

10.5.5 Subtype

The subtype for the Announce and Signaling messages is indicated by the majorSdoId field (see 10.6.2.2.1).

Table 10-5—EtherType for Announce and Signaling messages

EtherType
88-F7

10.5.6 Source port identity

The Announce message, Signaling message, and all time-synchronization messages contain a sourcePortIdentity field (see 10.6.2.2.11), which identifies the egress port (see 8.5) on which the respective message is sent.

10.5.7 Sequence number

Each PortSync entity of a PTP Instance maintains a separate sequenceId pool for each of the message types Announce and Signaling, respectively, transmitted by the MD entity of the PTP Port.

Each Announce and Signaling message contains a sequenceId field (see 10.6.2.2.12), which carries the message sequence number. The sequenceId of an Announce message shall be one greater than the sequenceId of the previous Announce message sent by the transmitting PTP Port, subject to the constraints of the rollover of the UInteger16 data type used for the sequenceId field. The sequenceId of a Signaling message shall be one greater than the sequenceId of the previous Signaling message sent by the transmitting PTP Port, subject to the constraints of the rollover of the UInteger16 data type used for the sequenceId field.

10.6 Message formats

10.6.1 General

The PTP messages Announce and Signaling each have a header, body, and, if present, a suffix that contains one or more TLVs (see 10.6.2, 10.6.3, and 10.6.4 of this standard and Clause 14 of IEEE Std 1588-2019). Reserved fields shall be transmitted with all bits of the field 0 and ignored by the receiver, unless otherwise specified. The data type of the field shall be the type indicated in brackets in the title of each subclause.

Subclause 10.6 defines the path trace TLV, which is carried by the Announce message (see 10.6.3.2.8), and the message interval request TLV, which is carried by the Signaling message (see 10.6.4.3).

PTP Management Messages are not used in this standard. They are specified in IEEE Std 1588-2019.

IEEE Std 1588-2019 specifies various optional features that have associated TLVs. These optional features, including the associated TLVs, may be supported by an implementation of this standard. IEEE Std 1588-2019 also specifies that certain TLVs are propagated by a Boundary Clock if they are attached to an Announce message and are not supported (see 14.2.2.2 and Table 52 of IEEE Std 1588-2019). These TLVs are listed in Table 10-6. The TLV Propagate requirement in IEEE Std 1588-2019 means that a Propagate TLV is propagated through a PTP Relay Instance (e.g., from an ingress PTP Port in the TimeReceiver state to an egress PTP Port in the TimeTransmitter state, even when the TLV is unsupported by the PTP Relay Instance). If the corresponding optional feature is not supported by the PTP Relay Instance, the PTP Relay Instance shall propagate the TLV unchanged.

If a PTP Instance cannot parse a non-forwarding TLV, it shall ignore it and attempt to parse the next TLV (see 14.1 of IEEE Std 1588-2019).

NOTE—Any overhead specific to the respective medium is added to each message.

Table 10-6—Propagate TLVs of IEEE Std 1588-2019

tlvType values	Value (hex)	TLV defined in
PATH_TRACE	0008	16.2 of IEEE Std 1588-2019, and required by 10.6.3.3 of this standard
ALTERNATE_TIME_OFFSET_INDICATOR	0009	16.3 of IEEE Std 1588-2019
ORGANIZATION_EXTENSION_PROPAGATE	4000	14.3 of IEEE Std 1588-2019
ENHANCED_ACCURACY_METRICS	4001	16.12 of IEEE Std 1588-2019
Reserved for assignment by the IEEE 1588 Working Group for TLVs that propagate	4002–7EFF	
Experimental values (see 4.2.9 of IEEE Std 1588-2019)	7FF0–7FFF	

1 10.6.2 Header

2 10.6.2.1 General header specifications

3 The common header for all PTP messages shall be as specified in Table 10-7 and 10.6.2.2.

Table 10-7—PTP message header

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
majorSdoId				messageType				1	0
minorVersionPTP				versionPTP				1	1
messageLength								2	2
domainNumber								1	4
minorSdoId								1	5
flags								2	6
correctionField								8	8
messageTypeSpecific								4	16
sourcePortIdentity								10	20
sequenceId								2	30
controlField								1	32
logMessageInterval								1	33

1 10.6.2.2 Header field specifications

2 10.6.2.2.1 majorSdold (Nibble)

3 The value is specified in 8.1 for all transmitted PTP messages of a gPTP domain. The value is specified in
4 11.2.17 for all transmitted PTP messages of the CMLDS Any PTP message received for which the value is
5 not one of the values specified in those subclauses shall be ignored.

6 10.6.2.2.2 messageType (Enumeration4)

7 The value indicates the type of the message, as defined in Table 10-8.

8 The most significant bit of the messageType field divides this field in half between event and general
9 messages, i.e., it is 0 for event messages and 1 for general messages.

Table 10-8—Values for messageType field

Message type	Message class	Value
Announce	General	0xB
Signaling	General	0xC
NOTE—Values for the messageType field for other PTP messages that are used only for specific media are defined in the respective media-dependent clause(s).		

10 10.6.2.2.3 minorVersionPTP (UInteger4)

11 For transmitted messages, the value shall be 1 (see 7.5.4 and 13.3.2.5 of IEEE Std 1588-2019). For received
12 messages, the value is ignored.

13 NOTE—minorVersionPTP indicates the minor version number of IEEE 1588 PTP used in the PTP profile contained in
14 this standard for information only.

15 10.6.2.2.4 versionPTP (UInteger4)

16 For transmitted messages, the value shall be 2 (see 7.5.4 and 13.3.2.4 of IEEE Std 1588-2019). For received
17 messages, if the value is not 2, the entire message shall be ignored.

18 NOTE—versionPTP indicates the version number of IEEE 1588 PTP used in the PTP profile contained in this standard.

19 10.6.2.2.5 messageLength (UInteger16)

20 The value is the total number of octets that form the PTP message. The counted octets start with, and
21 include, the first octet of the header and terminate with, and include, the last octet of the last TLV or, if there
22 are no TLVs, the last octet of the message as defined in this clause.

23 NOTE—For example, the Follow_Up message (see 11.4.4) contains a PTP header (34 octets), preciseOriginTimestamp
24 (10 octets), and Follow_Up information TLV (32 octets). The value of the messageLength field is $34+10+32 = 76$.

25 10.6.2.2.6 domainNumber (UInteger8)

26 The value is the gPTP domainNumber specified in 8.1.

1 10.6.2.2.7 minorSdold (UInteger8)

2 The value is specified in 8.1 for all transmitted PTP messages of a gPTP domain. The value is specified in
 3 11.2.17 for all transmitted PTP messages of the CMLDS. Any PTP message received for which the value is
 4 not one of the values specified in those subclauses shall be ignored.

5 10.6.2.2.8 flags (Octet2)

6 The value of the bits of the array are defined in Table 10-9. For message types where the bit is not defined in
 7 Table 10-9, the value of the bit is set to FALSE.

Table 10-9—Values of flag bits

Octet	Bit	Message types	Name	Value
0	0	All	alternateTimeTransmitterFlag in Announce, Sync, Follow_Up, and Delay_Resp messages	Not used in this standard; transmitted as FALSE and ignored on reception
0	1	Sync, Pdelay_Resp	twoStepFlag	<i>For Sync messages:</i> a) For a one-step transmitting PTP Port (see 11.1.3 and 11.2.13.10), the value is FALSE. b) For a two-step transmitting PTP Port, the value is TRUE. <i>For Pdelay_Resp messages:</i> The value is transmitted as TRUE and ignored on reception.
0	2	All	unicastFlag	Not used in this standard; transmitted as FALSE and ignored on reception
0	3	All	Reserved	Not used by IEEE Std 1588-2019; reserved as FALSE and ignored on reception
0	4	All	Reserved	Not used by IEEE Std 1588-2019; reserved as FALSE and ignored on reception
0	5	All	PTP profileSpecific 1	Not used in this standard; transmitted as FALSE and ignored on reception
0	6	All	PTP profileSpecific 2	Not used in this standard; transmitted as FALSE and ignored on reception
0	7	All	Reserved	Not used in this standard; transmitted as FALSE and ignored on reception
1	0	Announce	leap61	The value of the global variable leap61 (see 10.3.9.4)
1	1	Announce	leap59	The value of the global variable leap59 (see 10.3.9.5)
1	2	Announce	currentUtcOffsetValid	The value of the global variable currentUtcOffsetValid (see 10.3.9.6)

Table 10-9—Values of flag bits (continued)

Octet	Bit	Message types	Name	Value
1	3	Announce	ptpTimescale	For domain 0, transmitted as TRUE and ignored on receipt. For domains other than domain 0, the value of the global variable ptpTimescale (see 10.3.9.7)
1	4	Announce	timeTraceable	The value of the global variable timeTraceable (see 10.3.9.8)
1	5	Announce	frequencyTraceable	The value of the global variable frequencyTraceable (see 10.3.9.9)
1	6	All	Reserved	Not used by IEEE Std 1588-2019; reserved as FALSE and ignored on reception
1	7	All	Reserved	Not used in this standard; reserved as FALSE and ignored on reception

1 10.6.2.2.9 correctionField (Integer64)

2 The value is 0.

3 10.6.2.2.10 messageTypeSpecific (Octet4)

4 The value of the messageTypeSpecific field varies, based on the value of the messageType field, as
5 described in Table 10-10.

6 For Event messages only, the four octets of the messageTypeSpecific field may be used for internal
7 implementation of a PTP Instance and its ports. For example, if the clock consists of multiple hardware
8 components that are not synchronized, messageTypeSpecific can be used to transfer an internal timestamp
9 between components (e.g., a physical layer chip and the clock's processor).

10 The messageTypeSpecific field is not used for features of this standard, and it has no meaning from one
11 clock to another. In the on-the-wire format at each PTP Port, for all messageType values, the
12 messageTypeSpecific field is transmitted with all bits of the field 0 and ignored on receive.

Table 10-10—messageTypeSpecific semantics

Value of messageType	Description
Follow_Up, Pdelay_Resp_Follow_Up, Announce, Signaling, Management	For the General message class, this field is reserved; it is transmitted as 0 and ignored on reception.
Sync, Pdelay_Req, Pdelay_Resp	For the Event message class, this field may be used for internal implementation as specified in this subclause.

13 10.6.2.2.11 sourcePortIdentity (PortIdentity)

14 The value is the PTP Port identity attribute (see 8.5.2) of the PTP Port that transmits the PTP message.

1 10.6.2.2.12 sequenceId (UInteger16)

2 The sequenceId field is assigned as specified in 10.5.7.

3 10.6.2.2.13 controlField (UInteger8)

4 The value is 0.

5 10.6.2.2.14 logMessageInterval (Integer8)

6 For an Announce message, the value is the value of currentLogAnnounceInterval (see 10.3.10.6) for the PTP
7 Port that transmits the Announce message. For a Signaling message, the value is transmitted as 0x7F and
8 ignored on reception.

9 10.6.3 Announce message

10 10.6.3.1 General Announce message specifications

11 The fields of the body of the Announce message shall be as specified in Table 10-11 and 10.6.3.2.

Table 10-11—Announce message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 10.6.2)								34	0
reserved								10	34
currentUtcOffset								2	44
reserved								1	46
grandmasterPriority1								1	47
grandmasterClockQuality								4	48
grandmasterPriority2								1	52
grandmasterIdentity								8	53
stepsRemoved								2	61
timeSource								1	63
path trace TLV								4+8N	64

12 10.6.3.2 Announce message field specifications

13 10.6.3.2.1 currentUtcOffset (Integer16)

14 The value is the value of currentUtcOffset (see 10.3.9.10) for the PTP Instance that transmits the Announce
15 message.

1 **10.6.3.2.2 grandmasterPriority1 (UInteger8)**

2 The value is the value of the priority1 component of the rootSystemIdentity of the gmPriorityVector
3 (see 10.3.5) of the PTP Instance that transmits the Announce message.

4 **10.6.3.2.3 grandmasterClockQuality (ClockQuality)**

5 The value is the clockQuality formed by the clockClass, clockAccuracy, and offsetScaledLogVariance of the
6 rootSystemIdentity of the gmPriorityVector (see 10.3.5) of the PTP Instance that transmits the Announce
7 message.

8 **10.6.3.2.4 grandmasterPriority2 (UInteger8)**

9 The value is the value of the priority2 component of the rootSystemIdentity of the gmPriorityVector
10 (see 10.3.5) of the PTP Instance that transmits the Announce message.

11 **10.6.3.2.5 grandmasterIdentity (ClockIdentity)**

12 The value is the value of the clockIdentity component of the rootSystemIdentity of the gmPriorityVector
13 (see 10.3.5) of the PTP Instance that transmits the Announce message.

14 **10.6.3.2.6 stepsRemoved (UInteger16)**

15 The value is the value of timeTransmitterStepsRemoved (see 10.3.9.3) for the PTP Instance that transmits
16 the Announce message.

17 **10.6.3.2.7 timeSource (TimeSource)**

18 The value is the value of timeSource (see 8.6.2.7 and 10.3.9.11) for the PTP Instance that transmits the
19 Announce message.

20 **10.6.3.2.8 Path trace TLV**

21 The Announce message carries the path trace TLV, defined in 10.6.3.3.

22 **10.6.3.3 Path trace TLV definition**

23 **10.6.3.3.1 General**

24 The fields of the path-trace TLV shall be as specified in Table 10-12 and in 10.6.4.3.2 through 10.6.4.3.9.
25 This TLV and its use are defined in IEEE Std 1588-2019 (see 16.2 and Table 52 of IEEE Std 1588-2019).

26 **10.6.3.3.2 tlvType (Enumeration16)**

27 The value of the tlvType field is 0x8.

28 NOTE—This value indicates the TLV is a path trace TLV, as specified in 16.2.5.1 and Table 52 of IEEE Std 1588-2019.
29 The value 0x8 is specified in that standard as PATH_TRACE.

30 **10.6.3.3.3 lengthField (UInteger16)**

31 The value of the lengthField is 8N.

Table 10-12—Path trace TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
pathSequence								8N	4

1 10.6.3.3.4 pathSequence (ClockIdentity[N])

2 The value of pathSequence is a ClockIdentity array. The array elements are the clockIdentities of the
3 successive PTP Instances that receive and send an Announce message. The quantity N is the number of PTP
4 Instances, including the Grandmaster PTP Instance, that the Announce information has traversed.

5 NOTE—N is equal to stepsRemoved+1 (see 10.6.3.2.6). The size of the pathSequence array increases by 1 for each PTP
6 Instance that the Announce information traverses.

7 10.6.4 Signaling message

8 10.6.4.1 General Signaling message specifications

9 The fields of the body of the Signaling message shall be as specified in Table 10-13 and 10.6.4.2.

Table 10-13—Signaling message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 10.6.2)								34	0
targetPortIdentity								10	34
one or more TLVs								N	44

10 10.6.4.2 Signaling message field specifications

11 10.6.4.2.1 targetPortIdentity (PortIdentity)

12 The value of targetPortIdentity.clock identity is all ones, i.e., 0xFFFFFFFFFFFFFFFF. The value of
13 targetPortIdentity.portNumber is all ones, i.e., 0xFFFF.

14 10.6.4.2.2 TLVs carried in one Signaling message

15 The Signaling message carries either:

- 16 a) One or both interval request TLVs, defined in 10.6.4.3 and 10.6.4.5, or

- 1 b) The gPTP-capable TLV, defined in 10.6.4.4.

2 10.6.4.3 Message interval request TLV definition

3 10.6.4.3.1 General

4 The fields of the message interval request TLV shall be as specified in Table 10-14 and in 10.6.4.3.2 through
5 10.6.4.3.9. This TLV is a standard organization extension TLV for the Signaling message, as specified in
6 14.3 of IEEE Std 1588-2019.

Table 10-14—Message interval request TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
logLinkDelayInterval								1	10
logTimeSyncInterval								1	11
logAnnounceInterval								1	12
flags								1	13
reserved								2	14

7 10.6.4.3.2 tlvType (Enumeration16)

8 The value of the tlvType field is 0x3.

9 NOTE—This value indicates the TLV is a vendor and standard organization extension TLV, as specified in 14.3.2.1 and
10 Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION with a
11 value of 0x3.

12 10.6.4.3.3 lengthField (UInteger16)

13 The value of the lengthField is 12.

14 10.6.4.3.4 organizationId (Octet3)

15 The value of organizationId is 00-80-C2.

16 10.6.4.3.5 organizationSubType (Enumeration24)

17 The value of organizationSubType is 2.

1 10.6.4.3.6 logLinkDelayInterval (Integer8)

2 The value is the logarithm to base 2 of the mean time interval, desired by the port that sends this TLV,
3 between successive Pdelay_Req messages sent by the port at the other end of the link. The format and
4 allowed values of logLinkDelayInterval are the same as the format and allowed values of
5 initialLogPdelayReqInterval (see 11.5.2.2).

6 The values 127, 126, and –128 are interpreted as defined in Table 10-15.

Table 10-15—Interpretation of special values of logLinkDelayInterval

Value	Instruction to PTP Instance that receives this TLV
127	Instructs the port that receives this TLV to stop sending link delay measurement messages.
126	Instructs the port that receives this TLV to set currentLogPdelayReqInterval to the value of initialLogPdelayReqInterval (see 11.5.2.2).
–128	Instructs the port that receives this TLV not to change the mean time interval between successive Pdelay_Req messages.
All values in the ranges [–127, –25] and [25, 125] are reserved.	

7 10.6.4.3.7 logTimeSyncInterval (Integer8)

8 The value is the logarithm to base 2 of the mean time interval, desired by the PTP Port that sends this TLV,
9 between successive time-synchronization event messages sent by the PTP Port at the other end of the link.
10 The format and allowed values of logTimeSyncInterval are the same as the format and allowed values of
11 initialLogSyncInterval (see 10.7.2.3, 11.5.2.3, 12.8, and 13.9.2).

12 The values 127, 126, and –128 are interpreted as defined in Table 10-16.

Table 10-16—Interpretation of special values of logTimeSyncInterval

Value	Instruction to PTP Instance that receives this TLV
127	Instructs the PTP Port that receives this TLV to stop sending time-synchronization event messages.
126	Instructs the PTP Port that receives this TLV to set currentLogSyncInterval to the value of initialLogSyncInterval (see 10.7.2.3, 11.5.2.3, 12.8, and 13.9.2).
–128	Instructs the PTP Port that receives this TLV not to change the mean time interval between successive time-synchronization event messages.
All values in the ranges [–127, –25] and [25, 125] are reserved.	

13 When a Signaling message that contains this TLV is sent by a PTP Port, the value of
14 syncReceiptTimeoutTimeInterval for that PTP Port (see 10.2.5.3) shall be set equal to syncReceiptTimeout
15 (see 10.7.3.1) multiplied by the value of the interval, in seconds, reflected by logTimeSyncInterval.

1 10.6.4.3.8 logAnnounceInterval (Integer8)

2 The value is the logarithm to base 2 of the mean time interval, desired by the PTP Port that sends this TLV,
3 between successive Announce messages sent by the PTP Port at the other end of the link. The format and
4 allowed values of logAnnounceInterval are the same as the format and allowed values of
5 initialLogAnnounceInterval (see 10.7.2.2).

6 The values 127, 126, and –128 are interpreted as defined in Table 10-17.

Table 10-17—Interpretation of special values of logAnnounceInterval

Value	Instruction to PTP Instance that receives this TLV
127	Instructs the PTP Port that receives this TLV to stop sending Announce messages.
126	Instructs the PTP Port that receives this TLV to set currentLogAnnounceInterval to the value of initialLogAnnounceInterval (see 10.7.2.2).
–128	Instructs the PTP Port that receives this TLV not to change the mean time interval between successive Announce messages.
All values in the ranges [–127, –25] and [25, 125] are reserved.	

7 When a Signaling message that contains this TLV is sent by a PTP Port, the value of
8 announceReceiptTimeoutTimeInterval for that PTP Port (see 10.3.10.1) shall be set equal to
9 announceReceiptTimeout (see 10.7.3.2) multiplied by the value of the interval, in seconds, reflected by
10 logAnnounceInterval.

11 10.6.4.3.9 flags (Octet)

12 Bits 0 through 2 of the octet are defined in Table 10-18 and take on the values TRUE and FALSE. Bits not
13 defined in Table 10-18 are set to FALSE and ignored on receipt.

Table 10-18—Definitions of bits of flags field of message interval request TLV

Bit	Name
0	computeNeighborRateRatio
1	computeMeanLinkDelay
2	oneStepReceiveCapable

14 NOTE—For full-duplex point-to-point links (see Clause 11), it is expected that the PTP Port sending this TLV will set
15 bits 0 and/or 1 to FALSE if this PTP Port will not provide valid timing information in its subsequent responses
16 (Pdelay_Resp and Pdelay_Resp_Follow_Up) to Pdelay_Req messages. Similarly, it is expected that the PTP Port
17 sending this TLV will set bit 2 to TRUE if it is capable of receiving and correctly processing one-step Sync messages.

1 10.6.4.4 gPTP-capable TLV definition

2 10.6.4.4.1 General

3 The fields of the gPTP-capable TLV shall be as specified in Table 10-19 and in 10.6.4.4.2 through
4 10.6.4.4.7. This TLV is a standard organization extension TLV for the Signaling message, as specified in
5 14.3 of IEEE Std 1588-2019.

Table 10-19—gPTP-capable TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
logGptpCapableMessageInterval								1	10
flags								1	11
reserved								4	12

6 10.6.4.4.2 tlvType (Enumeration16)

7 The value of the tlvType field is 0x8000.

8 NOTE—This value indicates the TLV is a vendor and standard organization extension TLV that is not propagated, as
9 specified in 14.3.2.1 and Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as
10 ORGANIZATION_EXTENSION_DO_NOT_PROPAGATE with a value of 0x8000.

11 10.6.4.4.3 lengthField (UInteger16)

12 The value of the lengthField is 12.

13 10.6.4.4.4 organizationId (Octet3)

14 The value of organizationId is 00-80-C2.

15 10.6.4.4.5 organizationSubType (Enumeration24)

16 The value of organizationSubType is 4.

17 10.6.4.4.6 logGptpCapableMessageInterval (Integer8)

18 The value of logGptpCapableMessageInterval is the logarithm to base 2 of the mean gPTP-capable message
19 interval in seconds (see 10.7.2.1 and 10.7.2.5)

1 10.6.4.4.7 flags (Octet)

2 The flag bits shall be transmitted as FALSE and ignored on receipt.

3 10.6.4.5 gPTP-capable message interval request TLV definition

4 10.6.4.5.1 General

5 The fields of the gPTP-capable message interval request TLV shall be as specified in Table 10-20 and in
6 10.6.4.5.2 through 10.6.4.5.6. This TLV is a standard organization extension TLV for the Signaling message,
7 as specified in 14.3 of IEEE Std 1588-2019.

Table 10-20—gPTP-capable message interval request TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
logGptpCapableMessageInterval								1	10
reserved								3	11

8 10.6.4.5.2 tlvType (Enumeration16)

9 The value of the tlvType field is 0x8000.

10 NOTE—This value indicates the TLV is a vendor and standard organization extension TLV that is not propagated, as
11 specified in 14.3.2.1 and Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as
12 ORGANIZATION_EXTENSION_DO_NOT_PROPAGATE with a value of 0x8000.

13 10.6.4.5.3 lengthField (UInteger16)

14 The value of the lengthField is 10.

15 10.6.4.5.4 organizationId (Octet3)

16 The value of organizationId is 00-80-C2.

17 10.6.4.5.5 organizationSubType (Enumeration24)

18 The value of organizationSubType is 5.

1 10.6.4.5.6 logGptpCapableMessageInterval (Integer8)

2 The value is the logarithm to base 2 of the mean time interval, desired by the PTP Port that sends this TLV,
3 between successive Signaling messages that contain the gPTP-capable TLV (see 10.6.4.4), sent by the PTP
4 Port at the other end of the link. The format and allowed values of logGptpCapableMessageInterval are the
5 same as the format and allowed values of initialLogGptpCapableMessageInterval (see 10.7.2.5).

6 The values 127, 126, and –128 are interpreted as defined in Table 10-21.

Table 10-21—Interpretation of special values of logGptpCapableMessageInterval

Value	Instruction to PTP Instance that receives this TLV
127	Instructs the PTP Port that receives this TLV to stop sending Signaling messages that contain the gPTP-capable TLV.
126	Instructs the PTP Port that receives this TLV to set currentlogGptpCapableMessageInterval to the value of initialLogGptpCapableMessageInterval (see 10.7.2.4).
–128	Instructs the PTP Port that receives this TLV not to change the mean time interval between successive Signaling messages that contain the gPTP-capable TLV.
All values in the ranges [–127, –25] and [25, 125] are reserved.	

7 When a Signaling message that contains this TLV is sent by a PTP Port, the value of
8 gPtpCapableReceiptTimeoutTimeInterval for that PTP Port (see 10.3.10.1) shall be set equal to
9 gPtpCapableReceiptTimeout (see 10.7.3.3) multiplied by the value of the interval, in seconds, reflected by
10 logGptpCapableMessageInterval.

11 10.7 Protocol timing characterization

12 10.7.1 General

13 This subclause specifies timing and timeout attributes for the media-independent sublayer state machines.

14 10.7.2 Message transmission intervals

15 10.7.2.1 General interval specification

16 The mean time interval between the sending of successive Announce messages, known as the *announce*
17 *interval*, is specified in 10.7.2.2.

18 The mean time interval between the sending of successive time-synchronization event messages, known as
19 the *sync interval*. The sync interval is specified in 10.7.2.3.

20 The mean time interval between the sending of successive Signaling messages that contain the gPTP-
21 capable TLV, known as the *gPTP-capable message interval*, is specified in 10.7.2.5.

22 10.7.2.2 Announce message transmission interval

23 The logarithm to the base 2 of the announce interval (in seconds) is carried in the logMessageInterval field
24 of the Announce message.

1 When `useMgtSettableLogAnnounceInterval` (see 14.10.15) is FALSE, the `initialLogAnnounceInterval` specifies the announce interval when the PTP Port is initialized and the value to which the announce interval is set when a message interval request TLV is received with the `logAnnounceInterval` field set to 126 (see the `AnnounceIntervalSetting` state machine in 10.3.17). The `currentLogAnnounceInterval` specifies the current value of the announce interval. The default value of `initialLogAnnounceInterval` is 0. Every PTP Port supports the value 127; the PTP Port does not send Announce messages when `currentLogAnnounceInterval` has this value (see 10.3.17). A PTP Port may support other values, except for the reserved values indicated in Table 10-17. A PTP Port ignores requests for unsupported values (see 10.3.17). The `initialLogAnnounceInterval` and `currentLogAnnounceInterval` are per-PTP Port attributes.

10 When `useMgtSettableLogAnnounceInterval` is TRUE, `currentLogAnnounceInterval` is set equal to `mgtSettableLogAnnounceInterval` (see 14.10.16), and `initialLogAnnounceInterval` is ignored.

12 Announce messages shall be transmitted such that the value of the arithmetic mean of the intervals, in seconds, between message transmissions is within $\pm 30\%$ of $2^{\text{currentLogAnnounceInterval}}$. In addition, a PTP Port shall transmit Announce messages such that at least 90% of the inter-message intervals are within $\pm 30\%$ of the value of $2^{\text{currentLogAnnounceInterval}}$. The interval between successive Announce messages should not exceed twice the value of $2^{\text{currentLogAnnounceInterval}}$ in order to prevent causing an `announceReceiptTimeout` event. The `PortAnnounceTransmit` state machine (see 10.3.16) is consistent with these requirements, i.e., the requirements here and the requirements of the `PortAnnounceTransmit` state machine can be met simultaneously.

20 NOTE 1—In transient situations (i.e., during a synchronization tree reconfiguration), announce messages can be transmitted with a shorter interval than described above.

22 NOTE 2—A minimum number of inter-message intervals is necessary in order to verify that a PTP Port meets these requirements. The arithmetic mean is the sum of the inter-message interval samples divided by the number of samples. For more detailed discussion of statistical analyses, see Papoulis [B25].

25 NOTE 3—If `useMgtSettableLogAnnounceInterval` is FALSE, the value of `initialLogAnnounceInterval` is the value of the mean time interval between successive Announce messages when the PTP Port is initialized. The value of the mean time interval between successive Announce messages can be changed, e.g., if the PTP Port receives a Signaling message that carries a message interval request TLV (see 10.6.4.3) and the current value is stored in `currentLogAnnounceInterval`. The value of the mean time interval between successive Announce messages can be reset to the initial value, e.g., by a message interval request TLV for which the value of the field `logAnnounceInterval` is 126 (see 10.6.4.3.8).

31 NOTE 4—A PTP Port that requests (using a Signaling message that contains a message interval request TLV; see 10.6.4 and 10.3.17) that the PTP Port at the other end of the attached link set its `currentLogAnnounceInterval` to a specific value can determine if the request was honored by examining the `logMessageInterval` field of subsequent received Announce messages.

35 10.7.2.3 Time-synchronization event message transmission interval

36 The logarithm to the base 2 of the sync interval (in seconds) is carried in the `logMessageInterval` field of the time-synchronization messages.

38 When `useMgtSettableLogSyncInterval` (see 14.10.19) is FALSE, the `initialLogSyncInterval` specifies the sync interval when the PTP Port is initialized and the value to which the sync interval is set when a message interval request TLV is received with the `logTimeSyncInterval` field set to 126 (see the `SyncIntervalSetting` state machine in 10.3.18). The default value is media-dependent; the value is specified in the respective media-dependent clauses. The `initialLogSyncInterval` is a per-PTP Port attribute.

43 The `currentLogSyncInterval` specifies the current value of the sync interval and is a per-PTP Port attribute.

44 When `useMgtSettableLogSyncInterval` is TRUE, `currentLogSyncInterval` is set equal to `mgtSettableLogSyncInterval` (see 14.10.20), and `initialLogSyncInterval` is ignored.

1 When the value of `syncLocked` is `FALSE`, time-synchronization messages shall be transmitted such that the
 2 value of the arithmetic mean of the intervals, in seconds, between message transmissions is within $\pm 30\%$ of
 3 $2^{\text{currentLogSyncInterval}}$. In addition, a PTP Port shall transmit time-synchronization messages such that at least
 4 90% of the inter-message intervals are within $\pm 30\%$ of the value of $2^{\text{currentLogSyncInterval}}$. The interval
 5 between successive time-synchronization messages should not exceed twice the value of
 6 $2^{\text{currentLogSyncInterval}}$ in order to prevent causing a `syncReceiptTimeout` event. The `PortSyncSyncSend` state
 7 machine (see 10.2.12) is consistent with these requirements, i.e., the requirements here and the requirements
 8 of the `PortSyncSyncSend` state machine can be met simultaneously.

9 NOTE 1—A minimum number of inter-message intervals is necessary in order to verify that a PTP Port meets these
 10 requirements. The arithmetic mean is the sum of the inter-message interval samples divided by the number of samples.
 11 For more detailed discussion of statistical analyses, see Papoulis [B25].

12 NOTE 2—If `useMgtSettableLogSyncInterval` is `FALSE` the value of `initialLogSyncInterval` is the value of the sync
 13 interval when the PTP Port is initialized. The value of the sync interval can be changed, e.g., if the PTP Port receives a
 14 Signaling message that carries a message interval request TLV (see 10.6.4.3) and the current value is stored in
 15 `currentLogSyncInterval`. The value of the sync interval can be reset to the initial value, e.g., by a message interval
 16 request TLV for which the value of the field `logTimeSyncInterval` is 126 (see 10.6.4.3.7).

17 10.7.2.4 Interval for providing synchronization information by `ClockTimeTransmitter` entity

18 The `clockTimeTransmitterLogSyncInterval` specifies the mean time interval between successive instants at
 19 which the `ClockTimeTransmitter` entity provides time-synchronization information to the `SiteSync` entity.
 20 The value is less than or equal to the smallest `currentLogSyncInterval` (see 10.7.2.3) value for all the ports of
 21 the PTP Instance. The `clockTimeTransmitterLogSyncInterval` is an internal, per PTP Instance variable.

22 10.7.2.5 Interval for sending the gPTP-capable TLV Signaling message

23 The logarithm to the base 2 of the gPTP-capable message interval (in seconds) is carried in the
 24 `logGtpCapableMessageInterval` field of the gPTP-capable TLV. The default value shall be 0. The range
 25 shall be -24 through 24 . Other values in the range $[-128, 127]$ shall be reserved.

26 When `useMgtSettableLogGtpCapableMessageInterval` (see 14.10.15) is `FALSE`, the
 27 `initialLogGtpCapableMessageInterval` specifies the following:

- 28 a) The mean gPTP-capable message interval when the PTP Port is initialized, and
- 29 b) The value to which the gPTP-capable message interval is set when a gPTP-capable TLV is received
 30 with the `logGtpCapableMessageInterval` field set to 126 (see the
 31 `GtpCapableMessageIntervalSetting` state machine in 10.4.4).

32 The `currentLogGtpCapableMessageInterval` specifies the current value of the gPTP-capable message
 33 interval. Every PTP Port supports the value 127; the PTP Port does not send Signaling messages containing
 34 a gPTP-capable TLV when `currentLogGtpCapableMessageInterval` has this value (see 10.4.4). A PTP Port
 35 may support other values, except for the reserved values indicated in Table 10-21. A PTP Port shall ignore
 36 requests for unsupported values (see 10.4.4). The `initialLogGtpCapableMessageInterval` and
 37 `currentLogGtpCapableMessageInterval` are per-PTP Port attributes.

38 When `useMgtSettableLogGtpCapableMessageInterval` is `TRUE`, `currentLogGtpCapableMessageInterval`
 39 is set equal to `mgtSettableLogGtpCapableMessageInterval` (see 14.10.16), and
 40 `initialLogGtpCapableMessageInterval` is ignored.

41 NOTE 1—If `useMgtSettableLogGtpCapableMessageInterval` is `FALSE`, the value of
 42 `initialLogGtpCapableMessageInterval` is the value of the mean time interval between successive Signaling messages
 43 containing the gPTP-capable TLV when the PTP Port is initialized. The value of the mean time interval between
 44 successive Signaling messages containing the gPTP-capable TLV can be changed, e.g., if the PTP Port receives a
 45 Signaling message that carries a gPTP-capable TLV (see 10.6.4.4) and the current value is stored in

1 currentLogGtpCapableMessageInterval. The value of the mean time interval between successive Signaling messages
 2 containing the gPTP-capable TLV can be reset to the initial value, e.g., by a Signaling message containing a gPTP-
 3 capable message interval request TLV for which the value of the field logGtpCapableMessageInterval is 126
 4 (see 10.6.4.5.6).

5 NOTE 2—A PTP Port that requests (using a Signaling message that contains a gPTP-capable message interval request
 6 TLV; see 10.6.4 and 10.6.4.5) that the PTP Port at the other end of the attached link set its
 7 currentLogGtpCapableMessageInterval to a specific value can determine if the request was honored by examining the
 8 logGtpCapableMessageInterval field of subsequent received Signaling messages containing the gPTP-capable TLV.

9 NOTE 3—The GtpCapableTransmit state machine ensures that the times between transmission of successive Signaling
 10 messages, containing the gPTP-capable TLV, in seconds, are not smaller than $2^{\text{currentLogGtpCapableMessageInterval}}$. This is
 11 consistent with the manner in which Pdelay_Req messages are transmitted (see NOTE 3 of 11.5.2.2).

12 10.7.3 Timeouts

13 10.7.3.1 syncReceiptTimeout

14 The value of this attribute tells a timeReceiver port the number of sync intervals to wait without receiving
 15 synchronization information, before assuming that the timeTransmitter is no longer transmitting
 16 synchronization information and that the BTCA needs to be run, if appropriate. The condition of the
 17 timeReceiver port not receiving synchronization information for syncReceiptTimeout sync intervals is
 18 known as *sync receipt timeout*.

19 The default value shall be 3. The range shall be 2 through 255. The syncReceiptTimeout is a per-PTP Port
 20 attribute.

21 10.7.3.2 announceReceiptTimeout

22 The value of this attribute tells a timeReceiver port the number of announce intervals to wait without
 23 receiving an Announce message, before assuming that the timeTransmitter is no longer transmitting
 24 Announce messages, and that the BTCA needs to be run, if appropriate. The condition of the timeReceiver
 25 port not receiving an Announce message for announceReceiptTimeout announce intervals is known as
 26 *announce receipt timeout*.

27 The default value shall be 3. The range shall be 2 through 255. The announceReceiptTimeout is a per-PTP
 28 Port attribute.

29 10.7.3.3 gPtpCapableReceiptTimeout

30 The value of this attribute tells a PTP Port the number of gPTP-capable message intervals to wait without
 31 receiving from its neighbor a Signaling message containing a gPTP-capable TLV, before determining that its
 32 neighbor is no longer invoking gPTP.

33 NOTE—A determination that its neighbor is no longer invoking gPTP causes the PTP Port to set asCapable to FALSE.

34 The default value shall be 9. The range shall be 2 through 255.

1 11. Media-dependent layer specification for full-duplex point-to-point links

2 11.1 Overview

3 11.1.1 General

4 A port attached to a full-duplex point-to-point PTP Link uses the PTP peer delay protocol to measure
5 propagation delay on the PTP Link. An overview of the propagation delay measurement is given in 11.1.2.
6 Synchronization information is transported using the PTP messages Sync and Follow_Up. An overview of
7 the transport of synchronization information is given in 11.1.3. An overview of the MD entity model for a
8 full-duplex point-to-point medium is given in 11.1.4.

9 11.1.2 Propagation delay measurement

10 The measurement of propagation delay on a full-duplex point-to-point PTP Link using the peer-to-peer
11 delay mechanism is illustrated in Figure 11-1. The instance of the peer-to-peer delay mechanism that runs
12 only on domain 0 is referred to as the transport-specific peer-to-peer delay mechanism. The instance of the
13 peer-to-peer delay mechanism that runs as part of the CMLDS (see 11.2.17). The mechanism is the same as
14 the peer-to-peer delay mechanism described in IEEE Std 1588-2019, specialized to a two-step PTP Port¹⁶
15 and sending the requestReceiptTimestamp and the responseOriginTimestamp separately [see item c) 8) of
16 11.4.2 in IEEE Std 1588-2019]. The transport-specific peer-to-peer delay mechanism is mathematically
17 equivalent to CMLDS, though its computations are organized differently. The description in this subclause
18 focuses on the transport-specific peer-to-peer delay mechanism. In addition, when it is not important in this
19 standard to distinguish the transport-specific peer-to-peer delay mechanism from CMLDS, the mechanism is
20 simply referred to as the “peer-to-peer delay mechanism,” or more briefly as the “peer delay mechanism.”

21 The measurement is made by each port at the end of every full-duplex point-to-point PTP Link. Thus, both
22 ports sharing a PTP Link will independently make the measurement, and both ports will know the
23 propagation delay as a result. This allows the time-synchronization information described in 11.1.3 to be
24 transported irrespective of the direction taken by a Sync message. The propagation delay measurement is
25 made on ports otherwise blocked by non-PTP algorithms (e.g., Rapid Spanning Tree Protocol) used to
26 eliminate cyclic topologies. This enables either no loss of synchronization or faster resynchronization, after
27 a reconfiguration, because propagation delays are already known and do not have to be initially measured
28 when the reconfiguration occurs.

29 Since the propagation delay measurement is made using timestamps relative to the LocalClock entities at
30 each port at the ends of the PTP Link and the resulting mean link delay is expressed in the responder
31 timebase (see 11.2.19.3.4), there is no need to measure the mean link delay for the PTP Link in each domain
32 because the mean link delay is the same in each domain. In addition, the quantity nrrPdelay (see 11.2.13.14)
33 is the ratio of the responder to requester LocalClock frequency and is also the same in all domains.
34 Therefore, the propagation delay measurement and nrrPdelay measurement, made using the peer-to-peer
35 delay mechanism (i.e., nrrPdelay, see 11.2.13.14), are domain-independent. Single instances of the
36 respective state machines that cause these measurements to be made are invoked, rather than one instance
37 per domain, and the results are available to all domains. The PTP messages used for the measurements (i.e.,
38 Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up; see 11.4.5 through 11.4.7) carry 0 in the
39 domainNumber field, but this value is not used.

40 In Figure 11-1, the propagation delay measurement is initiated by a time-aware system at one end of a PTP
41 Link; this time-aware system is known as the *peer delay initiator*. For purposes of the measurement, the
42 other time-aware system is the *peer delay responder*. A similar measurement occurs in the opposite

¹⁶See 3.1.86 of IEEE Std 1588-2019 for the definition of a *two-step PTP Port*.

1 direction, with the initiator and responder interchanged and the directions of the messages in Figure 11-1
2 reversed.

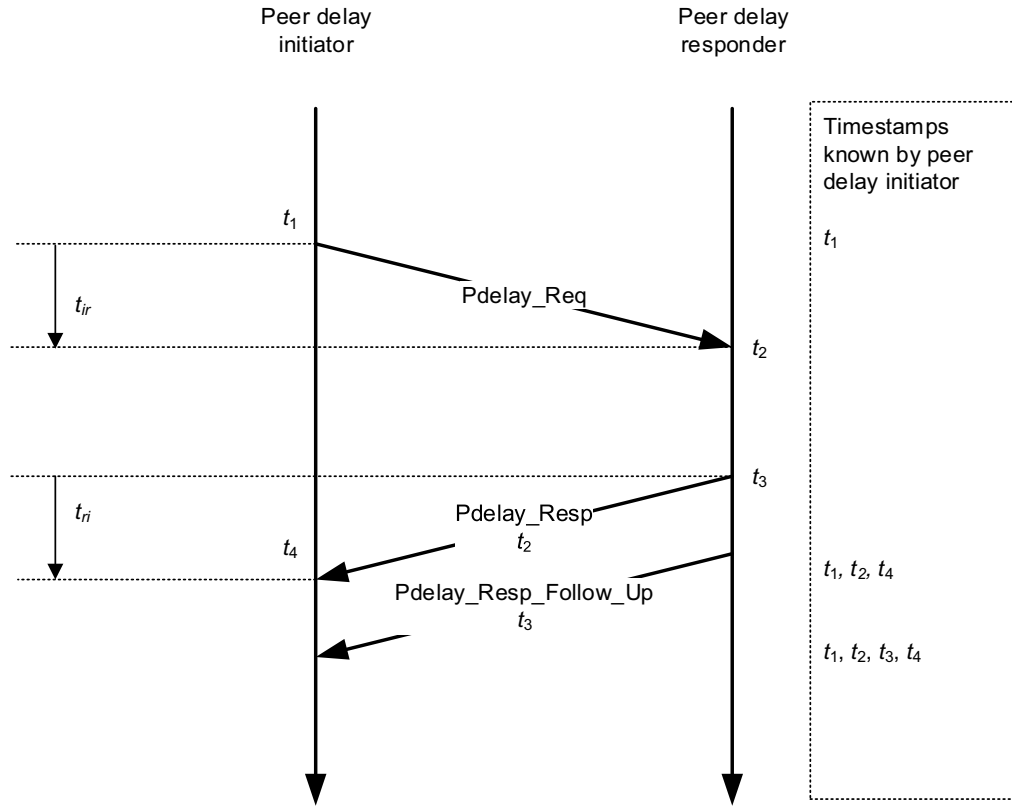


Figure 11-1—Propagation delay measurement using peer-to-peer delay mechanism

3 The propagation delay measurement starts with the initiator issuing a Pdelay_Req message and generating a
4 timestamp, t_1 . The responder receives this message and timestamps it with time t_2 . The responder returns a
5 Pdelay_Resp message and timestamps it with time t_3 . The responder returns the time t_2 in the Pdelay_Resp
6 message and the time t_3 in a Pdelay_Resp_Follow_Up message. The initiator generates a timestamp, t_4 ,
7 upon receiving the Pdelay_Resp message. The initiator then uses these four timestamps to compute the
8 mean propagation delay (i.e., meanLinkDelay; see 8.3) as shown in Equation (11-1).

$$\begin{aligned}
 t_{ir} &= t_2 - t_1 \\
 t_{ri} &= t_4 - t_3 \\
 D &= \frac{t_{ir} + t_{ri}}{2} = \frac{(t_4 - t_1) - (t_3 - t_2)}{2}
 \end{aligned}
 \tag{11-1}$$

10 where D is the measured mean propagation delay and the other quantities are defined in Figure 11-1.

11 NOTE—The equation 11-1 does not take into consideration neighbor rate ratio.

12 Note that it is the mean propagation delay that is measured here. Any PTP Link asymmetry is modeled as
13 described in 8.3. Any asymmetry that is not corrected for introduces an error in the transported synchronized
14 time value.

1 The accuracy of the mean propagation delay measurement depends on how accurately the times t_1 , t_2 , t_3 , and
 2 t_4 are measured. In addition, Equation (11-1) assumes that the initiator and responder timestamps are taken
 3 relative to clocks that have the same frequency. In practice, t_1 and t_4 are measured relative to the LocalClock
 4 entity of the initiator time-aware system, and t_2 and t_3 are measured relative to the LocalClock entity of the
 5 responder time-aware system. If the propagation delay measurement is desired relative to the responder time
 6 base, the term $(t_4 - t_1)$ in Equation (11-1) must be multiplied by the rate ratio of the responder relative to the
 7 initiator, otherwise there will be an error equal to $0.5y(t_4 - t_1)$, where y is the frequency offset of the
 8 responder relative to the initiator. Likewise, if the propagation delay measurement is desired relative to the
 9 initiator time base, the term $(t_3 - t_2)$ in Equation (11-1) must be multiplied by the rate ratio of the initiator
 10 relative to the responder, otherwise there will be an error equal to $0.5y(t_3 - t_2)$, where y is the frequency
 11 offset of the initiator relative to the responder. Finally, if the propagation delay measurement is desired
 12 relative to the Grandmaster Clock time base, each term must be multiplied by the rate ratio of the
 13 Grandmaster Clock relative to the time base in which the term is expressed.

14 The calculation of meanLinkDelay may result in negative values if very short cables are used and link delay
 15 compensation is not accurate, or because of random time stamping errors.

16 There can also be an error in measured propagation delay due to time measurement granularity (see B.1.2).
 17 For example, if the time measurement granularity is 40 ns (as specified in B.1.2), the timestamps t_1 , t_2 , t_3 ,
 18 and/or t_4 can undergo 40 ns step changes. When this occurs, the measured propagation delay, D , will change
 19 by 20 ns (or by a multiple of 20 ns if more than one of the timestamps has undergone a 40 ns step change).
 20 The actual propagation delay has not changed by 20 ns; the effect is due to time measurement granularity.
 21 The effect can be reduced, and the accuracy improved, by averaging successive measured propagation delay
 22 values. For example, an exponential averaging filter can be used, i.e., as shown in Equation (11-2).

$$23 \quad D_{avg,k} = aD_{avg,k-1} + (1-a)D_k \quad (11-2)$$

24 where

- 25 D_k is the k^{th} propagation delay measurement
- 26 $D_{avg,k}$ is the k^{th} computed average propagation delay
- 27 k is an index for the propagation delay measurements (i.e., peer delay message exchange)

28 The quantity a is the exponential weighting factor; it can be set so that the weight of a past propagation delay
 29 measurement is $1/e$ after M measurements, i.e., as shown in Equation (11-3).

$$30 \quad a = e^{-\frac{1}{M}} \quad (11-3)$$

31 The above averager must be initialized. One method is to use a simple average (i.e., the sum of the sample
 32 values divided by the number of samples) of the measurements made up to the current measurement until a
 33 window of M measurements has been accumulated. In this case, Equation (11-2) is used only for $k > M$.
 34 For $k \leq M$, the averaged propagation delay is given by Equation (11-4).

$$35 \quad D_{avg,k} = \frac{(k-1)D_{avg,k-1} + D_{k-1}}{k} \quad (11-4)$$

36 The rate ratio of the responder relative to the initiator is the quantity nrrPdelay (see 11.2.13.14). When
 37 computed using peer-to-peer delay messages, it is computed by the function computePdelayRateRatio() (see
 38 11.2.19.3.3) of the MDPdelayReq state machine (see 11.2.19) using successive values of t_3 and t_4 . As
 39 indicated in the description of computePdelayRateRatio(), any scheme that uses this information is
 40 acceptable as long as the performance requirements of B.2.4 are met. One example scheme is given in
 41 NOTE 1 of 11.2.19.3.3.

1 11.1.3 Transport of time-synchronization information

2 The transport of time-synchronization information by a PTP Instance, using Sync and Follow_Up (or just
3 Sync) messages, is illustrated in Figure 11-2. The mechanism is mathematically equivalent to the
4 mechanism described in IEEE Std 1588-2019 for a peer-to-peer Transparent Clock that is syntonized
5 (see 10.1, 10.3, 11.1, and 11.4 of IEEE Std 1588-2019). However, the processes of transporting
6 synchronization by a peer-to-peer Transparent Clock that is syntonized and by a Boundary Clock are
7 mathematically and functionally equivalent. The main functional difference between the two types of clocks
8 is that the Boundary Clock participates in best timeTransmitter selection and invokes the BTCA, while the
9 peer-to-peer Transparent Clock does not participate in best timeTransmitter selection and does not invoke
10 the BTCA (and implementations of the two types of clocks can be different).

11

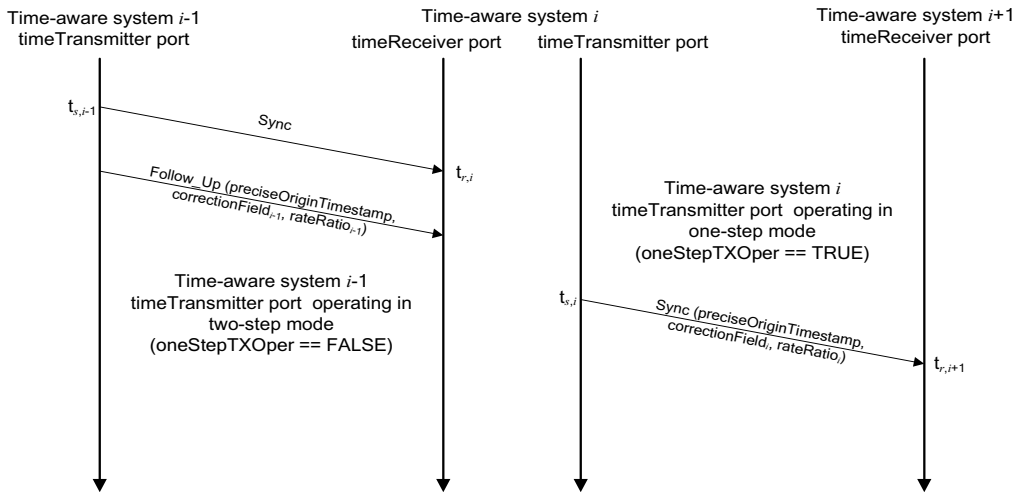


Figure 11-2—Transport of time-synchronization information

12 Figure 11-2 shows three adjacent PTP Instances, indexed $i-1$, i , and $i+1$. Synchronization is transported from
13 PTP Instance $i-1$ to PTP Instance i using two-step time transport, and then to PTP Instance $i+1$ using one-
14 step time transport. PTP Instance $i-1$ sends a Sync message to PTP Instance i at time $t_{s,i-1}$, relative to the
15 LocalClock entity of PTP Instance $i-1$. At a later time (since it is using two-step time transport), PTP
16 Instance $i-1$ sends an associated Follow_Up message to PTP Instance i , which contains a
17 preciseOriginTimestamp, $\text{correctionField}_{i-1}$, and rateRatio_{i-1} . The preciseOriginTimestamp contains the
18 time of the Grandmaster Clock when it originally sent this synchronization information. It is not indexed
19 here because it normally does not change as the Sync and Follow_Up messages traverse the network. The
20 quantity $\text{correctionField}_{i-1}$ contains the difference between the synchronized time when the Sync message is
21 sent (i.e., the synchronized time that corresponds to the local time $t_{s,i-1}$) and the preciseOriginTimestamp.
22 The sum of preciseOriginTimestamp and $\text{correctionField}_{i-1}$ gives the synchronized time that corresponds to
23 $t_{s,i-1}$. The quantity rateRatio_{i-1} is the ratio of the Grandmaster Clock frequency to the frequency of the
24 LocalClock entity of PTP Instance $i-1$.

1 PTP Instance i receives the Sync message from PTP Instance $i-1$ at time $t_{r,i}$, relative to its LocalClock entity.
 2 It timestamps the receipt of the Sync message, and the timestamp value is $t_{r,i}$. It receives the associated
 3 Follow_Up message some time later.

4 PTP Instance i will eventually send a new Sync message at time $t_{s,i}$, relative to its LocalClock entity. The
 5 time $t_{s,i}$ occurs after the Follow_Up message is received from PTP Instance $i-1$. It will have to compute
 6 correctionField_i , i.e., the difference between the synchronized time that corresponds to $t_{s,i}$ and the
 7 $\text{preciseOriginTimestamp}$. To do this calculation, it must compute the value of the time interval between $t_{s,i-1}$
 8 and $t_{s,i}$, expressed in the Grandmaster Clock time base. This interval is equal to the sum of the following
 9 quantities:

- 10 a) The propagation delay on the PTP Link between PTP Instances $i-1$ and i , expressed in the
 11 Grandmaster Clock time base, and
- 12 b) The difference between $t_{s,i}$ and $t_{r,i}$ (i.e., the residence time), expressed in the Grandmaster Clock
 13 time base.

14 The mean propagation delay on the PTP Link between PTP Instances $i-1$ and i , relative to the
 15 LocalClock entity of PTP Instance $i-1$, is equal to meanLinkDelay (see 10.2.5.8). This must be multiplied by
 16 rateRatio_{i-1} to express it in the Grandmaster Clock time base. The total propagation delay is equal to the
 17 mean propagation delay plus the quantity delayAsymmetry (see 8.3 and 10.2.5.9); delayAsymmetry is
 18 already expressed in the Grandmaster Clock time base. The residence time, $t_{s,i} - t_{r,i}$, must be multiplied by
 19 rateRatio_i to express it in the Grandmaster Clock time base.

20 The preceding computation is organized slightly differently in the state machines of 11.2.14 and 11.2.15.
 21 Rather than explicitly expressing the PTP Link propagation delay in the Grandmaster Clock time base, the
 22 local time at PTP Instance i that corresponds to $t_{s,i-1}$ is computed; this is the upstreamTxTime member of the
 23 MDSyncReceive structure (see 10.2.2.2.7; recall that $t_{s,i-1}$ is relative to the LocalClock entity of PTP
 24 Instance $i-1$). upstreamTxTime is equal to the quantity $t_{r,i}$ minus the PTP Link propagation delay expressed
 25 relative to the LocalClock entity of PTP Instance i . The PTP Link propagation delay expressed relative to
 26 the LocalClock entity of PTP Instance i is equal to the sum of the following:

- 27 c) The quantity meanLinkDelay (see 10.2.5.8) divided by nrPdelay (see 11.2.13.14), and
- 28 d) The quantity delayAsymmetry (see 10.2.5.9) divided by rateRatio_i .

29 The division of delayAsymmetry by rateRatio_i is performed after rateRatio_i has been updated. The
 30 computation of upstreamTxTime is done by the MDSyncReceiveSM state machine in the function
 31 $\text{setMDSyncReceiveMDSR}()$ (see 11.2.14.2.1). When PTP Instance i sends a Sync message to PTP Instance
 32 $i+1$, it computes the sum of the PTP Link propagation delay and residence time, expressed in the
 33 Grandmaster Clock time base, as shown in item e).

- 34 e) The quantity $(t_{s,i} - \text{upstreamTxTime})(\text{rateRatio}_i)$.

35 As in item d) in this subclause, this computation is performed after rateRatio_i has been updated. The quantity
 36 of item e) is added to $\text{correctionField}_{i-1}$ to obtain correctionField_i . The computation of item e) and
 37 correctionField_i is done by the MDSyncSendSM state machine in the function $\text{setFollowUp}()$ (see
 38 11.2.15.2.3). The quantity correctionField_i is inserted in the Sync message sent by PTP Instance i .

39 The difference between mean propagation delay relative to the Grandmaster Clock time base and relative to
 40 the time bases of the PTP Instance at the other end of the attached PTP Link or of the current PTP Instance is
 41 usually negligible. The former can be obtained from the latter by multiplying the latter by the ratio of the
 42 Grandmaster Clock frequency to the frequency of the LocalClock entity of the PTP Instance at the other end
 43 of the PTP Link attached to this PTP Port. This ratio differs from 1 by 200 ppm or less. For example, for a
 44 worst-case frequency offset of the LocalClock entity of the PTP Instance at the other end of the PTP Link,

1 relative to the Grandmaster Clock, of 200 ppm, and a measured propagation time of 100 ns, the difference in
2 $2D$ relative to the two time bases is 20 ps. The corresponding difference for PTP Link delay asymmetry in this
3 example is also negligible because the magnitude of the PTP Link delay asymmetry is of the same order of
4 magnitude as the mean propagation time, or less. However, the difference is usually not negligible for
5 residence time because residence time can be much larger (see B.2.2).

6 It was indicated in the first paragraph of this subclause that the processes of transporting synchronization by
7 a peer-to-peer Transparent Clock that is syntonized and by a Boundary Clock are mathematically and
8 functionally equivalent. The reason for this equivalency is that the computations described so far in this
9 subclause compute the synchronized time when the Sync message is sent by the PTP Instance. The same
10 computations are done if PTP Instance i sends a Sync message without having received a new Sync
11 message, i.e., if Sync receipt timeout occurs (see 10.7.3.1). In this case, PTP Instance i uses the most
12 recently received time-synchronization information from PTP Instance $i-1$, which would be prior to when
13 PTP Instance i sent its most recent Sync message. The synchronized time corresponding to the sending of a
14 Sync message is equal to the sum of the preciseOriginTimestamp and correctionField. Normally a Boundary
15 Clock places this entire value, except for any sub-nanosecond portion, in the preciseOriginTimestamp, while
16 a Transparent Clock retains the preciseOriginTimestamp and updates the correctionField. However, the sum
17 of the two fields is equal to the synchronized time when the Sync message is sent in both cases.

18 The ratio of the Grandmaster Clock frequency to the frequency of the LocalClock entity at PTP Instance i ,
19 $rateRatio_i$, is equal to the same quantity at PTP Instance $i-1$, $rateRatio_{i-1}$, multiplied by the ratio of the
20 frequency of the LocalClock entity at PTP Instance $i-1$ to the frequency of the LocalClock entity at PTP
21 Instance i , $nrrPdelay$ (see 11.2.13.14). If $nrrPdelay$ is sufficiently small, this is approximately equal to the
22 sum of $rateRatio_{i-1}$ and the quantity $nrrPdelay-1$, which is the frequency offset of PTP Instance $i-1$ relative
23 to PTP Instance i . This computation is done by the PortSyncSyncReceive state machine (see 10.2.8).

24 NOTE—The sending of time-synchronization information by the timeTransmitter ports of a PTP Instance might or
25 might not be tightly synchronized with the receipt of time-synchronization information by the timeReceiver port. If a
26 timeTransmitter port has the same logMessageInterval as the timeReceiver port, it will transmit timing event messages
27 as soon as possible after the timeReceiver port has received the corresponding timing event messages and the
28 timeTransmitter port is operating in “syncLocked” mode (see 10.2.5.15). If a timeTransmitter port and timeReceiver port
29 have different logMessageInterval values, then the timeTransmitter port can send timing event messages without any
30 synchronization with the timeReceiver port.

31 11.1.4 Model of operation

32 A PTP Instance contains one MD entity per PTP Instance, per PTP Port. This entity contains functions
33 generic to all media, which are described in Clause 10, and functions specific to the respective medium for
34 the PTP Link. Functions specific to full-duplex point-to-point links are described in the current clause.

35 NOTE—Full-duplex point-to-point IEEE 802.3 links are in the category of links specified in this clause.

36 The model for a PTP Instance of a time-aware system with full-duplex point-to-point links is shown in
37 Figure 11-3. It assumes the presence of one full-duplex point-to-point MD entity per PTP Port. The media-
38 independent entities shown in Figure 11-3 are described in 10.1.2.

39 A general, media-independent description of the generation of timestamps is given in 8.4.3. A more specific
40 description for PTP event messages is given in 11.3.2.1. A PTP event message is timestamped relative to the
41 LocalClock entity when the message timestamp point (see 3.19) crosses the timestamp measurement plane
42 (see 3.38). The timestamp is corrected for any ingressLatency or egressLatency (see 8.4.3) to produce a
43 timestamp relative to the reference plane (see 3.29). The corrected timestamp value is provided to the MD
44 entity.

45 The MD entity behavior and detailed state machines specific to full-duplex point-to-point links are described
46 in 11.2. The behavior of the MD entity that is generic to all media is described in Clause 10.

1

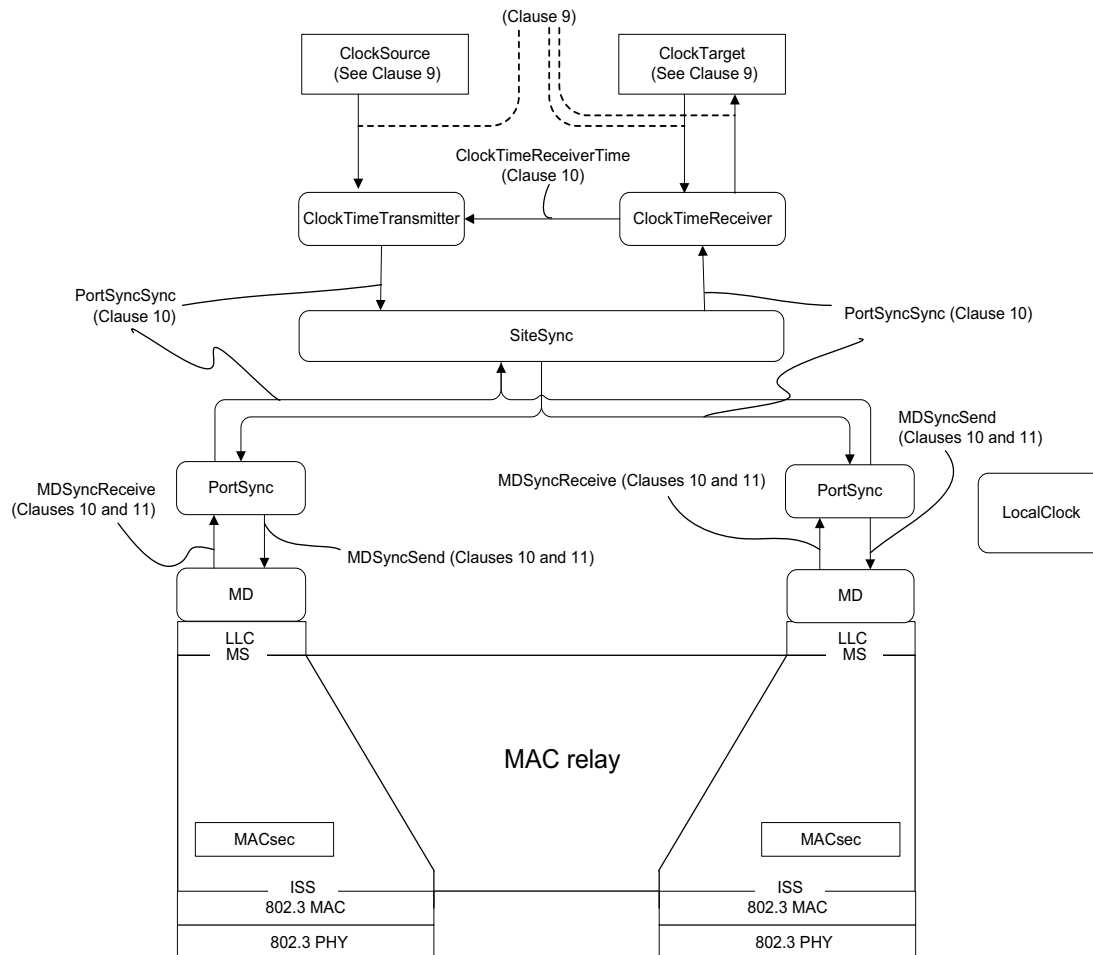


Figure 11-3—Model for a PTP Instance of a time-aware system with full-duplex point-to-point links

11.2 State machines for MD entity specific to full-duplex point-to-point links

11.2.1 General

This subclause describes the media-dependent state machines for an MD entity, for full-duplex point-to-point links. The state machines are all per port because an instance of each is associated with an MD entity. The state machines are as follows:

- a) **MDSyncReceiveSM** (shown in Figure 10-2, and in more detail in Figure 11-4): receives Sync and, if the received information is two-step, Follow_Up messages, and sends the time-synchronization information carried in these messages to the PortSync entity of the same PTP Port. In addition, if the global variable driftTrackingTlvSupport is TRUE, this state machine computes the ratio of the frequency of the LocalClock entity in the PTP Instance at the other end of the attached PTP Link to the frequency of the LocalClock entity in this PTP Instance using the information contained in successive Drift_Tracking TLVs, and places the result in the global variable nrrSync (see 11.2.13.15). The global variable neighborRateRatio (see 10.2.5.7) is set equal to nrrSync or nrrPdelay (see 11.2.13.14) depending on whether the value of the global variable nrrCompMethod (see 11.2.13.16) is equal to Sync or Pdelay, respectively. There is one instance of this state machine per PTP Port, per domain.
- b) **MDSyncSendSM** (shown in Figure 10-2, and in more detail in Figure 11-4): receives an MDSyncSend structure from the PortSync entity of the same PTP Port; transmits a Sync message; uses the syncEventEgressTimestamp, corrected for egressLatency, and information contained in the MDSyncSend structure to compute information needed for the Sync message if the PTP Port is currently operating as a one-step PTP Port and for the corresponding Follow_Up message if the PTP Port is currently operating as a two-step PTP Port; and transmits the Follow_Up message if the PTP Port is two-step. There is one instance of this state machine per PTP Port, per domain.

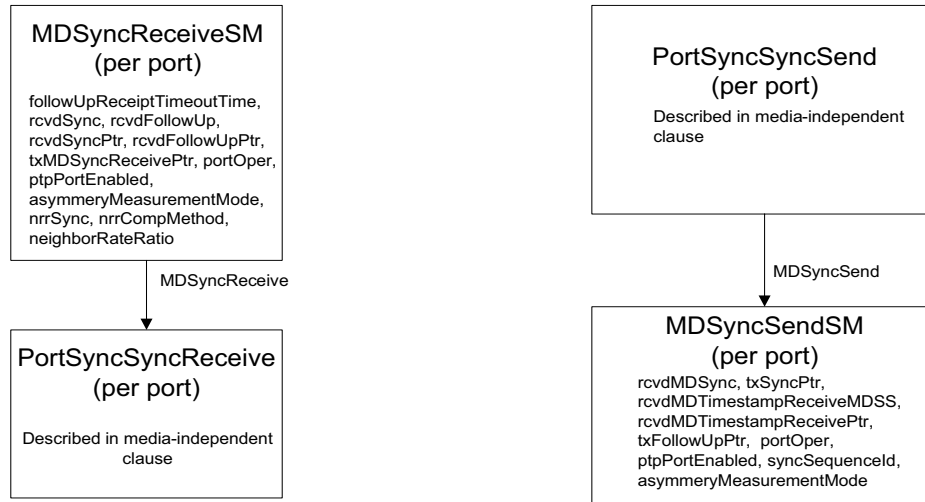
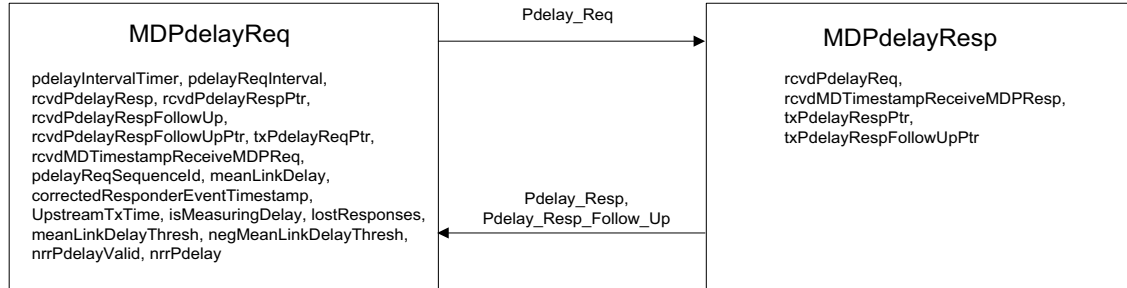


Figure 11-4—Detail of MD entity time-synchronization state machines for full-duplex point-to-point links

- c) **MDPdelayReq** (shown in Figure 11-5): transmits a Pdelay_Req message, receives Pdelay_Resp and Pdelay_Resp_Follow_Up messages corresponding to the transmitted Pdelay_Req message, uses the information contained in successive Pdelay_Resp and Pdelay_Resp_Follow_Up messages to compute the ratio of the frequency of the LocalClock entity in the PTP Instance at the other end of the attached PTP Link to the frequency of the LocalClock entity in this PTP Instance, places the

- 1 result in the global variable `nrrPdelay` (see 11.2.13.14), and uses the information obtained from the
 2 message exchange and the computed frequency ratio to compute propagation delay on the attached
 3 PTP Link. There is one instance of this state machine for all the domains, per port.
- 4 d) `MDPdelayResp` (shown in Figure 11-5): receives a `Pdelay_Req` message from the MD entity at the
 5 other end of the attached PTP Link, and responds with `Pdelay_Resp` and `Pdelay_Resp_Follow_Up`
 6 messages. There is one instance of this state machine for all the domains, per port.



**Figure 11-5—Peer-to-peer delay mechanism state machines—
overview and interrelationships**

- 7 e) `SyncIntervalSetting` state machine (not shown): receives a Signaling message that contains a
 8 Message Interval Request TLV (see 10.6.4.3), and sets the global variables that give the duration of
 9 the mean intervals between successive Sync messages. There is one instance of this state machine
 10 per PTP Port, per domain.
- 11 f) `LinkDelayIntervalSetting` state machine (not shown): receives a Signaling message that contains a
 12 Message Interval Request TLV (see 10.6.4.3), and sets the global variables that give the duration of
 13 the mean intervals between successive `Pdelay_Req` messages. There is one instance of this state
 14 machine for all the domains, per port.

15 Figure 10-2, Figure 11-4, and Figure 11-5 are not themselves state machines, but illustrate the machines,
 16 their interrelationships, the principle variables and messages used to communicate between them, their local
 17 variables, and performance parameters. The figures do not show the service interface primitives between the
 18 media-dependent layer and the logical link control (LLC). Figure 11-5 is analogous to Figure 10-2; while
 19 Figure 10-2 applies to the general time-synchronization protocol, Figure 11-5 is limited to the peer-to-peer
 20 delay mechanism for measurement of propagation delay in full-duplex point-to-point links. Figure 11-4
 21 shows greater detail of the `MDSyncReceiveSM` and `MDSyncSendSM` state machines for full-duplex point-
 22 to-point links than Figure 10-2.

23 The state machines described in 11.2 use some of the global per PTP Instance system variables specified in
 24 10.2.4, the global per-port variables specified in 10.2.5, and the functions specified in 10.2.6.

25 11.2.2 Determination of `asCapable` and `asCapableAcrossDomains`

26 There is one instance of the global variable `asCapable` (see 10.2.5.1) per PTP Port, per domain. There is one
 27 instance of the global variable `asCapableAcrossDomains` (see 11.2.13.13), per port, that is common across,
 28 and accessible by, all the domains.

29 The per-PTP Port global variable `asCapable` (see 10.2.5.1) indicates whether the IEEE 802.1AS protocol is
 30 operating, in this domain, on the PTP Link attached to this PTP Port, and can provide the time-

1 synchronization performance described in B.3. asCapable is used by the PortSync entity, which is media-
2 independent; however, the determination of asCapable is media-dependent.

3 The per-port global variable asCapableAcrossDomains is set by the MDPdelayReq state machine
4 (see 11.2.19 and Figure 11-9). For a port attached to a full-duplex point-to-point PTP Link,
5 asCapableAcrossDomains shall be set to TRUE if and only if it is determined, via the transport-specific
6 peer-to-peer delay mechanism or CMLDS, that the following conditions hold for the port:

- 7 a) The port is exchanging peer delay messages with its neighbor,
- 8 b) The measured delay does not exceed meanLinkDelayThresh or is below negMeanLinkDelayThresh,
- 9 c) The port does not receive multiple Pdelay_Resp or Pdelay_Resp_Follow_Up messages in response
10 to a single Pdelay_Req message, and
- 11 d) The port does not receive a response from itself or another PTP Port of the same PTP Instance.

12 NOTE 1—If a PTP Instance implements only domain 0 and the MDPdelayReq and MDPdelayResp state machines are
13 invoked on domain 0 (see 11.2.19), asCapableAcrossDomains is still set by the MDPdelayReq state machine.

14 The default value of meanLinkDelayThresh shall be set as specified in Table 11-1.

Table 11-1—Value of meanLinkDelayThresh for various links

Link	Value of meanLinkDelayThresh (ns) (see NOTE)
100BASE-TX, 1000BASE-T	800 ₁₀
100BASE-FX, 1000BASE-X	0x7FFFFFFF
NOTE—The actual propagation delay for 100BASE-TX and 1000BASE-T links is expected to be smaller than the above respective threshold. If the measured mean propagation delay (i.e., meanLinkDelay; see 10.2.5.8) exceeds this threshold, it is assumed that this is due to the presence of equipment that does not implement gPTP. For 100BASE-FX and 1000BASE-X links, the actual propagation delay can be on the order of, or larger than, the delay produced by equipment that does not implement gPTP; therefore, such equipment cannot be detected by comparing measured propagation delay with a threshold. In this case, meanLinkDelayThresh is set to the largest possible positive value (i.e., in binary notation a zero followed by all 1s). The default value for negMeanLinkDelayThresh is not defined and it is implementation specific.	

15 The per-PTP Port, per-domain global variable asCapable shall be set to TRUE if and only if the following
16 conditions hold:

- 17 e) The value of asCapableAcrossDomains is TRUE, and
- 18 f) One of the following conditions holds:
 - 19 1) The value of neighborGptpCapable for this PTP Port is TRUE, or
 - 20 2) The value of domainNumber is zero, and the value of sdoId for peer delay messages received
21 on this PTP Port is 0x100.

22 NOTE 2—Condition f) 2) ensures backward compatibility with the 2011 edition of this standard. A PTP Instance
23 compliant with the current edition of this standard that is attached, via a full-duplex point-to-point PTP Link, to a PTP
24 Instance compliant with the 2011 edition of this standard will not receive Signaling messages that contain the gPTP-
25 capable TLV and will not set neighborGptpCapable to TRUE. However, condition f) 2) ensures that asCapable for this
26 PTP Port and domain (i.e., domain 0) will still be set to TRUE if condition e) holds because the peer delay messages
27 received from the time-aware system compliant with the 2011 edition of this standard will have sdoId set to 0x100.

1 11.2.3 Use of MAC Control PAUSE operation

2 A PTP Instance shall not use the MAC Control PAUSE operation.

3 11.2.4 Use of priority-based flow control

4 A PTP Instance that implements priority-based flow control shall neither transmit nor honor upon receipt
5 priority-based flow control messages that act on the IEEE 802.1AS message priority code point (see 8.4.4).

6 11.2.5 Use of link aggregation

7 gPTP PDUs, when used with physical ports that are attached to an IEEE 802.1AX™ Link Aggregation
8 Group, shall be transmitted and received over the physical ports (Aggregation Ports), not the Aggregator
9 Port. Using the Parser/Multiplexer functions described in 6.1.3 and 6.2.7 of IEEE Std 802.1AX-2020,
10 received gPTP PDUs are recognized as control frames, separated from the incoming data stream, and
11 directed to the gPTP layers. gPTP PDUs output from the gPTP layers are integrated into the port's data
12 stream by the Parser/Multiplexer.

13 Assuming that ptpPortEnabled is true for all the physical links (Aggregation Links) in a Link Aggregation
14 Group and that all the physical links are connected to the same two systems, gPTP will measure the delay on
15 each physical link, and the IEEE 802.1AS protocol will choose one of the physical links for transmitting
16 time from the timeTransmitter clock.

17 11.2.6 Service interface primitives and data structures communicated between state 18 machines

19 The following subclauses describe the service primitives and data structures communicated between the
20 time-synchronization state machines of the MD entity. First the service primitives are described, followed by
21 the data structures.

22 11.2.7 M-UNITDATA.request

23 This service primitive is used by an MD entity to request to the associated LLC the transmission of a Sync,
24 Follow_Up, Pdelay_Req, Pdelay_Resp, or Pdelay_Resp_Follow_Up message. The primitive is described in
25 IEEE Std 802.1AC.

26 11.2.8 M-UNITDATA.indication

27 This service primitive is used by the LLC to indicate to the associated MD entity the receipt of a Sync,
28 Follow_Up, Pdelay_Req, Pdelay_Resp, or Pdelay_Resp_Follow_Up message. The primitive is described in
29 IEEE Std 802.1AC.

30 11.2.9 MDTimestampReceive

31 11.2.9.1 General

32 This structure provides the timestamp, relative to the timestamp measurement plane, of the event message
33 that was just sent or just received. The structure is received by the MD entity of the PTP Port. The MD entity
34 corrects this timestamp for any ingressLatency or egressLatency (see 8.4.3) before using it and/or passing it
35 to higher layer entities.

```
36 MDTimestampReceive{  
37     timestamp  
38 }
```


1 The member of the structure is defined in the following subclause.

2 11.2.9.2 timestamp (UScaledNs)

3 The timestamp is the value of the timestamp of the event message (i.e., Sync, Pdelay_Req, Pdelay_Resp)
4 that was just transmitted or received. This timestamp is taken relative to the timestamp measurement plane.

5 11.2.10 MDSyncReceive

6 This structure is specified in 10.2.2.2.

7 11.2.11 MDSyncSend

8 This structure is specified in 10.2.2.1.

9 11.2.12 Overview of MD entity global variables

10 Subclause 11.2.13 defines global variables used by the MD entity state machines whose scopes are as
11 follows:

- 12 — Per PTP Instance (i.e., per domain)
- 13 — Per PTP Instance, per PTP Port
- 14 — Instances used by CMLDS (see 11.2.17) (i.e., variable is common across all Link Ports)
- 15 — Instances used by CMLDS, per Link Port

16 Table 11-2 summarizes the scope of each global variable of 11.2.13.

Table 11-2 — Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5)

Variable name	Subclause of definition	Data Type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
currentLogPdelayReqInterval	11.2.13.1	Integer8	Per PTP Port	No	Yes
initialLogPdelayReqInterval	11.2.13.2	Integer8	Per PTP Port	No	Yes
pdelayReqInterval	11.2.13.3	UScaledNs	Per PTP Port	No	Yes
allowedLostResponses	11.2.13.4	UInteger8	Per PTP Port	No	Yes
allowedFaults	11.2.13.5	UInteger8	Per PTP Port	No	Yes
isMeasuringDelay	11.2.13.6	Boolean	Per PTP Port	No	Yes

Table 11-2 — Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5)

Variable name	Subclause of definition	Data Type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
meanLinkDelayThresh	11.2.13.7	ScaledNs	Per PTP Port	No	Yes
negMeanLinkDelayThresh	11.2.13.8	ScaledNs	Per PTP Port	No	Yes
syncSequenceId	11.2.13.9	UInteger16	Per PTP Port	No	No
oneStepReceive	11.2.13.10	Boolean	Per PTP Port	No	No
oneStepTransmit	11.2.13.11	Boolean	Per PTP Port	No	No
oneStepTxOper	11.2.13.12	Boolean	Per PTP Port	No	No
asCapableAcrossDomains	11.2.13.13	Boolean	Per PTP Port	No	Yes
nrrPdelay	11.2.13.14	Float64	Per PTP Port	No	Yes
nrrSync	11.2.13.15	Float64	Per PTP Port	No	No
nrrCompMethod	11.2.13.16	Enumeration1	Per PTP Port	No	No

11.2.13 MD entity global variables

2 11.2.13.1 currentLogPdelayReqInterval: The current value of the logarithm to base 2 of the mean time 3 interval, in seconds, between the sending of successive Pdelay_Req messages (see 11.5.2.2). This value is 4 set in the LinkDelayIntervalSetting state machine (see 11.2.21). There is one instance of this variable for all 5 the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if 6 domain 0 is implemented. The variable is accessible by all the domains.

7 11.2.13.2 initialLogPdelayReqInterval: The initial value of the logarithm to base 2 of the mean time 8 interval, in seconds, between the sending of successive Pdelay_Req messages (see 11.5.2.2). There is one 9 instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of 10 this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

11 11.2.13.3 pdelayReqInterval: A variable containing the mean Pdelay_Req message transmission interval 12 for the port corresponding to this MD entity. The value is set in the LinkDelayIntervalSetting state machine 13 (see 11.2.21). There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), 14 and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by 15 all the domains.

16 11.2.13.4 allowedLostResponses: The number of Pdelay_Req messages without valid responses above 17 which a port is considered to be not exchanging peer delay messages with its neighbor. The default value 18 and range of allowedLostResponses is given in 11.5.3. There is one instance of this variable for all the 19 domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 20 0 is implemented. The variable is accessible by all the domains.

21 11.2.13.5 allowedFaults: The number of faults [i.e., instances where meanLinkDelay (see 10.2.5.8) exceeds 22 meanLinkDelayThresh (see 11.2.13.7), or is below the negMeanLinkDelayThresh (see 11.2.13.8), and/or 23 the computation of nrrPdelay (see 11.2.13.14) is invalid (see 11.2.19.2.10)] above which 24 asCapableAcrossDomains is set to FALSE, i.e., the port is considered not capable of interoperating with its 25 neighbor via the IEEE 802.1AS protocol (see 10.2.5.1). The default value and range of allowedFaults is

1 given in 11.5.4. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per
2 port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is
3 accessible by all the domains.

4 **11.2.13.6 isMeasuringDelay:** A Boolean that is TRUE if the port is receiving Pdelay_Resp and
5 Pdelay_Resp_Follow_Up messages from the port at the other end of the PTP Link. For a full-duplex point-
6 to-point PTP Link, the port is measuring PTP Link propagation delay if it is receiving Pdelay_Resp and
7 Pdelay_Resp_Follow_Up messages from the port at the other end of the PTP Link (i.e., it performs the
8 measurement using the peer-to-peer delay mechanism). There is one instance of this variable for all the
9 domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain
10 is implemented. The variable is accessible by all the domains.

11 **11.2.13.7 meanLinkDelayThresh:** The propagation time threshold above which a port is considered not
12 capable of participating in the IEEE 802.1AS protocol. If meanLinkDelay (see 10.2.5.8) exceeds
13 meanLinkDelayThresh, then asCapableAcrossDomains (see 11.2.13.13) is set to FALSE. There is one
14 instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of
15 this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

16 **11.2.13.8 negMeanLinkDelayThresh:** The negative propagation time threshold above which a port is
17 considered not capable of participating in the IEEE 802.1AS protocol. If meanLinkDelay (see 10.2.5.8) is
18 below the negMeanLinkDelayThresh, then asCapableAcrossDomains (see 11.2.13.13) is set to FALSE.
19 There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one
20 instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the
21 domains.

22 NOTE—The variable meanLinkDelayThresh was named neighborPropDelayThresh in the 2011 edition of this standard.

23 **11.2.13.9 syncSequenceId:** The sequenceId (see 11.4.2.8) for the next Sync message to be sent by this MD
24 entity.

25 **11.2.13.10 oneStepReceive:** A Boolean variable that is TRUE if the PTP Port is capable of receiving one-
26 step Sync messages.

27 **11.2.13.11 oneStepTransmit:** A Boolean variable that is TRUE if the PTP Port is capable of transmitting
28 one-step Sync messages.

29 **11.2.13.12 oneStepTxOper:** A Boolean variable that is TRUE if the PTP Port will be transmitting one-step
30 Sync messages (see 11.1.3).

31 NOTE—Since a one-step port modifies Sync message fields (e.g., the correctionField) “on the fly” as the Sync message
32 is being transmitted, the modifying of the correctionField of the Sync message is often implemented in hardware. Each
33 distinct PTP Instance of a time-aware system runs in a distinct domain and tracks a distinct time. If a time-aware system
34 supports more than one PTP Instance, support for one-step often requires hardware for each PTP Instance (e.g., four
35 domains requires four hardware components), but support for two-step can share hardware among all PTP Instances.

36 **11.2.13.13 asCapableAcrossDomains:** A Boolean that is TRUE if and only if conditions a) through d) of
37 11.2.2 are satisfied. This Boolean is set by the MDPdelayReq state machine and is used in determining
38 asCapable for a port (see 11.2.2). There is one instance of this variable for all the domains (per port). The
39 variable is accessible by all the domains. When only one domain is active, asCapableAcrossDomains is
40 equivalent to the variable asCapable (see 10.2.5.1).

41 **11.2.13.14 nrrPdelay:** The neighborRateRatio (see 10.2.5.7) computed using the transport-specific
42 peer-to-peer delay mechanism or CMLDS (see 11.2.19).

1 **11.2.13.15 nrrSync:** The neighborRateRatio (see 10.2.5.7) computed using the Drift_Tracking TLV carried
2 by successive Sync or, if two-step, Follow_Up messages (see 11.2.14).

3 **11.2.13.16 nrrCompMethod:** An Enumeration1 that takes on the values Sync and Pdelay to indicate the
4 source of the value of neighborRateRatio (see 10.2.5.7):

- 5 a) If the value is Sync and the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE,
6 neighborRateRatio is populated with the value of nrrSync whenever a new value of nrrSync is
7 computed.
- 8 b) If the value is Pdelay or if the global variable driftTrackingTlvSupport (10.2.4.28) is FALSE,
9 neighborRateRatio is populated with the value of nrrPdelay whenever a new value of nrrPdelay is
10 computed.

11 **11.2.14 MDSyncReceiveSM state machine**

12 **11.2.14.1 State machine variables**

13 The following variables are used in the state diagram in Figure 11-6 (in 11.2.14.3):

14 **11.2.14.1.1 followUpReceiptTimeoutTime:** A variable used to save the time at which the information
15 conveyed by a received Sync message will be discarded if the associated Follow_Up message is not received
16 by then. The data type for followUpReceiptTimeoutTime is UScaledNs.

17 **11.2.14.1.2 rcvdSync:** A Boolean variable that is set to TRUE when a Sync message is received. This
18 variable is reset to FALSE by the current state machine.

19 **11.2.14.1.3 rcvdFollowUp:** A Boolean variable that notifies the current state machine when a Follow_Up
20 message is received. This variable is reset by the current state machine.

21 **11.2.14.1.4 rcvdSyncPtr:** A pointer to a structure whose members contain the values of the fields of the
22 Sync message whose receipt is indicated by rcvdSync (see 11.2.14.1.2).

23 **11.2.14.1.5 rcvdFollowUpPtr:** A pointer to a structure whose members contain the values of the fields of
24 the Follow_Up message whose receipt is indicated by rcvdFollowUp (see 11.2.14.1.3).

25 **11.2.14.1.6 txMDSyncReceivePtrMDSR:** A pointer to a structure whose members contain the values of the
26 parameters of an MDSyncReceive structure to be transmitted.

27 **11.2.14.1.7 upstreamSyncInterval:** The sync interval (see 10.7.2.1) for the upstream PTP Port that sent the
28 received Sync message. The data type for upstreamSyncInterval is UScaledNs.

29 **11.2.14.1.8 TEMP:** A temporary variable used to reduce clutter in the state diagram (see Figure 11-6). The
30 data type for TEMP is Integer16.

31 **11.2.14.2 State machine functions**

32 **11.2.14.2.1 setMDSyncReceiveMDSR():** Creates an MDSyncReceive structure, and returns a pointer to
33 this structure. The members of this structure are set as follows:

- 34 a) If twoStepFlag of the most recently received Sync message is TRUE, followUpCorrectionField is
35 set equal to the sum of the correctionField (see 11.4.2.6) of the most recently received Sync
36 message, plus the correctionField of the most recently received Follow_Up message. Otherwise,
37 followUpCorrectionField is set equal to the correctionField of the most recently received Sync
38 message.

- 1 b) sourcePortIdentity is set equal to the sourcePortIdentity (see 11.4.2.7) of the most recently received
- 2 Sync message (see 11.4.3).
- 3 c) logMessageInterval is set equal to the logMessageInterval (see 11.4.2.9) of the most recently
- 4 received Sync message (see 11.4.3).
- 5 d) If twoStepFlag of the most recently received Sync message is TRUE, preciseOriginTimestamp is set
- 6 equal to the preciseOriginTimestamp (see 11.4.4.2.1) of the most recently received Follow_Up
- 7 message. Otherwise, preciseOriginTimestamp is set equal to the originTimestamp (see 11.4.3.2.1) of
- 8 the most recently received Sync message.
- 9 e) rateRatio is set equal to the quantity $(\text{cumulativeScaledRateOffset} \times 2^{-41}) + 1.0$, where the
- 10 cumulativeScaledRateOffset field is for the most recently received Follow_Up information TLV
- 11 (see 11.4.4.3.6).
- 12 f) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the most
- 13 recently received Sync message (if twoStepFlag is FALSE) or Follow_Up message (if twoStepFlag
- 14 is TRUE) contains a Drift_Tracking TLV:
 - 15 1) syncGrandmasterIdentity (see 10.2.4.26) is set equal to the syncGrandmasterIdentity of the
 - 16 Drift_Tracking TLV.
 - 17 2) If the value of syncStepsRemoved of the Drift_Tracking TLV is not equal to 0xFFFF,
 - 18 syncStepsRemoved (see 10.2.4.27) is set equal to the value of syncStepsRemoved of the
 - 19 Drift_Tracking TLV plus one. If the value of syncStepsRemoved of the Drift_Tracking TLV is
 - 20 equal to 0xFFFF, syncStepsRemoved (see 10.2.4.27) is set equal to 0xFFFF.
 - 21 3) nrrSync (see 11.2.13.15) is computed using the following information conveyed by successive
 - 22 Sync and, if twoStepFlag is TRUE, Follow_Up messages; the specific algorithms are beyond
 - 23 the scope of this document:
 - 24 i) The syncEventIngressTimestamp values for the respective Sync messages.
 - 25 ii) The successive correctedSyncEgressTimestamp values, whose data type is UScaledNs,
 - 26 obtained by adding (A) the seconds field of the syncEgressTimestamp of the
 - 27 Drift_Tracking TLV, multiplied by 10^9 , (B) the nanoseconds field of the
 - 28 syncEgressTimestamp of the Drift_Tracking TLV, and (C) CMLDS delayAsymmetry
 - 29 (i.e., cmlDsLinkPort.delayAsymmetry, see 10.2.5.9) if CMLDS is provided, otherwise
 - 30 instance-specific delayAsymmetry (i.e., portDS.delayAsymmetry, see 14.10.11) divided
 - 31 by rateRatio (see 10.2.8.1.4).
 - 32 iii) The rateRatioDrift field of the Drift_Tracking TLV.

33 NOTE 1—If delayAsymmetry does not change during the time interval over which nrrSync is computed, it is not
 34 necessary to add it in f) 3) ii) (C) because in that case it will be canceled when computing the difference between earlier
 35 and later correctedSyncEgressTimestamps.

36 NOTE 2—The methodology in f) 3) for computing nrrSync follows the methodology described in 11.2.19.3.3 for
 37 computing nrrPdelay using information contained in peer delay messages. See the NOTES and description in 11.2.19.3.3
 38 for more detail. The variable correctedSyncEgressTimestamp is computed in a manner analogous to the computation of
 39 the variable correctedResponderEventTimestamp in 11.2.19.3.3.

40 NOTE 3—The use of the rateRatioDrift, e.g., in compensating for frequency drift when computing rateRatio (see the
 41 RECEIVED_SYNC state of the PortSyncSyncReceive state machine of Figure 10-4), is implementation specific or
 42 profile standard specific.

- 43 g) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the most
- 44 recently received Sync message (if twoStepFlag is FALSE) or Follow_Up message (if twoStepFlag
- 45 is TRUE) does not contain a Drift_Tracking TLV:
 - 46 1) syncGrandmasterIdentity (see 10.2.4.26) is set equal to 0xFFFF FFFF FFFF FFFF.
 - 47 2) syncStepsRemoved (see 10.2.4.27) is set equal to 0xFFFF.
- 48 h) The value of neighborRateRatio (see 10.2.5.7) is set equal to the value of nrrSync (see 11.2.13.15) or
- 49 nrrPdelay (see 11.2.13.14) based on the value of nrrCompMethod (see 11.2.13.16).

1 i) upstreamTxTime is set equal to the syncEventIngressTimestamp for the most recently received Sync
 2 message (see 11.4.3), minus the mean propagation time on the PTP Link attached to this PTP Port
 3 (meanLinkDelay; see 10.2.5.8) divided by nrrPdelay (see 11.2.13.14), minus delayAsymmetry (see
 4 10.2.5.9) for this PTP Port divided by the sum of rateRatio [see item e) in this subclause] and the
 5 quantity neighborRateRatio – 1. The syncEventIngressTimestamp is equal to the timestamp value
 6 measured relative to the timestamp measurement plane, minus any ingressLatency (see 8.4.3). The
 7 upstreamTxTime can be written as follows:

$$\begin{aligned} & \text{upstreamTxTime} = \text{syncEventIngressTimestamp} - (\text{meanLinkDelay}/\text{nrrPdelay}) - \\ & (\text{delayAsymmetry}/(\text{rateRatio} \times \text{neighborRateRatio})) \end{aligned}$$

10 NOTE 4—The syncEventIngressTimestamp is measured relative to the time base of the current PTP Instance. The mean
 11 propagation delay is divided by neighborRateRatio to convert it from the time base of the PTP Instance at the other end
 12 of the attached PTP Link to the time base of the current PTP Instance. delayAsymmetry (i.e., portDS.delayAsymmetry,
 13 see 14.10.11) is divided by the new rateRatio to convert it from the time base of the Grandmaster Clock to the time base
 14 of the current PTP Instance.

15 NOTE 5—meanLinkDelay is divided by nrrPdelay because the meanLinkDelay calculation is done using the
 16 peer-to-peer delay message timestamps. However, neighborRateRatio appears in the denominator of the
 17 delayAsymmetry term because delayAsymmetry, which is supplied via the managed object portDS.delayAsymmetry or
 18 cmlDsLinkPort.delayAsymmetry, is obtained via measurements outside of gPTP and is relative to the grandmaster time
 19 base. delayAsymmetry is converted to the local clock time base based on the respective method used to measure
 20 neighbor rate ratio.

21 NOTE 6—This represents a media-dependent modification to the calculation given in 10.2.2.2.7.

22 NOTE 7—The difference between the mean propagation time in the Grandmaster Clock time base, the time base of the
 23 PTP Instance at the other end of the PTP Link, and the time base of the current PTP Instance is usually negligible. The
 24 same is true of any delayAsymmetry (see NOTE 2 of 11.2.19.3.4).

25 j) gmTimeBaseIndicator is set equal to the gmTimeBaseIndicator of the most recently received
 26 Follow_Up information TLV (see 11.4.4.3.7).
 27 k) lastGmPhaseChange is set equal to the lastGmPhaseChange of the most recently received
 28 Follow_Up information TLV (see 11.4.4.3.8).
 29 l) lastGmFreqChange is set equal to the scaledLastGmFreqChange of the most recently received
 30 Follow_Up information TLV (see 11.4.4.3.9), multiplied by 2^{-41} .
 31 m) domainNumber is set equal to the domainNumber of the most recently received Sync message
 32 (see 11.4.3).

33 **11.2.14.2.2 txMDSyncReceive (txMDSyncReceivePtrMDSR):** Transmits an MDSyncReceive structure to
 34 the PortSyncSyncReceive state machine of the PortSync entity of this PTP Port.

35 11.2.14.3 State diagram

36 The MDSyncReceiveSM state machine shall implement the function specified by the state diagram in
 37 Figure 11-6, the local variables specified in 11.2.14.1, the functions specified in 11.2.14.2, the structure
 38 specified in 10.2.2.2, the messages specified in 11.4, the MD entity global variables specified in 11.2.13, and
 39 the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives
 40 Sync and, if the received information is two-step, Follow_Up messages, places the time-synchronization
 41 information in an MDSyncReceive structure, and sends the structure to the PortSyncSyncReceive state
 42 machine of the PortSync entity of this PTP Port.

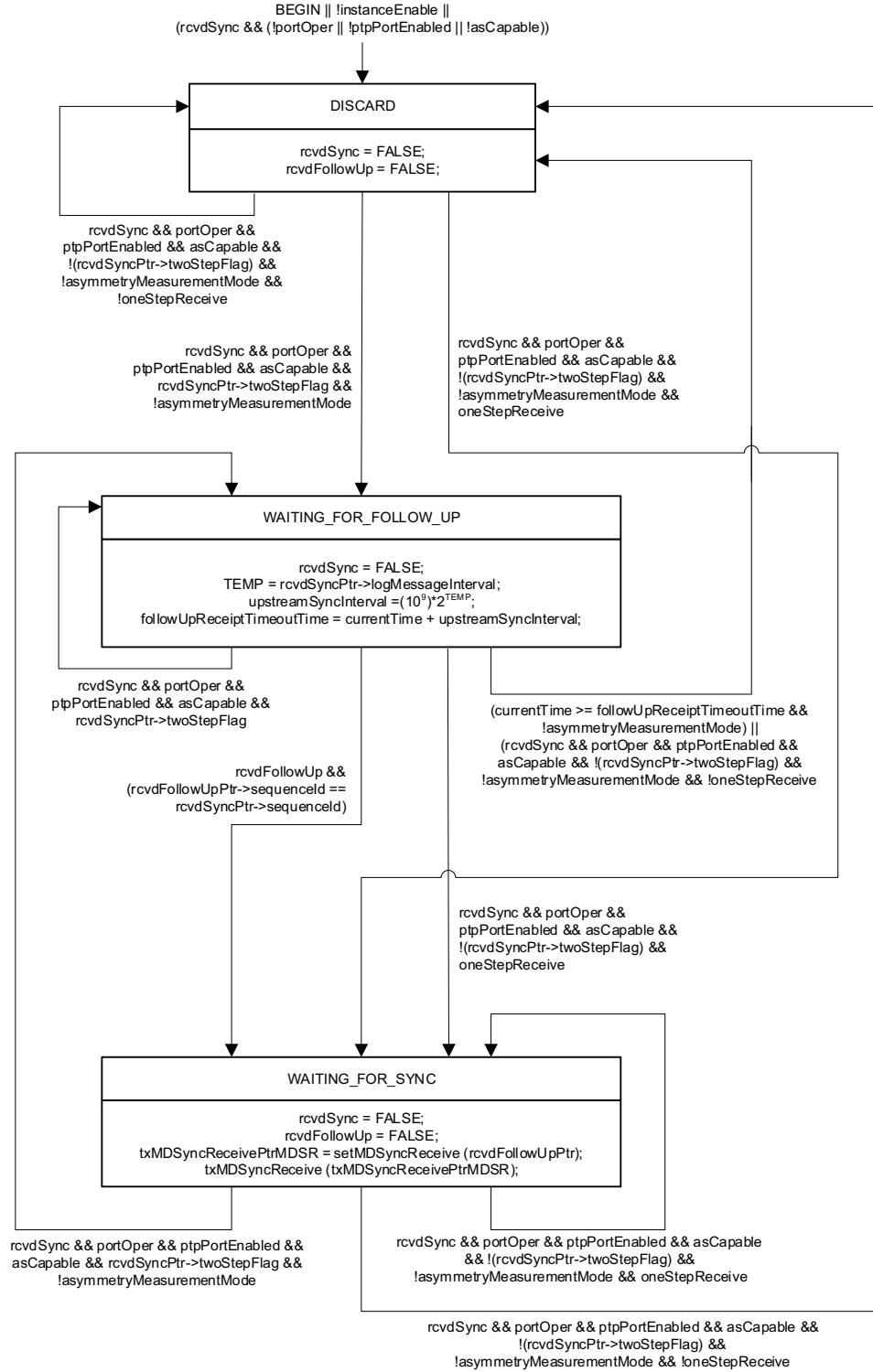


Figure 11-6—MDSyncReceiveSM state machine

1 11.2.15 MDSyncSendSM state machine

2 11.2.15.1 State machine variables

3 The following variables are used in the state diagram in Figure 11-7 (in 11.2.15.3):

4 **11.2.15.1.1 rcvdMDSyncMDSS:** A Boolean variable that notifies the current state machine when an
5 MDSyncSend structure is received. This variable is reset by the current state machine.

6 **11.2.15.1.2 txSyncPtr:** A pointer to a structure whose members contain the values of the fields of a Sync
7 message to be transmitted.

8 **11.2.15.1.3 rcvdMDTimestampReceiveMDSS:** A Boolean variable that notifies the current state machine
9 when the syncEventEgressTimestamp (see 11.3.2.1) for a transmitted Sync message is received. This
10 variable is reset by the current state machine.

11 **11.2.15.1.4 rcvdMDTimestampReceivePtr:** A pointer to the received MDTimestampReceive structure (see
12 11.2.9).

13 **11.2.15.1.5 txFollowUpPtr:** A pointer to a structure whose members contain the values of the fields of a
14 Follow_Up message to be transmitted.

15 11.2.15.2 State machine functions

16 **11.2.15.2.1 setSyncTwoStep():** Creates a structure whose parameters contain the fields (see 11.4) of a Sync
17 message to be transmitted, and returns a pointer to this structure. The parameters are set as follows:

- 18 a) correctionField is set equal to 0.
- 19 b) sourcePortIdentity is set equal to the sourcePortIdentity member of the most recently received
20 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
- 21 c) sequenceId is set equal to syncSequenceId (see 11.2.13.9).
- 22 d) logMessageInterval is set equal to the logMessageInterval member of the most recently received
23 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
- 24 e) The remaining fields are set as specified in 11.4.2 and 11.4.3 when twoStepFlag is TRUE.

25 **11.2.15.2.2 txSync (txSyncPtr):** Transmits a Sync message from this MD entity, whose fields contain the
26 parameters in the structure pointed to by txSyncPtr (see 11.2.15.1.2).

27 **11.2.15.2.3 setFollowUp():** Creates a structure whose parameters contain the fields (see 11.4) of a
28 Follow_Up message to be transmitted, and returns a pointer to this structure. The parameters are set
29 as follows:

- 30 a) followUpCorrectionField is set equal to the sum of the following:
 - 31 1) The followUpCorrectionField member of the most recently received MDSyncSend structure
32 (see 10.2.2.1 and 11.2.11)
 - 33 2) The quantity

$$\text{rateRatio} \times (\text{syncEventEgressTimestamp} - \text{upstreamTxTime})$$
- 34 where
 - 35 rateRatio is the rateRatio member of the most recently received
36 MDSyncSend structure (see 10.2.2.1 and 11.2.11)
 - 37 upstreamTxTime is the upstreamTxTime member of the most recently received
38 MDSyncSend structure (see 10.2.2.1 and 11.2.11)
 - 39 syncEventEgressTimestamp is the timestamp pointed to by rcvdMDTimestampReceivePtr,
40 corrected for egressLatency (see 8.4.3)
 - 41

- 1 NOTE—If the PTP Instance that contains this PortSync entity is a PTP Relay Instance, the quantity
2 $\text{syncEventEgressTimestamp} - \text{upstreamTxTime}$
3 is the sum of the residence time and PTP Link propagation delay on the upstream PTP Link, relative to the LocalClock
4 entity, and the quantity
5 $\text{rateRatio} \times (\text{syncEventEgressTimestamp} - \text{upstreamTxTime})$
6 is the sum of the residence time and PTP Link propagation delay on the upstream PTP Link, relative to the Grandmaster
7 Clock (see 11.2.14.2.1 for details on the setting of upstreamTxTime).
- 8 b) sourcePortIdentity is set equal to the sourcePortIdentity member of the most recently received
9 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
 - 10 c) sequenceId is set equal to syncSequenceId (see 11.2.13.9).
 - 11 d) logMessageInterval is set equal to the logMessageInterval member of the most recently received
12 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
 - 13 e) preciseOriginTimestamp is set equal to the preciseOriginTimestamp member of the most recently
14 received MDSyncSend structure (see 10.2.2.1 and 11.2.11).
 - 15 f) rateRatio is set equal to the rateRatio member of the most recently received MDSyncSend structure
16 (see 10.2.2.1 and 11.2.11).
 - 17 g) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the PTP
18 Instance is the Grandmaster PTP Instance, a Drift_Tracking TLV is attached to the transmitted
19 Follow_Up message with the syncGrandmasterIdentity, syncStepsRemoved, syncEgressTimestamp,
20 and rateRatioDrift fields set as follows:
 - 21 1) syncGrandmasterIdentity is set equal to the defaultDS.clockIdentity.
 - 22 2) syncStepsRemoved is set equal to 0.
 - 23 3) syncEgressTimestamp is set equal to the syncEventEgressTimestamp of the associated Sync
24 message.
 - 25 4) rateRatioDrift (see 11.4.4.4.9) is the measured estimate of the rate of change per second of the
26 frequency of the Grandmaster Clock relative to the frequency of the local clock at the
27 Grandmaster PTP Instance. If the Grandmaster Clock and the local clock at the Grandmaster
28 PTP Instance are the same, rateRatioDrift is zero.
 - 29 h) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the PTP
30 Instance is not the Grandmaster PTP Instance, a Drift_Tracking TLV is attached to the transmitted
31 Follow_Up message with the syncGrandmasterIdentity, syncStepsRemoved, syncEgressTimestamp,
32 and rateRatioDrift fields set as follows:
 - 33 1) syncGrandmasterIdentity is set equal to the value of the global variable
34 syncGrandmasterIdentity (see 10.2.4.26).
 - 35 2) syncStepsRemoved is set equal to the value of the global variable syncStepsRemoved
36 (see 10.2.4.27).
 - 37 3) syncEgressTimestamp is set equal to the syncEventEgressTimestamp of the associated Sync
38 message.
 - 39 4) rateRatioDrift (see 11.4.4.4.9) is the measured estimate of the rate of change per second of the
40 frequency of the Grandmaster Clock relative to the frequency of the local clock of the PTP
41 Instance that sends the message.
 - 42 i) gmTimeBaseIndicator is set equal to the gmTimeBaseIndicator member of the most recently
43 received MDSyncSend structure (see 10.2.2.1 and 11.2.11).
 - 44 j) lastGmPhaseChange is set equal to the lastGmPhaseChange member of the most recently received
45 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
 - 46 k) scaledLastGmFreqChange is set equal to the lastGmFreqChange member of the most recently
47 received MDSyncSend structure (see 10.2.2.1 and 11.2.11), multiplied by 2^{41} .

- 1 l) domainNumber is set equal to the domainNumber of the most recently received MDSyncSend
- 2 structure (see 10.2.2.1 and 11.2.11).
- 3 m) The remaining fields are set as specified in 11.4.2 and 11.4.4.

4 **11.2.15.2.4 txFollowUp (txFollowUpPtr):** Transmits a Follow_Up message from this MD entity, whose

5 fields contain the parameters in the structure pointed to by txFollowUpPtr (see 11.2.15.1.5).

6 **11.2.15.2.5 setSyncOneStep():** Creates a structure whose parameters contain the fields (see 11.4) of a Sync

7 message to be transmitted, and returns a pointer to this structure. The parameters are set as follows:

- 8 a) correctionField is set equal to the followUpCorrectionField member of the most recently received
- 9 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
- 10 b) sourcePortIdentity is set equal to the sourcePortIdentity member of the most recently received
- 11 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
- 12 c) sequenceId is set equal to syncSequenceId (see 11.2.13.9).
- 13 d) logMessageInterval is set equal to the logMessageInterval member of the most recently received
- 14 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
- 15 e) originTimestamp is set equal to the preciseOriginTimestamp member of the most recently received
- 16 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
- 17 f) rateRatio is set equal to the rateRatio member of the most recently received MDSyncSend structure
- 18 (see 10.2.2.1 and 11.2.11).
- 19 g) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the PTP
- 20 Instance is the Grandmaster PTP Instance, a Drift_Tracking TLV is attached to the transmitted Sync
- 21 message with the syncGrandmasterIdentity, syncStepsRemoved, syncEgressTimestamp, and
- 22 rateRatioDrift fields set as follows:
 - 23 1) syncGrandmasterIdentity is set equal to the defaultDS.clockIdentity.
 - 24 2) syncStepsRemoved is set equal to 0.
 - 25 3) syncEgressTimestamp is set equal to the syncEventEgressTimestamp of the associated Sync
 - 26 message.
 - 27 4) rateRatioDrift (see 11.4.4.4.9) is the measured estimate of the rate of change per second of the
 - 28 frequency of the Grandmaster Clock relative to the frequency of the local clock at the
 - 29 Grandmaster PTP Instance. If the Grandmaster Clock and the local clock at the Grandmaster
 - 30 PTP Instance are the same, rateRatioDrift is zero.
- 31 h) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the PTP
- 32 Instance is not the Grandmaster PTP Instance, a Drift_Tracking TLV is attached to the transmitted
- 33 Sync message with the syncGrandmasterIdentity, syncStepsRemoved, syncEgressTimestamp, and
- 34 rateRatioDrift fields set as follows:
 - 35 1) syncGrandmasterIdentity is set equal to the value of the global variable
 - 36 syncGrandmasterIdentity (see 10.2.4.26).
 - 37 2) syncStepsRemoved is set equal to the value of the global variable syncStepsRemoved
 - 38 (see 10.2.4.27).
 - 39 3) syncEgressTimestamp is set equal to the syncEventEgressTimestamp of the associated Sync
 - 40 message.
 - 41 4) rateRatioDrift (see 11.4.4.4.9) is the measured estimate of the rate of change per second of the
 - 42 frequency of the Grandmaster Clock relative to the frequency of the local clock of the PTP
 - 43 Instance that sends the message.
- 44 i) gmTimeBaseIndicator is set equal to the gmTimeBaseIndicator member of the most recently
- 45 received MDSyncSend structure (see 10.2.2.1 and 11.2.11).

- 1 j) lastGmPhaseChange is set equal to the lastGmPhaseChange member of the most recently received
- 2 MDSyncSend structure (see 10.2.2.1 and 11.2.11).
- 3 k) scaledLastGmFreqChange is set equal to the lastGmFreqChange member of the most recently
- 4 received MDSyncSend structure (see 10.2.2.1 and 11.2.11), multiplied by 2^{41} .
- 5 l) domainNumber is set equal to the domainNumber of the most recently received MDSyncSend
- 6 structure (see 10.2.2.1 and 11.2.11).
- 7 m) The remaining fields are set as specified in 11.4.2 and 11.4.3 when twoStepFlag is FALSE.

8 **11.2.15.2.6 modifySync():** Adds to the correctionField of the transmitted Sync message, after the

9 syncEventEgressTimestamp is taken and known, the quantity

$$10 \quad \text{rateRatio} \times (\text{syncEventEgressTimestamp} - \text{upstreamTxTime})$$

11 where

12	rateRatio	is the rateRatio member of the most recently received MDSyncSend
13		structure (see 10.2.2.1 and 11.2.11)
14	upstreamTxTime	is the upstreamTxTime member of the most recently received
15		MDSyncSend structure (see 10.2.2.1 and 11.2.11)
16	syncEventEgressTimestamp	is the timestamp pointed to by rcvdMDTimestampReceivePtr, corrected
17		for egressLatency (see 8.4.3)

18 NOTE—If the PTP Instance that contains this PortSync entity is a PTP Relay Instance, the quantity

19 $\text{syncEventEgressTimestamp} - \text{upstreamTxTime}$

20 is the sum of the residence time and PTP Link propagation delay on the upstream PTP Link, relative to the LocalClock

21 entity, and the quantity

22 $\text{rateRatio} \times (\text{syncEventEgressTimestamp} - \text{upstreamTxTime})$

23 is the sum of the residence time and PTP Link propagation delay on the upstream PTP Link, relative to the Grandmaster

24 Clock (see 11.2.14.2.1 for details on the setting of upstreamTxTime).

25 **11.2.15.3 State diagram**

26 The MDSyncSendSM state machine shall implement the function specified by the state diagram in

27 Figure 11-7; the local variables specified in 11.2.15.1; the functions specified in 11.2.15.2; the structures

28 specified in 11.2.9 and 10.2.2.1; the messages specified in 11.4; the MD entity global variables specified in

29 11.2.13; and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state

30 machine receives an MDSyncSend structure from the PortSyncSyncSend state machine of the PortSync

31 entity of this PTP Port and transmits a Sync and corresponding Follow_Up message.

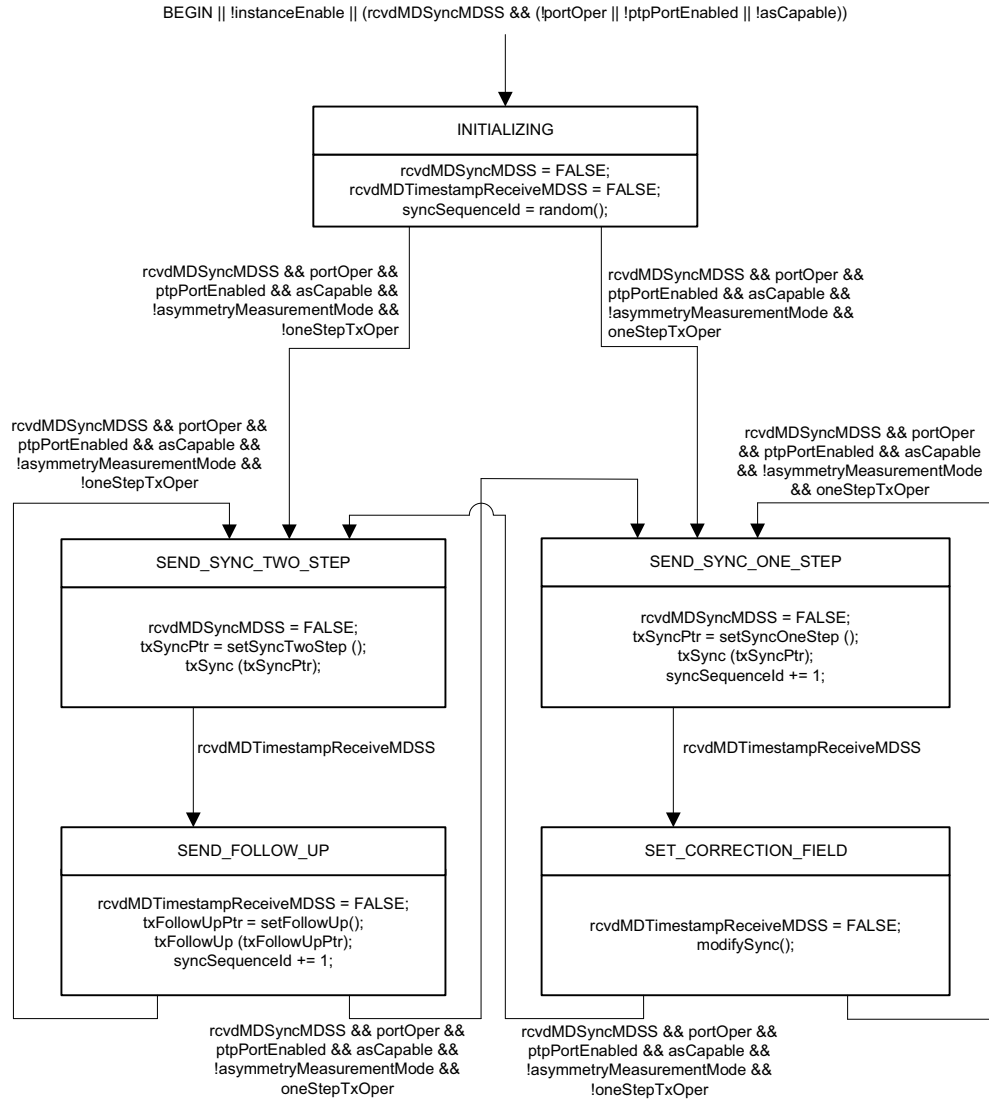


Figure 11-7—MDSyncSendSM state machine

1 11.2.16 OneStepTxOperSetting state machine

2 11.2.16.1 State machine variables

3 The following variables are used in the state diagram in Figure 11-8 (in 11.2.16.2):

4 **11.2.16.1.1 rcvdSignalingMsg4:** A Boolean variable that notifies the current state machine when a
5 Signaling message that contains a Message Interval Request TLV (see 10.6.4.3) is received. This variable is
6 reset by the current state machine.

7 **11.2.16.1.2 rcvdSignalingPtrOSTOS:** A pointer to a structure whose members contain the values of the
8 fields of the received Signaling message that contains a Message Interval Request TLV (see 10.6.4.3).

9 11.2.16.2 State diagram

10 The OneStepTxOperSetting state machine shall implement the function specified by the state diagram in
11 Figure 11-8, the local variables specified in 11.2.16.1, the messages specified in 10.6 and 11.4, the relevant
12 global variables specified in 10.2.5 and 11.2.13, and the relevant managed objects specified in 14.10. This
13 state machine is responsible for setting the relevant global variables and managed objects pertaining to one-
14 step/two-step operation.

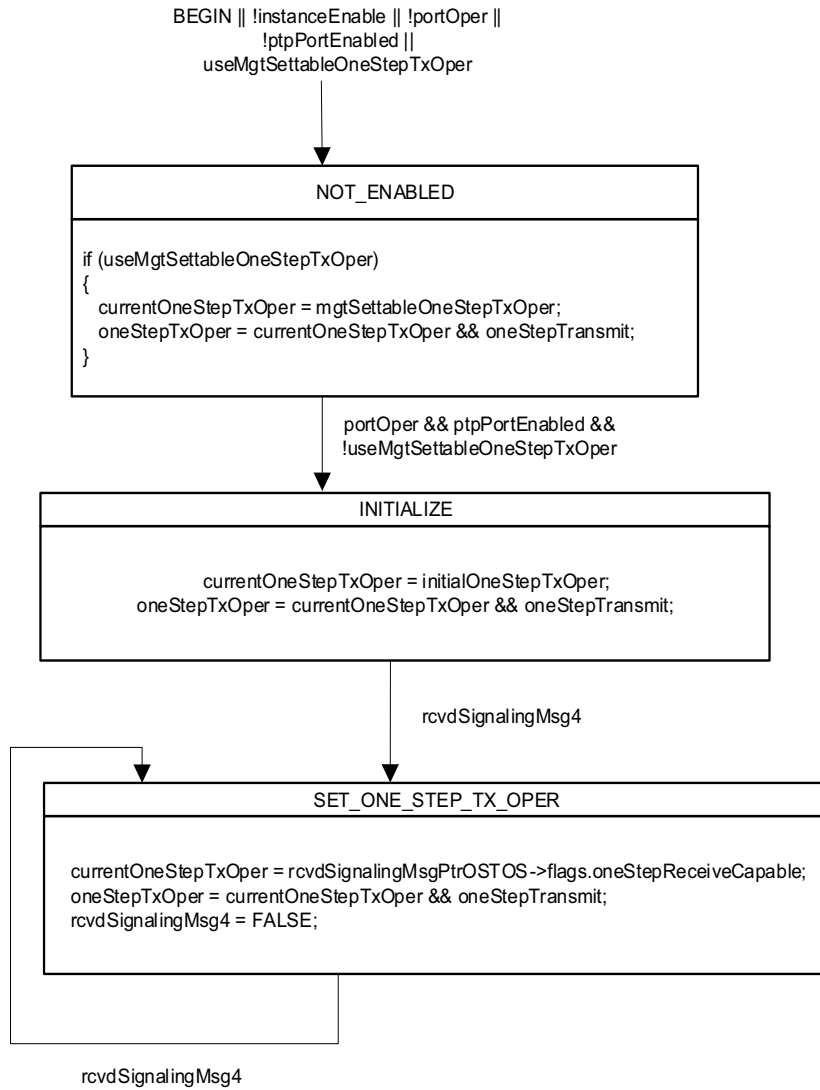


Figure 11-8—OneStepTxOperSetting state machine

1 11.2.17 Common Mean Link Delay Service (CMLDS)

2 11.2.17.1 General

3 Each PTP Port or Link Port of a time-aware system invokes a single instance of the MDPdelayReq state
4 machine (see 11.2.19) and the MDPdelayResp state machine (see 11.2.20). If the time-aware system
5 implements more than one domain or if domainNumber 0 is not present, these two state machines shall
6 provide a Common Mean Link Delay Service (CMLDS), as described in this subclause, that measures mean
7 propagation delay on the PTP Link attached to the port and the neighbor rate ratio for the port (i.e., the ratio
8 of the frequency of the LocalClock entity of the time-aware system at the other end of the PTP Link attached
9 to this port, to the frequency of the LocalClock entity of this time-aware system). The CMLDS makes the
10 mean propagation delay and neighbor rate ratio available to all active domains. If the time-aware system
11 implements one domain, and if the domainNumber of this domain is 0 (see 8.1), these two state machines
12 may provide the CMLDS; however, if the state machines do not provide the CMLDS (i.e., if only the
13 transport-specific peer delay mechanism is provided), they shall be invoked on domain 0. In other words, if
14 the domainNumber is not 0, portDS.delayMechanism (see Table 14-11 in 14.10.5) must not be P2P. If
15 CMLDS is used, the LocalClock entity for CMLDS and the LocalClock entity for each PTP Instance shall
16 be the same LocalClock.

17 NOTE 1—In the above, the case where the time-aware system implements only one domain implicitly assumes that
18 IEEE 802.1AS is the only PTP profile present on the respective port of the time-aware system, i.e., no other PTP profiles
19 are implemented on that port. If another PTP profile (see IEEE Std 1588) besides IEEE Std 802.1AS is active on the
20 port, the CMLDS provides the mean propagation delay and neighbor rate ratio to all of them.

21 In accordance with IEEE Std 1588-2019, the term *Link Port* refers to a port of the CMLDS. A PTP Port for
22 which portDS.delayMechanism is COMMON_P2P uses the CMLDS provided by the Link Port
23 whose cmlDsLinkPortDS.portIdentity.portNumber (see 14.18.2) is equal to the
24 commonServicesPortDS.cmlDsLinkPortPortNumber (see 14.16.2) for this PTP Port.

25 The value of majorSdoId for the CMLDS shall be 0x2. The value of minorSdoId for the CMLDS shall be
26 0x00. As a result, the value of sdoId for the CMLDS is 0x200.

27 NOTE 2—The above requirements for majorSdoId and minorSdoId are for the CMLDS. The requirements for gPTP
28 domains, including transport-specific peer delay messages, are given in 8.1.

29 NOTE 3—The requirements of this subclause for CMLDS are consistent with the requirements of IEEE Std 1588-2019.

30 If a PTP Port on a physical interface that also supports CMLDS and a Link Port receives a Pdelay_Req
31 message with majorSdoId value of 0x1, minorSdoId value of 0x00, and domainNumber value of 0, the PTP
32 Port shall respond with PTP transport-specific peer delay messages (i.e., the Pdelay_Resp and
33 Pdelay_Resp_Follow_Up corresponding to this Pdelay_Req) using the transport-specific peer-to-peer delay
34 mechanism if domain 0 is enabled. These transport-specific messages have majorSdoId value of 0x1,
35 minorSdoId value of 0x00, and domainNumber value of 0.

36 NOTE 4—The above requirement ensures:

- 37 a) Backward compatibility with time-aware systems that comply with the 2011 version of this standard and
- 38 b) Compatibility with time-aware systems that implement only one domain and invoke the MDPdelayReq and
- 39 MDPdelayResp state machines on domain 0.

40 NOTE 5—In general, a port can receive:

- 41 a) Peer delay messages of the CMLDS (with sdoId of 0x200),
- 42 b) Transport-specific peer delay messages of domain 0 (with sdoId of 0x100), and
- 43 c) If there are other PTP profiles on the neighbor port that use transport-specific peer delay, peer delay messages
- 44 of those profiles (i.e., with an sdoId of neither 0x100 nor 0x200).

1 The port responds to the messages of type a) if it invokes CMLDS, the messages of type b) if it invokes gPTP domain 0,
2 and the messages of type c) if it invokes the respective other PTP profiles.

3 The CMLDS shall be enabled on a Link Port if the value of portDS.delayMechanism (see 14.10.5) is
4 COMMON_P2P for at least one PTP Port that is enabled (i.e., for which portOper and ptpPortEnabled are
5 both TRUE) and corresponds to the same physical port as the Link Port (see 14.1). The value of
6 cmldsLinkPortEnabled is TRUE if the CMLDS is enabled on the Link Port and FALSE if the CMLDS is not
7 enabled on the Link Port.

8 11.2.17.2 Differences between transport-specific peer-to-peer delay mechanism and 9 CMLDS in computations of meanLinkDelay and nrrPdelay

10 The MDPdelayReq state machine (see 11.2.19), and the MDPdelay_Resp state machine (see 11.2.20), are
11 invoked either by the transport-specific peer-to-peer delay mechanism or by CMLDS. The resulting values
12 of meanLinkDelay and nrrPdelay are the same for both transport-specific peer delay and CMLDS; however,
13 the organization of the computations for transport-specific peer delay and CMLDS are different. Some of the
14 computations are done at the Initiator and some of the computations are done at the Responder. It is
15 necessary for the Initiator and the Responder to both perform the transport-specific peer delay computations
16 or both perform the CMLDS computations in order to obtain the correct results for meanLinkDelay and
17 nrrPdelay. The differences are:

- 18 a) Both transport-specific peer delay and CMLDS compute meanLinkDelay averaged over the two
19 directions, and add delayAsymmetry separately when computing upstreamTxTime by the
20 setMDSyncReceiveMDSR() function of the MDSyncReceive state machine. However, in the
21 computation of meanLinkDelay, CMLDS subtracts delayAsymmetry from the correctionField when
22 sending the pdelayReq message at the Initiator (see Figure 11-1) and adds delayAsymmetry back
23 when computing the quantity t_3 in the function computePropTime() (at the Initiator), while
24 transport-specific peer delay does not subtract and add back delayAsymmetry.
- 25 b) Transport-specific peer delay sets the correctionField of a transmitted Pdelay_Req message to 0,
26 while CMLDS sets it to $-\text{delayAsymmetry}$ (see 11.2.19.3.1).
- 27 c) Transport-specific peer delay sets the correctionField of Pdelay_Resp equal to the fractional
28 nanoseconds portion of the pdelayReqEventIngressTimestamp of the corresponding Pdelay_Req,
29 while CMLDS sets the correctionField of Pdelay_Resp equal to minus the fractional nanoseconds
30 portion of the pdelayReqEventIngressTimestamp of the corresponding Pdelay_Req
31 (see 11.2.20.3.1).
- 32 d) Transport-specific peer delay sets the correctionField of Pdelay_Resp_Follow_Up equal to
33 the fractional nanoseconds portion of the pdelayRespEventEgressTimestamp, while CMLDS sets
34 the correctionField of Pdelay_Resp_Follow_Up equal to the sum of the correctionField of
35 the corresponding Pdelay_Req and the fractional nanoseconds portion of the
36 pdelayRespEventEgressTimestamp (see 11.2.20.3.3).
- 37 e) When computing the quantity t_2 for the meanLinkDelay computation [i.e., the quantity D in
38 Equation (11-5), see 11.2.19.3.4], the correctionField of the Pdelay_Resp message, divided by 2^{16} ,
39 is added if transport-specific peer delay is used, while it is subtracted if CMLDS is used (see
40 11.2.19.3.4), and the quantity $-\text{delayAsymmetry}$, divided by 2^{16} , is added if CMLDS is used (i.e.,
41 delayAsymmetry is subtracted in the case of CMLDS).
- 42 f) When computing nrrPdelay, the computation of the correctedResponderEventTimestamp is
43 corrected for delayAsymmetry if, and only if, CMLDS is used. The reason for this correction is that,
44 with CMLDS, delayAsymmetry was subtracted from the Pdelay_Req correctionField, and then the
45 Pdelay_Resp_Follow_Up correctionField was set equal to the sum of the Pdelay_Req
46 correctionField and the fractional nanoseconds portion of the PdelayRespEventEgressTimestamp,
47 while with transport-specific peer delay, the correctionField of Pdelay_Req was set equal to 0 (see
48 11.2.19.3.1, 11.2.19.3.3, and 11.2.20.3.3).

1 The computations in this standard for the transport-specific peer-to-peer delay mechanism are the same as in
2 IEEE Std 802.1AS-2011, for backward compatibility. However, the computations in this standard for
3 CMLDS need to be consistent with IEEE Std 1588-2019 because CMLDS can be used by other PTP
4 profiles, in addition to the PTP profile included in IEEE Std 802.1AS, that might be present in a gPTP node.
5 Therefore, the computations for the transport-specific peer-to-peer delay mechanism and CMLDS are
6 different (i.e., are organized differently), even though they produce the same results.

7 11.2.18 Common Mean Link Delay Service (CMLDS) global variables

8 **11.2.18.1 cmlDsLinkPortEnabled:** A per-Link-Port Boolean that is TRUE if both the value of
9 portDS.delayMechanism is Common_P2P and the value of portDS.ptpPortEnabled is TRUE, for at least one
10 PTP Port that uses the CMLDS that is invoked on the Link Port; otherwise, the value is FALSE.

1 11.2.19 MDPdelayReq state machine

2 11.2.19.1 General

3 This state machine is either invoked as part of the CCMLDS or by the transport-specific peer-to-peer delay
4 mechanism. There is one instance of this state machine for all the domains (per port). As a result, there also
5 is one instance of each of the state machine variables of 11.2.19.2, state machine functions of 11.2.19.3, and
6 relevant global variables of 10.2.5, 11.2.13, and 11.2.18 for all the domains (per port). None of the variables
7 used or functions invoked in this state machine are specific to a single domain. However, the single instances
8 of all of these objects or entities are accessible to all the domains.

9 NOTE—This state machine uses the variable asCapableAcrossDomains (see 11.2.13.13). When only one domain is active,
10 asCapableAcrossDomains is equivalent to the variable asCapable.

11 11.2.19.2 State machine variables

12 The following variables are used in the state diagram in Figure 11-9 (in 11.2.19.4):

13 **11.2.19.2.1 pdelayIntervalTimer:** A variable used to save the time at which the Pdelay_Req interval timer
14 is started. A Pdelay_Req message is sent when this timer expires. The data type for pdelayIntervalTimer is
15 UScaledNs.

16 **11.2.19.2.2 rcvdPdelayResp:** A Boolean variable that notifies the current state machine when a
17 Pdelay_Resp message is received. This variable is reset by the current state machine.

18 **11.2.19.2.3 rcvdPdelayRespPtr:** A pointer to a structure whose members contain the values of the fields of
19 the Pdelay_Resp message whose receipt is indicated by rcvdPdelayResp (see 11.2.19.2.2).

20 **11.2.19.2.4 rcvdPdelayRespFollowUp:** A Boolean variable that notifies the current state machine when a
21 Pdelay_Resp_Follow_Up message is received. This variable is reset by the current state machine.

22 **11.2.19.2.5 rcvdPdelayRespFollowUpPtr:** A pointer to a structure whose members contain the values of
23 the fields of the Pdelay_Resp_Follow_Up message whose receipt is indicated by rcvdPdelayRespFollowUp
24 (see 11.2.19.2.4).

25 **11.2.19.2.6 txPdelayReqPtr:** A pointer to a structure whose members contain the values of the fields of a
26 Pdelay_Req message to be transmitted.

27 **11.2.19.2.7 rcvdMDTimestampReceiveMDPReq:** A Boolean variable that notifies the current state
28 machine when the pdelayReqEventEgressTimestamp (see 11.3.2.1) for a transmitted Pdelay_Req message is
29 received. This variable is reset by the current state machine.

30 **11.2.19.2.8 pdelayReqSequenceId:** A variable that holds the sequenceId for the next Pdelay_Req message
31 to be transmitted by this MD entity. The data type for pdelayReqSequenceId is UInteger16.

32 **11.2.19.2.9 lostResponses:** A count of the number of consecutive Pdelay_Req messages sent by the port, for
33 which Pdelay_Resp and/or Pdelay_Resp_Follow_Up messages are not received. The data type for
34 lostResponses is UInteger16.

35 **11.2.19.2.10 nrrPdelayValid:** A Boolean variable that indicates whether the function
36 computePdelayRateRatio() (see 11.2.19.3.3) successfully computed nrrPdelay (see 11.2.13.14). The default
37 value of nrrPdelayValid is TRUE.

1 **11.2.19.2.11 detectedFaults:** A count of the number of consecutive faults (see 11.5.4 for the definition of a
2 fault).

3 **11.2.19.2.12 portEnabled0:** A Boolean variable whose value is equal to ptpPortEnabled (see 10.2.5.13) if
4 this state machine is invoked by the transport-specific peer-to-peer delay mechanism and is equal to
5 cmlDsLinkPortEnabled (see 11.2.18.1) if this state machine is invoked by CMLDS.

6 **11.2.19.2.13 s:** A variable whose value is +1 if this state machine is invoked by the transport-specific peer-
7 to-peer delay mechanism and –1 if this state machine is invoked by CMLDS. The data type for s is Integer8.

8 **11.2.19.3 State machine functions**

9 **11.2.19.3.1 setPdelayReq():** Creates a structure containing the parameters (see 11.4) of a Pdelay_Req
10 message to be transmitted, and returns a pointer, txPdelayReqPtr (see 11.2.19.2.6), to this structure. The
11 parameters are set as follows:

- 12 a) sourcePortIdentity is set equal to the port identity of the port corresponding to this MD entity
13 (see 8.5.2).
- 14 b) sequenceId is set equal to pdelayReqSequenceId (see 11.2.19.2.8).
- 15 c) correctionField is set to
 - 16 1) 0 if this state machine is invoked by the transport-specific peer-to-peer delay mechanism, and
 - 17 2) –delayAsymmetry (i.e., the negative of delayAsymmetry) if this state machine is invoked by
18 the CMLDS.
- 19 d) The remaining parameters are set as specified in 11.4.2 and 11.4.5.

20 **11.2.19.3.2 txPdelayReq(txPdelayReqPtr):** Transmits a Pdelay_Req message from the MD entity,
21 containing the parameters in the structure pointed to by txPdelayReqPtr (see 11.2.19.2.6).

22 **11.2.19.3.3 computePdelayRateRatio():** Computes nrrPdelay (see 11.2.13.14) using the following
23 information conveyed by successive Pdelay_Resp and Pdelay_Resp_Follow_Up messages:

- 24 a) The pdelayRespEventIngressTimestamp (see 11.3.2.1) values for the respective Pdelay_Resp
25 messages
- 26 b) The correctedResponderEventTimestamp values, whose data type is UScaledNs, obtained by adding
27 the following fields of the received Pdelay_Resp_Follow_Up message:
 - 28 1) The seconds field of the responseOriginTimestamp field, multiplied by 10^9
 - 29 2) The nanoseconds field of the responseOriginTimestamp field
 - 30 3) The correctionField, divided by 2^{16}
 - 31 4) delayAsymmetry (see 10.2.5.9), if and only if this state machine is invoked by CMLDS

32 NOTE 1—If delayAsymmetry does not change during the time interval over which nrrPdelay is computed, it is not
33 necessary to add it if this state machine is invoked by CMLDS because in that case it will be canceled when computing
34 the difference between earlier and later correctedResponderEventTimestamps.

35 Any scheme that uses the preceding information, along with any other information conveyed by the
36 successive Pdelay_Resp and Pdelay_Resp_Follow_Up messages, to compute nrrPdelay is acceptable as
37 long as the performance requirements specified in B.2.4 are met. If nrrPdelay is successfully computed, the
38 Boolean nrrPdelayValid (see 11.2.19.2.10) is set to TRUE. If nrrPdelay is not successfully computed (e.g., if
39 the MD entity has not yet exchanged a sufficient number of peer delay messages with its peer), the Boolean
40 nrrPdelayValid is set to FALSE.

1 NOTE 2—As one example, $nrrPdelay$ can be estimated as the ratio of the elapsed time of the LocalClock entity of the
 2 time-aware system at the other end of the PTP Link attached to this port, to the elapsed time of the LocalClock entity of the
 3 this time-aware system. This ratio can be computed for the time interval between a set of received Pdelay_Resp and
 4 Pdelay_Resp_Follow_Up messages and a second set of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages
 5 some number of Pdelay_Req message transmission intervals later, i.e.,

$$6 \quad \frac{\text{correctedResponderEventTimestamp}_N - \text{correctedResponderEventTimestamp}_0}{\text{pdelayRespEventIngressTimestamp}_N - \text{pdelayRespEventIngressTimestamp}_0}$$

7 where N is the number of Pdelay_Req message transmission intervals separating the first set of received Pdelay_Resp
 8 and Pdelay_Resp_Follow_Up messages and the second set, and the successive sets of received Pdelay_Resp and
 9 Pdelay_Resp_Follow_Up messages are indexed from 0 to N with the first set indexed 0.

10 NOTE 3—This function must account for non-receipt of Pdelay_Resp and/or Pdelay_Resp_Follow_Up for a
 11 Pdelay_Req message and also for receipt of multiple Pdelay_Resp messages within one Pdelay_Req message
 12 transmission interval.

13 **11.2.19.3.4 computePropTime():** Computes the mean propagation delay on the PTP Link attached to this
 14 MD entity, D , and returns this value. D is given by Equation (11-5).

$$15 \quad D = \frac{r \cdot (t_4 - t_1) - (t_3 - t_2)}{2} \quad (11-5)$$

16 where

- 17 t_4 is pdelayRespEventIngressTimestamp (see 11.3.2.1) for the Pdelay_Resp message received in
 18 response to the Pdelay_Req message sent by the MD entity, in nanoseconds; the
 19 pdelayRespEventIngressTimestamp is equal to the timestamp value measured relative to the
 20 timestamp measurement plane, minus any ingressLatency (see 8.4.3)
- 21 t_1 is pdelayReqEventEgressTimestamp (see 11.3.2.1) for the Pdelay_Req message sent by the
 22 P2PPort entity, in nanoseconds
- 23 t_2 is the sum of (1) the ns field of the requestReceiptTimestamp, (2) the seconds field of the
 24 requestReceiptTimestamp multiplied by 10^9 , (3) the correctionField multiplied by s
 25 (see 11.2.19.2.13) and then divided by 2^{16} (i.e., the correctionField is expressed in nanoseconds
 26 plus fractional nanoseconds), of the Pdelay_Resp message received in response to the Pdelay_Req
 27 message sent by the MD entity, and (4) $-\text{delayAsymmetry}$ divided by 2^{16} if this state machine is
 28 invoked by CMLDS (i.e., delayAsymmetry is subtracted in the case of CMLDS)
- 29 t_3 is the sum of (1) the ns field of the responseOriginTimestamp, (2) the seconds field of the
 30 responseOriginTimestamp multiplied by 10^9 , and (3) the correctionField divided by 2^{16} (i.e., the
 31 correctionField is expressed in nanoseconds plus fractional nanoseconds), of the
 32 Pdelay_Resp_Follow_Up message received in response to the Pdelay_Req message sent by the
 33 MD entity
- 34 r is the current value of $nrrPdelay$ for this MD entity (see 11.2.13.14)

35 When CMLDS is used, Equation (11-5) can be rewritten as shown in Equation (11-6) (see NOTE 3 below)
 36 using the definitions of t_2 and t_3 above for the CMLDS case.

37 Propagation delay averaging may be performed, as described in 11.1.2 by Equation (11-2), Equation (11-3),
 38 and Equation (11-4). In this case, the successive values of propagation delay computed using Equation (11-
 39 5) are input to either Equation (11-2) or Equation (11-4), and the computed average propagation delay is
 40 returned by this function.

41 NOTE 1—Equation (11-5) defines D as the mean propagation delay relative to the time base of the time-aware system at
 42 the other end of the attached PTP Link. It is divided by $nrrPdelay$ (see 11.2.13.14) to convert it to the time base of the
 43 current time-aware system when subtracting from syncEventIngressTimestamp to compute upstreamTxTime [see item i)
 44 in 11.2.14.2.1].

1 NOTE 2—The difference between mean propagation delay relative to the Grandmaster Clock time base and relative to
 2 the time base of the time-aware system at the other end of the attached PTP Link is usually negligible. To see this, note
 3 that the former can be obtained from the latter by multiplying the latter by the ratio of the Grandmaster Clock frequency
 4 to the frequency of the LocalClock entity of the time-aware system at the other end of the PTP Link attached to this port.
 5 This ratio differs from 1 by 200 ppm or less. For example, for a worst-case frequency offset of the LocalClock entity of
 6 the time-aware system at the other end of the PTP Link, relative to the Grandmaster Clock, of 200 ppm, and a measured
 7 propagation time of 100 ns, the difference in D relative to the two time bases is 20 ps.

8 NOTE 3—In IEEE Std 1588-2019, the computation of meanLinkDelay is organized differently from the organization
 9 used for transport-specific peer delay in the present standard. Using the definitions of t_2 and t_3 above, Equation (11-5)
 10 can be rewritten as shown in Equation (11-6).

$$11 \quad D = [r \cdot (t_4 - t_1) - (\text{responseOriginTimestamp} - \text{requestReceiptTimestamp}) + s \times (\text{correctionField of Pdelay_Resp}) \\ - (\text{correctionField of Pdelay_Resp_Follow_Up}) - \text{delayAsymmetry}] / 2 \quad (11-6)$$

13 where each term is expressed in units of nanoseconds as described in the definitions of t_1 , t_2 , t_3 , and t_4 above.
 14 In IEEE Std 1588-2019, the fractional nanoseconds portion of t_2 is subtracted from the correctionField of
 15 Pdelay_Resp, rather than added as in the present standard for transport-specific peer delay [see 11.2.20.3.1 d) 1)];
 16 however, the correctionField of Pdelay_Resp is then subtracted in Equation (11-6) rather than added [in Equation (11-6)
 17 for transport-specific peer delay, where $s=1$], and the two minus signs result in the same sign for the fractional
 18 nanoseconds portion of t_2 and the seconds and nanoseconds portion of t_2 (requestReceiptTimestamp). The computations
 19 of D in this standard and IEEE Std 1588-2019 are mathematically equivalent. The organization of the computation with
 20 CMLDS needs to be used in the present standard for interoperability with IEEE Std 1588-2019 (see 11.2.17.2). This
 21 organization in the present standard, for the case where CMLDS is used, is consistent with the organization in IEEE Std
 22 1588-2019.

23 11.2.19.4 State diagram

24 The MDPdelayReq state machine shall implement the function specified by the state diagram in Figure 11-9,
 25 the local variables specified in 11.2.19.2, the functions specified in 11.2.19.3, the messages specified in 11.4,
 26 and the relevant global variables and functions specified in 11.2.13 and 11.2.18 and 10.2.4 through 10.2.6.
 27 This state machine is responsible for the following:

- 28 a) Sending Pdelay_Req messages and restarting the pdelayIntervalTimer.
- 29 b) Detecting that the peer mechanism is running.
- 30 c) Detecting if Pdelay_Resp and/or Pdelay_Resp_Follow_Up messages corresponding to a
 31 Pdelay_Req message sent are not received.
- 32 d) Detecting whether more than one Pdelay_Resp is received within one Pdelay_Req message
 33 transmission interval (see 11.5.2.2).
- 34 e) Computing propagation time on the attached PTP Link when Pdelay_Resp and
 35 Pdelay_Resp_Follow_Up messages are received.
- 36 f) Computing the ratio of the frequency of the LocalClock entity of the time-aware system at the other
 37 end of the attached PTP Link to the frequency of the LocalClock entity of the current time-aware
 38 system.

39 NOTE 1—The ratio of the frequency of the LocalClock entity of the time-aware system at the other end of the attached
 40 PTP Link to the frequency of the LocalClock entity of the current time-aware system, nrrPdelay, retains its most recent
 41 value when a Pdelay_Resp and/or Pdelay_Resp_Follow_Up message is lost.

42 NOTE 2—Normally, Pdelay_Resp should be received within a time interval after sending Pdelay_Req that is less than or
 43 equal to the Pdelay turnaround time plus twice the mean propagation delay. However, while receiving Pdelay_Resp after
 44 a time interval that is longer than this can result in worse time-synchronization performance (see 11.1.2 and B.2.3), the
 45 peer delay protocol will still operate. It is expected that a peer delay initiator can receive and process Pdelay_Resp and
 46 Pdelay_Resp_Follow_Up messages within a time interval after sending Pdelay_Req that is as large as the current Pdelay
 47 Req message transmission interval (see 11.5.2.2).

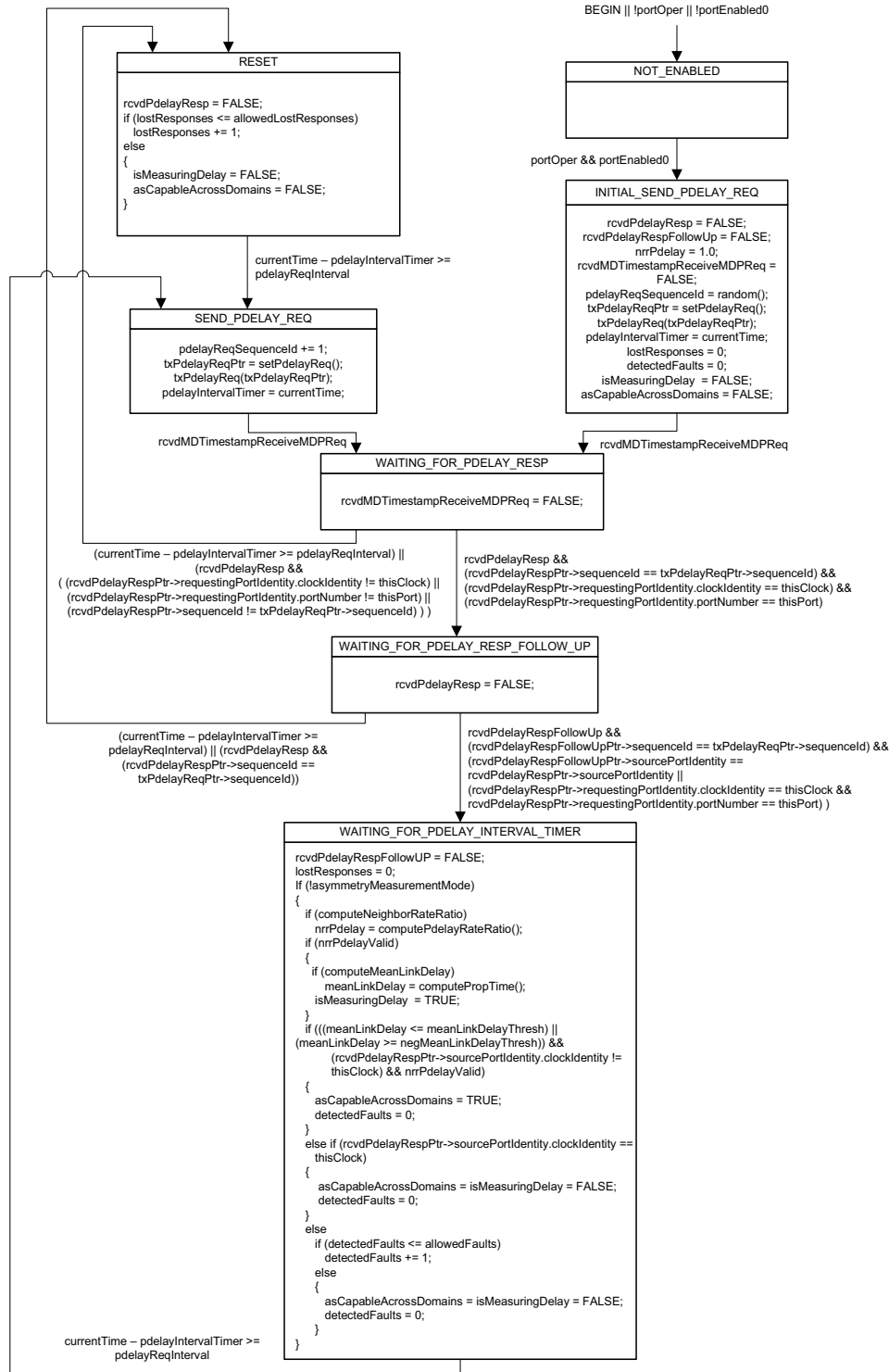


Figure 11-9—MDPdelayReq state machine

1 11.2.20 MDPdelayResp state machine

2 11.2.20.1 General

3 This state machine is either invoked as part of the CMLDS or by the transport-specific peer-to-peer delay
4 mechanism. There is one instance of this state machine for all the domains (per port). As a result, there also
5 is one instance of each of the state machine variables of 11.2.20.2, state machine functions of 11.2.20.3, and
6 relevant global variables of 10.2.5, 11.2.13, and 11.2.18 for all the domains (per port). None of the variables
7 used or functions invoked in this state machine are specific to a single domain. However, the single instances
8 of all of these objects or entities are accessible to all the domains.

9 11.2.20.2 State machine variables

10 The following variables are used in the state diagram in Figure 11-10 (in 11.2.20.4):

11 **11.2.20.2.1 rcvdPdelayReq:** A Boolean variable that notifies the current state machine when a Pdelay_Req
12 message is received. This variable is reset by the current state machine.

13 **11.2.20.2.2 rcvdMDTimestampReceiveMDPResp:** A Boolean variable that notifies the current state
14 machine when the pdelayRespEventEgressTimestamp (see 11.3.2.1) for a transmitted Pdelay_Resp message
15 is received. This variable is reset by the current state machine.

16 **11.2.20.2.3 txPdelayRespPtr:** A pointer to a structure whose members contain the values of the fields of a
17 Pdelay_Resp message to be transmitted.

18 **11.2.20.2.4 txPdelayRespFollowUpPtr:** A pointer to a structure whose members contain the values of the
19 fields of a Pdelay_Resp_Follow_Up message to be transmitted.

20 **11.2.20.2.5 portEnabled1:** A Boolean variable whose value is equal to ptpPortEnabled (see 10.2.5.13) if
21 this state machine is invoked by the transport-specific peer-to-peer delay mechanism and is equal to
22 cmlDsLinkPortEnabled (see 11.2.18.1) if this state machine is invoked by the CMLDS.

23 11.2.20.3 State machine functions

24 **11.2.20.3.1 setPdelayResp():** Creates a structure containing the parameters (see 11.4) of a Pdelay_Resp
25 message to be transmitted, and returns a pointer, txPdelayRespPtr (see 11.2.20.2.3), to this structure. The
26 parameters are set as follows:

- 27 a) sourcePortIdentity is set equal to the port identity of the port corresponding to this MD entity
28 (see 8.5.2).
- 29 b) sequenceId is set equal to the sequenceId field of the corresponding Pdelay_Req message.
- 30 c) requestReceiptTimestamp is set equal to the pdelayReqEventIngressTimestamp (see 11.3.2) of the
31 corresponding Pdelay_Req message, with any fractional nanoseconds portion truncated.
- 32 d) correctionField is set equal to the following:
 - 33 1) The fractional nanoseconds portion of the pdelayReqEventIngressTimestamp of the
34 corresponding Pdelay_Req message if this state machine is invoked by the transport-specific
35 peer-to-peer delay mechanism and
 - 36 2) Minus the fractional nanoseconds portion of the pdelayReqEventIngressTimestamp of the
37 corresponding Pdelay_Req message if this state machine is invoked by CMLDS.
- 38 e) requestingPortIdentity is set equal to the sourcePortIdentity field of the corresponding Pdelay_Req
39 message.
- 40 f) The remaining parameters are set as specified in 11.4.2 and 11.4.6.

1 **11.2.20.3.2 txPdelayResp(txPdelayRespPtr):** Transmits a Pdelay_Resp message from the MD entity,
 2 containing the parameters in the structure pointed to by txPdelayRespPtr (see 11.2.20.2.3).

3 **11.2.20.3.3 setPdelayRespFollowUp():** Creates a structure containing the parameters (see 11.4) of a
 4 Pdelay_Resp_Follow_Up message to be transmitted, and returns a pointer, txPdelayRespFollowUpPtr
 5 (see 11.2.20.2.4), to this structure. The parameters are set as follows:

- 6 a) sourcePortIdentity is set equal to the port identity of the port corresponding to this MD entity
 7 (see 8.5.2).
- 8 b) sequenceId is set equal to the sequenceId field of the corresponding Pdelay_Req message.
- 9 c) responseOriginTimestamp is set equal to the pdelayRespEventEgressTimestamp (see 11.3.2) of the
 10 corresponding Pdelay_Resp message, with any fractional nanoseconds truncated.
- 11 d) correctionField is set equal to the following:
 - 12 1) The fractional nanoseconds portion of the pdelayRespEventEgressTimestamp of the
 13 corresponding Pdelay_Resp message if this state machine is invoked by the transport-specific
 14 peer-to-peer delay mechanism and
 - 15 2) The sum of the correctionField of the corresponding Pdelay_Req message and the fractional
 16 nanoseconds portion of the pdelayRespEventEgressTimestamp of the corresponding
 17 Pdelay_Resp message if this state machine is invoked by CMLDS,
- 18 e) requestingPortIdentity is set equal to the sourcePortIdentity field of the corresponding Pdelay_Req
 19 message.
- 20 f) The remaining parameters are set as specified in 11.4.2 and 11.4.6.

21 **11.2.20.3.4 txPdelayRespFollowUp(txPdelayRespFollowUpPtr):** Transmits a Pdelay_Resp_Follow_Up
 22 message from the P2PPort entity containing the parameters in the structure pointed to by
 23 txPdelayRespFollowUpPtr (see 11.2.20.2.4).

24 **11.2.20.4 State diagram**

25 The MDPdelayResp state machine shall implement the function specified by the state diagram in
 26 Figure 11-10, the local variables specified in 11.2.20.2, the functions specified in 11.2.20.3, the messages
 27 specified in 11.4, and the relevant global variables and functions specified in 10.2.4 through 10.2.6, 11.2.13,
 28 and 11.2.18. This state machine is responsible for responding to Pdelay_Req messages, received from the MD
 29 entity at the other end of the attached PTP Link, with Pdelay_Resp and Pdelay_Resp_Follow_Up messages.

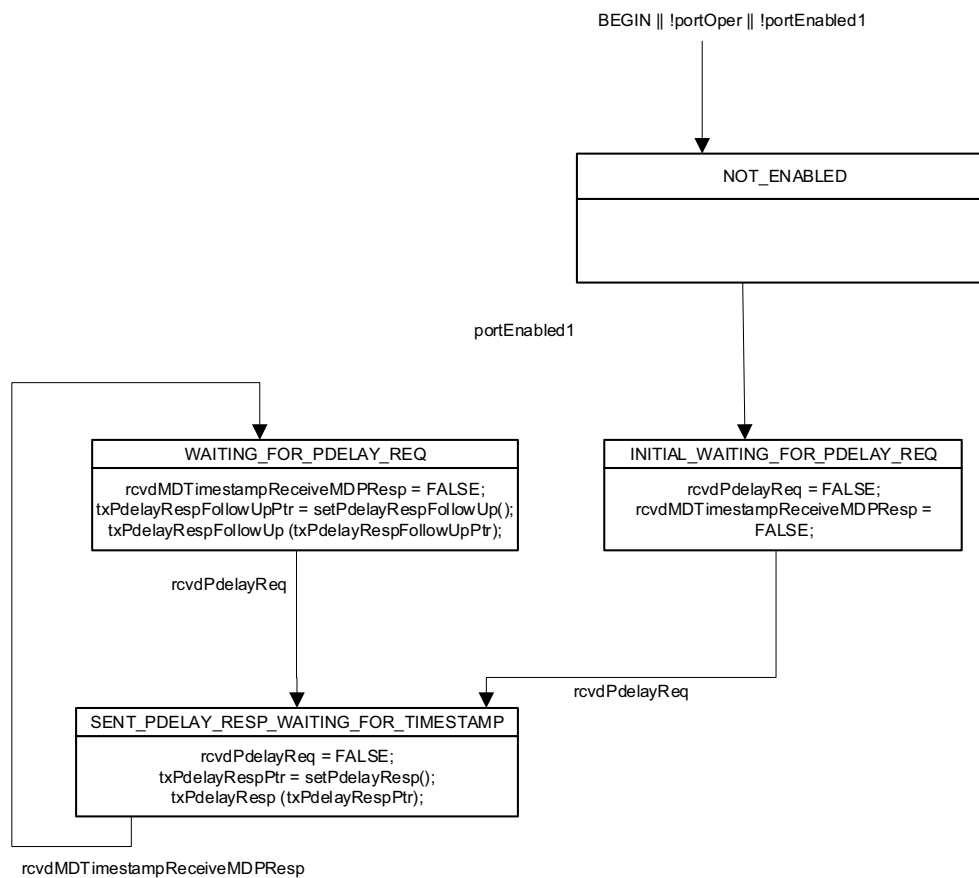


Figure 11-10—MDPdelayResp state machine

1 11.2.21 LinkDelayIntervalSetting state machine

2 11.2.21.1 General

3 This state machine is either invoked as part of the CMLDS or by the transport-specific peer-to-peer delay
4 mechanism. There is one instance of this state machine per port, for the Common Service of the time-aware
5 system.

6 11.2.21.2 State machine variables

7 The following variables are used in the state diagram in Figure 11-11 (in 11.2.21.4):

8 **11.2.21.2.1 rcvdSignalingMsg1:** A Boolean variable that notifies the current state machine when a
9 Signaling message that contains a Message Interval Request TLV (see 10.6.4.3) is received. This variable is
10 reset by the current state machine.

11 **11.2.21.2.2 rcvdSignalingPtrLDIS:** A pointer to a structure whose members contain the values of the fields
12 of the received Signaling message that contains a Message Interval Request TLV (see 10.6.4.3).

13 **11.2.21.2.3 portEnabled3:** A Boolean variable whose value is equal to ptpPortEnabled (see 10.2.5.13) if
14 this state machine is invoked by the transport-specific peer-to-peer delay mechanism and is equal to
15 cmlDsLinkPortEnabled (see 11.2.18.1) if this state machine is invoked by the CMLDS.

16 **11.2.21.2.4 logSupportedPdelayReqIntervalMax:** The maximum supported logarithm to base 2 of the
17 Pdelay_Req interval. The data type for logSupportedPdelayReqIntervalMax is Integer8.

18 **11.2.21.2.5 logSupportedClosestLongerPdelayReqInterval:** The logarithm to base 2 of the Pdelay_Req
19 interval, such that logSupportedClosestLongerPdelayReqInterval > logRequestedPdelayReqInterval, that is
20 numerically closest to logRequestedPdelayReqInterval, where logRequestedPdelayReqInterval is the
21 argument of the function computeLogPdelayReqInterval() (see 11.2.21.3.2). The data type for
22 logSupportedClosestLongerPdelayReqInterval is Integer8.

23 **11.2.21.2.6 computedLogPdelayReqInterval:** A variable used to hold the result of the function
24 computeLogPdelayReqInterval(). The data type for computedLogPdelayReqInterval is Integer8.

25 **11.2.21.2.7 TEMP:** A temporary variable used to reduce clutter in the state diagram (see Figure 11-11). The
26 data type for TEMP is Integer16.

27 11.2.21.3 State machine functions

28 **11.2.21.3.1 isSupportedLogPdelayReqInterval (logPdelayReqInterval):** A Boolean function that returns
29 TRUE if the Pdelay_Req interval given by the argument logPdelayReqInterval is supported by the PTP Port
30 and FALSE if the Pdelay_Req interval is not supported by the PTP Port. The argument
31 logPdelayReqInterval has the same data type and format as the field logLinkDelayInterval of the message
32 interval request TLV (see 10.6.4.3.6).

33 **11.2.21.3.2 computeLogPdelayReqInterval (logRequestedPdelayReqInterval):** An Integer8 function
34 that computes and returns the logPdelayReqInterval, based on the logRequestedPdelayReqInterval. This
35 function is defined as indicated below. It is defined here so that the detailed code that it invokes does not
36 need to be placed into the state machine diagram.

```
37 Integer8 computeLogPdelayReqInterval (logRequestedPdelayReqInterval)
38 Integer8 logRequestedPdelayReqInterval;
39 {
```

```

1      Integer8 logSupportedPdelayReqIntervalMax,
2          logSupportedClosestLongerPdelayReqInterval;
3      if (isSupportedLogPdelayReqInterval (logRequestedPdelayReqInterval))
4          // The requested Pdelay_Req Interval is supported and returned
5          return (logRequestedPdelayReqInterval)
6      else
7          {
8              if (logRequestedPdelayReqInterval > logSupportedPdelayReqIntervalMax)
9                  // Return the largest supported logPdelayReqInterval, even if smaller than
10 the requested interval
11                  return (logSupportedPdelayReqIntervalMax);
12              else
13                  // Return the smallest supported logPdelayReqInterval that is still larger
14 than
15                  // the requested interval.
16                  return (logSupportedClosestLongerPdelayReqInterval);
17          }
18 }

```

19 11.2.21.4 State diagram

20 The LinkDelayIntervalSetting state machine shall implement the function specified by the state diagram in
21 Figure 11-11, the local variables specified in 11.2.21.2, the functions specified in 11.2.21.3, the messages
22 specified in 10.6 and 11.4, the relevant global variables specified in 10.2.5, 11.2.13, and 11.2.18, the relevant
23 managed objects specified in 14.10 and 14.14, and the relevant timing attributes specified in 10.7 and 11.5.
24 This state machine is responsible for setting the global variables that give the duration of the mean interval
25 between successive Pdelay_Req messages and also the global variables that control whether
26 meanLinkDelay and nrrPdelay are computed, both at initialization and in response to the receipt of a
27 Signaling message that contains a Message Interval Request TLV (see 10.6.4.3).

28 NOTE—A Signaling message received by this state machine that carries the message interval request TLV
29 (see 10.6.4.3) is ignored if multiple profiles are present and the Signaling message is directed to the CMLDS (i.e., if the
30 value of sdoId of the Signaling message is 0x200).

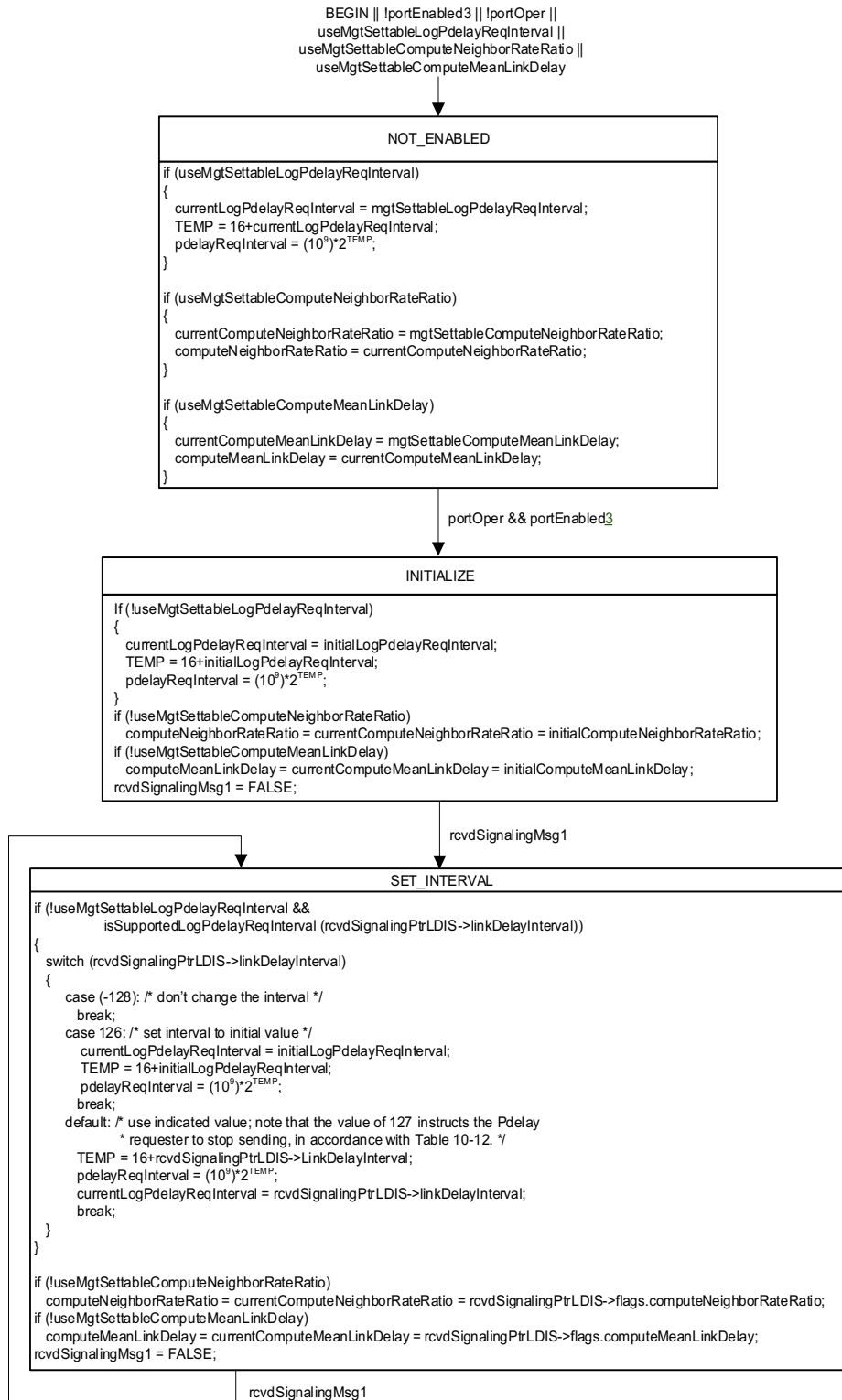


Figure 11-11—LinkDelayIntervalSetting state machine

1 11.3 Message attributes

2 11.3.1 General

3 This subclause describes attributes of the Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and
4 Pdelay_Resp_Follow_Up messages that are not described in 8.4.2.

5 11.3.2 Message types contained in each message class

6 11.3.2.1 Event message class

7 The event message class contains the following message types:

- 8 a) Sync: A Sync message contains time-synchronization information that originates at a
9 ClockTimeTransmitter entity. The appearance of a Sync message at the reference plane of the PTP
10 Port corresponding to an MD entity is an event to which the LocalClock assigns a timestamp, the
11 syncEventIngressTimestamp or syncEventEgressTimestamp, based on the time of the LocalClock.
12 The syncEventIngressTimestamp and syncEventEgressTimestamp are measured relative to the
13 timestamp measurement plane; the MD entity corrects them for ingress and egress latencies,
14 respectively (see 8.4.3). The Sync message is followed by a Follow_Up message containing
15 synchronization information that is based in part on the sum of the syncEventEgressTimestamp and
16 any egressLatency (see 8.4.3).
- 17 b) Pdelay_Req: A Pdelay_Req message is transmitted by an MD entity to another MD entity as part of
18 the peer-to-peer delay mechanism (see 11.2.19 and Figure 11-9) to determine the delay on the PTP
19 Link between them. The appearance of a Pdelay_Req message at the reference plane of the port of
20 an MD entity is an event to which the LocalClock assigns a timestamp, the
21 pdelayReqEventIngressTimestamp or pdelayReqEventEgressTimestamp, based on the time of the
22 LocalClock. The pdelayReqEventIngressTimestamp and pdelayReqEventEgressTimestamp are
23 measured relative to the timestamp measurement plane; the MD entity corrects them for ingress and
24 egress latencies, respectively (see 8.4.3).
- 25 c) Pdelay_Resp: A Pdelay_Resp message is transmitted by an MD entity to another MD entity in
26 response to the receipt of a Pdelay_Req message. The Pdelay_Resp message contains the
27 pdelayReqEventIngressTimestamp of the Pdelay_Req message to which it is transmitted in
28 response. The appearance of a Pdelay_Resp message at the reference plane of the port of an MD
29 entity is an event to which the LocalClock assigns a timestamp, the
30 pdelayRespEventIngressTimestamp or pdelayRespEventEgressTimestamp, based on the time of the
31 LocalClock. The pdelayRespEventIngressTimestamp and pdelayRespEventEgressTimestamp are
32 measured relative to the timestamp measurement plane; the MD entity corrects them for ingress and
33 egress latencies, respectively (see 8.4.3). The Pdelay_Resp message is followed by a
34 Pdelay_Resp_Follow_Up message containing the sum of the pdelayRespEventEgressTimestamp
35 and any egressLatency (see 8.4.3).

36 Event messages shall be assigned the timestamps previously defined, in accordance with 8.4.3.

37 11.3.2.2 General message class

38 The general message class contains the following message types:

- 39 a) Follow_Up: A Follow_Up message communicates the value of the syncEventEgressTimestamp for
40 the associated Sync message.
- 41 b) Pdelay_Resp_Follow_Up: A Pdelay_Resp_Follow_Up message communicates the value of the
42 PdelayRespEventEgressTimestamp for the associated Pdelay_Resp message.

1 General messages are not required to be timestamped.

2 **11.3.3 VLAN tag**

3 A frame that carries an IEEE 802.1AS message shall not have a VLAN tag nor a priority tag (see 3.190 of
4 IEEE Std 802.1Q-2022).

5 **11.3.4 Addresses**

6 The destination address of Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up
7 messages shall be the reserved multicast address given in Table 11-3.

**Table 11-3—Destination address for Sync, Follow_Up, Pdelay_Req, Pdelay_Resp,
and Pdelay_Resp_Follow_Up messages**

Destination address
01-80-C2-00-00-0E
NOTE—This address is taken from Table 8-1, Table 8-2, and Table 8-3 of IEEE Std 802.1Q-2020.

8 All Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall use the
9 MAC address of the respective egress physical port as the source address.

10 **11.3.5 EtherType**

11 The EtherType of Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages
12 shall be the EtherType given in Table 11-4.

**Table 11-4—EtherType for Sync, Follow_Up, Pdelay_Req, Pdelay_Resp,
and Pdelay_Resp_Follow_Up messages**

EtherType
88-F7

13 **11.3.6 Subtype**

14 The subtype for the Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages
15 is indicated by the majorSdoId field (see 10.6.2.2.1).

16 **11.3.7 Source port identity**

17 The Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages each contain a
18 sourcePortIdentity field (see 11.4.2.7) that identifies the egress port (see 8.5) on which the respective
19 message is sent.

1 11.3.8 Sequence number

2 Each MD entity shall maintain a separate sequenceId pool for each of the message types Sync and
3 Pdelay_Req.

4 Each Sync and Pdelay_Req message contains a sequenceId field (see 11.4.2.8), which carries the message
5 sequence number. The sequenceId of a Sync message shall be one greater than the sequenceId of the
6 previous Sync message sent by the transmitting port, subject to the constraints of the rollover of the
7 UInteger16 data type used for the sequenceId field. The sequenceId of a Pdelay_Req message shall be one
8 greater than the sequenceId of the previous Pdelay_Req message sent by the transmitting port, subject to the
9 constraints of the rollover of the UInteger16 data type used for the sequenceId field.

10 Separate pools of sequenceId are not maintained for the following message types:

- 11 a) Pdelay_Resp
- 12 b) Follow_Up
- 13 c) Pdelay_Resp_Follow_Up

14 For these exceptions, the sequenceId value is specified in Table 11-7 (in 11.4.2.8).

15 11.3.9 Event message timestamp point

16 The message timestamp point for a PTP event message shall be the beginning of the first symbol following
17 the start of frame delimiter.

18 11.4 Message formats

19 11.4.1 General

20 The PTP messages Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up shall each
21 have a header, body, and if present, a suffix that contains one or more TLVs (see 10.6.2, 11.4.3, 11.4.4,
22 11.4.5, 11.4.6, and 11.4.7 of this standard and Clause 14 of IEEE Std 1588-2019). Reserved fields shall be
23 transmitted with all bits of the field 0 and ignored by the receiver, unless otherwise specified. The data type
24 of the field shall be the type indicated in brackets in the title of each subclause.

25 Subclause 11.4.4.3 defines the Follow_Up information TLV, which is carried by the Follow_Up message if
26 the corresponding Sync message is two-step (i.e., twoStepFlag of the Sync message is TRUE;
27 see 10.6.2.2.8) and by the Sync message if the message is one-step (i.e., twoStepFlag is FALSE). The
28 Follow_Up information TLV is the first TLV of the Follow_Up message or Sync message. The requirements
29 for the parsing and forwarding of TLVs are given in 10.6.1.

30 NOTE—The standard Ethernet header and FCS (18 bytes total) must be added to each message.

31 11.4.2 Header

32 11.4.2.1 General

33 The common header for the PTP messages Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and
34 Pdelay_Resp_Follow_Up shall be as specified in 10.6.2, except as noted in the following subclauses.

1 11.4.2.2 messageType (Enumeration4)

2 The value indicates the type of the message, as defined in Table 11-5.

Table 11-5—Values for messageType field

Message type	Message class	Value
Sync	Event	0x0
Pdelay_Req	Event	0x2
Pdelay_Resp	Event	0x3
Follow_Up	General	0x8
Pdelay_Resp_Follow_Up	General	0xA
NOTE—Other values for the messageType field, except for 0xB that is used for the Announce message and 0xC that is used for the Signaling message (see 10.6.2.2.2), are not used in this standard.		

3 The most significant bit of the message ID field divides this field in half between event and general
4 messages, i.e., it is 0 for event messages and 1 for general messages.

5 NOTE—The reserved nibble immediately following messageType is reserved for future expansion of the messageType
6 field.

7 11.4.2.3 messageLength (UInteger16)

8 The value is the total number of octets that form the PTP message. The counted octets start with and include
9 the first octet of the header and terminate with and include the last octet of the last TLV or, if there are no
10 TLVs, with the last octet of the message as defined in this subclause.

11 NOTE—See 10.6.2.2.5 for an example.

12 11.4.2.4 domainNumber (UInteger8)

13 The domainNumber for Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be 0. The
14 domainNumber for all other PTP messages is as specified in 10.6.2.2.6.

15 11.4.2.5 flags (Octet2)

16 The value of the bits of the array are defined in Table 10-9.

17 11.4.2.6 correctionField (Integer64)

18 The correctionField is the value of the correction as specified in Table 11-6, measured in nanoseconds and
19 multiplied by 2^{16} . For example, 2.5 ns is represented as 0x00000000000028000.

20 A value of one in all bits, except the most significant, of the field, indicates that the correction is too big to
21 be represented.

Table 11-6—Value of correctionField

Message type	Value
Follow_Up Sync (sent by a one-step PTP Port; see 11.1.3 and 11.2.13.10)	Corrections for fractional nanoseconds (see 10.2.9 and Figure 10-5), difference between preciseOriginTimestamp field (if sent by a two-step PTP Port) or originTimestamp field (if sent by a one-step PTP Port) and current synchronized time (see 11.2.15.2.3 and Figure 11-7), and asymmetry (see 8.3, 11.2.14.2.1, and 11.2.15.2.3. The quantity delayAsymmetry is used in the computation of upstreamTxTime in 11.2.14.2.1, and upstreamTxTime is used in computing an addition to the correctionField in 11.2.15.2.3).
Pdelay_Resp, Pdelay_Resp_Follow_Up	Corrections for fractional nanoseconds (see Figure 11-9 and Figure 11-10) if the message is sent by the transport-specific peer-to-peer delay mechanism; or For Pdelay_Resp, minus the corrections for fractional nanoseconds (see 11.2.20.3.1, Figure 11-9, and Figure 11-10), and for Pdelay_Resp_Follow_Up, the sum of the correctionField of the corresponding Pdelay_Req message and the fractional nanoseconds portion of the pdelayRespEventEgressTimestamp of the corresponding Pdelay_Resp message, if this state machine is invoked by CMLDS.
Sync (sent by a two-step PTP Port), Pdelay_Req, Announce, Signaling	The value is 0 (see 10.6.2.2.9) if the message is sent by the transport-specific peer-to-peer delay mechanism, or The value is 0 for Sync (sent by a two-step PTP Port), Announce, and Signaling, and –delayAsymmetry for Pdelay_Req (see 11.2.19.3.1), if the message is sent by CMLDS.
NOTE—IEEE Std 1588-2019 describes asymmetry corrections for the Pdelay_Req and Pdelay_Resp messages. However, the peer-to-peer delay mechanism computes the mean propagation delay. Here, where the gPTP communication path is a full-duplex point-to-point PTP Link, these corrections cancel in the mean propagation delay computation and therefore are not needed.	

1 11.4.2.7 sourcePortIdentity (PortIdentity)

2 The value is the portIdentity of the egress port (see 8.5.2) on which the respective message is sent.

3 11.4.2.8 sequenceId (UInteger16)

4 The value is assigned by the originator of the message in conformance with 11.3.8, except for Follow_Up,
5 Pdelay_Resp, and Pdelay_Resp_Follow_Up messages. The sequenceId field values for these exceptions are
6 defined in the state diagrams given in the figures referenced in Table 11-7.

Table 11-7—References for sequenceId value exceptions

Message type	Reference
Follow_Up	See 11.2.15 and Figure 11-7
Pdelay_Resp	See 11.2.20 and Figure 11-10
Pdelay_Resp_Follow_Up	See 11.2.20 and Figure 11-10

1 11.4.2.9 logMessageInterval (Integer8)

2 For Sync and Follow_Up messages, the value is the value of currentLogSyncInterval (see 10.2.5.4 and
3 10.7.2.3). For Pdelay_Req messages, the value is the value of currentLogPdelayReqInterval. For
4 Pdelay_Resp and Pdelay_Resp_Follow_Up messages, the value is transmitted as 0x7F and ignored on
5 reception.

6 11.4.3 Sync message

7 11.4.3.1 General Sync message specifications

8 If the twoStep flag of the PTP common header (see Table 10-9) of the Sync message is TRUE, the fields of
9 the Sync message shall be as specified in Table 11-8. If the twoStep flag of the PTP common header of the
10 Sync message is FALSE, the fields of the Sync message shall be as specified in Table 11-9 and 11.4.3.2.
11 Carrying the Drift_Tracking TLV is optional

Table 11-8—Sync message fields if twoStep flag is TRUE

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
reserved								10	34

Table 11-9—Sync message fields if twoStep flag is FALSE

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
originTimestamp								10	34
Follow_Up information TLV								32	44
Drift_Tracking TLV (optional)								36	76

12 11.4.3.2 Sync message field specifications if twoStep flag is FALSE

13 11.4.3.2.1 originTimestamp (Timestamp)

14 The value of the originTimestamp field is the sourceTime of the ClockTimeTransmitter entity of the
15 Grandmaster PTP Instance, when the Sync message was sent by that Grandmaster PTP Instance, with any
16 fractional nanoseconds truncated (see 10.2.9).

17 The sum of the correctionField and the originTimestamp field of the Sync message is the value of the
18 synchronized time corresponding to the syncEventEgressTimestamp at the PTP Instance that sent the Sync
19 message, including any fractional nanoseconds.

1 11.4.3.2.2 Drift_Tracking TLV

2 The Sync message may carry the Drift_Tracking TLV, defined in 11.4.4.4.

3 11.4.3.2.3 Follow_Up information TLV

4 The Sync message carries the Follow_Up information TLV, defined in 11.4.4.3. This TLV shall be the first
5 TLV after the fixed fields.

6 11.4.4 Follow_Up message

7 11.4.4.1 General Follow_Up message specifications

8 The fields of the Follow_Up message shall be as specified in Table 11-10 and 11.4.4.2. Carrying the
9 Drift_Tracking TLV is optional

Table 11-10—Follow_Up message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
preciseOriginTimestamp								10	34
Follow_Up information TLV								32	44
Drift_Tracking TLV (optional)								36	76

10 11.4.4.2 Follow_Up message field specifications

11 11.4.4.2.1 preciseOriginTimestamp (Timestamp)

12 The value of the preciseOriginTimestamp field is the sourceTime of the ClockTimeTransmitter entity of the
13 Grandmaster PTP Instance, when the associated Sync message was sent by that Grandmaster PTP Instance,
14 with any fractional nanoseconds truncated (see 10.2.9).

15 The sum of the correctionFields in the Follow_Up and associated Sync messages, added to the
16 preciseOriginTimestamp field of the Follow_Up message, is the value of the synchronized time
17 corresponding to the syncEventEgressTimestamp at the PTP Instance that sent the associated Sync message,
18 including any fractional nanoseconds.

19 11.4.4.2.2 Follow_Up information TLV

20 The Follow_Up message carries the Follow_Up information TLV, defined in 11.4.4.3. This TLV shall be the
21 first TLV after the fixed fields.

22 11.4.4.2.3 Drift_Tracking TLV

23 The Follow_Up message may carry the Drift_Tracking TLV, defined in 11.4.4.4.

24

1 11.4.4.3 Follow_Up information TLV definition

2 11.4.4.3.1 General

3 The fields of the Follow_Up information TLV shall be as specified in Table 11-11 and in 11.4.4.3.2 through
4 11.4.4.3.9. This TLV is a standard organization extension TLV for the Follow_Up message, as specified in
5 14.3 of IEEE Std 1588-2019.

Table 11-11—Follow_Up information TLV

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
cumulativeScaledRateOffset								4	10
gmTimeBaseIndicator								2	14
lastGmPhaseChange								12	16
scaledLastGmFreqChange								4	28

6 NOTE—The Follow_Up information TLV is different from the CUMULATIVE_RATE_RATIO TLV of IEEE Std 1588-
7 2019 (see 16.10 and Table 52 of IEEE Std 1588-2019). While both TLVs carry cumulative rate offset information, the
8 Follow_Up information TLV also carries information on the Grandmaster Clock time base, most recent phase change,
9 and most recent frequency change. The CUMULATIVE_RATE_RATIO TLV is not used by gPTP.

10 11.4.4.3.2 tlvType (Enumeration16)

11 The value of the tlvType field is 0x3.

12 NOTE—This value indicates the TLV is a vendor and standard organization extension TLV, as specified in 14.3.2.1 and
13 Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION with a
14 value of 0x3.

15 11.4.4.3.3 lengthField (UInteger16)

16 The value of the lengthField is 28.

17 11.4.4.3.4 organizationId (Octet3)

18 The value of organizationId is 00-80-C2.

19 11.4.4.3.5 organizationSubType (Enumeration24)

20 The value of organizationSubType is 1.

1 11.4.4.3.6 cumulativeScaledRateOffset (Integer32)

2 The value of cumulativeScaledRateOffset is equal to $(\text{rateRatio} - 1.0) \times (2^{41})$, truncated to the next smaller
3 signed integer, where rateRatio is the ratio of the frequency of the Grandmaster Clock to the frequency of the
4 LocalClock entity in the PTP Instance that sends the message.

5 NOTE—The above scaling allows the representation of fractional frequency offsets in the range $[-(2^{-10} - 2^{-41}), 2^{-10} - 2^{-41}]$,
6 with granularity of 2^{-41} . This range is approximately $[-9.766 \times 10^{-4}, 9.766 \times 10^{-4}]$.

7 11.4.4.3.7 gmTimeBaseIndicator (UInteger16)

8 The value of gmTimeBaseIndicator is the timeBaseIndicator of the ClockSource entity for the current
9 Grandmaster PTP Instance (see 9.2.2.3).

10 NOTE—The timeBaseIndicator is supplied by the ClockSource entity to the ClockTimeTransmitter entity via the
11 ClockSourceTime.invoke function (see 9.2.2.3).

12 11.4.4.3.8 lastGmPhaseChange (ScaledNs)

13 The value of lastGmPhaseChange is the time of the current Grandmaster Clock minus the time of the
14 previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the
15 Grandmaster PTP Instance. The value is copied from the lastGmPhaseChange member of the MDSyncSend
16 structure whose receipt causes the MD entity to send the Follow_Up message (see 11.2.11).

17 11.4.4.3.9 scaledLastGmFreqChange (Integer32)

18 The value of scaledLastGmFreqChange is the fractional frequency offset of the current Grandmaster Clock
19 relative to the previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became
20 the Grandmaster PTP Instance, or relative to itself prior to the last change in gmTimeBaseIndicator,
21 multiplied by 2^{41} and truncated to the next smaller signed integer. The value is obtained by multiplying the
22 lastGmFreqChange member of MDSyncSend whose receipt causes the MD entity to send the Follow_Up
23 message (see 11.2.11) by 2^{41} , and truncating to the next smaller signed integer.

24 NOTE—The above scaling allows the representation of fractional frequency offsets in the range $[-(2^{-10} - 2^{-41}), 2^{-10} - 2^{-41}]$,
25 with granularity of 2^{-41} . This range is approximately $[-9.766 \times 10^{-4}, 9.766 \times 10^{-4}]$.

26 11.4.4.4 Drift_Tracking TLV definition

27 11.4.4.4.1 General

28 The fields of the Drift_Tracking TLV shall be as specified in Table 11-11 and in 11.4.4.4.2 through
29 11.4.4.4.9. This TLV is a standard organization extension TLV for the Sync or Follow_Up message, as
30 specified in 14.3 of IEEE Std 1588-2019.

31 11.4.4.4.2 tlvType (Enumeration16)

32 The value of the tlvType is 0x3.

33 NOTE—This value indicates the TLV is a vendor and standard-organization extension TLV, as specified in 14.3.2.1 and
34 Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION with a
35 value of 0x3.

36 11.4.4.4.3 lengthField (UInteger16)

37 The value of lengthField is 32.

Table 11-12—Drift_Tracking TLV

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
syncEgressTimestamp								12	10
syncGrandmasterIdentity								8	22
syncStepsRemoved								2	30
rateRatioDrift								4	32

1 11.4.4.4.4 organizationId (Octet3)

2 The value of organizationId is 00-80-C2.

3 11.4.4.4.5 organizationSubType (Enumeration24)

4 The value of organizationSubType is 6.

5 11.4.4.4.6 syncEgressTimestamp (ExtendedTimestamp)

6 The value of the syncEgressTimestamp field is the timestamp, based on the local clock, when the Sync
7 message was sent by that PTP Instance (see 11.4.3.2).

8 11.4.4.4.7 syncGrandmasterIdentity (ClockIdentity)

9 The value of syncGrandmasterIdentity as determined by the MDSyncSend state machine (see 11.2.15.2.3
10 and 11.2.15.2.5).

11 11.4.4.4.8 syncStepsRemoved (UInteger16)

12 The value of syncStepsRemoved as determined by the MDSyncSend state machine (see 11.2.15.2.3 and
13 11.2.15.2.5).

14 11.4.4.4.9 rateRatioDrift (Integer32)

15 The value of rateRatioDrift is equal to $(RRdrift - 1.0) \times (2^{41})$, truncated to the next smaller signed integer,
16 where RRdrift is the measured estimate of the rate of change per second of the ratio of the frequency of the
17 Grandmaster Clock to the frequency of the Local Clock entity in the PTP Instance that sends the message.

18 NOTE—The above scaling allows the representation of rates of change of fractional frequency offset in the range $[-(2^{-10} - 2^{-41}), 2^{-10} - 2^{-41}] s^{-1}$, with granularity of 2^{-41} . This range is approximately $[-9.766 \times 10^{-4}, 9.766 \times 10^{-4}] s^{-1}$.

20 11.4.5 Pdelay_Req message

21 The fields of the Pdelay_Req message shall be as specified in Table 11-13.

Table 11-13—Pdelay_Req message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
reserved								10	34
reserved								10	44

1 11.4.6 Pdelay_Resp message

2 11.4.6.1 General Pdelay_Resp message specifications

3 The fields of the Pdelay_Resp message shall be as specified in Table 11-14 and 11.4.6.2.

Table 11-14—Pdelay_Resp message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
requestReceiptTimestamp								10	34
requestingPortIdentity								10	44

4 11.4.6.2 Pdelay_Resp message field specifications

5 11.4.6.2.1 requestReceiptTimestamp (Timestamp)

6 The value is the seconds and nanoseconds portion of the pdelayReqEventIngressTimestamp of the
7 associated Pdelay_Req message (see 11.2.19).

8 11.4.6.2.2 requestingPortIdentity (PortIdentity)

9 The value is the value of the sourcePortIdentity field of the associated Pdelay_Req message (see 11.4.5).

10 11.4.7 Pdelay_Resp_Follow_Up message

11 11.4.7.1 General Pdelay_Resp_Follow_Up message specifications

12 The fields of the Pdelay_Resp_Follow_Up message shall be as specified in Table 11-15 and 11.4.7.2.

13 11.4.7.2 Pdelay_Resp_Follow_Up message field specifications

14 11.4.7.2.1 responseOriginTimestamp (Timestamp)

15 The value is the seconds and nanoseconds portion of the pdelayRespEventEgressTimestamp of the
16 associated Pdelay_Resp message (see 11.4.6.2).

Table 11-15—Pdelay_Resp_Follow_Up message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
responseOriginTimestamp								10	34
requestingPortIdentity								10	44

1 11.4.7.2.2 requestingPortIdentity (PortIdentity)

2 The value is the value of the sourcePortIdentity field of the associated Pdelay_Req message (see 11.4.5).

3 11.5 Protocol timing characterization

4 11.5.1 General

5 This subclause specifies timing attributes for the media-dependent sublayer specified in this clause.

6 11.5.2 Message transmission intervals

7 11.5.2.1 General interval specification

8 The mean time interval between successive Pdelay_Req messages is represented as the logarithm to the
9 base 2 of this time interval measured in seconds. The value of this logarithmic attribute shall be as specified
10 in 11.5.2.2.

11 The mean time interval between successive Sync messages shall be as specified in 10.7.2.1, 10.7.2.3, and
12 11.5.2.3.

13 11.5.2.2 Pdelay_Req message transmission interval

14 When useMgtSettableLogPdelayReqInterval (see 14.18.13) is FALSE, the initialLogPdelayReqInterval
15 specifies the following:

- 16 a) The mean time interval between successive Pdelay_Req messages sent over a PTP Link when the
17 port is initialized, and
- 18 b) The value to which the mean time interval between successive Pdelay_Req messages is set when a
19 message interval request TLV is received with the logLinkDelayIntervalField set to 126
20 (see 11.2.21).

21 The currentLogPdelayReqInterval specifies the current value of the mean time interval between successive
22 Pdelay_Req messages. The default value of initialLogPdelayReqInterval is 0. Every port supports the
23 value 127; the port does not send Pdelay_Req messages when currentLogPdelayReqInterval has this value
24 (see 11.2.21). A port may support other values, except for the reserved values indicated in Table 10-15. A
25 port shall ignore requests for unsupported values (see 11.2.21). The initialLogPdelayReqInterval and
26 currentLogPdelayReqInterval are per-port attributes.

27 When useMgtSettableLogPdelayReqInterval is TRUE, currentLogPdelayReqInterval is set equal to
28 mgtSettableLogPdelayReqInterval (see 14.18.14), and initialLogPdelayReqInterval is ignored.

1 Pdelay_Req messages shall be transmitted such that the value of the arithmetic mean of the intervals, in
2 seconds, between Pdelay_Req message transmissions is not less than the value of
3 $0.9 \times 2^{\text{currentLogPdelayReqInterval}}$.

4 NOTE 1—A minimum number of inter-message intervals is necessary to verify that a PTP Port meets these
5 requirements. The arithmetic mean is the sum of the inter-message interval samples divided by the number of samples.
6 For more detailed discussion of statistical analyses, see Papoulis [B25].

7 NOTE 2—If useMgtSettableLogPdelayReqInterval is FALSE, the value of initialLogPdelayReqInterval is the value of
8 the mean time interval between successive Pdelay_Req messages when the port is initialized. The value of the mean
9 time interval between successive Pdelay_Req messages can be changed, e.g., if the port receives a Signaling message
10 that carries a message interval request TLV (see 10.6.4.3) and the current value is stored in
11 currentLogPdelayReqInterval. The value of the mean time interval between successive Pdelay_Req messages can be
12 reset to the initial value, e.g., by a message interval request TLV for which the value of the field logLinkDelayInterval is
13 126 (see 10.6.4.3.6).

14 NOTE 3—A port that requests (using a Signaling message that contains a message interval request TLV; see 10.6.4 and
15 11.2.21) that the port at the other end of the attached PTP Link set its currentLogPdelayReqInterval to a specific value
16 can determine if the request was honored by examining the logMessageInterval field of subsequent received Pdelay_Req
17 messages.

18 NOTE 4—The MDPdelayReq state machine ensures that the times between transmission of successive Pdelay_Req messages,
19 in seconds, are not smaller than $2^{\text{currentLogPdelayReqInterval}}$. This is consistent with the requirements of the current subclause and
20 IEEE Std 1588-2019, which require that the value of the arithmetic mean of the intervals, in seconds, between Pdelay_Req
21 message transmissions is not less than the value of $0.9 \times 2^{\text{currentLogPdelayReqInterval}}$. The sending of Pdelay_Req messages
22 is governed by the LocalClock and not the synchronized time (i.e., the estimate of the Grandmaster Clock time). Since
23 the LocalClock frequency can be slightly larger than the Grandmaster Clock frequency (e.g., by 100 ppm, which is the
24 specified frequency accuracy of the LocalClock; see B.1.1), it is possible for the time intervals between successive
25 Pdelay_Req messages to be slightly less than $2^{\text{currentLogPdelayReqInterval}}$ when measured relative to the synchronized time.
26 The factor 0.9 allows a margin of 10% for the arithmetic mean of the successive intervals between Pdelay_Req
27 messages.

28 11.5.2.3 Sync message transmission interval default value

29 The default value of initialLogSyncInterval (see 10.7.2.3) is –3. Every PTP Port supports the value 127; the
30 PTP Port does not send Sync messages when currentLogSyncInterval has this value (see 10.3.18). A PTP
31 Port may support other values, except for the reserved values indicated in Table 10-16. A PTP Port ignores
32 requests for unsupported values (see 10.3.18).

33 NOTE—A PTP Port that requests (using a Signaling message that contains a message interval request TLV; see 10.6.4
34 and 10.3.18) that the PTP Port at the other end of the attached PTP Link set its currentLogSyncInterval to a specific
35 value can determine if the request was honored by examining the logMessageInterval field of subsequent received Sync
36 messages.

37 11.5.3 allowedLostResponses

38 The variable allowedLostResponses (see 11.2.13.4) is the number of Pdelay_Req messages without valid
39 responses above which a port is considered to be not exchanging peer delay messages with its neighbor. The
40 default value of allowedLostResponses shall be 9. The range shall be 1 through 255.

41 11.5.4 allowedFaults

42 The variable allowedFaults (see 11.2.13.5) is the number of faults above which asCapableAcrossDomains is
43 set to FALSE, i.e., the port is considered not capable of interoperating with its neighbor via the IEEE
44 802.1AS protocol (see 10.2.5.1). In this context, the term *faults* refers to instances where

- 1 a) The computed mean propagation delay, i.e., meanLinkDelay (see 10.2.5.8), exceeds the threshold,
- 2 meanLinkDelayThresh (see 11.2.13.7), or is below the negMeanLinkDelayThresh (see 11.2.13.8),
- 3 and/or
- 4 b) The computation of nrrPdelay is invalid (see 11.2.19.2.10).

5 The default value of allowedFaults shall be 9. The range shall be 1 through 255.

6 NOTE—The above description of allowedFaults uses the variable asCapableAcrossDomains (see 11.2.13.13). When only
7 one domain is active, asCapableAcrossDomains is equivalent to the variable asCapable.

8 11.6 Control of computation of neighborRateRatio

9 If the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and nrrCompMethod (see
10 11.2.13.16) is equal to Sync, the value of neighborRateRatio (see 10.2.5.7) is set equal to nrrSync (see
11 11.2.13.15).

12 If the global variable driftTrackingTlvSupport (see 10.2.4.28) is FALSE or nrrCompMethod (see 11.2.13.16)
13 is equal to Pdelay, the value of neighborRateRatio (see 10.2.5.7) is set equal to nrrPdelay (see 11.2.13.14).
14 The computation of nrrPdelay is controlled as described below. The description below applies only to the
15 computation of nrrPdelay, and not to the computation of nrrSync.

16 The variable computeNeighborRateRatio (see 10.2.5.10) indicates whether nrrPdelay is to be computed by
17 this port when the peer-to-peer delay mechanism is invoked.

18 When useMgtSettableComputeNeighborRateRatio (see 14.18.17) is FALSE, computeNeighborRateRatio is
19 initialized to the value of initialComputeNeighborRateRatio.

20 The currentComputeNeighborRateRatio specifies the current value of computeNeighborRateRatio. The
21 default value of initialComputeNeighborRateRatio is TRUE. The initialComputeNeighborRateRatio and
22 currentComputeNeighborRateRatio are per-port attributes.

23 When useMgtSettableComputeNeighborRateRatio is TRUE, currentComputeNeighborRateRatio is set
24 equal to mgtSettableComputeNeighborRateRatio (see 14.18.18), and initialComputeNeighborRateRatio is
25 ignored.

26 NOTE—If useMgtSettableComputeNeighborRateRatio is FALSE, the value of initialComputeNeighborRateRatio
27 determines whether nrrPdelay is computed by the peer delay mechanism when the port is initialized. The value of
28 computeNeighborRateRatio can be changed, e.g., if the port receives a Signaling message that carries a message interval
29 request TLV (see 10.6.4.3) and the current value is stored in currentComputeNeighborRateRatio.

30 11.7 Control of computation of meanLinkDelay

31 The variable computeMeanLinkDelay (see 10.2.5.10) indicates whether meanLinkDelay is to be computed
32 by this port when the peer-to-peer delay mechanism is invoked.

33 When useMgtSettableComputeMeanLinkDelay (see 14.18.21) is FALSE, computeMeanLinkDelay is
34 initialized to the value of initialComputeMeanLinkDelay.

35 The currentComputeMeanLinkDelay specifies the current value of computeMeanLinkDelay. The default
36 value of initialComputeMeanLinkDelay is TRUE. The initialComputeMeanLinkDelay and
37 currentComputeMeanLinkDelay are per-port attributes.

1 When `useMgtSettableComputeMeanLinkDelay` is `TRUE`, `currentComputeMeanLinkDelay` is set equal to
2 `mgtSettableComputeMeanLinkDelay` (see 14.18.22), and `initialComputeMeanLinkDelay` is ignored.

3 NOTE—If `useMgtSettableComputeMeanLinkDelay` is `FALSE`, the value of `initialComputeMeanLinkDelay` determines
4 whether `meanLinkDelay` is computed by the peer delay mechanism when the port is initialized. The value of
5 `computeMeanLinkDelay` can be changed, e.g., if the port receives a Signaling message that carries a message interval
6 request TLV (see 10.6.4.3) and the current value is stored in `currentComputeMeanLinkDelay`.

1 12. Media-dependent layer specification for IEEE 802.11 links

2 12.1 Overview

3 12.1.1 General

4 Accurate synchronized time is distributed across a domain through time measurements between adjacent
5 PTP Instances in a packet network. Time is communicated from the root of the synchronization tree (i.e., the
6 Grandmaster PTP Instance) to the leaves of the synchronization tree, by recursively propagating time from a
7 leaf-facing “timeTransmitter” port to some number of root-facing “timeReceiver” ports in PTP Instances at
8 the next level of the synchronization tree through measurements made across the links connecting the PTP
9 Instances. While the time semantics are consistent across the time-aware packet network, the method for
10 communicating synchronized time from a timeTransmitter port to the immediate downstream link partner
11 varies depending on the type of link interconnecting the two systems.

12 This clause specifies the interface primitives and state machines that provide accurate synchronized time
13 across wireless IEEE 802.11 links as part of a packet network. This clause builds upon time measurement
14 features defined in IEEE Std 802.11-2016 and makes no distinction between stations with an access point
15 function and stations without an access point function.

16 12.1.2 IEEE 802.11 Timing Measurement and Fine Timing Measurement procedures

17 12.1.2.1 General

18 IEEE Std 802.11-2016 defines a family of wireless measurements, including both “Timing Measurement”
19 (TM) and “Fine Timing Measurement” (FTM), which captures timestamps of the transmit time and receive
20 time of a round-trip message exchange between associated wireless local area network (WLAN) stations.

21 In contrast to the protocol defined for full-duplex point-to-point links, this clause does not define any new
22 frames nor the transmission of any frames. Rather, it makes use of a MAC layer management entity
23 (MLME) interface, which causes the IEEE 802.11 layer to not only take timestamps of measurement frames
24 as they are transmitted and received, but to also *generate* and *consume* the measurement frames, all within
25 the IEEE 802.11 MLME layer, and then to provide timestamp information from the MLME to this media-
26 dependent layer through a set of well-defined service primitives. However, as an aid to the reader, the
27 protocol and frames used by the IEEE 802.11 MLME for both Timing Measurement and Fine Timing
28 Measurement are described briefly as follows and illustrated in Figure 12-1 and Figure 12-2, respectively.

29 Both Timing Measurement and Fine Timing Measurement are accomplished through a round-trip frame
30 exchange. For Timing Measurement, the first frame of the round-trip measurement is generated by the
31 timeTransmitter within the IEEE 802.11 MLME when the MLME-TIMINGMSMT.request primitive is
32 invoked. For Fine Timing Measurement, an initial Fine Timing Measurement request frame is generated by
33 the timeReceiver within the IEEE 802.11 MLME when the MLME-FINETIMINGMSMT.request
34 primitive is invoked. After this frame is successfully received by the timeTransmitter, the first frame of the
35 round-trip measurement is generated by the TimeTransmitter within the IEEE 802.11 MLME when the
36 MLME-FINETIMINGMSMT.request primitive is invoked. As defined by IEEE Std 802.11-2016, upon
37 receipt of the resulting Timing Measurement or Fine Timing Measurement frame, the timeReceiver station
38 transmits an IEEE 802.11 Ack control frame to the timeTransmitter station. Four timestamps are captured
39 during this two-frame exchange, as follows:

- 40 a) t1 is when (in the timeTransmitter station’s time base) the request frame is transmitted
- 41 b) t2 is when (in the timeReceiver station’s time base) the request frame is received
- 42 c) t3 is when (in the timeReceiver station’s time base) the Ack control frame is transmitted
- 43 d) t4 is when (in the timeTransmitter station’s time base) the Ack control frame is received

1 When the timeTransmitter sends either a Fine Timing Measurement or a Timing Measurement frame, it
2 passes the t1 and t4 timestamps (and other end-to-end synchronization information) and
3 FollowUpInformation, from the previous measurement to the timeReceiver. A pair of tokens is passed in
4 each timing or Fine Timing Measurement frame, one to identify the current measurement and the other to
5 allow the timeReceiver to associate the timestamp information with the previous measurement.

6 NOTE 1—TM also can include a Timing Measurement Request Frame; however, this frame type is not used by this
7 standard.

8

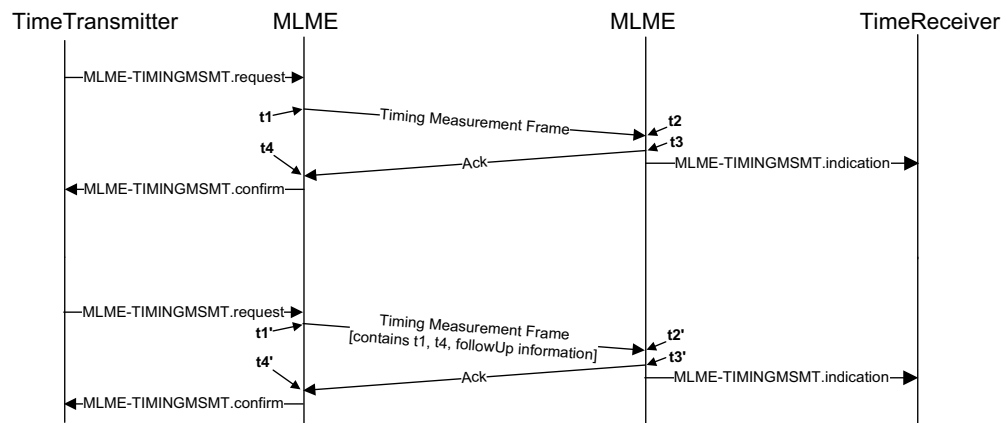


Figure 12-1—Timing measurement procedure for IEEE 802.11 links

1

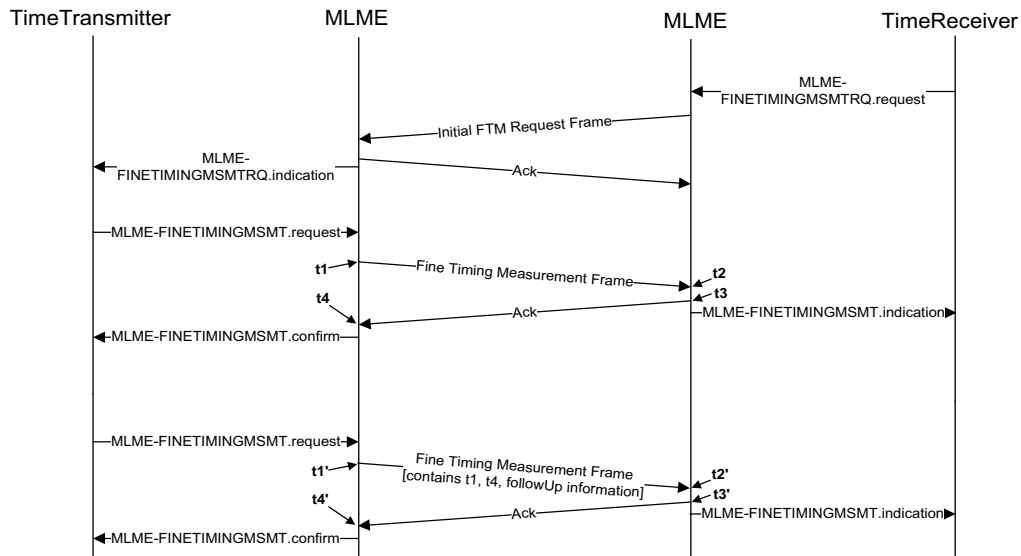


Figure 12-2—Fine Timing Measurement procedure for IEEE 802.11 links

2 Note that, unlike full-duplex point-to-point ports, IEEE 802.11 ports do not compute the link delay
3 measurements in both directions since only a PTP Port in the TimeReceiver state makes use of that
4 information. A PTP Port that transitions from the TimeTransmitter state to the TimeReceiver state (e.g., due
5 to selection of a new Grandmaster PTP Instance) can collect a number of link delay measurements and
6 perform averaging or other filtering to achieve the desired accuracy.

7 NOTE 2—Fine Timing Measurement can be used for time synchronization as described in this standard; however, it can
8 also be used for location services as defined in IEEE Std 802.11-2016. Since FTM supports only one configuration at a
9 time, it is possible that setting that single configuration for time synchronization might disable its previous configuration
10 for use by another application for location services or, conversely, an application that configures FTM for location
11 services could disable this standard's use for time synchronization. Implementations that use FTM for time
12 synchronization and other applications need to coordinate usage of the FTM protocol.

13 The timeTransmitter generates MLME-TIMINGMSMT.request primitives for Timing Measurement, as
14 described in this standard, in a manner such that the requirements of 10.7.2.3 and 12.8 for the time
15 synchronization message interval are satisfied. Timing measurement frames are then sent from the
16 timeTransmitter to the timeReceiver continuously and at a rate that satisfies the requirements of those two
17 subclauses. It is not necessary for the timeReceiver to continually request timing information from the
18 timeTransmitter. In contrast, the timeReceiver must request timing information from the timeTransmitter for
19 Fine Timing Measurement. A Fine Timing Measurement frame carries timestamp information for a previous
20 measurement. The timeReceiver requests a burst of Fine Timing Measurement frames from the
21 timeTransmitter. Figure 12-2 shows an example of a burst of Fine Timing Measurement frames (there are
22 two frames in that example). The Fine Timing Measurement process is described in more detail in 12.1.2.2
23 and is illustrated in Figure 12-3. In that discussion, the focus is on the transmission of the frames. In the
24 simplified example, service primitives are omitted from Figure 12-3.

25 12.1.2.2 Detailed description of Fine Timing Measurement (FTM)

26 Figure 12-3 is adapted from Figure 11-37 of IEEE Std 802.11-2016. Additional details on the FTM
27 procedure are given in 11.24.6 of IEEE Std 802.11-2016. The example of this figure is for when the

1 initiating station (STA), i.e., the timeReceiver, requests a single burst of three FTM frames from the
 2 responding STA, i.e., the timeTransmitter, as soon as possible. The timeReceiver makes this request by
 3 sending an initial FTM Request to the timeTransmitter with respective parameters set to appropriate values.
 4 The FTM parameters that are relevant to time synchronization in this standard are described in 12.6, and all
 5 the FTM parameters are described in more detail in 9.4.2.168 of IEEE Std 802.11-2016. However, in the
 6 example here, the parameter ASAP is set to 1 to indicate to the timeTransmitter that the FTM frames are
 7 desired as soon as possible, and the Number of Bursts Exponent parameter is set to 0 to indicate a single
 8 burst. Figure 12-3 is a simplified view; the timeReceiver causes the frame to be sent by invoking the
 9 MLME-FINETIMINGMSMTRQ.request primitive, which includes the FTM parameter values. The
 10 timeTransmitter sends an acknowledgment (Ack) frame to the timeReceiver to indicate it received the initial
 11 FTM request. The timeTransmitter then sends an initial FTM frame at a time that is recommended to be no
 12 more than 10 ms later than the receipt of the initial FTM request. The initial FTM frame indicates to the
 13 timeReceiver whether the timeTransmitter was able to grant the values of the FTM parameters that the
 14 timeReceiver requested. If the requested parameters are granted, the procedure continues (Figure 12-3
 15 illustrates this case). If the requested parameters are not granted, the timeReceiver sends a new initial FTM
 16 request for a burst of two FTM frames. If the new request is granted, the procedure continues. If the new
 17 request is not granted, the timeReceiver and timeTransmitter use TM if they both support TM. If, at this
 18 point, the timeReceiver or the timeTransmitter, or both, do not support TM, the procedure terminates and
 19 asCapable is set to FALSE (see 12.4).

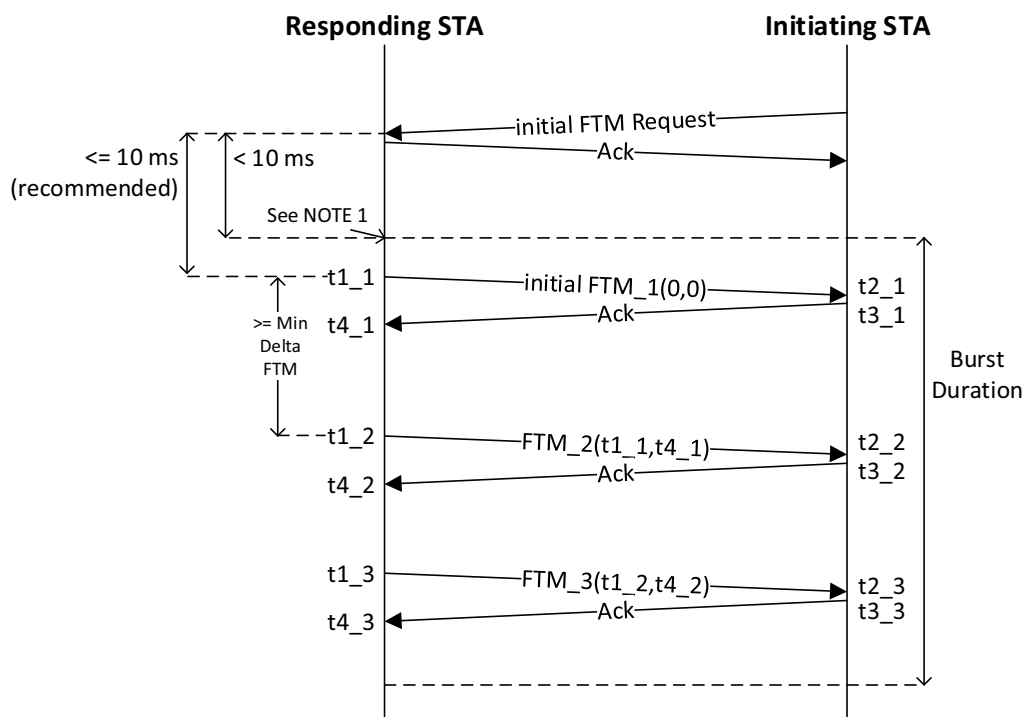


Figure 12-3—Illustration of Fine Timing Measurement burst

20 NOTE—IEEE Std 802.11-2016 allows various options in case the timeTransmitter does not grant the request. The above
 21 procedure is used in this subclause.

22 The initial FTM frame (initial FTM_1 in Figure 12-3) sent by the timeTransmitter is timestamped with the
 23 value $t1_1$ on transmission from the timeTransmitter and timestamped with the value $t2_1$ on receipt by the
 24 timeReceiver. The initial FTM frame has fields that carry the $t1$ and $t4$ timestamps of the previous FTM
 25 frame and corresponding Ack; however, since this is the first FTM frame of the burst, these fields are set to

1 zero. The timeReceiver responds to the timeTransmitter with an Ack, which is timestamped with the value
2 $t3_1$ on transmission from the timeReceiver and with the value $t4_1$ on receipt by the timeTransmitter.

3 The timeTransmitter sends the second FTM frame (FTM_2 in Figure 12-3) after a time interval since
4 sending the initial FTM frame that is greater than or equal to the Min Delta FTM parameter that was
5 requested by the timeReceiver. This FTM frame is timestamped with $t1_2$ on transmission and $t2_2$ on
6 reception. This FTM frame also carries the values of the timestamps $t1_1$ and $t4_1$ of the initial FTM frame
7 and corresponding Ack. On receipt of the second FTM frame, the timeReceiver sends an Ack frame to the
8 timeTransmitter; this frame is timestamped with $t3_2$ on transmission and $t4_2$ on reception. Finally, the
9 timeTransmitter sends the third FTM frame (FTM_3 in Figure 12-3) to the timeReceiver, also at a time
10 interval since sending FTM_2 that is greater than or equal to the Min Delta FTM parameter that was
11 requested by the timeReceiver. As with FTM_2, this frame is timestamped with $t1_3$ on transmission and
12 $t2_3$ on reception. This FTM frame also carries the values of the timestamps $t1_2$ and $t4_2$ of the second
13 FTM frame and corresponding Ack. On receipt of the third FTM frame, the timeReceiver sends an Ack
14 frame to the timeTransmitter; this frame is timestamped with $t3_3$ on transmission and $t4_3$ on reception.

15 On completion of the above exchanges of FTM frames and corresponding acknowledgments, the
16 timeReceiver knows the transmission and reception times for the initial FTM frame ($t1_1$, $t2_1$, $t3_1$, and
17 $t4_1$) and second FTM frame ($t1_2$, $t2_2$, $t3_2$, and $t4_2$) and the reception time for the third FTM frame
18 ($t2_3$) and transmission time for the corresponding Ack ($t3_3$). The timeReceiver can use this information
19 (along with FollowUpInformation contained in the VendorSpecific information element; see 12.7) to
20 synchronize to the timeTransmitter. In this standard, timestamps for the minimum delay FTM frames are
21 used. Specifically, the timeReceiver computes the quantities $D1 = t2_1 - t1_1$, and $D2 = t2_2 - t1_2$, and
22 uses the timestamps $t1_i$ and $t2_i$, where $i = 1$ if $D1 < D2$ and $i = 2$ if $D1 \geq D2$, to compute the respective
23 members of the MDSyncReceive structure. The timestamps of FTM_3 and its corresponding Ack are not
24 used; FTM_3 is used only to convey the timestamps of FTM_2 and its corresponding Ack.

25 If the timeTransmitter does not grant the parameters requested initially by the timeReceiver, i.e., for a burst
26 of three FTM frames, but it does grant the subsequent request for a burst of two FTM frames, the
27 timeReceiver has a full set of timestamps for only the initial FTM_1 frame. In this case, the timeReceiver
28 uses the timestamps $t1_1$, $t2_1$, $t3_1$, and $t4_1$ to compute the respective members of the MDSyncReceive
29 structure.

30 With the above procedure for FTM, the timeReceiver controls the rate at which time synchronization
31 information is sent from the timeTransmitter. Therefore, the timeReceiver requests time synchronization
32 information from the timeTransmitter at an average rate equal to the inverse of the current synchronization
33 message interval `currentLogSyncInterval` (see 12.8 and 14.10.18). In addition, the actual intervals between
34 successive requests by the timeReceiver for time synchronization information meet the requirements of
35 10.7.2.3. Also, the value of `syncLocked` at the timeTransmitter port will not affect the sending of time
36 synchronization information from the timeTransmitter to the timeReceiver; the requests for time
37 synchronization information from the timeReceiver are asynchronous to the receipt of time synchronization
38 information from upstream at the node that contains the timeTransmitter port.

39 NOTE—The rate at which time synchronization information is sent from the timeTransmitter can be controlled by the
40 timeTransmitter instead of the timeReceiver for non-FTM cases (e.g., TM).

41 12.1.3 Layering for IEEE 802.11 links

42 The *media-dependent* (MD) entity is tailored to the link technology and is responsible for translating the
43 PortSync entity's media-independent actions to media-dependent PDUs or primitives as necessary for
44 communicating synchronized time from the timeTransmitter port over the link to a single timeReceiver port.
45 For an IEEE 802.11 link, this one-to-one relationship between the MD entities of the timeTransmitter and
46 timeReceiver implies that if the one physical IEEE 802.11 port is associated with multiple stations, each
47 association requires its own instantiation of the IEEE 802.1AS PortSync entity and MD entity. The MLME-
48 TIMINGMSMT and MLME-FINETIMINGMSMT service primitives defined in IEEE Std 802.11-2016 are
49 used to perform Timing Measurements and Fine Timing Measurements, respectively, between a

1timeTransmitter IEEE 802.11 station and associated IEEE 802.11 timeReceiver stations. Figure 12-4
2 illustrates how the MD entity interacts with the higher and lower layers.

3

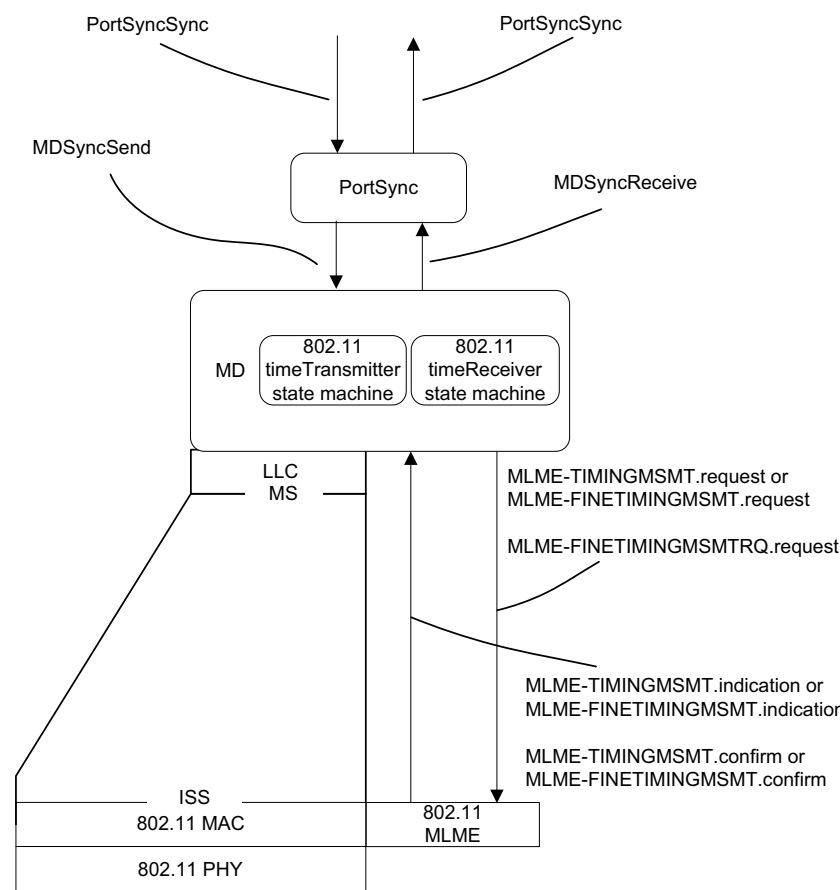


Figure 12-4—Media-dependent and lower entities in stations with IEEE 802.11 links

4

5 **12.2 Messages**

6 All media-dependent frames are generated and consumed by the lower-layer IEEE 802.11 MLME and thus
7 none are defined here. Also, since the IEEE 802.11 event messages are timestamped by the MAC/PHY, the
8 timestamp point is defined in IEEE Std 802.11-2016 as well. Media-independent messages, i.e., Announce
9 and Signaling messages, are transmitted using the unicast address of the WLAN station instead of the group
10 address defined in 10.5.3.

12.3 Determination of Timing Measurement and Fine Timing Measurement capability

The bits of the per-PTP Port global variable tmFtmSupport (see 12.5.1.5) shall be set as indicated in Table 12-1.

Table 12-1—Values of bits of tmFtmSupport

Bit	Value
0	TRUE if: a) The port supports Timing Measurement and b) The Timing Measurement bit in the Extended Capabilities information element defined in Table 9-135 of IEEE Std 802.11-2016 indicates that the peer IEEE 802.11 station is capable of participating in the Timing Measurement protocol. FALSE otherwise.
1	TRUE if: a) The port supports Fine Timing Measurement and b) The Fine Timing Measurement responder and initiator bits in the Extended Capabilities information element defined in Table 9-135 of IEEE Std 802.11-2016 indicate that the peer IEEE 802.11 station is capable of participating in the Fine Timing Measurement protocol. FALSE otherwise.
2–7	Reserved as FALSE.

12.4 Determination of asCapable

The per-PTP Port, per-domain instance of the global variable asCapable (see 10.2.5.1) is set to TRUE if the following conditions hold (see 12.5.1 and 12.5.2):

- a) The value of tmFtmSupport is not zero.
- b) neighborGtpCapable is TRUE.
- c) At least one of the following conditions hold:
 - 1) Bit 0 of tmFtmSupport is TRUE.
 - 2) Bit 1 of tmFtmSupport is TRUE and, if the PTP Port is a timeTransmitter port, it can support (i.e., grant) the parameters requested by the timeReceiver with either FTMs per burst equal to 3 or FTMs per burst equal to 2.
 - 3) Bit 1 of tmFtmSupport is TRUE and, if the PTP Port is a timeReceiver port, the timeTransmitter port at the other end of the link can support (i.e., grant) the parameters requested by the timeReceiver with either FTMs per burst equal to 3 or FTMs per burst equal to 2.

If the value of domainNumber is zero (which is required for support of the 2011 edition of this standard) and bit 0 of tmFtmSupport is TRUE, asCapable can be set to TRUE. In all other instances, asCapable shall be set to FALSE.

1 NOTE—The above conditions ensure backward compatibility with the 2011 edition of this standard. A time-aware
 2 system that is compliant with the 2011 edition of this standard will not process the gPTP capable TLV, and asCapable
 3 will be determined as specified in the 2011 edition. A PTP Instance of a time-aware system compliant with the current
 4 edition of this standard that is attached, via an IEEE 802.11 link, to a node compliant with the 2011 edition of this
 5 standard will not receive Signaling messages that contain the gPTP capable TLV and will not set neighborGtpCapable
 6 to TRUE. However, item c) 1) in this subclause ensures that asCapable for this PTP Port and domain (i.e., domain 0) will
 7 still be set in a manner consistent with the 2011 edition of this standard.

8 12.5 State machines

9 12.5.1 Media-dependent timeTransmitter state machines

10 12.5.1.1 Overview

11 The MD entity of an IEEE 802.11 port whose port state is TimeTransmitterPort (see Table 10-2) shall
 12 behave in a manner that is indistinguishable, relative to an observer external to a system, from a strict
 13 implementation of the timeTransmitter state machines in Figure 12-5 and Figure 12-6 (denoted as
 14 timeTransmitter state machine A and timeTransmitter state machine B, respectively, in 12.5.1.2), the local
 15 variables specified in 12.5.1.3, the functions specified in 12.5.1.4, the shared variables specified in 12.5.1.5,
 16 and the primitives defined in 12.5.1.6.

17 For Timing Measurement, timeTransmitter state machine A is responsible for initiating a time measurement
 18 whenever the PortSync entity requests it do so, as indicated by the rcvdMDSyncDot11TimeTransmitterA
 19 Boolean (see 12.5.1.3.8). TimeTransmitter state machine A invokes the IEEE 802.11 MLME-
 20 TIMINGMSMT.request primitive and waits for the subsequent MLME-TIMINGMSMT.confirm primitive.
 21 It collects local timestamp information from the measurement (t1 and t4, provided by the confirm primitive)
 22 and includes the information in the subsequent request. See 8.4.3 for more information on timestamps.
 23 TimeTransmitter state machine B is not used for Timing Measurement.

24 For Fine Timing Measurement, timeTransmitter state machine A receives and stores information from the
 25 PortSync entity. TimeTransmitter state machine B receives the MLME-FINETIMINGMSMTRQ.indication
 26 caused by the initial FTM request from the timeReceiver. It sets asCapable as specified in 12.4. It then
 27 generates successive MLME-FINETIMINGMSMT.request primitives to indicate to the timeReceiver
 28 whether it can grant the requested parameters and also to cause information saved by timeTransmitter state
 29 machine A to be sent to the timeReceiver. It receives MLME-FINETIMINGMSMT.confirm primitives
 30 caused by Acks received from the timeReceiver. It collects local timestamp information from the current
 31 measurement (t1 and t4, provided by the confirm primitive) and includes the information in the subsequent
 32 MLME-FINETIMINGMSMT.request.

33 12.5.1.2 State diagrams

34 NOTE—In the computation of the burstDuration in timeTransmitter state machine B, the burst duration parameter from
 35 IEEE Std 802.11-2016 is converted to UScaledNs (i.e., units of 2^{-16} ns; see 6.4.3.2). The burst duration in UScaledNs
 36 (see 6.4.3.2) is related to the quantity $A = \text{initReqParamsDot11TimeTransmitterB.burstDuration} - 2$ by:

$$37 \quad \text{burst duration in UScaledNs} = 1000 \times 2^{16} \times 250 \times 2^A$$

38 i.e., A is the logarithm to base 2 of the burst duration, in microseconds, divided by 250. Also, it is assumed that the burst
 39 duration starts when the initial FTM request is received. In actuality, the timer begins by the partial TSF timer value
 40 indicated in the initial FTM frame, which is slightly after the initial FTM request is received.

1

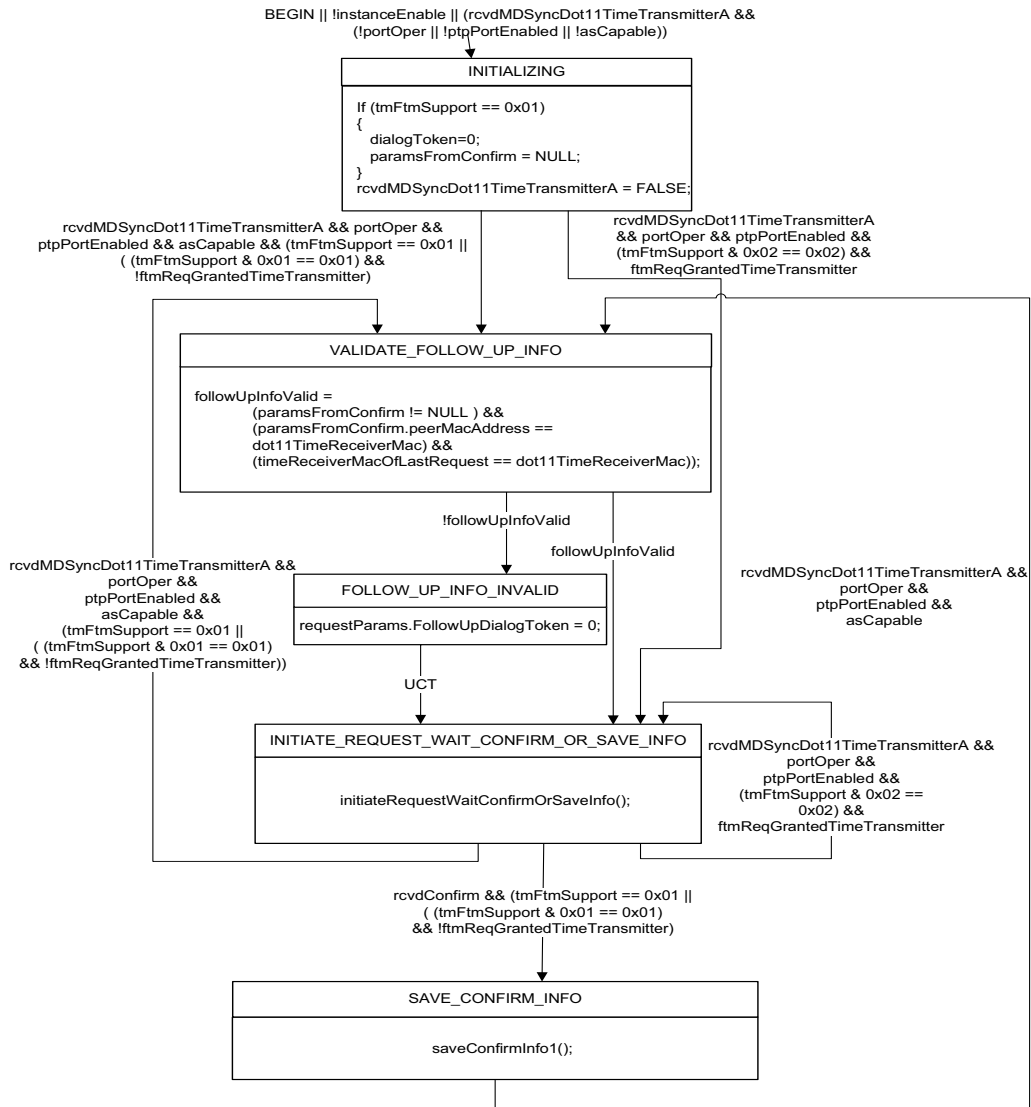


Figure 12-5—TimeTransmitter state machine A
(a) For TM, receives information from the PortSync entity and sends to timeReceiver, and
(b) for FTM, receives and stores information from the PortSync entity

1

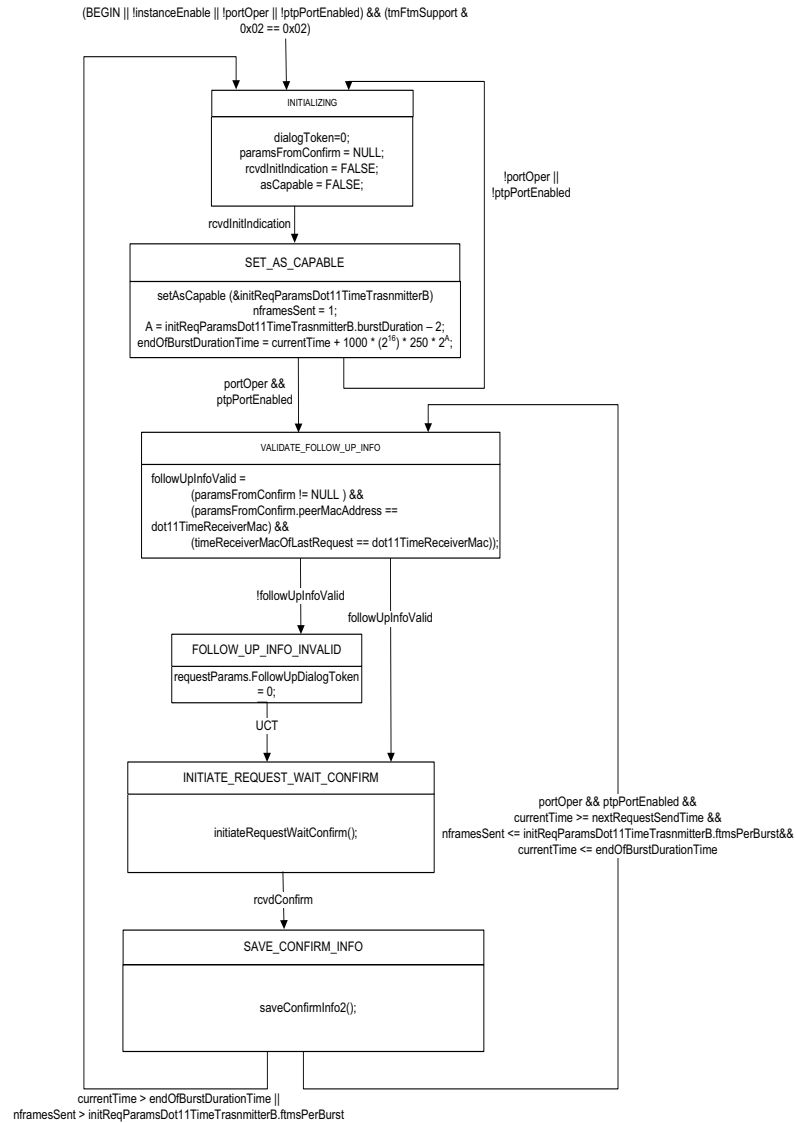


Figure 12-6—TimeTransmitter state machine B

(a) For TM, not invoked and

(b) for FTM, receives initial FTM request from timeReceiver and sends information received from upstream to timeReceiver in successive FTM frames

2 12.5.1.3 State machine local variables

3 **12.5.1.3.1 dialogToken:** An unsigned 8-bit integer used to identify a measurement from among those
4 preceding and following it.

5 **12.5.1.3.2 followUpInfoValid:** A Boolean variable indicating whether the FollowUpInformation (e.g.,
6 timestamps and rateRatio; see 12.7) and the link partner are unchanged since the last Timing Measurement
7 or Fine Timing Measurement.

1 **12.5.1.3.3 requestParams:** A structure whose members contain the values of the fields of either the
2 MLME-TIMINGMSMT.request or MLME-FINETIMINGMSMT.request primitive.

3 **12.5.1.3.4 paramsFromConfirm:** A structure whose members contain the values of the fields of either the
4 MLME-TIMINGMSMT.confirm or MLME-FINETIMINGMSMT.confirm primitive.

5 **12.5.1.3.5 dot11TimeReceiverMac:** The MAC address of the station associated with the current port.

6 **12.5.1.3.6 timeReceiverMacOfLastRequest:** The MAC address of the station of the previous request, used
7 to validate FollowUpInformation.

8 **12.5.1.3.7 residenceTime:** A temporary variable that holds the computation of the time between receipt of
9 the last synchronization information and transmission of synchronization information.

10 **12.5.1.3.8 rcvdMDSyncDot11TimeTransmitterA:** A Boolean variable that is set to TRUE when an
11 MDSyncSend structure is provided by the PortSync entity.

12 **12.5.1.3.9 rcvdConfirm:** A Boolean variable that is set to TRUE when either the MLME-
13 TIMINGMSMT.confirm or MLME-FINETIMINGMSMT.confirm primitive is received.

14 **12.5.1.3.10 nframesSent:** An unsigned 8-bit integer used to count the number of frames sent by the
15 timeReceiver (the number of indications received is counted).

16 **12.5.1.3.11 rcvdInitIndication:** A Boolean variable that is set to TRUE when the initial MLME-
17 FINETIMINGMSMTRQ.indication primitive is received.

18 **12.5.1.3.12 initReqParamsDot11TimeTransmitterB:** A structure whose members contain the values of
19 the fields of an MLME-FINETIMINGMSMTRQ.indication primitive.

20 **12.5.1.3.13 endOfBurstDurationTime:** A UScaledNs variable whose value is the time at which the current
21 burst ends.

22 **12.5.1.3.14 nextRequestSendTime:** A UScaledNs variable whose value is the expected time that the next
23 MLME-FINETIMINGMSMTRQ.indication primitive, for the next request from the timeReceiver, will be
24 received.

25 **12.5.1.4 State machine functions**

26 **12.5.1.4.1 setRequestParams(&requestParams, MDSyncSend):** Assigns values to the parameters of the
27 request primitive of either MLME-TIMINGMSMT or MLME-FINETIMINGMSMT (see 12.5.1.6) as
28 follows:

- 29 a) Members of the FollowUpInformation member of the VendorSpecific information element, as
30 defined in 12.7, are assigned as defined in 11.4.4 with the exception of the correctionField, which is
31 assigned, for TM, by the function saveConfirmInfo1() in TimeTransmitter state machine A (see
32 Figure 12-5) and, for FTM, by the function saveConfirmInfo2() in TimeTransmitter state machine B
33 (see Figure 12-6).
- 34 b) The other fields of the VendorSpecific information element are assigned as follows:
 - 35 1) ElementID is assigned the value 221 as defined in Table 9-77 (Element IDs) of
36 IEEE Std 802.11-2016, indicating that the information element is of type Vendor Specific.
 - 37 2) The Length field is set to 80.

38 NOTE—This is equal to the length of the Follow_Up payload defined in 11.4.4 (including the common header) plus the
39 length of the OUI or CID field and the Type field (see Figure 12-8).

- 1 3) The OUI or CID field is set to 00-80-C2.
- 2 4) The Type field is set to 0.
- 3 c) Max t1 Error, Maxt4 Error are set to zero.
- 4 d) For Fine Timing Measurement frames, the location configuration information (LCI) Report and
- 5 Location Civic Report are not present. All other members are left unchanged.

6 **12.5.1.4.2 setAsCapable (&initReqParamsDot11TimeTransmitterB):** Determines the value of asCapable
7 consistent with 12.4 and whether the timeTransmitter is able to grant the parameters requested by the
8 timeReceiver. This function is used only for FTM.

9 **12.5.1.4.3 initiateRequestWaitConfirmOrSaveInfo():** This function is defined as indicated below. It is
10 used in TimeTransmitter state machine A. It is defined here so that the detailed code that it invokes does not
11 need to be placed into the state machine diagram.

```

12     initiateRequestWaitConfirmOrSaveInfo()
13     {
14         rcvdMDSyncDot11TimeTransmitterA = FALSE;
15
16         If (tmFtmSupport == 0x01)
17         {
18             if ((++dialogToken % 256) == 0) dialogToken++;
19             requestParams.DialogToken=dialogToken;
20             requestParams.PeerMACAddress = dot11TimeReceiverMac;
21             setRequestParams(&requestParams, MDSyncSend);
22             MLME-TIMINGMSMT.request(requestParams);
23             requestParams.FollowUpDialogToken = 0;
24             //In case no confirm is received
25             timeReceiverMacOfLastRequest = dot11TimeReceiverMac;
26         }
27     }

```

28 **12.5.1.4.4 saveConfirmInfo1():** This function is defined as indicated below. It is used in TimeTransmitter
29 state machine A. It is defined here so that the detailed code that it invokes does not need to be placed into the
30 state machine diagram.

```

31     saveConfirmInfo1()
32     {
33         MLME-TIMINGMSMT.confirm(&paramsFromConfirm);
34
35         requestParams.FollowUpDialogToken = paramsFromConfirm.DialogToken;
36         requestParams.T1 = paramsFromConfirm.T1;
37         requestParams.T4 = paramsFromConfirm.T4;
38
39         // NOTE: In Timing Measurement, T1 is in units of 10 ns.
40         // upstreamTxTime is units of 2-16 ns.
41
42         K = 1;
43         // K is 1 for Timing Measurement.
44         residenceTime = MDSyncSend.rateRatio *
45             (paramsFromConfirm.T1 * 10K*(216) - MDSyncSend.upstreamTxTime);
46
47         requestParams.VendorSpecific.correctionField =
48             residenceTime + MDSyncSend.followUpCorrectionField;
49         // NOTE: T1 and T4 are timestamps from a single
50         // local clock source. The roll-over of the 32-bit timestamps returned by
51         // MLME-TIMINGMSMT.request and MLME-TIMINGMSMT.indication

```

```

1          // must be accounted for.
2      }

```

3 **12.5.1.4.5 initiateRequestWaitConfirm():** This function is defined as indicated below. It is used in
4 TimeTransmitter state machine B. It is defined here so that the detailed code that it invokes does not need to
5 be placed into the state machine diagram.

```

6      initiateRequestWaitConfirm()
7      {
8          If ((++dialogToken % 256) == 0) dialogToken++;
9          If (nframesSent == initReqParamsDot11TimeTransmitterB.ftmsPerBurst)
10             dialogToken = 0;
11
12             requestParams.DialogToken=dialogToken;
13             requestParams.PeerMACAddress = dot11TimeReceiverMac;
14             setRequestParams(&requestParams, MDSyncSend);
15             // In the following statement, MinDeltaFTM, which is in units of 100
16             // microseconds, is converted to UScaledNs (i.e., units of 2-16 ns; see 6.3.3.2)
17             nextRequestSendTime = currentTime +
18                 initReqParamsDot11TimeTransmitterB.MinDeltaFTM * (65536 x 105);
19             MLME-FINETIMINGMSMT.request(requestParams);
20             requestParams.FollowUpDialogToken = 0; //In case no confirm is received
21             timeReceiverMacOfLastRequest = dot11TimeReceiverMac;
22     }

```

23 **12.5.1.4.6 saveConfirmInfo2():** This function is defined as indicated below. It is used in TimeTransmitter
24 state machine B. It is defined here so that the detailed code that it invokes does not need to be placed into the
25 state machine diagram.

```

26      saveConfirmInfo2()
27      {
28          MLME-FINETIMINGMSMT.confirm(&paramsFromConfirm);
29
30          requestParams.FollowUpDialogToken = paramsFromConfirm.DialogToken;
31          requestParams.T1 = paramsFromConfirm.T1;
32          requestParams.T4 = paramsFromConfirm.T4;
33
34          // NOTE: In Fine Timing Measurement, T1 is in units of 0.1 ns.
35          // upstreamTxTime is units of 2-16 ns.
36
37          K = -3;
38          // K is 1 for Timing Measurement and -3 for Fine Timing Measurement.
39          residenceTime = MDSyncSend.rateRatio *
40              (paramsFromConfirm.T1 * 10K*(216) - MDSyncSend.upstreamTxTime);
41
42          requestParams.VendorSpecific.correctionField =
43              residenceTime + MDSyncSend.followUpCorrectionField;
44          // NOTE: T1 and T4 are timestamps from a single
45          // local clock source. The roll-over of the 48-bit timestamps returned by
46          // MLME-FINETIMINGMSMT.request and MLME-FINETIMINGMSMT.indication
47          // must be accounted for.
48
49          // A frame is only counted as being sent, for purposes of number of frames in a
50          // burst, if a confirm is received. It is up to IEEE Std 802.11-2016 to handle the
51          // case where a confirm is not received.
52          nframesSent += 1;
53      }

```


1 12.5.1.5 Shared variables

2 12.5.1.5.1 **MDSyncSend**: A structure as defined in 10.2.2.1.

3 12.5.1.5.2 **portOper**: A Boolean as defined in 10.2.5.12.

4 12.5.1.5.3 **ptpPortEnabled**: A Boolean as defined in 10.2.5.13.

5 12.5.1.5.4 **asCapable**: A Boolean whose value is specified in 12.4 and 10.2.5.1.

6 12.5.1.5.5 **tmFtmSupport**: An Octet whose bits are interpreted as Booleans and whose values are specified
7 in Table 12-1.

8 12.5.1.5.6 **ftmReqGrantedTimeTransmitter**: A Boolean whose value is TRUE if the timeTransmitter
9 grants the current initial FTM request and FALSE otherwise.

10 12.5.1.6 TimeTransmitter primitives

11 12.5.1.6.1 MLME-TIMINGMSMT.request

12 The MLME-TIMINGMSMT.request primitive is used by a timeTransmitter station to initiate a Timing
13 Measurement and also communicates timestamps t1 and t4 captured by the timeTransmitter during a
14 previous measurement. The primitive and its parameters are specified in 6.3.57.2 of IEEE Std 802.11-2016.

15 12.5.1.6.2 MLME-TIMINGMSMT.confirm

16 The MLME-TIMINGMSMT.confirm primitive indicates that a Timing Measurement request has completed.
17 The primitive and its parameters are specified in 6.3.57.3 of IEEE Std 802.11-2016.

18 12.5.1.6.3 MLME-FINETIMINGMSMT.request

19 The MLME-FINETIMINGMSMT request primitive is used by a timeTransmitter station to initiate a Fine
20 Timing Measurement and also communicates timestamps t1 and t4 captured by the timeTransmitter during a
21 previous measurement. The primitive and its parameters are specified in 6.3.58.2 of IEEE Std 802.11-2016.

22 12.5.1.6.4 MLME-FINETIMINGMSMT.confirm

23 The MLME-FINETIMINGMSMT request primitive indicates that a Fine Timing Measurement request has
24 completed. The primitive and its parameters are specified in 6.3.58.3 of IEEE Std 802.11-2016.

25 12.5.1.6.5 MLME-FINETIMINGMSMTRQ.indication

26 The MLME-FINETIMINGMSMTRQ.indication primitive indicates to a timeTransmitter station that the
27 timeReceiver is requesting a burst of FTM frames with the indicated parameters. The primitive and its
28 parameters are specified in 6.3.70.3 of IEEE Std 802.11-2016.

29 12.5.2 Media-dependent timeReceiver state machine

30 12.5.2.1 Overview

31 The MD entity of an IEEE 802.11 port whose PTP Port state is TimeReceiverPort or PassivePort (see 10.3.6)
32 shall behave in a manner that is indistinguishable, relative to an observer external to a system, from a strict
33 implementation of the timeReceiver state machine in 12.5.2.2, the local variables specified in 12.5.2.3, the

1 functions specified in 12.5.2.4, the shared variables specified in 12.5.2.5, and the primitives defined in
2 12.5.2.6.

3 The timeReceiver state machine is responsible for collecting information from the Timing measurement or
4 Fine Timing measurement indications, constructing an MDSyncReceive structure with the relevant
5 information, and passing the structure to the PortSync entity for further processing. In order to do this, the
6 state machine saves locally captured timestamps (i.e., t_2 and t_3) received in the indication and associates
7 them with the timestamps sent from the timeTransmitter port in a future indication (i.e., t_1 and t_4). In
8 addition, for Fine Timing measurement, the timeReceiver state machine is responsible for generating the
9 MLME-FINETIMINGMSMTRQ.request primitive, which causes the initial FTM request frame to be sent
10 to the timeTransmitter.

11 12.5.2.2 State diagram

12 Figure 12-7 presents the timeReceiver state machine. While quantities are shown to be computed from
13 information in consecutive indications, an implementation can choose to compute over longer intervals as
14 long as the clock performance requirements of Annex B are met.

1

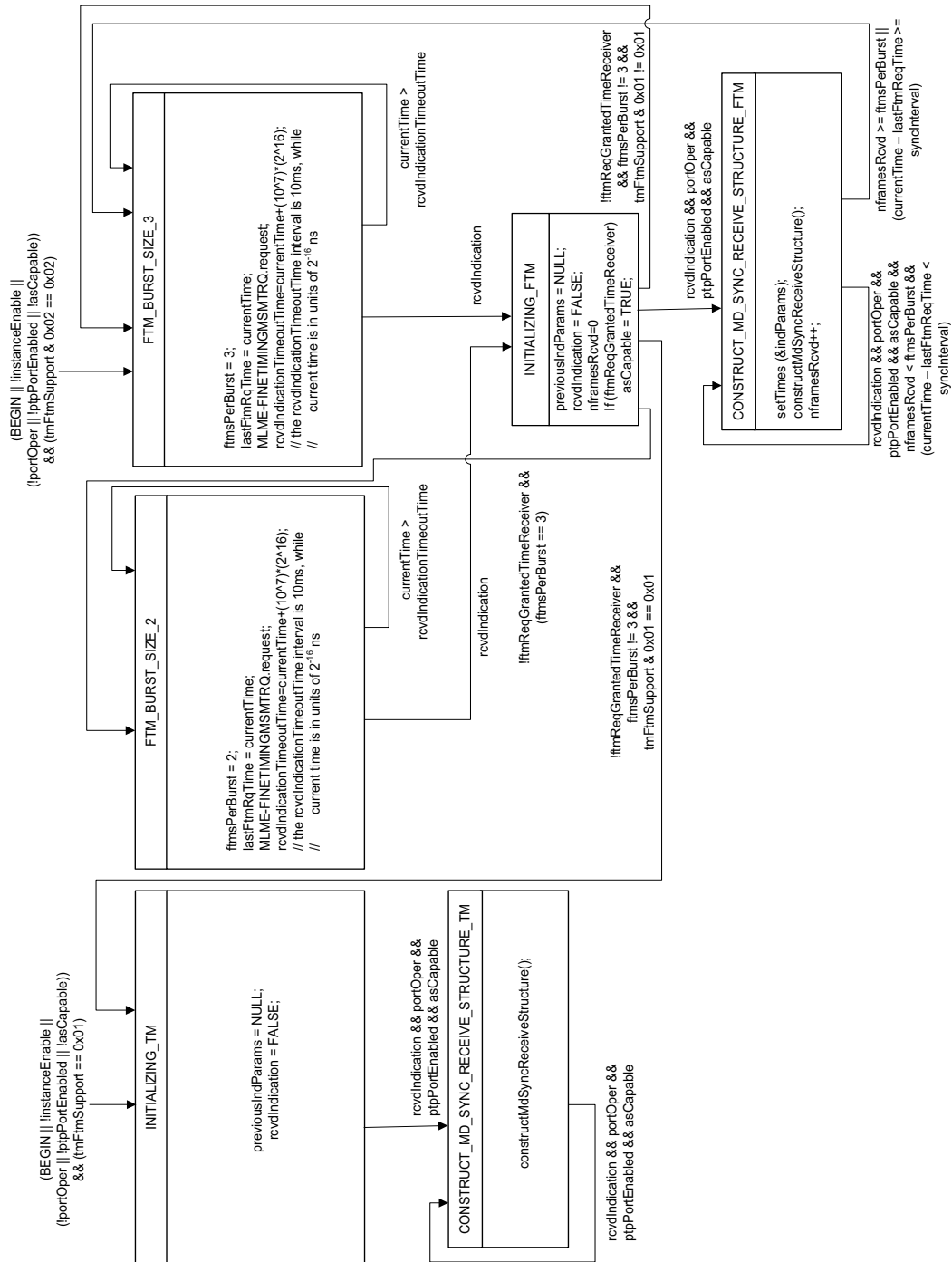


Figure 12-7—TimeReceiver state machine

1 12.5.2.3 State machine local variables

2 **12.5.2.3.1 indParams:** A structure whose members contain the values of the fields of the MLME-
3 TIMINGMSMT.indication primitive or MLME-FINETIMINGMSMT.indication primitive, as defined in
4 12.5.2.6, depending on whether TM or FTM, respectively, is used.

5 **12.5.2.3.2 previousIndParams:** A structure with members identical to those of indParams, used to save
6 parameters from the previous indication.

7 **12.5.2.3.3 rcvdIndication:** A Boolean that is set to TRUE when either the MLME-
8 TIMINGMSMT.indication or MLME-FINETIMINGMSMT.indication primitive is received.

9 **12.5.2.3.4 rcvdIndicationTimeoutTime:** A UScaledNs variable whose value is the time after which the
10 state machine will not wait any longer for the next MLME-FINETIMINGMSMT.indication, and a new burst
11 is initiated.

12 **12.5.2.3.5 nframesRcvd:** An unsigned 8-bit integer used to count the number of frames received by the
13 timeTransmitter in the burst (the number of indications received from the timeTransmitter are counted).

14 **12.5.2.3.6 initReqParamsDot11TimeReceiver:** A structure whose members contain the values of the fields
15 of an MLME-FINETIMINGMSMTRQ.request primitive.

16 **12.5.2.3.7 ftmsPerBurst:** The value of the FTM parameter ‘FTMs per burst’ (see 12.6), i.e., the number of
17 FTM frames in the burst granted by the timeTransmitter.

18 **12.5.2.3.8 ftmReqGrantedTimeReceiver:** A Boolean that is TRUE if the timeTransmitter has granted the
19 respective request for a burst and FALSE otherwise.

20 **12.5.2.3.9 t1_1, t1_2, t2_1, t2_2, t2_3, t3_1, t3_2, t3_3, t4_1, t4_2:** Temporary local variables used to hold
21 the values of the timestamps t1, t2, t3, and t4 returned by the MLME-FINETIMINGMSMT.indication
22 primitive in the structure indParams.

23 **12.5.2.3.11 D1, D2:** Temporary local variables used to hold the values of delay computed for the first two
24 FTM frames and corresponding Acks when the timeTransmitter grants the request for three FTM frames.

25 12.5.2.4 State machine functions

26 **12.5.2.4.1 setMDSyncReceiveDot11TimeReceiver(indParams):** Creates an MDSyncReceive structure
27 and returns the structure. All fields are assigned from FollowUpInformation (contained in the
28 VendorSpecific information element) of indParams as in 11.2.14.2.1.

29 **12.5.2.4.2 passMDSyncReceiveToPortSync():** Passes an MDSyncReceive structure to the PortSync entity
30 of this PTP Port.

31 **12.5.2.4.3 setTimes (&indParams):** extracts the timestamp values from the successive indParams
32 structures returned by the multiple FTM frame exchanges of a burst, and places the correct times
33 (corresponding to minimum delay frames and Acks) in the final indParams structure. This final indParams
34 structure is then used in the function constructMdSyncReceiveStructure(). This procedure is needed when
35 the timeTransmitter grants three FTM frames for the burst. If the timeTransmitter grants only two FTM
36 frames for the burst, the timestamp values returned in the indication primitive of the second frame are used.
37 The function setTimes is used in the TimeReceiver state machine and is defined here so that the detailed
38 code that it invokes does not need to be placed into the state machine diagram.

39 setTimes (&indParams)

```

1      {
2          if (nframesRcvd == 1)
3          {
4              t2_1 = indParams.T2;
5              t3_1 = indParams.T3;
6          }
7          if (nframesRcvd == 2)
8          {
9              t1_1 = indParams.T1;
10             t2_2 = indParams.T2;
11             t3_2 = indParams.T3;
12             t4_1 = indParams.T4
13         }
14         if (nframesRcvd == 3)
15         {
16             t1_2 = indParams.T1;
17             t2_3 = indParams.T2;
18             t3_3 = indParams.T3;
19             t4_2 = indParams.T4;
20             if (ftmsPerBurst == 3)
21             {
22                 D1 = t2_1 - t1_1;
23                 D2 = t2_2 - t1_2
24                 if (D2 <= D1)
25                 {
26                     indParams.T2 = t2_2;
27                     indParams.T1 = t1_2;
28                 }
29                 else
30                 {
31                     indParams.T2 = t2_1;
32                     indParams.T1 = t1_1;
33                 }
34                 D1 = t4_1 - t3_1;
35                 D2 = t4_2 - t3_2
36                 if (D2 <= D1)
37                 {
38                     indParams.T4 = t4_2;
39                     indParams.T3 = t3_2;
40                 }
41                 else
42                 {
43                     indParams.T4 = t4_1;
44                     indParams.T3 = t3_1;
45                 }
46             }
47         }
48     }

```

49 **12.5.2.4.4 constructMdSyncReceiveStructure():** This function constructs the MD SyncReceive structure
50 and is defined as indicated below. It is used in the TimeReceiver state machine. It is defined here so that the
51 detailed code that it invokes does not need to be placed into the state machine diagram.

```

52     constructMdSyncReceiveStructure()
53     {
54         if (tmFtmSupport == 0x01)
55             MLME-TIMINGMSMT.indication(&indParams);
56         else if (tmFtmSupport & 0x02 == 0x02)
57         {

```

```

1      MLME-FINETIMINGMSMT.indication(&indParams);
2      }
3
4      if ((previousIndParams != NULL) &&
5          (previousIndParams.PeerMacAddress == dot11TimeReceiverMac) &&
6          (indParams.FollowUpDialogToken != 0))
7      {
8
9          neighborRateRatio =
10             (indParams.T1-previousIndParams.T1) /
11             (indParams.T2-previousIndParams.T2);
12          //NOTE: Other methods of computing neighborRateRatio
13             can be used.
14
15          if (tmFtmSupport == 0x01)
16              K = 1;
17          else if (tmFtmSupport & 0x02 == 0x02)
18              K = -3;
19          //K = 1 for Timing Measurement and K = -3 for Fine Timing Measurement
20          meanLinkDelay =
21              (((indParams.T4 - indParams.T1) -
22               neighborRateRatio * (indParams.T3 - indParams.T2)) /
23               (2.0)) * (10K);
24
25          //NOTE: Other methods of computing meanLinkDelay
26             can be used.
27
28          MDSyncReceive = setMDSyncReceiveDot11TimeReceiver(indParams);
29          MDSyncReceive.VendorSpecific.rateRatio +=
30              (neighborRateRatio - 1);
31          MDSyncReceive.VendorSpecific.upstreamTxTime =
32              indParams.T2 * (216) * (10K) -
33              meanLinkDelay * (216) / neighborRateRatio;
34          //NOTE: Actions performed with the timestampError
35             parameters of indParams are implementation independent.
36
37          passMDSyncReceiveToPortSync(&MDSyncReceive);
38      }
39      previousIndParams = indParams;
40      rcvdIndication = FALSE;
41  }

```

42 12.5.2.5 State machine shared variables

43 **12.5.2.5.1 MDSyncReceive:** A structure used for passing information between MD and PortSync, as
44 defined in 10.2.2.2.

45 **12.5.2.5.2 portOper:** A Boolean as defined in 10.2.5.12.

46 **12.5.2.5.3 ptpPortEnabled:** A Boolean as defined in 10.2.5.

47 **12.5.2.5.4 asCapable:** A Boolean as defined in 12.4 and 10.2.5.1.

48 **12.5.2.5.5 meanLinkDelay:** The delay over the link to the associated WLAN station as defined in 10.2.5.8.

1 **12.5.2.5.6 neighborRateRatio:** The measured ratio of the frequency of the LocalClock entity of the time-
2 aware system at the other end of the link attached to this PTP Port, to the frequency of the LocalClock entity
3 of this time-aware system, as defined in 10.2.5.7.

4 **12.5.2.5.7 tmFtmSupport:** An Octet whose value is specified in 12.3.

5 **12.5.2.6 TimeReceiver primitives**

6 **12.5.2.6.1 MLME-TIMINGMSMT.indication**

7 The MLME-TIMINGMSMT.indication primitive is received by a timeReceiver station as the natural result
8 of the peer timeTransmitter station issuing the corresponding request primitive and carries the same
9 parameters plus local timestamp information. The primitive and its parameters are specified in 6.3.57.4 of
10 IEEE Std 802.11-2016.

11 **12.5.2.6.2 MLME-FINETIMINGMSMT.indication**

12 The MLME-FINETIMINGMSMT.indication primitive is received by a timeReceiver station as the natural
13 result of the peer timeTransmitter station issuing the corresponding request primitive and carries the same
14 parameters plus local timestamp information. The primitive and its parameters are specified in 6.3.58.4 of
15 IEEE Std 802.11-2016.

16 **12.5.2.6.3 MLME-FINETIMINGMSMTRQ.request**

17 The MLME-FINETIMINGMSMTRQ.request primitive is used by the timeReceiver to request a burst of
18 FTM frames from the timeTransmitter, with respective FTM parameters. The primitive and its parameters
19 are specified in 6.3.70.2 of IEEE Std 802.11-2016.

20 **12.6 FTM parameters**

21 The values of the FTM parameters that are relevant to time synchronization transport in this standard are
22 specified in Table 12-2, along with a brief description of each parameter. These parameter values are carried
23 in the initial FTM Request. More detailed descriptions of these and other FTM parameters are given in
24 9.4.2.168 of IEEE Std 802.11-2016. The parameters Burst Duration and Min Delta FTM depend on the time
25 synchronization message interval, i.e., on currentLogSyncInterval (see 10.7.2.3 and 12.8). The values for
26 these parameters are specified in Table 12-3. In this table, the Burst Duration encoding (i.e., value and
27 corresponding duration in ms) is taken from Table 9-257 of IEEE Std 802.11-2016. The Min Delta FTM
28 encoding values are in multiples of 100 μ s (see 9.4.2.168 of IEEE Std 802.11-2016).

29 The values of the FTM parameters given in Table 12-2 shall be used in the MLME-
30 FINETIMINGMSMTRQ.request invoked by the timeReceiver STA (i.e., in the initial FTM Request). The
31 values for Burst Duration and Min Delta FTM given in Table 12-3 shall be used in the MLME-
32 FINETIMINGMSMTRQ.request invoked by the timeReceiver STA.

33 Background on the derivation of the FTM parameters of Table 12-2 and Table 12-3 is given in Garner [B4].

1

Table 12-2—FTM parameters relevant to time-synchronization transport

Parameter	Value	Description
Number of Bursts Exponent	0	Log to base 2 of the number of bursts requested by the timeReceiver (value of 0 indicates that one burst is requested)
Burst Duration	See Table 12-3	Duration of the burst of FTM frames and their corresponding Acks
Min Delta FTM	See Table 12-3	Minimum time between consecutive FTM frames
Partial TSF Timer	1	See 9.4.2.168 of IEEE Std 802.11-2016
Partial TSF Timer No Preference	Reserved in the initial FTM request	See 9.4.2.168 of IEEE Std 802.11-2016
ASAP	1	ASAP = 1 indicates that the timeReceiver would like the timeTransmitter to respond as soon as possible
ASAP Capable	Reserved in the initial FTM request	See 9.4.2.168 of IEEE Std 802.11-2016
FTMs per burst	3 in the first initial FTM Request 2 in the first retry, if the first initial FTM Request is not granted	Desired number of FTM frames and corresponding Acks in the requested burst
Burst Period	Reserved when Number of Bursts Exponent is zero	See 9.4.2.168 of IEEE Std 802.11-2016

2

Table 12-3—Values of Burst Duration and Min Delta FTM, for each value of currentLogSyncInterval

currentLogSyncInterval	Nominal message rate (messages/s)	Burst duration	Min delta FTM
–24 through –6, inclusive	64 through 16 777 216	6 (4 ms)	6 (0.6 ms)
–5	32	8 (16 ms)	25 (2.5 ms)
–4	16	9 (32ms)	50 (5 ms)
–3	8	10 (64 ms)	100 (10 ms)
–2 through 24, inclusive	4	11 (128 ms)	200 (20 ms)

1 **12.7 Format of VendorSpecific information element**

2 The IEEE 802.11 MLME request and indication primitives for Timing Measurement and Fine Timing
3 Measurement support an ability to carry data transparently between stations using the VendorSpecific
4 information element. The Type field within the VendorSpecific Content identifies the type of information
5 that follows the Type field. See Figure 12-8 and Table 12-4.

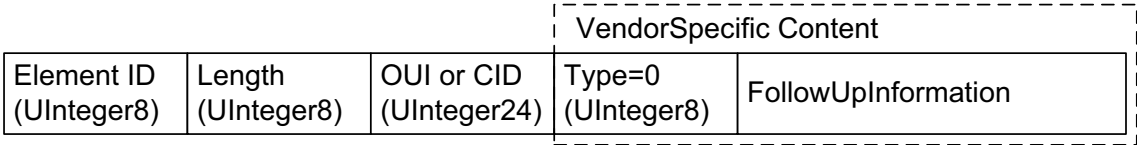


Figure 12-8—Format of VendorSpecific information element when Type = 0

Table 12-4—Values of the Type field in the VendorSpecific information element

Value	Description
0	The Type field is followed by FollowUp-Information
1–255	Reserved

6 This mechanism shall be used to carry end-to-end link-independent timing information from the
7 timeTransmitter port to the associated timeReceiver port, including preciseOriginTimestamp, rateRatio,
8 correctionField, and other fields of the Follow-Up message, as described in 12.5.1.4. For consistency, all of
9 these fields are packed into the FollowUpInformation field using exactly the same format as used for full-
10 duplex point-to-point links. In other words, the timeTransmitter state machine communicates an entire
11 Follow_Up message [i.e., including all the fields of the common header (see 11.4.2 and 10.6.2), the
12 preciseOriginTimestamp, and all the fields of the Follow_Up information TLV (see 11.4.4)] using this
13 mechanism. The Type field, illustrated in Figure 12-8, identifies this use of the OUI or CID within the
14 VendorSpecific information element. Table 12-4 lists values for the Type field.

15 **12.8 Synchronization message interval**

16 **12.8.1 General synchronization message interval specification**

17 The mean time interval between successive synchronization messages shall be as specified in 10.7.2.1,
18 10.7.2.3, and 12.8.2.

19 **12.8.2 Synchronization message interval default value**

20 The default value of initialLogSyncInterval (see 10.7.2.3) is –3. Every PTP Port supports the value 127; the
21 PTP Port does not send Sync messages when currentLogSyncInterval has this value (see 10.3.18). A PTP
22 Port may support other values, except for the reserved values indicated in Table 10-16. A PTP Port ignores
23 requests (see 10.3.18) for unsupported values.

24 Processing of the message interval request TLV carried in a Signaling message (see 10.6.4) shall be
25 supported, as specified by the SyncIntervalSetting state machine of 10.3.18 (see Figure 10-20), except that

1 the logLinkDelayInterval (which is not relevant to IEEE 802.11 ports) is set to –128 by the sender of the
2 Signaling message, the logLinkDelayInterval is ignored by the receiver, and unsupported values of
3 logTimeSyncInterval are ignored by the receiver.

4 NOTE 1—For TM, a timeReceiver port that requests (using a Signaling message that contains a message interval
5 request TLV; see 10.6.4 and 10.3.18) that the PTP Port at the other end of the attached link set its
6 currentLogSyncInterval to a specific value can determine if the request was honored by examining the
7 logMessageInterval field of a FollowUpInformation contained in the VendorSpecific information element of a
8 subsequent MLME indication primitive.

9 NOTE 2—The time interval between every pair of adjacent timing or Fine Timing Measurements is not guaranteed to be
10 precisely the same. Some variation is expected, due to factors such as the MAC protocol (e.g., delay in accessing the
11 medium and/or packet retries) and the power state of the associated station. However, timestamp t1 is not captured when
12 either TIMINGMSMT.request or FINETIMINGMSMT.request is invoked, but only after the frame resulting from the
13 request is actually transmitted.

1 13. Media-dependent layer specification for interface to IEEE 802.3 Ethernet 2 passive optical network link

3 13.1 Overview

4 13.1.1 General

5 This clause specifies the service interface primitives, state machines, and message formats that provide
6 accurate synchronized time across IEEE 802.3 Ethernet passive optical network (EPON) links, through the
7 use of the timing process and measurements specified in 64.2.1.1, 64.3.2.4, 77.2.1.1, and 76.1.2 of
8 IEEE Std 802.3-2022. For purposes of this clause, an EPON link is an EPON that contains one optical line
9 terminal (OLT) and associated optical network units (ONUs).

10 A time-aware system may contain more than one OLT and/or ONU. Each PTP Instance of a time-aware
11 system uses at most one ONU port, but may serve, i.e., provide timing to, more than one OLT port (i.e., each
12 PTP Instance of a time-aware system is a clock timeReceiver to at most one EPON link, but can be clock
13 timeTransmitter to more than one EPON link). Two different PTP Instances of a time-aware system may use
14 different ONU ports.

15 13.1.2 Description of the EPON timing process

16 The timing process in EPON relies on the 32-bit counters (see 64.2.2.2 and 77.2.2.2 of IEEE Std
17 802.3-2022) at both the OLT and the ONU. The 32-bit counter used by EPON is the LocalClock entity of the
18 PTP Instance that uses the respective OLT or ONU. These counters increment every time_quantum, which is
19 equal to 16 ns (see 64.2.2.1 and 77.2.2.1 of IEEE Std 802.3-2022). IEEE Std 802.3-2022 defines multipoint
20 control protocol (MPCP), which is one of the protocols that enable MAC clients to communicate over a
21 point-to-multipoint optical network. When either the clock timeTransmitter (OLT) or the clock timeReceiver
22 (ONU) transmits an MPCP data unit (MPCPDU), its counter value is mapped into the timestamp field.
23 Clause 64 and Clause 77 of IEEE Std 802.3-2022 specify the EPON timing mechanism.

24 13.1.3 Best timeTransmitter selection

25 13.1.3.1 General

26 An EPON link contains one OLT and the associated ONUs. The OLT is the clock timeTransmitter and the
27 associated ONUs are clock timeReceivers. The OLT initiates the time synchronization as a requester. The
28 ONUs are the responders of the time synchronization. In other words, the invocation of the BTCA results in
29 the OLT having the PTP Port state TimeTransmitterPort and the ONU having the PTP Port state
30 TimeReceiverPort (see 10.3.1.1 and Table 10-2), for all PTP Instances using these PTP Ports, regardless of
31 the attributes of PTP Instances downstream from the ONU. This behavior is achieved using the acceptable
32 timeTransmitter table feature defined in 17.5 of IEEE Std 1588-2019.

33 A PTP Instance that contains an ONU port shall maintain a configured table, the
34 acceptableTimeTransmitterTable, and a per-PTP Port Boolean variable
35 acceptableTimeTransmitterTableEnabled. The data type of acceptableTimeTransmitterTable is
36 AcceptableTimeTransmitterTable (see 13.1.3.2).

1 13.1.3.2 AcceptableTimeTransmitterTable

2 The AcceptableTimeTransmitterTable type represents a table of AcceptableTimeTransmitter entries.

```
3 struct AcceptableTimeTransmitterTable{
4     UInteger16 maxTableSize;
5     UInteger16 actualTableSize;
6     AcceptableTimeTransmitter[actualTableSize] acceptableTimeTransmitter;
7 }
```

8 The maxTableSize member is the maximum size of the AcceptableTimeTransmitterTable. The
9 actualTableSizeMember is the actual size of the AcceptableTimeTransmitterTable. The
10 AcceptableTimeTransmitter array contains a list of AcceptableTimeTransmitter PTP Ports. The value of
11 maxTableSize is implementation specific. actualTableSize shall be less than or equal to maxTableSize.

12 An AcceptableTimeTransmitterTable is configurable and may contain a number of
13 AcceptableTimeTransmitter entries up to maxTableSize.

14 13.1.3.3 AcceptableTimeTransmitter

15 The AcceptableTimeTransmitter type represents a PTP Port that can be considered, in the execution of the
16 BTCA, as a candidate for timeTransmitter.

```
17 struct AcceptableTimeTransmitter{
18     PortIdentity acceptablePortIdentity;
19     UInteger8 alternatePriority1;
20 }
```

21 The acceptablePortIdentity member is the PortIdentity of an acceptable timeTransmitter port. The
22 alternatePriority1 member contains an alternate value for the priority1 attribute of the acceptable
23 timeTransmitter port (see 13.1.3.4).

24 13.1.3.4 Acceptable timeTransmitter table feature

25 The acceptable timeTransmitter table feature shall modify the operation of the BTCA (see 10.3) as follows:

- 26 a) If acceptableTimeTransmitterTableEnabled for a PTP Port is FALSE, the BTCA operates as
27 described in 10.3.
- 28 b) If acceptableTimeTransmitterTableEnabled for a PTP Port is TRUE, then the following apply:
 - 29 1) The function qualifyAnnounce() of the PortAnnounceReceive state machine (see 10.3.11.2.1)
30 is replaced by the following:
31
32 **qualifyAnnounce (rcvdAnnouncePtr):** qualifies the received Announce message pointed to
33 by rcvdAnnouncePtr as follows:
 - 34 i) if the Announce message was sent by the current PTP Instance, i.e., if
35 sourcePortIdentity.clockIdentity (see 10.6.2.2.11 and 8.5.2) is equal to thisClock (see
36 10.2.4.22), the Announce message is not qualified, and FALSE is returned;
 - 37 ii) if the stepsRemoved field is greater than or equal to 255, the Announce message is not
38 qualified, and FALSE is returned;
 - 39 iii) if the sourcePortIdentity of the Announce message is not equal to the sourcePortIdentity
40 of one of the entries of the acceptableTimeTransmitterTable, FALSE is returned;
 - 41 iv) if a path trace TLV is present and one of the elements of the pathSequence array field of
42 the path trace TLV is equal to thisClock (i.e., the clockIdentity of the current PTP
43 Instance; see 10.2.4.22), the Announce message is not qualified, and FALSE is returned;

- 1 otherwise, the Announce message is qualified, and TRUE is returned. If a path trace
 2 TLV is present, it is saved in the per port global variable receivedPathTrace. If a path
 3 trace TLV is not present, the per port global variable receivedPathTrace is set to the
 4 empty array.
- 5 2) If the alternatePriority1 member of the AcceptableTimeTransmitter array element that
 6 corresponds to the sourcePortIdentity of a received Announce message is 0, the
 7 alternatePriority1 member has no effect on the operation of the BTCA.
- 8 3) If the alternatePriority1 member of the AcceptableTimeTransmitter array element that
 9 corresponds to the sourcePortIdentity of a received Announce message is greater than 0, the
 10 value of the grandmasterPriority1 field of the Announce message is replaced by the value of
 11 alternatePriority1 of this AcceptableTimeTransmitter array element for use in the invocation of
 12 the BTCA.

13 13.1.3.5 Default configuration of acceptable timeTransmitter table feature

14 The default configuration of the acceptable timeTransmitter table feature for a PTP Instance that is attached
 15 to an IEEE 802.3 EPON link shall be as follows:

- 16 a) If the PTP Instance does not contain an ONU port, the default acceptableTimeTransmitterTable is
 17 empty, i.e., the member actualTableSize is 0 and there are no AcceptableTimeTransmitter array
 18 entries. The variable acceptableTimeTransmitterTableEnabled for each PTP Port is set to FALSE.
- 19 b) If the PTP Instance contains an ONU port, the default acceptableTimeTransmitterTable contains one
 20 element in the AcceptableTimeTransmitter array. The member actualTableSize is 1. The
 21 acceptablePortIdentity of that element is set equal to the portIdentity of the OLT port that the ONU
 22 port is attached to, and alternatePriority1 set equal to 244. The variable
 23 acceptableTimeTransmitterTableEnabled for each PTP Port is set to TRUE.

24 NOTE—These default settings ensure that, with the default priority1 values of 8.6.2.1, Table 8-1, used for all PTP
 25 Instances, the PTP Instance that contains the ONU port will consider Announce messages only from the OLT that the
 26 ONU port is attached to when invoking the BTCA. The alternatePriority1 value of 244 ensures that the OLT will be
 27 considered better than the ONU in the sense of the BTCA, which will cause the OLT port state to be set to
 28 TimeTransmitterPort and the ONU port state to be set to TimeReceiverPort. All other PTP Ports of this PTP Instance that
 29 are not disabled and for which asCapable is TRUE will have PTP Port states of either TimeTransmitterPort or
 30 PassivePort. If all PTP Instances downstream from the ONU have priority1 greater than 244, then the PTP Port at the
 31 other end of each link attached to each non-ONU port that is not disabled and for which asCapable is TRUE will have
 32 PTP Port states of either TimeReceiverPort or PassivePort; in this case, the downstream network portions will get their
 33 timing through the EPON. However, if a downstream PTP Instance has priority1 less than 244, or priority1 equal to 244
 34 and is better than the Grandmaster PTP Instance information contained in the Announce message received by the ONU
 35 based on other attributes, then the portion of the network that is downstream of the ONU and includes that better PTP
 36 Instance will get its timing from that better downstream PTP Instance. In this case, the endpoints of the link of that
 37 network portion attached to the PTP Instance that contains the ONU will both have PTP Port states of
 38 TimeTransmitterPort, and the PTP Ports at each end of the link will send Announce messages. However, the Announce
 39 messages sent by the downstream PTP Instance will be ignored by the PTP Instance that contains the ONU because the
 40 sourcePortIdentity of those Announce messages will not be contained in the acceptableTimeTransmitterTable. The
 41 Announce messages sent by the PTP Instance that contains the ONU will be used in the invocation of the BTCA at the
 42 downstream PTP Instance; however, those Announce messages will not reflect the best timeTransmitter because one of
 43 the downstream PTP Instances is better.

44 13.1.4 Time synchronization in EPON

45 Transmission in the EPON downstream direction (from OLT to ONUs) utilizes time division multiplexing
 46 (TDM). In the upstream direction (from ONUs to OLT), time division multiple access (TDMA) is employed.
 47 Due to the frame queuing in TDMA, the downstream delay is different from the upstream delay.
 48 Asymmetric delay also occurs in the EPON physical layer due to upstream and downstream transmission
 49 using different wavelengths. The index of refraction is frequency dependent, which results in the upstream
 50 and downstream delays being asymmetric. The accurate time synchronization across the EPON links is

1 operated as follows. It is assumed that the clock timeTransmitter (the OLT) has an accurate synchronized
2 time. The clock timeTransmitter informs the clock timeReceiver (the ONU) what the accurate synchronized
3 time will be when the counter of the clock timeReceiver reaches a certain value. The information transfer
4 can be accomplished using the organization-specific slow protocol (OSSP) message (see Clause 57 of IEEE
5 Std 802.3-2022).

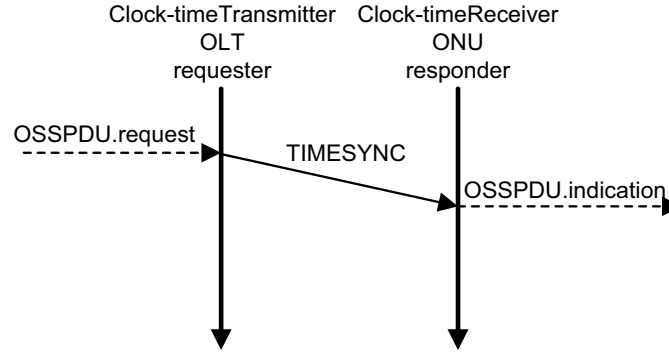


Figure 13-1—IEEE 802.3 EPON time-synchronization interfaces

6 The following reference process, illustrated schematically in Figure 13-1, will result in the clock
7 timeReceiver of an ONU being synchronized to the clock timeTransmitter of the OLT:

- 8 a) The clock timeTransmitter selects a value X of the local MPCP counter that is used as the timing
9 reference. Any value can be chosen, provided it is relative to the current epoch of the MPCP counter.
- 10 b) The clock timeTransmitter calculates the $ToD_{X,i}$ based on $ToD_{X,o}$ using Equation (13-1).

$$11 \quad ToD_{X,i} = ToD_{X,o} + RTT_i \times \frac{ndown}{(nup + ndown)} \times rateRatio \quad (13-1)$$

12 where $ToD_{X,i}$ is the synchronized time when the MPCP counter at the clock timeReceiver i reaches a
13 value equal to the timestamp X minus the *onuLatencyFactor*; $ToD_{X,o}$ is the synchronized time when
14 the MPCP counter at the clock timeTransmitter reaches a value equal to the timestamp X plus the
15 *oltLatencyFactor*; RTT_i is the round-trip time measured by the clock timeTransmitter for clock
16 timeReceiver i , i.e., ONU i ; nup is the effective refraction index of the light propagating in the
17 upstream channel; $ndown$ is the effective refraction index of the light propagating in the downstream
18 channel; and $rateRatio$ is the $rateRatio$ member of the most recently received *MDSyncSend*
19 structure. The *onuLatencyFactor* and *oltLatencyFactor* are given in Equation (13-2) and
20 Equation (13-3), respectively. The impact of the worst-case variation in the transmission wavelength
21 for the clock timeTransmitter and clock timeReceiver transmitters is examined in appendix VII of
22 ITU-T G.984.3, Amendment 2 (11/2009).

$$23 \quad onuLatencyFactor = onuIngressLatency - \quad (13-2)$$

$$(onuIngressLatency + onuEgressLatency) \times \frac{ndown}{(nup + ndown)} \times rateRatio$$

$$24 \quad oltLatencyFactor = oltEgressLatency - \quad (13-3)$$

$$(oltIngressLatency + oltEgressLatency) \times \frac{ndown}{(nup + ndown)} \times rateRatio$$

- 1 c) The clock timeTransmitter sends the pair of values $(X, ToD_{X,i})$ to clock timeReceiver i via the
2 downstream TIMESYNC message.

3 NOTE—After the clock timeReceiver receives the downstream TIMESYNC message, it can compute the synchronized
4 time, ToD , when the value of the local MPCP counter is equal to S ; ToD is given by the following equation:

5
$$ToD = ToD_{X,i} + [(S - X) \bmod (2^{32})](16 \text{ ns})(\text{rateRatio})$$

6 where $(A) \bmod (B)$ is A modulo B .

7 The OSSP message is a general message (see 3.12), analogous to Follow_Up. Note that the preceding
8 synchronized time values correspond to timestamps that are referenced to the MAC control sublayer. Both
9 the clock timeTransmitter and clock timeReceiver are responsible for compensating their processing delays
10 (e.g., the ingressLatency and egressLatency, as described in 8.4.3). RTT_i is measured using MPCPDU
11 timestamps, inserted into the frame structure as specified by 64.2.1.1 and 77.2.1.1 of IEEE Std 802.3-2022.

12 13.2 Message attributes

13 13.2.1 Message class

14 The TIMESYNC message is a general message (see 3.12 and 8.4.2.2). It is transmitted in the downstream
15 direction, from OLT to ONU.

16 13.3 Message format

17 13.3.1 TIMESYNC message

18 13.3.1.1 General TIMESYNC message specifications

19 The fields of the body of the TIMESYNC message shall be as specified in Table 13-1 and 13.3.1.2.

20 13.3.1.2 TIMESYNC message field specifications

21 13.3.1.2.1 Destination address (Octet6)

22 The destination address field is equal to 01-80-C2-00-00-02 (see 57A.3 of IEEE Std 802.3-2022).

23 13.3.1.2.2 Source address (Octet6)

24 The source address field is the individual MAC address associated with the port through which the
25 TIMESYNC message is transmitted (see 57B.1.1 of IEEE Std 802.3-2022).

26 13.3.1.2.3 Length/Type (Octet2)

27 The value of this field is equal to 0x8809 (see 57A.4 of IEEE Std 802.3-2022).

Table 13-1—TIMESYNC message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
Destination Address								6	0
Source Address								6	6
Length/Type								2	12
Subtype								1	14
OUI or CID								3	15
Message Identifier								2	18
<i>X</i>								4	20
<i>ToD_{X,i}</i>								10	24
sourcePortIdentity								10	34
logMessageInterval								1	44
rateRatio								8	45
gmTimeBaseIndicator								2	53
lastGmPhaseChange								12	55
scaledLastGmFreqChange								4	67
domainNumber								1	71
majorSdoId				reserved				1	72
minorSdoId								1	73
reserved								0	74
FCS								4	

1 13.3.1.2.4 Subtype (Octet)

2 The value of this field is equal to 0x0A (see 57A.4 of IEEE Std 802.3-2022).

3 13.3.1.2.5 OUI or CID (Octet3)

4 This field contains the OUI or CID that identifies the Organization-Specific Data. The value is 00-80-C2, 5 i.e., the OUI assigned to IEEE 802.1.

6 13.3.1.2.6 Message identifier (Octet2)

7 This field is the TIMESYNC message identifier. The value of this field is 1.

8 13.3.1.2.7 *X* (UInteger32)

9 The *X* field is the selected timestamp that will be used as the timing reference as specified in 13.1.4.

1 **13.3.1.2.8 $ToD_{X,i}$ (Timestamp)**

2 $ToD_{X,i}$ is the synchronized time when the MPCP counter at the clock timeReceiver i reaches a value equal to
3 X minus the *onuLatencyFactor* (see 13.1.4). X is carried in the respective TIMESYNC message.
4 Synchronization of the MPCP clock is described in detail in 64.2.1.1 and 77.2.1.1 in IEEE Std 802.3-2022,
5 for 1G-EPON and 10G-EPON, respectively.

6 NOTE—Any subnanosecond portion of synchronized time (in this case, time of day), normally transported in a
7 correction field (see 10.2.2.1.2, 10.2.2.2.2, and 10.2.2.3.5), is not transported over EPON.

8 **13.3.1.2.9 sourcePortIdentity (PortIdentity)**

9 This field is specified as the sourcePortIdentity member of the MDSyncSend structure most recently
10 received from the PortSync entity of the OLT (see 10.2.2.1.4).

11 **13.3.1.2.10 logMessageInterval (Integer8)**

12 This field is specified as the logMessageInterval member of the MDSyncSend structure most recently
13 received from the PortSync entity of the OLT (see 10.2.2.1.5). It is the value of the currentLogSyncInterval
14 for this PTP Port (see 10.7.2.3).

15 **13.3.1.2.11 rateRatio (Float64)**

16 This field is specified as the rateRatio member of the MDSyncSend structure most recently received from
17 the PortSync entity of the OLT (see 10.2.2.1.8).

18 **13.3.1.2.12 gmTimeBaseIndicator (UInteger16)**

19 This field is specified as the gmTimeBaseIndicator member of the MDSyncSend structure most recently
20 received from the PortSync entity of the OLT (see 10.2.2.1.9).

21 **13.3.1.2.13 lastGmPhaseChange (ScaledNs)**

22 This field is specified as the lastGmPhaseChange member of the MDSyncSend structure most recently
23 received from the PortSync entity of the OLT (see 10.2.2.1.10).

24 **13.3.1.2.14 scaledLastGmFreqChange (Integer32)**

25 The value of scaledLastGmFreqChange is the fractional frequency offset of the current Grandmaster Clock
26 relative to the previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became
27 the Grandmaster PTP Instance, or relative to itself prior to the last change in gmTimeBaseIndicator,
28 multiplied by 2^{41} and truncated to the next smaller signed integer. The value is obtained by multiplying the
29 lastGmFreqChange member of MDSyncSend (see 10.2.2.1) whose receipt causes the MD entity to send the
30 TIMESYNC message by 2^{41} , and truncating to the next smaller signed integer.

31 NOTE—The above scaling allows the representation of fractional frequency offsets in the range $[-(2^{-10} - 2^{-41}), 2^{-10} -$
32 $2^{-41}]$, with granularity of 2^{-41} . This range is approximately $[-9.766 \times 10^{-4}, 9.766 \times 10^{-4}]$.

33 **13.3.1.2.15 domainNumber (UInteger8)**

34 This field is specified as the gPTP domainNumber (see 8.1).

1 13.3.1.2.16 majorSdoId (Nibble)

2 The value is the same as the value specified in 8.1 for all transmitted PTP messages of a gPTP domain. Any
 3 TIMESYNC message received for which the value is not one of the values specified in 8.1 shall be ignored.

4 NOTE—The nibble that immediately follows majorSdoId is reserved (see 10.6.1).

5 13.3.1.2.17 minorSdoId (UInteger8)

6 The value is the same as the value specified in 8.1 for all transmitted PTP messages of a gPTP domain. Any
 7 TIMESYNC message received for which the value is not one of the values specified in 8.1 shall be ignored.

8 13.3.1.2.18 reserved

9 The reserved field that follows minorSdoId has variable length. The field shall have zero length on
 10 transmission. Any bytes between the minorSdoId and the FCS field shall be ignored on reception.

11 13.3.1.2.19 FCS (Octet4)

12 This field is the frame check sequence (see 57B.1.1 of IEEE Std 802.3-2022).

13 13.4 Determination of asCapable

14 The default value of the per-PTP Port, per-domain global variable asCapable shall be TRUE.

15 The per-PTP Port, per-domain global variable asCapable shall be set to TRUE if the value of
 16 neighborGtpCapable for this PTP Port is TRUE.

17 NOTE—The above conditions ensure backward compatibility with the 2011 edition of this standard. A time-aware
 18 system that is compliant with the 2011 edition of this standard will not process the gPTP capable TLV, and asCapable
 19 will be determined as specified in the 2011 edition. A PTP Instance of a time-aware system compliant with the current
 20 edition of this standard that is attached, via an EPON link, to a node compliant with the 2011 edition of this standard will
 21 not receive Signaling messages that contain the gPTP capable TLV and will not set neighborGtpCapable to TRUE.
 22 However, the above ensures that asCapable for this PTP Port and domain (i.e., domain 0) will still be set in a manner
 23 consistent with that of the 2011 edition of this standard because the default value of asCapable is TRUE in that edition.

24 13.5 Layering for IEEE 802.3 EPON links

25 The MD entity is media-dependent and is responsible for translating the media-independent layer to media-
 26 dependent PDUs or primitives as necessary for communicating synchronized time over the EPON link from
 27 the OLT to a single ONU. This implies that if one OLT port is associated with multiple ONUs, it will require
 28 one IEEE 802.1AS PortSync entity and one MD entity per associated ONU. The OSSPDU primitives are
 29 used to communicate synchronized time information. Figure 13-2 illustrates how the MD entity interacts
 30 with the OSSP sublayer.

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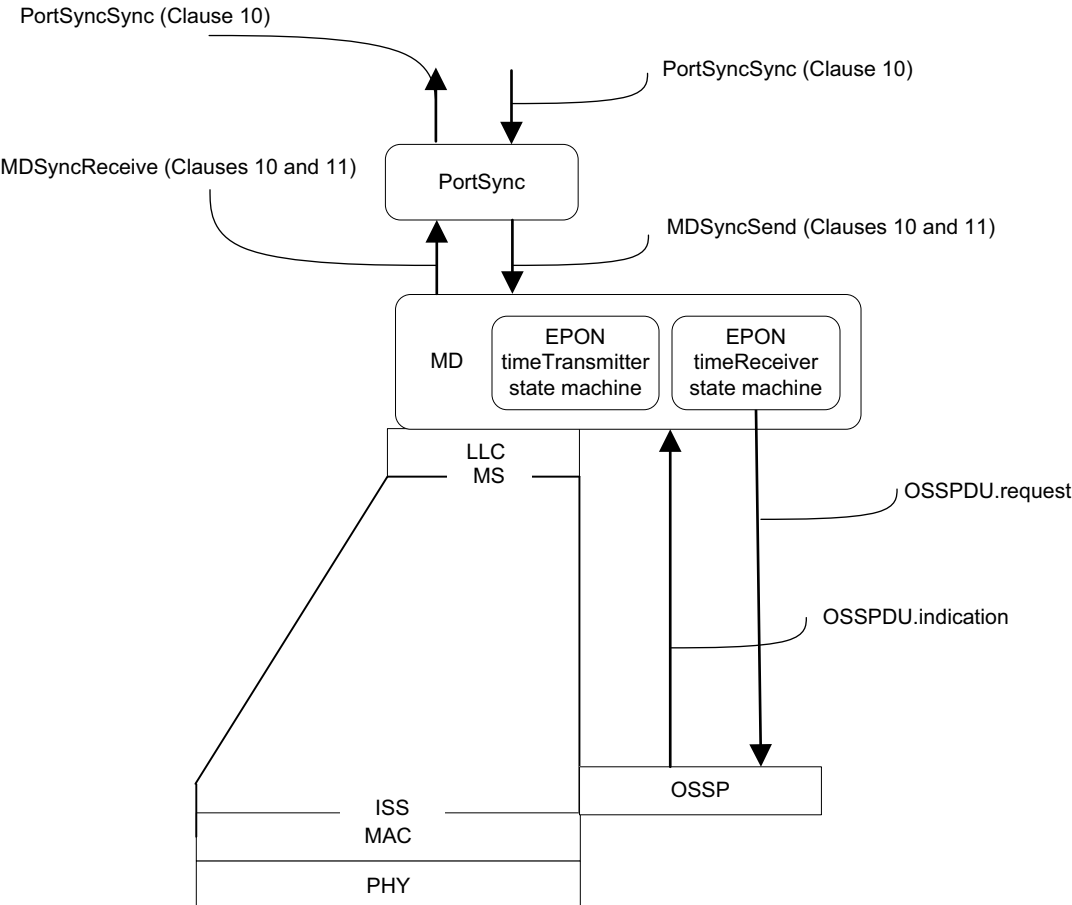


Figure 13-2—IEEE 802.3 EPON interface model

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2

3 **13.6 Service interface definitions**

4 **13.6.1 OSSPDU.request**

5 **13.6.1.1 General**

6 This service interface primitive is generated periodically by the MD entity of the clock timeTransmitter
7 every sync interval (see 10.7.2.1). It triggers transmission of a TIMESYNC message from the clock
8 timeTransmitter to the clock timeReceiver. The values of the parameters of the primitive are sent to the
9 clock timeReceiver via the TIMESYNC message.

1 13.6.1.2 OSSPDU.request parameters**2 13.6.1.2.1 Syntax of the primitive**

```

3 OSSPDU.request {
4     X
5     ToDX,i
6     sourcePortIdentity
7     logMessageInterval
8     rateRatio
9     gmTimeBaseIndicator
10    lastGmPhaseChange
11    scaledLastGmFreqChange
12    domainNumber
13    sdold
14 }

```

15 13.6.1.2.2 *X* (Integer32)

16 The *X* field is the selected timestamp that will be used as the timing reference as specified in 13.1.4.

17 13.6.1.2.3 *ToD_{X,i}* (Timestamp)

18 *ToD_{X,i}* is the synchronized time when the MPCP counter at the clock timeReceiver *i* reaches a value equal to
19 *X* minus the *onuLatencyFactor* (see 13.1.4). *X* is carried in the respective TIMESYNC message.
20 Synchronization of the MPCP clock is described in detail in 64.2.1.1 and 77.2.1.1 in IEEE Std 802.3-2022,
21 for 1G-EPON and 10G-EPON, respectively.

22 13.6.1.2.4 sourcePortIdentity (PortIdentity)

23 This parameter identifies the sourcePortIdentity value for this PTP Port (see 13.3.1.2.9).

24 13.6.1.2.5 logMessageInterval (Integer8)

25 This parameter identifies the currentLogSyncInterval value for this PTP Port (see 13.3.1.2.10).

26 13.6.1.2.6 rateRatio (Float64)

27 This parameter identifies the rateRatio value for this PTP Port (see 13.3.1.2.11).

28 13.6.1.2.7 gmTimeBaseIndicator (UInteger16)

29 This parameter identifies the gmTimeBaseIndicator value for this PTP Port (see 13.3.1.2.12).

30 13.6.1.2.8 lastGmPhaseChange (ScaledNs)

31 This parameter identifies the lastGmPhaseChange value for this PTP Port (see 13.3.1.2.13).

32 13.6.1.2.9 scaledLastGmFreqChange (Integer32)

33 This parameter identifies the scaledLastGmFreqChange value for this PTP Port (see 13.3.1.2.14).

1 13.6.1.2.10 domainNumber

2 This parameter identifies the domainNumber for this instance of gPTP (see 13.3.1.2.15).

3 13.6.1.2.11 sdoId

4 This parameter identifies the sdoId for this instance of gPTP (see 13.3.1.2.16 and 13.3.1.2.17).

5 13.6.1.3 When generated

6 This primitive is generated by the clock timeTransmitter every $2^{\text{currentLogSyncInterval}}$ seconds when it is in the
7 TIME_TRANSMITTER state, as the first phase of synchronized time information transfer.

8 13.6.1.4 Effect of receipt

9 Upon receipt of this primitive, a TIMESYNC message is enqueued for transmission.

10 NOTE—Arrival of the TIMESYNC message at the ONU after the selected time X does not impede proper operation of
11 the synchronization mechanism defined in this clause.

12 13.6.2 OSSPDU.indication

13 13.6.2.1 General

14 This service interface primitive is generated on receipt of a TIMESYNC message by the responder, and
15 provides the values contained in the corresponding OSSPDU.request primitive to the clock timeReceiver.

16 13.6.2.2 OSSPDU.indication parameters

17 OSSPDU.indication {
18 X
19 $ToD_{X,i}$
20 sourcePortIdentity
21 logMessageInterval
22 rateRatio
23 gmTimeBaseIndicator
24 lastGmPhaseChange
25 scaledLastGmFreqChange
26 domainNumber
27 sdoId
28 }

29 The parameters of the OSSPDU.indication are set equal to the corresponding fields of the most recently
30 received TIMESYNC message. Their definitions are given in 13.3.1.2.7 through 13.3.1.2.17, respectively.

31 13.6.2.3 When generated

32 This primitive is generated by the receipt of a TIMESYNC message during the phase of synchronized time
33 information transfer.

34 13.6.2.4 Effect of receipt

35 Upon receipt, the OSSPDU.indication parameters are used by the MD entity to compute the parameters of
36 the MDSyncReceive structure that will be transmitted to the PortSync entity of this PTP Port.

1 13.7 MD entity global variables

2 **13.7.1 RTT_i :** Is used only by the OLT MD entity. RTT_i is the RTT between the clock timeTransmitter and
3 clock timeReceiver. The data type for RTT_i is UInteger32.

4 NOTE—RTT is measured and updated by the MPCP using the mechanism specified in IEEE Std 802.3-2022 and stored
5 in RTT_i when measured and updated. RTT_i is not used by the ONU and is set to zero in an ONU MD entity.

6 13.8 State machines

7 13.8.1 Requester state machine

8 13.8.1.1 Function

9 This state machine generates and consumes primitives, at the requester, used to provide accurate
10 synchronized time across EPON links to the responder.

11 13.8.1.2 State machine variables

12 The following variables are used in the state diagram in Figure 13-3 (in 13.8.1.4):

13 **13.8.1.2.1 $ndown$:** The effective index of the light propagating in the downstream channel. The data type for
14 $ndown$ is Float64.

15 **13.8.1.2.2 nup :** The effective index of the light propagating in the upstream channel. The data type for
16 $ndown$ is Float64.

17 **13.8.1.2.3 $rcvdMDSyncEponReq$:** A Boolean variable that notifies the current state machine when an
18 MDSyncSend structure is received. This variable is reset by the current state machine.

19 **13.8.1.2.4 $rcvdMDSyncPtrEponReq$:** A pointer to the received MDSyncSend structure.

20 **13.8.1.2.5 $registered$:** A Boolean variable that indicates an ONU has registered to EPON.

21 **13.8.1.2.6 $ToD_{X,i}$:** The synchronized time when the MPCP counter at the clock timeReceiver i reaches a
22 value equal to X (see 13.8.1.2.8) minus the $onuLatencyFactor$ (see 13.1.4). The data type for $ToD_{X,i}$ is
23 Timestamp.

24 **13.8.1.2.7 $ToD_{X,o}$:** The synchronized time when the MPCP counter at the clock timeTransmitter reaches a
25 value equal to X (see 13.8.1.2.8) plus the $oltLatencyFactor$ (see 13.1.4). The data type for $ToD_{X,o}$ is
26 Timestamp.

27 **13.8.1.2.8 X :** The value of the timestamp [see item a) of 13.1.4] that is selected as the reference time. The
28 data type for X is UInteger32.

29 13.8.1.3 State machine functions

30 **13.8.1.3.1 $setToDXo()$:** Computes the state machine variable $ToD_{X,o}$ (see 13.8.1.2.7) as the sum of the
31 following:

- 32 a) The preciseOriginTimestamp member of the most recently received MDSyncSend structure
- 33 b) The followUpCorrectionField of the most recently received MDSyncSend structure

1 c) The quantity in Equation (13-4)

$$2 \quad \text{rateRatio} \times (X \times (16 \text{ ns}) - \text{upstreamTxTime}) \quad (13-4)$$

3 where

4 rateRatio is the rateRatio member of the most recently received MDSyncSend structure
5 upstreamTxTime is the upstreamTxTime member of the most recently received MDSyncSend
6 structure
7 X is defined in 13.8.1.2.8

8 13.8.1.4 State diagram

9 The requester state machine shall implement the function specified by the state diagram in Figure 13-3, the
10 local variables specified in 13.8.1.2, the function specified in 13.8.1.3, the service interface primitives
11 specified in 13.6, the structure specified in 10.2.2.1, the message specified in 13.3, and the relevant global
12 variables specified in 10.2.5 and 13.7. The state machine receives an MDSyncSend structure from the
13 PortSyncSyncSend state machine of the PortSync entity of this PTP Port and transmits an OSSPDU.request
14 primitive to cause a TIMESYNC message to be sent to the responder (ONU).

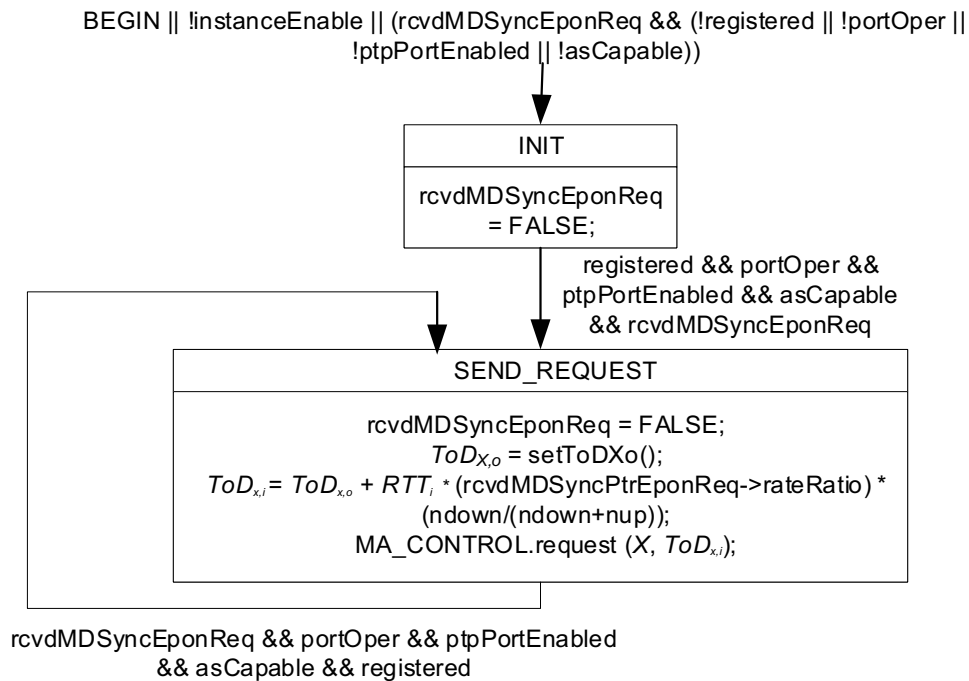


Figure 13-3—State machine for IEEE 802.3 EPON requester

15 13.8.2 Responder state machine

16 13.8.2.1 Function

17 This state machine responds to EPON-specific primitives generated by receipt of a TIMESYNC message
18 from the requester.

1 13.8.2.2 State machine variables

2 The following variables are used in the state diagram in Figure 13-4 (in 13.8.2.4):

3 **13.8.2.2.1 rcvdOSSPDUind:** a Boolean variable that notifies the responder state machine when a
4 TIMESYNC message is received and the OSSPDU.indication primitive is generated.

5 **13.8.2.2.2 txMDSyncReceivePtrEponResp:** a pointer to a structure whose members contain the values of
6 the parameters of an MDSyncReceive structure to be transmitted.

7 **13.8.2.2.3 rcvdOSSPDUptr:** a pointer to a structure whose members contain the values of the parameters of
8 the OSSPDU.indication primitive whose receipt is indicated by rcvdOSSPDUind (see 13.8.2.2.1).

9 13.8.2.3 State machine functions

10 **13.8.2.3.1 setMDSyncReceiveEponResp():** creates an MDSyncReceive structure (see 10.2.2.2) using
11 members of the structure pointed to by rcvdOSSPDUptr (see 13.8.2.2.3), and returns a pointer to this
12 structure. The members of this structure are set as follows:

- 13 a) followUpCorrectionField is set equal to 0.
- 14 b) sourcePortIdentity is set equal to the sourcePortIdentity field of the most recently received
15 TIMESYNC message (see 13.3.1.2.9).
- 16 c) logMessageInterval is set equal to the logMessageInterval of the most recently received
17 TIMESYNC message (see 13.3.1.2.10).
- 18 d) preciseOriginTimestamp is set equal to the $ToD_{X,i}$ field of the most recently received TIMESYNC
19 message (see 13.3.1.2.8).
- 20 e) rateRatio is set to the rateRatio of the most recently received TIMESYNC message (see
21 13.3.1.2.11).
- 22 f) upstreamTxTime is set equal to X multiplied by 16 ns, where X is the value of the X field of the most
23 recently received TIMESYNC message (see 13.3.1.2.7).
- 24 g) gmTimeBaseIndicator is set equal to the gmTimeBaseIndicator of the most recently received
25 TIMESYNC message (see 13.3.1.2.12).
- 26 h) lastGmPhaseChange is set equal to the lastGmPhaseChange of the most recently received
27 TIMESYNC message (see 13.3.1.2.13).
- 28 i) lastGmFreqChange is set equal to the scaledLastGmFreqChange of the most recently received
29 TIMESYNC message (see 13.3.1.2.14), divided by 2^{41} .
- 30 j) domainNumber is set equal to the domainNumber of the most recently received TIMESYNC
31 message (see 13.3.1.2.15).

32 **13.8.2.3.2 txMDSyncReceive (txMDSyncReceivePtrEponResp):** transmits an MDSyncReceive structure
33 to the PortSyncSyncReceive state machine of the PortSync entity of this PTP Port.

34 13.8.2.4 State diagram

35 The responder state machine shall implement the function specified by the state diagram in Figure 13-4, the
36 local variables specified in 13.8.2.2, the functions specified in 13.8.2.3, the service interface primitives
37 specified in 13.6, the structure specified in 10.2.2.2, the message specified in 13.3, and the relevant global
38 variables specified in 10.2.5 and 13.7. The state machine receives an OSSPDU.indication primitive in
39 response to its having received a TIMESYNC message from the requester (OLT), and transmits an
40 MDSyncReceive structure to the PortSync entity of this PTP Port.

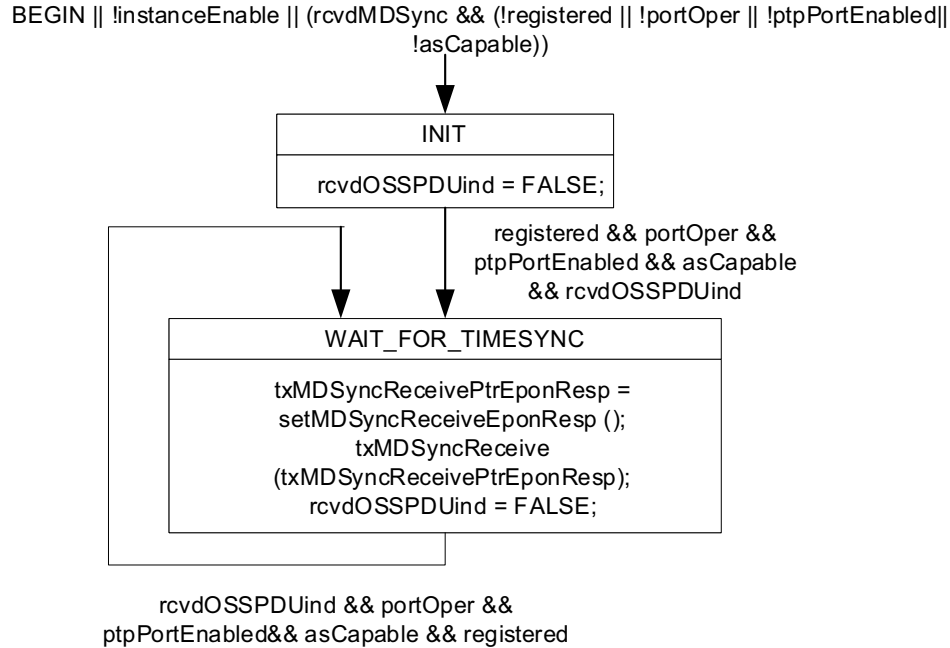


Figure 13-4—State machine for IEEE 802.3 EPON responder

1 13.9 Message transmission intervals

2 13.9.1 General interval specification

3 The mean time interval between successive TIMESYNC messages shall be as specified in 10.7.2.1, 10.7.2.3,
4 and 13.9.2.

5 13.9.2 TIMESYNC message transmission interval default value

6 The default value of initialLogSyncInterval (see 10.7.2.4) is –3. Every PTP Port supports the value 127; the
7 PTP Port does not send TIMESYNC messages when currentLogSyncInterval has this value. A PTP Port
8 may support other values, except for the reserved values indicated in Table 10-16.

9 Processing of the message interval request TLV carried in a Signaling message (see 10.6.4) shall be
10 supported, as specified by the SyncIntervalSetting state machine of 10.3.18 (see Figure 10-20), except that
11 the logLinkDelayInterval (which is not relevant to IEEE 802.3 EPON ports) is set to 128 by the sender of the
12 Signaling message, the logLinkDelayInterval is ignored by the receiver, and unsupported values of
13 logTimeSyncInterval are ignored by the receiver.

1 14. Timing and synchronization management

2 14.1 General

3 14.1.1 Data set hierarchy

4 This clause defines the set of managed objects, and their functionality, that allow administrative
5 configuration of clock parameters and timing and synchronization protocols.

6 Management data models typically represent data for the physical device (i.e., time-aware system). The
7 specifications for discovery, management address, and security for the physical device are typically covered
8 by standards of the management mechanism, which are outside the scope of this standard. For the
9 management information model of this standard, the scope of work is the data contained within a time-aware
10 system. From a management perspective, the time-aware system contains a list of one or more PTP
11 Instances. Each entry in the list is a set of managed data sets for the respective PTP Instance.

12 Conformance for each managed object is optional. This standard operates correctly using default values;
13 therefore, management is not essential. Since the management mechanism is not limited to remote protocols
14 (e.g., SNMP, NETCONF), management can use a local mechanism with a simple interface (e.g., DIP
15 switches). Therefore, each product can determine the support of managed objects as appropriate for its
16 management mechanism.

17 The following hierarchy summarizes the managed data sets within a gPTP Node:

- 18 a) instanceList[]
 - 19 1) defaultDS (see 14.2)
 - 20 2) currentDS (see 14.3)
 - 21 3) parentDS (see 14.4)
 - 22 4) timePropertiesDS (see 14.5)
 - 23 5) pathTraceDS (see 14.6)
 - 24 6) acceptableTimeTransmitterTableDS (see 14.7)
 - 25 7) ptpInstanceSyncDS (see 14.8)
 - 26 8) driftTrackingDS (see 14.9)
 - 27 9) portList[]
 - 28 i) portDS (see 14.10)
 - 29 ii) descriptionPortDS (see 14.11)
 - 30 iii) portStatisticsDS (see 14.12)
 - 31 iv) acceptableTimeTransmitterPortDS (see 14.13)
 - 32 v) externalPortConfigurationPortDS (see 14.14)
 - 33 vi) asymmetryMeasurementModeDS (see 14.15)
 - 34 vii) commonServicesPortDS (see 14.16)
- 35 b) commonServices
 - 36 1) commonMeanLinkDelayService
 - 37 i) cmlDsDefaultDS (see 14.17)
 - 38 ii) cmlDsLinkPortList[]
 - 39 — cmlDsLinkPortDS (see 14.18)
 - 40 — cmlDsLinkPortStatisticsDS (see 14.19)
 - 41 — cmlDsAsymmetryMeasurementModeDS (see 14.20)
 - 42 2) hotStandbyService

- 1 i) hotStandbySystemList[]
- 2 — hotStandbySystemDS (see 14.21)
- 3 — hotStandbySystemDescriptionDS (see 14.22)

4 The instanceList is a structure for the gPTP datasets, and it is indexed using a number that is unique per PTP
5 Instance within the time-aware system, applicable to the management context only (i.e., not used in PTP
6 messages). The domainNumber of the PTP Instance must not be used as the index to instanceList since it is
7 possible for a time-aware system to contain multiple PTP Instances using the same domainNumber. The
8 portList is indexed using a number that is unique per logical port (i.e., PTP Port) in the PTP Instance (see
9 8.5.1). Since the portNumber of a logical port can have any value in the range 1, 2, 3, ..., 0xFFFFE (see
10 8.5.2.3), the portList index and portNumber values for a logical port are not necessarily the same. PTP
11 Instances and logical ports may be created or deleted dynamically in implementations that support dynamic
12 create/delete of devices. Unless otherwise indicated, the data sets and managed objects under the
13 instanceList[] are maintained separately for each PTP Instance supported by the time-aware system.

14 The commonServices is an overall structure containing commonMeanLinkDelayService structure for the
15 data sets used for CMLDS, and the hotStandbyService structure for the data sets used for Hot Standby
16 service.

17 The commonMeanLinkDelayService structure contains the cmldsLinkPortList, which is a list of CMLDS
18 Link Ports (see 11.2.17), of the time-aware system that runs the common service. The CMLDS must be
19 implemented on every physical port for which there is a PTP Port of a PTP Instance that can use the
20 CMLDS. Therefore, the cmldsLinkPortList[] must include Link Ports that correspond to all such physical
21 ports. As is the case for the portList of a PTP Instance, the cmldsLinkPortList is indexed using a number that
22 is unique per Link Port that invokes the CMLDS (see 8.5.1). Since the portNumber of a logical port (i.e.,
23 PTP Port or CMLDS Link Port) can have any value in the range 1, 2, 3, ..., 0xFFFFE (see 8.5.2.3), the
24 cmldsLinkPortList index and cmldsLinkPortDS.portIdentity.portNumber values for a logical port of the
25 CMLDS are not necessarily the same. CMLDS Link Ports may be created or deleted dynamically in
26 implementations that support dynamic create/delete of devices.

27 The CMLD Data Sets are not maintained separately for each PTP Instance. Rather, a single copy of the
28 commonServices.cmldsDefaultDS is maintained for the time-aware system, and a single copy of each data
29 set under the cmldsLinkPortList[] is maintained per Link Port of the time-aware system.

30 A PTP Instance can use the commonServicesPortDS to determine which Link Port it must use when it
31 obtains information provided by the CMLD (see 14.16).

32 The hotStandbyService structure contains the hotStandbySystemList, which is a list of instances of the Hot
33 Standby Service. Since a Hot Standby Service Instance is associated with two PTP Instances, but a
34 time-aware system can have more than two PTP Instances, there can, in general, be multiple instances of the
35 Hot Standby Service (i.e., one Hot Standby Service Instance associated with each set of two distinct PTP
36 Instances). A hotStandbySystemDS and a hotStandbySystemDescriptionDS are maintained for each
37 instance of the Hot Standby Service.

38 Each instance of the Hot Standby Service (i.e., each hotStandbySystemList member) is associated with two
39 PTP Instances, i.e., the primary PTP Instance and the secondary PTP Instance (see 18.1 and 18.2). The
40 indices of these PTP Instances in the instanceList are contained in the members primaryPtpInstanceIndex
41 and secondaryPtpInstanceIndex, respectively, of the hotStandbySystemDS for this instance of the Hot
42 Standby Service.

1 14.2 Default Parameter Data Set (defaultDS)

2 14.2.1 General

3 The defaultDS represents the native capabilities of a PTP Instance, i.e., a PTP Relay Instance or a PTP End
4 Instance.

5 The parameters of defaultDS are summarized in Table 14-1 and subsequently specified.

Table 14-1—defaultDS

Name	Data type	Operations supported ^a	References
clockIdentity	ClockIdentity	R	14.2.2
numberPorts	UInteger16	R	14.2.3
clockQuality.clockClass	UInteger8	R	14.2.4.2; 7.6.2.5 of IEEE Std 1588-2019
clockQuality.clockAccuracy	Enumeration8	R	14.2.4.3; 7.6.2.6 of IEEE Std 1588-2019
clockQuality.offsetScaledLogVariance	UInteger16	R	14.2.4.4
priority1	UInteger8	RW	14.2.5
priority2	UInteger8	RW	14.2.6
gmCapable	Boolean	R	14.2.7
currentUtcOffset	Integer16	R	14.2.8
currentUtcOffsetValid	Boolean	R	14.2.9
leap59	Boolean	R	14.2.10
leap61	Boolean	R	14.2.11
timeTraceable	Boolean	R	14.2.12
frequencyTraceable	Boolean	R	14.2.13
ptpTimescale	Boolean	R	14.2.14
timeSource	TimeSource	R	14.2.15 and Table 8-2
domainNumber	UInteger8	RW	14.2.16
sdoId	UInteger16	R	14.2.17
externalPortConfigurationEnabled	Boolean	RW	14.2.18
instanceEnable	Boolean	RW	14.2.19

^a R = Read only access; RW = Read/write access.

1 14.2.2 clockIdentity

2 The value is the clockIdentity (see 8.5.2.2) of the PTP Instance.

3 14.2.3 numberPorts

4 The value is the number of PTP Ports of the PTP Instance (see 8.6.2.8). For an end station the value is 1.

5 14.2.4 clockQuality

6 14.2.4.1 General

7 This is a structure whose data type is ClockQuality (see 6.4.3.8).

8 14.2.4.2 clockQuality.clockClass

9 The value is the clockClass of the PTP Instance, which implements the clockClass specifications of 8.6.2.2.

10 14.2.4.3 clockQuality.clockAccuracy

11 The value is the clockAccuracy of the PTP Instance, which implements the clockAccuracy specifications of
12 8.6.2.3.

13 14.2.4.4 clockQuality.offsetScaledLogVariance

14 The value is the offsetScaledLogVariance of the PTP Instance, which implements the
15 offsetScaledLogVariance specifications of 8.6.2.4.

16 14.2.5 priority1

17 The value is the priority1 attribute of the PTP Instance (see 8.6.2.1).

18 14.2.6 priority2

19 The value is the priority2 attribute of the PTP Instance (see 8.6.2.5).

20 14.2.7 gmCapable

21 The value is TRUE if the PTP Instance is capable of being a Grandmaster PTP Instance and FALSE if the
22 PTP Instance is not capable of being a Grandmaster PTP Instance.

23 14.2.8 currentUtcOffset

24 The value is the offset between TAI and UTC, relative to the ClockTimeTransmitter entity of this PTP
25 Instance. It is equal to the global variable sysCurrentUtcOffset (see 10.3.9.18). The value is in units of
26 seconds.

27 The default value is selected as follows:

- 28 a) The value is the value obtained from a primary reference if the value is known at the time of
29 initialization, else
- 30 b) The value is the current IERS defined value of TAI – UTC (see IERS Bulletin C) when the PTP
31 Instance is designed.

1 14.2.9 currentUtcOffsetValid

2 The value is TRUE if the currentUtcOffset, relative to the ClockTimeTransmitter entity of this PTP Instance,
3 is known to be correct. It is equal to the global variable sysCurrentUtcOffsetValid (see 10.3.9.14).

4 The default value is TRUE if the value of currentUtcOffset is known to be correct; otherwise, it is set to
5 FALSE.

6 14.2.10 leap59

7 A TRUE value indicates that the last minute of the current UTC day, relative to the ClockTimeTransmitter
8 entity of this PTP Instance, will contain 59 s. It is equal to the global variable sysLeap59 (see 10.3.9.13).

9 The value is selected as follows:

- 10 a) The value is obtained from a primary reference if known at the time of initialization, else
- 11 b) The value is set to FALSE.

12 14.2.11 leap61

13 A TRUE value indicates that the last minute of the current UTC day, relative to the ClockTimeTransmitter
14 entity of this PTP Instance, will contain 61 s. It is equal to the global variable sysLeap61 (see 10.3.9.12).

15 The value is selected as follows:

- 16 a) The value is obtained from a primary reference if known at the time of initialization, else
- 17 b) The value is set to FALSE.

18 14.2.12 timeTraceable

19 The value is set to TRUE if the timescale and the value of currentUtcOffset, relative to the
20 ClockTimeTransmitter entity of this PTP Instance, are traceable to a primary reference standard; otherwise
21 the value is set to FALSE. It is equal to the global variable sysTimeTraceable (see 10.3.9.16).

22 The value is selected as follows:

- 23 a) If the time and the value of currentUtcOffset are traceable to a primary reference standard at the time
24 of initialization, the value is set to TRUE, else
- 25 b) The value is set to FALSE.

26 14.2.13 frequencyTraceable

27 The value is set to TRUE if the frequency determining the timescale of the ClockTimeTransmitter Entity of
28 this PTP Instance is traceable to a primary standard; otherwise the value is set to FALSE. It is equal to the
29 global variable sysFrequencyTraceable (see 10.3.9.17).

30 The value is selected as follows:

- 31 a) If the frequency is traceable to a primary reference standard at the time of initialization the value is
32 set to TRUE, else
- 33 b) The value is set to FALSE.

1 **14.2.14 ptpTimescale**

2 The value is set to TRUE if the clock timescale of the ClockTimeTransmitter Entity of this PTP Instance is
3 PTP (see 8.2) and FALSE otherwise.

4 **14.2.15 timeSource**

5 The value is the source of time used by the Grandmaster Clock (see 8.6.2.7).

6 **14.2.16 domainNumber**

7 The value is the domainNumber of the gPTP domain for this instance of gPTP supported by the time-aware
8 system (see 8.1).

9 NOTE—The PTP Instance for which domainNumber is 0 has constraints applied to it, e.g., timescale (see 8.2.1).

10 **14.2.17 sdoId**

11 The value is the sdoId of the gPTP domain for this instance of gPTP supported by the time-aware system
12 (see 8.1).

13 NOTE—The attribute sdoId is specified as a 12-bit unsigned integer in 8.1. The data type for the managed object sdoId
14 is UInteger16 in Table 14-1, for compatibility with IEEE Std 1588-2019. The range of the managed object is limited to
15 12 bits; in addition, only the single value 0x100 is specified in this standard for the gPTP domain of a PTP Instance.

16 **14.2.18 externalPortConfigurationEnabled**

17 The value is the externalPortConfigurationEnabled attribute of the PTP Instance (see 10.3.9.24).

18 **14.2.19 instanceEnable**

19 The value is the instanceEnable attribute of the PTP Instance (see 10.2.4.24).

20 **14.3 Current Parameter Data Set (currentDS)**

21 **14.3.1 General**

22 The currentDS represents the position of a local system and other information, relative to the Grandmaster
23 PTP Instance.

1 The parameters of currentDS are summarized in Table 14-2 and subsequently specified.

Table 14-2—currentDS

Name	Data type	Operations supported ^a	References
stepsRemoved	UInteger16	R	14.3.2
offsetFromTimeTransmitter	TimeInterval	R	14.3.3
lastGmPhaseChange	ScaledNs	R	14.3.4
lastGmFreqChange	Float64	R	14.3.5
gmTimebaseIndicator	UInteger16	R	14.3.6
gmChangeCount	UInteger32	R	14.3.7
timeOfLastGmChangeEvent	UInteger32 (sysUp Time, IETF RFC 3418)	R	14.3.8
timeOfLastGmPhaseChangeEvent	UInteger32 (sysUp Time, IETF RFC 3418)	R	14.3.9
timeOfLastGmFreqChangeEvent	UInteger32 (sysUp Time, IETF RFC 3418)	R	14.3.10

^a R = Read only access; RW = Read/write access.

2 14.3.2 stepsRemoved

3 The value is the number of gPTP communication paths traversed between this PTP Instance and the
4 Grandmaster PTP Instance, as specified in 10.3.3.

5 14.3.3 offsetFromTimeTransmitter

6 The value is an implementation-specific or profile standard-specific representation of the current value of
7 the time difference between a timeReceiver and the Grandmaster Clock, as computed by the timeReceiver,
8 and as specified in 10.2.10. The value is computed by an algorithm that is implementation-specific or profile
9 standard-specific. The data type shall be TimeInterval. The default value is implementation-specific.

10 NOTE—For example, the inputs to this implementation-specific algorithm could be the successive values of
11 clockSourcePhaseOffset (see 10.2.4.7) of the ClockSyncOffset state machine (see 10.2.10 and Figure 10-6).

12 14.3.4 lastGmPhaseChange

13 The value (see 10.2.4.16) is the phase change that occurred on the most recent change in either Grandmaster
14 PTP Instance or gmTimeBaseIndicator (see 9.2.2.3).

15 14.3.5 lastGmFreqChange

16 The value (see 10.2.4.17) is the frequency change that occurred on the most recent change in either
17 Grandmaster PTP Instance or gmTimeBaseIndicator (see 9.2.2.3).

1 14.3.6 gmTimebaseIndicator

2 The value is the value of timeBaseIndicator of the current Grandmaster PTP Instance (see 9.2.2.3 and
3 9.6.2.3).

4 14.3.7 gmChangeCount

5 This statistics counter tracks the number of times the Grandmaster PTP Instance has changed in a gPTP
6 domain. This counter increments when the PortAnnounceInformation state machine enters the
7 SUPERIOR_TIME_TRANSMITTER_PORT state or the
8 INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state (see 10.3.12 and Figure 10-14).

9 14.3.8 timeOfLastGmChangeEvent

10 This timestamp takes the value of sysUpTime (see IETF RFC 3418) when the most recent Grandmaster PTP
11 Instance change occurred in a gPTP domain. This timestamp is updated when the PortAnnounceInformation
12 state machine enters the SUPERIOR_TIME_TRANSMITTER_PORT state or the
13 INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state (see 10.3.12 and Figure 10-14).

14 14.3.9 timeOfLastGmPhaseChangeEvent

15 This timestamp takes the value of sysUpTime (see IETF RFC 3418) when the most recent change in
16 Grandmaster Clock phase occurred due to a change of either the Grandmaster PTP Instance or the
17 Grandmaster Clock time base. This timestamp is updated when one of the following occurs:

- 18 a) The PortAnnounceInformation state machine enters the
19 SUPERIOR_TIME_TRANSMITTER_PORT state or the
20 INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state (see 10.3.12 and Figure 10-14), or
21 b) The gmTimebaseIndicator managed object (see 14.3.6) changes and the lastGmPhaseChange field
22 of the most recently received Follow_Up information TLV is nonzero.

23 14.3.10 timeOfLastGmFreqChangeEvent

24 This timestamp takes the value of sysUpTime (see IETF RFC 3418) when the most recent change in
25 Grandmaster Clock frequency occurred due to a change of either the Grandmaster PTP Instance or the
26 Grandmaster Clock time base. This timestamp is updated when one of the following occurs:

- 27 a) The PortAnnounceInformation state machine enters the
28 SUPERIOR_TIME_TRANSMITTER_PORT state or the
29 INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state (see 10.3.12 and Figure 10-14), or
30 b) The gmTimebaseIndicator managed object (see 14.3.6) changes, and the lastGmFreqChange field of
31 the most recently received Follow_Up information TLV is nonzero.

32 14.4 Parent Parameter Data Set (parentDS)

33 14.4.1 General

34 The parentDS represents capabilities of the upstream system, toward the Grandmaster PTP Instance, as
35 measured at a local system.

36 The parameters of parentDS are summarized in Table 14-3 and subsequently specified.

37

Table 14-3—parentDS

Name	Data type	Operations supported ^a	References
parentPortIdentity	PortIdentity (see 6.4.3.7)	R	14.4.2
cumulativeRateRatio	Integer32	R	14.4.3
grandmasterIdentity	ClockIdentity	R	14.4.4
grandmasterClockQuality.clock Class	UInteger8	R	14.4.5.2; 7.6.2.5 of IEEE Std 1588-2019
grandmasterClockQuality.clock Accuracy	Enumeration8	R	14.4.5.3; 7.6.2.6 of IEEE Std 1588-2019
grandmasterClockQuality.offset ScaledLogVariance	UInteger16	R	14.4.5.4
grandmasterPriority1	UInteger8	R	14.4.6
grandmasterPriority2	UInteger8	R	14.4.7
gmPresent	Boolean	R	14.4.8

^a R = Read only access; RW = Read/write access.

1

2 14.4.2 parentPortIdentity

3 If this PTP Instance is the Grandmaster PTP Instance, the value is a portIdentity whose clockIdentity is the
4 clockIdentity of this PTP Instance, and whose portNumber is 0.

5 If this PTP Instance is not the Grandmaster PTP Instance, the value is the portIdentity of the
6 TimeTransmitterPort (see Table 10-7) of the gPTP communication path attached to the single timeReceiver
7 port of this PTP Instance.

8 The default value is a portIdentity for which the following apply:

- 9 a) The clockIdentity member is the value of the clockIdentity member of the default data set.
- 10 b) The portNumber member is 0.

11 14.4.3 cumulativeRateRatio

12 The value is an estimate of the ratio of the frequency of the Grandmaster Clock to the frequency of the
13 LocalClock entity of this PTP Instance. cumulativeRateRatio is expressed as the fractional frequency offset
14 multiplied by 2^{41} , i.e., the quantity $(\text{rateRatio} - 1.0)(2^{41})$, where rateRatio is computed by the
15 PortSyncSyncReceive state machine (see 10.2.8.1.4).

16 14.4.4 grandmasterIdentity

17 The value is the clockIdentity attribute (see 8.5.2.2) of the Grandmaster PTP Instance.

18 The default value is the value of defaultDS.clockIdentity (14.2.2).

1 **14.4.5 grandmasterClockQuality**

2 **14.4.5.1 General**

3 This is a structure whose data type is ClockQuality (see 6.4.3.8).

4 **14.4.5.2 grandmasterClockQuality.clockClass**

5 The value is the clockClass (see 8.6.2.2) of the Grandmaster PTP Instance.

6 The default value is the clockClass member of the default data set.

7 **14.4.5.3 grandmasterClockQuality.clockAccuracy**

8 The value is the clockAccuracy (see 8.6.2.3) of the Grandmaster PTP Instance.

9 The default value is the clock accuracy member of the default data set.

10 **14.4.5.4 grandmasterClockQuality.offsetScaledLogVariance**

11 The value is the offsetScaledLogVariance (see 8.6.2.4) of the Grandmaster PTP Instance.

12 The default value is the offsetScaledLogVariance member of the default data set.

13 **14.4.6 grandmasterPriority1**

14 The value is the priority1 attribute (see 8.6.2.1) of the Grandmaster PTP Instance.

15 The default value is the priority1 value of the default data set.

16 **14.4.7 grandmasterPriority2**

17 The value is the priority2 attribute (see 8.6.2.5) of the Grandmaster PTP Instance.

18 The default value is the priority2 value of the default data set.

19 **14.4.8 gmPresent**

20 The value is the value of the per PTP Instance global variable gmPresent (see 10.2.4.13).

1 **14.5 Time Properties Parameter Data Set (timePropertiesDS)**

2 **14.5.1 General**

3 The timePropertiesDS represents capabilities of the Grandmaster PTP Instance, as measured at a local
4 system.

5 The parameters of timePropertiesDS are summarized in Table 14-4 and subsequently specified.

Table 14-4—timePropertiesDS

Name	Data type	Operations supported ^a	References
currentUtcOffset	Integer16	R	14.5.2
currentUtcOffsetValid	Boolean	R	14.5.3
leap59	Boolean	R	14.5.4
leap61	Boolean	R	14.5.5
timeTraceable	Boolean	R	14.5.6
frequencyTraceable	Boolean	R	14.5.7
ptpTimescale	Boolean	R	14.5.8
timeSource	TimeSource	R	14.5.9 and Table 8-2

^a R = Read only access; RW = Read/write access.

6 **14.5.2 currentUtcOffset**

7 The value is currentUtcOffset for the current Grandmaster PTP Instance (see 14.2.8). It is equal to the value
8 of the global variable currentUtcOffset (see 10.3.9.10). The value is in units of seconds.

9 **14.5.3 currentUtcOffsetValid**

10 The value is currentUtcOffsetValid for the current Grandmaster PTP Instance (see 14.2.9). It is equal to the
11 global variable currentUtcOffsetValid (see 10.3.9.6).

12 **14.5.4 leap59**

13 The value is leap59 for the current Grandmaster PTP Instance (see 14.2.10). It is equal to the global variable
14 leap59 (see 10.3.9.5).

15 **14.5.5 leap61**

16 The value is leap61 for the current Grandmaster PTP Instance (see 14.2.11). It is equal to the global variable
17 leap61 (see 10.3.9.4).

18 **14.5.6 timeTraceable**

19 The value is timeTraceable for the current Grandmaster PTP Instance (see 14.2.12). It is equal to the global
20 variable timeTraceable (see 10.3.9.8).

1 **14.5.7 frequencyTraceable**

2 The value is frequencyTraceable for the current Grandmaster PTP Instance (see 14.2.13). It is equal to the
3 global variable frequencyTraceable (see 10.3.9.9).

4 **14.5.8 ptpTimescale**

5 The value is ptpTimescale for the current Grandmaster PTP Instance (see 14.2.14).

6 **14.5.9 timeSource**

7 The value is timeSource for the current Grandmaster PTP Instance (see 14.2.15). It is equal to the global
8 variable timeSource (see 10.3.9.11).

9 **14.6 Path Trace Parameter Data Set (pathTraceDS)**

10 **14.6.1 General**

11 The pathTraceDS represents the current path trace information available at the PTP Instance.

12 The parameters of pathTraceDS are summarized in Table 14-5 and subsequently specified.

Table 14-5—pathTraceDS

Name	Data type	Operations supported ^a	References
list	ClockIdentity[N], where N is defined in 10.3.9.23	R	14.6.2
enable	Boolean	R	14.6.3

^a R = Read only access; RW = Read/write access.

13 **14.6.2 list**

14 The value is the array of ClockIdentity values contained in the pathTrace array (see 10.3.9.23), which
15 represents the current path trace information and which is carried in the path trace TLV (see 10.6.3.3).

16 The initialization value shall be the empty list (i.e., an array of length 0).

17 **14.6.3 enable**

18 The value is TRUE.

19 NOTE—This member is included for compatibility with IEEE Std 1588-2019. In IEEE Std 1588-2019, the path trace
20 mechanism is optional, and the pathTraceDS.enable member is configurable (its value in IEEE Std 1588-2019 is TRUE
21 or FALSE, depending on whether the path trace mechanism is operational or not operational, respectively. However, the
22 pathTrace mechanism is mandatory in this standard, and the value of enable is always TRUE.

1 **14.7 Acceptable TimeTransmitter Table Parameter Data Set**
2 **(acceptableTimeTransmitterTableDS)**

3 **14.7.1 General**

4 The acceptableTimeTransmitterTableDS represents the acceptable timeTransmitter table used when an
5 EPON port is used by a PTP Instance of a time-aware system.

6 The parameters of acceptableTimeTransmitterTableDS are summarized in Table 14-6 and subsequently
7 specified.

Table 14-6—acceptableTimeTransmitterTableDS

Name	Data type	Operations supported ^a	References
maxTableSize	UInteger16	R	14.7.2
actualTableSize	UInteger16	RW	14.7.3
acceptableTimeTransmitterArray	AcceptableTimeTransmitter[actualTableSize] (see 13.1.3.3)	RW	14.7.4

^a R = Read only access; RW = Read/write access.

8 **14.7.2 maxTableSize**

9 The value is the maximum size of the AcceptableTimeTransmitterTable. It is equal to the maxTableSize
10 member of the AcceptableTimeTransmitterTable structure (see 13.1.3.2).

11 **14.7.3 actualTableSize**

12 The value is the actual size of the AcceptableTimeTransmitterTable. It is equal to the actualTableSize
13 member of the AcceptableTimeTransmitterTable structure (see 13.1.3.2 and 13.1.3.5), i.e., the current
14 number of elements in the acceptable timeTransmitter array. The actual table size is less than or equal to the
15 maxTableSize.

16 **14.7.4 acceptableTimeTransmitterArray**

17 Each element of this array is an AcceptableTimeTransmitter structure (see 13.1.3.3 and 13.1.3.5).

18 **14.8 PTP Instance Synchronization Parameter Data Set (ptpInstanceSyncDS)**

19 **14.8.1 General**

20 The ptpInstanceSyncDS describes the synchronization status of the PTP Instance. The ptpInstanceSyncDS
21 shall be implemented if the optional hot standby feature (see Clause 18) is implemented and may be
22 implemented otherwise.

23 The parameters of ptpInstanceSyncDS are summarized in Table 14-7. and subsequently specified.

24

Table 14-7—ptpInstanceSyncDS

Name	Data type	Operations supported ^a	References
isSynced	Boolean	R	14.8.2
offsetFromTimeTransmitterMax	TimeInterval	RW	14.8.3
rxSyncCountTimeReceiverPThresh	UInteger32	RW	14.8.4
offsetMaxExceededCountThresh	UInteger32	RW	14.8.5
offsetMaxMetCountThresh	UInteger32	RW	14.8.6

^a R = Read only access; RW = Read/write access.

1 14.8.2 isSynced

2 The value of the global variable isSynced (see 18.4.1.1).

3 14.8.3 offsetFromTimeTransmitterMax

4 The value is the threshold for offsetFromTimeTransmitter (see 18.4.1.2), below which the PTP Instance is
5 considered to be synchronized. For values less than or equal to zero, the PTP Instance is considered
6 synchronized if and only if offsetFromTimeTransmitter is zero. For values greater than zero, the PTP
7 Instance is considered synchronized if and only if:

$$8 \quad -\text{offsetFromTimeTransmitterThreshold} \leq \text{offsetFromTimeTransmitter} \leq \text{offsetFromTimeTransmitterThreshold}$$

9 is satisfied.

10 14.8.4 rxSyncCountTimeReceiverPThresh

11 The value of rxSyncCountTimeReceiverPThresh is the threshold for rxSyncCountTimeReceiverP
12 (see 18.4.1.4), above which the PTP Instance is considered to be synchronized.

13 14.8.5 offsetMaxExceededCountThresh

14 The value of offsetMaxExceededCountThresh (see 18.4.1.7) is the threshold for the number of consecutive
15 exceedances of offsetFromTimeTransmitterMax (see 18.4.1.3) by offsetFromTimeTransmitter
16 (see 18.4.1.2), at which isSynced (see 18.4.1.1) is no longer TRUE.

17 14.8.6 offsetMaxMetCountThresh

18 The value of offsetMaxMetCountThresh (18.4.1.9) is the threshold for the number of consecutive
19 occurrences of offsetFromTimeTransmitter (see 18.4.1.2) being within offsetFromTimeTransmitterMax
20 (see 18.4.1.3), at which isSynced (see 18.4.1.1) is changed to TRUE if it currently is FALSE.

1 14.9 Drift Tracking Parameter Data Set (driftTrackingDS)

2 14.9.1 General

3 The driftTrackingDS contains a managed object that is used to enable or disable the optional Drift_Tracking
4 TLV.

5 The parameters of driftTrackingDS are summarized in Table 14-8 and subsequently specified.

Table 14-8—driftTrackingDS

Name	Data type	Operations supported ^a	References
driftTrackingTlvSupport	Boolean	RW	14.9.2
syncGrandmasterIdentity	ClockIdentity	R	14.9.3
syncStepsRemoved	UInteger16	R	14.9.4
rateRatioDrift	Integer32	R	14.9.5

^a R = Read only access; RW = Read/write access.

6 14.9.2 driftTrackingTlvSupport

7 The value of driftTrackingTlvSupport indicates whether the Drift_Tracking TLV is enabled or disabled. If
8 the value is TRUE, the TLV is enabled, i.e., the global variable driftTrackingTlvSupport (see 10.2.4.28) is
9 set to TRUE. If the value is FALSE, the TLV is disabled, i.e., the global variable driftTrackingTlvSupport is
10 set to FALSE.

11 NOTE—Setting the value of the managed object to TRUE only has an effect if the instance supports the
12 Drift_Tracking TLV. If the instance does not support the Drift_Tracking TLV, reading the managed object
13 will always return FALSE.

14 14.9.3 syncGrandMasterIdentity

15 The value of syncGrandmasterIdentity as determined by the MDSyncSend state machine (see 11.2.15.2.3
16 and 11.2.15.2.5).

17 14.9.4 syncStepsRemoved

18 The value of syncStepsRemoved as determined by the MDSyncSend state machine (see 11.2.15.2.3 and
19 11.2.15.2.5).

20 14.9.5 RateRatioDrift

21 The value of rateRatioDrift is equal to $(RR_{drift} - 1.0) \times (2^{41})$, truncated to the next smaller signed integer,
22 where RR_{drift} is the measured estimate of the rate of change per second of the ratio of the frequency of the
23 Grandmaster Clock to the frequency of the Local Clock entity in the PTP Instance that sends the message.

1 14.10 Port Parameter Data Set (portDS)

2 14.10.1 General

3 The portDS represents PTP Port time-aware capabilities for a PTP Instance of a time-aware system.

4 For the single PTP Port of a PTP End Instance and for each PTP Port of a PTP Relay Instance, the portDS is
5 maintained as the basis for making protocol decisions and providing values for message fields. The number
6 of such data sets is the same as the value of defaultDS.numberPorts.

7 There is one set of portDS parameters per PTP Port, per PTP Instance of a time-aware system. Each portDS
8 contains a set of parameters for each PTP Port that supports the time-synchronization capability, as detailed
9 in Table 14-9. Each set of portDS can be created or removed dynamically in implementations that support
10 dynamic configuration of PTP Ports and components.

11 The parameters of portDS are summarized in Table 14-9 and subsequently specified.

Table 14-9—portDS

Name	Data type	Operations supported ^a	References
portIdentity	PortIdentity (see 6.4.3.7)	R	14.10.2
portState	Enumeration8	R	14.10.3, Table 14-10
ptpPortEnabled	Boolean	RW	14.10.4
delayMechanism	Enumeration8	RW	14.10.5
isMeasuringDelay	Boolean	R	14.10.6
asCapable	Boolean	R	14.10.7
meanLinkDelay	TimeInterval	R	14.10.8
meanLinkDelayThresh	TimeInterval	RW	14.10.9
negMeanLinkDelayThresh	TimeInterval	RW	14.10.10
delayAsymmetry	TimeInterval	RW	14.10.11
neighborRateRatio	Integer32	R	14.10.12
initialLogAnnounceInterval	Integer8	RW	14.10.13
currentLogAnnounceInterval	Integer8	R	14.10.14
useMgtSettableLogAnnounceInterval	Boolean	RW	14.10.15
mgtSettableLogAnnounceInterval	Integer8	RW	14.10.16
announceReceiptTimeout	UInteger8	RW	14.10.16
initialLogSyncInterval	Integer8	RW	14.10.17
currentLogSyncInterval	Integer8	R	14.10.18
useMgtSettableLogSyncInterval	Boolean	RW	14.10.19
mgtSettableLogSyncInterval	Integer8	RW	14.10.20
syncReceiptTimeout	UInteger8	RW	14.10.21

Table 14-9—portDS

Name	Data type	Operations supported ^a	References
syncReceiptTimeoutTimeInterval	UScaledNs	R	14.10.22
initialLogPdelayReqInterval	Integer8	RW	14.10.23
currentLogPdelayReqInterval	Integer8	R	14.10.24
useMgtSettableLogPdelayReqInterval	Boolean	RW	14.10.25
mgtSettableLogPdelayReqInterval	Integer8	RW	14.10.26
initialLogGtpCapableMessageInterval	Integer8	RW	14.10.27
currentLogGtpCapableMessageInterval	Integer8	R	14.10.28
useMgtSettableLogGtpCapableMessageInterval	Boolean	RW	14.10.29
mgtSettableLogGtpCapableMessageInterval	Integer8	RW	14.10.30
initialComputeNeighborRateRatio	Boolean	RW	14.10.31
currentComputeNeighborRateRatio	Boolean	R	14.10.32
useMgtSettableComputeNeighborRateRatio	Boolean	RW	14.10.33
mgtSettableComputeNeighborRateRatio	Boolean	RW	14.10.34
initialComputeMeanLinkDelay	Boolean	RW	14.10.35
currentComputeMeanLinkDelay	Boolean	R	14.10.36
useMgtSettableComputeMeanLinkDelay	Boolean	RW	14.10.37
mgtSettableComputeMeanLinkDelay	Boolean	RW	14.10.38
allowedLostResponses	UInteger8	RW	14.10.39
allowedFaults	UInteger8	RW	14.10.40
gPtpCapableReceiptTimeout	UInteger8	RW	14.10.41
versionNumber	UInteger4	R	14.10.42
nup (NOTE)	Float64	RW	14.10.43
ndown (NOTE)	Float64	RW	14.10.44
oneStepTxOper	Boolean	R	14.10.45
oneStepReceive	Boolean	R	14.10.46
oneStepTransmit	Boolean	R	14.10.47
initialOneStepTxOper	Boolean	RW	14.10.48
currentOneStepTxOper	Boolean	R	14.10.49
useMgtSettableOneStepTxOper	Boolean	RW	14.10.50

Table 14-9—portDS

Name	Data type	Operations supported ^a	References
mgtSettableOneStepTxOper	Boolean	RW	14.10.51
syncLocked	Boolean	R	14.10.52
pdelayTruncatedTimestampsArray	UInteger48[4]	R	14.10.53
minorVersionNumber	UInteger4	R	14.10.54
gptpCapableStateMachinesEnabled	Boolean	RW	14.10.55
nrrPdelay	Integer32	R	14.10.56
nrrSync	Integer32	R	14.10.57
nrrCompMethod	Enumeration2	RW	14.10.58
asCapableAcrossDomains	Boolean	R	14.10.59
NOTE—The values of nup and ndown in Table 14-9 depend on the particular PHY used.			

^a R = Read only access; RW = Read/write access.

1 14.10.2 portIdentity

2 The value is the portIdentity attribute of the local PTP Port (see 8.5.2).

3 14.10.3 portState

4 The value is the value of the PTP Port state of this PTP Port (see Table 10-2) and is taken from the
5 enumeration in Table 14-10. It is equal to the value of the global variable selectedState (see 10.2.4.20) for
6 this PTP Port.

Table 14-10—portState enumeration

State	Value
DisabledPort	3
TimeTransmitterPort	6
PassivePort	7
TimeReceiverPort	9
	All other values reserved
NOTE—The enumeration values are consistent with Table 20 in IEEE Std 1588-2019.	

7 The default value is 3 (DisabledPort).

1 14.10.4 ptpPortEnabled

2 The value is equal to the value of the Boolean ptpPortEnabled (see 10.2.5.13). Setting the managed object
3 ptpPortEnabled causes the Boolean ptpPortEnabled to have the same value.

4 14.10.5 delayMechanism

5 The value of delayMechanism indicates the mechanism for measuring mean propagation delay and neighbor
6 rate ratio on the link attached to this PTP Port and is taken from the enumeration in Table 14-11.
7 portDS.delayMechanism may be P2P (see 11.2.17) only if the only domainNumber active on the physical
8 port is 0.

Table 14-11—delayMechanism enumeration

Delay mechanism	Value	Specification
P2P	02	The PTP Port uses the transport-specific peer-to-peer delay mechanism
COMMON_P2P	03	The PTP Port uses the CMLDS
SPECIAL	04	The PTP Port uses a transport that has a native time transfer mechanism and, therefore, does not use the peer-to-peer delay mechanism (e.g., IEEE 802.11, IEEE 802.3 EPON)
	All other values reserved	
NOTE—The enumeration values are consistent with Table 21 in IEEE Std 1588-2019.		

9 14.10.6 isMeasuringDelay

10 The value is equal to the value of the per-port, per PTP Instance global variable isMeasuringDelay (see
11 11.2.13.6 and 16.4.3.3).

12 14.10.7 asCapable

13 The value is equal to the value of the Boolean asCapable (see 10.2.5.1).

14 14.10.8 meanLinkDelay

15 The value is equal to the value of the per-PTP Port global variable meanLinkDelay (see 10.2.5.8). It is an
16 estimate of the current one-way propagation time on the link attached to this PTP Port, measured as
17 specified for the respective medium (see 11.2.17, 12.5.2, and 16.4). The value is zero for PTP Ports attached
18 to IEEE 802.3 EPON links and for the timeTransmitter port of an IEEE 802.11 link, because one-way
19 propagation delay is not measured on the latter and not directly measured on the former. The data type shall
20 be TimeInterval. The default value is zero.

21 14.10.9 meanLinkDelayThresh

22 The value is equal to the value of the per-PTP Port global variable meanLinkDelayThresh (see 11.2.13.7). It
23 is the propagation time threshold above which a PTP Port is considered not capable of participating in the
24 IEEE 802.1AS protocol. Setting the managed object meanLinkDelayThresh causes the per PTP Port global
25 variable meanLinkDelayThresh to have the same value.

1 14.10.10 negMeanLinkDelayThresh

2 The value is equal to the value of the per-PTP Port global variable negMeanLinkDelayThresh (see
3 11.2.13.8). It is the negative propagation time threshold below which a PTP Port is considered not capable of
4 participating in the IEEE 802.1AS protocol. Setting the managed object negMeanLinkDelayThresh causes
5 the per PTP Port global variable negMeanLinkDelayThresh to have the same value.

6 14.10.11 delayAsymmetry

7 The value is the asymmetry in the propagation delay on the link attached to this PTP Port relative to the
8 Grandmaster Clock time base, as defined in 10.2.5.9 and 8.3. If propagation delay asymmetry is not
9 modeled, then delayAsymmetry is 0.

10 NOTE—The underlying per-port global variable delayAsymmetry is of type ScaledNS, which is a 96-bit value
11 (see 6.4.3.1). delayAsymmetry values that are larger than the maximum value that can be represented by the
12 TimeInterval data type, i.e., 0x7FFF FFFF FFFF FFFF (where the units are 2^{-16} ns; see 6.4.3.3), used for this
13 managed object are set to this largest value. delayAsymmetry values that are less than the minimum value that can be
14 represented by the TimeInterval data type, i.e., 0x8000 0000 0000 0001 written in twos complement form (where the
15 units are 2^{-16} ns; see 6.4.3.3), used for this managed object are set to this smallest value.

16 14.10.12 neighborRateRatio

17 The value is an estimate of the ratio of the frequency of the LocalClock entity of the PTP Instance at the
18 other end of the link attached to this PTP Port, to the frequency of the LocalClock entity of this PTP Instance
19 (see 10.2.5.7). neighborRateRatio is expressed as the fractional frequency offset multiplied by 2^{41} , i.e., the
20 quantity $(\text{neighborRateRatio} - 1.0)(2^{41})$.

21 14.10.13 initialLogAnnounceInterval

22 If useMgtSettableLogAnnounceInterval is FALSE, the value is the logarithm to base 2 of the announce
23 interval used when:

- 24 a) The PTP Port is initialized or
- 25 b) A message interval request TLV is received with the logAnnounceInterval field set to 126
26 (see 10.7.2.2 and the AnnounceIntervalSetting state machine in 10.3.17).

27 14.10.14 currentLogAnnounceInterval

28 The value is the logarithm to the base 2 of the current announce interval (see 10.7.2.2).

29 14.10.15 useMgtSettableLogAnnounceInterval

30 The managed object is a Boolean that determines the source of the announce interval. If the value is TRUE,
31 the value of currentLogAnnounceInterval is set equal to the value of mgtSettableLogAnnounceInterval (see
32 14.10.16). If the value is FALSE, the value of currentLogAnnounceInterval is determined by the
33 AnnounceIntervalSetting state machine (see 10.3.17). The default value of
34 useMgtSettableLogAnnounceInterval is FALSE for domain 0 and TRUE for domains other than domain 0.

35 14.10.16 mgtSettableLogAnnounceInterval

36 The value is the logarithm to base 2 of the announce interval used if useMgtSettableLogAnnounceInterval is
37 TRUE. The value is not used if useMgtSettableLogAnnounceInterval is FALSE.

1 **14.10.16 announceReceiptTimeout**

2 The value is the number of Announce message transmission intervals that a timeReceiver port waits without
3 receiving an Announce message before assuming that the timeTransmitter is no longer transmitting
4 Announce messages and the BTCA needs to be run, if appropriate (see 10.7.3.2).

5 **14.10.17 initialLogSyncInterval**

6 If useMgtSettableLogSyncInterval is FALSE, the value is the logarithm to base 2 of the sync interval used
7 when

- 8 a) The PTP Port is initialized or
- 9 b) A message interval request TLV is received with the logTimeSyncInterval field set to 126
10 (see 10.7.2.3, 11.5.2.3, 12.8.2, 13.9.2, and the SyncIntervalSetting state machine in 10.3.18).

11 **14.10.18 currentLogSyncInterval**

12 The value is the logarithm to the base 2 of the current time-synchronization transmission interval
13 (see 10.7.2.3).

14 **14.10.19 useMgtSettableLogSyncInterval**

15 The managed object is a Boolean that determines the source of the sync interval. If the value is TRUE, the
16 value of currentLogSyncInterval is set equal to the value of mgtSettableLogSyncInterval (see 14.10.20). If
17 the value of the managed object is FALSE, the value of currentLogSyncInterval is determined by the
18 SyncIntervalSetting state machine (see 10.3.18). The default value of useMgtSettableLogSyncInterval is
19 FALSE for domain 0 and TRUE for domains other than domain 0.

20 **14.10.20 mgtSettableLogSyncInterval**

21 The value is the logarithm to base 2 of the sync interval if useMgtSettableLogSyncInterval is TRUE. The
22 value is not used if useMgtSettableLogSyncInterval is FALSE.

23 **14.10.21 syncReceiptTimeout**

24 The value is the number of time-synchronization transmission intervals that a timeReceiver port waits
25 without receiving synchronization information before assuming that the timeTransmitter is no longer
26 transmitting synchronization information and that the BTCA needs to be run, if appropriate (see 10.7.3.1).

27 **14.10.22 syncReceiptTimeoutTimeInterval**

28 The value is equal to the value of the per-PTP Port global variable syncReceiptTimeoutTimeInterval
29 (see 10.2.5.3). It is the time interval after which sync receipt timeout occurs if time-synchronization
30 information has not been received during the interval.

31 **14.10.23 initialLogPdelayReqInterval**

32 For full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure
33 path delay (see 16.4.3.2), the value is the logarithm to base 2 of the Pdelay_Req message transmission
34 interval used when:

- 35 a) The PTP Port is initialized or
- 36 b) A message interval request TLV is received with the logLinkDelayInterval field set to 126 (see
37 11.5.2.2 and the LinkDelayIntervalSetting state machine in 11.2.21).

1 For all other media, the value is 127.

2 **14.10.24 currentLogPdelayReqInterval**

3 For full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure
4 path delay (see 16.4.3.2), the value is the logarithm to the base 2 of the current Pdelay_Req message
5 transmission interval (see 11.5.2.2).

6 For all other media, the value is 127.

7 **14.10.25 useMgtSettableLogPdelayReqInterval**

8 The managed object is a Boolean that determines the source of the mean time interval between successive
9 Pdelay_Req messages. If the value is TRUE, the value of currentLogPdelayReqInterval is set equal to the
10 value of mgtSettableLogPdelayReqInterval (see 14.10.26). If the value of the managed object is FALSE,
11 the value of currentLogPdelayReqInterval is determined by the LinkDelayIntervalSetting state machine
12 (see 11.2.21). The default value of useMgtSettableLogPdelayReqInterval is FALSE.

13 **14.10.26 mgtSettableLogPdelayReqInterval**

14 The value is the logarithm to base 2 of the mean time interval between successive Pdelay_Req
15 messages if useMgtSettableLogPdelayReqInterval is TRUE. The value is not used if
16 useMgtSettableLogPdelayReqInterval is FALSE.

17 **14.10.27 initialLogGtpCapableMessageInterval**

18 The value is the logarithm to base 2 of the gTP capable message interval used when:

- 19 a) The PTP Port is initialized or
- 20 b) A gPtpCapableMessage interval request TLV is received with the logGtpCapableMessageInterval
21 field set to 126 (see 10.6.4.5 and the GtpCapableIntervalSetting state machine in 10.4.4).

22 **14.10.28 currentLogGtpCapableMessageInterval**

23 The value is the logarithm to the base 2 of the current gTP capable message interval (see 10.7.2.5).

24 **14.10.29 useMgtSettableLogGtpCapableMessageInterval**

25 The managed object is a Boolean that determines the source of the gTP capable message interval. If the
26 value is TRUE, the value of currentLogGtpCapableMessageInterval is set equal to the value of
27 mgtSettableLogGtpCapableMessageInterval (see 14.10.30). If the value of the managed object is FALSE,
28 the value of currentLogGtpCapableMessageInterval is determined by the
29 GtpCapableMessageIntervalSetting state machine (see 10.4.4). The default value of
30 useMgtSettableLogGtpCapableMessageInterval is FALSE.

31 **14.10.30 mgtSettableLogGtpCapableMessageInterval**

32 The value is the logarithm to base 2 of the gPtpCapableMessageInterval
33 if useMgtSettableLogGtpCapableMessageInterval is TRUE. The value is not used if
34 useMgtSettableLogGtpCapableMessageInterval is FALSE.

1 **14.10.31 initialComputeNeighborRateRatio**

2 If useMgtSettableComputeNeighborRateRatio is FALSE, then for full-duplex IEEE 802.3 media and for
3 CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the
4 initial value of computeNeighborRateRatio (see 10.2.5.10).

5 For all other media, the value is TRUE.

6 **14.10.32 currentComputeNeighborRateRatio**

7 For full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure
8 path delay (see 16.4.3.2), the value is the current value of computeNeighborRateRatio.

9 For all other media, the value is TRUE.

10 **14.10.33 useMgtSettableComputeNeighborRateRatio**

11 The managed object is a Boolean that determines the source of the value of computeNeighborRateRatio. If
12 the value is TRUE, the value of computeNeighborRateRatio is set equal to the value of
13 mgtSettableComputeNeighborRateRatio (see 14.18.18). If the value of the managed object is FALSE, the
14 value of currentComputeNeighborRateRatio is determined by the LinkDelayIntervalSetting state machine
15 (see 11.2.21). The default value of useMgtSettableComputeNeighborRateRatio is FALSE.

16 **14.10.34 mgtSettableComputeNeighborRateRatio**

17 computeNeighborRateRatio is configured to this value if useMgtSettableComputeNeighborRateRatio is
18 TRUE. The value is not used if useMgtSettableComputeNeighborRateRatio is FALSE.

19 **14.10.35 initialComputeMeanLinkDelay**

20 If useMgtSettableComputeMeanLinkDelay is FALSE, then for full-duplex IEEE 802.3 media and for CSN
21 media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the initial
22 value of computeMeanLinkDelay (see 10.2.5.10).

23 For all other media, the value is TRUE.

24 **14.10.36 currentComputeMeanLinkDelay**

25 For full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure
26 path delay (see 16.4.3.2), the value is the current value of computeMeanLinkDelay.

27 For all other media, the value is TRUE.

28 **14.10.37 useMgtSettableComputeMeanLinkDelay**

29 The managed object is a Boolean that determines the source of the value of computeMeanLinkDelay. If the
30 value is TRUE, the value of computeMeanLinkDelay is set equal to the value of
31 mgtSettableComputeMeanLinkDelay (see 14.10.38). If the value of the managed object is FALSE, the value
32 of currentComputeMeanLinkDelay is determined by the LinkDelayIntervalSetting state machine
33 (see 11.2.21). The default value of useMgtSettableComputeMeanLinkDelay is FALSE.

1 **14.10.38 mgtSettableComputeMeanLinkDelay**

2 computeMeanLinkDelay is configured to this value if useMgtSettableComputeMeanLinkDelay is TRUE.
3 The value is not used if useMgtSettableComputeMeanLinkDelay is FALSE.

4 **14.10.39 allowedLostResponses**

5 The value is equal to the value of the per-PTP Port global variable allowedLostResponses (see 11.5.3 and
6 11.2.13.4). It is the number of Pdelay_Req messages without valid responses above which a PTP Port is
7 considered to be not exchanging peer delay messages with its neighbor. Setting the managed object
8 allowedLostResponses causes the per PTP Port global variable allowedLostResponses to have the same
9 value.

10 **14.10.40 allowedFaults**

11 The value is equal to the value of the per-PTP Port global variable allowedFaults (see 11.5.4 and 11.2.13.5).
12 It is the number of faults (see 11.5.4) above which asCapable is set to FALSE, i.e., a PTP Port is considered
13 not capable of interoperating with its neighbor via the IEEE 802.1AS protocol (see 10.2.5.1). Setting the
14 managed object allowedLostResponses causes the per PTP Port global variable allowedFaults to have the
15 same value.

16 **14.10.41 gPtpCapableReceiptTimeout**

17 The value is the number of transmission intervals that a PTP Port waits without receiving the gPTP capable
18 TLV before assuming that the neighbor PTP Port is no longer invoking gPTP (see 10.7.3.3).

19 **14.10.42 versionNumber**

20 This value is set to versionPTP as specified in 10.6.2.2.4.

21 **14.10.43 nup**

22 For an OLT port of an IEEE 802.3 EPON link, the value is the effective index of refraction for the EPON
23 upstream wavelength light of the optical path (see 13.1.4 and 13.8.1.2.2). The default value is 1.46770 for
24 1 Gb/s upstream links and 1.46773 for 10 Gb/s upstream links.

25 For all other PTP Ports, the value is 0.

26 **14.10.44 ndown**

27 For an OLT port of an IEEE 802.3 EPON link, the value is the effective index of refraction for the EPON
28 downstream wavelength light of the optical path (see 13.1.4 and 13.8.1.2.1). The default value is 1.46805 for
29 1 Gb/s downstream links and 1.46851 for 10 Gb/s downstream links.

30 For all other PTP Ports, the value is 0.

31 **14.10.45 oneStepTxOper**

32 The value is equal to the value of the per-PTP Port global variable oneStepTxOper (see 11.2.13.12). Its value
33 is TRUE if the PTP Port is sending one-step Sync messages and FALSE if the PTP Port is sending two-step
34 Sync and Follow-Up messages.

1 14.10.46 oneStepReceive

2 The value is equal to the value of the per-PTP Port global variable oneStepReceive (see 11.2.13.10). Its
 3 value is TRUE if the PTP Port is capable of receiving and processing one-step Sync messages.

4 14.10.47 oneStepTransmit

5 The value is equal to the value of the per-PTP Port global variable oneStepTransmit (see 11.2.13.11). Its
 6 value is TRUE if the PTP Port is capable of transmitting one-step Sync messages.

7 14.10.48 initialOneStepTxOper

8 If useMgtSettableOneStepTxOper is FALSE, the value is used to initialize currentOneStepTxOper when the
 9 PTP Port is initialized. If useMgtSettableOneStepTxOper is TRUE, the value of initialOneStepTxOper is not
 10 used. The default value of initialOneStepTxOper shall be FALSE.

11 14.10.49 currentOneStepTxOper

12 The value is TRUE if it is desired, either via management or via a received Signaling message, that the PTP
 13 Port transmit one-step Sync messages. The value is FALSE if it is not desired, either via management or via
 14 a received Signaling message, that the PTP Port transmit one-step Sync messages.

15 NOTE—The PTP Port will send one-step Sync messages only if currentOneStepTxOper and oneStepTransmit
 16 (see 14.10.47) are both TRUE (see 11.2.16 and Figure 11-8).

17 14.10.50 useMgtSettableOneStepTxOper

18 The managed object is a Boolean that determines the source of currentOneStepTxOper. If the value is
 19 TRUE, the value of currentOneStepTxOper is set equal to the value of mgtSettableOneStepTxOper
 20 (see 14.10.51). If the value is FALSE, the value of currentOneStepTxOper is determined by the
 21 OneStepTxOperSetting state machine (see 11.2.16 and Figure 11-8). The default value of
 22 useMgtSettableOneStepTxOper is TRUE.

23 14.10.51 mgtSettableOneStepTxOper

24 If useMgtSettableOneStepTxOper is TRUE, currentOneStepTxOper is set equal to the value of
 25 mgtSettableOneStepTxOper. The value of mgtSettableOneStepTxOper is not used if
 26 useMgtSettableOneStepTxOper is FALSE. The default value of mgtSettableOneStepTxOper is FALSE for
 27 domains other than domain 0.

28 14.10.52 syncLocked

29 The value is equal to the value of the per-PTP Port global variable syncLocked (see 10.2.5.15). Its value is
 30 TRUE if the PTP Port will transmit a Sync as soon as possible after the timeReceiver PTP Port receives a
 31 Sync.

32 14.10.53 pdelayTruncatedTimestampsArray

33 For full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure
 34 path delay (see 16.4.3.2), the values of the four elements of this array are as described in Table 14-12. For all
 35 other media, the values are zero. Array elements 0, 1, 2, and 3 correspond to the timestamps t1, t2, t3, and t4,
 36 respectively, in Figure 11-1 and are expressed in units of 2^{-16} ns (i.e., the value of each array element is
 37 equal to the remainder obtained upon dividing the respective timestamp, expressed in units of 2^{-16} ns,
 38 by 2^{48}). At any given time, the timestamp values stored in the array are for the same, and most recently
 39 completed, peer delay message exchange.

Table 14-12—Description of `pdelayTruncatedTimestampsArray`

Array element	Timestamp description	Corresponding timestamp of Figure 11-1	Units
0	<code>pdelayReqEventEgressTimestamp</code> for <code>Pdelay_Req</code> message, of most recently completed peer delay message exchange, transmitted by this PTP Instance (NOTE 1)	t1	2^{-16} ns
1	<code>pdelayReqEventIngressTimestamp</code> for <code>Pdelay_Req</code> message received at peer delay responder to which this Link Port sends <code>Pdelay_Req</code> , of most recently completed peer delay message exchange; it is equal to the sum of the following: a) The ns field of the <code>requestReceiptTimestamp</code> (see Table 11-14), multiplied by 2^{16} b) The <code>correctionField</code> (see Table 11-6) (NOTE 2)	t2	2^{-16} ns
2	<code>pdelayRespEventEgressTimestamp</code> for <code>Pdelay_Resp</code> message, of most recently completed peer delay message exchange, transmitted by peer delay responder to which this Link Port sends <code>Pdelay_Req</code> ; it is equal to the sum of the following: a) The ns field of the <code>responseOriginTimestamp</code> (see Table 11-15), multiplied by 2^{16} b) The <code>correctionField</code> (see Table 11-6) (NOTE 3)	t3	2^{-16} ns
3	<code>pdelayRespEventIngressTimestamp</code> for <code>Pdelay_Resp</code> message, of most recently completed peer delay message exchange, received by this PTP Instance (NOTE 4)	t4	2^{-16} ns
NOTE 1—This quantity is not simply the nanoseconds plus fractional nanoseconds portion of the <code>pdelayReqEventEgressTimestamp</code>. Rather, it is equal to $[\text{pdelayReqEventEgressTimestamp.seconds} \times (10^9)(2^{16}) + \text{pdelayReqEventEgressTimestamp.fractionalNanoseconds}] \bmod 2^{48}$, where <code>pdelayReqEventEgressTimestamp</code> is expressed as an <code>ExtendedTimestamp</code> (see 6.4.3.5). Its units are 2^{-16} ns. NOTE 2—This quantity is not simply the nanoseconds plus fractional nanoseconds portion of the <code>pdelayReqEventIngressTimestamp</code>. Rather, it is equal to $[\text{pdelayReqEventIngressTimestamp.seconds} \times (10^9)(2^{16}) + \text{pdelayReqEventIngressTimestamp.fractionalNanoseconds}] \bmod 2^{48}$, where <code>pdelayReqEventIngressTimestamp</code> is expressed as an <code>ExtendedTimestamp</code> (see 6.4.3.5). Its units are 2^{-16} ns. NOTE 3—This quantity is not simply the nanoseconds plus fractional nanoseconds portion of the <code>pdelayRespEventEgressTimestamp</code>. Rather, it is equal to $[\text{pdelayRespEventEgressTimestamp.seconds} \times (10^9)(2^{16}) + \text{pdelayRespEventEgressTimestamp.fractionalNanoseconds}] \bmod 2^{48}$, where <code>pdelayRespEventEgressTimestamp</code> is expressed as an <code>ExtendedTimestamp</code> (see 6.4.3.5). Its units are 2^{-16} ns. NOTE 4—This quantity is not simply the nanoseconds plus fractional nanoseconds portion of the <code>pdelayRespEventIngressTimestamp</code>. Rather, it is equal to $[\text{pdelayRespEventIngressTimestamp.seconds} \times (10^9)(2^{16}) + \text{pdelayRespEventIngressTimestamp.fractionalNanoseconds}] \bmod 2^{48}$, where <code>pdelayRespEventIngressTimestamp</code> is expressed as an <code>ExtendedTimestamp</code> (see 6.4.3.5). Its units are 2^{-16} ns.			

1 14.10.54 `minorVersionNumber`

2 This value is set to `minorVersionPTP` as specified in 10.6.2.2.3.

3 14.10.55 `gptpCapableStateMachinesEnabled`

4 A Boolean that is used to enable or disable the `GptpCapableTransmit`, `GptpCapableReceive`, and
5 `GptpCapableIntervalSetting` state machines. If the value is TRUE, the `GptpCapableTransmit` and

1 GtpCapableReceive state machines are enabled, and the GtpCapableIntervalSetting state machine is
2 enabled if it is implemented. If the value is FALSE, the GtpCapableTransmit and GtpCapableReceive
3 state machines are disabled, and the GtpCapableIntervalSetting state machine is disabled if it is
4 implemented. The default value is TRUE.

5 **14.10.56 nrrPdelay**

6 The value of the global variable nrrPdelay (see 11.2.13.14), expressed as Integer32.

7 The value is an estimate of the ratio of the frequency of the LocalClock entity of the time-aware system at
8 the other end of the link attached to this Link Port, to the frequency of the LocalClock entity of this time-
9 aware system (see 10.2.5.7). The managed object nrrPdelay is expressed as the fractional frequency offset
10 stored in the global variable nrrPdelay (see 11.2.13.14) multiplied by 2^{41} , i.e., the quantity $(\text{nrrPdelay}(\text{global}$
11 $\text{variable}) - 1.0)(2^{41})$. The default value of the managed object nrrPdelay is 0.0, corresponding to a rate ratio
12 of 1.0.

13 **14.10.57 nrrSync**

14 The value of the global variable nrrSync (see 11.2.13.15), expressed as Integer32.

15 The value is an estimate of the ratio of the frequency of the LocalClock entity of the time-aware system at
16 the other end of the link attached to this Link Port, to the frequency of the LocalClock entity of this time-
17 aware system (see 10.2.5.7). The managed object nrrSync is expressed as the fractional frequency offset
18 stored in the global variable nrrSync (see 11.2.13.15) multiplied by 2^{41} , i.e., the quantity $(\text{nrrSync}(\text{global}$
19 $\text{variable}) - 1.0)(2^{41})$.

20 **14.10.58 nrrCompMethod**

21 The value of the global variable nrrCompMethod (see 11.2.13.16).

22 **14.10.59 asCapableAcrossDomains**

23 The value is equal to the value of the Boolean asCapableAcrossDomains (see 11.2.2 and 11.2.13.13).

24 **14.11 Description Port Parameter Data Set (descriptionPortDS)**

25 **14.11.1 General**

26 The descriptionPortDS contains the profileIdentifier for this PTP profile, as specified in F.2.

27 The parameter of descriptionPortDS is summarized in Table 14-13 and subsequently specified.

Table 14-13—descriptionPortDS

Name	Data type	Operations supported ^a	References
profileIdentifier	Octet6	R	14.11.2

^a R= Read only access.

1 14.11.2 profileIdentifier

2 The value is the profileIdentifier for this PTP profile (see F.2).

3 14.12 Port Parameter Statistics Data Set (portStatisticsDS)

4 14.12.1 General

5 For the single PTP Port of a PTP End Instance and for each PTP Port of a PTP Relay Instance, the
6 portStatisticsDS provides counters associated with PTP Port capabilities at a given PTP Instance. The
7 number of such statistics sets is the same as the value of defaultDS.numberPorts.

8 There is one set of portStatisticsDS parameters per PTP Port, per PTP Instance of a time-aware system. Each
9 portStatisticsDS contains a set of counters for each PTP Port that supports the time-synchronization
10 capability, as detailed in Table 14-14. Each set of portStatisticsDS can be created or removed dynamically in
11 implementations that support dynamic configuration of PTP Ports and components

12 The parameters of portStatisticsDS are summarized in Table 14-14 and subsequently specified.

Table 14-14—portStatisticsDS

Name	Data type	Operations supported ^a	References
rxSyncCount	UInteger32	R	14.12.2
rxOneStepSyncCount	UInteger32	R	14.12.3
rxFollowUpCount	UInteger32	R	14.12.4
rxPdelayRequestCount	UInteger32	R	14.12.5
rxPdelayResponseCount	UInteger32	R	14.12.6
rxPdelayResponseFollowUpCount	UInteger32	R	14.12.7
rxAnnounceCount	UInteger32	R	14.12.8
rxPtpPacketDiscardCount	UInteger32	R	14.12.9
syncReceiptTimeoutCount	UInteger32	R	14.12.10
announceReceiptTimeoutCount	UInteger32	R	14.12.11
pdelayAllowedLostResponsesExceededCount	UInteger32	R	14.12.12
txSyncCount	UInteger32	R	14.12.13
txOneStepSyncCount	UInteger32	R	14.12.14
txFollowUpCount	UInteger32	R	14.12.15
txPdelayRequestCount	UInteger32	R	14.12.16
txPdelayResponseCount	UInteger32	R	14.12.17
txPdelayResponseFollowUpCount	UInteger32	R	14.12.18
txAnnounceCount	UInteger32	R	14.12.19
rxSyncCountTimeReceiverP	UInteger32	R	14.12.20

^aR= Read only access.

1 14.12.2 rxSyncCount

2 This counter increments every time synchronization information is received, denoted by one of the following
3 events:

- 4 — A transition to TRUE from FALSE of the rcvdSync variable of the MDSyncReceiveSM state
5 machine (see 11.2.14.1.2 and Figure 11-6) when in the DISCARD, WAITING_FOR_SYNC, or
6 WAITING_FOR_FOLLOW_UP states; or
- 7 — rcvdIndication transitions to TRUE (see Figure 12-7).

8 14.12.3 rxOneStepSyncCount

9 This counter increments every time a one-step Sync message is received, denoted by a transition to TRUE
10 from FALSE of the rcvdSync variable of the MDSyncReceiveSM state machine (see 11.2.14.1.2 and
11 Figure 11-6) with the variable rcvdSyncPtr->twoStepFlag FALSE when in the DISCARD or
12 WAITING_FOR_SYNC states.

13 14.12.4 rxFollowUpCount

14 This counter increments every time a Follow_Up message is received, denoted by a transition to TRUE from
15 FALSE of the rcvdFollowUp variable of the MDSyncReceiveSM state machine (see 11.2.14.1.3 and
16 Figure 11-6) when in the WAITING_FOR_FOLLOW_UP state.

17 14.12.5 rxPdelayRequestCount

18 This counter increments every time a Pdelay_Req message is received, denoted by a transition to TRUE
19 from FALSE of the rcvdPdelayReq variable of the MDPdelayResp state machine (see 11.2.20.2.1 and
20 Figure 11-10) when in the WAITING_FOR_PDELAY_REQ or INITIAL_WAITING_FOR_PDELAY_REQ
21 states.

22 14.12.6 rxPdelayResponseCount

23 This counter increments every time a Pdelay_Resp message is received, denoted by a transition to TRUE
24 from FALSE of the rcvdPdelayResp variable of the MDPdelayReq state machine (see 11.2.19.2.2 and
25 Figure 11-9) when in the WAITING_FOR_PDELAY_RESP state.

26 14.12.7 rxPdelayResponseFollowUpCount

27 This counter increments every time a Pdelay_Resp_Follow_Up message is received, denoted by a transition
28 to TRUE from FALSE of the rcvdPdelayRespFollowUp variable of the MDPdelayReq state machine
29 (see 11.2.19.2.4 and Figure 11-9) when in the WAITING_FOR_PDELAY_RESP_FOLLOW_UP state.

30 14.12.8 rxAnnounceCount

31 This counter increments every time an Announce message is received, denoted by a transition to TRUE from
32 FALSE of the rcvdAnnounce variable of the PortAnnounceReceive state machine (see 10.3.11 and
33 Figure 10-13) when in the DISCARD or RECEIVE states.

34 14.12.9 rxPtpPacketDiscardCount

35 This counter increments every time a PTP message of the respective PTP Instance is discarded, caused by
36 the occurrence of any of the following conditions:

- 1 a) A received Announce message is not qualified, denoted by the function qualifyAnnounce (see
2 10.3.11.2.1 and 13.1.3.4) of the PortAnnounceReceive state machine (see 10.3.11 and Figure 10-13)
3 returning FALSE;
- 4 b) A Follow_Up message corresponding to a received Sync message is not received, denoted by a
5 transition of the condition (currentTime >= followUpReceiptTimeoutTime) to TRUE from FALSE
6 when in the WAITING_FOR_FOLLOW_UP state of the MDSyncReceiveSM state machine (see
7 11.2.14 and Figure 11-6);
- 8 c) A Pdelay_Resp message corresponding to a transmitted Pdelay_Req message is not received,
9 denoted by a transition from the WAITING_FOR_PDELAY_RESP state to the RESET state of the
10 MDPdelayReq state machine (see 11.2.19 and Figure 11-9);
- 11 d) A Pdelay_Resp_Follow_Up message corresponding to a transmitted Pdelay_Req message is not
12 received, denoted by a transition from the WAITING_FOR_PDELAY_RESP_FOLLOW_UP state
13 to the RESET state of the MDPdelayReq state machine (see 11.2.19 and Figure 11-9).

14 14.12.10 syncReceiptTimeoutCount

15 This counter increments every time sync receipt timeout occurs, denoted by entering the AGED state of the
16 PortAnnounceInformation state machine (see 10.3.12 and Figure 10-14) with the condition (currentTime>=
17 syncReceiptTimeoutTime && gmPresent) evaluating to TRUE.

18 14.12.11 announceReceiptTimeoutCount

19 This counter increments every time announce receipt timeout occurs, denoted by entering the AGED state of
20 the PortAnnounceInformation state machine from the CURRENT state of the PortAnnounceInformation
21 state machine (see 10.3.12 and Figure 10-14) with the condition (currentTime >=
22 announceReceiptTimeoutTime) evaluating to TRUE.

23 14.12.12 pdelayAllowedLostResponsesExceededCount

24 This counter increments every time the value of the variable lostResponses (see 11.2.19.2.9) exceeds the
25 value of the variable allowedLostResponses (see 11.2.13.4), in the RESET state of the MDPdelayReq state
26 machine (see 11.2.19 and Figure 11-9).

27 14.12.13 txSyncCount

28 This counter increments every time synchronization information is transmitted, denoted by one of the
29 following events:

- 30 — A transition to TRUE from FALSE of the rcvdMDSyncMDSS variable of the MDSyncSendSM
31 state machine (see 11.2.15.1.1 and Figure 11-7) when in the INITIALIZING, SEND_FOLLOW_UP,
32 or SET_CORRECTION_FIELD states; or
- 33 — The INITIATE_REQUEST_WAIT_CONFIRM_OR_SAVE_INFO state is entered in Figure 12-5
34 and TM is being used [i.e., (tmFtmSupport == 0x01 || (tmFtmSupport & 0x01 == 0x01)&&
35 ftmReqGrantedTimeTransmitter)) in timeTransmitter state machine A of Figure 12-5]; or
- 36 — The INITIATE_REQUEST_WAIT_CONFIRM state is entered in Figure 12-6 (in this case FTM is
37 being used).

38 14.12.14 txOneStepSyncCount

39 This counter increments every time a one-step Sync message is transmitted, denoted by a transition to TRUE
40 from FALSE of the rcvdMDSyncMDSS variable of the MDSyncSendSM state machine (see 11.2.15.1.1 and
41 Figure 11-7) with the variable oneStepTxOper TRUE when in the INITIALIZING,
42 SEND_SYNC_ONE_STEP, or SET_CORRECTION_FIELD states.

1 14.12.15 txFollowUpCount

2 This counter increments every time a Follow_Up message is transmitted, denoted by a transition to TRUE
3 from FALSE of the rcvdMDTimestampReceive variable of the MDSyncSendSM state machine (see
4 11.2.15.1.3 and Figure 11-7) when in the SEND_SYNC state.

5 14.12.16 txPdelayRequestCount

6 This counter increments every time a Pdelay_Req message is transmitted, denoted by entering the
7 INITIAL_SEND_PDELAY_REQ or SEND_PDELAY_REQ states of the MDPdelayReq state machine (see
8 11.2.19 and Figure 11-9).

9 14.12.17 txPdelayResponseCount

10 This counter increments every time a Pdelay_Resp message is transmitted, denoted by the following events:

- 11 — A transition to TRUE from FALSE of the rcvdPdelayReq variable of the MDPdelayResp state
12 machine (see 11.2.20.2.1 and Figure 11-10) when in the WAITING_FOR_PDELAY_REQ or
13 INITIAL_WAITING_FOR_PDELAY_REQ state and
- 14 — The resulting entry to the SENT_PDELAY_RESP_WAITING_FOR_TIMESTAMP state.

15 14.12.18 txPdelayResponseFollowUpCount

16 This counter increments every time a Pdelay_Resp_Follow_Up message is transmitted, denoted by the
17 following events:

- 18 — A transition to TRUE from FALSE of the rcvdMDTimestampReceive variable of the
19 MDPdelayResp state machine (see 11.2.20.2.2 and Figure 11-10) when in the
20 SENT_PDELAY_RESP_WAITING_FOR_TIMESTAMP states and
- 21 — The resulting entry to the WAITING_FOR_PDELAY_REQ state.

22 14.12.19 txAnnounceCount

23 This counter increments every time an Announce message is transmitted, denoted by entering the
24 TRANSMIT_ANNOUNCE state of the PortAnnounceReceive state machine (see 10.3.16 and Figure 10-18).

25 14.12.20 rxSyncCountTimeReceiverP

26 This counter increments whenever time synchronization information is received on a PTP Port when its port
27 state is TimeReceiverPort. The receipt of time synchronization information is denoted by one of the following
28 events:

- 29 — A transition to TRUE from FALSE of the rcvdSync variable of the MDSyncReceiveSM state
30 machine (see 11.2.14.1.2 and Figure 11-6) when in the DISCARD, WAITING_FOR_SYNC, or
31 WAITING_FOR_FOLLOW_UP states; or
- 32 — rcvdIndication transitions to TRUE (see Figure 12-7).

33 This counter is initialized to zero, and resets to zero when the port state is not TimeReceiverPort.

14.13 Acceptable TimeTransmitter Port Parameter Data Set (acceptableTimeTransmitterPortDS)

14.13.1 General

The acceptableTimeTransmitterPortDS represents the capability to enable/disable the acceptable timeTransmitter table feature on a PTP Port. For the single PTP Port of a PTP End Instance and for each PTP Port of a PTP Relay Instance, this data set contains the single member acceptableTimeTransmitterTableEnabled, which is used to enable/disable the Acceptable TimeTransmitter Table Feature. The number of such data sets is the same as the value of defaultDS.numberPorts.

The parameter of acceptableTimeTransmitterPortDS is summarized in Table 14-15 and subsequently specified.

Table 14-15—acceptableTimeTransmitterPortDS

Name	Data type	Operations supported ^a	References
acceptableTimeTransmitterTableEnabled	Boolean	RW	14.13.2

^a R = Read only access; RW = Read/write access.

14.13.2 acceptableTimeTransmitterTableEnabled

The value is equal to the value of the Boolean acceptableTimeTransmitterTableEnabled (see 13.1.3.2 and 13.1.3.5).

14.14 External Port Configuration Port Parameter Data Set (externalPortConfigurationPortDS)

14.14.1 General

The externalPortConfigurationPortDS is used with the external port configuration option to indicate the desired state for the PTP Port. This data set contains the single member desiredState, which indicates the desired state for the PTP Port. The number of such data sets is the same as the value of defaultDS.numberPorts.

The parameter of externalPortConfigurationPortDS is summarized in Table 14-16 and subsequently specified.

Table 14-16—externalPortConfigurationPortDS

Name	Data type	Operations supported ^a	References
desiredState	Enumeration8 (see 6.4.2)	RW	14.14.2, Table 14-10

^a R = Read only access; RW = Read/write access.

1 **14.14.2 desiredState**

2 When the value of defaultDS.externalPortConfigurationEnabled is TRUE, the value of
3 externalPortConfigurationPortDS.desiredState is the desired state of the PTP Port. This member sets the
4 value of the variable portStateInd (see 10.3.15.1.5). When a new value is written to the member by
5 management, the variable rcvdPortStateInd (see 10.3.15.1.4) is set to TRUE.

6 **14.15 Asymmetry Measurement Mode Parameter Data Set**
7 **(asymmetryMeasurementModeDS)**

8 **14.15.1 General**

9 The asymmetryMeasurementModeDS represents the capability to enable/disable the Asymmetry
10 Compensation Measurement Procedure on a PTP Port (see Annex G). This data set is used instead of the
11 cmlDsAsymmetryMeasurementModeDS when only domain 0 is present and CMLDS is not used.

12 For the single PTP Port of a PTP End Instance and for each PTP Port of a PTP Relay Instance, the
13 asymmetryMeasurementModeDS contains the single member asymmetryMeasurementMode, which is used
14 to enable/disable the Asymmetry Compensation Measurement Procedure. The number of such data sets is
15 the same as the value of defaultDS.numberPorts.

16 There is one asymmetryMeasurementModeDS for the single PTP Instance whose domainNumber is 0, per
17 PTP Port, as detailed in Table 14-17. This data set is used only when there is a single gPTP domain and
18 CMLDS is not used.

19 The parameter of asymmetryMeasurementModeDS is summarized in Table 14-17 and subsequently
20 specified.

Table 14-17—asymmetryMeasurementModeDS

Name	Data type	Operations supported ^a	References
asymmetryMeasurementMode	Boolean	RW	14.15.2

^a R = Read only access; RW = Read/write access.

21 **14.15.2 asymmetryMeasurementMode**

22 The value is equal to the value of the Boolean asymmetryMeasurementMode (see G.3). For full-duplex
23 IEEE 802.3 media, the value is TRUE if an asymmetry measurement is being performed for the link
24 attached to this PTP Port and FALSE otherwise. For all other media, the value shall be FALSE
25 (see 10.2.5.2). Setting the managed object asymmetryMeasurementMode causes the Boolean
26 asymmetryMeasurementMode to have the same value.

27 NOTE—If an asymmetry measurement is being performed for a link, asymmetryMeasurementMode must be TRUE for
28 the PTP Ports at each end of the link.

1 14.16 Common Services Port Parameter Data Set (commonServicesPortDS)

2 14.16.1 General

3 The commonServicesPortDS enables a PTP Port of a PTP Instance to determine which port of the respective
4 common service corresponds to that PTP Port.

5 This member contains the port number of the CMLDS Link Port that corresponds to this PTP Port.

6 The parameter of commonServicesPortDS is summarized in Table 14-18 and subsequently specified.

Table 14-18—commonServicesPortDS

Name	Data type	Operations supported ^a	References
cmldsLinkPortPortNumber	UInteger16	R	14.16.2

^a R = Read only access.

7 14.16.2 cmldsLinkPortPortNumber

8 The value is the portNumber attribute of the cmldsLinkPortDS.portIdentity (see 14.18.2) of the Link Port
9 that corresponds to this PTP Port.

10 14.17 Common Mean Link Delay Service Default Parameter Data Set (cmldsDefaultDS)

11 14.17.1 General

12 The cmldsDefaultDS describes the per-time-aware-system attributes of the CMLD.

13 NOTE—The value of sdoId for the CMLD is fixed as 0x200 (see 11.2.17) and cannot be changed. Therefore, a
14 corresponding data set member for sdoId is not needed.

15 The parameters of cmldsDefaultDS are summarized in Table 14-19 and subsequently specified.

Table 14-19—cmldsDefaultDS

Name	Data type	Operations supported ^a	References
clockIdentity	ClockIdentity	R	14.17.2
numberLinkPorts	UInteger16	R	14.17.3

^a R = Read only access; RW = Read/write access.

16 14.17.2 clockIdentity

17 The value is the clockIdentity (see 8.5.2.2) that will be used to identify the CMLD.

1 14.17.3 numberLinkPorts

2 The value is the number of Link Ports of the time-aware system on which the CMLD is implemented. For an
3 end station the value is 1.

4 14.18 Common Mean Link Delay Service Link Port Parameter Data Set 5 (cmldsLinkPortDS)

6 14.18.1 General

7 The cmldsLinkPortDS represents time-aware Link Port capabilities for the CMLD of a Link Port of a time-
8 aware system.

9 For every Link Port of the CMLD of a time-aware system, the cmldsLinkPortDS is maintained as the basis
10 for making protocol decisions and providing values for message fields. The number of such data sets is the
11 same as the value of cmldsDefaultDS.numberLinkPorts.

12 There is one set of cmldsLinkPortDS parameters per Link Port, for the PTP Instance of a time-aware system.
13 Each cmldsLinkPortDS contains a set of parameters for each Link Port that supports the time-
14 synchronization capability, as detailed in Table 14-20. Each set of cmldsLinkPortDS can be created or
15 removed dynamically in implementations that support dynamic configuration of Link Ports and
16 components.

17 The parameters of cmldsLinkPortDS are summarized in Table 14-20 and subsequently specified.

Table 14-20—cmldsLinkPortDS

Name	Data type	Operations supported ^a	References
portIdentity	PortIdentity (see 6.4.3.7)	R	14.18.2
cmldsLinkPortEnabled	Boolean	R	14.18.3
isMeasuringDelay	Boolean	R	14.18.4
asCapableAcrossDomains	Boolean	R	14.18.5
meanLinkDelay	TimeInterval	R	14.18.6
meanLinkDelayThresh	TimeInterval	RW	14.18.7
negMeanLinkDelayThresh	TimeInterval	RW	14.18.8
delayAsymmetry	TimeInterval	RW	14.18.9
nrrPdelay	Integer32	R	14.18.10
initialLogPdelayReqInterval	Integer8	RW	14.18.11
currentLogPdelayReqInterval	Integer8	R	14.18.12
useMgtSettableLogPdelayReqInterval	Boolean	RW	14.18.13
mgtSettableLogPdelayReqInterval	Integer8	RW	14.18.14
initialComputeNeighborRateRatio	Boolean	RW	14.18.15
currentComputeNeighborRateRatio	Boolean	R	14.18.16

Table 14-20—cmldsLinkPortDS

Name	Data type	Operations supported ^a	References
useMgtSettableComputeNeighborRateRatio	Boolean	RW	14.18.17
mgtSettableComputeNeighborRateRatio	Boolean	RW	14.18.18
initialComputeMeanLinkDelay	Boolean	RW	14.18.19
currentComputeMeanLinkDelay	Boolean	R	14.18.20
useMgtSettableComputeMeanLinkDelay	Boolean	RW	14.18.21
mgtSettableComputeMeanLinkDelay	Boolean	RW	14.18.22
allowedLostResponses	UInteger8	RW	14.18.23
allowedFaults	UInteger8	RW	14.18.24
versionNumber	UInteger4	R	14.18.25
pdelayTruncatedTimestampsArray	UInteger48[4]	R	14.18.26
minorVersionNumber	UInteger4	R	14.18.27

^a R = Read only access; RW = Read/write access.

1 14.18.2 portIdentity

2 The value is the portIdentity attribute of the local port (see 8.5.2).

3 14.18.3 cmldsLinkPortEnabled

4 The value is equal to the value of the Boolean cmldsLinkPortEnabled (see 11.2.18.1).

5 14.18.4 isMeasuringDelay

6 The value is equal to the value of the instance of the Boolean isMeasuringDelay (see 11.2.13.6 and 16.4.3.3)
7 that is per Link Port and across all domains.

8 14.18.5 asCapableAcrossDomains

9 The value is equal to the value of the Boolean asCapableAcrossDomains (see 11.2.2 and 11.2.13.13).

10 14.18.6 meanLinkDelay

11 The value is equal to the value of the per-port global variable meanLinkDelay (see 10.2.5.8). It is an estimate
12 of the current one-way propagation time on the link attached to this Link Port, measured as specified for the
13 respective medium (see 11.2.17, 12.5.2, and 16.4). The value is zero for Link Ports attached to IEEE 802.3
14 EPON links and for the timeTransmitter port of an IEEE 802.11 link because one-way propagation delay is
15 not measured on the latter and not directly measured on the former. The data type shall be TimeInterval. The
16 default value is zero.

1 14.18.7 meanLinkDelayThresh

2 The value is equal to the value of the per-Link-Port global variable meanLinkDelayThresh (see 11.2.13.7). It
 3 is the propagation time threshold above which a Link Port (and therefore any PTP Ports that use the CMLDS
 4 on this Link Port) is considered not capable of participating in the IEEE 802.1AS protocol. Setting the
 5 managed object meanLinkDelayThresh causes the per-Link-Port global variable meanLinkDelayThresh to
 6 have the same value.

7 14.18.8 negMeanLinkDelayThresh

8 The value is equal to the value of the per-Link-Port global variable negMeanLinkDelayThresh (see
 9 11.2.13.7). It is the negative propagation time threshold below which a Link Port (and therefore any PTP
 10 Ports that use the CMLDS on this Link Port) is considered not capable of participating in the IEEE 802.1AS
 11 protocol. Setting the managed object negMeanLinkDelayThresh causes the per-Link-Port global variable
 12 negMeanLinkDelayThresh to have the same value.

13 14.18.9 delayAsymmetry

14 The value is the asymmetry in the propagation delay on the link attached to this Link Port relative to the
 15 local clock, as defined in 10.2.5.9 and 8.3. If propagation delay asymmetry is not modeled, then
 16 delayAsymmetry is 0.

17 NOTE—The underlying per-port global variable delayAsymmetry is of type ScaledNS, which is a 96-bit value
 18 (see 6.4.3.1). delayAsymmetry values that are larger than the maximum value that can be represented by the
 19 TimeInterval data type, i.e., 0x7FFF FFFF FFFF FFFF (where the units are 2^{-16} ns; see 6.4.3.3), used for this
 20 managed object are set to this largest value. delayAsymmetry values that are less than the minimum value that can be
 21 represented by the TimeInterval data type, i.e., 0x8000 0000 0000 0001 written in twos complement form (where the
 22 units are 2^{-16} ns; see 6.4.3.3), used for this managed object are set to this smallest value.

23 14.18.10 nrrPdelay

24 The value is an estimate of the ratio of the frequency of the LocalClock entity of the time-aware system at
 25 the other end of the link attached to this Link Port, to the frequency of the LocalClock entity of this time-
 26 aware system (see 10.2.5.7). The managed object nrrPdelay is expressed as the fractional frequency offset
 27 stored in the global variable nrrPdelay (see 11.2.13.14) multiplied by 2^{41} , i.e., the quantity $(\text{nrrPdelay}(\text{global}$
 28 $\text{variable}) - 1.0)(2^{41})$. The default value of the managed object nrrPdelay is 0.0, corresponding to a rate ratio
 29 of 1.0.

30 NOTE—This data set member corresponds to the scaledNeighborRateRatio member of the
 31 CommonMeanLinkDelayInformation Structure in 16.6.3.2 of IEEE Std 1588-2019.

32 14.18.11 initialLogPdelayReqInterval

33 If useMgtSettableLogPdelayReqInterval is FALSE, then for full-duplex IEEE 802.3 media and for CSN
 34 media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the
 35 logarithm to base 2 of the Pdelay_Req message transmission interval used when:

- 36 a) The Link Port is initialized, or
- 37 b) A message interval request TLV is received with the logLinkDelayInterval field set to 126 (see
 38 11.5.2.2 and the LinkDelayIntervalSetting state machine in 11.2.21).

39 For all other media, the value is 127.

1 **14.18.12 currentLogPdelayReqInterval**

2 For full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure
3 path delay (see 16.4.3.2), the value is the logarithm to the base 2 of the current Pdelay_Req message
4 transmission interval (see 11.5.2.2).

5 For all other media, the value is 127.

6 **14.18.13 useMgtSettableLogPdelayReqInterval**

7 The managed object is a Boolean that determines the source of the sync interval and mean time interval
8 between successive Pdelay_Req messages. If the value is TRUE, the value of currentLogPdelayReqInterval
9 is set equal to the value of mgtSettableLogPdelayReqInterval (see 14.18.14). If the value of the managed
10 object is FALSE, the value of currentLogPdelayReqInterval is determined by the LinkDelayIntervalSetting
11 state machine (see 11.2.21). The default value of useMgtSettableLogPdelayReqInterval is FALSE.

12 **14.18.14 mgtSettableLogPdelayReqInterval**

13 The value is the logarithm to base 2 of the mean time interval between successive Pdelay_Req messages
14 if useMgtSettableLogPdelayReqInterval is TRUE. The value is not used if
15 useMgtSettableLogPdelayReqInterval is FALSE.

16 **14.18.15 initialComputeNeighborRateRatio**

17 If useMgtSettableComputeNeighborRateRatio is FALSE, then for full-duplex IEEE 802.3 media and for
18 CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the
19 initial value of computeNeighborRateRatio (see 10.2.5.10).

20 For all other media, the value is TRUE.

21 **14.18.16 currentComputeNeighborRateRatio**

22 For full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure
23 path delay (see 16.4.3.2), the value is the current value of computeNeighborRateRatio.

24 For all other media, the value is TRUE.

25 **14.18.17 useMgtSettableComputeNeighborRateRatio**

26 The managed object is a Boolean that determines the source of the value of computeNeighborRateRatio. If
27 the value is TRUE, the value of computeNeighborRateRatio is set equal to the value of
28 mgtSettablecomputeNeighborRateRatio (see 14.18.18). If the value of the managed object is FALSE, the
29 value of currentComputeNeighborRateRatio is determined by the LinkDelayIntervalSetting state machine
30 (see 11.2.21). The default value of useMgtSettableComputeNeighborRateRatio is FALSE.

31 **14.18.18 mgtSettableComputeNeighborRateRatio**

32 computeNeighborRateRatio is configured to this value if useMgtSettableComputeNeighborRateRatio is
33 TRUE. The value is not used if useMgtSettableComputeNeighborRateRatio is FALSE.

1 **14.18.19 initialComputeMeanLinkDelay**

2 If useMgtSettableComputeMeanLinkDelay is FALSE, then for full-duplex IEEE 802.3 media and for CSN
3 media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the initial
4 value of computeMeanLinkDelay (see 10.2.5.10).

5 For all other media, the value is TRUE.

6 **14.18.20 currentComputeMeanLinkDelay**

7 For full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure
8 path delay (see 16.4.3.2), the value is the current value of computeMeanLinkDelay.

9 For all other media, the value is TRUE.

10 **14.18.21 useMgtSettableComputeMeanLinkDelay**

11 The managed object is a Boolean that determines the source of the value of computeMeanLinkDelay. If the
12 value is TRUE, the value of computeMeanLinkDelay is set equal to the value of
13 mgtSettableComputeMeanLinkDelay (see 14.18.18). If the value of the managed object is FALSE, the value
14 of currentComputeMeanLinkDelay is determined by the LinkDelayIntervalSetting state machine
15 (see 11.2.21). The default value of useMgtSettableComputeMeanLinkDelay is FALSE.

16 **14.18.22 mgtSettableComputeMeanLinkDelay**

17 computeMeanLinkDelay is configured to this value if useMgtSettableComputeMeanLinkDelay is TRUE.
18 The value is not used if useMgtSettableComputeMeanLinkDelay is FALSE.

19 **14.18.23 allowedLostResponses**

20 The value is equal to the value of the per-Link-Port global variable allowedLostResponses (see 11.5.3 and
21 11.2.13.4). It is the number of Pdelay_Req messages without valid responses above which a Link Port is
22 considered to be not exchanging peer delay messages with its neighbor. Setting the managed object
23 allowedLostResponses causes the per-Link-Port global variable allowedLostResponses to have the same value.

24 **14.18.24 allowedFaults**

25 The value is equal to the value of the per-Link-Port global variable allowedFaults (see 11.5.4 and 11.2.13.5).
26 It is the number of faults (see 11.5.4) above which asCapableAcrossDomains is set to FALSE, i.e., a Link
27 Port is considered not capable of interoperating with its neighbor via the IEEE 802.1AS protocol
28 (see 10.2.5.1). Setting the managed object allowedFaults causes the per-Link-Port global variable
29 allowedFaults to have the same value.

30 **14.18.25 versionNumber**

31 This value is set to versionPTP as specified in 10.6.2.2.4.

32 **14.18.26 pdelayTruncatedTimestampsArray**

33 For full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure
34 path delay (see 16.4.3.2), the values of the four elements of this array are as described in Table 14-12. For all
35 other media, the values are zero. Array elements 0, 1, 2, and 3 correspond to the timestamps t1, t2, t3, and t4,
36 modulo 2^{32} , respectively, in Figure 11-1 and are expressed in units of 2^{-16} ns (i.e., the value of each
37 array element is equal to the remainder obtained upon dividing the respective timestamp, expressed in units

1 of 2^{-16} ns, by 2^{48}). At any given time, the timestamp values stored in the array are for the same, and most
2 recently completed, peer delay message exchange.

3 NOTE—This managed object is used with the asymmetry measurement compensation procedure, which is based on
4 line-swapping.

5 14.18.27 minorVersionNumber

6 This value is set to minorVersionPTP as specified in 10.6.2.2.3.

7 14.19 Common Mean Link Delay Service Link Port Parameter Statistics Data Set 8 (cmldsLinkPortStatisticsDS)

9 14.19.1 General

10 For every Link Port of the CMLD of a time-aware system, the cmldsLinkPortStatisticsDS provides counters
11 associated with Link Port capabilities at a given time-aware system. The number of such statistics sets is the
12 same as the value of cmldsDefaultDS.numberLinkPorts.

13 There is one set of cmldsLinkPortStatisticsDS parameters per Link Port of a time-aware system. The
14 cmldsLinkPortStatisticsDS contains a set of counters for each Link Port that supports the time-
15 synchronization capability, as detailed in Table 14-21. Each set of parameters can be created or removed
16 dynamically in implementations that support dynamic configuration of Link Ports and components

17 The parameters of cmldsLinkPortStatisticsDS are summarized in Table 14-21 and subsequently specified.

Table 14-21—cmldsLinkPortStatisticsDS

Name	Data type	Operations supported ^a	References
rxPdelayRequestCount	UInteger32	R	14.19.2
rxPdelayResponseCount	UInteger32	R	14.19.3
rxPdelayResponseFollowUpCount	UInteger32	R	14.19.4
rxPtpPacketDiscardCount	UInteger32	R	14.19.5
pdelayAllowedLostResponsesExceededCount	UInteger32	R	14.19.6
txPdelayRequestCount	UInteger32	R	14.19.7
txPdelayResponseCount	UInteger32	R	14.19.8
txPdelayResponseFollowUpCount	UInteger32	R	14.19.9

^a R= Read only access.

18 14.19.2 rxPdelayRequestCount

19 This counter increments every time a Pdelay_Req message is received, denoted by a transition to TRUE
20 from FALSE of the rcvdPdelayReq variable of the MDPdelayResp state machine (see 11.2.20.2.1 and
21 Figure 11-10) when in the WAITING_FOR_PDELAY_REQ or INITIAL_WAITING_FOR_PDELAY_REQ
22 states.

1 14.19.3 rxPdelayResponseCount

2 This counter increments every time a Pdelay_Resp message is received, denoted by a transition to TRUE
3 from FALSE of the rcvdPdelayResp variable of the MDPdelayReq state machine (see 11.2.19.2.2 and
4 Figure 11-9) when in the WAITING_FOR_PDELAY_RESP state.

5 14.19.4 rxPdelayResponseFollowUpCount

6 This counter increments every time a Pdelay_Resp_Follow_Up message is received, denoted by a transition
7 to TRUE from FALSE of the rcvdPdelayRespFollowUp variable of the MDPdelayReq state machine (see
8 11.2.19.2.4 and Figure 11-9) when in the WAITING_FOR_PDELAY_RESP_FOLLOW_UP state.

9 14.19.5 rxPtpPacketDiscardCount

10 This counter increments every time a PTP message of the CMLD is discarded, caused by the occurrence of
11 any of the following conditions:

- 12 — A Pdelay_Resp message corresponding to a transmitted Pdelay_Req message is not received,
13 denoted by a transition from the WAITING_FOR_PDELAY_RESP state to the RESET state of the
14 MDPdelayReq state machine (see 11.2.19 and Figure 11-9).
- 15 — A Pdelay_Resp_Follow_Up message corresponding to a transmitted Pdelay_Req message is not
16 received, denoted by a transition from the WAITING_FOR_PDELAY_RESP_FOLLOW_UP state
17 to the RESET state of the MDPdelayReq state machine (see 11.2.19 and Figure 11-9).

18 14.19.6 pdelayAllowedLostResponsesExceededCount

19 This counter increments every time the value of the variable lostResponses (see 11.2.19.2.9) exceeds the
20 value of the variable allowedLostResponses (see 11.2.13.4) in the RESET state of the MDPdelayReq state
21 machine (see 11.2.19 and Figure 11-9).

22 14.19.7 txPdelayRequestCount

23 This counter increments every time a Pdelay_Req message is transmitted, denoted by entering the
24 INITIAL_SEND_PDELAY_REQ or SEND_PDELAY_REQ states of the MDPdelayReq state machine
25 (see 11.2.19 and Figure 11-9).

26 14.19.8 txPdelayResponseCount

27 This counter increments every time a Pdelay_Resp message is transmitted, denoted by the following events:

- 28 — A transition to TRUE from FALSE of the rcvdPdelayReq variable of the MDPdelayResp state
29 machine (see 11.2.20.2.1 and Figure 11-10) when in the WAITING_FOR_PDELAY_REQ or
30 INITIAL_WAITING_FOR_PDELAY_REQ states and
- 31 — The resulting entry to the SENT_PDELAY_RESP_WAITING_FOR_TIMESTAMP state.

32 14.19.9 txPdelayResponseFollowUpCount

33 This counter increments every time a Pdelay_Resp_Follow_Up message is transmitted, denoted by the
34 following events:

- 35 — A transition to TRUE from FALSE of the rcvdMDTimestampReceiveMDPResp variable of the
36 MDPdelayResp state machine (see 11.2.20.2.2 and Figure 11-10) when in the
37 SENT_PDELAY_RESP_WAITING_FOR_TIMESTAMP state and
- 38 — The resulting entry to the WAITING_FOR_PDELAY_REQ state.

1 **14.20 Common Mean Link Delay Service Asymmetry Measurement Mode Parameter Data**
2 **Set (cmldsAsymmetryMeasurementModeDS)**

3 **14.20.1 General**

4 The cmldsAsymmetryMeasurementModeDS represents the capability to enable/disable the Asymmetry
5 Compensation Measurement Procedure on a Link Port (see Annex G).

6 For every Link Port of the CMLD of a time-aware system, the cmldsAsymmetryMeasurementModeDS
7 contains the single member asymmetryMeasurementMode, which is used to enable/disable the Asymmetry
8 Compensation Measurement Procedure. The number of such data sets is the same as the numberLinkPorts
9 value of the cmldsDefaultDS.

10 The parameter of cmldsAsymmetryMeasurementModeDS is summarized in Table 14-22 and subsequently
11 specified.

Table 14-22—cmldsAsymmetryMeasurementModeDS

Name	Data type	Operations supported ^a	References
asymmetryMeasurementMode	Boolean	RW	14.20.2

^a R = Read only access; RW = Read/write access.

12 **14.20.2 asymmetryMeasurementMode**

13 The value is equal to the value of the Boolean asymmetryMeasurementMode (see G.3). For full-duplex
14 IEEE 802.3 media, the value is TRUE if an asymmetry measurement is being performed for the link
15 attached to this Link Port and FALSE otherwise. For all other media, the value shall be FALSE
16 (see 10.2.5.2). Setting the managed object asymmetryMeasurementMode causes the Boolean
17 asymmetryMeasurementMode to have the same value.

18 NOTE—If an asymmetry measurement is being performed for a link, asymmetryMeasurementMode must be TRUE for
19 the Link Ports at each end of the link.

20 **14.21 Hot Standby System Parameter Data Set (hotStandbySystemDS)**

21 **14.21.1 General**

22 The hotStandbySystemDS describes the attributes of the respective instance of the Hot Standby Service.

23 The parameters of hotStandbyDS are summarized in Table 14-19 and subsequently specified.

24

25

26

Table 14-23—hotStandbySystemDS

Name	Data type	Operations supported ^a	References
primaryPtpInstanceIndex	UInteger32	RW	14.21.2
secondaryPtpInstanceIndex	UInteger32	RW	14.21.3
hotStandbySystemEnable	Boolean	RW	14.21.4
hotStandbySystemState	Enumeration8	R	14.21.5
hotStandbySystemSplitFunctionality	Boolean	RW	14.21.6
primarySecondaryOffset	ScaledNs	R	14.21.7
primarySecondaryOffsetThresh	ScaledNs	RW	14.21.8
hotStandbySystemLogSyncTimeThresh	Integer8	RW	14.21.9

^a R = Read only access; RW = Read/write access.

1 14.21.2 primaryPtpInstanceIndex

2 The value of primaryPtpInstanceIndex is the index (see 14.1.1) of the primary PTP Instance associated with
3 this hotStandbySystem instance.

4 14.21.3 secondaryPtpInstanceIndex

5 The value of secondaryPtpInstanceIndex is the index (see 14.1.1) of the secondaryPTP Instance associated
6 with this hotStandbySystem instance.

7 14.21.4 hotStandbySystemEnable

8 The value is the hotStandbySystemEnable attribute of the HotStandbySystem entity (see 18.5.1.2).

9 14.21.5 hotStandbySystemState

10 The value of hotStandbySystemState is the state of the hotStandbySystem, i.e., the value of the global
11 variable hotStandbySystemState (see 18.5.1.1).

12 14.21.6 hotStandbySystemSplitFunctionality

13 If the value is TRUE, the optional split functionality (see 18.6.4) is used. If the value is FALSE, the optional
14 split functionality is not used.

15 14.21.7 primarySecondaryOffset

16 The absolute value of the difference between the clockTimeReceiverTimes (see 10.2.4.3) of the primary and
17 secondary PTP Instances.

18 14.21.8 primarySecondaryOffsetThresh

19 The threshold for hotStandbySystemDS.primarySecondaryOffset (see 14.21.7), above which the
20 hotStandbySystemState transitions from REDUNDANT to NOT_REDUNDANT, or does not transition from

1 NOT_REDUNDANT or OUT_OF_SYNC to REDUNDANT even if other conditions for these transitions
2 are satisfied.

3 **14.21.9 hotStandbySystemLogSyncTimeThresh**

4 The value of hotStandbySystemLogSyncTimeThresh is the logarithm to base 2 of the time interval, in
5 seconds, after which the hotStandbySystem transitions from the OUT_OF_SYNC state to either the NOT-
6 REDUNDANT or REDUNDANT state, or from the NOT_REDUNDANT to the REDUNDANT state, if all
7 other conditions for the respective transition are met. The value −128 means that the transition time is zero,
8 i.e., the transition occurs immediately.

9 **14.22 Hot Standby System Description Parameter Data Set**
10 **(hotStandbySystemDescriptionDS)**

11 **14.22.1 General**

12 The hotStandbySystemDescriptionDS contains descriptive information for the respective instance of the Hot
13 Standby Service.

14 The parameter of hotStandbySystemDescriptionDS is summarized in Table 14-24 and subsequently
15 specified.

Table 14-24—hotStandbySystemDescriptionDS

Name	Data type	Operations supported ^a	References
userDescription	Octet128	RW	14.22.2

^a RW= Read/write access.

16 **14.22.2 userDescription**

17 The user description is a character string whose maximum length is 128.

18

1 15. Management Information Base (MIB)

2 15.1 Internet Standard Management Framework

3 For a detailed overview of the documents that describe the current Internet Standard Management
4 Framework, refer to section 7 of IETF RFC 3410 (Dec. 2002).

5 Managed objects are accessed via a virtual information store, termed the *Management Information Base*
6 (MIB). MIB objects are generally accessed through the Simple Network Management Protocol (SNMP).
7 Objects in the MIB are defined using the mechanisms defined in the Structure of Management Information
8 (SMI). This clause specifies a MIB module that is compliant to the SMIV2, which is described in
9 IETF STD 58, comprising IETF RFC 2578 [B11], IETF RFC 2579 [B12], and IETF RFC 2580 .

10 This clause contains a complete SMIV2 MIB set for all features of this standard, except for hot standby (see
11 Clause 18), the drift tracking feature (see 11.4.4.4), and new managed objects included in this revision of
12 the standard.

13 15.2 Structure of the MIB

14 The IEEE 802.1AS MIB provides objects to configure and manage the IEEE 802.1AS timing and
15 synchronization for time-sensitive applications.

16 The MIB contains a set of textual conventions and is additionally subdivided into the following subtrees,
17 each of which is organized as a set of related objects:

- 18 a) The Default Parameter Data Set (defaultDS) represents the native capabilities of a PTP Instance.
- 19 b) The Current Parameter Data Set (currentDS) represents topological position of a local PTP Instance
20 relative to the Grandmaster PTP Instance.
- 21 c) The Parent Parameter Data Set (parentDS) represents capabilities of the upstream PTP Instance
22 toward the Grandmaster PTP Instance, as measured at a local PTP Instance.
- 23 d) The Time Properties Parameter Data Set (timePropertiesDS) represents capabilities of the
24 Grandmaster PTP Instance, as measured at a local PTP Instance.
- 25 e) The Path Trace Parameter Data Set (pathTraceDS) represents the current path trace information
26 (see 10.3.9.23) available at the PTP Instance.
- 27 f) The Acceptable TimeTransmitter Table Parameter Data Set (acceptableTimeTransmitterTableDS)
28 represents the acceptable timeTransmitter table used when the media-dependent PTP Port type of
29 EPON is present in a PTP Instance.
- 30 g) The Port Parameter Data Set (portDS) represents time-aware capabilities at a given PTP Port, as a
31 set of augmentation to the interface table entry (ifEntry).
- 32 h) The Description Port Parameter Data Set (descriptionPortDS) contains the profileIdentifier for this
33 PTP profile as specified in F.2.
- 34 i) The Port Parameter Statistics Data Set (portStatisticsDS) represents statistics and counters
35 associated with time-aware capabilities at a given PTP Relay Instance or PTP End Instance port.
- 36 j) The Acceptable TimeTransmitter Port Parameter Data Set (acceptableTimeTransmitterPortDS)
37 represents the capability to enable/disable the acceptable timeTransmitter table feature on a PTP
38 Port.
- 39 k) The External Port Configuration Port Parameter Data Set (externalPortConfigurationPortDS) is used
40 with the external port configuration option to indicate the desired state of a PTP Port.
- 41 l) The Asymmetry Measurement Mode Parameter Data Set (asymmetryMeasurementModeDS)
42 represents the capability to enable/disable the Asymmetry Compensation Measurement Procedure

- 1 on a port (see Annex G) and is used instead of the `cmldsAsymmetryMeasurementModeDS` when
2 CMLDS is not used and there is a single gPTP domain.
- 3 m) The Common Services Port Parameter Data Set (`commonServicesPortDS`) enables a PTP Port of a
4 PTP Instance to determine which port of the respective common service corresponds to that
5 PTP Port.
- 6 n) The CMLD Default Parameter Data Set (`cmldsDefaultDS`) describes the per-time-aware-system
7 attributes of the CMLD.
- 8 o) The CMLD Link Port Parameter Data Set (`cmldsLinkPortDS`) represents time-aware Link Port
9 capabilities for the CMLD of a time-aware system.
- 10 p) The CMLD Link Port Parameter Statistics Data Set (`cmldsLinkPortStatisticsDS`) represents
11 statistics and counters associated with Link Port capabilities at a given time-aware system.
- 12 q) The CMLD Asymmetry Measurement Mode Parameter Data Set
13 (`cmldsAsymmetryMeasurementModeDS`) represents the capability to enable/disable the
14 Asymmetry Compensation Measurement Procedure on a Link Port (see Annex G).

15 Table 15-1 shows the structure of the MIB and the relationship of the MIB objects to the above data sets.

Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference

MIB table	MIB object	Reference
ieee8021AsV3DefaultDS		defaultDS (Table 14-1)
	ieee8021AsV3DefaultDSClockIdentity	14.2.2
	ieee8021AsV3DefaultDSNumberPorts	14.2.3
	ieee8021AsV3DefaultDSClockQualityClockClass	14.2.4.2
	ieee8021AsV3DefaultDSClockQualityClockAccuracy	14.2.4.3
	ieee8021AsV3DefaultDSClockQualityOffsetScaledLogVariance	14.2.4.4
	ieee8021AsV3DefaultDSPriority1	14.2.5
	ieee8021AsV3DefaultDSPriority2	14.2.6
	ieee8021AsV3DefaultDSGmCapable	14.2.7
	ieee8021AsV3DefaultDSCurrentUtcOffset	14.2.8
	ieee8021AsV3DefaultDSCurrentUtcOffsetValid	14.2.9
	ieee8021AsV3DefaultDSLLeap59	14.2.10
	ieee8021AsV3DefaultDSLLeap61	14.2.11
	ieee8021AsV3DefaultDSTimeTraceable	14.2.12
	ieee8021AsV3DefaultDSFrequencyTraceable	14.2.13
	ieee8021AsV3DefaultDSPtpTimescale	14.2.14
	ieee8021AsV3DefaultDSTimeSource	14.2.15
	ieee8021AsV3DefaultDSDomainNumber	14.2.16
	ieee8021AsV3DefaultDSSdoId	14.2.17
	ieee8021AsV3DefaultDSExternalPortConfigurationEnabled	14.2.18

Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference (continued)

MIB table	MIB object	Reference
	ieee8021AsV3DefaultDSInstanceEnable	14.2.19
ieee8021AsV3CurrentDS		currentDS (Table 14-2)
	ieee8021AsV3CurrentDSStepsRemoved	14.3.2
	ieee8021AsV3CurrentDSOffsetFromTimeTransmitter	14.3.3
	ieee8021AsV3CurrentDSLstGmPhaseChange	14.3.4
	ieee8021AsV3CurrentDSLstGmFreqChange	14.3.5
	ieee8021AsV3CurrentDSGmTimebaseIndicator	14.3.6
	ieee8021AsV3CurrentDSGmChangeCount	14.3.7
	ieee8021AsV3CurrentDSTimeOfLstGmChangeEvent	14.3.8
	ieee8021AsV3CurrentDSTimeOfLstGmPhaseChangeEvent	14.3.9
	ieee8021AsV3CurrentDSTimeOfLstGmFreqChangeEvent	14.3.10
ieee8021AsV3ParentDS		parentDS (Table 14-3)
	ieee8021AsV3ParentDSParentClockIdentity	14.4.2
	ieee8021AsV3ParentDSParentPortNumber	14.4.2
	ieee8021AsV3ParentDSCumulativeRateRatio	14.4.3
	ieee8021AsV3ParentDSGrandmasterIdentity	14.4.4
	ieee8021AsV3ParentDSGrandmasterClockQualityclockClass	14.4.5.2
	ieee8021AsV3ParentDSGrandmasterClockQualityclockAccuracy	14.4.5.3
	ieee8021AsV3ParentDSGrandmasterClockQualityoffsetScaledLogVar	14.4.5.4
	ieee8021AsV3ParentDSGrandmasterPriority1	14.4.6
	ieee8021AsV3ParentDSGrandmasterPriority2	14.4.7
ieee8021AsV3TimePropertiesDS		timePropertiesDS (Table 14-4)
	ieee8021AsV3TimePropertiesDSCurrentUtcOffset	14.5.2
	ieee8021AsV3TimePropertiesDSCurrentUtcOffsetValid	14.5.3
	ieee8021AsV3TimePropertiesDSLeap59	14.5.4
	ieee8021AsV3TimePropertiesDSLeap61	14.5.5
	ieee8021AsV3TimePropertiesDSTimeTraceable	14.5.6
	ieee8021AsV3TimePropertiesDSFrequencyTraceable	14.5.7
	ieee8021AsV3TimePropertiesDSPtpTimescale	14.5.8
	ieee8021AsV3TimePropertiesDSTimeSource	14.5.9
ieee8021AsV3PathTraceDS		pathTraceDS (Table 14-5)
	ieee8021AsV3PathTraceDSEnable	14.6.3

Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference (continued)

MIB table	MIB object	Reference
ieee8021AsV3PathTraceDSArray		pathTraceDS (Table 14-5)
	ieee8021AsV3PathTraceDSArrayList	14.6.2
ieee8021AsV3AcceptableTimeTransmitterTableDS		acceptableTimeTransmitterTable DS (Table 14-6)
	ieee8021AsV3AcceptableTimeTransmitterTableDSMaxTableSize	14.7.2
	ieee8021AsV3AcceptableTimeTransmitterTableDSActualTableSize	14.7.3
ieee8021AsV3AcceptableTimeTransmitterTableDSArray		acceptableTimeTransmitterTable DS (Table 14-6)
	ieee8021AsV3AcceptableTimeTransmitterTableDSArrayPortIdentity	14.7.4
	ieee8021AsV3AcceptableTTTableDSArrayAlternatePriority1	14.7.4
ieee8021AsV3PortDS		portDS (Table 14-9)
	ieee8021AsV3PortDSClockIdentity	14.10.2
	ieee8021AsV3PortDSPortNumber	14.10.2
	ieee8021AsV3PortDSPortState	14.10.3
	ieee8021AsV3PortDSPtpPortEnabled	14.10.4
	ieee8021AsV3PortDSdelayMechanism	14.10.5
	ieee8021AsV3PortDSIsMeasuringDelay	14.10.6
	ieee8021AsV3PortDSAsCapable	14.10.7
	ieee8021AsV3PortDSMeanLinkDelay	14.10.8
	ieee8021AsV3PortDSMeanLinkDelayThresh	14.10.9
	ieee8021AsV3PortDSDelayAsym	14.10.11
	ieee8021AsV3PortDSNbrRateRatio	14.10.12
	ieee8021AsV3PortDSInitialLogAnnounceInterval	14.10.13
	ieee8021AsV3PortDSCurrentLogAnnounceInterval	14.10.14
	ieee8021AsV3PortDSUseMgtSettableLogAnnounceInterval	14.10.15
	ieee8021AsV3PortDSMgtSettableLogAnnounceInterval	14.10.16
	ieee8021AsV3PortDSAnnounceReceiptTimeout	14.10.16
	ieee8021AsV3PortDSInitialLogSyncInterval	14.10.17
	ieee8021AsV3PortDSCurrentLogSyncInterval	14.10.18
	ieee8021AsV3PortDSUseMgtSettableLogSyncInterval	14.10.19
	ieee8021AsV3PortDSMgtSettableLogSyncInterval	14.10.20
	ieee8021AsV3PortDSSyncReceiptTimeout	14.10.21
	ieee8021AsV3PortDSSyncReceiptTimeoutTimeInterval	14.10.22

Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference (continued)

MIB table	MIB object	Reference
	ieee8021AsV3PortDSInitialLogPdelayReqInterval	14.10.23
	ieee8021AsV3PortDSCurrentLogPdelayReqInterval	14.10.24
	ieee8021AsV3PortDSUseMgtSettableLogPdelayReqInterval	14.10.25
	ieee8021AsV3PortDSMgtSettableLogPdelayReqInterval	14.10.26
	ieee8021AsV3PortDSInitialLogGtpCapableMessageInterval	14.10.27
	ieee8021AsV3PortDSCurrentLogGtpCapableMessageInterval	14.10.28
	ieee8021AsV3PortDSUseMgtSettableLogGtpCapableMessageInterval	14.10.29
	ieee8021AsV3PortDSMgtSettableLogGtpCapableMessageInterval	14.10.30
	ieee8021AsV3PortDSInitialComputeNbrRateRatio	14.10.31
	ieee8021AsV3PortDSCurrentComputeNbrRateRatio	14.10.32
	ieee8021AsV3PortDSUseMgtSettableComputeNbrRateRatio	14.10.33
	ieee8021AsV3PortDSMgtSettableComputeNbrRateRatio	14.10.34
	ieee8021AsV3PortDSInitialComputeMeanLinkDelay	14.10.35
	ieee8021AsV3PortDSCurrentComputeMeanLinkDelay	14.10.36
	ieee8021AsV3PortDSUseMgtSettableComputeMeanLinkDelay	14.10.37
	ieee8021AsV3PortDSMgtSettableComputeMeanLinkDelay	14.10.38
	ieee8021AsV3PortDSAllowedLostRsp	14.10.39
	ieee8021AsV3PortDSAllowedFaults	14.10.40
	ieee8021AsV3PortDSGtpCapableReceiptTimeout	14.10.41
	ieee8021AsV3PortDSVersionNumber	14.10.42
	ieee8021AsV3PortDSNup	14.10.43
	ieee8021AsV3PortDSNdown	14.10.44
	ieee8021AsV3PortDSOneStepTxOper	14.10.45
	ieee8021AsV3PortDSOneStepReceive	14.10.46
	ieee8021AsV3PortDSOneStepTransmit	14.10.47
	ieee8021AsV3PortDSInitialOneStepTxOper	14.10.48
	ieee8021AsV3PortDSCurrentOneStepTxOper	14.10.49
	ieee8021AsV3PortDSUseMgtSettableOneStepTxOper	14.10.50
	ieee8021AsV3PortDSMgtSettableOneStepTxOper	14.10.51
	ieee8021AsV3PortDSSyncLocked	14.10.52
	ieee8021AsV3PortDSPdelayTruncTST1	14.10.53
	ieee8021AsV3PortDSPdelayTruncTST2	14.10.53
	ieee8021AsV3PortDSPdelayTruncTST3	14.10.53

Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference (continued)

MIB table	MIB object	Reference
	ieee8021AsV3PortDSPdelayTruncTST4	14.10.53
	ieee8021AsV3PortDSMinorVersionNumber	14.10.54
ieee8021AsV3DescriptionPortDS		descriptionPortDS (Table 14-13)
	ieee8021AsV3DescriptionPortDSProfileIdentifier	14.11.2
ieee8021AsV3PortStatDS		portStatisticsDS (Table 14-14)
	ieee8021AsV3PortStatRxSyncCount	14.12.2
	ieee8021AsV3PortStatRxOneStepSyncCount	14.12.3
	ieee8021AsV3PortStatRxFollowUpCount	14.12.4
	ieee8021AsV3PortStatRxPdelayRequestCount	14.12.5
	ieee8021AsV3PortStatRxPdelayRspCount	14.12.6
	ieee8021AsV3PortStatRxPdelayRspFollowUpCount	14.12.7
	ieee8021AsV3PortStatRxAnnounceCount	14.12.8
	ieee8021AsV3PortStatRxPtpPacketDiscardCount	14.12.9
	ieee8021AsV3PortStatSyncReceiptTimeoutCount	14.12.10
	ieee8021AsV3PortStatAnnounceReceiptTimeoutCount	14.12.11
	ieee8021AsV3PortStatPdelayAllowedLostRspExceededCount	14.12.12
	ieee8021AsV3PortStatTxSyncCount	14.12.13
	ieee8021AsV3PortStatTxOneStepSyncCount	14.12.14
	ieee8021AsV3PortStatTxFollowUpCount	14.12.15
	ieee8021AsV3PortStatTxPdelayRequestCount	14.12.16
	ieee8021AsV3PortStatTxPdelayRspCount	14.12.17
	ieee8021AsV3PortStatTxPdelayRspFollowUpCount	14.12.18
	ieee8021AsV3PortStatTxAnnounceCount	14.12.19
ieee8021AsV3AcceptableTimeTransmitterPortDS		acceptableTimeTransmitterTable DS (Table 14-15)
	ieee8021AsV3AcceptableTTPortDSAcceptableTTTableEnabled	14.13.2
ieee8021AsV3ExternalPortConfigurationPortDS		externalPortConfigurationPortDS (Table 14-16)
	ieee8021AsV3ExternalPortConfigurationPortDSDesiredState	14.14.2
ieee8021AsV3AsymMeasurementModeDS		asymmetryMeasurementModeDS (Table 14-17)
	ieee8021AsV3AsymMeasurementModeDSAsymMeasurementMode	14.15.2
ieee8021AsV3CommonServicesPortDS		commonServicesPortDS (Table 14-18)

Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference (continued)

MIB table	MIB object	Reference
	ieee8021AsV3CommonServicesPortDSCmlDsLinkPortPortNumber	14.16.2
ieee8021AsV3CommonMeanLinkDelayServiceDefaultDS		cmlDsDefaultDS (Table 14-19)
	ieee8021AsV3CmlDsDefaultDSClockIdentity	14.17.2
	ieee8021AsV3CmlDsDefaultDSNumberLinkPorts	14.17.3
ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDS		cmlDsLinkPortDS (Table 14-20)
	ieee8021AsV3CmlDsLinkPortDSClockIdentity	14.18.2
	ieee8021AsV3CmlDsLinkPortDSPortNumber	14.18.2
	ieee8021AsV3CmlDsLinkPortDSCmlDsLinkPortEnabled	14.18.3
	ieee8021AsV3CmlDsLinkPortDSIsMeasuringDelay	14.18.4
	ieee8021AsV3CmlDsLinkPortDSAsCapableAcrossDomains	14.18.5
	ieee8021AsV3CmlDsLinkPortDSMeanLinkDelay	14.18.6
	ieee8021AsV3CmlDsLinkPortDSMeanLinkDelayThresh	14.18.7
	ieee8021AsV3CmlDsLinkPortDSDelayAsym	14.18.9
	ieee8021AsV3CmlDsLinkPortDSNbrRateRatio	14.18.10
	ieee8021AsV3CmlDsLinkPortDSInitialLogPdelayReqInterval	14.18.11
	ieee8021AsV3CmlDsLinkPortDSCurrentLogPdelayReqInterval	14.18.12
	ieee8021AsV3CmlDsLinkPortDSUseMgtSettableLogPdelayReqInterval	14.18.13
	ieee8021AsV3CmlDsLinkPortDSMgtSettableLogPdelayReqInterval	14.18.14
	ieee8021AsV3CmlDsLinkPortDSInitialComputeNbrRateRatio	14.18.15
	ieee8021AsV3CmlDsLinkPortDSCurrentComputeNbrRateRatio	14.18.16
	ieee8021AsV3CmlDsLinkPortDSUseMgtSettableComputeNbrRateRatio	14.18.17
	ieee8021AsV3CmlDsLinkPortDSMgtSettableComputeNbrRateRatio	14.18.18
	ieee8021AsV3CmlDsLinkPortDSInitialComputeMeanLinkDelay	14.18.19
	ieee8021AsV3CmlDsLinkPortDSCurrentComputeMeanLinkDelay	14.18.20
	ieee8021AsV3CmlDsLinkPortDSUseMgtSettableComputeMeanLinkDelay	14.18.21
	ieee8021AsV3CmlDsLinkPortDSMgtSettableComputeMeanLinkDelay	14.18.22
	ieee8021AsV3CmlDsLinkPortDSAllowedLostRsp	14.18.23
	ieee8021AsV3CmlDsLinkPortDSAllowedFaults	14.18.24
	ieee8021AsV3CmlDsLinkPortDSVersionNumber	14.18.25
	ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST1	14.18.26
	ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST2	14.18.26

Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference (continued)

MIB table	MIB object	Reference
	ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST3	14.18.26
	ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST4	14.18.26
	ieee8021AsV3CmlDsLinkPortDSMinorVersionNumber	14.18.27
ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDS		cmlDsLinkPortStatisticsDS (Table 14-21)
	ieee8021AsV3CmlDsLinkPortStatDSRxDelayRequestCount	14.19.2
	ieee8021AsV3CmlDsLinkPortStatDSRxDelayRspCount	14.19.3
	ieee8021AsV3CmlDsLinkPortStatDSRxDelayRspFollowUpCount	14.19.4
	ieee8021AsV3CmlDsLinkPortStatDSRxDelayRspPacketDiscardCount	14.19.5
	ieee8021AsV3CmlDsLinkPortStatDSPdelayAllowedLostRspExceededCount	14.19.6
	ieee8021AsV3CmlDsLinkPortStatDSTxDelayRequestCount	14.19.7
	ieee8021AsV3CmlDsLinkPortStatDSTxDelayRspCount	14.19.8
	ieee8021AsV3CmlDsLinkPortStatDSTxDelayRspFollowUpCount	14.19.9
ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDS		cmlDsAsymmetryMeasurementModeDS (Table 14-22)
	ieee8021AsV3CmlDsAsymMeasurementModeDSAsymMeasurementMode	14.20.2

1 15.3 Relationship to MIB in IEEE Std 802.1AS-2011

2 The version 1 MIB module (IEEE8021-AS MIB) that was published in IEEE Std 802.1AS-2011 has been
3 superseded by the version 2 MIB module (IEEE8021-AS-V2 MIB) specified in 15.6 of IEEE Std
4 802.1AS-2020. The version 3 MIB module (IEEE8021-AS-V3 MIB) specified in the current standard
5 remains unchanged from the IEEE8021-AS-V2 MIB except updating terminology. Support of the version 3
6 module is a requirement for conformance to the required or optional capabilities (Clause 5) in the current
7 standard.

8 For an implementation that supports a single PTP Instance, version 1, version 2, and version 3
9 implementations can successfully co-exist and interoperate.

10 15.4 Security considerations

11 SNMP versions prior to SNMPv3 did not include adequate security. Even if the network itself is secure
12 (for example by using IPsec), there is no control as to who on the secure network is allowed to access and
13 GET/SET (read/change/create/delete) the objects in these MIB module.

14 It is recommended that implementers consider the security features as provided by the SNMPv3 framework
15 [see section 8 in IETF RFC 3410 (Dec. 2002)], including full support for the SNMPv3 cryptographic
16 mechanisms (for authentication and privacy).

1 Further, deployment of SNMP versions prior to SNMPv3 is not recommended. Instead, it is recommended
 2 to deploy SNMPv3 and to enable cryptographic security. It is then a customer/operator responsibility to
 3 ensure that the SNMP entity giving access to an instance of these MIB modules is properly configured to
 4 give access to the objects only to the principals (users) that have legitimate rights to indeed GET or SET
 5 (change/create/delete) them.

6 A number of management objects defined in the IEEE8021-AS-V3 MIB module have a MAX-ACCESS
 7 clause of read-write and/or read-create. Such objects might be considered sensitive or vulnerable in some
 8 network environments. The support for SET operations in a non-secure environment without proper
 9 protection can have a negative effect on network operations.

10 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
 11 “not-accessible”) might be considered sensitive or vulnerable in some network environments. It is thus
 12 important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to
 13 even encrypt the values of these objects when sending them over the network via SNMP.

14 The following objects in the IEEE8021-AS-V2 MIB can be manipulated to interfere with the operation of
 15 timing synchronization. This could, for example, be used to force a reinitialization of state machines to
 16 cause timing synchronization and network instability. Another possibility would be for an attacker to
 17 override Grandmaster PTP Instance status to give a user (or an attacker) unauthorized control over the
 18 network time.

19 Improper manipulation of the following writable objects could result in an unintended Grandmaster PTP
 20 Instance to be elected when a system is grandmaster-capable in a gPTP domain. It could also be used
 21 maliciously to cause frequent Grandmaster PTP Instance changes that could affect network stability.

22 ieee8021AsV3DefaultDSPriority1
 23 ieee8021AsV3DefaultDSPriority2

24 Improper manipulation of the following writable objects could result in a segmented time-aware network,
 25 could compromise the expected accuracy, and could interrupt paths of the gPTP domain.

26 ieee8021AsV3PortDSPTpPortEnabled
 27 ieee8021AsV3PortDSDelayAsymmetry

28 Unintended access to any of the readable tables or variables in the IEEE8021-AS-V3 MIB alerts the reader
 29 that timing synchronization in gPTP domain is configured, and on which values timing parameters are
 30 configured, and which system is current Grandmaster PTP Instance. This information can suggest to an
 31 attacker what applications are being run, and thus suggest application-specific attacks, or can enable the
 32 attacker to detect whether their attacks are being successful. It is thus important to control even GET access
 33 to these objects and possibly to even encrypt the values of these objects when sending them over the network
 34 via SNMP.

35 15.5 Textual conventions defined in this MIB

36 The following textual conventions are defined in this MIB:

- 37 a) Ieee8021AsV3ClockIdentity. IEEE 802 MAC address represented in “canonical” order defined by
 38 IEEE Std 802-2024, 64-bit Network Unique Identifier (NUI-64) as described in IEEE Std 802-2024.
- 39 b) Ieee8021AsV3GPTpProfileIdentifier. Profile identifier (see 14.11.2).
- 40 c) Ieee8021AsV3ClockClassValue. Clock class value (see 8.6.2.2).
- 41 d) Ieee8021AsV3ClockAccuracyValue. Clock accuracy value (see 8.6.2.3).

- 1 e) Ieee8021ASV3TimeSourceValue. Source of time used by Grandmaster PTP Instance (see 8.6.2.7).
- 2 f) Ieee8021ASV3PtpTimeInterval. Time intervals in units of 2^{-16} ns (see 6.4.3.3).
- 3 g) Ieee8021ASV3PtpPortIdentity. Identifies a port of a PTP Instance (see 6.4.3.7).
- 4 h) Ieee8021ASV3ScaledNs. Represents signed values of time and time interval in units of 2^{-16} ns
5 (see 6.4.3.1).
- 6 i) Ieee8021ASV3UScaledNs. Represents unsigned values of time and time interval in units of 2^{-16} ns
7 (see 6.4.3.2).
- 8 j) Ieee8021ASV3PTPInstanceIdentifier. Entity of a single time-aware system that executes gPTP in
9 one gPTP domain (see 7.2.1 and 8.1).
- 10 k) Ieee8021ASV3Timestamp. Value of Ieee8021ASV2Timestamp is equal to the remainder obtained
11 upon dividing the respective timestamp, expressed in units of 2^{-16} ns, by 2^{48} (see 14.10.53).

12 **15.6 IEEE 802.1AS MIB module^{17,18}**

13 In the following MIB modules definitions, if any discrepancy between the DESCRIPTION text and the
14 corresponding definition in any other part of this standard occurs, the definitions outside this subclause take
15 precedence.

¹⁷ Copyright release for MIBs: Users of this standard may freely reproduce the MIBs contained in this subclause so that they can be used for their intended purpose.

¹⁸ An ASCII version of this MIB module can be obtained from the IEEE 802.1 website at <https://www.ieee802.org/1/pages/MIBS.html>.

```

1 IEEE8021-AS-V3-MIB DEFINITIONS ::= BEGIN
2 -- =====
3 -- MIB for support of 802.1AS Timing and Synchronization in
4 -- IEEE 802.1Q Bridged Local Area Networks
5 -- =====
6
7 IMPORTS
8     MODULE-IDENTITY, OBJECT-TYPE, Unsigned32, Integer32, Counter32
9         FROM SNMPv2-SMI -- [RFC2578]
10    TEXTUAL-CONVENTION, TruthValue, RowStatus, TimeStamp
11        FROM SNMPv2-TC -- [RFC2579]
12    MODULE-COMPLIANCE, OBJECT-GROUP -- [RFC2580]
13        FROM SNMPv2-CONF
14        SnmpAdminString
15        FROM SNMP-FRAMEWORK-MIB -- [RFC3411]
16    InterfaceIndexOrZero
17        FROM IF-MIB -- [RFC2863]
18    Float64TC
19        FROM FLOAT-TC-MIB -- [RFC6340]
20    IEEE8021BridgePortNumber
21        FROM IEEE8021-TC-MIB
22    ;
23
24 ieee8021AsV3TimeSyncMib MODULE-IDENTITY
25     LAST-UPDATED "202402140000Z" -- February 14, 2024
26     ORGANIZATION "IEEE 802.1 Working Group"
27     CONTACT-INFO
28         "WG-URL: http://ieee802.org/1/
29         WG-EMail: stds-802-1-l@ieee.org
30
31     Contact: IEEE 802.1 Working Group Chair
32     Postal: C/O IEEE 802.1 Working Group
33             IEEE Standards Association
34             445 Hoes Lane
35             Piscataway, NJ 08854
36             USA
37
38     E-mail: stds-802-1-chairs@ieee.org"
39
40     DESCRIPTION
41         "The Management Information Base module for
42         IEEE 802.1AS time-synchronization protocol."
43
44     REVISION "202402140000Z" -- February 14, 2024
45     DESCRIPTION
46         "This MIB module remains unchanged from
47         IEEE8021-AS-V2-MIB except updating the terminology.
48
49         Published as part of IEEE Std 802.1ASdr-2024
50         Copyright (C) IEEE (2024).
51         This version of this MIB module is part of IEEE Std
52         802.1ASdr-2024; see the standard itself for full legal
53         notices."
54
55 ::= { iso(1) org(3) ieee(111)
56     standards-association-numbers-series-standards (2)
57     lan-man-stds (802) ieee802dot1 (1) ieee802dot1mibs (1) 40 }
58
59 ieee8021AsV3MIBObjects OBJECT IDENTIFIER ::= {ieee8021AsV3TimeSyncMib 1}

```



```

1 ieee8021AsV3Conformance    OBJECT IDENTIFIER ::= {ieee8021AsV3TimeSyncMib 2}
2
3 -- =====
4 -- Textual Conventions
5 -- =====
6
7 Ieee8021AsV3ClockIdentity ::= TEXTUAL-CONVENTION
8     DISPLAY-HINT
9         "1x:"
10    STATUS current
11    DESCRIPTION
12        "The Ieee8021AsV3ClockIdentity type identifies a PTP Instance.
13        The clockIdentity attribute shall be as specified in
14        IEEE Std 1588-2019."
15    REFERENCE      "6.4.3.6, 8.5.2.2 and IEEE Std 1588-2019 7.5.2.2"
16    SYNTAX OCTET STRING (SIZE (8))
17
18 Ieee8021AsV3GptpProfileIdentifier ::= TEXTUAL-CONVENTION
19     DISPLAY-HINT
20         "1x:"
21    STATUS current
22    DESCRIPTION
23        "The Ieee8021AsV3GptpProfileIdentifier attribute is the
24        profileIdentifier for this PTP profile."
25    REFERENCE      "14.9.2, F.2 "
26    SYNTAX OCTET STRING (SIZE (6))
27
28 Ieee8021AsV3ClockClassValue ::= TEXTUAL-CONVENTION
29     STATUS      current
30     DESCRIPTION
31         "The Ieee8021AsV3ClockClassValue attribute denotes the traceability
32         of the synchronized time distributed by a ClockTimeTransmitter when it is
33         the Grandmaster PTP Instance.
34         A more detailed description of clockClass can be found in
35         IEEE Std 1588-2019."
36     REFERENCE      "8.6.2.2 and IEEE Std 1588-2019 7.6.2.5"
37     SYNTAX      INTEGER {
38         primarySync(6),
39         primarySyncLost(7),
40         applicationSpecificSync(13),
41         applicationSpecificSyncLost(14),
42         primarySyncAlternativeA(52),
43         applicationSpecificAlternativeA(58),
44         primarySyncAlternativeB(187),
45         applicationSpecificAlternativeB(193),
46         defaultClock(248),
47         timeReceiverOnlyClock(255)
48     }
49
50 Ieee8021AsV3ClockAccuracyValue ::= TEXTUAL-CONVENTION
51     STATUS      current
52     DESCRIPTION
53         "The Ieee8021AsV3ClockAccuracyValue attribute indicates the
54         expected time accuracy of a ClockTimeTransmitter.
55         A more detailed description of clockAccuracy can be found in
56         IEEE Std 1588-2019."
57     REFERENCE      "8.6.2.3 and IEEE Std 1588-2019 7.6.2.6"
58     SYNTAX      INTEGER {
59         timeAccurateTo25ns(32),

```

```

1      timeAccurateTo100ns(33),
2      timeAccurateTo250ns(34),
3      timeAccurateTo1us(35),
4      timeAccurateTo2dot5us(36),
5      timeAccurateTo10us(37),
6      timeAccurateTo25us(38),
7      timeAccurateTo100us(39),
8      timeAccurateTo250us(40),
9      timeAccurateTo1ms(41),
10     timeAccurateTo2dot5ms(42),
11     timeAccurateTo10ms(43),
12     timeAccurateTo25ms(44),
13     timeAccurateTo100ms(45),
14     timeAccurateTo250ms(46),
15     timeAccurateTo1s(47),
16     timeAccurateTo10s(48),
17     timeAccurateToGT10s(49),
18     timeAccurateToUnknown(254)
19 }
20
21 Ieee8021AsV3TimeSourceValue ::= TEXTUAL-CONVENTION
22     STATUS      current
23     DESCRIPTION
24         "The Ieee8021AsV3TimeSourceValue is an information only
25         attribute indicating the type of source of time used by a
26         ClockTimeTransmitter. The value is not used in the selection of the
27         Grandmaster PTP Instance. The values of TimeSource are
28         given below and are specified in Table 8-2. These represent
29         categories. For example, the GPS entry includes not only the
30         GPS system of the U.S. Department of Defense but the European
31         Galileo system and other present and future GNSSs.
32
33         In the absence of a default value set by a user of this standard,
34         the default value of timeSource shall be INTERNAL_OSCILLATOR.
35
36         A more detailed description of timeSource can be found in
37         IEEE Std 1588-2019.
38
39         The following interpretation is placed on the value:
40             0x10: Atomic Clock,
41             0x20: GPS,
42             0x30: Terrestrial Radio,
43             0x40: PTP,
44             0x50: NTP,
45             0x60: Hand Set,
46             0x90: Other,
47             0xA0: Internal Oscillator "
48     REFERENCE "8.6.2.7, Table 8-2 and IEEE Std 1588-2019 7.6.2.8"
49     SYNTAX      INTEGER {
50         atomicClock(16),
51         gps(32),
52         terrestrialRadio(48),
53         ptp(64),
54         ntp(80),
55         handSet(96),
56         other(144),
57         internalOscillator(160)
58     }
59

```

```

1 Ieee8021ASV3PtpTimeInterval ::= TEXTUAL-CONVENTION
2   STATUS current
3   DESCRIPTION
4     "The Ieee8021ASV3PtpTimeInterval type represents time intervals
5     in units of  $2^{-16}$  ns. Positive or negative time
6     intervals outside the maximum range of this data type shall
7     be encoded as the largest positive and negative values of
8     the data type respectively.
9     For example: 2.5 ns is expressed as:
10    (hex) 0x0000 0000 0002 8000"
11   REFERENCE "6.4.3.3"
12 SYNTAX OCTET STRING (SIZE (8))
13
14 Ieee8021ASV3PtpPortIdentity ::= TEXTUAL-CONVENTION
15 STATUS current
16 DESCRIPTION
17   "The Ieee8021ASV3PtpPortIdentity type identifies a port of a
18   PTP Instance.
19   The first 8 octets within this value specifies the
20   ClockIdentity.
21   The last 2 octets within this value specifies the port number."
22   REFERENCE "6.4.3.7"
23 SYNTAX OCTET STRING (SIZE (10))
24
25 Ieee8021ASV3ScaledNs ::= TEXTUAL-CONVENTION
26 STATUS current
27 DESCRIPTION
28   "The Ieee8021ASV3ScaledNs type represents signed values of
29   time and time interval in units of  $2^{-16}$  ns.
30   Positive or negative values of time or time interval outside the
31   maximum range of this data type are encoded as the largest
32   positive or negative value of the data type, respectively.
33   For example: -2.5 ns is expressed as:
34   (hex) 0xFFFF FFFF FFFF FFFF FFFD 8000"
35   REFERENCE "6.4.3.1"
36 SYNTAX OCTET STRING (SIZE (12))
37
38 Ieee8021ASV3UScaledNs ::= TEXTUAL-CONVENTION
39 STATUS current
40 DESCRIPTION
41   "The Ieee8021ASV3UScaledNs type represents unsigned values of
42   time and time interval in units of  $2^{-16}$  ns.
43   Positive or negative values of time or time interval outside
44   the maximum range of this data type are encoded as the largest
45   positive or negative value of the data type, respectively.
46   For example: 2.5 ns is expressed as:
47   (hex) 0x0000 0000 0000 0000 0002 8000"
48   REFERENCE "6.4.3.2"
49 SYNTAX OCTET STRING (SIZE (12))
50
51 Ieee8021ASV3PTPInstanceIdentifier ::= TEXTUAL-CONVENTION
52 DISPLAY-HINT "d"
53 STATUS current
54 DESCRIPTION
55   "The entity of a single time-aware system that executes gPTP in
56   one gPTP domain is called a PTP Instance. A time-aware system
57   can contain multiple PTP Instances, which are each associated
58   with a different gPTP domain. There are two types of
59   PTP Instances, a PTP End Instance and a PTP Relay Instance."

```

```

1  REFERENCE    "7.2.1"
2  SYNTAX    Unsigned32
3
4  Ieee8021ASV3Timestamp ::= TEXTUAL-CONVENTION
5  STATUS current
6  DESCRIPTION
7      "The value of Ieee8021ASV3Timestamp is equal to the remainder
8      obtained upon dividing the respective timestamp, expressed
9      in units of 2-16 ns, by 248)."
10 REFERENCE    "14.8.53, 14.16.25 and Table 14-9"
11 SYNTAX    OCTET STRING (SIZE (6))
12
13 -- =====
14 -- subtrees in the IEEE8021-AS-MIB
15 --
16 -- System Time-Aware Parameters/Capabilities for each instance of
17 -- gPTP domain. ieee8021AsV3InstanceListIndex that is of
18 -- ieee8021AsV3DomainIdentificationNumber object-type is used as Index.
19 --
20 -- =====
21
22 -- =====
23 -- The PTP Instance set is used to allow for dynamic creation and
24 -- deletion of PTP Instances and logical ports implementations that
25 -- support dynamic create/delete of devices.
26 -- =====
27
28 ieee8021AsV3PtpInstanceTable  OBJECT-TYPE
29     SYNTAX      SEQUENCE OF Ieee8021AsV3PtpInstanceEntry
30     MAX-ACCESS  not-accessible
31     STATUS      current
32     DESCRIPTION
33         "This table is used to allow for dynamic creation and deletion
34         of PTP Instances and logical ports implementations that support
35         dynamic create/delete of devices."
36     REFERENCE   "14.1"
37     ::= { ieee8021AsV3MIBObjects 1 }
38
39 ieee8021AsV3PtpInstanceEntry  OBJECT-TYPE
40     SYNTAX      Ieee8021AsV3PtpInstanceEntry
41     MAX-ACCESS  not-accessible
42     STATUS      current
43     DESCRIPTION
44         "An entry that specifies a PTP Instance."
45     INDEX { ieee8021AsV3PtpInstance }
46     ::= { ieee8021AsV3PtpInstanceTable 1 }
47
48 Ieee8021AsV3PtpInstanceEntry ::=
49     SEQUENCE {
50         ieee8021AsV3PtpInstance          Ieee8021ASV3PTPInstanceIdentifier,
51         ieee8021AsV3PtpInstanceName      SnmpAdminString,
52         ieee8021AsV3PtpInstanceRowStatus RowStatus
53     }
54
55 ieee8021AsV3PtpInstance  OBJECT-TYPE
56     SYNTAX Ieee8021ASV3PTPInstanceIdentifier
57     MAX-ACCESS not-accessible
58     STATUS current
59     DESCRIPTION

```

```

1      "The entity of a single time-aware system that executes gPTP in
2      one gPTP domain is called a PTP Instance. A time-aware system can
3      contain multiple PTP Instances, which are each associated with
4      a different gPTP domain. There are two types of PTP Instances,
5      a PTP End Instance and a PTP Relay Instance."
6  REFERENCE "7.2.1"
7  ::= { ieee8021AsV3PtpInstanceEntry 1 }
8
9  ieee8021AsV3PtpInstanceName OBJECT-TYPE
10   SYNTAX SnmpAdminString
11   MAX-ACCESS read-create
12   STATUS current
13   DESCRIPTION
14   "Name for identification of a PTP Instance."
15   DEFVAL { "" }
16   ::= { ieee8021AsV3PtpInstanceEntry 2 }
17
18  ieee8021AsV3PtpInstanceRowStatus OBJECT-TYPE
19   SYNTAX RowStatus
20   MAX-ACCESS read-create
21   STATUS current
22   DESCRIPTION
23   "This attribute is used to create and delete PTP Instances."
24   REFERENCE "14.1"
25   ::= { ieee8021AsV3PtpInstanceEntry 3 }
26
27  -- =====
28  -- The Default data set represents native time capability of a time-
29  -- aware system and is consistent with respective IEEE 1588 data set.
30  -- =====
31
32  ieee8021AsV3DefaultDSTable OBJECT-TYPE
33   SYNTAX SEQUENCE OF Ieee8021AsV3DefaultDSEntry
34   MAX-ACCESS not-accessible
35   STATUS current
36   DESCRIPTION
37   "The Default Parameter Data Set represents the native capabilities
38   of a PTP Instance, i.e., a PTP Relay Instance or a
39   PTP End Instance."
40   REFERENCE "14.2"
41   ::= { ieee8021AsV3MIBObjects 2 }
42
43  ieee8021AsV3DefaultDSEntry OBJECT-TYPE
44   SYNTAX Ieee8021AsV3DefaultDSEntry
45   MAX-ACCESS not-accessible
46   STATUS current
47   DESCRIPTION
48   "Default Data Set contains the profile Identifier for
49   this instance of gPTP domain."
50   INDEX { ieee8021AsV3PtpInstance }
51   ::= { ieee8021AsV3DefaultDSTable 1 }
52
53  Ieee8021AsV3DefaultDSEntry ::=
54   SEQUENCE {
55     ieee8021AsV3DefaultDSClockIdentity Ieee8021AsV3ClockIdentity,
56     ieee8021AsV3DefaultDSNumberPorts Unsigned32,
57     ieee8021AsV3DefaultDSClockQualityClockClass Ieee8021AsV3ClockClassValue,
58     ieee8021AsV3DefaultDSClockQualityClockAccuracy
59     Ieee8021AsV3ClockAccuracyValue,

```

```

1   ieee8021AsV3DefaultDSClockQualityOffsetScaledLogVariance Unsigned32,
2   ieee8021AsV3DefaultDSPriority1 Unsigned32,
3   ieee8021AsV3DefaultDSPriority2 Unsigned32,
4   ieee8021AsV3DefaultDSGmCapable TruthValue,
5   ieee8021AsV3DefaultDSCurrentUtcOffset Integer32,
6   ieee8021AsV3DefaultDSCurrentUtcOffsetValid TruthValue,
7   ieee8021AsV3DefaultDSLeap59 TruthValue,
8   ieee8021AsV3DefaultDSLeap61 TruthValue,
9   ieee8021AsV3DefaultDSTimeTraceable TruthValue,
10  ieee8021AsV3DefaultDSFrequencyTraceable TruthValue,
11  ieee8021AsV3DefaultDSPtpTimescale TruthValue,
12  ieee8021AsV3DefaultDSTimeSource Ieee8021AsV3TimeSourceValue,
13  ieee8021AsV3DefaultDSDomainNumber Unsigned32,
14  ieee8021AsV3DefaultDSSdoId Unsigned32,
15  ieee8021AsV3DefaultDSExternalPortConfigurationEnabled TruthValue,
16  ieee8021AsV3DefaultDSInstanceEnable TruthValue
17  }
18
19 ieee8021AsV3DefaultDSClockIdentity OBJECT-TYPE
20     SYNTAX      Ieee8021AsV3ClockIdentity
21     MAX-ACCESS  read-only
22     STATUS      current
23     DESCRIPTION
24         "The value is the clockIdentity of the PTP Instance.
25         The clockIdentity attribute shall be as specified in
26         IEEE Std 1588-2019."
27     REFERENCE   "14.2.2 and IEEE Std 1588-2019 7.5.2.2"
28     ::= { ieee8021AsV3DefaultDSEntry 1 }
29
30
31 ieee8021AsV3DefaultDSNumberPorts OBJECT-TYPE
32     SYNTAX      Unsigned32(1..65535)
33     MAX-ACCESS  read-only
34     STATUS      current
35     DESCRIPTION
36         "The number of ports of the PTP Instance. For an end
37         station the value is 1."
38     REFERENCE   "14.2.3"
39     ::= { ieee8021AsV3DefaultDSEntry 2 }
40
41
42 ieee8021AsV3DefaultDSClockQualityClockClass OBJECT-TYPE
43     SYNTAX      Ieee8021AsV3ClockClassValue
44     MAX-ACCESS  read-only
45     STATUS      current
46     DESCRIPTION
47         "The value is the clockClass of the PTP Instance, which
48         implements the clockClass specifications of 8.6.2.2."
49     REFERENCE   "14.2.4.2"
50     ::= { ieee8021AsV3DefaultDSEntry 3 }
51
52
53 ieee8021AsV3DefaultDSClockQualityClockAccuracy OBJECT-TYPE
54     SYNTAX      Ieee8021AsV3ClockAccuracyValue
55     MAX-ACCESS  read-only
56     STATUS      current
57     DESCRIPTION
58         "The value is the clockAccuracy of the PTP Instance, which
59         implements the clockAccuracy specifications of 8.6.2.3."

```

```

1  REFERENCE    "14.2.4.3"
2  ::= { ieee8021AsV3DefaultDSEntry 4 }
3
4
5  ieee8021AsV3DefaultDSClockQualityOffsetScaledLogVariance OBJECT-TYPE
6  SYNTAX      Unsigned32(0..65535)
7  MAX-ACCESS  read-only
8  STATUS      current
9  DESCRIPTION
10     "The value is the offsetScaledLogVariance of the PTP Instance,
11     which implements the offsetScaledLogVariance specifications
12     of 8.6.2.4."
13  REFERENCE    "14.2.4.4"
14  ::= { ieee8021AsV3DefaultDSEntry 5 }
15
16  ieee8021AsV3DefaultDSPriority1 OBJECT-TYPE
17  SYNTAX      Unsigned32(0..255)
18  MAX-ACCESS  read-write
19  STATUS      current
20  DESCRIPTION
21     "The value is the priority1 attribute of the PTP Instance."
22  REFERENCE    "14.2.5"
23  ::= { ieee8021AsV3DefaultDSEntry 6 }
24
25
26  ieee8021AsV3DefaultDSPriority2 OBJECT-TYPE
27  SYNTAX      Unsigned32(0..255)
28  MAX-ACCESS  read-write
29  STATUS      current
30  DESCRIPTION
31     "The value is the priority2 attribute of the PTP Instance."
32  REFERENCE    "14.2.6"
33  DEFVAL { 248 }
34  ::= { ieee8021AsV3DefaultDSEntry 7 }
35
36
37  ieee8021AsV3DefaultDSGmCapable OBJECT-TYPE
38  SYNTAX      TruthValue
39  MAX-ACCESS  read-only
40  STATUS      current
41  DESCRIPTION
42     "The value is TRUE (1) if the PTP Instance is capable of being a
43     Grandmaster PTP Instance, and FALSE (2) if the PTP Instance is
44     not capable of being a Grandmaster PTP Instance."
45  REFERENCE    "14.2.7"
46  ::= { ieee8021AsV3DefaultDSEntry 8 }
47
48
49  ieee8021AsV3DefaultDSCurrentUtcOffset OBJECT-TYPE
50  SYNTAX      Integer32(-32768..32767)
51  UNITS       "seconds"
52  MAX-ACCESS  read-only
53  STATUS      current
54  DESCRIPTION
55     "The value is the offset between TAI and UTC, relative to
56     the ClockTimeTransmitter entity of this PTP Instance. It is equal
57     to the global variable sysCurrentUtcOffset.
58     The value is in units of seconds.
59

```

```

1      The default value is selected as follows:
2      a)The value is the value obtained from a primary
3         reference if the value is known at the time of
4         initialization, else
5      b)The value is the current IERS defined value of
6         TAI - UTC (see IERS Bulletin C) when the PTP Instance
7         is designed.currentUtcOffsetValid"
8  REFERENCE    "14.2.8"
9  ::= { ieee8021AsV3DefaultDSEntry 9 }
10
11
12 ieee8021AsV3DefaultDSCurrentUtcOffsetValid OBJECT-TYPE
13     SYNTAX      TruthValue
14     MAX-ACCESS  read-only
15     STATUS      current
16     DESCRIPTION
17         "The default value is TRUE (1) if the value of
18         ieee8021AsV3DefaultDSCurrentUtcOffset is known to be
19         correct, otherwise it is set to FALSE (2)."
```

```

20 REFERENCE    "14.2.9"
21 ::= { ieee8021AsV3DefaultDSEntry 10 }
22
23
24 ieee8021AsV3DefaultDSLeap59 OBJECT-TYPE
25     SYNTAX      TruthValue
26     MAX-ACCESS  read-only
27     STATUS      current
28     DESCRIPTION
29         "A TRUE (1) value indicates that the last minute of the
30         current UTC day, relative to the ClockTimeTransmitter entity of
31         this PTP Instance, will contain 59 s. It is equal to the
32         global variable sysLeap59.
33
34         The value is selected as follows:
35         a)The value is obtained from a primary reference if
36            known at the time of initialization, else
37         b)The value is set to FALSE (2)."
```

```

38 REFERENCE    "14.2.10"
39 ::= { ieee8021AsV3DefaultDSEntry 11 }
40
41
42 ieee8021AsV3DefaultDSLeap61 OBJECT-TYPE
43     SYNTAX      TruthValue
44     MAX-ACCESS  read-only
45     STATUS      current
46     DESCRIPTION
47         "A TRUE (1) value indicates that the last minute of the
48         current UTC day, relative to the ClockTimeTransmitter entity of
49         this PTP Instance, will contain 61 s. It is equal to the global
50         variable sysLeap61.
51
52         The value is selected as follows:
53         a)The value is obtained from a primary reference if
54            known at the time of initialization, else
55         b)The value is set to FALSE (2)."
```

```

56 REFERENCE    "14.2.11"
57 ::= { ieee8021AsV3DefaultDSEntry 12 }
58
59

```



```

1 ieee8021AsV3DefaultDSTimeTraceable OBJECT-TYPE
2     SYNTAX      TruthValue
3     MAX-ACCESS  read-only
4     STATUS      current
5     DESCRIPTION
6         "The value is set to TRUE (1) if the timescale and the value
7         of currentUtcOffset, relative to the ClockTimeTransmitter entity of
8         this PTP Instance, are traceable to a primary reference
9         standard; otherwise the value is set to FALSE (2).
10        It is equal to the global variable sysTimeTraceable.
11
12        The value is selected as follows:
13        a)If the time and the value of currentUtcOffset are
14        traceable to a primary reference standard at the time of
15        initialization, the value is set to TRUE (1), else
16        b)The value is set to FALSE (2)."
```

REFERENCE "14.2.12"

```

17 ::= { ieee8021AsV3DefaultDSEntry 13 }
18
19
20
21 ieee8021AsV3DefaultDSFrequencyTraceable OBJECT-TYPE
22     SYNTAX      TruthValue
23     MAX-ACCESS  read-only
24     STATUS      current
25     DESCRIPTION
26         "The value is set to TRUE (1) if the frequency determining the
27         timescale of the ClockTimeTransmitter Entity of this PTP Instance is
28         traceable to a primary standard; otherwise the value is set
29         to FALSE (2). It is equal to the global variable
30         sysFrequencyTraceable.
31
32        The value is selected as follows:
33        a)If the frequency is traceable to a primary reference
34        standard at the time of initialization the value is set
35        to TRUE (1), else
36        b)The value is set to FALSE (2)."
```

REFERENCE "14.2.13"

```

37 ::= { ieee8021AsV3DefaultDSEntry 14 }
38
39
40 ieee8021AsV3DefaultDSptpTimescale OBJECT-TYPE
41     SYNTAX      TruthValue
42     MAX-ACCESS  read-only
43     STATUS      current
44     DESCRIPTION
45         "The value is set to TRUE (1) if the clock timescale of the
46         ClockTimeTransmitter Entity of this PTP Instance is PTP and
47         FALSE (2) otherwise."
48     REFERENCE   "14.2.14"
49     ::= { ieee8021AsV3DefaultDSEntry 15 }
50
51 ieee8021AsV3DefaultDSTimeSource OBJECT-TYPE
52     SYNTAX      Ieee8021AsV3TimeSourceValue
53     MAX-ACCESS  read-only
54     STATUS      current
55     DESCRIPTION
56         "The value is the source of time used by the
57         Grandmaster PTP Instance clock."
58     REFERENCE   "14.2.15"
59     ::= { ieee8021AsV3DefaultDSEntry 16 }

```

```

1
2 ieee8021AsV3DefaultDSDomainNumber OBJECT-TYPE
3     SYNTAX      Unsigned32(0..127)
4     MAX-ACCESS  read-write
5     STATUS      current
6     DESCRIPTION
7         "The value is the domain number of the gPTP domain for this
8         instance of gPTP supported by the time-aware system."
9     REFERENCE   "14.2.16"
10    ::= { ieee8021AsV3DefaultDSEntry 17 }
11
12 ieee8021AsV3DefaultDSSdoId OBJECT-TYPE
13     SYNTAX      Unsigned32(0..4095)
14     MAX-ACCESS  read-only
15     STATUS      current
16     DESCRIPTION
17         "The value is the sdoId of the gPTP domain for this instance
18         of gPTP supported by the time-aware system.
19         For compatibility with IEEE Std 1588, the range of the
20         managed object is limited to 12 bits; in addition, only the
21         single value 0x100 is specified in this standard for the
22         gPTP domain of a PTP Instance."
23     REFERENCE   "14.2.17"
24    ::= { ieee8021AsV3DefaultDSEntry 18 }
25
26 ieee8021AsV3DefaultDSEnabled OBJECT-TYPE
27     SYNTAX      TruthValue
28     MAX-ACCESS  read-write
29     STATUS      current
30     DESCRIPTION
31         "The value is the externalPortConfigurationEnabled attribute
32         of the PTP Instance."
33     REFERENCE   "14.2.18"
34    ::= { ieee8021AsV3DefaultDSEntry 19 }
35
36 ieee8021AsV3DefaultDSInstanceEnable OBJECT-TYPE
37     SYNTAX      TruthValue
38     MAX-ACCESS  read-write
39     STATUS      current
40     DESCRIPTION
41         "The value is the instanceEnable attribute of the PTP Instance."
42     REFERENCE   "14.2.19"
43    ::= { ieee8021AsV3DefaultDSEntry 20 }
44
45 -- =====
46 -- The Current data set represents this system's topological location
47 -- relative to the known Grandmaster PTP Instance.
48 -- This data set is consistent with respective IEEE 1588 data set.
49 -- =====
50
51 ieee8021AsV3CurrentDSTable OBJECT-TYPE
52     SYNTAX      SEQUENCE OF Ieee8021AsV3CurrentDSEntry
53     MAX-ACCESS  not-accessible
54     STATUS      current
55     DESCRIPTION
56         "The Current Parameter Data Set represents the position of a local
57         system and other information, relative to the
58         Grandmaster PTP Instance."
59     REFERENCE   "14.3"

```

```

1 ::= { ieee8021AsV3MIBObjects 3 }
2
3 ieee8021AsV3CurrentDSEntry OBJECT-TYPE
4     SYNTAX      Ieee8021AsV3CurrentDSEntry
5     MAX-ACCESS  not-accessible
6     STATUS      current
7     DESCRIPTION
8         "Current Data Set for a specific PTP Instance."
9     INDEX { ieee8021AsV3PtpInstance }
10    ::= { ieee8021AsV3CurrentDSTable 1 }
11
12 Ieee8021AsV3CurrentDSEntry ::=
13     SEQUENCE {
14         ieee8021AsV3CurrentDSStepsRemoved          Unsigned32,
15         ieee8021AsV3CurrentDSOffsetFromTimeTransmitter
16 Ieee8021ASV3PtpTimeInterval,
17         ieee8021AsV3CurrentDSLstGmPhaseChange      Ieee8021ASV3ScaledNs,
18         ieee8021AsV3CurrentDSLstGmFreqChange       Float64TC,
19         ieee8021AsV3CurrentDSGmTimebaseIndicator   Unsigned32,
20         ieee8021AsV3CurrentDSGmChangeCount         Counter32,
21         ieee8021AsV3CurrentDSTimeOfLastGmChangeEvent TimeStamp,
22         ieee8021AsV3CurrentDSTimeOfLastGmPhaseChangeEvent TimeStamp,
23         ieee8021AsV3CurrentDSTimeOfLastGmFreqChangeEvent TimeStamp
24     }
25
26 ieee8021AsV3CurrentDSStepsRemoved OBJECT-TYPE
27     SYNTAX      Unsigned32(0..65535)
28     MAX-ACCESS  read-only
29     STATUS      current
30     DESCRIPTION
31         "The value is the number of gPTP communication paths
32         traversed between this PTP Instance and the
33         Grandmaster PTP Instance, as specified in 10.3.3."
34     REFERENCE   "14.3.2"
35     ::= { ieee8021AsV3CurrentDSEntry 1 }
36
37
38 ieee8021AsV3CurrentDSOffsetFromTimeTransmitter OBJECT-TYPE
39     SYNTAX      Ieee8021ASV3PtpTimeInterval
40     MAX-ACCESS  read-only
41     STATUS      current
42     DESCRIPTION
43         "The value is an implementation-specific representation of
44         the current value of the time difference between a timeReceiver
45         and the Grandmaster Clock, as computed by the timeReceiver, and
46         as specified in 10.2.10."
47     REFERENCE   "14.3.3"
48     ::= { ieee8021AsV3CurrentDSEntry 2 }
49
50 ieee8021AsV3CurrentDSLstGmPhaseChange OBJECT-TYPE
51     SYNTAX      Ieee8021ASV3ScaledNs
52     MAX-ACCESS  read-only
53     STATUS      current
54     DESCRIPTION
55         "The value is the phase change that occurred on the most
56         recent change in either Grandmaster PTP Instance or
57         gmTimeBaseIndicator."
58     REFERENCE   "14.3.4"
59     ::= { ieee8021AsV3CurrentDSEntry 3 }

```

```

1
2 ieee8021AsV3CurrentDSLstGmFreqChange OBJECT-TYPE
3     SYNTAX      Float64TC
4     MAX-ACCESS  read-only
5     STATUS      current
6     DESCRIPTION
7         "The value is the frequency change that occurred on the most
8         recent change in either Grandmaster PTP Instance or
9         gmTimeBaseIndicator."
10    REFERENCE   "14.3.5"
11    ::= { ieee8021AsV3CurrentDSEntry 4 }
12
13 ieee8021AsV3CurrentDSGmTimebaseIndicator OBJECT-TYPE
14     SYNTAX      Unsigned32(0..65535)
15     MAX-ACCESS  read-only
16     STATUS      current
17     DESCRIPTION
18         "The value is the value of timeBaseIndicator of the
19         current Grandmaster PTP Instance."
20     REFERENCE   "14.3.6"
21     ::= { ieee8021AsV3CurrentDSEntry 5 }
22
23 ieee8021AsV3CurrentDSGmChangeCount OBJECT-TYPE
24     SYNTAX      Counter32
25     MAX-ACCESS  read-only
26     STATUS      current
27     DESCRIPTION
28         "This statistics counter tracks the number of times the
29         Grandmaster PTP Instance has changed in a gPTP domain.
30         This counter increments when the PortAnnounceInformation
31         state machine enters the SUPERIOR_TIME_TRANSMITTER_PORT state
32         or the INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state."
33     REFERENCE   "14.3.7"
34     ::= { ieee8021AsV3CurrentDSEntry 6 }
35
36 ieee8021AsV3CurrentDSTimeOfLastGmChangeEvent OBJECT-TYPE
37     SYNTAX      TimeStamp
38     UNITS       "0.01 seconds"
39     MAX-ACCESS  read-only
40     STATUS      current
41     DESCRIPTION
42         "This timestamp takes the value of sysUpTime (see RFC3418) when
43         the most recent Grandmaster PTP Instance change occurred in
44         a gPTP domain.
45         This timestamp is updated when the PortAnnounceInformation
46         state machine enters the SUPERIOR_TIME_TRANSMITTER_PORT state or
47         the INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state."
48     REFERENCE   "14.3.8"
49     ::= { ieee8021AsV3CurrentDSEntry 7 }
50
51
52 ieee8021AsV3CurrentDSTimeOfLastGmPhaseChangeEvent OBJECT-TYPE
53     SYNTAX      TimeStamp
54     UNITS       "0.01 seconds"
55     MAX-ACCESS  read-only
56     STATUS      current
57     DESCRIPTION
58         "This timestamp takes the value of sysUpTime (see RFC3418)
59         when the most recent change in Grandmaster Clock phase

```

```

1      occurred, due to a change of either the
2      Grandmaster PTP Instance or the Grandmaster Clock
3      time base. This timestamp is updated when one of the
4      following occurs:
5          a)The PortAnnounceInformation state machine enters the
6              SUPERIOR_TIME_TRANSMITTER_PORT state or the
7              INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state, or
8          b)The gmTimebaseIndicator managed object changes and the
9              lastGmPhaseChange field of the most recently received
10             Follow_Up information TLV is nonzero."
11  REFERENCE    "14.3.9"
12  ::= { ieee8021AsV3CurrentDSEntry 8 }
13
14 ieee8021AsV3CurrentDSTimeOfLastGmFreqChangeEvent OBJECT-TYPE
15     SYNTAX      TimeStamp
16     UNITS       "0.01 seconds"
17     MAX-ACCESS  read-only
18     STATUS      current
19     DESCRIPTION
20         "This timestamp takes the value of sysUpTime (see RFC3418)
21         when the most recent change in Grandmaster Clock frequency
22         occurred, due to a change of either the Grandmaster PTP
23         Instance or the Grandmaster Clock time base. This timestamp
24         is updated when one of the following occurs:
25             a)The PortAnnounceInformation state machine enters the
26                 SUPERIOR_TIME_TRANSMITTER_PORT state or the
27                 INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state, or
28             b)The gmTimebaseIndicator managed object changes and the
29                 lastGmFreqChange field of the most recently received
30                 Follow_Up information TLV is nonzero."
31  REFERENCE    "14.3.10"
32  ::= { ieee8021AsV3CurrentDSEntry 9 }
33
34 -- =====
35 -- The Parent data set represents the upstream (toward
36 -- Grandmaster PTP Instance) system's timing parameters as measured
37 -- at this system.
38 -- This data set is consistent with the respective IEEE 1588 data set.
39 -- =====
40 ieee8021AsV3ParentDSTable OBJECT-TYPE
41     SYNTAX      SEQUENCE OF Ieee8021AsV3ParentDSEntry
42     MAX-ACCESS  not-accessible
43     STATUS      current
44     DESCRIPTION
45         "The Parent Parameter Data Set represents capabilities of the
46         upstream system, toward the Grandmaster PTP Instance, as
47         measured at a local system."
48     REFERENCE    "14.4"
49     ::= { ieee8021AsV3MIBObjects 4 }
50
51 ieee8021AsV3ParentDSEntry OBJECT-TYPE
52     SYNTAX      Ieee8021AsV3ParentDSEntry
53     MAX-ACCESS  not-accessible
54     STATUS      current
55     DESCRIPTION
56         "Parent Data Set for a specific PTP Instance."
57     INDEX { ieee8021AsV3PtpInstance }
58     ::= { ieee8021AsV3ParentDSTable 1 }
59

```

```

1 Ieee8021AsV3ParentDSEntry ::=
2   SEQUENCE {
3     ieee8021AsV3ParentDSParentClockIdentity    Ieee8021AsV3ClockIdentity,
4     ieee8021AsV3ParentDSParentPortNumber       Unsigned32,
5     ieee8021AsV3ParentDSCumulativeRateRatio    Integer32,
6     ieee8021AsV3ParentDSGrandmasterIdentity    Ieee8021AsV3ClockIdentity,
7     ieee8021AsV3ParentDSGrandmasterClockQualityclockClass
8     Ieee8021AsV3ClockClassValue,
9     ieee8021AsV3ParentDSGrandmasterClockQualityclockAccuracy
10    Ieee8021AsV3ClockAccuracyValue,
11    ieee8021AsV3ParentDSGrandmasterClockQualityoffsetScaledLogVar
12    Unsigned32,
13    ieee8021AsV3ParentDSGrandmasterPriority1    Unsigned32,
14    ieee8021AsV3ParentDSGrandmasterPriority2    Unsigned32
15  }
16
17 ieee8021AsV3ParentDSParentClockIdentity OBJECT-TYPE
18   SYNTAX      Ieee8021AsV3ClockIdentity
19   MAX-ACCESS  read-only
20   STATUS      current
21   DESCRIPTION
22     "The value is the first of the parentPortIdentity attribute
23     for this instance of gPTP domain, which is a set made of
24     Ieee8021AsV3ClockIdentity and portNumber."
25   REFERENCE   "14.4.2"
26   ::= { ieee8021AsV3ParentDSEntry 1 }
27
28 ieee8021AsV3ParentDSParentPortNumber OBJECT-TYPE
29   SYNTAX      Unsigned32(0..65535)
30   MAX-ACCESS  read-only
31   STATUS      current
32   DESCRIPTION
33     "The value is the second of the parentPortIdentity attribute
34     for this instance of gPTP domain, which is a set made of
35     Ieee8021AsV3ClockIdentity and portNumber."
36   REFERENCE   "14.4.2"
37   ::= { ieee8021AsV3ParentDSEntry 2 }
38
39 ieee8021AsV3ParentDSCumulativeRateRatio OBJECT-TYPE
40   SYNTAX      Integer32
41   MAX-ACCESS  read-only
42   STATUS      current
43   DESCRIPTION
44     "The value is an estimate of the ratio of the frequency of
45     the Grandmaster Clock to the frequency of the LocalClock
46     entity of this PTP Instance.
47     CumulativeRateRatio is expressed as the fractional
48     frequency offset multiplied by 2^41, i.e., the quantity
49     (rateRatio - 1.0)(2^41), where rateRatio is computed by
50     the PortSyncSyncReceive state machine."
51   REFERENCE   "14.4.3"
52   ::= { ieee8021AsV3ParentDSEntry 3 }
53
54 ieee8021AsV3ParentDSGrandmasterIdentity OBJECT-TYPE
55   SYNTAX      Ieee8021AsV3ClockIdentity
56   MAX-ACCESS  read-only
57   STATUS      current
58   DESCRIPTION
59     "The value is the clockIdentity attribute of the

```

```

1      Grandmaster PTP Instance."
2  REFERENCE    "14.4.4"
3  ::= { ieee8021AsV3ParentDSEntry 4 }
4
5  ieee8021AsV3ParentDSGrandmasterClockQualityclockClass OBJECT-TYPE
6  SYNTAX      Ieee8021AsV3ClockClassValue
7  MAX-ACCESS  read-only
8  STATUS      current
9  DESCRIPTION
10     "The value is the clockClass of the Grandmaster PTP Instance."
11  REFERENCE    "14.4.5.2"
12  ::= { ieee8021AsV3ParentDSEntry 5 }
13
14  ieee8021AsV3ParentDSGrandmasterClockQualityclockAccuracy OBJECT-TYPE
15  SYNTAX      Ieee8021AsV3ClockAccuracyValue
16  MAX-ACCESS  read-only
17  STATUS      current
18  DESCRIPTION
19     "The value is the clockAccuracy of the Grandmaster PTP Instance."
20  REFERENCE    "14.4.5.3"
21  ::= { ieee8021AsV3ParentDSEntry 6 }
22
23  ieee8021AsV3ParentDSGrandmasterClockQualityoffsetScaledLogVar
24  OBJECT-TYPE
25  SYNTAX      Unsigned32(0..65535)
26  MAX-ACCESS  read-only
27  STATUS      current
28  DESCRIPTION
29     "The value is the offsetScaledLogVariance of the
30     Grandmaster PTP Instance."
31  REFERENCE    "14.4.5.4"
32  ::= { ieee8021AsV3ParentDSEntry 7 }
33
34  ieee8021AsV3ParentDSGrandmasterPriority1 OBJECT-TYPE
35  SYNTAX      Unsigned32(0..255)
36  MAX-ACCESS  read-only
37  STATUS      current
38  DESCRIPTION
39     "The value is the priority1 attribute of the
40     Grandmaster PTP Instance."
41  REFERENCE    "14.4.6"
42  ::= { ieee8021AsV3ParentDSEntry 8 }
43
44  ieee8021AsV3ParentDSGrandmasterPriority2 OBJECT-TYPE
45  SYNTAX      Unsigned32(0..255)
46  MAX-ACCESS  read-only
47  STATUS      current
48  DESCRIPTION
49     "The value is the priority2 attribute of the
50     Grandmaster PTP Instance."
51  REFERENCE    "14.4.7"
52  ::= { ieee8021AsV3ParentDSEntry 9 }
53
54 -- =====
55 -- TimePropertiesDS represents the Grandmaster PTP Instance's
56 -- parameters, as measured at this system and are derived from
57 -- IEEE 802.1AS protocol.
58 -- This data set is consistent with respective IEEE 1588 data set.
59 -- =====

```

```

1 ieee8021AsV3TimePropertiesDSTable    OBJECT-TYPE
2     SYNTAX        SEQUENCE OF Ieee8021AsV3TimePropertiesDSEntry
3     MAX-ACCESS    not-accessible
4     STATUS        current
5     DESCRIPTION
6         "The Time Properties Parameter Data Set represents capabilities of
7         the Grandmaster PTP Instance, as measured at a local system"
8     REFERENCE     "14.5"
9         ::= { ieee8021AsV3MIBObjects 5 }
10
11 ieee8021AsV3TimePropertiesDSEntry    OBJECT-TYPE
12     SYNTAX        Ieee8021AsV3TimePropertiesDSEntry
13     MAX-ACCESS    not-accessible
14     STATUS        current
15     DESCRIPTION
16         "Time Properties Data Set contains the profile Identifier for
17         this instance of gPTP domain."
18     INDEX { ieee8021AsV3PtpInstance }
19     ::= { ieee8021AsV3TimePropertiesDSTable 1 }
20
21 Ieee8021AsV3TimePropertiesDSEntry ::=
22     SEQUENCE {
23         ieee8021AsV3TimePropertiesDSCurrentUtcOffset      Integer32,
24         ieee8021AsV3TimePropertiesDSCurrentUtcOffsetValid TruthValue,
25         ieee8021AsV3TimePropertiesDSLeap59                 TruthValue,
26         ieee8021AsV3TimePropertiesDSLeap61                 TruthValue,
27         ieee8021AsV3TimePropertiesDSTimeTraceable          TruthValue,
28         ieee8021AsV3TimePropertiesDSFrequencyTraceable     TruthValue,
29         ieee8021AsV3TimePropertiesDSPtpTimescale           TruthValue,
30         ieee8021AsV3TimePropertiesDSTimeSource             Ieee8021AsV3TimeSourceValue
31     }
32
33 ieee8021AsV3TimePropertiesDSCurrentUtcOffset OBJECT-TYPE
34     SYNTAX        Integer32(-32768..32767)
35     UNITS         "seconds"
36     MAX-ACCESS    read-only
37     STATUS        current
38     DESCRIPTION
39         "The value is currentUtcOffset for the current
40         Grandmaster PTP Instance. It is equal to the value of
41         the global variable currentUtcOffset. The value is in
42         units of seconds."
43     REFERENCE     "14.5.2"
44     ::= { ieee8021AsV3TimePropertiesDSEntry 1 }
45
46 ieee8021AsV3TimePropertiesDSCurrentUtcOffsetValid OBJECT-TYPE
47     SYNTAX        TruthValue
48     MAX-ACCESS    read-only
49     STATUS        current
50     DESCRIPTION
51         "The value is currentUtcOffsetValid for the current
52         Grandmaster PTP Instance. It is equal to the global
53         variable currentUtcOffsetValid."
54     REFERENCE     "14.5.3"
55     ::= { ieee8021AsV3TimePropertiesDSEntry 2 }
56
57 ieee8021AsV3TimePropertiesDSLeap59 OBJECT-TYPE
58     SYNTAX        TruthValue
59     MAX-ACCESS    read-only

```



```

1  STATUS      current
2  DESCRIPTION
3      "The value is leap59 for the current Grandmaster PTP Instance.
4      It is equal to the global variable leap59."
5  REFERENCE   "14.5.4"
6  ::= { ieee8021AsV3TimePropertiesDSEntry 3 }
7
8  ieee8021AsV3TimePropertiesDSLeap61 OBJECT-TYPE
9      SYNTAX      TruthValue
10     MAX-ACCESS   read-only
11     STATUS       current
12     DESCRIPTION
13         "The value is leap61 for the current Grandmaster PTP Instance.
14         It is equal to the global variable leap61."
15     REFERENCE    "14.5.5"
16     ::= { ieee8021AsV3TimePropertiesDSEntry 4 }
17
18  ieee8021AsV3TimePropertiesDSTimeTraceable OBJECT-TYPE
19      SYNTAX      TruthValue
20      MAX-ACCESS   read-only
21      STATUS       current
22      DESCRIPTION
23          "The value is timeTraceable for the current
24          Grandmaster PTP Instance. It is equal to the global
25          variable timeTraceable."
26      REFERENCE    "14.5.6"
27      ::= { ieee8021AsV3TimePropertiesDSEntry 5 }
28
29  ieee8021AsV3TimePropertiesDSFrequencyTraceable OBJECT-TYPE
30      SYNTAX      TruthValue
31      MAX-ACCESS   read-only
32      STATUS       current
33      DESCRIPTION
34          "The value is frequencyTraceable for the current
35          Grandmaster PTP Instance. It is equal to the global
36          variable frequencyTraceable."
37      REFERENCE    "14.5.7"
38      ::= { ieee8021AsV3TimePropertiesDSEntry 6 }
39
40  ieee8021AsV3TimePropertiesDSptpTimescale OBJECT-TYPE
41      SYNTAX      TruthValue
42      MAX-ACCESS   read-only
43      STATUS       current
44      DESCRIPTION
45          "The value is ptpTimescale for the current
46          Grandmaster PTP Instance."
47      REFERENCE    "14.5.8"
48      ::= { ieee8021AsV3TimePropertiesDSEntry 7 }
49
50  ieee8021AsV3TimePropertiesDSTimeSource OBJECT-TYPE
51      SYNTAX      Ieee8021AsV3TimeSourceValue
52      MAX-ACCESS   read-only
53      STATUS       current
54      DESCRIPTION
55          "The value is timeSource for the current
56          Grandmaster PTP Instance. It is equal to the global
57          variable timeSource"
58      REFERENCE    "14.5.9"
59      ::= { ieee8021AsV3TimePropertiesDSEntry 8 }

```

```

1
2 -- =====
3 -- The Path Trace Parameter Data set represents the current path
4 -- trace information available at the PTP Instance.
5 -- =====
6
7 ieee8021AsV3PathTracedSTable    OBJECT-TYPE
8     SYNTAX          SEQUENCE OF Ieee8021AsV3PathTracedSEntry
9     MAX-ACCESS      not-accessible
10    STATUS           current
11    DESCRIPTION
12        "The pathTracedS represents the current path trace information
13        available at the PTP Instance."
14    REFERENCE        "14.6"
15    ::= { ieee8021AsV3MIBObjects 6 }
16
17 ieee8021AsV3PathTracedSEntry    OBJECT-TYPE
18     SYNTAX          Ieee8021AsV3PathTracedSEntry
19     MAX-ACCESS      not-accessible
20     STATUS           current
21     DESCRIPTION
22        "Path Trace Data Set for a specific PTP Instance."
23     INDEX { ieee8021AsV3PtpInstance }
24     ::= { ieee8021AsV3PathTracedSTable 1 }
25
26 Ieee8021AsV3PathTracedSEntry ::=
27 SEQUENCE {
28     ieee8021AsV3PathTraceDSEnable          TruthValue
29 }
30
31 ieee8021AsV3PathTraceDSEnable    OBJECT-TYPE
32     SYNTAX          TruthValue
33     MAX-ACCESS      read-only
34     STATUS           current
35     DESCRIPTION
36        "The value is TRUE.
37        NOTE: This member is included for compatibility with
38        IEEE Std 1588. In IEEE Std 1588, the path trace mechanism
39        is optional, and the pathTracedS.enable member is
40        configurable (its value in IEEE Std 1588 is TRUE (1) or
41        FALSE (2), depending on whether the path trace mechanism is
42        operational or not operational, respectively. However, the
43        pathTrace mechanism is mandatory in this standard, and the
44        value of enable is always TRUE (1)."

```

```

1 ieee8021AsV3PathTraceDSArrayEntry OBJECT-TYPE
2   SYNTAX      Ieee8021AsV3PathTraceDSArrayEntry
3   MAX-ACCESS  not-accessible
4   STATUS      current
5   DESCRIPTION
6       "Path Trace Data Set Table Array for a specific PTP Instance."
7   INDEX { ieee8021AsV3PtpInstance, ieee8021AsV3PathTraceDSArrayIndex }
8   ::= { ieee8021AsV3PathTraceDSArrayTable 1 }
9
10 Ieee8021AsV3PathTraceDSArrayEntry ::=
11   SEQUENCE {
12       ieee8021AsV3PathTraceDSArrayIndex Unsigned32,
13       ieee8021AsV3PathTraceDSArrayList Ieee8021AsV3ClockIdentity
14   }
15
16 ieee8021AsV3PathTraceDSArrayIndex OBJECT-TYPE
17   SYNTAX      Unsigned32 (1..179)
18   MAX-ACCESS  not-accessible
19   STATUS      current
20   DESCRIPTION
21       "Index of the Path Trace Data Set Array."
22   REFERENCE   "10.3.9.23"
23   ::= { ieee8021AsV3PathTraceDSArrayEntry 1 }
24
25 ieee8021AsV3PathTraceDSArrayList OBJECT-TYPE
26   SYNTAX      Ieee8021AsV3ClockIdentity
27   MAX-ACCESS  read-only
28   STATUS      current
29   DESCRIPTION
30       "The value is the array of ClockIdentity values contained
31       in the pathTrace array, which represents the current
32       path trace information, and which is carried in the path
33       trace TLV."
34   REFERENCE   "14.6.2"
35   ::= { ieee8021AsV3PathTraceDSArrayEntry 2 }
36
37
38 -- *****
39 -- The Acceptable TimeTransmitter Table Parameter Data Set represents the
40 -- acceptable timeTransmitter table used when an EPON port is used by a PTP
41 -- Instance of a time-aware system.
42 -- *****
43
44 ieee8021AsV3AcceptableTimeTransmitterTableDSTable OBJECT-TYPE
45   SYNTAX      SEQUENCE OF Ieee8021AsV3AcceptableTimeTransmitterTableDSEntry
46   MAX-ACCESS  not-accessible
47   STATUS      current
48   DESCRIPTION
49       "The acceptableTimeTransmitterTableDS represents the acceptable
50   timeTransmitter
51       table used when an EPON port is used by a PTP Instance of a
52       time-aware system."
53   REFERENCE   "14.7"
54   ::= { ieee8021AsV3MIBObjects 8 }
55
56 ieee8021AsV3AcceptableTimeTransmitterTableDSEntry OBJECT-TYPE
57   SYNTAX      Ieee8021AsV3AcceptableTimeTransmitterTableDSEntry
58   MAX-ACCESS  not-accessible
59   STATUS      current

```

```

1  DESCRIPTION
2  "Acceptable TimeTransmitter Table Data Set represents the acceptable
3 timeTransmitter
4  table used when an EPON port is used by a PTP Instance of a
5  time-aware system."
6  INDEX { ieee8021AsV3PtpInstance }
7  ::= { ieee8021AsV3AcceptableTimeTransmitterTabledSTable 1 }
8
9 Ieee8021AsV3AcceptableTimeTransmitterTabledSEntry ::=
10 SEQUENCE {
11     ieee8021AsV3AcceptableTimeTransmitterTabledSMaxTableSize
12 Unsigned32,
13     ieee8021AsV3AcceptableTimeTransmitterTabledSActualTableSize
14 Unsigned32
15 }
16
17 ieee8021AsV3AcceptableTimeTransmitterTabledSMaxTableSize  OBJECT-TYPE
18 SYNTAX      Unsigned32(0..65535)
19 MAX-ACCESS  read-only
20 STATUS      current
21 DESCRIPTION
22     "The value is the maximum size of the AcceptableTimeTransmitterTable.
23     It is equal to the maxTableSize member of the
24     AcceptableTimeTransmitterTable structure."
25 REFERENCE   "14.7.2"
26 ::= { ieee8021AsV3AcceptableTimeTransmitterTabledSEntry 1 }
27
28 ieee8021AsV3AcceptableTimeTransmitterTabledSActualTableSize  OBJECT-TYPE
29 SYNTAX      Unsigned32(0..65535)
30 MAX-ACCESS  read-write
31 STATUS      current
32 DESCRIPTION
33     "The value is the actual size of the AcceptableTimeTransmitterTable.
34     It is equal to the actualTableSize member of the
35     AcceptableTimeTransmitterTable structure, i.e., the current number
36     of elements in the acceptable timeTransmitter array. The actual
37     table size is less than or equal to the max table size."
38 REFERENCE   "14.7.3"
39 ::= { ieee8021AsV3AcceptableTimeTransmitterTabledSEntry 2 }
40
41 ieee8021AsV3AcceptableTimeTransmitterTabledSArrayTable  OBJECT-TYPE
42 SYNTAX      SEQUENCE OF
43 Ieee8021AsV3AcceptableTimeTransmitterTabledSArrayEntry
44 MAX-ACCESS  not-accessible
45 STATUS      current
46 DESCRIPTION
47     "The acceptableTimeTransmitterTabledS represents the acceptable
48 timeTransmitter table
49     used when an EPON port is used by a PTP Instance of a time-aware
50     system."
51 REFERENCE   "14.7"
52 ::= { ieee8021AsV3MIBObjects 9 }
53
54 ieee8021AsV3AcceptableTimeTransmitterTabledSArrayEntry  OBJECT-TYPE
55 SYNTAX      Ieee8021AsV3AcceptableTimeTransmitterTabledSArrayEntry
56 MAX-ACCESS  not-accessible
57 STATUS      current
58 DESCRIPTION
59     "Each element of this array is an AcceptableTimeTransmitter structure per

```

```

1      PTP Instance."
2      INDEX { ieee8021AsV3PtpInstance,
3 ieee8021AsV3AcceptableTimeTransmitterTableDSArrayIndex }
4      ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSArrayTable 1 }
5
6 Ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry ::=
7 SEQUENCE {
8     ieee8021AsV3AcceptableTimeTransmitterTableDSArrayIndex
9 Unsigned32,
10    ieee8021AsV3AcceptableTimeTransmitterTableDSArrayPortIdentity
11 Ieee8021ASV3PtpPortIdentity,
12    ieee8021AsV3AcceptableTTTableDSArrayAlternatePriority1      Unsigned32
13 }
14
15 ieee8021AsV3AcceptableTimeTransmitterTableDSArrayIndex OBJECT-TYPE
16 SYNTAX      Unsigned32(0..65535)
17 MAX-ACCESS  not-accessible
18 STATUS      current
19 DESCRIPTION
20     "Index of the Acceptable TimeTransmitter Table Data Set Array."
21 REFERENCE   "14.7.4"
22 ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry 1 }
23
24 ieee8021AsV3AcceptableTimeTransmitterTableDSArrayPortIdentity OBJECT-TYPE
25 SYNTAX      Ieee8021ASV3PtpPortIdentity
26 MAX-ACCESS  read-write
27 STATUS      current
28 DESCRIPTION
29     "The acceptablePortIdentity member is the PortIdentity of
30     an acceptable timeTransmitter port."
31 REFERENCE   "14.7.4"
32 ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry 2 }
33
34 ieee8021AsV3AcceptableTTTableDSArrayAlternatePriority1 OBJECT-TYPE
35 SYNTAX      Unsigned32(0..255)
36 MAX-ACCESS  read-write
37 STATUS      current
38 DESCRIPTION
39     "The alternatePriority1 member contains an alternate value
40     for the priority1 attribute of the acceptable timeTransmitter port."
41 REFERENCE   "14.7.4"
42 ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry 3 }
43
44 -- =====
45 -- The Port Parameter Data Set (portDS) represents PTP Port
46 -- time-aware capabilities for a PTP Instance of a time-aware
47 -- system.
48 -- =====
49 ieee8021AsV3PortDSTable OBJECT-TYPE
50 SYNTAX      SEQUENCE OF Ieee8021AsV3PortDSEntry
51 MAX-ACCESS  not-accessible
52 STATUS      current
53 DESCRIPTION
54     "For the single PTP Port of a PTP End Instance and for each
55     PTP Port of a PTP Relay Instance , the portDS is maintained
56     as the basis for making protocol decisions and providing
57     values for message fields.
58     The number of such data sets is the same as the value of
59     defaultDS.numberPorts."

```

```

1  REFERENCE    "14.8"
2  ::= { ieee8021AsV3MIBObjects 10 }
3
4  ieee8021AsV3PortDSEntry  OBJECT-TYPE
5      SYNTAX      Ieee8021AsV3PortDSEntry
6      MAX-ACCESS  not-accessible
7      STATUS      current
8      DESCRIPTION
9          "A list of objects pertaining to a PTP Port of a PTP Instance."
10     INDEX { ieee8021AsV3PtpInstance,
11             ieee8021AsV3PortDSIndex }
12     ::= { ieee8021AsV3PortDSTable 1 }
13
14  Ieee8021AsV3PortDSEntry ::=
15      SEQUENCE {
16          ieee8021AsV3BridgeBasePort      IEEE8021BridgePortNumber,
17          ieee8021AsV3PortDSIndex         InterfaceIndexOrZero,
18          ieee8021AsV3PortDSClockIdentity Ieee8021AsV3ClockIdentity,
19          ieee8021AsV3PortDSPortNumber    Unsigned32,
20          ieee8021AsV3PortDSPortState     INTEGER,
21          ieee8021AsV3PortDSPtpPortEnabled TruthValue,
22          ieee8021AsV3PortDSDelayMechanism INTEGER,
23          ieee8021AsV3PortDSIsMeasuringDelay TruthValue,
24          ieee8021AsV3PortDSAsCapable     TruthValue,
25          ieee8021AsV3PortDSMeanLinkDelay Ieee8021ASV3PtpTimeInterval,
26          ieee8021AsV3PortDSMeanLinkDelayThresh Ieee8021ASV3PtpTimeInterval,
27          ieee8021AsV3PortDSDelayAsym     Ieee8021ASV3PtpTimeInterval,
28          ieee8021AsV3PortDSNbrRateRatio  Integer32,
29          ieee8021AsV3PortDSInitialLogAnnounceInterval Integer32,
30          ieee8021AsV3PortDSCurrentLogAnnounceInterval Integer32,
31          ieee8021AsV3PortDSUseMgtSettableLogAnnounceInterval TruthValue,
32          ieee8021AsV3PortDSMgtSettableLogAnnounceInterval Integer32,
33          ieee8021AsV3PortDSAnnounceReceiptTimeout Unsigned32,
34          ieee8021AsV3PortDSInitialLogSyncInterval Integer32,
35          ieee8021AsV3PortDSCurrentLogSyncInterval Integer32,
36          ieee8021AsV3PortDSUseMgtSettableLogSyncInterval TruthValue,
37          ieee8021AsV3PortDSMgtSettableLogSyncInterval Integer32,
38          ieee8021AsV3PortDSSyncReceiptTimeout Unsigned32,
39          ieee8021AsV3PortDSSyncReceiptTimeoutTimeInterval Ieee8021ASV3UScaledNs,
40          ieee8021AsV3PortDSInitialLogPdelayReqInterval Integer32,
41          ieee8021AsV3PortDSCurrentLogPdelayReqInterval Integer32,
42          ieee8021AsV3PortDSUseMgtSettableLogPdelayReqInterval TruthValue,
43          ieee8021AsV3PortDSMgtSettableLogPdelayReqInterval Integer32,
44          ieee8021AsV3PortDSInitialLogGptpCapableMessageInterval Integer32,
45          ieee8021AsV3PortDSCurrentLogGptpCapableMessageInterval Integer32,
46          ieee8021AsV3PortDSUseMgtSettableLogGptpCapableMessageInterval
47      TruthValue,
48          ieee8021AsV3PortDSMgtSettableLogGptpCapableMessageInterval
49      Integer32,
50          ieee8021AsV3PortDSInitialComputeNbrRateRatio TruthValue,
51          ieee8021AsV3PortDSCurrentComputeNbrRateRatio TruthValue,
52          ieee8021AsV3PortDSUseMgtSettableComputeNbrRateRatio TruthValue,
53          ieee8021AsV3PortDSMgtSettableComputeNbrRateRatio TruthValue,
54          ieee8021AsV3PortDSInitialComputeMeanLinkDelay TruthValue,
55          ieee8021AsV3PortDSCurrentComputeMeanLinkDelay TruthValue,
56          ieee8021AsV3PortDSUseMgtSettableComputeMeanLinkDelay TruthValue,
57          ieee8021AsV3PortDSMgtSettableComputeMeanLinkDelay TruthValue,
58          ieee8021AsV3PortDSAllowedLostRsp Unsigned32,
59          ieee8021AsV3PortDSAllowedFaults Unsigned32,

```

```

1      ieee8021AsV3PortDSGPTpCapableReceiptTimeout      Unsigned32,
2      ieee8021AsV3PortDSVersionNumber                  Unsigned32,
3      ieee8021AsV3PortDSNup                             Float64TC,
4      ieee8021AsV3PortDSNdown                           Float64TC,
5      ieee8021AsV3PortDSOneStepTxOper                   TruthValue,
6      ieee8021AsV3PortDSOneStepReceive                  TruthValue,
7      ieee8021AsV3PortDSOneStepTransmit                 TruthValue,
8      ieee8021AsV3PortDSInitialOneStepTxOper            TruthValue,
9      ieee8021AsV3PortDSCurrentOneStepTxOper            TruthValue,
10     ieee8021AsV3PortDSUseMgtSettableOneStepTxOper      TruthValue,
11     ieee8021AsV3PortDSMgtSettableOneStepTxOper         TruthValue,
12     ieee8021AsV3PortDSSyncLocked                       TruthValue,
13     ieee8021AsV3PortDSPdelayTruncTST1                  Ieee8021ASV3Timestamp,
14     ieee8021AsV3PortDSPdelayTruncTST2                  Ieee8021ASV3Timestamp,
15     ieee8021AsV3PortDSPdelayTruncTST3                  Ieee8021ASV3Timestamp,
16     ieee8021AsV3PortDSPdelayTruncTST4                  Ieee8021ASV3Timestamp,
17     ieee8021AsV3PortDSMinorVersionNumber               Unsigned32
18 }
19
20 ieee8021AsV3BridgeBasePort OBJECT-TYPE
21     SYNTAX      IEEE8021BridgePortNumber
22     MAX-ACCESS  not-accessible
23     STATUS      current
24     DESCRIPTION
25         "This object identifies the bridge port number of the port for
26         which this entry contains bridge management information.
27         For end stations, this port number shall be (1)."
28     ::= { ieee8021AsV3PortDSEntry 1 }
29
30 ieee8021AsV3PortDSIndex OBJECT-TYPE
31     SYNTAX      InterfaceIndexOrZero
32     MAX-ACCESS  not-accessible
33     STATUS      current
34     DESCRIPTION
35         "This object identifies the gPTP interface group within
36         the system for which this entry contains information. It
37         is the value of the instance of the IfIndex object,
38         defined in the IF-MIB, for the gPTP interface group
39         corresponding to this port, or the value 0 if the port
40         has not been bound to an underlying frame source and
41         sink.
42
43         For a given media port of a Bridge or an end station,
44         there can be one or more PTP Port, and depends whether
45         a media port supports point to point link (e.g. IEEE
46         802.3 Ethernet) or point to multi-point (e.g. CSN, IEEE
47         802.3 EPON) links on the media port."
48     ::= { ieee8021AsV3PortDSEntry 2 }
49
50 ieee8021AsV3PortDSClockIdentity OBJECT-TYPE
51     SYNTAX      Ieee8021AsV3ClockIdentity
52     MAX-ACCESS  read-only
53     STATUS      current
54     DESCRIPTION
55         "The value is the first of the portIdentity attribute
56         of the local port, which is a set made of
57         Ieee8021AsV3ClockIdentity and portNumber."
58     REFERENCE   "14.8.2"
59     ::= { ieee8021AsV3PortDSEntry 3 }

```

```

1
2
3 ieee8021AsV3PortDSPortNumber OBJECT-TYPE
4     SYNTAX      Unsigned32(0..65535)
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "The value is the second of the portIdentity attribute
9         of the local port, which is a set made of
10        Ieee8021AsV3ClockIdentity and portNumber."
11     REFERENCE   "14.8.2"
12     ::= { ieee8021AsV3PortDSEntry 4 }
13
14 ieee8021AsV3PortDSPortState OBJECT-TYPE
15     SYNTAX      INTEGER {
16         disabledPort(3),
17         timeTransmitterPort(6),
18         passivePort(7),
19         timeReceiverPort(9)
20     }
21     MAX-ACCESS  read-only
22     STATUS      current
23     DESCRIPTION
24         "The value is the value of the PTP Port state of this
25         PTP Port (see Table 10-2) and is taken from the enumeration
26         in Table 14-7. It is equal to the value of the global
27         variable selectedState."
28     REFERENCE   "14.8.3"
29     ::= { ieee8021AsV3PortDSEntry 5 }
30
31 ieee8021AsV3PortDSPtpPortEnabled OBJECT-TYPE
32     SYNTAX      TruthValue
33     MAX-ACCESS  read-write
34     STATUS      current
35     DESCRIPTION
36         "The value is equal to the value of the Boolean ptpPortEnabled.
37         Setting this managed object causes the Boolean ptpPortEnabled
38         to have the same value."
39     REFERENCE   "14.8.4"
40     ::= { ieee8021AsV3PortDSEntry 6 }
41
42 ieee8021AsV3PortDSDelayMechanism OBJECT-TYPE
43     SYNTAX      INTEGER {
44         p2p(2),
45         commonp2p(3),
46         special(4)
47     }
48     MAX-ACCESS  read-write
49     STATUS      current
50     DESCRIPTION
51         "The value indicates the mechanism for measuring mean
52         propagation delay and neighbor rate ratio on the link
53         attached to this PTP Port, and is taken from the enumeration
54         in Table 14-8. If the domain number is not 0, portDS.delay
55         mechanism must not be P2P."
56     REFERENCE   "14.8.5"
57     ::= { ieee8021AsV3PortDSEntry 7 }
58
59 ieee8021AsV3PortDSIsMeasuringDelay OBJECT-TYPE

```



```

1  SYNTAX      TruthValue
2  MAX-ACCESS  read-only
3  STATUS      current
4  DESCRIPTION
5      "The value is equal to the value of the Boolean isMeasuringDelay."
6  REFERENCE   "14.8.6"
7  ::= { ieee8021AsV3PortDSEntry 8 }
8
9
10 ieee8021AsV3PortDSAsCapable OBJECT-TYPE
11     SYNTAX      TruthValue
12     MAX-ACCESS  read-only
13     STATUS      current
14     DESCRIPTION
15         "The value is equal to the value of the Boolean asCapable."
16     REFERENCE   "14.8.7"
17     ::= { ieee8021AsV3PortDSEntry 9 }
18
19 ieee8021AsV3PortDSMeanLinkDelay OBJECT-TYPE
20     SYNTAX      Ieee8021ASV3PtpTimeInterval
21     MAX-ACCESS  read-only
22     STATUS      current
23     DESCRIPTION
24         "The value is equal to the value of the per-PTP Port global
25         variable meanLinkDelay. It is an estimate of the current
26         one-way propagation time on the link attached to this
27         PTP Port, measured as specified for the respective medium.
28         The value is zero for PTP Port attached to IEEE 802.3
29         EPON links and for the timeTransmitter port of an IEEE 802.11 link,
30         because one-way propagation delay is not measured on the
31         latter and not directly measured on the former.
32         NOTE: The underlying per-PTP Port, global variable
33         meanLinkDelay is of type UScaledNS, which is a 96-Bit value.
34         meanLinkDelay values that are larger than the maximum value
35         that can be represented by the TimeInterval data type, i.e.,
36         0xFFFF FFFF FFFF FFFF (where the units are 2sup -16 ns), used
37         for this managed object are set to this largest value."
38     REFERENCE   "14.8.8"
39     ::= { ieee8021AsV3PortDSEntry 10 }
40
41
42 ieee8021AsV3PortDSMeanLinkDelayThresh OBJECT-TYPE
43     SYNTAX      Ieee8021ASV3PtpTimeInterval
44     MAX-ACCESS  read-write
45     STATUS      current
46     DESCRIPTION
47         "The value is equal to the value of the per-PTP Port global
48         variable meanLinkDelayThresh. It is the propagation time
49         threshold above which a PTP Port is considered not capable of
50         participating in the IEEE 802.1AS protocol.
51         Setting this managed object causes the per PTP Port global
52         variable meanLinkDelayThresh to have the same value.
53         NOTE: The underlying per-PTP Port, global variable
54         meanLinkDelayThresh is of type UScaledNS, which is a 96-Bit
55         value. meanLinkDelayThresh values that are larger than the
56         maximum value that can be represented by the TimeInterval
57         data type, i.e., 0xFFFF FFFF FFFF FFFF (where the units are
58         2sup -16 ns), used for this managed object are set to this
59         largest value."

```

```

1  REFERENCE    "14.8.9"
2  ::= { ieee8021AsV3PortDSEntry 11 }
3
4  ieee8021AsV3PortDSDelayAsym OBJECT-TYPE
5      SYNTAX      Ieee8021ASV3PtpTimeInterval
6      MAX-ACCESS  read-write
7      STATUS      current
8      DESCRIPTION
9          "The value is the asymmetry in the propagation delay on
10         the link attached to this PTP Port relative to the
11         Grandmaster Clock time base, as defined in 10.2.5.9 and
12         8.3. If propagation delay asymmetry is not modeled, then
13         delayAsymmetry is 0.
14         NOTE: The underlying per-port global variable delayAsymmetry
15         is of type ScaledNS, which is a 96-Bit value.
16         delayAsymmetry values that are larger than the maximum value
17         that can be represented by the TimeInterval data type, i.e.,
18         0x7FFF FFFF FFFF FFFF, (where the units are 2sup -16 ns),
19         used for this managed object are set to this largest value.
20         delayAsymmetry values that are less than the minimum value
21         that can be represented by the TimeInterval data type, i.e.,
22         0x8000 0000 0000 0001 written in twos complement form (where
23         the units are 2sup -16 ns), used for this managed object are
24         set to this smallest value."
25  REFERENCE    "14.8.10 and 8.3"
26  ::= { ieee8021AsV3PortDSEntry 12 }
27
28  ieee8021AsV3PortDSNbrRateRatio OBJECT-TYPE
29      SYNTAX      Integer32
30      MAX-ACCESS  read-only
31      STATUS      current
32      DESCRIPTION
33          "The value is an estimate of the ratio of the frequency of
34         the LocalClock entity of the PTP Instance at the other end
35         of the link attached to this PTP Port, to the frequency of
36         the LocalClock entity of this PTP Instance. neighborRateRatio
37         is expressed as the fractional frequency offset multiplied
38         by 2sup41, i.e., the quantity (neighborRateRatio -1.0)(2sup41)."
39  REFERENCE    "14.8.11"
40  ::= { ieee8021AsV3PortDSEntry 13 }
41
42
43  ieee8021AsV3PortDSInitialLogAnnounceInterval OBJECT-TYPE
44      SYNTAX      Integer32(-128..127)
45      MAX-ACCESS  read-write
46      STATUS      current
47      DESCRIPTION
48          "If useMgtSettableLogAnnounceInterval is FALSE (2), the
49         value is the logarithm to base 2 of the announce interval
50         used when (a) the PTP Port is initialized, or (b) a message
51         interval request TLV is received with the logAnnounceInterval
52         field set to 126."
53  REFERENCE    "14.8.12"
54  DEFVAL { 0 }
55  ::= { ieee8021AsV3PortDSEntry 14 }
56
57  ieee8021AsV3PortDSCurrentLogAnnounceInterval OBJECT-TYPE
58      SYNTAX      Integer32(-128..127)
59      MAX-ACCESS  read-only

```

```

1  STATUS      current
2  DESCRIPTION
3      "The value is the logarithm to the base 2 of the
4      current announce interval."
5  REFERENCE   "14.8.13"
6  ::= { ieee8021AsV3PortDSEntry 15 }
7
8  ieee8021AsV3PortDSUseMgtSettableLogAnnounceInterval OBJECT-TYPE
9      SYNTAX      TruthValue
10     MAX-ACCESS   read-write
11     STATUS      current
12     DESCRIPTION
13         "The managed object is a Boolean that determines the
14         source of the announce interval. If the value is TRUE (1),
15         the value of currentLogAnnounceInterval is set equal to the
16         value of mgtSettableLogAnnounceInterval. If the value is
17         FALSE (2), the value of currentLogAnnounceInterval is
18         determined by the AnnounceIntervalSetting state machine. The
19         default value of useMgtSettableLogAnnounceInterval is
20         FALSE (2) for domain 0 and TRUE (1) for domains other than
21         domain 0."
22     REFERENCE   "14.8.14"
23     ::= { ieee8021AsV3PortDSEntry 16 }
24
25  ieee8021AsV3PortDSMgtSettableLogAnnounceInterval OBJECT-TYPE
26      SYNTAX      Integer32(-128..127)
27      MAX-ACCESS   read-write
28      STATUS      current
29      DESCRIPTION
30          "The value is the logarithm to base 2 of the announce interval
31          used if useMgtSettableLogAnnounceInterval is TRUE (1).
32          The value is not used if useMgtSettableLogAnnounceInterval is
33          FALSE (2)."

```

```

1      interval request TLV is received with the logTimeSyncInterval
2      field set to 126."
3      REFERENCE    "14.8.17"
4      ::= { ieee8021AsV3PortDSEntry 19 }
5
6 ieee8021AsV3PortDSCurrentLogSyncInterval OBJECT-TYPE
7     SYNTAX      Integer32(-128..127)
8     MAX-ACCESS  read-only
9     STATUS      current
10    DESCRIPTION
11        "The value is the logarithm to the base 2 of the current
12        time-synchronization transmission interval."
13    REFERENCE    "14.8.18"
14    ::= { ieee8021AsV3PortDSEntry 20 }
15
16 ieee8021AsV3PortDSUseMgtSettableLogSyncInterval OBJECT-TYPE
17     SYNTAX      TruthValue
18     MAX-ACCESS  read-write
19     STATUS      current
20     DESCRIPTION
21        "The managed object is a Boolean that determines the source
22        of the sync interval. If the value is TRUE (1), the value
23        of currentLogSyncInterval is set equal to the value of
24        mgtSettableLogSyncInterval. If the value of the managed
25        object is FALSE (2), the value of currentLogSyncInterval is
26        determined by the SyncIntervalSetting state machine. The
27        default value of useMgtSettableLogSyncInterval is FALSE (2)
28        for domain 0 and TRUE (1) for domains other than domain 0."
29    REFERENCE    "14.8.19"
30    ::= { ieee8021AsV3PortDSEntry 21 }
31
32 ieee8021AsV3PortDSMgtSettableLogSyncInterval OBJECT-TYPE
33     SYNTAX      Integer32(-128..127)
34     MAX-ACCESS  read-write
35     STATUS      current
36     DESCRIPTION
37        "The value is the logarithm to base 2 of the sync interval
38        if useMgtSettableLogSyncInterval is TRUE (1). The value is
39        not used if useMgtSettableLogSyncInterval is FALSE (2)."

```

```

1  UNITS      "2**-16 ns"
2  MAX-ACCESS read-only
3  STATUS     current
4  DESCRIPTION
5      "The value is equal to the value of the per-PTP Port global
6      variable syncReceiptTimeoutTimeInterval. It is the time
7      interval after which sync receipt timeout occurs if
8      time-synchronization information has not been received during
9      the interval."
10 REFERENCE  "14.8.22"
11 ::= { ieee8021AsV3PortDSEntry 24 }
12
13 ieee8021AsV3PortDSInitialLogPdelayReqInterval OBJECT-TYPE
14     SYNTAX      Integer32(-128..127)
15     MAX-ACCESS  read-write
16     STATUS      current
17     DESCRIPTION
18         "For full-duplex IEEE 802.3 media and for CSN media that
19         use the peer-to-peer delay mechanism to measure path delay,
20         the value is the logarithm to base 2 of the Pdelay_Req
21         message transmission interval used when (a) the PTP Port is
22         initialized, or (b) a message interval request TLV is
23         received with the logLinkDelayInterval field set to 126.
24         For all other media, the value is 127."
25     REFERENCE  "14.8.23"
26     ::= { ieee8021AsV3PortDSEntry 25 }
27
28 ieee8021AsV3PortDSCurrentLogPdelayReqInterval OBJECT-TYPE
29     SYNTAX      Integer32(-128..127)
30     MAX-ACCESS  read-only
31     STATUS      current
32     DESCRIPTION
33         "For full-duplex IEEE 802.3 media and for CSN media that
34         use the peer-to-peer delay mechanism to measure path delay,
35         the value is the logarithm to the base 2 of the current
36         Pdelay_Req message transmission interval.
37         For all other media, the value is 127."
38     REFERENCE  "14.8.24"
39     ::= { ieee8021AsV3PortDSEntry 26 }
40
41 ieee8021AsV3PortDSUseMgtSettableLogPdelayReqInterval OBJECT-TYPE
42     SYNTAX      TruthValue
43     MAX-ACCESS  read-write
44     STATUS      current
45     DESCRIPTION
46         "The managed object is a Boolean that determines the source
47         of the mean time interval between successive Pdelay_Req
48         messages. If the value is TRUE (1), the value of
49         currentLogPdelayReqInterval is set equal to the value of
50         mgtSettableLogPdelayReqInterval. If the value of the managed
51         object is FALSE (2), the value of currentLogPdelayReqInterval
52         is determined by the LinkDelayIntervalSetting state machine.
53         The default value of useMgtSettableLogPdelayReqInterval is
54         FALSE (2)."
55     REFERENCE  "14.8.25"
56     DEFVAL { false }
57     ::= { ieee8021AsV3PortDSEntry 27 }
58
59 ieee8021AsV3PortDSMgtSettableLogPdelayReqInterval OBJECT-TYPE

```

```

1  SYNTAX      Integer32 (-128..127)
2  MAX-ACCESS  read-write
3  STATUS      current
4  DESCRIPTION
5      "The value is the logarithm to base 2 of the mean time
6      interval between successive Pdelay_Req messages if
7      useMgtSettableLogPdelayReqInterval is TRUE (1). The
8      value is not used if useMgtSettableLogPdelayReqInterval
9      is FALSE (2)."
```

10 REFERENCE "14.8.26"

11 ::= { ieee8021AsV3PortDSEntry 28 }

12

13 ieee8021AsV3PortDSInitialLogGtpCapableMessageInterval OBJECT-TYPE

```

14  SYNTAX      Integer32 (-128..127)
15  MAX-ACCESS  read-write
16  STATUS      current
17  DESCRIPTION
18      "The value is the logarithm to base 2 of the gTP capable
19      message interval used when (a) the PTP Port is initialized,
20      or (b) a gPtpCapableMessage interval request TLV is received
21      with the logGtpCapableMessageInterval field set to 126."
22  REFERENCE  "14.8.27"
23  ::= { ieee8021AsV3PortDSEntry 29 }
```

24

25 ieee8021AsV3PortDSCurrentLogGtpCapableMessageInterval OBJECT-TYPE

```

26  SYNTAX      Integer32 (-128..127)
27  MAX-ACCESS  read-only
28  STATUS      current
29  DESCRIPTION
30      "The value is the logarithm to the base 2 of the current
31      gTP capable message interval."
32  REFERENCE  "14.8.28"
33  ::= { ieee8021AsV3PortDSEntry 30 }
```

34

35 ieee8021AsV3PortDSUseMgtSettableLogGtpCapableMessageInterval OBJECT-TYPE

```

36  SYNTAX      TruthValue
37  MAX-ACCESS  read-write
38  STATUS      current
39  DESCRIPTION
40      "The managed object is a Boolean that determines the source
41      of the gTP capable message interval. If the value is
42      TRUE (1), the value of currentLogGtpCapableMessageInterval
43      is set equal to the value of
44      mgtSettableLogGtpCapableMessageInterval. If the value of
45      the managed object is FALSE (2), the value of
46      currentLogGtpCapableMessageInterval is determined by the
47      GtpCapableMessageIntervalSetting state machine.
48      The default value of
49      useMgtSettableLogGtpCapableMessageInterval is FALSE (2)."
```

50 REFERENCE "14.8.29"

51 DEFVAL { false }

52 ::= { ieee8021AsV3PortDSEntry 31 }

53

54 ieee8021AsV3PortDSMgtSettableLogGtpCapableMessageInterval OBJECT-TYPE

```

55  SYNTAX      Integer32 (-128..127)
56  MAX-ACCESS  read-write
57  STATUS      current
58  DESCRIPTION
59      "The value is the logarithm to base 2 of the
```

```

1      gPtpCapableMessageInterval if
2      useMgtSettableLogGptpCapableMessageInterval is TRUE (1).
3      The value is not used if
4      useMgtSettableLogGptpCapableMessageInterval is FALSE (2).
5  REFERENCE    "14.8.30"
6  ::= { ieee8021AsV3PortDSEntry 32 }
7
8  ieee8021AsV3PortDSInitialComputeNbrRateRatio OBJECT-TYPE
9      SYNTAX      TruthValue
10     MAX-ACCESS  read-write
11     STATUS      current
12     DESCRIPTION
13         "If useMgtSettableComputeNeighborRateRatio is FALSE (2),
14         then for full-duplex IEEE 802.3 media and for CSN media that
15         use the peer-to-peer delay mechanism to measure path delay,
16         the value is the initial value of computeNeighborRateRatio."
17     REFERENCE    "14.8.31"
18     ::= { ieee8021AsV3PortDSEntry 33 }
19
20  ieee8021AsV3PortDSCurrentComputeNbrRateRatio OBJECT-TYPE
21     SYNTAX      TruthValue
22     MAX-ACCESS  read-only
23     STATUS      current
24     DESCRIPTION
25         "For full-duplex IEEE 802.3 media and for CSN media that
26         use the peer-to-peer delay mechanism to measure path delay,
27         the value is the current value of computeNeighborRateRatio."
28     REFERENCE    "14.8.32"
29     ::= { ieee8021AsV3PortDSEntry 34 }
30
31  ieee8021AsV3PortDSUseMgtSettableComputeNbrRateRatio OBJECT-TYPE
32     SYNTAX      TruthValue
33     MAX-ACCESS  read-write
34     STATUS      current
35     DESCRIPTION
36         "The managed object is a Boolean that determines the source
37         of the value of computeNeighborRateRatio. If the value is
38         TRUE (1), the value of computeNeighborRateRatio is set equal
39         to the value of mgtSettableComputeNeighborRateRatio. If the
40         value of the managed object is FALSE (2), the value of
41         currentComputeNeighborRateRatio is determined by the
42         LinkDelayIntervalSetting state machine.
43         The default value of useMgtSettableComputeNbrRateRatio is
44         FALSE (2)."
45     REFERENCE    "14.8.33"
46     DEFVAL { false }
47     ::= { ieee8021AsV3PortDSEntry 35 }
48
49  ieee8021AsV3PortDSMgtSettableComputeNbrRateRatio OBJECT-TYPE
50     SYNTAX      TruthValue
51     MAX-ACCESS  read-write
52     STATUS      current
53     DESCRIPTION
54         "ComputeNeighborRateRatio is configured to this value if
55         useMgtSettableComputeNeighborRateRatio is TRUE (1). The
56         value is not used if useMgtSettableComputeNeighborRateRatio
57         is FALSE (2)."
58     REFERENCE    "14.8.34"
59     ::= { ieee8021AsV3PortDSEntry 36 }

```

```

1
2 ieee8021AsV3PortDSInitialComputeMeanLinkDelay OBJECT-TYPE
3     SYNTAX      TruthValue
4     MAX-ACCESS  read-write
5     STATUS      current
6     DESCRIPTION
7         "If useMgtSettableComputeMeanLinkDelay is FALSE (2) then,
8         for full-duplex IEEE 802.3 media and for CSN media that use
9         the peer-to-peer delay mechanism to measure path delay,
10        the value is the initial value of computeMeanLinkDelay."
11     REFERENCE   "14.8.35"
12     ::= { ieee8021AsV3PortDSEntry 37 }
13
14 ieee8021AsV3PortDSCurrentComputeMeanLinkDelay OBJECT-TYPE
15     SYNTAX      TruthValue
16     MAX-ACCESS  read-only
17     STATUS      current
18     DESCRIPTION
19         "For full-duplex IEEE 802.3 media and for CSN media that
20         use the peer-to-peer delay mechanism to measure path delay,
21         the value is the current value of computeMeanLinkDelay."
22     REFERENCE   "14.8.36"
23     ::= { ieee8021AsV3PortDSEntry 38 }
24
25 ieee8021AsV3PortDSUseMgtSettableComputeMeanLinkDelay OBJECT-TYPE
26     SYNTAX      TruthValue
27     MAX-ACCESS  read-write
28     STATUS      current
29     DESCRIPTION
30         "The managed object is a Boolean that determines the source
31         of the value of computeMeanLinkDelay. If the value is
32         TRUE (1), the value of computeMeanLinkDelay is set equal to
33         the value of mgtSettableComputeMeanLinkDelay. If the value
34         of the managed object is FALSE (2), the value of
35         currentComputeMeanLinkDelay is determined by the
36         LinkDelayIntervalSetting state machine.
37         The default value of useMgtSettableComputeMeanLinkDelay
38         is FALSE (2)."
39     REFERENCE   "14.8.37"
40     DEFVAL { false }
41     ::= { ieee8021AsV3PortDSEntry 39 }
42
43 ieee8021AsV3PortDSMgtSettableComputeMeanLinkDelay OBJECT-TYPE
44     SYNTAX      TruthValue
45     MAX-ACCESS  read-write
46     STATUS      current
47     DESCRIPTION
48         "ComputeMeanLinkDelay is configured to this value if
49         useMgtSettableComputeMeanLinkDelay is TRUE (1). The
50         value is not used if useMgtSettableComputeMeanLinkDelay
51         is FALSE (2)."
52     REFERENCE   "14.8.38"
53     ::= { ieee8021AsV3PortDSEntry 40 }
54
55 ieee8021AsV3PortDSAllowedLostRsp OBJECT-TYPE
56     SYNTAX      Unsigned32(1..255)
57     MAX-ACCESS  read-write
58     STATUS      current
59     DESCRIPTION

```



```

1      "The value is equal to the value of the per-PTP Port global
2      variable allowedLostResponses. It is the number of Pdelay_Req
3      messages without valid responses above which a PTP Port
4      is considered to be not exchanging peer delay messages with
5      its neighbor.
6      Setting this managed object causes the per-PTP Port global
7      variable allowedLostResponses to have the same value."
8  REFERENCE   "14.8.39 and 11.5.3"
9  DEFVAL { 9 }
10 ::= { ieee8021AsV3PortDSEntry 41 }
11
12 ieee8021AsV3PortDSAllowedFaults OBJECT-TYPE
13     SYNTAX      Unsigned32(1..255)
14     MAX-ACCESS  read-write
15     STATUS      current
16     DESCRIPTION
17         "The value is equal to the value of the per-PTP-Port global
18         variable allowedFaults. It is the number of faults above
19         which asCapable is set to FALSE (1), i.e., a PTP Port is
20         considered not capable of interoperating with its
21         neighbor via the IEEE 802.1AS protocol.
22         Setting this managed object causes the per-PTP Port global
23         variable allowedFaults to have the same value."
24     REFERENCE   "14.8.40"
25     DEFVAL { 9 }
26     ::= { ieee8021AsV3PortDSEntry 42 }
27
28 ieee8021AsV3PortDSGtpCapableReceiptTimeout OBJECT-TYPE
29     SYNTAX      Unsigned32(1..255)
30     MAX-ACCESS  read-write
31     STATUS      current
32     DESCRIPTION
33         "The value is the number of transmission intervals that a
34         PTP Port waits without receiving the gPTP capable TLV, before
35         assuming that the neighbor PTP Port is no longer invoking
36         the gPTP protocol."
37     REFERENCE   "14.8.41"
38     DEFVAL { 9 }
39     ::= { ieee8021AsV3PortDSEntry 43 }
40
41 ieee8021AsV3PortDSVersionNumber OBJECT-TYPE
42     SYNTAX      Unsigned32(0..16)
43     MAX-ACCESS  read-only
44     STATUS      current
45     DESCRIPTION
46         "This value is set to versionPTP as specified in 10.6.2.2.4."
47     REFERENCE   "14.8.42"
48     ::= { ieee8021AsV3PortDSEntry 44 }
49
50 ieee8021AsV3PortDSNup OBJECT-TYPE
51     SYNTAX      Float64TC
52     MAX-ACCESS  read-write
53     STATUS      current
54     DESCRIPTION
55         "For an OLT port of an IEEE 802.3 EPON link, the value is
56         the effective index of refraction for the EPON upstream
57         wavelength light of the optical path. The default value is
58         1.46770 for 1 Gb/s upstream links, and 1.46773 for
59         10 Gb/s upstream links."

```

```

1      For all other PTP Ports, the value is 0."
2  REFERENCE  "14.8.43"
3  ::= { ieee8021AsV3PortDSEntry 45 }
4
5  ieee8021AsV3PortDSNdown OBJECT-TYPE
6      SYNTAX      Float64TC
7      MAX-ACCESS  read-write
8      STATUS      current
9      DESCRIPTION
10         "For an OLT port of an IEEE 802.3 EPON link, the value is
11         the effective index of refraction for the EPON downstream
12         wavelength light of the optical path. The default value is
13         1.46805 for 1 Gb/s downstream links, and 1.46851 for
14         10 Gb/s downstream links.
15         For all other PTP Ports, the value is 0."
16  REFERENCE  "14.8.44"
17  ::= { ieee8021AsV3PortDSEntry 46 }
18
19  ieee8021AsV3PortDSOneStepTxOper OBJECT-TYPE
20      SYNTAX      TruthValue
21      MAX-ACCESS  read-only
22      STATUS      current
23      DESCRIPTION
24         "The value is equal to the value of the per-PTP Port global
25         variable oneStepTxOper. Its value is TRUE (1) if the
26         PTP Port is sending one-step Sync messages, and FALSE (2)
27         if the PTP Port is sending two-step Sync and Follow-Up
28         messages."
29  REFERENCE  "14.8.45"
30  ::= { ieee8021AsV3PortDSEntry 47 }
31
32  ieee8021AsV3PortDSOneStepReceive OBJECT-TYPE
33      SYNTAX      TruthValue
34      MAX-ACCESS  read-only
35      STATUS      current
36      DESCRIPTION
37         "The value is equal to the value of the per-PTP Port global
38         variable oneStepReceive. Its value is TRUE (1) if the
39         PTP Port is capable of receiving and processing one-step
40         Sync messages."
41  REFERENCE  "14.8.46"
42  ::= { ieee8021AsV3PortDSEntry 48 }
43
44  ieee8021AsV3PortDSOneStepTransmit OBJECT-TYPE
45      SYNTAX      TruthValue
46      MAX-ACCESS  read-only
47      STATUS      current
48      DESCRIPTION
49         "The value is equal to the value of the per-PTP Port global
50         variable oneStepTransmit. Its value is TRUE (1) if the
51         PTP Port is capable of transmitting one-step Sync messages."
52  REFERENCE  "14.8.47"
53  ::= { ieee8021AsV3PortDSEntry 49 }
54
55  ieee8021AsV3PortDSInitialOneStepTxOper OBJECT-TYPE
56      SYNTAX      TruthValue
57      MAX-ACCESS  read-write
58      STATUS      current
59      DESCRIPTION

```

```

1      "If useMgtSettableOneStepTxOper is FALSE (2), the value is
2      used to initialize currentOneStepTxOper when the PTP Port is
3      initialized. If useMgtSettableOneStepTxOper is TRUE (1),
4      the value of initialOneStepTxOper is not used."
5  REFERENCE    "14.8.48"
6  DEFVAL { false }
7  ::= { ieee8021AsV3PortDSEntry 50 }
8
9  ieee8021AsV3PortDSCurrentOneStepTxOper OBJECT-TYPE
10 SYNTAX      TruthValue
11 MAX-ACCESS  read-only
12 STATUS      current
13 DESCRIPTION
14     "The value is TRUE (1) if it is desired, either via
15     management or via a received Signaling message, that the
16     PTP Port transmit one-step Sync messages. The value is
17     FALSE (2) if it is not desired, either via management or via
18     a received Signaling message, that the PTP Port transmit
19     one-step Sync messages.
20     NOTE: The PTP Port will send one-step Sync messages only if
21     currentOneStepTxOper and oneStepTransmit are both TRUE (1)."
```

REFERENCE "14.8.49"

```

22 ::= { ieee8021AsV3PortDSEntry 51 }
23
24
25 ieee8021AsV3PortDSUseMgtSettableOneStepTxOper OBJECT-TYPE
26 SYNTAX      TruthValue
27 MAX-ACCESS  read-write
28 STATUS      current
29 DESCRIPTION
30     "The managed object is a Boolean that determines the source
31     of currentOneStepTxOper. If the value is TRUE (1), the
32     value of currentOneStepTxOper is set equal to the value of
33     mgtSettableOneStepTxOper. If the value is FALSE (2), the
34     value of currentOneStepTxOper is determined by the
35     OneStepTxOperSetting state machine.
36     The default value of useMgtSettableOneStepTxOper is TRUE (1)."
```

REFERENCE "14.8.50"

```

37 DEFVAL      { true }
38 ::= { ieee8021AsV3PortDSEntry 52 }
39
40
41 ieee8021AsV3PortDSMgtSettableOneStepTxOper OBJECT-TYPE
42 SYNTAX      TruthValue
43 MAX-ACCESS  read-write
44 STATUS      current
45 DESCRIPTION
46     "If useMgtSettableOneStepTxOper is TRUE (1),
47     currentOneStepTxOper is set equal to the value of
48     mgtSettableOneStepTxOper. The value of mgtSettableOneStepTxOper
49     is not used if useMgtSettableOneStepTxOper is FALSE (2).
50     The default value of mgtSettableOneStepTxOper is FALSE (2)
51     for domains other than domain 0."
```

REFERENCE "14.8.51"

```

52 ::= { ieee8021AsV3PortDSEntry 53 }
53
54
55 ieee8021AsV3PortDSSyncLocked OBJECT-TYPE
56 SYNTAX      TruthValue
57 MAX-ACCESS  read-only
58 STATUS      current
59 DESCRIPTION
```

1 "The value is equal to the value of the per-PTP Port global
2 variable syncLocked. Its value is TRUE (1) if the PTP Port
3 will transmit a Sync as soon as possible after the timeReceiver
4 PTP Port receives a Sync."
5 REFERENCE "14.8.52"
6 ::= { ieee8021AsV3PortDSEntry 54 }
7
8 ieee8021AsV3PortDSPdelayTruncTST1 OBJECT-TYPE
9 SYNTAX Ieee8021ASV3Timestamp
10 MAX-ACCESS read-only
11 STATUS current
12 DESCRIPTION
13 "For full-duplex IEEE 802.3 media and for CSN media that use
14 the peer-to-peer delay mechanism to measure path delay, the
15 first value, T1, of the four elements of this array is as
16 described in Table 14-9. For all other media, the values are
17 zero. This object corresponds to the timestamp t1 in
18 Figure 11-1, and expressed in units of 2^{-16} ns (i.e., the
19 value of this array element is equal to the remainder obtained
20 upon dividing the respective timestamp , expressed in units
21 of 2^{-16} ns, by 2^{48}).
22 At any given time, the timestamp values stored in the T1, T2,
23 T3, T4 PdelayTruncTS are for the same, and most recently
24 completed, peer delay message exchange."
25 REFERENCE "14.8.53"
26 ::= { ieee8021AsV3PortDSEntry 55 }
27
28 ieee8021AsV3PortDSPdelayTruncTST2 OBJECT-TYPE
29 SYNTAX Ieee8021ASV3Timestamp
30 MAX-ACCESS read-only
31 STATUS current
32 DESCRIPTION
33 "For full-duplex IEEE 802.3 media and for CSN media that use
34 the peer-to-peer delay mechanism to measure path delay, the
35 second value, T2, of the four elements of this array is as
36 described in Table 14-9. For all other media, the values are
37 zero. This object corresponds to the timestamp t2 in
38 Figure 11-1, and expressed in units of 2^{-16} ns (i.e., the
39 value of this array element is equal to the remainder obtained
40 upon dividing the respective timestamp , expressed in units
41 of 2^{-16} ns, by 2^{48}).
42 At any given time, the timestamp values stored in the T1, T2,
43 T3, T4 PdelayTruncTS are for the same, and most recently
44 completed, peer delay message exchange."
45 REFERENCE "14.8.53"
46 ::= { ieee8021AsV3PortDSEntry 56 }
47
48 ieee8021AsV3PortDSPdelayTruncTST3 OBJECT-TYPE
49 SYNTAX Ieee8021ASV3Timestamp
50 MAX-ACCESS read-only
51 STATUS current
52 DESCRIPTION
53 "For full-duplex IEEE 802.3 media and for CSN media that use
54 the peer-to-peer delay mechanism to measure path delay, the
55 third value, T3, of the four elements of this array is as
56 described in Table 14-9. For all other media, the values are
57 zero. This object corresponds to the timestamp t3 in
58 Figure 11-1, and expressed in units of 2^{-16} ns (i.e., the
59 value of this array element is equal to the remainder obtained

```

1      upon dividing the respective timestamp , expressed in units
2      of 2-16 ns, by 248).
3      At any given time, the timestamp values stored in the T1, T2,
4      T3, T4 PdelayTruncTS are for the same, and most recently
5      completed, peer delay message exchange."
6  REFERENCE    "14.8.53"
7  ::= { ieee8021AsV3PortDSEntry 57 }
8
9  ieee8021AsV3PortDSPdelayTruncTST4 OBJECT-TYPE
10 SYNTAX      Ieee8021ASV3Timestamp
11 MAX-ACCESS  read-only
12 STATUS      current
13 DESCRIPTION
14     "For full-duplex IEEE 802.3 media and for CSN media that use
15     the peer-to-peer delay mechanism to measure path delay, the
16     fourth value, T4, of the four elements of this array is as
17     described in Table 14-9. For all other media, the values are
18     zero. This object corresponds to the timestamp t4 in
19     Figure 11-1, and expressed in units of 2-16 ns (i.e., the
20     value of this array element is equal to the remainder obtained
21     upon dividing the respective timestamp , expressed in units
22     of 2-16 ns, by 248).
23     At any given time, the timestamp values stored in the T1, T2,
24     T3, T4 PdelayTruncTS are for the same, and most recently
25     completed, peer delay message exchange."
26 REFERENCE    "14.8.53"
27 ::= { ieee8021AsV3PortDSEntry 58 }
28
29 ieee8021AsV3PortDSMinorVersionNumber OBJECT-TYPE
30 SYNTAX      Unsigned32 (0..15)
31 MAX-ACCESS  read-only
32 STATUS      current
33 DESCRIPTION
34     "This value is set to minorVersionPTP as specified in 10.6.2.2.3."
35 REFERENCE    "14.8.54"
36 ::= { ieee8021AsV3PortDSEntry 59 }
37
38 -- =====
39 -- The Description Port Parameter Data Set contains the
40 -- profileIdentifier for this PTP profile, as specified in F.2
41 -- =====
42
43 ieee8021AsV3DescriptionPortDSTable OBJECT-TYPE
44 SYNTAX      SEQUENCE OF Ieee8021AsV3DescriptionPortDSEntry
45 MAX-ACCESS  not-accessible
46 STATUS      current
47 DESCRIPTION
48     "The descriptionPortDS contains the profileIdentifier for
49     this PTP profile, as specified in F.2."
50 REFERENCE    "14.9"
51 ::= { ieee8021AsV3MIBObjects 11 }
52
53 ieee8021AsV3DescriptionPortDSEntry OBJECT-TYPE
54 SYNTAX      Ieee8021AsV3DescriptionPortDSEntry
55 MAX-ACCESS  not-accessible
56 STATUS      current
57 DESCRIPTION
58     "The descriptionPortDS contains the profileIdentifier for
59     this PTP profile"

```

```

1  INDEX { ieee8021AsV3PtpInstance,
2          ieee8021AsV3DescriptionPortDSAsIndex }
3  ::= { ieee8021AsV3DescriptionPortDSTable 1 }
4
5  Ieee8021AsV3DescriptionPortDSEntry ::=
6      SEQUENCE {
7          ieee8021AsV3DescriptionPortDSAsIndex
8              InterfaceIndexOrZero,
9          ieee8021AsV3DescriptionPortDSProfileIdentifier
10             Ieee8021AsV3GtpProfileIdentifier }
11
12 ieee8021AsV3DescriptionPortDSAsIndex OBJECT-TYPE
13     SYNTAX      InterfaceIndexOrZero
14     MAX-ACCESS  not-accessible
15     STATUS      current
16     DESCRIPTION
17         "This object identifies the gPTP interface group within
18         the system for which this entry contains information. It
19         is the value of the instance of the IfIndex object,
20         defined in the IF-MIB, for the gPTP interface group
21         corresponding to this port, or the value 0 if the port
22         has not been bound to an underlying frame source and
23         sink.
24
25         For a given media port of a Bridge or an end station,
26         there can be one or more PTP Port, and depends whether
27         a media port supports point to point link (e.g. IEEE
28         802.3 Ethernet) or point to multi-point (e.g. CSN, IEEE
29         802.3 EPON) links on the media port."
30     REFERENCE   "IEEE Std 802.1AS Description Port Parameter DS Group
31                 PTP Port Index"
32     ::= { ieee8021AsV3DescriptionPortDSEntry 1 }
33
34 ieee8021AsV3DescriptionPortDSProfileIdentifier OBJECT-TYPE
35     SYNTAX      Ieee8021AsV3GtpProfileIdentifier
36     MAX-ACCESS  read-only
37     STATUS      current
38     DESCRIPTION
39         "The value is the profileIdentifier for this PTP profile."
40     REFERENCE   "14.9.2 and F.2"
41     ::= { ieee8021AsV3DescriptionPortDSEntry 2 }
42
43 -- =====
44 -- The Port Parameter Statistics Data Set provides counters
45 -- associated with PTP Port capabilities at a given PTP Instance.
46 -- =====
47
48 ieee8021AsV3PortStatDSTable OBJECT-TYPE
49     SYNTAX      SEQUENCE OF Ieee8021AsV3PortStatDSEntry
50     MAX-ACCESS  not-accessible
51     STATUS      current
52     DESCRIPTION
53         "The portStatisticsDS provides counters associated with PTP Port
54         capabilities at a given PTP Instance."
55     REFERENCE   "14.10"
56     ::= { ieee8021AsV3MIBObjects 12 }
57
58 ieee8021AsV3PortStatDSEntry OBJECT-TYPE
59     SYNTAX      Ieee8021AsV3PortStatDSEntry

```

```

1  MAX-ACCESS  not-accessible
2  STATUS      current
3  DESCRIPTION
4      "Port Statistics Data Set provides counters associated with
5      PTP Port capabilities at a given PTP Instance."
6  INDEX { ieee8021AsV3PtpInstance,
7          ieee8021AsV3PortDSIndex }
8  ::= { ieee8021AsV3PortStatDSTable 1 }
9
10 ieee8021AsV3PortStatDSEntry ::=
11     SEQUENCE {
12         ieee8021AsV3PortStatRxSyncCount          Counter32,
13         ieee8021AsV3PortStatRxOneStepSyncCount    Counter32,
14         ieee8021AsV3PortStatRxFollowUpCount        Counter32,
15         ieee8021AsV3PortStatRxPdelayRequestCount   Counter32,
16         ieee8021AsV3PortStatRxPdelayRspCount       Counter32,
17         ieee8021AsV3PortStatRxPdelayRspFollowUpCount Counter32,
18         ieee8021AsV3PortStatRxAnnounceCount        Counter32,
19         ieee8021AsV3PortStatRxPtpPacketDiscardCount Counter32,
20         ieee8021AsV3PortStatSyncReceiptTimeoutCount Counter32,
21         ieee8021AsV3PortStatAnnounceReceiptTimeoutCount Counter32,
22         ieee8021AsV3PortStatPdelayAllowedLostRspExceededCount Counter32,
23         ieee8021AsV3PortStatTxSyncCount            Counter32,
24         ieee8021AsV3PortStatTxOneStepSyncCount      Counter32,
25         ieee8021AsV3PortStatTxFollowUpCount         Counter32,
26         ieee8021AsV3PortStatTxPdelayRequestCount    Counter32,
27         ieee8021AsV3PortStatTxPdelayRspCount        Counter32,
28         ieee8021AsV3PortStatTxPdelayRspFollowUpCount Counter32,
29         ieee8021AsV3PortStatTxAnnounceCount         Counter32
30     }
31
32 ieee8021AsV3PortStatRxSyncCount OBJECT-TYPE
33     SYNTAX      Counter32
34     MAX-ACCESS  read-only
35     STATUS      current
36     DESCRIPTION
37         "A counter that increments every time synchronization
38         information is received."
39     REFERENCE   "14.10.2"
40     ::= { ieee8021AsV3PortStatDSEntry 1 }
41
42 ieee8021AsV3PortStatRxOneStepSyncCount OBJECT-TYPE
43     SYNTAX      Counter32
44     MAX-ACCESS  read-only
45     STATUS      current
46     DESCRIPTION
47         "A counter that increments every time a one-step Sync
48         message is received."
49     REFERENCE   "14.10.3"
50     ::= { ieee8021AsV3PortStatDSEntry 2 }
51
52 ieee8021AsV3PortStatRxFollowUpCount OBJECT-TYPE
53     SYNTAX      Counter32
54     MAX-ACCESS  read-only
55     STATUS      current
56     DESCRIPTION
57         "A counter that increments every time a Follow_Up message
58         is received."
59     REFERENCE   "14.10.4"

```

```

1      ::= { ieee8021AsV3PortStatDSEntry 3 }
2
3 ieee8021AsV3PortStatRxPdelayRequestCount OBJECT-TYPE
4     SYNTAX      Counter32
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "A counter that increments every time a Pdelay_Req message
9         is received."
10    REFERENCE   "14.10.5"
11    ::= { ieee8021AsV3PortStatDSEntry 4 }
12
13 ieee8021AsV3PortStatRxPdelayRspCount OBJECT-TYPE
14    SYNTAX      Counter32
15    MAX-ACCESS  read-only
16    STATUS      current
17    DESCRIPTION
18        "A counter that increments every time a Pdelay_Rsp message
19        is received."
20    REFERENCE   "14.10.6"
21    ::= { ieee8021AsV3PortStatDSEntry 5 }
22
23 ieee8021AsV3PortStatRxPdelayRspFollowUpCount OBJECT-TYPE
24    SYNTAX      Counter32
25    MAX-ACCESS  read-only
26    STATUS      current
27    DESCRIPTION
28        "A counter that increments every time a Pdelay_Rsp_Follow_Up
29        message is received."
30    REFERENCE   "14.10.7"
31    ::= { ieee8021AsV3PortStatDSEntry 6 }
32
33 ieee8021AsV3PortStatRxAnnounceCount OBJECT-TYPE
34    SYNTAX      Counter32
35    MAX-ACCESS  read-only
36    STATUS      current
37    DESCRIPTION
38        "A counter that increments every time an Announce message
39        is received."
40    REFERENCE   "14.10.8"
41    ::= { ieee8021AsV3PortStatDSEntry 7 }
42
43 ieee8021AsV3PortStatRxPtpPacketDiscardCount OBJECT-TYPE
44    SYNTAX      Counter32
45    MAX-ACCESS  read-only
46    STATUS      current
47    DESCRIPTION
48        "A counter that increments every time a PTP message of the
49        respective PTP Instance is discarded."
50    REFERENCE   "14.10.9"
51    ::= { ieee8021AsV3PortStatDSEntry 8 }
52
53 ieee8021AsV3PortStatSyncReceiptTimeoutCount OBJECT-TYPE
54    SYNTAX      Counter32
55    MAX-ACCESS  read-only
56    STATUS      current
57    DESCRIPTION
58        "A counter that increments every time sync receipt timeout
59        occurs."

```



```

1  REFERENCE    "14.10.10"
2  ::= { ieee8021AsV3PortStatDSEntry 9 }
3
4  ieee8021AsV3PortStatAnnounceReceiptTimeoutCount OBJECT-TYPE
5      SYNTAX      Counter32
6      MAX-ACCESS  read-only
7      STATUS      current
8      DESCRIPTION
9          "A counter that increments every time announce receipt timeout
10         occurs."
11     REFERENCE    "14.10.11"
12     ::= { ieee8021AsV3PortStatDSEntry 10 }
13
14  ieee8021AsV3PortStatPdelayAllowedLostRspExceededCount OBJECT-TYPE
15      SYNTAX      Counter32
16      MAX-ACCESS  read-only
17      STATUS      current
18      DESCRIPTION
19          "A counter that increments every time the value of the
20          variable lostResponses exceeds the value of the variable
21          allowedLostResponses."
22     REFERENCE    "14.10.12"
23     ::= { ieee8021AsV3PortStatDSEntry 11 }
24
25  ieee8021AsV3PortStatTxSyncCount OBJECT-TYPE
26      SYNTAX      Counter32
27      MAX-ACCESS  read-only
28      STATUS      current
29      DESCRIPTION
30          "A counter that increments every time synchronization
31          information is transmitted."
32     REFERENCE    "14.10.13"
33     ::= { ieee8021AsV3PortStatDSEntry 12 }
34
35  ieee8021AsV3PortStatTxOneStepSyncCount OBJECT-TYPE
36      SYNTAX      Counter32
37      MAX-ACCESS  read-only
38      STATUS      current
39      DESCRIPTION
40          "A counter that increments every time a one-step Sync
41          message is transmitted."
42     REFERENCE    "14.10.14"
43     ::= { ieee8021AsV3PortStatDSEntry 13 }
44
45  ieee8021AsV3PortStatTxFollowUpCount OBJECT-TYPE
46      SYNTAX      Counter32
47      MAX-ACCESS  read-only
48      STATUS      current
49      DESCRIPTION
50          "A counter that increments every time a Follow_Up message
51          is transmitted."
52     REFERENCE    "14.10.15"
53     ::= { ieee8021AsV3PortStatDSEntry 14 }
54
55  ieee8021AsV3PortStatTxPdelayRequestCount OBJECT-TYPE
56      SYNTAX      Counter32
57      MAX-ACCESS  read-only
58      STATUS      current
59      DESCRIPTION

```

```

1      "A counter that increments every time a Pdelay_Req message
2      is transmitted."
3      REFERENCE    "14.10.16"
4      ::= { ieee8021AsV3PortStatDSEntry 15 }
5
6 ieee8021AsV3PortStatTxPdelayRspCount OBJECT-TYPE
7     SYNTAX      Counter32
8     MAX-ACCESS  read-only
9     STATUS      current
10    DESCRIPTION
11        "A counter that increments every time a Pdelay_Resp message
12        is transmitted."
13    REFERENCE    "14.10.17"
14    ::= { ieee8021AsV3PortStatDSEntry 16 }
15
16 ieee8021AsV3PortStatTxPdelayRspFollowUpCount OBJECT-TYPE
17     SYNTAX      Counter32
18     MAX-ACCESS  read-only
19     STATUS      current
20     DESCRIPTION
21        "A counter that increments every time a
22        Pdelay_Resp_Follow_Up message is transmitted."
23    REFERENCE    "14.10.18"
24    ::= { ieee8021AsV3PortStatDSEntry 17 }
25
26 ieee8021AsV3PortStatTxAnnounceCount OBJECT-TYPE
27     SYNTAX      Counter32
28     MAX-ACCESS  read-only
29     STATUS      current
30     DESCRIPTION
31        "A counter that increments every time an Announce message is
32        transmitted."
33    REFERENCE    "14.10.19"
34    ::= { ieee8021AsV3PortStatDSEntry 18 }
35
36 -- =====
37 -- The Acceptable TimeTransmitter Port Parameter Data Ser represents the
38 -- capability to enable/disable the acceptable timeTransmitter table
39 -- feature on a PTP Port.
40 -- =====
41
42 ieee8021AsV3AcceptableTimeTransmitterPortDSTable OBJECT-TYPE
43     SYNTAX      SEQUENCE OF Ieee8021AsV3AcceptableTimeTransmitterPortDSEntry
44     MAX-ACCESS  not-accessible
45     STATUS      current
46     DESCRIPTION
47        "For the single PTP Port of a PTP End Instance and for each
48        PTP Port of a PTP Relay Instance, the acceptableTimeTransmitterPortDS
49        contains the single member acceptableTimeTransmitterTableEnabled, which
50        is used to enable/disable the Acceptable TimeTransmitter Table Feature.
51        The number of such data sets is the same as the value of
52        defaultDS.numberPorts."
53     REFERENCE    "14.11"
54     ::= { ieee8021AsV3MIBObjects 13 }
55
56 ieee8021AsV3AcceptableTimeTransmitterPortDSEntry OBJECT-TYPE
57     SYNTAX      Ieee8021AsV3AcceptableTimeTransmitterPortDSEntry
58     MAX-ACCESS  not-accessible
59     STATUS      current

```

```

1  DESCRIPTION
2      "The Acceptable TimeTransmitter Port Data Set represents the capability
3      to enable/disable the acceptable timeTransmitter table feature on a
4      PTP Port.
5      For the single PTP Port of a PTP End Instance and for each
6      PTP Port of a PTP Relay Instance, the acceptableTimeTransmitterPortDS
7      contains the single member acceptableTimeTransmitterTableEnabled, which
8      is used to enable/disable the Acceptable TimeTransmitter Table Feature.
9      The number of such data sets is the same as the value of
10     defaultDS.numberPorts."
11  INDEX { ieee8021AsV3PtpInstance,
12          ieee8021AsV3AcceptableTimeTransmitterPortDSAsIndex }
13  ::= { ieee8021AsV3AcceptableTimeTransmitterPortDSTable 1 }
14
15  Ieee8021AsV3AcceptableTimeTransmitterPortDSEntry ::=
16  SEQUENCE {
17      ieee8021AsV3AcceptableTimeTransmitterPortDSAsIndex      InterfaceIndexOrZero,
18      ieee8021AsV3AcceptableTTPortDSTableEnabled              TruthValue
19  }
20
21  ieee8021AsV3AcceptableTimeTransmitterPortDSAsIndex OBJECT-TYPE
22  SYNTAX      InterfaceIndexOrZero
23  MAX-ACCESS  not-accessible
24  STATUS      current
25  DESCRIPTION
26      "An index to identify an entry in the Acceptable TimeTransmitter
27      Port Table Data Set."
28  REFERENCE   "14.11"
29  ::= { ieee8021AsV3AcceptableTimeTransmitterPortDSEntry 1 }
30
31  ieee8021AsV3AcceptableTTPortDSTableEnabled OBJECT-TYPE
32  SYNTAX      TruthValue
33  MAX-ACCESS  read-write
34  STATUS      current
35  DESCRIPTION
36      "The value is equal to the value of the Boolean
37      acceptableTimeTransmitterTableEnabled."
38  REFERENCE   "14.11.2"
39  ::= { ieee8021AsV3AcceptableTimeTransmitterPortDSEntry 2 }
40
41  -- =====
42  -- The External Port Configuration Port Data Set is used with
43  -- the external port configuration option to indicate the
44  -- desired state for the PTP Port.
45  -- =====
46  ieee8021AsV3ExternalPortConfigurationPortDSTable OBJECT-TYPE
47  SYNTAX      SEQUENCE OF Ieee8021AsV3ExternalPortConfigurationPortDSEntry
48  MAX-ACCESS  not-accessible
49  STATUS      current
50  DESCRIPTION
51      "The externalPortConfigurationPortDS contains the single member
52      desiredState, which indicates the desired state for the PTP Port.
53      The number of such data sets is the same as the value of
54      defaultDS.numberPorts."
55  REFERENCE   "14.12"
56  ::= { ieee8021AsV3MIBObjects 14 }
57
58  ieee8021AsV3ExternalPortConfigurationPortDSEntry OBJECT-TYPE
59  SYNTAX      Ieee8021AsV3ExternalPortConfigurationPortDSEntry

```

```

1   MAX-ACCESS    not-accessible
2   STATUS        current
3   DESCRIPTION
4       "The externalPortConfigurationPortDS contains the single member
5       desiredState, which indicates the desired state for the PTP Port.
6       The number of such data sets is the same as the value of
7       defaultDS.numberPorts."
8   INDEX { ieee8021AsV3PtpInstance,
9           ieee8021AsV3ExternalPortConfigurationPortDSAsIndex }
10  ::= { ieee8021AsV3ExternalPortConfigurationPortDSTable 1 }
11
12  Ieee8021AsV3ExternalPortConfigurationPortDSEntry ::=
13  SEQUENCE {
14      ieee8021AsV3ExternalPortConfigurationPortDSAsIndex      InterfaceIndexOrZero,
15      ieee8021AsV3ExternalPortConfigurationPortDSDesiredState  INTEGER
16  }
17
18  ieee8021AsV3ExternalPortConfigurationPortDSAsIndex OBJECT-TYPE
19  SYNTAX      InterfaceIndexOrZero
20  MAX-ACCESS  not-accessible
21  STATUS      current
22  DESCRIPTION
23      "An index to identify an entry in the External Port
24      Configuration Port Table Data Set."
25  REFERENCE   "14.12"
26  ::= { ieee8021AsV3ExternalPortConfigurationPortDSEntry 1 }
27
28  ieee8021AsV3ExternalPortConfigurationPortDSDesiredState OBJECT-TYPE
29  SYNTAX      INTEGER {
30      disabledPort(3),
31      timeTransmitterPort(6),
32      passivePort(7),
33      timeReceiverPort(9)
34  }
35  MAX-ACCESS  read-write
36  STATUS      current
37  DESCRIPTION
38      "When the value of defaultDS.externalPortConfigurationEnabled
39      is TRUE (1), the value of
40      externalPortConfigurationPortDS.desiredState is the desired
41      state of the PTP Port. This member sets the value of the
42      variable portStateInd. When a new value is written to the
43      member by management, the variable rcvdPortStateInd is set
44      to TRUE (1)."

```

```

1      on a PTP Port (see Annex G). This data set is used instead of
2      the cmlDsAsymmetryMeasurementModeDS, when only domain 0 is
3      present and CMLDS is not used."
4  REFERENCE "14.13"
5  ::= { ieee8021AsV3MIBObjects 15 }
6
7  ieee8021AsV3AsymMeasurementModeDSEntry OBJECT-TYPE
8  SYNTAX      Ieee8021AsV3AsymMeasurementModeDSEntry
9  MAX-ACCESS  not-accessible
10 STATUS      current
11 DESCRIPTION
12     "The asymmetryMeasurementModeDS represents the capability to
13     enable/disable the Asymmetry Compensation Measurement Procedure
14     on a PTP Port (see Annex G). This data set is used instead of
15     the cmlDsAsymmetryMeasurementModeDS, when only domain 0 is
16     present and CMLDS is not used. "
17 INDEX { ieee8021AsV3PtpInstance,
18         ieee8021AsV3AsymMeasurementModeDSAsIndex }
19 ::= { ieee8021AsV3AsymMeasurementModeDSTable 1 }
20
21 Ieee8021AsV3AsymMeasurementModeDSEntry ::=
22     SEQUENCE {
23         ieee8021AsV3AsymMeasurementModeDSAsIndex      InterfaceIndexOrZero,
24         ieee8021AsV3AsymMeasurementModeDSAsymMeasurementMode
25 TruthValue
26     }
27
28 ieee8021AsV3AsymMeasurementModeDSAsIndex OBJECT-TYPE
29 SYNTAX      InterfaceIndexOrZero
30 MAX-ACCESS  not-accessible
31 STATUS      current
32 DESCRIPTION
33     "An index to identify an entry in the Asymmetry Measurement
34     Mode Data Set."
35 REFERENCE "14.13"
36 ::= { ieee8021AsV3AsymMeasurementModeDSEntry 1 }
37
38 ieee8021AsV3AsymMeasurementModeDSAsymMeasurementMode OBJECT-TYPE
39 SYNTAX      TruthValue
40 MAX-ACCESS  read-write
41 STATUS      current
42 DESCRIPTION
43     "The value is equal to the value of the Boolean
44     asymmetryMeasurementMode. For full-duplex IEEE 802.3
45     media, the value is TRUE (1) if an asymmetry measurement
46     is being performed for the link attached to this PTP Port,
47     and FALSE (2) otherwise. For all other media, the value
48     shall be FALSE (2). Setting this managed object causes the
49     Boolean asymmetryMeasurementMode to have the same value.
50     NOTE: If an asymmetry measurement is being performed for a
51     link, asymmetryMeasurementMode must be TRUE (1) for the
52     PTP Ports at each end of the link."
53 REFERENCE "14.13.2"
54 ::= { ieee8021AsV3AsymMeasurementModeDSEntry 2 }
55
56 -- =====
57 -- The Common Services Port Parameter Data Set enables a
58 -- PTP Port of a PTP Instance to determine which port of the
59 -- respective common service corresponds to that PTP Port.

```

```

1 -- =====
2 ieee8021AsV3CommonServicesPortDSTable OBJECT-TYPE
3     SYNTAX      SEQUENCE OF Ieee8021AsV3CommonServicesPortDSEntry
4     MAX-ACCESS  not-accessible
5     STATUS      current
6     DESCRIPTION
7         "This member contains the port number
8         of the CMLDS Link Port that corresponds to this PTP Port."
9     REFERENCE   "14.14"
10    ::= { ieee8021AsV3MIBObjects 16 }
11
12 ieee8021AsV3CommonServicesPortDSEntry OBJECT-TYPE
13     SYNTAX      Ieee8021AsV3CommonServicesPortDSEntry
14     MAX-ACCESS  not-accessible
15     STATUS      current
16     DESCRIPTION
17         "This member contains the port number
18         of the CMLDS Link Port that
19         corresponds to this PTP Port."
20     INDEX { ieee8021AsV3PtpInstance,
21             ieee8021AsV3CommonServicesPortDSAsIndex }
22    ::= { ieee8021AsV3CommonServicesPortDSTable 1 }
23
24 Ieee8021AsV3CommonServicesPortDSEntry ::=
25     SEQUENCE {
26         ieee8021AsV3CommonServicesPortDSAsIndex      InterfaceIndexOrZero,
27         ieee8021AsV3CommonServicesPortDSCmlldsLinkPortPortNumber      Unsigned32
28     }
29
30 ieee8021AsV3CommonServicesPortDSAsIndex OBJECT-TYPE
31     SYNTAX      InterfaceIndexOrZero
32     MAX-ACCESS  not-accessible
33     STATUS      current
34     DESCRIPTION
35         "An index to identify an entry in the Common Services Port
36         Data Set."
37     REFERENCE   "14.14"
38    ::= { ieee8021AsV3CommonServicesPortDSEntry 1 }
39
40 ieee8021AsV3CommonServicesPortDSCmlldsLinkPortPortNumber OBJECT-TYPE
41     SYNTAX      Unsigned32 (0..65535)
42     MAX-ACCESS  read-only
43     STATUS      current
44     DESCRIPTION
45         "The value is the portNumber attribute of the
46         cmlldsLinkPortDS.portIdentity of the Link Port that
47         corresponds to this PTP Port."
48     REFERENCE   "14.14.2"
49    ::= { ieee8021AsV3CommonServicesPortDSEntry 2 }
50
51 -- =====
52 -- The Common Mean Link Delay Service Default Parameter Data Set
53 -- describes the per-time-aware-system attributes of the Common
54 -- Mean Link Delay Service.
55 -- =====
56
57 ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSTable OBJECT-TYPE
58     SYNTAX      SEQUENCE OF Ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry
59     MAX-ACCESS  not-accessible

```

```

1  STATUS      current
2  DESCRIPTION
3      "The cmlDsDefaultDS describes the per-time-aware-system attributes
4      of the Common Mean Link Delay Service."
5  REFERENCE   "14.15"
6  ::= { ieee8021AsV3MIBObjects 17 }
7
8  ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry  OBJECT-TYPE
9  SYNTAX      Ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry
10 MAX-ACCESS   not-accessible
11 STATUS      current
12 DESCRIPTION
13     "The cmlDsDefaultDS describes the per-time-aware-system attributes
14     of the Common Mean Link Delay Service."
15 INDEX { ieee8021AsV3CmlDsDefaultDSAsIndex }
16 ::= { ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSTable 1 }
17
18 Ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry ::=
19 SEQUENCE {
20     ieee8021AsV3CmlDsDefaultDSAsIndex  InterfaceIndexOrZero,
21     ieee8021AsV3CmlDsDefaultDSClockIdentity  Ieee8021AsV3ClockIdentity,
22     ieee8021AsV3CmlDsDefaultDSNumberLinkPorts  Unsigned32
23 }
24
25
26 ieee8021AsV3CmlDsDefaultDSAsIndex  OBJECT-TYPE
27 SYNTAX      InterfaceIndexOrZero
28 MAX-ACCESS   not-accessible
29 STATUS      current
30 DESCRIPTION
31     "An index to identify an entry in the Common Mean Link
32     Delay Default Data Set."
33 REFERENCE   "14.15"
34 ::= { ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry 1 }
35
36 ieee8021AsV3CmlDsDefaultDSClockIdentity  OBJECT-TYPE
37 SYNTAX      Ieee8021AsV3ClockIdentity
38 MAX-ACCESS   read-only
39 STATUS      current
40 DESCRIPTION
41     "The value is the clockIdentity that will be used to
42     identify the Common Mean Link Delay Service."
43 REFERENCE   "14.15.2"
44 ::= { ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry 2 }
45
46 ieee8021AsV3CmlDsDefaultDSNumberLinkPorts  OBJECT-TYPE
47 SYNTAX      Unsigned32 (0..65535)
48 MAX-ACCESS   read-only
49 STATUS      current
50 DESCRIPTION
51     "The value is the number of Link Ports of the time-aware
52     system on which the Common Mean Link Delay Service is
53     implemented. For an end station the value is 1."
54 REFERENCE   "14.15.3"
55 ::= { ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry 3 }
56
57 -- =====
58 -- The Common Mean Link Delay Service Link Port Parameter Data Set
59 -- represents time-aware Link Port capabilities for the Common Mean

```

```

1 -- Link Delay Service of a Link Port of a time-aware system.
2 -- =====
3
4 ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSTable OBJECT-TYPE
5     SYNTAX      SEQUENCE OF Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry
6     MAX-ACCESS  not-accessible
7     STATUS      current
8     DESCRIPTION
9         "For every Link Port of the Common Mean Link Delay Service of a
10         time-aware system, the cmldsLinkPortDS is maintained as the
11         basis for making protocol decisions and providing values for
12         message fields. The number of such data sets is the same as
13         the value of cmldsDefaultDS.numberLinkPorts."
14     REFERENCE   "14.16"
15     ::= { ieee8021AsV3MIBObjects 18 }
16
17 ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry OBJECT-TYPE
18     SYNTAX      Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry
19     MAX-ACCESS  not-accessible
20     STATUS      current
21     DESCRIPTION
22         "For every Link Port of the Common Mean Link Delay Service of a
23         time-aware system, the cmldsLinkPortDS is maintained as the
24         basis for making protocol decisions and providing values for
25         message fields. The number of such data sets is the same as
26         the value of cmldsDefaultDS.numberLinkPorts."
27     INDEX { ieee8021AsV3BridgeBasePort,
28             ieee8021AsV3CmldsLinkPortDSAsIndex }
29     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSTable 1 }
30
31 Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry ::=
32     SEQUENCE {
33         ieee8021AsV3CmldsLinkPortDSAsIndex      InterfaceIndexOrZero,
34         ieee8021AsV3CmldsLinkPortDSClockIdentity Ieee8021AsV3ClockIdentity,
35         ieee8021AsV3CmldsLinkPortDSPORTNumber   Unsigned32,
36         ieee8021AsV3CmldsLinkPortDSCmldsLinkPortEnabled TruthValue,
37         ieee8021AsV3CmldsLinkPortDSIsMeasuringDelay TruthValue,
38         ieee8021AsV3CmldsLinkPortDSAsCapableAcrossDomains TruthValue,
39         ieee8021AsV3CmldsLinkPortDSMeanLinkDelay Ieee8021ASV3PtpTimeInterval,
40         ieee8021AsV3CmldsLinkPortDSMeanLinkDelayThresh Ieee8021ASV3PtpTimeInterval,
41         ieee8021AsV3CmldsLinkPortDSDelayAsym     Ieee8021ASV3PtpTimeInterval,
42         ieee8021AsV3CmldsLinkPortDSNbrRateRatio   Integer32,
43         ieee8021AsV3CmldsLinkPortDSInitialLogPdelayReqInterval Integer32,
44         ieee8021AsV3CmldsLinkPortDSCurrentLogPdelayReqInterval Integer32,
45         ieee8021AsV3CmldsLinkPortDSUseMgtSettableLogPdelayReqInterval TruthValue,
46         ieee8021AsV3CmldsLinkPortDSMgtSettableLogPdelayReqInterval Integer32,
47         ieee8021AsV3CmldsLinkPortDSInitialComputeNbrRateRatio TruthValue,
48         ieee8021AsV3CmldsLinkPortDSCurrentComputeNbrRateRatio TruthValue,
49         ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeNbrRateRatio TruthValue,
50         ieee8021AsV3CmldsLinkPortDSMgtSettableComputeNbrRateRatio TruthValue,
51         ieee8021AsV3CmldsLinkPortDSInitialComputeMeanLinkDelay TruthValue,
52         ieee8021AsV3CmldsLinkPortDSCurrentComputeMeanLinkDelay TruthValue,
53         ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeMeanLinkDelay
54         TruthValue,
55         ieee8021AsV3CmldsLinkPortDSMgtSettableComputeMeanLinkDelay TruthValue,
56         ieee8021AsV3CmldsLinkPortDSAllowedLostRsp Unsigned32,
57         ieee8021AsV3CmldsLinkPortDSAllowedFaults Unsigned32,
58         ieee8021AsV3CmldsLinkPortDSVersionNumber Unsigned32,
59         ieee8021AsV3CmldsLinkPortDSPdelayTruncTST1 Ieee8021ASV3Timestamp,

```



```

1      ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST2 Ieee8021ASV3Timestamp,
2      ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST3 Ieee8021ASV3Timestamp,
3      ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST4 Ieee8021ASV3Timestamp,
4      ieee8021AsV3CmlDsLinkPortDSMinorVersionNumber      Unsigned32
5      }
6
7 ieee8021AsV3CmlDsLinkPortDSAsIndex OBJECT-TYPE
8     SYNTAX      InterfaceIndexOrZero
9     MAX-ACCESS  not-accessible
10    STATUS      current
11    DESCRIPTION
12        "An index to identify an entry in the Comon Mean Link
13        Delay Link Port Data Set."
14    REFERENCE   "14.16"
15    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 1 }
16
17 ieee8021AsV3CmlDsLinkPortDSClockIdentity OBJECT-TYPE
18    SYNTAX      Ieee8021AsV3ClockIdentity
19    MAX-ACCESS  read-only
20    STATUS      current
21    DESCRIPTION
22        "The value is the first of the portIdentity attribute
23        of the local port, which is a set made of
24        Ieee8021AsV3ClockIdentity and portNumber."
25    REFERENCE   "14.16.2"
26    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 2 }
27
28 ieee8021AsV3CmlDsLinkPortDSPortNumber OBJECT-TYPE
29    SYNTAX      Unsigned32(0..65535)
30    MAX-ACCESS  read-only
31    STATUS      current
32    DESCRIPTION
33        "The value is the second of the portIdentity attribute
34        of the local port, which is a set made of
35        Ieee8021AsV3ClockIdentity and portNumber."
36    REFERENCE   "14.16.2"
37    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 3 }
38
39 ieee8021AsV3CmlDsLinkPortDSCmlDsLinkPortEnabled
40 OBJECT-TYPE
41    SYNTAX      TruthValue
42    MAX-ACCESS  read-only
43    STATUS      current
44    DESCRIPTION
45        "The value is equal to the value of the Boolean
46        cmlDsLinkPortEnabled."
47    REFERENCE   "14.16.3"
48    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 4 }
49
50 ieee8021AsV3CmlDsLinkPortDSIsMeasuringDelay
51 OBJECT-TYPE
52    SYNTAX      TruthValue
53    MAX-ACCESS  read-only
54    STATUS      current
55    DESCRIPTION
56        "The value is equal to the value of the Boolean
57        isMeasuringDelay."
58    REFERENCE   "14.16.4"
59    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 5 }

```

```

1
2 ieee8021AsV3CmlDsLinkPortDSAsCapableAcrossDomains
3 OBJECT-TYPE
4     SYNTAX      TruthValue
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "The value is equal to the value of the Boolean
9         asCapableAcrossDomains."
10    REFERENCE   "14.16.5"
11    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 6 }
12
13 ieee8021AsV3CmlDsLinkPortDSMeanLinkDelay
14 OBJECT-TYPE
15     SYNTAX      Ieee8021ASV3PtpTimeInterval
16     MAX-ACCESS  read-only
17     STATUS      current
18     DESCRIPTION
19         "The value is equal to the value of the per-port global
20         variable meanLinkDelay. It is an estimate of the current
21         one-way propagation time on the link attached to this Link
22         Port, measured as specified for the respective medium. The
23         value is zero for Link Ports attached to IEEE 802.3 EPON
24         links and for the timeTransmitter port of an IEEE 802.11 link,
25         because one-way propagation delay is not measured on the
26         latter and not directly measured on the former.
27         NOTE: The underlying per-port global variable meanLinkDelay is
28         of type UScaledNS, which is a 96-Bit value. meanLinkDelay
29         values that are larger than the maximum value that can be
30         represented by the TimeInterval data type, i.e.,
31         0xFFFF FFFF FFFF FFFF (where the units are 2 sup -16 ns), used
32         for this managed object are set to this largest value."
33     REFERENCE   "14.16.6"
34     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 7 }
35
36 ieee8021AsV3CmlDsLinkPortDSMeanLinkDelayThresh
37 OBJECT-TYPE
38     SYNTAX      Ieee8021ASV3PtpTimeInterval
39     MAX-ACCESS  read-write
40     STATUS      current
41     DESCRIPTION
42         "The value is equal to the value of the per-Link-Port global
43         variable meanLinkDelayThresh. It is the propagation time
44         threshold above which a Link Port (and therefore any PTP Ports
45         that use the CMLDS on this Link Port) is considered not
46         capable of participating in the IEEE 802.1AS protocol.
47         Setting this managed object causes the per-Link-Port global
48         variable meanLinkDelayThresh to have the same value.
49         NOTE: The underlying per-port global variable
50         meanLinkDelayThresh is of type UScaledNS, which is a 96-Bit
51         value. meanLinkDelayThresh values that are larger than the
52         maximum value that can be represented by the TimeInterval
53         data type, i.e., 0xFFFF FFFF FFFF FFFF (where the units are
54         2 sup -16 ns), used for this managed object are set to this
55         largest value."
56     REFERENCE   "14.16.7"
57     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 8 }
58
59 ieee8021AsV3CmlDsLinkPortDSDelayAsym

```

```

1 OBJECT-TYPE
2     SYNTAX      Ieee8021ASV3PtpTimeInterval
3     MAX-ACCESS  read-write
4     STATUS      current
5     DESCRIPTION
6         "The value is the asymmetry in the propagation delay on the
7         link attached to this Link Port relative to the local clock.
8         If propagation delay asymmetry is not modeled, then
9         delayAsymmetry is 0.
10        NOTE: The underlying per-port global variable delayAsymmetry
11        is of type ScaledNS, which is a 96-Bit value.
12        delayAsymmetry values that are larger than the maximum value
13        that can be represented by the TimeInterval data type, i.e.,
14        0x7FFF FFFF FFFF FFFF, (where the units are 2 sup -16 ns),
15        used for this managed object are set to this largest value.
16        delayAsymmetry values that are less than the minimum value
17        that can be represented by the TimeInterval data type, i.e.,
18        0x8000 0000 0000 0001 written in twos complement form (where
19        the units are 2 sup -16 ns), used for this managed object are
20        set to this smallest value."
21     REFERENCE   "14.16.8"
22     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 9 }
23
24 ieee8021AsV3CmldsLinkPortDSNbrRateRatio
25 OBJECT-TYPE
26     SYNTAX      Integer32
27     MAX-ACCESS  read-only
28     STATUS      current
29     DESCRIPTION
30         "The value is an estimate of the ratio of the frequency of the
31         LocalClock entity of the time-aware system at the other end
32         of the link attached to this Link Port, to the frequency of
33         the LocalClock entity of this time-aware system.
34         neighborRateRatio is expressed as the fractional frequency
35         offset multiplied by 2^41, i.e., the quantity
36         (neighborRateRatio -1.0)(2^41)."
37     REFERENCE   "14.16.9"
38     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 10 }
39
40 ieee8021AsV3CmldsLinkPortDSInitialLogPdelayReqInterval OBJECT-TYPE
41     SYNTAX      Integer32(-128..127)
42     MAX-ACCESS  read-write
43     STATUS      current
44     DESCRIPTION
45         "If useMgtSettableLogPdelayReqInterval is FALSE (2) then,
46         for full-duplex IEEE 802.3 media and for CSN media that use
47         the peer-to-peer delay mechanism to measure path delay, the
48         value is the logarithm to base 2 of the Pdelay_Req message
49         transmission interval used when (a) the Link Port is
50         initialized, or (b) a message interval request TLV is
51         received with the logLinkDelayInterval field set to 126.
52         For all other media, the value is 127."
53     REFERENCE   "14.16.10"
54     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 11 }
55
56 ieee8021AsV3CmldsLinkPortDSCurrentLogPdelayReqInterval OBJECT-TYPE
57     SYNTAX      Integer32(-128..127)
58     MAX-ACCESS  read-only
59     STATUS      current

```

```

1  DESCRIPTION
2      "For full-duplex IEEE 802.3 media and for CSN media that use
3      the peer-to-peer delay mechanism to measure path delay,
4      the value is the logarithm to the base 2 of the current
5      Pdelay_Req message transmission interval.
6      For all other media, the value is 127."
7  REFERENCE    "14.16.11"
8  ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 12 }
9
10 ieee8021AsV3CmlDsLinkPortDSUseMgtSettableLogPdelayReqInterval OBJECT-TYPE
11     SYNTAX      TruthValue
12     MAX-ACCESS  read-write
13     STATUS      current
14     DESCRIPTION
15         "The managed object is a Boolean that determines the source
16         of the sync interval and mean time interval between
17         successive Pdelay_Req messages. If the value is TRUE (1),
18         the value of currentLogPdelayReqInterval is set equal to
19         the value of mgtSettableLogPdelayReqInterval. If the value
20         of the managed object is FALSE (2), the value of
21         currentLogPdelayReqInterval is determined by the
22         LinkDelayIntervalSetting state machine."
23     REFERENCE    "14.16.12"
24     DEFVAL { false }
25     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 13 }
26
27 ieee8021AsV3CmlDsLinkPortDSMgtSettableLogPdelayReqInterval OBJECT-TYPE
28     SYNTAX      Integer32 (-128..127)
29     MAX-ACCESS  read-write
30     STATUS      current
31     DESCRIPTION
32         "The value is the logarithm to base 2 of the mean time
33         interval between successive Pdelay_Req messages if
34         useMgtSettableLogPdelayReqInterval is TRUE (1). The
35         value is not used if useMgtSettableLogPdelayReqInterval
36         is FALSE (2)."
37     REFERENCE    "14.16.13"
38     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 14 }
39
40 ieee8021AsV3CmlDsLinkPortDSInitialComputeNbrRateRatio OBJECT-TYPE
41     SYNTAX      TruthValue
42     MAX-ACCESS  read-write
43     STATUS      current
44     DESCRIPTION
45         "If useMgtSettableComputeNeighborRateRatio is FALSE (2),
46         then for full-duplex IEEE 802.3 media and for CSN media
47         that use the peer-to-peer delay mechanism to measure path
48         delay, the value is the initial value of
49         computeNeighborRateRatio.
50         For all other media, the value is TRUE."
51     REFERENCE    "14.16.14"
52     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 15 }
53
54 ieee8021AsV3CmlDsLinkPortDSCurrentComputeNbrRateRatio OBJECT-TYPE
55     SYNTAX      TruthValue
56     MAX-ACCESS  read-only
57     STATUS      current
58     DESCRIPTION
59         "For full-duplex IEEE 802.3 media and for CSN media that use

```

```

1      the peer-to-peer delay mechanism to measure path delay,
2      the value is the current value of computeNeighborRateRatio.
3      For all other media, the value is TRUE (1).
4  REFERENCE    "14.16.15"
5  ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 16 }
6
7  ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeNbrRateRatio OBJECT-TYPE
8      SYNTAX      TruthValue
9      MAX-ACCESS  read-write
10     STATUS      current
11     DESCRIPTION
12         "The managed object is a Boolean that determines the source
13         of the value of computeNeighborRateRatio. If the value is
14         TRUE (1), the value of computeNeighborRateRatio is set equal
15         to the value of mgtSettablecomputeNeighborRateRatio. If
16         the value of the managed object is FALSE (2), the
17         value of currentComputeNeighborRateRatio is determined by
18         the LinkDelayIntervalSetting state machine."
19     REFERENCE    "14.16.16"
20     DEFVAL { false }
21     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 17 }
22
23  ieee8021AsV3CmldsLinkPortDSMgtSettableComputeNbrRateRatio OBJECT-TYPE
24      SYNTAX      TruthValue
25      MAX-ACCESS  read-write
26      STATUS      current
27      DESCRIPTION
28          "computeNeighborRateRatio is configured to this value if
29          useMgtSettableComputeNeighborRateRatio is TRUE (1). The
30          value is not used if useMgtSettableComputeNeighborRateRatio
31          is FALSE (2)."
32      REFERENCE    "14.16.17"
33      ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 18 }
34
35  ieee8021AsV3CmldsLinkPortDSInitialComputeMeanLinkDelay OBJECT-TYPE
36      SYNTAX      TruthValue
37      MAX-ACCESS  read-write
38      STATUS      current
39      DESCRIPTION
40          "If useMgtSettableComputeMeanLinkDelay is FALSE (2) then,
41          for full-duplex IEEE 802.3 media and for CSN media that use
42          the peer-to-peer delay mechanism to measure path delay,
43          the value is the initial value of computeMeanLinkDelay.
44          For all other media, the value is TRUE (1)."
45      REFERENCE    "14.16.18"
46      ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 19 }
47
48  ieee8021AsV3CmldsLinkPortDSCurrentComputeMeanLinkDelay OBJECT-TYPE
49      SYNTAX      TruthValue
50      MAX-ACCESS  read-only
51      STATUS      current
52      DESCRIPTION
53          "For full-duplex IEEE 802.3 media and for CSN media that
54          use the peer-to-peer delay mechanism to measure path delay,
55          the value is the current value of computeMeanLinkDelay.
56          For all other media, the value is TRUE."
57      REFERENCE    "14.16.19"
58      ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 20 }
59

```

```

1 ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeMeanLinkDelay OBJECT-TYPE
2     SYNTAX      TruthValue
3     MAX-ACCESS  read-write
4     STATUS      current
5     DESCRIPTION
6         "The managed object is a Boolean that determines the source
7         of the value of computeMeanLinkDelay. If the value is
8         TRUE (1), the value of computeMeanLinkDelay is set equal
9         to the value of mgtSettableComputeMeanLinkDelay. If the
10        value of the managed object is FALSE (2), the value of
11        currentComputeMeanLinkDelay is determined by the
12        LinkDelayIntervalSetting state machine."
13    REFERENCE   "14.16.20"
14    DEFVAL { false }
15    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 21 }
16
17 ieee8021AsV3CmldsLinkPortDSMgtSettableComputeMeanLinkDelay OBJECT-TYPE
18     SYNTAX      TruthValue
19     MAX-ACCESS  read-write
20     STATUS      current
21     DESCRIPTION
22         "computeMeanLinkDelay is configured to this value if
23         useMgtSettableComputeMeanLinkDelay is TRUE (1). The value
24         is not used if useMgtSettableComputeMeanLinkDelay is
25         FALSE (2)."

```

```

1  DEFVAL { 9 }
2  ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 24 }
3
4  ieee8021AsV3CmlDsLinkPortDSVersionNumber OBJECT-TYPE
5      SYNTAX      Unsigned32(0..15)
6      MAX-ACCESS  read-only
7      STATUS      current
8      DESCRIPTION
9          "This value is set to versionPTP as specified in 10.6.2.2.4."
10     REFERENCE   "14.16.24"
11     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 25 }
12
13  ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST1
14  OBJECT-TYPE
15      SYNTAX      Ieee8021ASV3Timestamp
16      MAX-ACCESS  read-only
17      STATUS      current
18      DESCRIPTION
19          "For full-duplex IEEE 802.3 media and for CSN media that use
20          the peer-to-peer delay mechanism to measure path delay, the
21          first value, T1, of the four elements of this array is as
22          described in Table 14-9. For all other media, the values are
23          zero. This object corresponds to the timestamp t1 modulo 2^32
24          in Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
25          value of this array element is equal to the remainder obtained
26          upon dividing the respective timestamp, expressed in units
27          of 2^-16 ns, by 2^48).
28          At any given time, the timestamp values stored in the T1, T2,
29          T3, T4 PdelayTruncTS are for the same, and most recently
30          completed, peer delay message exchange.
31          NOTE: This managed object is used with the asymmetry
32          measurement compensation procedure, which is based on
33          line-swapping."
34      REFERENCE   "14.16.25"
35      ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 26 }
36
37  ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST2
38  OBJECT-TYPE
39      SYNTAX      Ieee8021ASV3Timestamp
40      MAX-ACCESS  read-only
41      STATUS      current
42      DESCRIPTION
43          "For full-duplex IEEE 802.3 media and for CSN media that use
44          the peer-to-peer delay mechanism to measure path delay, the
45          second value, T2, of the four elements of this array is as
46          described in Table 14-9. For all other media, the values are
47          zero. This object corresponds to the timestamp t1 modulo 2^32
48          in Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
49          value of this array element is equal to the remainder obtained
50          upon dividing the respective timestamp, expressed in units
51          of 2^-16 ns, by 2^48).
52          At any given time, the timestamp values stored in the T1, T2,
53          T3, T4 PdelayTruncTS are for the same, and most recently
54          completed, peer delay message exchange.
55          NOTE: This managed object is used with the asymmetry
56          measurement compensation procedure, which is based on
57          line-swapping."
58      REFERENCE   "14.16.25"
59      ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 27 }

```

```

1
2 ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST3
3 OBJECT-TYPE
4     SYNTAX      Ieee8021ASV3Timestamp
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "For full-duplex IEEE 802.3 media and for CSN media that use
9         the peer-to-peer delay mechanism to measure path delay, the
10        third value, T3, of the four elements of this array is as
11        described in Table 14-9. For all other media, the values are
12        zero. This object corresponds to the timestamp t1 modulo 2^32
13        in Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
14        value of this array element is equal to the remainder obtained
15        upon dividing the respective timestamp, expressed in units
16        of 2^-16 ns, by 2^48).
17        At any given time, the timestamp values stored in the T1, T2,
18        T3, T4 PdelayTruncTS are for the same, and most recently
19        completed, peer delay message exchange.
20        NOTE: This managed object is used with the asymmetry
21        measurement compensation procedure, which is based on
22        line-swapping."
23     REFERENCE   "14.16.25"
24     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 28 }
25
26 ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST4
27 OBJECT-TYPE
28     SYNTAX      Ieee8021ASV3Timestamp
29     MAX-ACCESS  read-only
30     STATUS      current
31     DESCRIPTION
32         "For full-duplex IEEE 802.3 media and for CSN media that use
33         the peer-to-peer delay mechanism to measure path delay, the
34         fourth value, T4, of the four elements of this array is as
35         described in Table 14-9. For all other media, the values are
36         zero. This object corresponds to the timestamp t1 modulo 2^32
37         in Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
38         value of this array element is equal to the remainder obtained
39         upon dividing the respective timestamp, expressed in units
40         of 2^-16 ns, by 2^48).
41        At any given time, the timestamp values stored in the T1, T2,
42        T3, T4 PdelayTruncTS are for the same, and most recently
43        completed, peer delay message exchange.
44        NOTE: This managed object is used with the asymmetry
45        measurement compensation procedure, which is based on
46        line-swapping."
47     REFERENCE   "14.16.25"
48     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 29 }
49
50 ieee8021AsV3CmlDsLinkPortDSMinorVersionNumber OBJECT-TYPE
51     SYNTAX      Unsigned32 (0..15)
52     MAX-ACCESS  read-only
53     STATUS      current
54     DESCRIPTION
55         "This value is set to minorVersionPTP as specified in
56         10.6.2.2.3."
57     REFERENCE   "14.16.26"
58     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 30 }
59

```



```

1 -- =====
2 -- The Common Mean Link Delay Service Link Port Parameter
3 -- Statistics Data Set provides counters associated with Link
4 -- Port capabilities at a given time-aware system.
5 -- =====
6
7 ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSTable OBJECT-TYPE
8     SYNTAX      SEQUENCE OF
9     Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry
10    MAX-ACCESS  not-accessible
11    STATUS      current
12    DESCRIPTION
13        "For every Link Port of the Common Mean Link Delay Service of a
14        time-aware system, the following cmlDsLinkPortStatisticsDS
15        provides counters. The number of such statistics sets is the
16        same as the value of cmlDsDefaultDS.numberLinkPorts."
17    REFERENCE   "14.17"
18    ::= { ieee8021AsV3MIBObjects 19 }
19
20 ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry OBJECT-TYPE
21     SYNTAX      Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry
22     MAX-ACCESS  not-accessible
23     STATUS      current
24     DESCRIPTION
25         "For every Link Port of the Common Mean Link Delay Service of a
26         time-aware system, the following cmlDsLinkPortStatisticsDS
27         provides counters. The number of such statistics sets is the
28         same as the value of cmlDsDefaultDS.numberLinkPorts."
29     INDEX { ieee8021AsV3BridgeBasePort,
30             ieee8021AsV3CmlDsLinkPortStatDSIndex }
31     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSTable 1 }
32
33 Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry ::=
34     SEQUENCE {
35         ieee8021AsV3CmlDsLinkPortStatDSIndex          InterfaceIndexOrZero,
36         ieee8021AsV3CmlDsLinkPortStatDSRxPdelayRequestCount Counter32,
37         ieee8021AsV3CmlDsLinkPortStatDSRxPdelayRspCount Counter32,
38         ieee8021AsV3CmlDsLinkPortStatDSRxPdelayRspFollowUpCount Counter32,
39         ieee8021AsV3CmlDsLinkPortStatDSRxPtpPacketDiscardCount Counter32,
40         ieee8021AsV3CmlDsLinkPortStatDSPdelayAllowedLostRspExceededCount Counter32,
41         ieee8021AsV3CmlDsLinkPortStatDSTxPdelayRequestCount Counter32,
42         ieee8021AsV3CmlDsLinkPortStatDSTxPdelayRspCount Counter32,
43         ieee8021AsV3CmlDsLinkPortStatDSTxPdelayRspFollowUpCount Counter32
44     }
45
46 ieee8021AsV3CmlDsLinkPortStatDSIndex OBJECT-TYPE
47     SYNTAX      InterfaceIndexOrZero
48     MAX-ACCESS  not-accessible
49     STATUS      current
50     DESCRIPTION
51         "An index to identify an entry in the Common Mean Link
52         Port Statistics Data Set."
53     REFERENCE   "14.17"
54     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 1 }
55
56 ieee8021AsV3CmlDsLinkPortStatDSRxPdelayRequestCount OBJECT-TYPE
57     SYNTAX      Counter32
58     MAX-ACCESS  read-only
59     STATUS      current

```

```

1  DESCRIPTION
2      "A counter that increments every time a Pdelay_Req message is
3      received."
4  REFERENCE  "14.17.2"
5      ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 2 }
6
7  ieee8021AsV3CmlDsLinkPortStatDSRxPdelayRspCount OBJECT-TYPE
8      SYNTAX      Counter32
9      MAX-ACCESS  read-only
10     STATUS      current
11     DESCRIPTION
12         "A counter that increments every time a Pdelay_Resp message is
13         received."
14     REFERENCE  "14.17.3"
15     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 3 }
16
17  ieee8021AsV3CmlDsLinkPortStatDSRxPdelayRspFollowUpCount OBJECT-TYPE
18     SYNTAX      Counter32
19     MAX-ACCESS  read-only
20     STATUS      current
21     DESCRIPTION
22         "A counter that increments every time a Pdelay_Resp_Follow_Up
23         message is received."
24     REFERENCE  "14.17.4"
25     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 4 }
26
27  ieee8021AsV3CmlDsLinkPortStatDSRxPtpPacketDiscardCount
28  OBJECT-TYPE
29     SYNTAX      Counter32
30     MAX-ACCESS  read-only
31     STATUS      current
32     DESCRIPTION
33         "A counter that increments every time a PTP message of the
34         Common Mean Link Delay Service is discarded, caused by the
35         occurrence of any of the conditions given in 14.17.5."
36     REFERENCE  "14.17.5"
37     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 5 }
38
39  ieee8021AsV3CmlDsLinkPortStatDSPdelayAllowedLostRspExceededCount
40  OBJECT-TYPE
41     SYNTAX      Counter32
42     MAX-ACCESS  read-only
43     STATUS      current
44     DESCRIPTION
45         "A counter that increments every time the value of the variable
46         lostResponses exceeds the value of the variable
47         allowedLostResponses, in the RESET state of the
48         MDPdelayReq state machine."
49     REFERENCE  "14.17.6"
50     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 6 }
51
52  ieee8021AsV3CmlDsLinkPortStatDSTxPdelayRequestCount
53  OBJECT-TYPE
54     SYNTAX      Counter32
55     MAX-ACCESS  read-only
56     STATUS      current
57     DESCRIPTION
58         "A counter that increments every time a Pdelay_Req message is
59         transmitted."

```

```

1  REFERENCE    "14.17.7"
2  ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 7 }
3
4  ieee8021AsV3CmlDsLinkPortStatDSTxPdelayRspCount
5  OBJECT-TYPE
6      SYNTAX      Counter32
7      MAX-ACCESS  read-only
8      STATUS      current
9      DESCRIPTION
10         "A counter that increments every time a Pdelay_Resp message is
11         transmitted."
12     REFERENCE    "14.17.8"
13     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 8 }
14
15  ieee8021AsV3CmlDsLinkPortStatDSTxPdelayRspFollowUpCount
16  OBJECT-TYPE
17      SYNTAX      Counter32
18      MAX-ACCESS  read-only
19      STATUS      current
20      DESCRIPTION
21         "A counter that increments every time a Pdelay_Resp_Follow_Up
22         message is transmitted."
23     REFERENCE    "14.17.9"
24     ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 9 }
25
26  -- =====
27  -- The Common Mean Link Delay Service Asymmetry Measurement Mode
28  -- Parameter Data Set represents the capability to enable/disable
29  -- the Asymmetry Compensation Measurement Procedure on a Link Port
30  -- (see Annex G).
31  -- =====
32
33  ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSTable  OBJECT-TYPE
34      SYNTAX      SEQUENCE OF
35  Ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry
36      MAX-ACCESS  not-accessible
37      STATUS      current
38      DESCRIPTION
39         "The Common Mean Link Delay Service Asymmetry Measurement Mode
40         Parameter Data Set represents the capability to enable/disable
41         the Asymmetry Compensation Measurement Procedure on a Link Port
42         (see Annex G)."
43     REFERENCE    "14.18"
44     ::= { ieee8021AsV3MIBObjects 20 }
45
46  ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry  OBJECT-TYPE
47      SYNTAX      Ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry
48      MAX-ACCESS  not-accessible
49      STATUS      current
50      DESCRIPTION
51         "This table uses
52         ieee8021AsV3CmlDsAsymmetryMeasurementModeDSAsIndex,
53         and corresponds to
54         ieee8021AsV3CommonMeanLinkDelayServiceAsymmetryMeasurementModeDSTable
55         entry."
56     INDEX { ieee8021AsV3BridgeBasePort,
57             ieee8021AsV3CmlDsAsymMeasurementModeDSAsIndex }
58     ::= { ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSTable 1 }
59

```

```

1 Ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry ::=
2     SEQUENCE {
3     ieee8021AsV3CmldsAsymMeasurementModeDSAsIndex InterfaceIndexOrZero,
4     ieee8021AsV3CmldsAsymMeasurementModeDSAsymMeasurementMode TruthValue
5     }
6
7 ieee8021AsV3CmldsAsymMeasurementModeDSAsIndex OBJECT-TYPE
8     SYNTAX      InterfaceIndexOrZero
9     MAX-ACCESS  not-accessible
10    STATUS      current
11    DESCRIPTION
12        "This object identifies the gPTP interface group within
13        the system for which this entry contains information. It
14        is the value of the instance of the IfIndex object,
15        defined in the IF-MIB, for the gPTP interface group
16        corresponding to this port, or the value 0 if the port
17        has not been bound to an underlying frame source and
18        sink.
19
20        For a given media port of a Bridge or an end station,
21        there can be one or more PTP Port, and depends whether
22        a media port supports point to point link (e.g. IEEE
23        802.3 Ethernet) or point to multi-point (e.g. CSN, IEEE
24        802.3 EPON) links on the media port."
25    REFERENCE   "IEEE Std 802.1AS
26 CommonMeanLinkDelaySvcAsymMeasurementModeParamDS Group PTP Port Index"
27 ::= { ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry 1 }
28
29 ieee8021AsV3CmldsAsymMeasurementModeDSAsymMeasurementMode
30     OBJECT-TYPE
31     SYNTAX      TruthValue
32     MAX-ACCESS  read-write
33     STATUS      current
34     DESCRIPTION
35         "The value is equal to the value of the Boolean
36         asymmetryMeasurementMode (see G.3). For full-duplex
37         IEEE 802.3 media, the value is TRUE (1) if an asymmetry
38         measurement is being performed for the link attached to
39         this Link Port, and FALSE (2) otherwise. For all other
40         media, the value shall be FALSE (2) (see 10.2.5.2).
41         Setting this managed object causes the Boolean
42         allowedFaults to have the same value.
43         NOTE: If an asymmetry measurement is being performed
44         for a link, asymmetryMeasurementMode must be TRUE (1)
45         for the Link Ports at each end of the link.
46         There is one Common Mean Link Delay Service Asymmetry
47         Measurement Mode Parameter Data Set Table for all PTP
48         Instances, per Link Port."
49     REFERENCE   "14.18.2"
50 ::= { ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry 2 }
51
52 -- *****
53 -- IEEE 802.1ASV3 MIB - Conformance Information
54 -- *****
55 ieee8021AsV3Groups          OBJECT IDENTIFIER ::= { ieee8021AsV3Conformance 1 }
56 ieee8021AsV3Compliances     OBJECT IDENTIFIER ::= { ieee8021AsV3Conformance 2 }
57
58 -- =====
59 -- units of conformance

```

```

1 -- =====
2
3 ieee8021AsV3PtpInstanceGroup OBJECT-GROUP
4     OBJECTS {
5         ieee8021AsV3PtpInstanceName,
6         ieee8021AsV3PtpInstanceRowStatus
7     }
8     STATUS      current
9     DESCRIPTION
10        "A collection of objects providing information for dynamic
11        creation and deletion of PTP Instances and logical ports."
12    ::= { ieee8021AsV3Groups 1 }
13
14 ieee8021AsV3DefaultDSGroup OBJECT-GROUP
15     OBJECTS {
16         ieee8021AsV3DefaultDSClockIdentity,
17         ieee8021AsV3DefaultDSNumberPorts,
18         ieee8021AsV3DefaultDSClockQualityClockClass,
19         ieee8021AsV3DefaultDSClockQualityClockAccuracy,
20         ieee8021AsV3DefaultDSClockQualityOffsetScaledLogVariance,
21         ieee8021AsV3DefaultDSPriority1,
22         ieee8021AsV3DefaultDSPriority2,
23         ieee8021AsV3DefaultDSGmCapable,
24         ieee8021AsV3DefaultDSCurrentUtcOffset,
25         ieee8021AsV3DefaultDSCurrentUtcOffsetValid,
26         ieee8021AsV3DefaultDSLeap59,
27         ieee8021AsV3DefaultDSLeap61,
28         ieee8021AsV3DefaultDSTimeTraceable,
29         ieee8021AsV3DefaultDSFrequencyTraceable,
30         ieee8021AsV3DefaultDSPTPTimescale,
31         ieee8021AsV3DefaultDSTimeSource,
32         ieee8021AsV3DefaultDSDomainNumber,
33         ieee8021AsV3DefaultDSSdoId,
34         ieee8021AsV3DefaultDSExternalPortConfigurationEnabled,
35         ieee8021AsV3DefaultDSInstanceEnable
36     }
37     STATUS      current
38     DESCRIPTION
39        "A collection of objects providing information on the Default
40        Parameter Data Set representing the native capabilities of a
41        PTP Instance, i.e., a PTP Relay Instance or a PTP End Instance."
42    ::= { ieee8021AsV3Groups 2 }
43
44 ieee8021AsV3CurrentDSGroup OBJECT-GROUP
45     OBJECTS {
46         ieee8021AsV3CurrentDSStepsRemoved,
47         ieee8021AsV3CurrentDSOffsetFromTimeTransmitter,
48         ieee8021AsV3CurrentDSLstGmPhaseChange,
49         ieee8021AsV3CurrentDSLstGmFreqChange,
50         ieee8021AsV3CurrentDSGmTimebaseIndicator,
51         ieee8021AsV3CurrentDSGmChangeCount,
52         ieee8021AsV3CurrentDSTimeOfLastGmChangeEvent,
53         ieee8021AsV3CurrentDSTimeOfLastGmPhaseChangeEvent,
54         ieee8021AsV3CurrentDSTimeOfLastGmFreqChangeEvent
55     }
56     STATUS      current
57     DESCRIPTION
58        "A collection of objects providing information on the Current
59        Parameter Data Set representing the position of a local system

```

```

1      and other information, relative to the Grandmaster PTP Instance."
2      ::= { ieee8021AsV3Groups 3 }
3
4 ieee8021AsV3ParentDSGroup OBJECT-GROUP
5     OBJECTS {
6         ieee8021AsV3ParentDSParentClockIdentity,
7         ieee8021AsV3ParentDSParentPortNumber,
8         ieee8021AsV3ParentDSCumulativeRateRatio,
9         ieee8021AsV3ParentDSGrandmasterIdentity,
10        ieee8021AsV3ParentDSGrandmasterClockQualityclockClass,
11        ieee8021AsV3ParentDSGrandmasterClockQualityclockAccuracy,
12        ieee8021AsV3ParentDSGrandmasterClockQualityoffsetScaledLogVar,
13        ieee8021AsV3ParentDSGrandmasterPriority1,
14        ieee8021AsV3ParentDSGrandmasterPriority2
15    }
16    STATUS      current
17    DESCRIPTION
18        "A collection of objects providing information on the Parent
19        Parameter Data Set representing capabilities of the upstream
20        system, toward the Grandmaster PTP Instance, as measured at
21        a local system."
22    ::= { ieee8021AsV3Groups 4 }
23
24 ieee8021AsV3TimePropertiesDSGroup OBJECT-GROUP
25     OBJECTS {
26         ieee8021AsV3TimePropertiesDSCurrentUtcOffset,
27         ieee8021AsV3TimePropertiesDSCurrentUtcOffsetValid,
28         ieee8021AsV3TimePropertiesDSLeap59,
29         ieee8021AsV3TimePropertiesDSLeap61,
30         ieee8021AsV3TimePropertiesDSTimeTraceable,
31         ieee8021AsV3TimePropertiesDSFrequencyTraceable,
32         ieee8021AsV3TimePropertiesDSPtpTimescale,
33         ieee8021AsV3TimePropertiesDSTimeSource
34     }
35     STATUS      current
36     DESCRIPTION
37         "A collection of objects providing information on the Time
38         Properties Parameter Data Set representing capabilities of
39         the Grandmaster PTP Instance, as measured at a local system."
40     ::= { ieee8021AsV3Groups 5 }
41
42 ieee8021AsV3PathTraceDSGroup OBJECT-GROUP
43     OBJECTS {
44         ieee8021AsV3PathTraceDSEnable
45     }
46     STATUS      current
47     DESCRIPTION
48         "A collection of objects providing information on the Path Trace
49         Data Set representing the current path trace information
50         available at the PTP Instance."
51     ::= { ieee8021AsV3Groups 6 }
52
53 ieee8021AsV3PathTraceDSArrayTableGroup OBJECT-GROUP
54     OBJECTS {
55         ieee8021AsV3PathTraceDSArrayList
56     }
57     STATUS      current
58     DESCRIPTION
59         "A collection of objects providing information of an array of

```

```

1      ClockIdentity values contained in the pathTrace array,
2      representing the current path trace information, and which is
3      carried in the path trace TLV per PTP Instance."
4      ::= { ieee8021AsV3Groups 7 }
5
6 ieee8021AsV3AcceptableTimeTransmitterTableDSGroup OBJECT-GROUP
7     OBJECTS {
8         ieee8021AsV3AcceptableTimeTransmitterTableDSMaxTableSize,
9         ieee8021AsV3AcceptableTimeTransmitterTableDSActualTableSize
10    }
11    STATUS      current
12    DESCRIPTION
13        "A collection of objects providing information on the
14        Acceptable TimeTransmitter Table Data Set representing the acceptable
15        timeTransmitter table used when an EPON port is used by a PTP Instance
16        of a time-aware system."
17    ::= { ieee8021AsV3Groups 8 }
18
19 ieee8021AsV3AcceptableTimeTransmitterTableDSArrayGroup OBJECT-GROUP
20     OBJECTS {
21         ieee8021AsV3AcceptableTimeTransmitterTableDSArrayPortIdentity,
22         ieee8021AsV3AcceptableTTTableDSArrayAlternatePriority1
23    }
24    STATUS      current
25    DESCRIPTION
26        "A collection of objects providing information on the
27        Acceptable TimeTransmitter Table Array Data Set representing the
28        acceptable timeTransmitter table used when an EPON port is used by a
29        PTP Instance of a time-aware system."
30    ::= { ieee8021AsV3Groups 9 }
31
32 ieee8021AsV3PortDSGroup OBJECT-GROUP
33     OBJECTS {
34         ieee8021AsV3PortDSClockIdentity,
35         ieee8021AsV3PortDSPortNumber,
36         ieee8021AsV3PortDSPortState,
37         ieee8021AsV3PortDSPtpPortEnabled,
38         ieee8021AsV3PortDSDelayMechanism,
39         ieee8021AsV3PortDSIsMeasuringDelay,
40         ieee8021AsV3PortDSAsCapable,
41         ieee8021AsV3PortDSMeanLinkDelay,
42         ieee8021AsV3PortDSMeanLinkDelayThresh,
43         ieee8021AsV3PortDSDelayAsym,
44         ieee8021AsV3PortDSNbrRateRatio,
45         ieee8021AsV3PortDSInitialLogAnnounceInterval,
46         ieee8021AsV3PortDSCurrentLogAnnounceInterval,
47         ieee8021AsV3PortDSUseMgtSettableLogAnnounceInterval,
48         ieee8021AsV3PortDSMgtSettableLogAnnounceInterval,
49         ieee8021AsV3PortDSAnnounceReceiptTimeout,
50         ieee8021AsV3PortDSInitialLogSyncInterval,
51         ieee8021AsV3PortDSCurrentLogSyncInterval,
52         ieee8021AsV3PortDSUseMgtSettableLogSyncInterval,
53         ieee8021AsV3PortDSMgtSettableLogSyncInterval,
54         ieee8021AsV3PortDSSyncReceiptTimeout,
55         ieee8021AsV3PortDSSyncReceiptTimeoutTimeInterval,
56         ieee8021AsV3PortDSInitialLogPdelayReqInterval,
57         ieee8021AsV3PortDSCurrentLogPdelayReqInterval,
58         ieee8021AsV3PortDSUseMgtSettableLogPdelayReqInterval,
59         ieee8021AsV3PortDSMgtSettableLogPdelayReqInterval,

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1      ieee8021AsV3PortDSInitialLogGtpCapableMessageInterval,
2      ieee8021AsV3PortDSCurrentLogGtpCapableMessageInterval,
3      ieee8021AsV3PortDSUseMgtSettableLogGtpCapableMessageInterval,
4      ieee8021AsV3PortDSMgtSettableLogGtpCapableMessageInterval,
5      ieee8021AsV3PortDSInitialComputeNbrRateRatio,
6      ieee8021AsV3PortDSCurrentComputeNbrRateRatio,
7      ieee8021AsV3PortDSUseMgtSettableComputeNbrRateRatio,
8      ieee8021AsV3PortDSMgtSettableComputeNbrRateRatio,
9      ieee8021AsV3PortDSInitialComputeMeanLinkDelay,
10     ieee8021AsV3PortDSCurrentComputeMeanLinkDelay,
11     ieee8021AsV3PortDSUseMgtSettableComputeMeanLinkDelay,
12     ieee8021AsV3PortDSMgtSettableComputeMeanLinkDelay,
13     ieee8021AsV3PortDSAllowedLostRsp,
14     ieee8021AsV3PortDSAllowedFaults,
15     ieee8021AsV3PortDSGTPCapableReceiptTimeout,
16     ieee8021AsV3PortDSVersionNumber,
17     ieee8021AsV3PortDSNup,
18     ieee8021AsV3PortDSNdown,
19     ieee8021AsV3PortDSOneStepTxOper,
20     ieee8021AsV3PortDSOneStepReceive,
21     ieee8021AsV3PortDSOneStepTransmit,
22     ieee8021AsV3PortDSInitialOneStepTxOper,
23     ieee8021AsV3PortDSCurrentOneStepTxOper,
24     ieee8021AsV3PortDSUseMgtSettableOneStepTxOper,
25     ieee8021AsV3PortDSMgtSettableOneStepTxOper,
26     ieee8021AsV3PortDSSyncLocked,
27     ieee8021AsV3PortDSPdelayTruncTST1,
28     ieee8021AsV3PortDSPdelayTruncTST2,
29     ieee8021AsV3PortDSPdelayTruncTST3,
30     ieee8021AsV3PortDSPdelayTruncTST4,
31     ieee8021AsV3PortDSMinorVersionNumber
32 }
33 STATUS      current
34 DESCRIPTION
35     "A collection of objects providing information on PTP Port
36     related variables in a time-aware Bridge or for a time-aware
37     end station."
38 ::= { ieee8021AsV3Groups 10 }
39
40 ieee8021AsV3DescriptionPortDSGroup OBJECT-GROUP
41     OBJECTS {
42         ieee8021AsV3DescriptionPortDSProfileIdentifier
43     }
44     STATUS      current
45     DESCRIPTION
46         "A collection of objects providing information on the
47         Description Port Data Set containing the profileIdentifier for
48         this PTP profile, as specified in F.2."
49     ::= { ieee8021AsV3Groups 11 }
50
51 ieee8021AsV3PortStatIfGroup OBJECT-GROUP
52     OBJECTS {
53         ieee8021AsV3PortStatRxSyncCount,
54         ieee8021AsV3PortStatRxOneStepSyncCount,
55         ieee8021AsV3PortStatRxFollowUpCount,
56         ieee8021AsV3PortStatRxPdelayRequestCount,
57         ieee8021AsV3PortStatRxPdelayRspCount,
58         ieee8021AsV3PortStatRxPdelayRspFollowUpCount,
59         ieee8021AsV3PortStatRxAnnounceCount,

```



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1      ieee8021AsV3PortStatRxPtpPacketDiscardCount,
2      ieee8021AsV3PortStatSyncReceiptTimeoutCount,
3      ieee8021AsV3PortStatAnnounceReceiptTimeoutCount,
4      ieee8021AsV3PortStatPdelayAllowedLostRspExceededCount,
5      ieee8021AsV3PortStatTxSyncCount,
6      ieee8021AsV3PortStatTxOneStepSyncCount,
7      ieee8021AsV3PortStatTxFollowUpCount,
8      ieee8021AsV3PortStatTxPdelayRequestCount,
9      ieee8021AsV3PortStatTxPdelayRspCount,
10     ieee8021AsV3PortStatTxPdelayRspFollowUpCount,
11     ieee8021AsV3PortStatTxAnnounceCount
12 }
13 STATUS      current
14 DESCRIPTION
15     "A collection of objects providing information on the Port
16     Statistics Data Set providing counters associated with PTP Port
17     capabilities at a given PTP Instance."
18 ::= { ieee8021AsV3Groups 12 }
19
20 ieee8021AsV3AcceptableTimeTransmitterPortDSGroup OBJECT-GROUP
21     OBJECTS {
22         ieee8021AsV3AcceptableTTPortDSAcceptableTTTableEnabled
23     }
24     STATUS      current
25     DESCRIPTION
26         "A collection of objects providing information for the single
27         PTP Port of a PTP End Instance and for each PTP Port of a
28         PTP Relay Instance."
29     ::= { ieee8021AsV3Groups 13 }
30
31 ieee8021AsV3ExternalPortConfigurationPortDSGroup OBJECT-GROUP
32     OBJECTS {
33         ieee8021AsV3ExternalPortConfigurationPortDSDesiredState
34     }
35     STATUS      current
36     DESCRIPTION
37         "A collection of objects providing information on the
38         External Port Configuration Port Data Set containing the
39         single member desiredState, which indicates the desired state
40         for the PTP Port."
41     ::= { ieee8021AsV3Groups 14 }
42
43 ieee8021AsV3AsymMeasurementModeDSGroup OBJECT-GROUP
44     OBJECTS {
45         ieee8021AsV3AsymMeasurementModeDSAsymMeasurementMode
46     }
47     STATUS      current
48     DESCRIPTION
49         "A collection of objects providing information on the
50         Asymmetry Measurement Mode Data Set representing the capability
51         to enable/disable the Asymmetry Compensation Measurement
52         Procedure on a Link Port (see Annex G)."
53     ::= { ieee8021AsV3Groups 15 }
54
55 ieee8021AsV3CommonServicesPortDSGroup OBJECT-GROUP
56     OBJECTS {
57         ieee8021AsV3CommonServicesPortDSCmlDsLinkPortPortNumber
58     }
59     STATUS      current

```

```

1  DESCRIPTION
2      "A collection of objects providing information on the
3      Common Services Port Data Set."
4      ::= { ieee8021AsV3Groups 16 }
5
6  ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSGroup OBJECT-GROUP
7      OBJECTS {
8          ieee8021AsV3CmlDsDefaultDSClockIdentity,
9          ieee8021AsV3CmlDsDefaultDSNumberLinkPorts
10     }
11     STATUS      current
12     DESCRIPTION
13         "A collection of objects providing information on the
14         CMLDs Default Data Set describing the per-time-aware-system
15         attributes of the Common Mean Link Delay Service."
16         ::= { ieee8021AsV3Groups 17 }
17
18  ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSGroup OBJECT-GROUP
19     OBJECTS {
20         ieee8021AsV3CmlDsLinkPortDSClockIdentity,
21         ieee8021AsV3CmlDsLinkPortDSPortNumber,
22         ieee8021AsV3CmlDsLinkPortDSCmlDsLinkPortEnabled,
23         ieee8021AsV3CmlDsLinkPortDSIsMeasuringDelay,
24         ieee8021AsV3CmlDsLinkPortDSAsCapableAcrossDomains,
25         ieee8021AsV3CmlDsLinkPortDSMeanLinkDelay,
26         ieee8021AsV3CmlDsLinkPortDSMeanLinkDelayThresh,
27         ieee8021AsV3CmlDsLinkPortDSDelayAsym,
28         ieee8021AsV3CmlDsLinkPortDSNbrRateRatio,
29         ieee8021AsV3CmlDsLinkPortDSInitialLogPdelayReqInterval,
30         ieee8021AsV3CmlDsLinkPortDSCurrentLogPdelayReqInterval,
31         ieee8021AsV3CmlDsLinkPortDSUseMgtSettableLogPdelayReqInterval,
32         ieee8021AsV3CmlDsLinkPortDSMgtSettableLogPdelayReqInterval,
33         ieee8021AsV3CmlDsLinkPortDSInitialComputeNbrRateRatio,
34         ieee8021AsV3CmlDsLinkPortDSCurrentComputeNbrRateRatio,
35         ieee8021AsV3CmlDsLinkPortDSUseMgtSettableComputeNbrRateRatio,
36         ieee8021AsV3CmlDsLinkPortDSMgtSettableComputeNbrRateRatio,
37         ieee8021AsV3CmlDsLinkPortDSInitialComputeMeanLinkDelay,
38         ieee8021AsV3CmlDsLinkPortDSCurrentComputeMeanLinkDelay,
39         ieee8021AsV3CmlDsLinkPortDSUseMgtSettableComputeMeanLinkDelay,
40         ieee8021AsV3CmlDsLinkPortDSMgtSettableComputeMeanLinkDelay,
41         ieee8021AsV3CmlDsLinkPortDSAllowedLostRsp,
42         ieee8021AsV3CmlDsLinkPortDSAllowedFaults,
43         ieee8021AsV3CmlDsLinkPortDSVersionNumber,
44         ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST1,
45         ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST2,
46         ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST3,
47         ieee8021AsV3CmlDsLinkPortDSPdelayTruncTST4,
48         ieee8021AsV3CmlDsLinkPortDSMinorVersionNumber
49     }
50     STATUS      current
51     DESCRIPTION
52         "A collection of objects providing information for every
53         Link Port of the Common Mean Link Delay Service of a
54         time-aware system."
55         ::= { ieee8021AsV3Groups 18 }
56
57  ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSGroup OBJECT-GROUP
58     OBJECTS {
59         ieee8021AsV3CmlDsLinkPortStatDSRxDelayRequestCount,

```

```

1      ieee8021AsV3CmlDsLinkPortStatDSRxPdelayRspCount,
2      ieee8021AsV3CmlDsLinkPortStatDSRxPdelayRspFollowUpCount,
3      ieee8021AsV3CmlDsLinkPortStatDSRxPtpPacketDiscardCount,
4      ieee8021AsV3CmlDsLinkPortStatDSPdelayAllowedLostRspExceededCount,
5      ieee8021AsV3CmlDsLinkPortStatDSTxPdelayRequestCount,
6      ieee8021AsV3CmlDsLinkPortStatDSTxPdelayRspCount,
7      ieee8021AsV3CmlDsLinkPortStatDSTxPdelayRspFollowUpCount
8  }
9  STATUS      current
10 DESCRIPTION
11     "A collection of objects providing information for every
12     Link Port Statistics of the Common Mean Link Delay Service of a
13     time-aware system."
14 ::= { ieee8021AsV3Groups 19 }
15
16 ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSGroup OBJECT-GROUP
17     OBJECTS {
18         ieee8021AsV3CmlDsAsymMeasurementModeDSAsymMeasurementMode
19     }
20     STATUS      current
21     DESCRIPTION
22         "A collection of objects providing information on the
23         Common Mean Link Delay Service Asymmetry Measurement Mode
24         Parameter Data Set representing the capability to enable/disable
25         the Asymmetry Compensation Measurement Procedure on a Link Port
26         (see Annex G)."
27 ::= { ieee8021AsV3Groups 20 }
28
29 -- =====
30 -- compliance statements
31 -- =====
32
33 ieee8021AsV3Compliance MODULE-COMPLIANCE
34     STATUS      current
35     DESCRIPTION
36         "The compliance statement for devices supporting
37         IEEE Std 802.1AS-2020."
38
39     MODULE -- this module
40
41     GROUP ieee8021AsV3PtpInstanceGroup
42     DESCRIPTION
43         "Implementation of this group is optional."
44
45     GROUP ieee8021AsV3DefaultDSGroup
46     DESCRIPTION
47         "Implementation of this group is optional."
48
49     GROUP ieee8021AsV3CurrentDSGroup
50     DESCRIPTION
51         "Implementation of this group is optional."
52
53     GROUP ieee8021AsV3ParentDSGroup
54     DESCRIPTION
55         "Implementation of this group is optional."
56
57     GROUP ieee8021AsV3TimePropertiesDSGroup
58     DESCRIPTION
59         "Implementation of this group is optional."

```

1
2 GROUP ieee8021AsV3PathTracedDSGroup
3 DESCRIPTION
4 "Implementation of this group is optional."
5
6 GROUP ieee8021AsV3PathTracedDSArrayTableGroup
7 DESCRIPTION
8 "Implementation of this group is optional."
9
10 GROUP ieee8021AsV3AcceptableTimeTransmitterTableDSGroup
11 DESCRIPTION
12 "Implementation of this group is optional."
13
14 GROUP ieee8021AsV3AcceptableTimeTransmitterTableDSArrayGroup
15 DESCRIPTION
16 "Implementation of this group is optional."
17
18 GROUP ieee8021AsV3PortDSGroup
19 DESCRIPTION
20 "Implementation of this group is optional."
21
22 GROUP ieee8021AsV3DescriptionPortDSGroup
23 DESCRIPTION
24 "Implementation of this group is optional."
25
26 GROUP ieee8021AsV3PortStatIfGroup
27 DESCRIPTION
28 "Implementation of this group is optional."
29
30 GROUP ieee8021AsV3AcceptableTimeTransmitterPortDSGroup
31 DESCRIPTION
32 "Implementation of this group is optional."
33
34 GROUP ieee8021AsV3ExternalPortConfigurationPortDSGroup
35 DESCRIPTION
36 "Implementation of this group is optional."
37
38 GROUP ieee8021AsV3AsymMeasurementModeDSGroup
39 DESCRIPTION
40 "Implementation of this group is optional."
41
42 GROUP ieee8021AsV3CommonServicesPortDSGroup
43 DESCRIPTION
44 "Implementation of this group is optional."
45
46 GROUP ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSGroup
47 DESCRIPTION
48 "Implementation of this group is optional."
49
50 GROUP ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSGroup
51 DESCRIPTION
52 "Implementation of this group is optional."
53
54 GROUP ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSGroup
55 DESCRIPTION
56 "Implementation of this group is optional."
57
58 GROUP ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSGroup
59 DESCRIPTION

```
1      "Implementation of this group is optional."  
2  
3      ::= { ieee8021AsV3Compliances 1 }  
4  
5 END  
6
```

16. Media-dependent layer specification for CSN

16.1 Overview

Accurate synchronized time is distributed throughout a gPTP domain through time measurements between adjacent PTP Relay Instances or PTP End Instances in a packet network. Time is communicated from the root of the synchronization tree (i.e., the Grandmaster PTP Instance) toward the leaves of the synchronization tree (i.e., from leaf-facing “timeTransmitter” ports to root-facing “timeReceiver” ports) through measurements made across the links connecting the PTP Instances. While the semantics of time transfer are consistent across the time-aware packet network, the method for communicating synchronized time from a timeTransmitter port to its immediate downstream link partner varies depending on the type of link interconnecting the two PTP Instances.

This clause specifies the protocol that provides accurate synchronized time across links of a *coordinated shared network* (CSN) as part of a packet network.

16.2 Coordinated Shared Network characteristics

A CSN is a contention-free, time-division, multiplexed-access network of devices sharing a common medium and supporting reserved bandwidth based on priority or flow (QoS). One of the nodes of the CSN acts as the network coordinator, granting transmission opportunities to the other CSN nodes of the network. A CSN physically is a shared medium, in that a CSN node has a single physical port connected to the half-duplex medium, but is logically a fully connected one-hop mesh network, in that every CSN node can transmit frames to every other CSN node over the shared medium.

A CSN supports two types of transmission: unicast transmission for point-to-point (CSN node-to-node) transmission and multicast/broadcast transmission for point-to-multipoint (CSN node-to-other/all-nodes) transmission. Figure 16-1 illustrates a CSN acting as a backbone for PTP Instances.

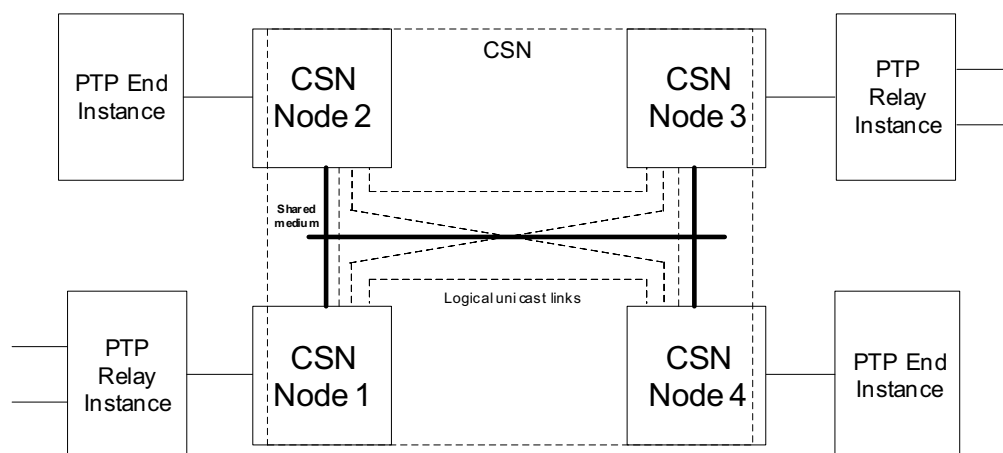


Figure 16-1—Example of CSN backbone

NOTE—In this clause, the term *node* is used to refer to a CSN node (i.e., it does not refer to a PTP Relay Instance or PTP End Instance). A CSN node is a 2-port PTP Relay Instance that forwards data packets between a segment external to the CSN (which can connect to an upstream or downstream PTP Instance) and the CSN, all at the data link layer. Nonetheless, to avoid confusion the term *node* is usually preceded by *CSN*, except when it is obvious that CSN nodes are being referenced.

1 16.3 Layering for CSN links

2 One PortSync entity and one MD entity are together associated with each CSN logical port (CSN node-to-
3 node link) as illustrated in Figure 16-2. The PortSync entities is described in 10.1.2. The MD entity
4 translates media-independent primitives to MD primitives as necessary for communicating synchronized
5 time over the CSN links. The CSN MD entity shall implement the MDSyncSendSM and
6 MDSyncReceiveSM states machines of 11.2.14 and 11.2.15.

7 The CSN MD entity either implements the MDPdelayReq and MDPdelayResp state machines of 11.2.19
8 and 11.2.20 to measure the propagation delay on a CSN link, or measures it through a CSN-native method
9 and populates the variables described in 16.4.3.3.

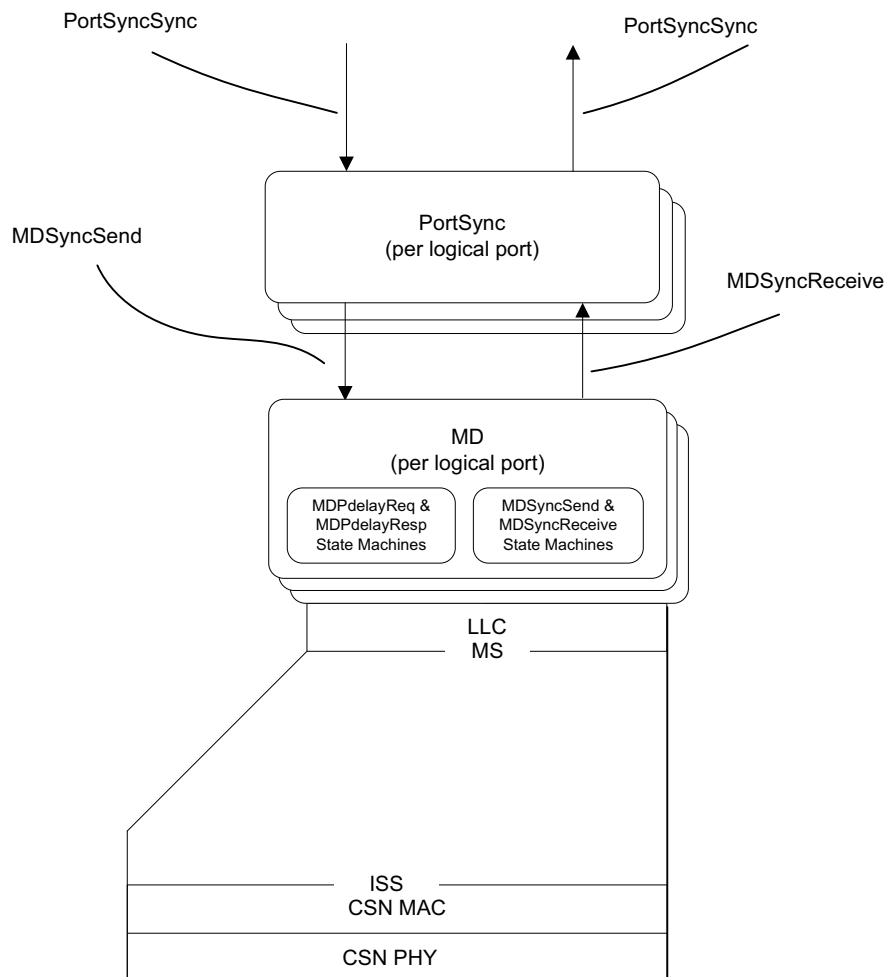


Figure 16-2—Media-dependent and lower entities in CSN nodes

1 16.4 Path delay measurement over a CSN backbone

2 16.4.1 General

3 The Path Delay over a CSN backbone is calculated for the following path types:

- 4 a) Between the upstream PTP Relay Instance and the ingress CSN node
- 5 b) Between the ingress and egress CSN nodes
- 6 c) Between the egress CSN node and the downstream PTP Instance (PTP Relay Instance or PTP End Instance)

8 To maintain the synchronization, residence time on each PTP Instance and the propagation delay between
9 PTP Instances is measured, requiring precise timestamping on both CSN node ingress and egress ports as
10 illustrated in Figure 16-4 (the numbered paths in Figure 16-4 refer to the numbered paths in Figure 16-3). In
11 Figure 16-4, ti_1 is the syncEventEgressTimestamp for the Sync message at the upstream PTP Relay
12 Instance, ti_2 is the syncEventIngressTimestamp for the Sync message at the ingress CSN time-aware node,
13 te_1 is the syncEventEgressTimestamp for the Sync message at the egress CSN time-aware node, and te_2 is
14 the syncEventIngressTimestamp for the Sync message at the downstream PTP Relay Instance or PTP End
15 Instance.

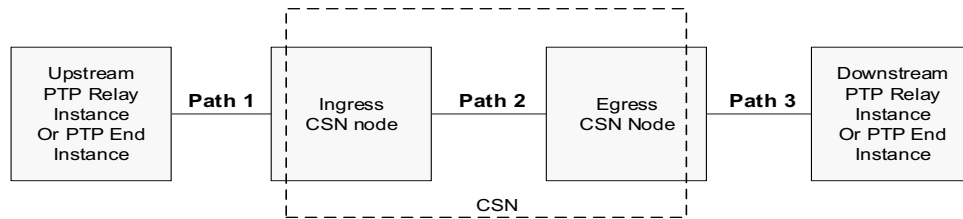


Figure 16-3—Path types over CSN as IEEE 802.1AS backbone

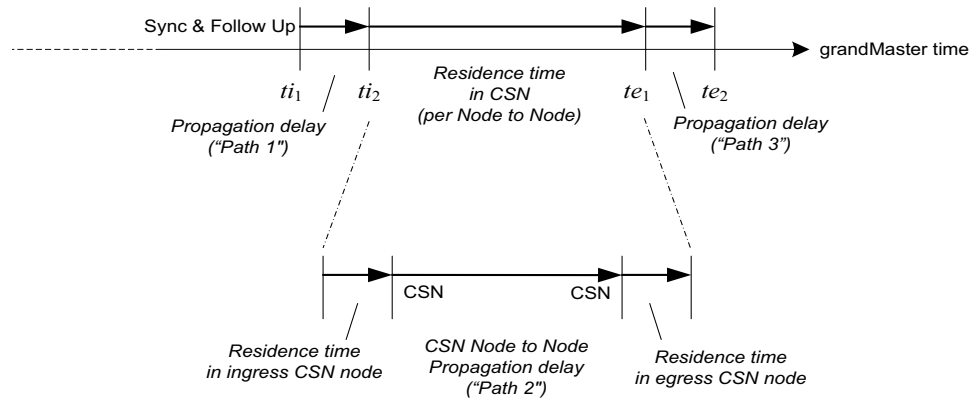


Figure 16-4—Propagation delay and residence time over a CSN backbone

16 16.4.2 Path delay measurement between CSN node and neighbor PTP Instance

17 The path delay measurement between a CSN node and a neighbor PTP Instance is made as specified for
18 the respective medium. This path delay measurement is made for the link between the CSN node and the
19 neighbor PTP Instance.

1 16.4.3 Path delay measurement between CSN nodes

2 16.4.3.1 General

3 The path delay between the two nodes of a CSN is the propagation delay for the logical link that connects
4 those two nodes. The method of measuring the path delay between two CSN nodes has two variations,
5 which are described in 16.4.3.2 and 16.4.3.3. The specific method to be used for a specific link technology is
6 specified in 16.6.

7 16.4.3.2 Path delay measurement without network clock reference

8 Each CSN node has a free-running local clock. The path delay measurement uses the peer-to-peer delay
9 mechanism protocol, messages, and state machines described in Clause 11 for full-duplex point-to-point
10 links, as illustrated by Figure 16-5. The criteria of 11.2.17 for determining whether the peer-to-peer delay
11 mechanism is transport-specific peer-to-peer delay mechanism or the CMLDS apply here.

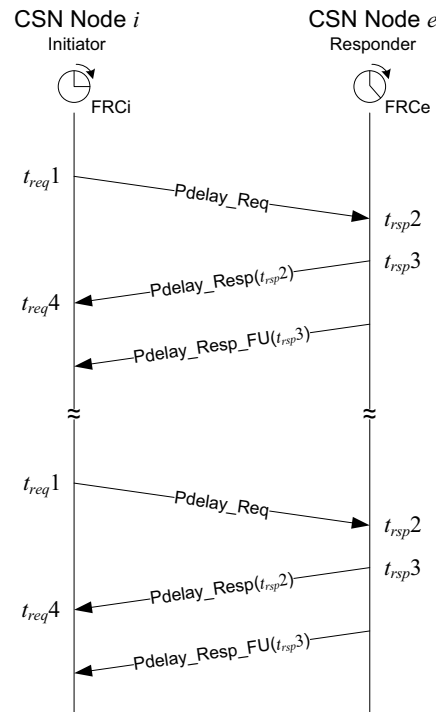


Figure 16-5—CSN node-to-node path delay measurement

12 The computation of the neighborRateRatio and meanLinkDelay between two CSN nodes is done using the
13 timestamps at the initiator and information conveyed in the successive Pdelay_Resp and
14 Pdelay_Resp_Follow_Up messages. Any scheme that uses this information is acceptable, as long as the
15 performance requirements of B.2.4 are met. As one example, the neighborRateRatio is computed as the ratio
16 between a time interval measured by the local clock of the responder and its associated time interval
17 measured by the local clock of the initiator, using a set of received Pdelay_Resp and
18 Pdelay_Resp_Follow_Up messages and a second set of received Pdelay_Resp and
19 Pdelay_Resp_Follow_Up messages some number of Pdelay_Req message transmission intervals later, i.e.,
20 as show in Equation (16-1).

$$\frac{(t_{rsp3})_N - (t_{rsp3})_0}{(t_{req4})_N - (t_{req4})_0} \quad (16-1)$$

2 where

- 3 $(t_{rsp3})_k$ is the time relative to the local clock of the responder that the k^{th} Pdelay_Resp message is sent
- 4 $(t_{req4})_k$ is the time relative to the local clock of the initiator that the k^{th} Pdelay_Resp message is
- 5 received
- 6 N is the number of Pdelay_Req message transmission intervals separating the first set of received
- 7 Pdelay_Resp and Pdelay_Resp_Follow_Up messages and the second set

8 The successive sets of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages are indexed from 0 to

9 N with the first set indexed 0. The meanLinkDelay between the PTP Instance and the CSN node is computed

10 as shown in Equation (16-2).

$$\frac{(t_{req4} - t_{req1})r - (t_{rsp3} - t_{rsp2})}{2} \quad (16-2)$$

12 where

- 13 r is equal to neighborRateRatio
- 14 t_{req1} is the time relative to the local clock of the initiator that the Pdelay_Req message for this message
- 15 exchange is sent
- 16 t_{rsp2} is the time relative to the local clock of the responder that the Pdelay_Req message for this message
- 17 exchange is received
- 18 t_{rsp3} is the time relative to the local clock of the responder that the Pdelay_Resp message for this
- 19 message exchange is sent
- 20 t_{req4} is the time relative to the local clock of the initiator that the Pdelay_Resp message for this message
- 21 exchange is received

22 NOTE—The difference between mean propagation delay relative to the Grandmaster Clock time base and relative to the

23 time base of the CSN node at the other end of the attached link (i.e., the responder CSN node) is usually negligible. To

24 see this, note that the former can be obtained from the latter by multiplying the latter by the ratio of the Grandmaster

25 Clock frequency to the frequency of the LocalClock entity of the CSN node at the other end of the link. This ratio differs

26 from 1 by 200 ppm or less. For example, for a worst-case frequency offset of the LocalClock entity of the CSN node at

27 the other end of the link, relative to the Grandmaster Clock, of 200 ppm, and a measured propagation time of 100 ns, the

28 difference in mean propagation delay relative to the two time bases is 20 ps.

29 Although the propagation delay between two CSN nodes is constant, a Pdelay_Req message is still sent

30 periodically by each CSN node to each other active CSN node to measure the neighborRateRatio between

31 the node and each other node. Each CSN node shall implement the state machines described in 11.2.19 and

32 11.2.20.

33 16.4.3.3 Native CSN path delay measurement

34 Some CSN technologies feature a native mechanism that provides a path delay measurement with accuracy

35 similar to the accuracy the peer delay protocol provides. For these CSNs, the path delay can be provided

36 using the native measurement method rather than using the Pdelay protocol defined in 11.2.19 and 11.2.20.

37 Such a situation is described in more detail as follows. The CSN MD entity populates the following per-PTP

38 Port and MD-entity global variables (described respectively in 10.2.5 and 11.2.13) as indicated:

- 39 — asCapable (10.2.5.1) is set to TRUE.
- 40 — neighborRateRatio (10.2.5.7) is set to the value provided by the native CSN measurement.
- 41 — meanLinkDelay (10.2.5.8) is set to the value provided by the native CSN measurement.
- 42 — computeNeighborRateRatio (10.2.5.10) is set to FALSE.

- 1 — computeMeanLinkDelay (10.2.5.11) is set to FALSE.
- 2 — isMeasuringDelay (11.2.13.6) is set to TRUE to indicate that the CSN MD entity is measuring path
- 3 delay (in this case, using its internal mechanism).
- 4 — domainNumber (8.1) is set to the domainNumber of this gPTP domain.

5 16.4.3.4 Intrinsic CSN path delay measurement

6 If the CSN features a native mechanism that causes each CSN node's local clock to be fully synchronized to
7 the local clocks of other nodes of the CSN such that the synchronized CSN time complies with the
8 requirements specified in B.1, the CSN nodes need not implement the path delay mechanism but rather can
9 treat the path delay as part of the residence time of the distributed system. The propagation of the Sync
10 messages in this case is described in 16.5.3.

11 16.5 Synchronization messages

12 16.5.1 General

13 The CSN shall propagate synchronized time over the CSN to CSN end stations and to downstream non-CSN
14 links, using Sync (and, if the message is two-step, the associated Follow_Up) messages, as illustrated in
15 Figure 16-6.

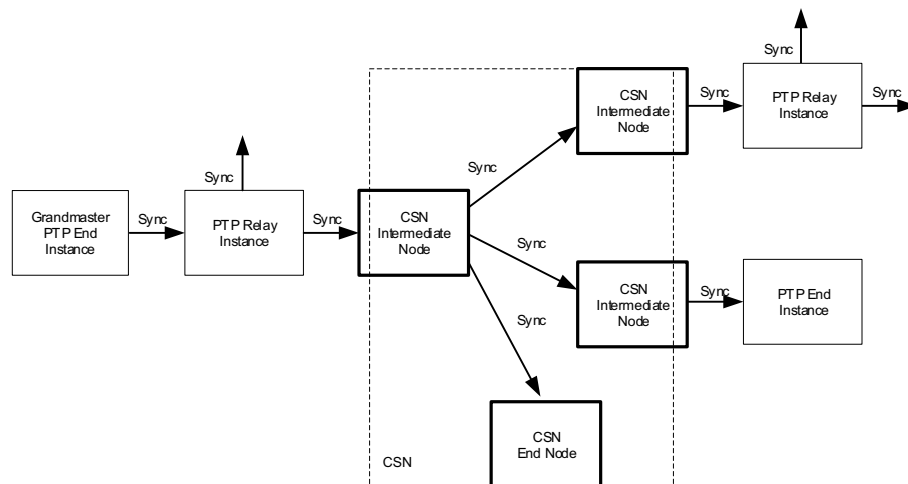


Figure 16-6—IEEE 802.1AS Sync message propagation over the CSN backbone

16 Once the path delays have been measured (a) between the upstream PTP Relay Instance or PTP End
17 Instance and the ingress CSN node, (b) between the CSN nodes, and (c) between the egress CSN node and
18 the downstream PTP Relay Instance or PTP End Instance, the CSN backbone can propagate the
19 synchronization information received at its boundary nodes.

20 As with path delay measurements, various CSN technologies choose various methods for propagating time.
21 These methods are described in 16.5.2 and 16.5.3.

22 16.5.2 Synchronization message propagation on CSN without network reference clock

23 If the CSN does not feature a native mechanism that synchronizes the CSN node local clocks to each other
24 or to a reference (i.e., to synchronize the CSN time in compliance with the requirements specified in B.1),

1 the CSN local clocks at CSN ingress and CSN egress nodes are considered independent, free-running
2 clocks.

3 In this case, synchronization over the CSN links uses the Sync and Follow_Up protocol (or only Sync if
4 optional one-step processing is used), messages, and state machines specified for full-duplex point-to-point
5 links in 10.2.8, 10.2.12, 11.2.14, and 11.2.15. Individual CSN link technologies may specify media-
6 dependent encapsulation of gPTP event messages. See Table 16-2 for selection of options per link
7 technology.

8 One PortSync and one MD entity is instantiated per logical port (i.e., per CSN link). A CSN node behaves
9 equivalently to a PTP Instance. Sync and, in the two-step case, Follow_Up messages are either transmitted
10 using unicast on each link or broadcasted. However, if Sync and Follow_Up messages are broadcasted, the
11 following apply:

- 12 — The Sync and Follow_Up messages are broadcast with the same port number used to broadcast
- 13 Announce messages.
- 14 — All PortSync/MD entity pairs except one set their logSyncInterval attribute (see 10.7.2.3) to 127 so
- 15 that they do not generate any Sync messages.
- 16 — A dynamic selection of the MD entity that broadcasts the Sync message is needed (a CSN node can
- 17 dynamically leave the CSN). The dynamic selection algorithm is implementation specific and out of
- 18 the scope of this standard.

19 **16.5.3 Synchronization message propagation on a CSN with network reference clock**

20 **16.5.3.1 General**

21 If the CSN features a native mechanism that allows the CSN node's local clocks to be fully synchronized to
22 each other in a way that complies with the requirements specified in B.1, it is possible to simplify the path
23 delay mechanism, as described below. This method is an alternative to 16.5.2. Individual CSN link
24 technologies may specify media-dependent encapsulation of gPTP event messages. See Table 16-2 (in
25 16.6.1) for selection of options per link technology.

26 Sync messages are timestamped when they are:

- 27 — Received at the ingress CSN node's PTP Port (syncEventIngressTimestamp) and
- 28 — Transmitted at the egress CSN node's PTP Port (syncEventEgressTimestamp).

29 The elapsed time between the egress and ingress timestamps is computed as the CSN residence time. In this
30 scheme, the Sync message handling is split between the MDSyncSendSM state machine in the CSN ingress
31 node and the MDSyncReceiveSM state machine in the CSN egress node as described in 16.5.3.2 and
32 16.5.3.3.

33 The reference plane for a CSN port and the path delay measurement method are specific to the type of CSN
34 technology, and are defined in Table 16-2.

35 **16.5.3.2 CSN ingress node**

36 **16.5.3.2.1 General**

37 The CSN ingress node timestamps Sync messages received from the upstream PTP Instance and compute
38 the upstreamTxTime as described in item f) of 11.2.14.2.1.

39 In addition, the setFollowUp function of the MDSyncSendSM state machine [see item a) in 11.2.15.2.3] is
40 modified as follows:

- 1 a) The quantity $rateRatio \times (syncEventEgressTimestamp - upstreamTxTime)$ is not added to the
- 2 followUpCorrectionField of the Follow_Up message (or the Sync message in the one-step case) to
- 3 be transmitted by the ingress to the egress CSN node
- 4 b) The CSN TLV (see 16.5.3.2.2) is appended to the Follow_Up message (or to the Sync message in
- 5 the one-step case) transmitted by the ingress to the egress CSN node.

6 **16.5.3.2.2 CSN TLV**

7 **16.5.3.2.2.1 General**

8 The fields of the CSN TLV are specified in Table 16-1 and 16.5.3.2.2.2 through 16.5.3.2.2.10. This TLV is a
9 standard organization extension TLV for the Follow_Up message (or the Sync message in the one-step
10 case), as specified in 14.3 of IEEE Std 1588-2019. This TLV is not allowed to occur before the Follow_Up
11 information TLV (see 11.4.4.3).

Table 16-1—CSN TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
upstreamTXTime								12	10
neighborRateRatio								4	22
meanLinkDelay								12	26
delayAsymmetry								12	38
domainNumber								1	50

12 **16.5.3.2.2.2 tlvType (Enumeration16)**

13 The value of the tlvType field is 0x3.

14 NOTE—This value indicates the TLV is a vendor and standard organization extension TLV, as specified in 14.3.2.1 and
15 Table 50 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION with a
16 value of 0x3.

17 **16.5.3.2.2.3 lengthField (UInteger16)**

18 The value of the length is 47.

1 16.5.3.2.2.4 organizationId (Octet3)

2 The value of organizationId is 00-80-C2.

3 16.5.3.2.2.5 organizationSubType (Enumeration24)

4 The value of organizationSubType is 3.

5 16.5.3.2.2.6 upstreamTxTime (UScaledNs)

6 The computed upstreamTxTime value is described in item f) of 11.2.14.2.1.

7 16.5.3.2.2.7 neighborRateRatio (Integer32)

8 The neighborRateRatio value is described in 10.2.5.7.

9 16.5.3.2.2.8 meanLinkDelay (UScaledNs)

10 The meanLinkDelay value is described in 10.2.5.8.

11 16.5.3.2.2.9 delayAsymmetry (ScaledNs)

12 The delayAsymmetry value is described in 10.2.5.9.

13 16.5.3.2.2.10 domainNumber (UInteger8)

14 This parameter is the domainNumber of this gPTP domain.

15 16.5.3.3 CSN egress node

16 The CSN egress port sets neighborRateRatio to 1 and meanLinkDelay to 0 for its CSN port.

17 The CSN egress port modifies the function setMDSyncReceive of the MDSyncReceiveSM state machine
18 [item f) in 11.2.14.2.1] for the port attached to the CSN link entity to extract upstreamTxTime,
19 meanLinkDelay, and neighborRateRatio from the respective fields of the CSN TLV in the Follow_Up
20 message (or Sync message in the one-step case) received from the CSN ingress node.

21 The CSN egress port also modifies the ClockTimeReceiverSync state machine (see 10.2.13) to get the
22 upstreamTxTime, meanLinkDelay, neighborRateRatio, and delayAsymmetry values from the respective
23 fields of the CSN TLV in the Follow_Up message (or Sync message in the one-step case) received from the
24 CSN ingress node.

25 The CSN egress node removes the CSN TLV from the Follow_Up message (or Sync message in the one-
26 step case) received from the CSN ingress node and transmits the message to the downstream PTP Instance.

1 16.6 Specific CSN requirements

2 16.6.1 General

3 The reference plane for a CSN port is specific to the type of CSN technology and is defined in Table 16-2.

Table 16-2—Definitions and option selections per link technology

CSN link technology	Reference plane	Path Delay measurement	gPTP event message encapsulation
Multimedia over Coax Alliance (MoCA) v2.0	The first bit of an Event message crossing to and from the MoCA CTC domain.	MoCA Ranging Protocol (method of either 16.4.3.3 or 16.4.3.4)	Encapsulated in control frames as described in the MoCA MAC/PHY Specification v2.0.
ITU-T G.hn (SG15)	The first bit of an Event message crossing the A-interface (see 16.6.3).	peer-to-peer mechanism as defined in 16.4.3.2.	None

4 16.6.2 MoCA-specific behavior

5 The MoCA network is a CSN. The non-MoCA port of a CSN time-aware node behaves as a PTP Port, which might or might not use the MoCA channel time clock (CTC) for timestamping event messages. If the non-MoCA port of the CSN time-aware node uses a different clock than the MoCA CTC, then the CSN time-aware node reconciles the non-CTC timestamp with the CTC time.

9 Sync messages shall be timestamped using the CTC when they cross:

- 10 — The MoCA ingress node's timestamp reference plane (syncEventIngressTimestamp) and
- 11 — The egress CSN node's reference plane (syncEventEgressTimestamp).

12 The elapsed time between the egress and ingress timestamps, syncEventIngressTimestamp – syncEventEgressTimestamp, is computed as the MoCA residence time.

14 The MoCA port whose PTP Port state is TimeTransmitterPort propagates the Sync and Follow_Up messages (or only the Sync message in the one-step case) as described in 16.5.3. The CSN TLV values of the Follow_Up message sent over the MoCA network are computed using the LocalClock, i.e., the MoCA CTC.

17 IEEE 802.1AS frames shall be transmitted over the MoCA network as MoCA control frames as described in the MoCA MAC/PHY Specification v2.0.

19 NOTE—The CTC is specified in the MoCA MAC/PHY Specification v2.0.

1 16.6.3 ITU-T G.hn-specific behavior

2 A port of a PTP Instance that includes one or more ITU-T G.hn ports shall behave as defined in 16.3, 16.4,
3 and 16.5; however, for aspects where more than one behavior or option is described, the system behaves as
4 defined in Table 16-2.

5 ITU-T G.hn defines a 32-bit timestamp (which is placed in the TSMP field of an ITU-T G.hn encapsulation
6 header). This timestamp, as described in Table 16-2, is captured each time an Event message is transmitted
7 or received by an ITU-T G.hn port and is used for all gPTP event messages, including SYNC and PDELAY
8 messages.

9 16.7 Grandmaster PTP Instance capability

10 Each CSN Node may be grandmaster capable to allow a CSN node to act as the Grandmaster PTP Instance
11 either for a homogeneous CSN or for a heterogeneous network.

12 The Announce messages are either sent via unicast on each link or broadcasted. However, if Announce
13 messages are broadcasted, the Announce message shall use the same port number as used by the Sync and
14 Follow_Up messages (or only the Sync message in the one-step case) by a single PortSync/MD entity pair
15 on the PTP Port. This is accomplished by setting the logAnnounceInterval attribute (see 10.7.2.2) to 127 for
16 all but one PortSync/MD entity pair so that they do not generate any Announce messages.

17 16.8 CSN clock and node performance requirements

18 The CSN clock performance complies with the requirements specified in B.1. The CSN node performance
19 complies with the requirements specified in B.2.4 and should follow the recommendations of B.2.2 and
20 B.2.3.

1 17. YANG data model

2 YANG (IETF RFC 7950) is a data modeling language used to model configuration data and state data for
3 remote network management protocols. YANG-based remote network management protocols include the
4 Network Configuration Protocol (NETCONF) (IETF RFC 6242 [B13a]) and RESTCONF (IETF RFC 8040
5 [B13e]). Each remote network management protocol uses a specific encoding on the wire, such as XML or
6 JSON. A YANG module specifies the organization and rules for the management data, and a mapping from
7 YANG to the specific encoding enables the data to be understood correctly by both client (e.g., network
8 manager) and server (e.g., PTP Instances).

9 This clause specifies the YANG data model for IEEE Std 802.1AS.

10 This clause:

- 11 a) Introduces the organization of the data models, including the relationship with other standards (17.1)
- 12 b) Provides an overview of the hierarchy of the data models using a representation similar to the
13 Unified Modeling Language (UML) (17.2)
- 14 c) Summarizes the structure of the YANG data model (17.3)
- 15 d) Reviews security considerations (17.4)
- 16 e) Provides a schema tree as an overview of the YANG module (17.5)
- 17 f) Specifies the YANG module ()

18 17.1 YANG framework

19 The YANG framework applies hierarchy in the following areas:

- 20 a) The Uniform Resource Name (URN), as specified in IEEE Std 802.
- 21 b) The YANG objects form a hierarchy of configuration and operational data structures that define the
22 YANG model.

23 Clause 14 specifies the information model for management of this standard. The data model for a specific
24 management mechanism is derived from the information model. Since YANG-based protocols are an
25 example of a management mechanism, the YANG data model of this clause is derived from Clause 14.

26 NOTE—The MIB modules specified in Clause 15 were also derived from Clause 14. Consequently, the capabilities and
27 structure of the YANG data models are aligned with those represented by the MIB. However the YANG data model has
28 not been derived from the MIB, and there has been no attempt to include data or modeling constructs that might appear
29 in the MIB but not in the information model.

30 The information model in Clause 14 is organized as a hierarchy of data sets. Each data set contains one or
31 more related members (items of data that can be read or written). In the context of YANG, each data set is
32 represented as a YANG “container,” and each member is represented as a YANG “leaf.”

33 17.1.1 Relationship to the IEEE Std 1588 data model

34 The YANG data models specified in this standard are based on, and augment, those specified in IEEE
35 Std 1588. In particular the `ieee802-dot1as-gtp.yang` module imports the `ieee1588-ptp-tt` module as a whole,
36 augmenting that module as necessary to meet the requirements of this standard. In addition, the
37 `ieee802-dot1as-hs.yang` module imports the `ieee1588-ptp-tt` and `ieee802-dot1as-gtp` modules as a whole,
38 augmenting those modules as necessary to meet the requirements of this standard.

1 Some of the data sets in Clause 14 (e.g., defaultDS) are derived from IEEE Std 1588, and some of the data
2 sets are unique to IEEE Std 802.1AS (i.e., not derived from IEEE Std 1588). For each data set in Clause 14
3 that is derived from IEEE Std 1588, a portion of the members are derived from IEEE Std 1588, and the
4 remaining members are unique to IEEE Std 802.1AS. For the members that are derived from IEEE Std 1588,
5 the specifications in both standards are analogous (same name, data type, semantics, etc.).

6 The YANG data model for IEEE Std 1588-2019 is published as amendment IEEE Std 1588e. The YANG
7 module of IEEE Std 1588e (ieee1588-ptp-tt.yang) contains the hierarchy (tree) of data sets and their
8 members.

9 The YANG module of this clause (ieee802-dot1as-gtp.yang and ieee802-dot1as-hs.yang) use the YANG
10 “import” statement to import the YANG module of IEEE Std 1588e. This effectively uses the IEEE Std 1588
11 YANG tree as the foundation of the IEEE Std 802.1AS YANG tree. By importing the tree and its data set
12 containers, all members from Clause 14 that are derived from IEEE Std 1588 are also imported.

13 The core of the YANG module for IEEE Std 802.1AS consists of YANG “augment” statements, used to add
14 members to the tree that are unique for IEEE Std 802.1AS.

15 NOTE— IETF RFC 8575 [B13g] is the standard YANG data model for IEEE Std 1588-2008. The YANG data model of
16 IEEE Std 1588e is effectively a newer version of RFC 8575. Therefore, the YANG module of RFC 8575 is not imported
17 by the YANG module of this clause.

18 **17.2 IEEE 802.1AS YANG data model**

19 This clause uses a UML-like representation to provide an overview of the hierarchy of the IEEE
20 Std 802.1AS YANG data model.

21 A representation of the management model is provided in Figure 17-1 through Figure 17-4. The purpose of
22 the diagrams is to express the model design in a concise manner. The structure of the representation shows
23 the name of the object followed by a list of properties for the object. The properties indicate their type and
24 accessibility. The representation is meant to express simplified semantics for the properties. It is not meant
25 to provide the specific datatype used to encode the object in either MIB or YANG. In the representation, a
26 box with a white background represents information that comes from sources outside of this IEEE standard.
27 A box with a gray background represents objects that are defined by this IEEE standard.

28 NOTE 1—OMG® UML 2.5¹⁹ [B24a] conventions together with C++ language constructs are used in this clause as a
29 representation to convey model structure and relationships.

30 NOTE 2—This standard specifies YANG for Clause 14 of this standard. There are optional features in the YANG
31 module of IEEE Std 1588 that are not specified in Clause 14, and are therefore not shown in the figures of this subclause.
32 If optional IEEE Std 1588 YANG features are implemented, conformance is specified by IEEE Std 1588.

33 For all figures, Clause 14 data that is imported from the ieee1588-ptp-tt.yang module is shown in white, and
34 Clause 14 data in augments of ieee802-dot1as-gtp.yang is shown in gray.

35 Figure 17-1 provides an overview of the IEEE Std 802.1AS YANG tree. The top level instance list provides
36 the list of one or more PTP Instances, each with data sets. For each PTP Instance, port-ds-list provides the
37 list of one or more PTP Ports, each with data sets. The common-services apply to all PTP Instances,
38 including the CMLDS.

¹⁹ OMG® is a registered trademark of the Object Management Group, Inc.

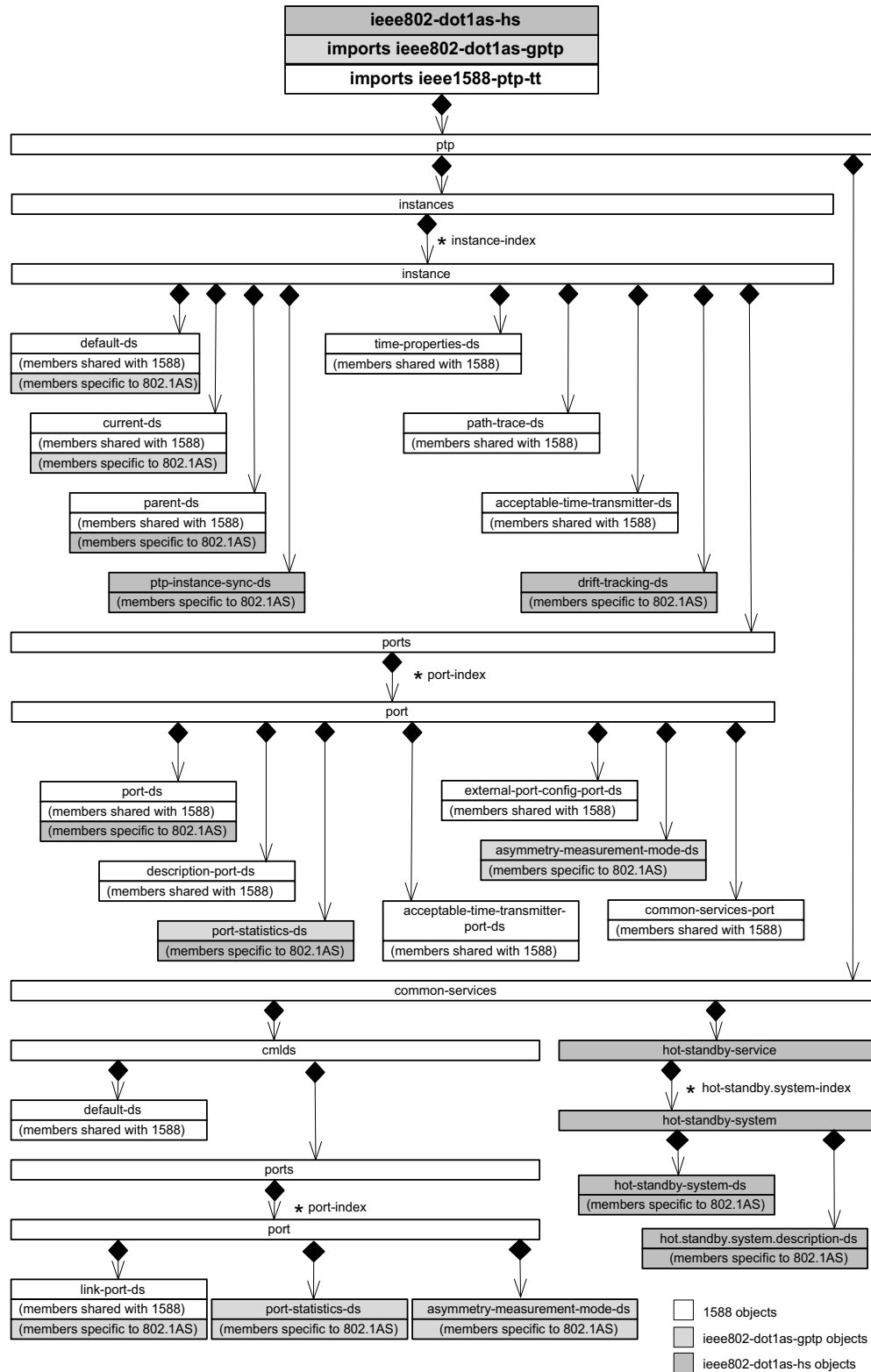


Figure 17-1—Overview of YANG tree

1 Figure 17-2 provides detail for the data sets of each PTP Instance, including each data set member.

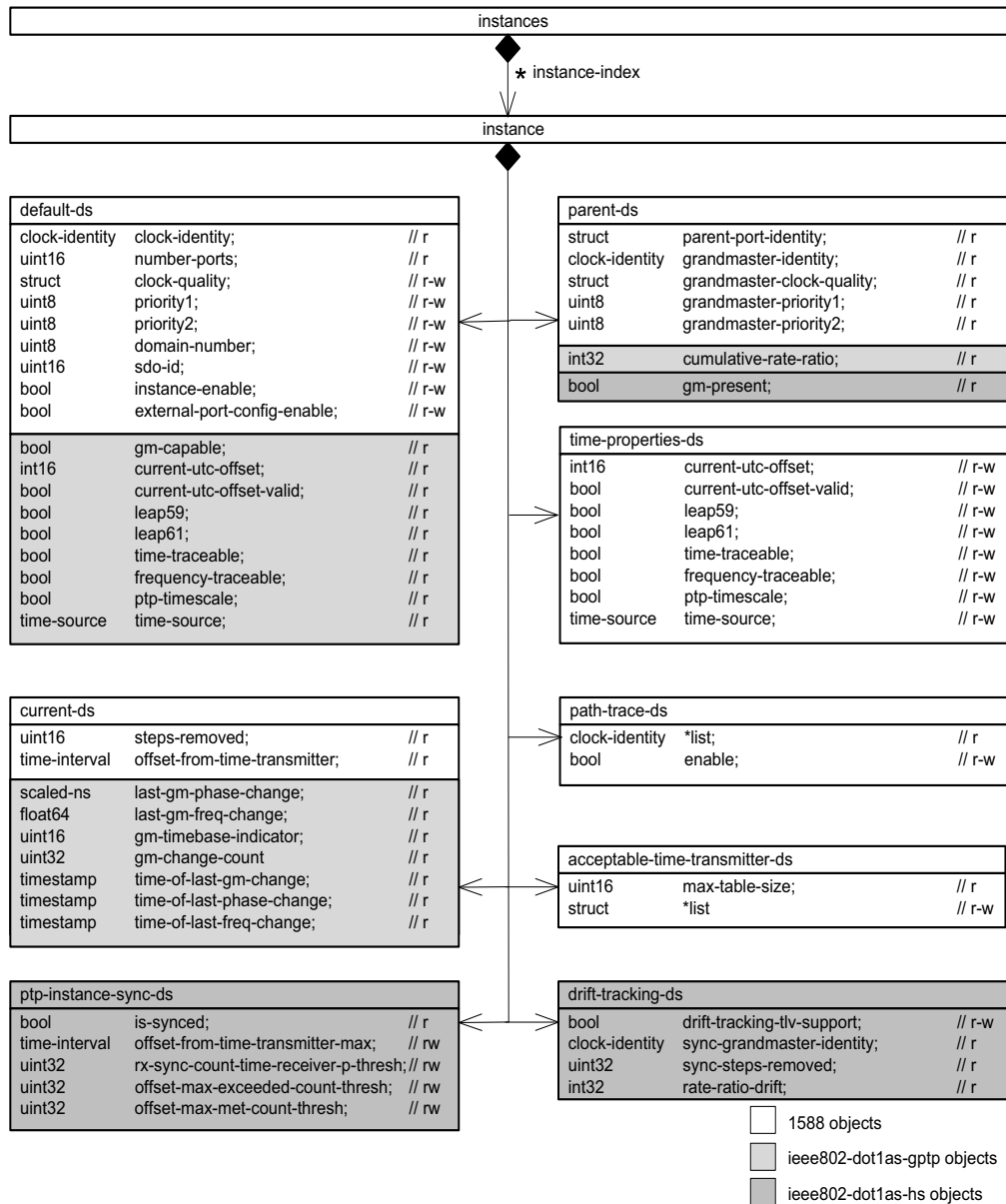


Figure 17-2—PTP Instance detail

2 Figure 17-3 provides detail for the data sets of each PTP Port, including each data set member.

3 NOTE 3—Subclause 14.8.4 specifies ptpPortEnabled (ptp-port-enabled), which is provided in YANG as the
4 semantically equivalent node in ieee1588-ptp-tt named port-enable (in port-ds of Figure 17-3). Subclause 14.8.15
5 specifies mgtSettableLogAnnounceInterval (mgt-settable-log-announce-interval), which is provided in YANG as the
6 semantically equivalent node in ieee1588-ptp-tt named log-announce-interval (in port-ds of Figure 17-3). Subclause
7 14.8.20 specifies mgtSettableLogSyncInterval (mgt-settable-log-sync-interval), which is provided in YANG as the
8 semantically equivalent node in ieee1588-ptp-tt named log-sync-interval (in port-ds of Figure 17-3).

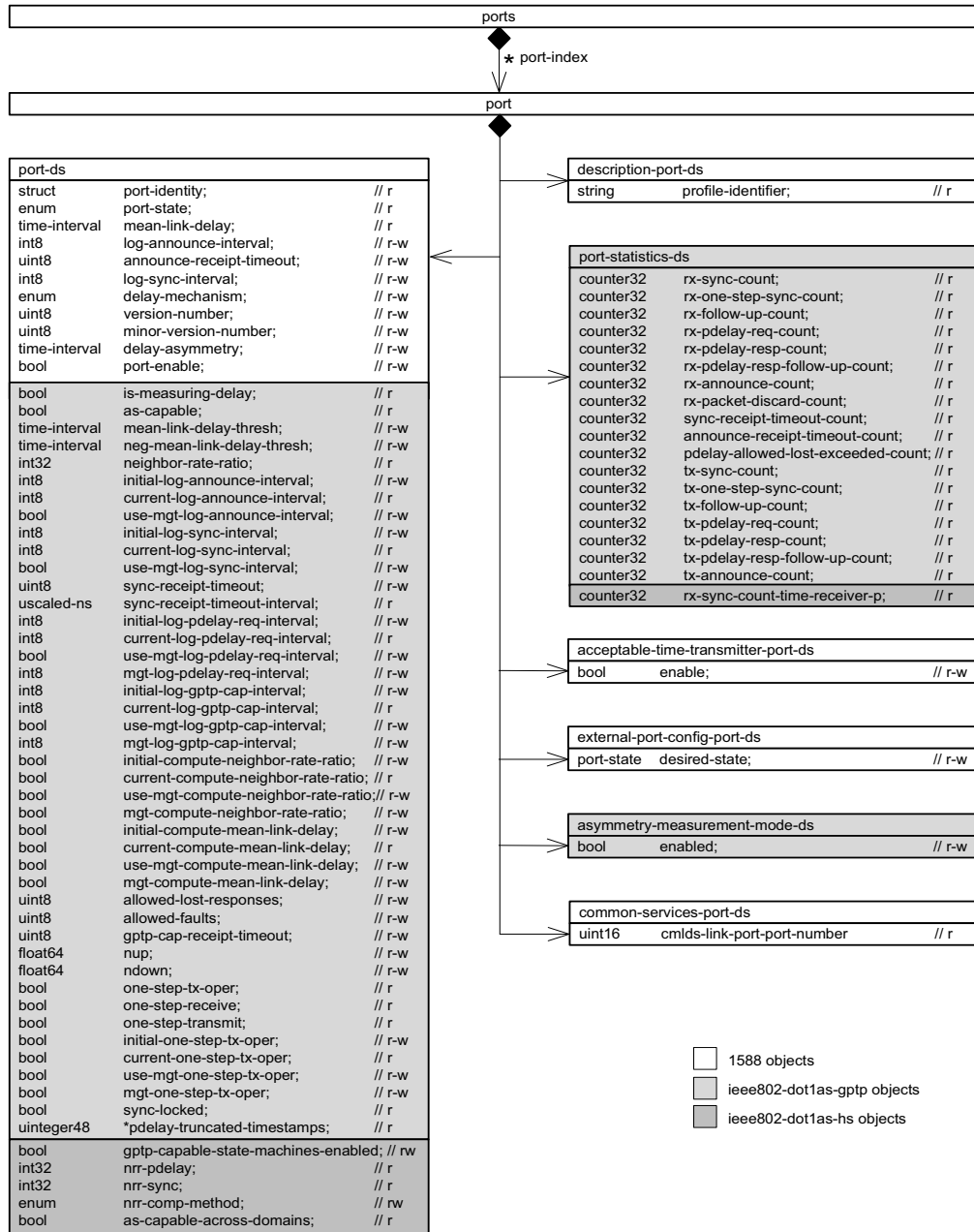


Figure 17-3—PTP Port detail

1 Figure 17-4 provides detail for the common services, including each data set member. The CMLDS has a
2 data set for the service itself (default-ds), and data sets for each Link Port. The Hot Standby Service has data
3 sets for each HotStandbySystem.

4 NOTE 4—Subclause 14.16.9 specifies neighborRateRatio (neighbor-rate-ratio), which is provided in YANG as the
5 semantically equivalent node in ieee1588-ptp-tt named scaled-neighbor-rate-ratio (in link-port-ds of Figure 17-4).

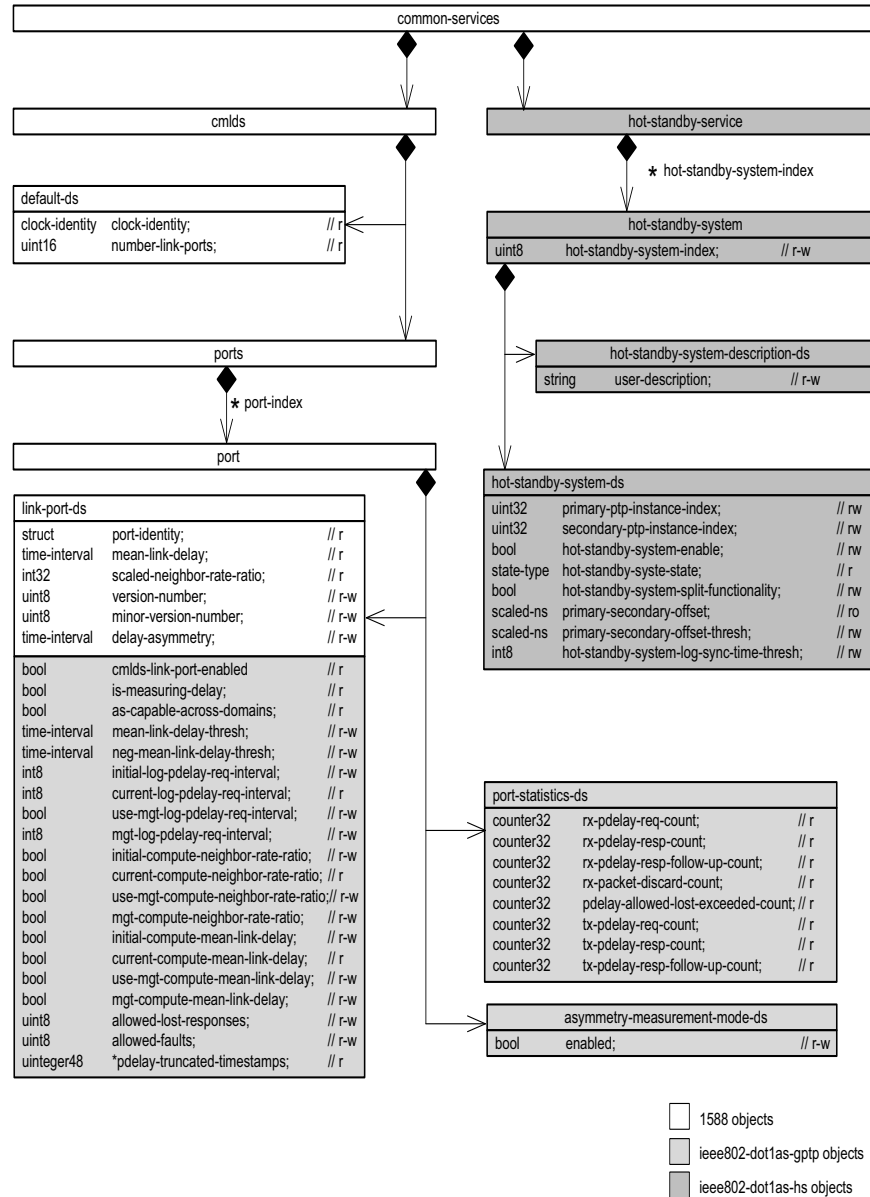


Figure 17-4—Common services detail

1 17.3 Structure of the YANG data model

2 The YANG data model specified by this standard uses the YANG modules summarized in Table 17-1.

3 In the YANG module definitions, if any discrepancy between the “description” text and the corresponding
4 definition in any other part of this standard occurs, the definitions outside this clause (Clause 17) take
5 precedence.

Table 17-1—Summary of the YANG modules

Module	Managed functionality	YANG specification notes
ietf-yang-types	Type definitions	IETF RFC 6991—Common YANG Data Types.
ieee1588-ptp-tt	Clause 14	IEEE Std 1588e—MIB and YANG Data Models. it imports this YANG module as its foundational tree, including a subset of members from Clause 14.
ieee802-dot1as-gptp	Clause 14	The YANG module of this clause uses YANG augments to add members from Clause 14 that are unique to IEEE Std 802.1AS.
ieee802-dot1as-hs	Clause 14	Hot standby - YANG Data Model. The YANG module of this clause uses YANG augments to add members from Clause 14 that are unique to Hot standby and drift tracking feature.

6 17.4 Security considerations

7 The YANG module specified in this document defines a schema for data that is designed to be accessed via
8 network management protocols such as NETCONF (IETF RFC 6242 [B13a]) and RESTCONF (IETF
9 RFC 8040 [B13e]). NETCONF and RESTCONF protocols provide the means to secure communication
10 between client and server, using secure transport layers such as Secure Shell (SSH) (IETF RFC 6242
11 [B13b]) and Transport Layer Security (TLS) (IETF RFC 7589 [B13d]).

12 It is the responsibility of a system’s implementor and administrator to ensure that the protocol entities in the
13 system that support NETCONF, and any other remote configuration protocols that make use of these YANG
14 modules, are properly configured to allow access only to those principals (users) that have legitimate rights
15 to read or write data nodes. This standard does not specify how the credentials of those users are to be stored
16 or validated.

17 The Network Configuration Access Control Model (IETF RFC 8341 [B13c]) provides the means to restrict
18 access for particular NETCONF or RESTCONF users to a pre-configured subset of all available NETCONF
19 or RESTCONF protocol operations and content.

20 There are a number of data sets in this YANG module that contain writable data nodes (IETF RFC 8040
21 [B13e]), such as:

```

22     /ptp-tt/instances/instance/default-ds
23     /ptp-tt/instances/instance/path-trace-ds
24     /ptp-tt/instances/instance/acceptable-time-transmitter-ds
25     /ptp-tt/instances/instance/ports/port/port-ds
26     /ptp-tt/instances/instance/ports/port/acceptable-time-transmitter-port-ds
27     /ptp-tt/instances/instance/ports/port/external-port-config-port-ds
28     /ptp-tt/instances/instance/ports/port/asymmetry-measurement-mode-ds

```

1 /ptp-tt/common-services/cmlds/ports/port/link-port-ds
2 /ptp-tt/common-services/cmlds/ports/port/asymmetry-measurement-mode-ds

3 Write operations (e.g., edit-config) to these data nodes without proper protection can have a negative effect
4 on network operations. Specifically, an inappropriate configuration of them can adversely impact a PTP
5 synchronization network. For example, loss of synchronization on a clock, accuracy degradation on a set of
6 clocks, or even breakdown of a whole synchronization network.

7 17.5 YANG schema tree definitions

8 The schema tree in this clause is provided as an overview of the YANG module in . The symbols and their
9 meaning are specified in YANG Tree Diagrams (IETF RFC 8340 [B13f]).

10 17.5.1 Tree diagram for ieee802-dot1as-gptp.yang

```

11 module: ieee802-dot1as-gptp
12
13 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:default-ds:
14   +--ro gm-capable?                boolean
15   +--ro current-utc-offset?        int16
16   +--ro current-utc-offset-valid?  boolean
17   +--ro leap59?                    boolean
18   +--ro leap61?                    boolean
19   +--ro time-traceable?            boolean
20   +--ro frequency-traceable?       boolean
21   +--ro ptp-timescale?             boolean
22   +--ro time-source?               identityref
23 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:current-ds:
24   +--ro last-gm-phase-change?       scaled-ns
25   +--ro last-gm-freq-change?        float64
26   +--ro gm-timebase-indicator?      uint16
27   +--ro gm-change-count?            yang:counter32
28   +--ro time-of-last-gm-change?     yang:timestamp
29   +--ro time-of-last-phase-change?  yang:timestamp
30   +--ro time-of-last-freq-change?   yang:timestamp
31 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:parent-ds:
32   +--ro cumulative-rate-ratio?      int32
33 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port/
34 ptp-tt:port-ds:
35   +--ro is-measuring-delay?          boolean
36   +--ro as-capable?                 boolean
37   +--rw mean-link-delay-thresh?      ptp-tt:time-interval
38   +--rw neg-mean-link-delay-thresh?  ptp-tt:time-interval
39   +--ro neighbor-rate-ratio?         int32
40   +--rw initial-log-announce-interval? int8
41   +--ro current-log-announce-interval? int8
42   +--rw use-mgt-log-announce-interval? boolean
43   +--rw initial-log-sync-interval?   int8
44   +--ro current-log-sync-interval?   int8
45   +--rw use-mgt-log-sync-interval?   boolean
46   +--rw sync-receipt-timeout?        uint8
47   +--ro sync-receipt-timeout-interval? uscaled-ns
48   +--rw initial-log-pdelay-req-interval? int8
49   +--ro current-log-pdelay-req-interval? int8
50   +--rw use-mgt-log-pdelay-req-interval? boolean
51   +--rw mgt-log-pdelay-req-interval? int8
52   +--rw initial-log-gptp-cap-interval? int8

```



```

1  +--ro current-log-gtp-cap-interval?      int8
2  +--rw use-mgt-log-gtp-cap-interval?      boolean
3  +--rw mgt-log-gtp-cap-interval?          int8
4  +--rw initial-compute-neighbor-rate-ratio? boolean
5  +--ro current-compute-neighbor-rate-ratio? boolean
6  +--rw use-mgt-compute-neighbor-rate-ratio? boolean
7  +--rw mgt-compute-neighbor-rate-ratio?   boolean
8  +--rw initial-compute-mean-link-delay?   boolean
9  +--ro current-compute-mean-link-delay?   boolean
10 +--rw use-mgt-compute-mean-link-delay?   boolean
11 +--rw mgt-compute-mean-link-delay?       boolean
12 +--rw allowed-lost-responses?            uint8
13 +--rw allowed-faults?                   uint8
14 +--rw gtp-cap-receipt-timeout?           uint8
15 +--rw nup?                              float64
16 +--rw ndown?                            float64
17 +--ro one-step-tx-oper?                  boolean
18 +--ro one-step-recv?                     boolean
19 +--ro one-step-transmit?                  boolean
20 +--rw initial-one-step-tx-oper?           boolean
21 +--ro current-one-step-tx-oper?           boolean
22 +--rw use-mgt-one-step-tx-oper?           boolean
23 +--rw mgt-one-step-tx-oper?               boolean
24 +--ro sync-locked?                       boolean
25 +--ro pdelay-truncated-timestamps*       uinteger48
26 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
27   +--rw port-statistics-ds
28     +--ro rx-sync-count?                  yang:counter32
29     +--ro rx-one-step-sync-count?          yang:counter32
30     +--ro rx-follow-up-count?              yang:counter32
31     +--ro rx-pdelay-req-count?             yang:counter32
32     +--ro rx-pdelay-resp-count?            yang:counter32
33     +--ro rx-pdelay-resp-follow-up-count?  yang:counter32
34     +--ro rx-announce-count?               yang:counter32
35     +--ro rx-packet-discard-count?         yang:counter32
36     +--ro sync-receipt-timeout-count?      yang:counter32
37     +--ro announce-receipt-timeout-count?  yang:counter32
38     +--ro pdelay-allowed-lost-exceeded-count? yang:counter32
39     +--ro tx-sync-count?                   yang:counter32
40     +--ro tx-one-step-sync-count?          yang:counter32
41     +--ro tx-follow-up-count?              yang:counter32
42     +--ro tx-pdelay-req-count?             yang:counter32
43     +--ro tx-pdelay-resp-count?            yang:counter32
44     +--ro tx-pdelay-resp-follow-up-count?  yang:counter32
45     +--ro tx-announce-count?               yang:counter32
46 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
47   +--rw asymmetry-measurement-mode-ds
48     +--rw enabled?                        boolean
49   augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmls/ptp-tt:ports/ptp-
50 tt:port/ptp-tt:link-port-ds:
51     +--ro cmls-link-port-enabled?         boolean
52     +--ro is-measuring-delay?              boolean
53     +--ro as-capable-across-domains?       boolean
54     +--rw mean-link-delay-thresh?          ptp-tt:time-interval
55     +--rw neg-mean-link-delay-thresh?      ptp-tt:time-interval
56     +--rw initial-log-pdelay-req-interval? int8
57     +--ro current-log-pdelay-req-interval? int8
58     +--rw use-mgt-log-pdelay-req-interval? boolean
59     +--rw mgt-log-pdelay-req-interval?     int8

```

```

1  +--rw initial-compute-neighbor-rate-ratio?  boolean
2  +--ro current-compute-neighbor-rate-ratio?  boolean
3  +--rw use-mgt-compute-neighbor-rate-ratio?  boolean
4  +--rw mgt-compute-neighbor-rate-ratio?      boolean
5  +--rw initial-compute-mean-link-delay?      boolean
6  +--ro current-compute-mean-link-delay?      boolean
7  +--rw use-mgt-compute-mean-link-delay?      boolean
8  +--rw mgt-compute-mean-link-delay?          boolean
9  +--rw allowed-lost-responses?                uint8
10 +--rw allowed-faults?                        uint8
11 +--ro pdelay-truncated-timestamps*          uinteger48
12   augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlDs/ptp-tt:ports/ptp-
13 tt:port:
14   +--rw port-statistics-ds
15     +--ro rx-pdelay-req-count?                yang:counter32
16     +--ro rx-pdelay-resp-count?              yang:counter32
17     +--ro rx-pdelay-resp-follow-up-count?    yang:counter32
18     +--ro rx-packet-discard-count?          yang:counter32
19     +--ro pdelay-allowed-lost-exceeded-count? yang:counter32
20     +--ro tx-pdelay-req-count?                yang:counter32
21     +--ro tx-pdelay-resp-count?              yang:counter32
22     +--ro tx-pdelay-resp-follow-up-count?    yang:counter32
23   augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlDs/ptp-tt:ports/ptp-
24 tt:port:
25   +--rw asymmetry-measurement-mode-ds
26   +--rw enabled?  boolean
27

```

28 17.5.2 Tree diagram for ieee802-dot1as-hs.yang

```

29 module: ieee802-dot1as-hs
30
31   augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:parent-ds:
32     +--ro gm-present?  boolean
33   augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance:
34     +--rw ptp-instance-sync-ds
35       | +--ro is-synced?                boolean
36       | +--rw offset-from-time-transmitter-max?  ptp-tt:time-interval
37       | +--rw rx-sync-count-time-receiver-p-thresh?  uint32
38       | +--rw offset-max-exceeded-count-thresh?    uint32
39       | +--rw offset-max-met-count-thresh?         uint32
40     +--rw drift-tracking-ds
41       +--rw drift-tracking-tlv-support?  boolean
42       +--ro sync-grandmaster-identity?  ptp-tt:clock-identity
43       +--ro sync-steps-removed?         uint32
44       +--ro rate-ratio-drift?           int32
45   augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port/
46 ptp-tt:port-ds:
47     +--rw gptp-capable-state-machines-enabled?  boolean
48     +--ro nrr-pdelay?                            int32
49     +--ro nrr-sync?                              int32
50     +--rw nrr-comp-method?                        nrr-comp-method-type
51     +--ro as-capable-across-domains?             boolean
52   augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port/
53 dot1as-gptp:port-statistics-ds:
54     +--ro rx-sync-count-time-receiver-p?  yang:counter32
55   augment /ptp-tt:ptp/ptp-tt:common-services:
56     +--rw hot-standby-service {hot-standby}?

```

```

1      +--rw hot-standby-system* [hot-standby-system-index]
2      +--rw hot-standby-system-index          uint8
3      +--rw hot-standby-system-ds
4      | +--rw primary-ptp-instance-index?      uint32
5      | +--rw secondary-ptp-instance-index?    uint32
6      | +--rw hot-standby-system-enable?      boolean
7      | +--ro hot-standby-system-state?        hot-standby-system-
8 state-type
9      | +--rw hot-standby-system-split-functionality? boolean
10     | +--ro primary-secondary-offset?        dot1as-gptp:scaled-ns
11     | +--rw primary-secondary-offset-thresh? dot1as-gptp:scaled-
12 ns
13     | +--rw hot-standby-system-log-sync-time-thresh? int8
14     +--rw hot-standby-system-description-ds
15     +--rw user-description? string
16

```

17 17.6 YANG modules^{20, 21}

18 The ieee1588-ptp-tt.yang YANG module²² specified by IEEE Std 1588e serves as the foundation of the
19 YANG module specified in this clause.

20 17.6.1 Module ieee802-dot1as-gptp.yang

```

21 module ieee802-dot1as-gptp {
22   yang-version "1.1";
23   namespace urn:ieee:std:802.1AS:yang:ieee802-dot1as-gptp;
24   prefix dot1as-gptp;
25   import ietf-yang-types {
26     prefix yang;
27   }
28   import ieee1588-ptp-tt {
29     prefix ptp-tt;
30   }
31   organization
32     "IEEE 802.1 Working Group";
33   contact
34     "WG-URL: http://ieee802.org/1/
35     WG-Email: stds-802-1-l@ieee.org
36
37     Contact: IEEE 802.1 Working Group Chair
38             Postal: C/O IEEE 802.1 Working Group
39             IEEE Standards Association
40             445 Hoes Lane
41             Piscataway, NJ 08854
42             USA
43
44     E-mail: stds-802-1-chairs@ieee.org";
45   description
46     "Management objects that control timing and synchronization for time

```

²⁰ Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

²¹ ASCII versions of the YANG modules are attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

²² An ASCII version of the IEEE Std 1588 YANG module can be obtained from <https://github.com/YangModels/yang/tree/main/standard/ieee/published/1588>.

```
1 sensitive applications, as specified in clause 14 of IEEE Std
2 802.1AS-2020-REV.
3
4 Copyright (C) IEEE (2025). This version of this YANG module is part
5 of IEEE Std 802.1AS-2020-REV; see the standard itself for full legal
6 notices.";
7 revision 2025-02-04 {
8   description
9     "Published as part of IEEE Std 802.1AS-2020-REV.";
10  reference
11    "IEEE Std 802.1AS - Timing and Synchronization for Time-Sensitive
12    Applications: IEEE Std 802.1AS-2020-REV.
13    IEEE Std 1588 - IEEE Standard for a Precision Clock Synchronization
14    Protocol for Networked Measurement and Control Systems: IEEE Std
15    1588-2019, IEEE Std 1588g-2022, IEEE Std 1588e-2024.";
16 }
17 typedef scaled-ns {
18   type string {
19     pattern "[0-9A-F]{2}(-[0-9A-F]{2}){11}";
20   }
21   description
22     "The IEEE Std 802.1AS ScaledNs type represents signed values of
23     time and time interval in units of 2^16 ns, as a signed 96-bit
24     integer. Each of the 12 octets is represented as a pair of
25     hexadecimal characters, using uppercase for a letter. Octets are
26     separated by a dash character. The most significant octet is first.";
27   reference
28     "6.4.3.1 of IEEE Std 802.1AS";
29 }
30 typedef unscaled-ns {
31   type string {
32     pattern "[0-9A-F]{2}(-[0-9A-F]{2}){11}";
33   }
34   description
35     "The IEEE Std 802.1AS UScaledNs type represents unsigned values of
36     time and time interval in units of 2^16 ns, as an unsigned 96-bit
37     integer. Each of the 12 octets is represented as a pair of
38     hexadecimal characters, using uppercase for a letter. Octets are
39     separated by a dash character. The most significant octet is first.";
40   reference
41     "6.4.3.2 of IEEE Std 802.1AS";
42 }
43 typedef float64 {
44   type string {
45     pattern "[0-9A-F]{2}(-[0-9A-F]{2}){7}";
46   }
47   description
48     "The IEEE Std 802.1AS Float64 type represents IEEE Std 754
49     binary64. Each of the 8 octets is represented as a pair of
50     hexadecimal characters, using uppercase for a letter. Octets are
51     separated by a dash character. The most significant octet is first.";
52   reference
53     "6.4.2 of IEEE Std 802.1AS";
54 }
55 typedef uinteger48 {
56   type uint64 {
57     range "0..281474976710655";
58   }
59   description
```

```

1      "48-bit unsigned integer data type.";
2      reference
3      "6.4.2 of IEEE Std 802.1AS";
4  }
5  augment
6  "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:default-ds" {
7      description
8      "Augment IEEE Std 1588 defaultDS.";
9      leaf gm-capable {
10         type boolean;
11         config false;
12         description
13         "The value is true if the time-aware system is capable of being a
14         Grandmaster, and false if the time-aware system is not capable of
15         being a Grandmaster.";
16         reference
17         "14.2.7 of IEEE Std 802.1AS";
18     }
19     leaf current-utc-offset {
20         when
21         "../current-utc-offset-valid='true'";
22         type int16;
23         config false;
24         description
25         "Offset from UTC (TAI - UTC). The offset is in units of seconds.
26         This leaf applies to the ClockTimeTransmitter entity (i.e., local
27         only, unrelated to a remote GM).";
28         reference
29         "14.2.8 of IEEE Std 802.1AS";
30     }
31     leaf current-utc-offset-valid {
32         type boolean;
33         config false;
34         description
35         "The value of current-utc-offset-valid shall be true if the value
36         of current-utc-offset is known to be correct, otherwise it shall
37         be false. This leaf applies to the ClockTimeTransmitter entity
38         (i.e., local only, unrelated to a remote GM).";
39         reference
40         "14.2.9 of IEEE Std 802.1AS";
41     }
42     leaf leap59 {
43         type boolean;
44         config false;
45         description
46         "If the timescale is PTP, a true value for leap59 shall indicate
47         that the last minute of the current UTC day contains 59 seconds.
48         If the timescale is not PTP, the value shall be false. This leaf
49         applies to the ClockTimeTransmitter entity (i.e., local only,
50         unrelated to a remote GM).";
51         reference
52         "14.2.10 of IEEE Std 802.1AS";
53     }
54     leaf leap61 {
55         type boolean;
56         config false;
57         description
58         "If the timescale is PTP, a true value for leap61 shall indicate
59         that the last minute of the current UTC day contains 61 seconds.

```

```

1      If the timescale is not PTP, the value shall be false. This leaf
2      applies to the ClockTimeTransmitter entity (i.e., local only,
3      unrelated to a remote GM).";
4      reference
5      "14.2.11 of IEEE Std 802.1AS";
6  }
7  leaf time-traceable {
8      type boolean;
9      config false;
10     description
11         "The value of time-traceable shall be true if the timescale is
12         traceable to a primary reference; otherwise, the value shall be
13         false. This leaf applies to the ClockTimeTransmitter entity
14         (i.e., local only, unrelated to a remote GM).";
15     reference
16         "14.2.12 of IEEE Std 802.1AS";
17 }
18 leaf frequency-traceable {
19     type boolean;
20     config false;
21     description
22         "The value of frequency-traceable shall be true if the frequency
23         determining the timescale is traceable to a primary reference;
24         otherwise, the value shall be false. This leaf applies to the
25         ClockTimeTransmitter entity (i.e., local only, unrelated to a
26         remote GM).";
27     reference
28         "14.2.13 of IEEE Std 802.1AS";
29 }
30 leaf ptp-timescale {
31     type boolean;
32     config false;
33     description
34         "If ptp-timescale is true, the timescale of the
35         ClockTimeTransmitter entity is PTP, which is the elapsed time
36         since the PTP epoch measured using the second defined by
37         International Atomic Time (TAI). If ptp-timescale is false, the
38         timescale of the ClockTimeTransmitter entity is ARB, which is the
39         elapsed time since an arbitrary epoch. This leaf applies to the
40         ClockTimeTransmitter entity (i.e., local only, unrelated to a
41         remote GM).";
42     reference
43         "14.2.14 of IEEE Std 802.1AS";
44 }
45 leaf time-source {
46     type identityref {
47         base ptp-tt:time-source;
48     }
49     config false;
50     description
51         "The source of time used by the Grandmaster Clock. This leaf
52         applies to the ClockTimeTransmitter entity (i.e., local only,
53         unrelated to a remote GM).";
54     reference
55         "14.2.15 of IEEE Std 802.1AS";
56 }
57 }
58 augment
59     "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:current-ds" {

```

```

1  description
2  "Augment IEEE Std 1588 currentDS.";
3  leaf last-gm-phase-change {
4      type scaled-ns;
5      config false;
6      description
7          "Phase change that occurred on the most recent change in either
8          the Grandmaster PTP Instance or gm-timebase-indicator leaf.";
9      reference
10         "14.3.4 of IEEE Std 802.1AS";
11 }
12 leaf last-gm-freq-change {
13     type float64;
14     config false;
15     description
16         "Frequency change that occurred on the most recent change in
17         either the Grandmaster PTP Instance or gm-timebase-indicator
18         leaf.";
19     reference
20         "14.3.5 of IEEE Std 802.1AS";
21 }
22 leaf gm-timebase-indicator {
23     type uint16;
24     config false;
25     description
26         "The timeBaseIndicator of the current Grandmaster PTP Instance.";
27     reference
28         "14.3.6 of IEEE Std 802.1AS";
29 }
30 leaf gm-change-count {
31     type yang:counter32;
32     config false;
33     description
34         "This statistics counter tracks the number of times the
35         Grandmaster PTP Instance has changed in a gPTP domain.";
36     reference
37         "14.3.7 of IEEE Std 802.1AS";
38 }
39 leaf time-of-last-gm-change {
40     type yang:timestamp;
41     config false;
42     description
43         "System time when the most recent Grandmaster Clock change
44         occurred in a gPTP domain. This leaf's type is YANG timestamp,
45         which is based on system time. System time is an unsigned integer
46         in units of 10 milliseconds, using an epoch defined by the
47         implementation (typically time of boot-up).";
48     reference
49         "14.3.8 of IEEE Std 802.1AS";
50 }
51 leaf time-of-last-phase-change {
52     type yang:timestamp;
53     config false;
54     description
55         "System time when the most recent change in Grandmaster Clock
56         phase occurred. This leaf's type is YANG timestamp, which is
57         based on system time. System time is an unsigned integer in units
58         of 10 milliseconds, using an epoch defined by the implementation
59         (typically time of boot-up).";

```

```

1      reference
2      "14.3.9 of IEEE Std 802.1AS";
3  }
4  leaf time-of-last-freq-change {
5      type yang:timestamp;
6      config false;
7      description
8          "System time when the most recent change in Grandmaster Clock
9          frequency occurred. This leaf's type is YANG timestamp, which is
10         based on system time. System time is an unsigned integer in units
11         of 10 milliseconds, using an epoch defined by the implementation
12         (typically time of boot-up).";
13     reference
14         "14.3.10 of IEEE Std 802.1AS";
15 }
16 }
17 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:parent-ds" {
18     description
19         "Augment IEEE Std 1588 parentDS.";
20     leaf cumulative-rate-ratio {
21         type int32;
22         config false;
23         description
24             "Estimate of the ratio of the frequency of the Grandmaster Clock
25             to the frequency of the LocalClock entity of this PTP Instance.
26             cumulative-rate-ratio is expressed as the fractional frequency
27             offset multiplied by 2^41, i.e., the quantity (rateRatio -
28             1.0)(2^41).";
29         reference
30             "14.4.3 of IEEE Std 802.1AS";
31     }
32 }
33 augment
34     "/ptp-tt:ptp"+
35     "/ptp-tt:instances"+
36     "/ptp-tt:instance"+
37     "/ptp-tt:ports"+
38     "/ptp-tt:port"+
39     "/ptp-tt:port-ds" {
40     description
41         "Augment IEEE Std 1588 portDS.
42
43         14.10.4 of IEEE Std 802.1AS specifies ptpPortEnabled
44         (ptp-port-enabled), which is provided in YANG as the semantically
45         equivalent node in ieee1588-ptp-tt named port-enable (in port-ds).
46
47         14.10.15 of IEEE Std 802.1AS specifies
48         mgtSettableLogAnnounceInterval (mgt-log-announce-interval), which
49         is provided in YANG as the semantically equivalent node in
50         ieee1588-ptp-tt named log-announce-interval (in port-ds). In the
51         context of IEEE Std 802.1AS, log-announce-interval cannot be used
52         unless use-mgt-log-announce-interval is true.
53
54         14.10.20 of IEEE Std 802.1AS specifies mgtSettableLogSyncInterval
55         (mgt-log-sync-interval), which is provided in YANG as the
56         semantically equivalent node in ieee1588-ptp-tt named
57         log-sync-interval (in port-ds). In the context of IEEE Std 802.1AS,
58         log-sync-interval cannot be used unless use-mgt-log-sync-interval
59         is true.";

```



```

1  leaf is-measuring-delay {
2      type boolean;
3      config false;
4      description
5          "Boolean that is true if the port is measuring PTP Link
6          propagation delay.";
7      reference
8          "14.10.6 of IEEE Std 802.1AS";
9  }
10 leaf as-capable {
11     type boolean;
12     config false;
13     description
14         "Boolean that is true if and only if it is determined that this
15         PTP Instance and the PTP Instance at the other end of the link
16         attached to this port can interoperate with each other via the
17         IEEE Std 802.1AS protocol.";
18     reference
19         "10.2.5.1 of IEEE Std 802.1AS
20         14.10.7 of IEEE Std 802.1AS";
21 }
22 leaf mean-link-delay-thresh {
23     type ptp-tt:time-interval;
24     description
25         "Propagation time threshold for mean-link-delay, above which a
26         port is not considered capable of participating in the IEEE Std
27         802.1AS protocol.";
28     reference
29         "14.10.9 of IEEE Std 802.1AS";
30 }
31 leaf neg-mean-link-delay-thresh {
32     type ptp-tt:time-interval;
33     description
34         "The negative propagation time threshold for mean-link-delay,
35         below which a port is not considered capable of participating
36         in the IEEE Std 802.1AS protocol.";
37     reference
38         "14.10.10 of IEEE Std 802.1AS";
39 }
40 leaf neighbor-rate-ratio {
41     type int32;
42     config false;
43     description
44         "Estimate of the ratio of the frequency of the LocalClock entity
45         of the PTP Instance at the other end of the link attached to this
46         PTP Port, to the frequency of the LocalClock entity of this PTP
47         Instance. neighbor-rate-ratio is expressed as the fractional
48         frequency offset multiplied by 2^41, i.e., the quantity
49         (rateRatio - 1.0)(2^41).";
50     reference
51         "14.10.12 of IEEE Std 802.1AS";
52 }
53 leaf initial-log-announce-interval {
54     type int8;
55     description
56         "When use-mgt-log-announce-interval is false (i.e., change with
57         Signaling message), this is the logarithm to base 2 of the
58         announce interval used when the port is initialized.";
59     reference

```

```

1      "14.10.13 of IEEE Std 802.1AS";
2  }
3  leaf current-log-announce-interval {
4      type int8;
5      config false;
6      description
7          "Logarithm to base 2 of the current announce interval.";
8      reference
9          "14.10.14 of IEEE Std 802.1AS";
10 }
11 leaf use-mgt-log-announce-interval {
12     type boolean;
13     description
14         "Boolean that determines the source of the announce interval. If
15         the value is true, the announce interval
16         (current-log-announce-interval) is set equal to the value of
17         mgt-log-announce-interval. If the value is false, the announce
18         interval is determined by the AnnounceIntervalSetting state
19         machine (i.e., changed with Signaling message).";
20     reference
21         "14.10.15 of IEEE Std 802.1AS";
22 }
23 leaf initial-log-sync-interval {
24     type int8;
25     description
26         "When use-mgt-log-sync-interval is false (i.e., change with
27         Signaling message), this is the logarithm to base 2 of the
28         sync interval used when the port is initialized.";
29     reference
30         "14.10.17 of IEEE Std 802.1AS";
31 }
32 leaf current-log-sync-interval {
33     type int8;
34     config false;
35     description
36         "Logarithm to base 2 of the current sync interval.";
37     reference
38         "14.10.18 of IEEE Std 802.1AS";
39 }
40 leaf use-mgt-log-sync-interval {
41     type boolean;
42     description
43         "Boolean that determines the source of the sync interval. If the
44         value is true, the sync interval (current-log-sync-interval) is
45         set equal to the value of mgt-log-sync-interval. If the value is
46         false, the sync interval is determined by the SyncIntervalSetting
47         state machine (i.e., changed with Signaling message).";
48     reference
49         "14.10.19 of IEEE Std 802.1AS";
50 }
51 leaf sync-receipt-timeout {
52     type uint8;
53     description
54         "Number of sync intervals that a timeReceiver port waits without
55         receiving synchronization information, before assuming that the
56         timeTransmitter is no longer transmitting synchronization
57         information and that the BTCA needs to be run, if appropriate.";
58     reference
59         "14.10.21 of IEEE Std 802.1AS";

```

```

1   }
2   leaf sync-receipt-timeout-interval {
3       type uscaled-ns;
4       config false;
5       description
6           "Time interval after which sync receipt timeout occurs if
7           time-synchronization information has not been received during the
8           interval.";
9       reference
10          "14.10.22 of IEEE Std 802.1AS";
11   }
12   leaf initial-log-pdelay-req-interval {
13       type int8;
14       description
15           "When use-mgt-log-pdelay-req-interval is false (i.e., change with
16           Signaling message), this is the logarithm to base 2 of the
17           Pdelay_Req transmit interval used when the port is initialized.";
18       reference
19          "14.10.23 of IEEE Std 802.1AS";
20   }
21   leaf current-log-pdelay-req-interval {
22       type int8;
23       config false;
24       description
25           "Logarithm to base 2 of the current Pdelay_Req transmit interval.";
26       reference
27          "14.10.24 of IEEE Std 802.1AS";
28   }
29   leaf use-mgt-log-pdelay-req-interval {
30       type boolean;
31       description
32           "Boolean that determines the source of the Pdelay_Req transmit
33           interval. If the value is true, the Pdelay_Req transmit interval
34           (current-log-pdelay-req-interval) is set equal to the value of
35           mgt-log-pdelay-req-interval. If the value is false, the
36           Pdelay_Req transmit interval is determined by the
37           LinkDelayIntervalSetting state machine (i.e., changed with
38           Signaling message).";
39       reference
40          "14.10.25 of IEEE Std 802.1AS";
41   }
42   leaf mgt-log-pdelay-req-interval {
43       type int8;
44       description
45           "Logarithm to base 2 of the Pdelay_Req transmit interval, used if
46           use-mgt-log-pdelay-req-interval is true. This value is not used
47           if use-mgt-log-pdelay-req-interval is false.";
48       reference
49          "14.10.26 of IEEE Std 802.1AS";
50   }
51   leaf initial-log-gptp-cap-interval {
52       type int8;
53       description
54           "When use-mgt-log-gptp-cap-interval is false (i.e., change with
55           Signaling message), this is the logarithm to base 2 of the
56           gPTP capable message interval used when the port is initialized.";
57       reference
58          "14.10.27 of IEEE Std 802.1AS";
59   }

```

```

1  leaf current-log-gtp-cap-interval {
2      type int8;
3      config false;
4      description
5          "Logarithm to base 2 of the current gTP capable message
6          interval.";
7      reference
8          "14.10.28 of IEEE Std 802.1AS";
9  }
10 leaf use-mgt-log-gtp-cap-interval {
11     type boolean;
12     description
13         "Boolean that determines the source of the gTP capable message
14         interval. If the value is true, the gTP capable message interval
15         (current-log-gtp-cap-interval) is set equal to the value of
16         mgt-gtp-cap-req-interval. If the value is false, the gTP
17         capable message interval is determined by the
18         GtpCapableMessageIntervalSetting state machine (i.e., changed
19         with Signaling message).";
20     reference
21         "14.10.29 of IEEE Std 802.1AS";
22 }
23 leaf mgt-log-gtp-cap-interval {
24     type int8;
25     description
26         "Logarithm to base 2 of the gTP capable message interval, used
27         if use-mgt-log-gtp-cap-interval is true. This value is not used
28         if use-mgt-log-pdelay-req-interval is false.";
29     reference
30         "14.10.30 of IEEE Std 802.1AS";
31 }
32 leaf initial-compute-neighbor-rate-ratio {
33     type boolean;
34     description
35         "When use-mgt-compute-neighbor-rate-ratio is false (i.e., change
36         with Signaling message), this is the initial value of
37         computeNeighborRateRatio.";
38     reference
39         "14.10.31 of IEEE Std 802.1AS";
40 }
41 leaf current-compute-neighbor-rate-ratio {
42     type boolean;
43     config false;
44     description
45         "Current value of computeNeighborRateRatio.";
46     reference
47         "14.10.32 of IEEE Std 802.1AS";
48 }
49 leaf use-mgt-compute-neighbor-rate-ratio {
50     type boolean;
51     description
52         "Boolean that determines the source of computeNeighborRateRatio..
53         If the value is true, computeNeighborRateRatio is set equal to
54         the value of mgt-compute-neighbor-rate-ratio. If the value is
55         false, computeNeighborRateRatio is determined by the
56         LinkDelayIntervalSetting state machine (i.e., changed with
57         Signaling message).";
58     reference
59         "14.10.33 of IEEE Std 802.1AS";

```

```

1   }
2   leaf mgt-compute-neighbor-rate-ratio {
3       type boolean;
4       description
5           "Value of computeNeighborRateRatio, used if
6           use-mgt-compute-neighbor-rate-ratio is true. This value is not
7           used if use-mgt-compute-neighbor-rate-ratio is false.";
8       reference
9           "14.10.34 of IEEE Std 802.1AS";
10  }
11  leaf initial-compute-mean-link-delay {
12      type boolean;
13      description
14          "When use-mgt-compute-mean-link-delay is false (i.e., change with
15          Signaling message), this is the initial value of
16          computeMeanLinkDelay.";
17      reference
18          "14.10.35 of IEEE Std 802.1AS";
19  }
20  leaf current-compute-mean-link-delay {
21      type boolean;
22      config false;
23      description
24          "Current value of computeMeanLinkDelay.";
25      reference
26          "14.10.36 of IEEE Std 802.1AS";
27  }
28  leaf use-mgt-compute-mean-link-delay {
29      type boolean;
30      description
31          "Boolean that determines the source of computeMeanLinkDelay. If
32          the value is true, computeMeanLinkDelay is set equal to the value
33          of mgt-compute-mean-link-delay. If the value is false,
34          computeMeanLinkDelay is determined by the
35          LinkDelayIntervalSetting state machine (i.e., changed with
36          Signaling message).";
37      reference
38          "14.10.37 of IEEE Std 802.1AS";
39  }
40  leaf mgt-compute-mean-link-delay {
41      type boolean;
42      description
43          "Value of computeMeanLinkDelay, used if
44          use-mgt-compute-mean-link-delay is true. This value is not used
45          if use-mgt-compute-mean-link-delay is false.";
46      reference
47          "14.10.38 of IEEE Std 802.1AS";
48  }
49  leaf allowed-lost-responses {
50      type uint8;
51      description
52          "Number of Pdelay_Req messages for which a valid response is not
53          received, above which a port is considered to not be exchanging
54          peer delay messages with its neighbor.";
55      reference
56          "14.10.39 of IEEE Std 802.1AS";
57  }
58  leaf allowed-faults {
59      type uint8;

```

```

1      description
2      "Number of faults above which asCapable is set to false.";
3      reference
4      "14.10.40 of IEEE Std 802.1AS";
5  }
6  leaf gptp-cap-receipt-timeout {
7      type uint8;
8      description
9      "Number of transmission intervals that a port waits without
10     receiving the gPTP capable TLV, before assuming that the neighbor
11     port is no longer invoking the gPTP protocol.";
12     reference
13     "14.10.41 of IEEE Std 802.1AS";
14 }
15 leaf nup {
16     type float64;
17     description
18     "For an OLT port of an IEEE Std 802.3 EPON link, this value is
19     the effective index of refraction for the EPON upstream
20     wavelength light of the optical path.";
21     reference
22     "14.10.43 of IEEE Std 802.1AS";
23 }
24 leaf ndown {
25     type float64;
26     description
27     "For an OLT port of an IEEE 802.3 EPON link, this value is the
28     effective index of refraction for the EPON downstream wavelength
29     light of the optical path.";
30     reference
31     "14.10.44 of IEEE Std 802.1AS";
32 }
33 leaf one-step-tx-oper {
34     type boolean;
35     config false;
36     description
37     "This value is true if the port is sending one-step Sync
38     messages, and false if the port is sending two-step Sync and
39     Follow-Up messages.";
40     reference
41     "14.10.45 of IEEE Std 802.1AS";
42 }
43 leaf one-step-receive {
44     type boolean;
45     config false;
46     description
47     "This value is true if the port is capable of receiving and
48     processing one-step Sync messages.";
49     reference
50     "14.10.46 of IEEE Std 802.1AS";
51 }
52 leaf one-step-transmit {
53     type boolean;
54     config false;
55     description
56     "This value is true if the port is capable of transmitting
57     one-step Sync messages.";
58     reference
59     "14.10.47 of IEEE Std 802.1AS";

```

```

1   }
2   leaf initial-one-step-tx-oper {
3       type boolean;
4       description
5           "When use-mgt-one-step-tx-oper is false (i.e., change with
6           Signaling message), this is the initial value of
7           current-one-step-tx-oper.";
8       reference
9           "14.10.48 of IEEE Std 802.1AS";
10  }
11  leaf current-one-step-tx-oper {
12      type boolean;
13      config false;
14      description
15          "This value is true if the port is configured to transmit
16          one-step Sync messages, either via management
17          (mgt-one-step-tx-oper) or Signaling. If both
18          current-one-step-tx-oper and one-step-transmit are true, the port
19          transmits one-step Sync messages (i.e., one-step-tx-oper true).";
20      reference
21          "14.10.49 of IEEE Std 802.1AS";
22  }
23  leaf use-mgt-one-step-tx-oper {
24      type boolean;
25      description
26          "Boolean that determines the source of current-one-step-tx-oper.
27          If the value is true, current-one-step-tx-oper is set equal to
28          the value of mgt-one-step-tx-oper. If the value is false,
29          current-one-step-tx-oper is determined by the
30          OneStepTxOperSetting state machine (i.e., changed with Signaling
31          message).";
32      reference
33          "14.10.50 of IEEE Std 802.1AS";
34  }
35  leaf mgt-one-step-tx-oper {
36      type boolean;
37      description
38          "If use-mgt-one-step-tx-oper is true, current-one-step-tx-oper is
39          set equal to this value. This value is not used if
40          use-mgt-one-step-tx-oper is false.";
41      reference
42          "14.10.51 of IEEE Std 802.1AS";
43  }
44  leaf sync-locked {
45      type boolean;
46      config false;
47      description
48          "This value is true if the port will transmit a Sync as soon as
49          possible after the timeReceiver port receives a Sync message.";
50      reference
51          "14.10.52 of IEEE Std 802.1AS";
52  }
53  leaf-list pdelay-truncated-timestamps {
54      type uinteger48;
55      config false;
56      description
57          "For full-duplex IEEE Std 802.3 media, and CSN media that use the
58          peer-to-peer delay mechanism to measure path delay, the values of
59          the four elements of this leaf-list correspond to the timestamps

```

```

1      t1, t2, t3, and t4, listed in that order. Each timestamp is
2      expressed in units of 2-16 ns (i.e., the value of each array
3      element is equal to the remainder obtained upon dividing the
4      respective timestamp, expressed in units of 2-16 ns, by 248).
5      At any given time, the timestamp values stored in the array are
6      for the same, and most recently completed, peer delay message
7      exchange. For each timestamp, only 48-bits are valid (the upper
8      16-bits are always zero).";
9      reference
10     "14.10.53 of IEEE Std 802.1AS";
11 }
12 }
13 augment
14     "/ptp-tt:ptp"+
15     "/ptp-tt:instances"+
16     "/ptp-tt:instance"+
17     "/ptp-tt:ports"+
18     "/ptp-tt:port" {
19     description
20     "Augment to add port-statistics-ds to IEEE Std 1588 PTP Port.";
21     container port-statistics-ds {
22         description
23         "Provides counters associated with the port of the PTP Instance.";
24         reference
25         "14.12 of IEEE Std 802.1AS";
26         leaf rx-sync-count {
27             type yang:counter32;
28             config false;
29             description
30             "Counter that increments every time synchronization information
31              is received.";
32             reference
33             "14.12.2 of IEEE Std 802.1AS";
34         }
35         leaf rx-one-step-sync-count {
36             type yang:counter32;
37             config false;
38             description
39             "Counter that increments every time a one-step Sync message is
40              received.";
41             reference
42             "14.12.3 of IEEE Std 802.1AS";
43         }
44         leaf rx-follow-up-count {
45             type yang:counter32;
46             config false;
47             description
48             "Counter that increments every time a Follow_Up message is
49              received.";
50             reference
51             "14.12.4 of IEEE Std 802.1AS";
52         }
53         leaf rx-pdelay-req-count {
54             type yang:counter32;
55             config false;
56             description
57             "Counter that increments every time a Pdelay_Req message is
58              received.";
59             reference

```



```

1      "14.12.5 of IEEE Std 802.1AS";
2  }
3  leaf rx-pdelay-resp-count {
4      type yang:counter32;
5      config false;
6      description
7          "Counter that increments every time a Pdelay_Resp message is
8          received.";
9      reference
10         "14.12.6 of IEEE Std 802.1AS";
11 }
12 leaf rx-pdelay-resp-follow-up-count {
13     type yang:counter32;
14     config false;
15     description
16         "Counter that increments every time a Pdelay_Resp_Follow_Up
17         message is received.";
18     reference
19         "14.12.7 of IEEE Std 802.1AS";
20 }
21 leaf rx-announce-count {
22     type yang:counter32;
23     config false;
24     description
25         "Counter that increments every time an Announce message is
26         received.";
27     reference
28         "14.12.8 of IEEE Std 802.1AS";
29 }
30 leaf rx-packet-discard-count {
31     type yang:counter32;
32     config false;
33     description
34         "Counter that increments every time a PTP message of the
35         respective PTP Instance is discarded.";
36     reference
37         "14.12.9 of IEEE Std 802.1AS";
38 }
39 leaf sync-receipt-timeout-count {
40     type yang:counter32;
41     config false;
42     description
43         "Counter that increments every time a sync receipt timeout
44         occurs.";
45     reference
46         "14.12.10 of IEEE Std 802.1AS";
47 }
48 leaf announce-receipt-timeout-count {
49     type yang:counter32;
50     config false;
51     description
52         "Counter that increments every time an announce receipt timeout
53         occurs.";
54     reference
55         "14.12.11 of IEEE Std 802.1AS";
56 }
57 leaf pdelay-allowed-lost-exceeded-count {
58     type yang:counter32;
59     config false;

```

```

1      description
2          "Counter that increments every time the value of the variable
3          lostResponses exceeds the value of the variable
4          allowedLostResponses, in the RESET state of the MDPdelayReq
5          state machine.";
6      reference
7          "14.12.12 of IEEE Std 802.1AS";
8  }
9  leaf tx-sync-count {
10     type yang:counter32;
11     config false;
12     description
13         "Counter that increments every time synchronization information
14         is transmitted.";
15     reference
16         "14.12.13 of IEEE Std 802.1AS";
17 }
18 leaf tx-one-step-sync-count {
19     type yang:counter32;
20     config false;
21     description
22         "Counter that increments every time a one-step Sync message is
23         transmitted.";
24     reference
25         "14.12.14 of IEEE Std 802.1AS";
26 }
27 leaf tx-follow-up-count {
28     type yang:counter32;
29     config false;
30     description
31         "Counter that increments every time a Follow_Up message is
32         transmitted.";
33     reference
34         "14.12.15 of IEEE Std 802.1AS";
35 }
36 leaf tx-pdelay-req-count {
37     type yang:counter32;
38     config false;
39     description
40         "Counter that increments every time a Pdelay_Req message is
41         transmitted.";
42     reference
43         "14.12.16 of IEEE Std 802.1AS";
44 }
45 leaf tx-pdelay-resp-count {
46     type yang:counter32;
47     config false;
48     description
49         "Counter that increments every time a Pdelay_Resp message is
50         transmitted.";
51     reference
52         "14.12.17 of IEEE Std 802.1AS";
53 }
54 leaf tx-pdelay-resp-follow-up-count {
55     type yang:counter32;
56     config false;
57     description
58         "Counter that increments every time a Pdelay_Resp_Follow_Up
59         message is transmitted.";

```

```

1      reference
2      "14.12.18 of IEEE Std 802.1AS";
3  }
4  leaf tx-announce-count {
5      type yang:counter32;
6      config false;
7      description
8          "Counter that increments every time an Announce message is
9          transmitted.";
10     reference
11         "14.12.19 of IEEE Std 802.1AS";
12 }
13 }
14 }
15 augment
16     "/ptp-tt:ptp"+
17     "/ptp-tt:instances"+
18     "/ptp-tt:instance"+
19     "/ptp-tt:ports"+
20     "/ptp-tt:port" {
21     description
22         "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588 PTP
23         Port.";
24     container asymmetry-measurement-mode-ds {
25         description
26             "Represents the capability to enable/disable the Asymmetry
27             Compensation Measurement Procedure on a PTP Port. This data set
28             is used instead of the CMLDS asymmetry-measurement-mode-ds when
29             only a single PTP Instance is present (i.e., CMLDS is not used).";
30         reference
31             "14.15 of IEEE Std 802.1AS
32             Annex G of IEEE Std 802.1AS";
33         leaf enabled {
34             type boolean;
35             description
36                 "For full-duplex IEEE Std 802.3 media, the value is true if an
37                 asymmetry measurement is being performed for the link attached
38                 to this PTP Port, and false otherwise. For all other media, the
39                 value shall be false.";
40             reference
41                 "14.15.2 of IEEE Std 802.1AS";
42         }
43     }
44 }
45 augment
46     "/ptp-tt:ptp"+
47     "/ptp-tt:common-services"+
48     "/ptp-tt:cmlds"+
49     "/ptp-tt:ports"+
50     "/ptp-tt:port"+
51     "/ptp-tt:link-port-ds" {
52     description
53         "Augment IEEE Std 1588 cmldsLinkPortDS.";
54     leaf cmlds-link-port-enabled {
55         type boolean;
56         config false;
57         description
58             "Boolean that is true if both delay-mechanism is common-p2p and
59             the value of ptp-port-enabled is true, for at least one PTP Port

```

```
1      that uses the CMLDS; otherwise, the value is false.";
2      reference
3      "11.2.18.1 of IEEE Std 802.1AS
4      14.18.3 of IEEE Std 802.1AS";
5  }
6  leaf is-measuring-delay {
7      type boolean;
8      config false;
9      description
10     "This leaf is analogous to is-measuring-delay for a PTP Port, but
11     applicable to this Link Port.";
12     reference
13     "14.18.4 of IEEE Std 802.1AS";
14 }
15 leaf as-capable-across-domains {
16     type boolean;
17     config false;
18     description
19     "This leaf is true when all PTP Instances (domains) for this Link
20     Port detect proper exchange of Pdelay messages.";
21     reference
22     "11.2.2 of IEEE Std 802.1AS
23     14.18.5 of IEEE Std 802.1AS";
24 }
25 leaf mean-link-delay-thresh {
26     type ptp-tt:time-interval;
27     description
28     "Propagation time threshold for mean-link-delay, above which a
29     Link Port is not considered capable of participating in the IEEE
30     Std 802.1AS protocol.";
31     reference
32     "14.18.7 of IEEE Std 802.1AS";
33 }
34 leaf neg-mean-link-delay-thresh {
35     type ptp-tt:time-interval;
36     description
37     "The negative propagation time threshold for mean-link-delay,
38     below which a Link Port is not considered capable of
39     participating in the IEEE Std 802.1AS protocol.";
40     reference
41     "14.18.8 of IEEE Std 802.1AS";
42 }
43 leaf initial-log-pdelay-req-interval {
44     type int8;
45     description
46     "This leaf is analogous to initial-log-pdelay-req-interval for a
47     PTP Port, but applicable to this Link Port.";
48     reference
49     "14.18.11 of IEEE Std 802.1AS";
50 }
51 leaf current-log-pdelay-req-interval {
52     type int8;
53     config false;
54     description
55     "This leaf is analogous to current-log-pdelay-req-interval for a
56     PTP Port, but applicable to this Link Port.";
57     reference
58     "14.18.12 of IEEE Std 802.1AS";
59 }
```

```

1  leaf use-mgt-log-pdelay-req-interval {
2      type boolean;
3      description
4          "This leaf is analogous to use-mgt-log-pdelay-req-interval for a
5          PTP Port, but applicable to this Link Port.";
6      reference
7          "14.18.13 of IEEE Std 802.1AS";
8  }
9  leaf mgt-log-pdelay-req-interval {
10     type int8;
11     description
12         "This leaf is analogous to mgt-log-pdelay-req-interval for a PTP
13         Port, but applicable to this Link Port.";
14     reference
15         "14.18.14 of IEEE Std 802.1AS";
16 }
17 leaf initial-compute-neighbor-rate-ratio {
18     type boolean;
19     description
20         "This leaf is analogous to initial-compute-neighbor-rate-ratio
21         for a PTP Port, but applicable to this Link Port.";
22     reference
23         "14.18.15 of IEEE Std 802.1AS";
24 }
25 leaf current-compute-neighbor-rate-ratio {
26     type boolean;
27     config false;
28     description
29         "This leaf is analogous to current-compute-neighbor-rate-ratio
30         for a PTP Port, but applicable to this Link Port.";
31     reference
32         "14.18.16 of IEEE Std 802.1AS";
33 }
34 leaf use-mgt-compute-neighbor-rate-ratio {
35     type boolean;
36     description
37         "This leaf is analogous to use-mgt-compute-neighbor-rate-ratio
38         for a PTP Port, but applicable to this Link Port.";
39     reference
40         "14.18.17 of IEEE Std 802.1AS";
41 }
42 leaf mgt-compute-neighbor-rate-ratio {
43     type boolean;
44     description
45         "This leaf is analogous to mgt-compute-neighbor-rate-ratio for a
46         PTP Port, but applicable to this Link Port.";
47     reference
48         "14.18.18 of IEEE Std 802.1AS";
49 }
50 leaf initial-compute-mean-link-delay {
51     type boolean;
52     description
53         "This leaf is analogous to initial-compute-mean-link-delay for a
54         PTP Port, but applicable to this Link Port.";
55     reference
56         "14.18.19 of IEEE Std 802.1AS";
57 }
58 leaf current-compute-mean-link-delay {
59     type boolean;

```

```

1      config false;
2      description
3          "This leaf is analogous to current-compute-mean-link-delay for a
4          PTP Port, but applicable to this Link Port.";
5      reference
6          "14.18.20 of IEEE Std 802.1AS";
7  }
8  leaf use-mgt-compute-mean-link-delay {
9      type boolean;
10     description
11         "This leaf is analogous to use-mgt-compute-mean-link-delay for a
12         PTP Port, but applicable to this Link Port.";
13     reference
14         "14.18.21 of IEEE Std 802.1AS";
15 }
16 leaf mgt-compute-mean-link-delay {
17     type boolean;
18     description
19         "This leaf is analogous to mgt-compute-mean-link-delay for a PTP
20         Port, but applicable to this Link Port.";
21     reference
22         "14.18.22 of IEEE Std 802.1AS";
23 }
24 leaf allowed-lost-responses {
25     type uint8;
26     description
27         "This leaf is analogous to allowed-lost-responses for a PTP Port,
28         but applicable to this Link Port.";
29     reference
30         "14.18.23 of IEEE Std 802.1AS";
31 }
32 leaf allowed-faults {
33     type uint8;
34     description
35         "This leaf is analogous to allowed-faults for a PTP Port, but
36         applicable to this Link Port.";
37     reference
38         "14.18.24 of IEEE Std 802.1AS";
39 }
40 leaf-list pdelay-truncated-timestamps {
41     type uinteger48;
42     config false;
43     description
44         "This leaf is analogous to pdelay-truncated-timestamps for a PTP
45         Port, but applicable to this Link Port.";
46     reference
47         "14.18.26 of IEEE Std 802.1AS";
48 }
49 }
50 augment
51     "/ptp-tt:ptp"+
52     "/ptp-tt:common-services"+
53     "/ptp-tt:cmlds"+
54     "/ptp-tt:ports"+
55     "/ptp-tt:port" {
56     description
57         "Augment to add port-statistics-ds to IEEE Std 1588 Link Port.";
58     container port-statistics-ds {
59         description

```

```

1      "This container is analogous to port-statistics-ds for a PTP
2      Port, but applicable to this Link Port.";
3      reference
4      "14.19 of IEEE Std 802.1AS";
5      leaf rx-pdelay-req-count {
6          type yang:counter32;
7          config false;
8          description
9              "This leaf is analogous to rx-pdelay-req-count for a PTP Port,
10             but applicable to this Link Port.";
11          reference
12              "14.19.2 of IEEE Std 802.1AS";
13      }
14      leaf rx-pdelay-resp-count {
15          type yang:counter32;
16          config false;
17          description
18              "This leaf is analogous to rx-pdelay-resp-count for a PTP Port,
19             but applicable to this Link Port.";
20          reference
21              "14.19.3 of IEEE Std 802.1AS";
22      }
23      leaf rx-pdelay-resp-follow-up-count {
24          type yang:counter32;
25          config false;
26          description
27              "This leaf is analogous to rx-pdelay-resp-follow-up-count for a
28              PTP Port, but applicable to this Link Port.";
29          reference
30              "14.19.4 of IEEE Std 802.1AS";
31      }
32      leaf rx-packet-discard-count {
33          type yang:counter32;
34          config false;
35          description
36              "This leaf is analogous to rx-packet-discard-count for a PTP
37              Port, but applicable to this Link Port.";
38          reference
39              "14.19.5 of IEEE Std 802.1AS";
40      }
41      leaf pdelay-allowed-lost-exceeded-count {
42          type yang:counter32;
43          config false;
44          description
45              "This leaf is analogous to pdelay-allowed-lost-exceeded-count
46              for a PTP Port, but applicable to this Link Port.";
47          reference
48              "14.19.6 of IEEE Std 802.1AS";
49      }
50      leaf tx-pdelay-req-count {
51          type yang:counter32;
52          config false;
53          description
54              "This leaf is analogous to tx-pdelay-req-count for a PTP Port,
55             but applicable to this Link Port.";
56          reference
57              "14.19.7 of IEEE Std 802.1AS";
58      }
59      leaf tx-pdelay-resp-count {

```

```

1      type yang:counter32;
2      config false;
3      description
4          "This leaf is analogous to tx-pdelay-resp-count for a PTP Port,
5          but applicable to this Link Port.";
6      reference
7          "14.19.8 of IEEE Std 802.1AS";
8  }
9  leaf tx-pdelay-resp-follow-up-count {
10     type yang:counter32;
11     config false;
12     description
13         "This leaf is analogous to tx-pdelay-resp-follow-up-count for a
14         PTP Port, but applicable to this Link Port.";
15     reference
16         "14.19.10 of IEEE Std 802.1AS";
17 }
18 }
19 }
20 augment
21     "/ptp-tt:ptp"+
22     "/ptp-tt:common-services"+
23     "/ptp-tt:cmls"+
24     "/ptp-tt:ports"+
25     "/ptp-tt:port" {
26     description
27         "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588 Link
28         Port.";
29     container asymmetry-measurement-mode-ds {
30         description
31             "This container is analogous to asymmetry-measurement-mode-ds for
32             a PTP Port, but applicable to this Link Port.";
33         reference
34             "14.20 of IEEE Std 802.1AS";
35         leaf enabled {
36             type boolean;
37             description
38                 "This leaf is analogous to
39                 asymmetry-measurement-mode-ds.enabled for a PTP Port, but
40                 applicable to this Link Port.";
41             reference
42                 "14.20.2 of IEEE Std 802.1AS";
43         }
44     }
45 }
46 }
47

```

48 17.6.2 Module ieee802-dot1as-hs.yang

```

49 module ieee802-dot1as-hs {
50     yang-version 1.1;
51     namespace "urn:ieee:std:802.1AS:yang:ieee802-dot1as-hs";
52     prefix dot1as-hs;
53
54     import ietf-yang-types {
55         prefix yang;
56     }

```



```

1  import ieee1588-ptp-tt {
2      prefix ptp-tt;
3  }
4  import ieee802-dot1as-gtp {
5      prefix dot1as-gtp;
6  }
7
8  organization
9      "IEEE 802.1 Working Group";
10 contact
11     "WG-URL: http://ieee802.org/1/
12     WG-EMail: stds-802-1-l@ieee.org
13
14     Contact: IEEE 802.1 Working Group Chair
15             Postal: C/O IEEE 802.1 Working Group
16             IEEE Standards Association
17             445 Hoes Lane
18             Piscataway, NJ 08854
19             USA
20
21     E-mail: stds-802-1-chairs@ieee.org";
22 description
23     "Management objects that control hot standby systems as
24     specified in clause 18 of IEEE Std 802.1AS-2020-REV and
25     drift tracking features as specified in clause 11.4.4.4
26     of IEEE Std 802.1AS-2020-REV.
27
28     References in this YANG module to IEEE Std 802.1AS are to
29     IEEE Std 802.1AS-2020-REV.
30
31     Copyright (C) IEEE (2025).
32     This version of this YANG module is part of IEEE Std
33     802.1AS-2020-REV; see the standard itself for full legal
34     notices.";
35
36 revision 2025-02-04 {
37     description
38         "Published as part of IEEE Std 802.1AS-2020-REV.";
39     reference
40         "IEEE Std 802.1AS - Timing and Synchronization for Time-Sensitive
41         Applications: IEEE Std 802.1AS-2020-REV.";
42 }
43
44 feature hot-standby {
45     description
46         "This feature indicates that the device supports the
47         hot-standby functionality.";
48 }
49
50 typedef hot-standby-system-state-type {
51     type enumeration {
52         enum init {
53             value 0;
54             description
55                 "Initialization after the HotStandbySystem powers on and
56                 is enabled. In this state, the system is waiting for
57                 both PTP Instances to synchronize.";
58         }
59         enum redundant {

```

```

1      value 1;
2      description
3          "Both PTP Instances are synchronized according to the
4              requirements of the respective application or profile
5              standard. Time synchronization is redundant.";
6      }
7      enum not-redundant {
8          value 2;
9          description
10             "One PTP Instance is synchronized, and the other
11                 PTP Instance is faulted (not synchronized). Time
12                 synchronization continues to meet the requirements of
13                 the respective application or profile standard. Time
14                 synchronization is not redundant.";
15     }
16     enum out-of-sync {
17         value 3;
18         description
19             "The HotStandbySystem is adjusting phase/frequency of
20                 its local time using the data stored while the system
21                 was in the REDUNDANT or NOT_REDUNDANT state, but the
22                 local time will eventually drift relative to other
23                 time-aware systems. During OUT_OF_SYNC state, time
24                 synchronization might not meet the requirements of the
25                 respective application or profile standard.";
26     }
27 }
28 description
29     "Possible states of the hot-standby system.";
30 }
31
32 typedef nrr-comp-method-type {
33     type enumeration {
34         enum sync {
35             value 0;
36             description
37                 "If the value is Sync and driftTrackingTlvSupport is
38                     TRUE, neighborRateRatio is populated with the value of
39                     nrrSync whenever a new value of nrrSync is computed.";
40         }
41         enum pdelay {
42             value 1;
43             description
44                 "If the value is Pdelay or if driftTrackingTlvSupport
45                     is FALSE, neighborRateRatio is populated with the value
46                     of nrrPdelay whenever a new value of nrrPdelay is
47                     computed.";
48         }
49     }
50     description
51         "typedef for nrrCompMethod.";
52 }
53
54 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
55     + "/ptp-tt:parent-ds" {
56     description
57         "Augment IEEE Std 1588 parentDS.";
58     leaf gm-present {
59         type boolean;

```

```

1      config false;
2      description
3          "The value of gmPresent is set equal to the value of the
4          global variable gmPresent. This parameter indicates to the
5          ClockTarget whether a Grandmaster PTP Instance is
6          present.";
7      reference
8          "14.4.8 of IEEE Std 802.1AS";
9  }
10 }
11
12 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance" {
13     description
14         "Augment IEEE Std 1588 instanceList.";
15     container ptp-instance-sync-ds {
16         description
17             "The ptpInstanceSyncDS describes the synchronization status
18             of the PTP Instance.";
19         reference
20             "14.8 of IEEE Std 802.1AS";
21         leaf is-synced {
22             type boolean;
23             config false;
24             description
25                 "The value of the global variable isSynced.";
26             reference
27                 "14.8.2 of IEEE Std 802.1AS";
28         }
29         leaf offset-from-time-transmitter-max {
30             type ptp-tt:time-interval;
31             description
32                 "The value is the threshold for
33                 offsetFromTimeTransmitter, below which the PTP Instance
34                 is considered to be synchronized.";
35             reference
36                 "14.8.3 of IEEE Std 802.1AS";
37         }
38         leaf rx-sync-count-time-receiver-p-thresh {
39             type uint32;
40             description
41                 "The value of rxSyncCountTimeReceiverPThresh is the
42                 threshold for rxSyncCountTimeReceiverP, above which
43                 the PTP Instance is considered to be synchronized.";
44             reference
45                 "14.8.4 of IEEE Std 802.1AS";
46         }
47         leaf offset-max-exceeded-count-thresh {
48             type uint32;
49             description
50                 "The value of offsetMaxExceededCountThresh is the
51                 threshold for the number of consecutive exceedances of
52                 offsetFromTimeTransmitterMax by
53                 offsetFromTimeTransmitter, at which isSynced is no
54                 longer TRUE.";
55             reference
56                 "14.8.5 of IEEE Std 802.1AS";
57         }
58         leaf offset-max-met-count-thresh {
59             type uint32;

```

```

1      description
2          "The value of offsetMaxMetCountThresh is the threshold
3          for the number of consecutive occurrences of
4          offsetFromTimeTransmitter being within
5          offsetFromTimeTransmitterMax, at which isSynced is
6          changed to TRUE if it currently is FALSE.";
7      reference
8          "14.8.6 of IEEE Std 802.1AS";
9  }
10 }
11 container drift-tracking-ds {
12     description
13         "The driftTrackingDS contains managed objects that are used
14         to enable or disable the optional Drift_Tracking TLV.";
15     reference
16         "14.9 of IEEE Std 802.1AS";
17     leaf drift-tracking-tlv-support {
18         type boolean;
19         description
20             "The value of driftTrackingTlvSupport indicates whether
21             the Drift_Tracking TLV is enabled or disabled.";
22         reference
23             "14.9.2 of IEEE Std 802.1AS";
24     }
25     leaf sync-grandmaster-identity {
26         type ptp-tt:clock-identity;
27         config false;
28         description
29             "The value of syncGrandmasterIdentity as determined by
30             the MDSyncSend state machine.";
31         reference
32             "14.9.3 of IEEE Std 802.1AS";
33     }
34     leaf sync-steps-removed {
35         type uint32;
36         config false;
37         description
38             "The value of syncStepsRemoved as determined by the
39             MDSyncSend state machine.";
40         reference
41             "14.9.4 of IEEE Std 802.1AS";
42     }
43     leaf rate-ratio-drift {
44         type int32;
45         config false;
46         description
47             "This value is equal to (RRdrift - 1.0) x (2^41),
48             truncated to the next smaller signed integer, where
49             RRdrift is the measured estimate of the rate of change
50             per second of the ratio of the frequency of the
51             Grandmaster Clock to the frequency of the Local Clock
52             entity in the PTP instance that sends the message.";
53         reference
54             "14.9.5 of IEEE Std 802.1AS";
55     }
56 }
57 }
58
59 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"

```

```

1      + "/ptp-tt:ports/ptp-tt:port/ptp-tt:port-ds" {
2  description
3      "Augment IEEE Std 1588 commonServices.";
4  leaf gtp-capable-state-machines-enabled {
5      type boolean;
6      description
7          "A Boolean that is used to enable or disable the
8            GtpCapableTransmit, GtpCapableReceive, and
9            GtpCapableIntervalSetting state machines.";
10     reference
11         "14.10.55 of IEEE Std 802.1AS";
12 }
13 leaf nrr-pdelay {
14     type int32;
15     config false;
16     description
17         "The value is an estimate of the ratio of the frequency of
18           the LocalClock entity of the time-aware system at the
19           other end of the link attached to this Link Port, to the
20           frequency of the LocalClock entity of this time-aware
21           system. nrrPdelay is expressed as the fractional frequency
22           offset stored in the global variable nrrPdelay multiplied
23           by 2^41, i.e., the quantity (nrrPdelay - 1.0)(2^41).";
24     reference
25         "14.10.56 of IEEE Std 802.1AS";
26 }
27 leaf nrr-sync {
28     type int32;
29     config false;
30     description
31         "The value is an estimate of the ratio of the frequency of
32           the LocalClock entity of the time-aware system at the
33           other end of the link attached to this Link Port, to the
34           frequency of the LocalClock entity of this time-aware
35           system. nrrSync is expressed as the fractional frequency
36           offset stored in the global variable nrrSync multiplied by
37           2^41, i.e., the quantity (nrrSync - 1.0)(2^41).";
38     reference
39         "14.10.57 of IEEE Std 802.1AS";
40 }
41 leaf nrr-comp-method {
42     type nrr-comp-method-type;
43     description
44         "An Enumeration that takes on the values sync and pdelay to
45           indicate the source of the value of neighborRateRatio.";
46     reference
47         "14.10.58 of IEEE Std 802.1AS";
48 }
49 leaf as-capable-across-domains {
50     type boolean;
51     description
52         "This leaf is true when this PTP port detects proper
53           exchange of Pdelay messages.";
54     reference
55         "14.10.59 of IEEE Std 802.1AS";
56 }
57 }
58
59 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"

```

```

1      + "/ptp-tt:ports/ptp-tt:port"
2      + "/dot1as-gptp:port-statistics-ds" {
3  description
4      "Augment IEEE Std 802.1AS PortStatisticsDS.";
5  leaf rx-sync-count-time-receiver-p {
6      type yang:counter32;
7      config false;
8      description
9          "This counter increments whenever time synchronization
10             information is received on a PTP Port when its port
11             state is TimeReceiverPort.";
12      reference
13          "14.12.20 of IEEE Std 802.1AS";
14  }
15  }
16
17  augment "/ptp-tt:ptp/ptp-tt:common-services" {
18      description
19          "Augment IEEE Std 1588 commonServices.
20
21          IEEE Std 802.1ASdm-2024 specifies nrrPdelay
22          (nrr-pdelay), which is provided in YANG as the
23          semantically equivalent node in ieee1588-ptp-tt named
24          scaled-neighbor-rate-ratio (in link-port-ds).";
25      container hot-standby-service {
26          if-feature "hot-standby";
27          description
28              "The hotStandbyService structure contains the
29              hotStandbySystemList, which is a list of instances of the
30              Hot Standby Service.";
31          reference
32              "14.21 of IEEE Std 802.1AS";
33          list hot-standby-system {
34              key "hot-standby-system-index";
35              description
36                  "List of instances of the Hot Standby Service";
37              leaf hot-standby-system-index {
38                  type uint8;
39                  description
40                      "Index for the hot-standby system.";
41              }
42              container hot-standby-system-ds {
43                  description
44                      "The hotStandbySystemDS describes the attributes of the
45                      respective instance of the Hot Standby Service.";
46                  reference
47                      "14.21 of IEEE Std 802.1AS";
48                  leaf primary-ptp-instance-index {
49                      type uint32;
50                      description
51                          "The value of primaryPtpInstanceIndex is the index of
52                          the primary PTP Instance associated with this
53                          hotStandbySystem instance.";
54                      reference
55                          "14.21.2 of IEEE Std 802.1AS";
56                  }
57                  leaf secondary-ptp-instance-index {
58                      type uint32;
59                      description

```

```

1         "The value of secondaryPtpInstanceId is the index
2         of the secondaryPTP Instance associated with this
3         hotStandbySystem instance.";
4     reference
5         "14.21.3 of IEEE Std 802.1AS";
6 }
7 leaf hot-standby-system-enable {
8     type boolean;
9     description
10        "The value is the hotStandbySystemEnable attribute of
11        the HotStandbySystem entity.";
12    reference
13        "14.21.4 of IEEE Std 802.1AS";
14 }
15 leaf hot-standby-system-state {
16     type hot-standby-system-state-type;
17     config false;
18     description
19        "The value of hotStandbySystemState is the state of
20        the hotStandbySystem, i.e., the value of the global
21        variable hotStandbySystemState.";
22    reference
23        "14.21.5 of IEEE Std 802.1AS";
24 }
25 leaf hot-standby-system-split-functionality {
26     type boolean;
27     description
28        "If the value is TRUE, the optional split
29        functionality is used. If the value is FALSE, the
30        optional split functionality is not used.";
31    reference
32        "14.21.6 of IEEE Std 802.1AS";
33 }
34 leaf primary-secondary-offset {
35     type dot1as-gptp:scaled-ns;
36     config false;
37     description
38        "The absolute value of the difference between the
39        clockTimeTransmitterTimes of the primary and
40        secondary PTP Instances.";
41    reference
42        "14.21.7 of IEEE Std 802.1AS";
43 }
44 leaf primary-secondary-offset-thresh {
45     type dot1as-gptp:scaled-ns;
46     description
47        "The threshold for
48        hotStandbySystemDS.primarySecondaryOffset, above
49        which the hotStandbySystemState transitions from
50        REDUNDANT to NOT_REDUNDANT, or does not transition
51        from NOT_REDUNDANT or OUT_OF_SYNC to REDUNDANT even
52        if other conditions for these transitions are
53        satisfied.";
54    reference
55        "14.21.8 of IEEE Std 802.1AS";
56 }
57 leaf hot-standby-system-log-sync-time-thresh {
58     type int8;
59     description

```

```

1         "The value of hotStandbySystemLogSyncTimeThresh is
2         the logarithm to base 2 of the time interval, in
3         seconds, after which the hotStandbySystem
4         transitions from the OUT_OF_SYNC state to either the
5         NOT_REDUNDANT or REDUNDANT state, or from the
6         NOT_REDUNDANT to the REDUNDANT state, if all other
7         conditions for the respective transition are met.
8         The value -128 means that the transition time is
9         zero, i.e., the transition occurs immediately.";
10        reference
11        "14.21.9 of IEEE Std 802.1AS";
12    }
13 }
14 container hot-standby-system-description-ds {
15     description
16         "The hotStandbySystemDescriptionDS contains descriptive
17         information for the respective instance of the Hot
18         Standby Service.";
19     reference
20         "14.22 of IEEE Std 802.1AS";
21     leaf user-description {
22         type string {
23             length "0..128";
24         }
25         description
26             "Configurable description of the hot standby system.";
27         reference
28             "14.22.2 of IEEE Std 802.1AS";
29     }
30 }
31 }
32 }
33 }
34 }
35

```


1 18. Provisions for hot standby on the media-independent layer

2 18.1 General

3 Hot standby extends the media-dependent layer by the following:

- 4 — A function that transforms the synchronized times of two generalized Precision Time Protocol
- 5 (gPTP) domains into one synchronized time for use by applications;
- 6 — A function that directs the synchronized time of one gPTP domain into a different gPTP domain;
- 7 and
- 8 — Mechanisms that determine whether a gPTP domain has sufficient quality to be used for hot standby.

9 For time synchronization using hot standby, two distinct domains are statically configured in the network
10 and the best timeTransmitter clock algorithm (BTCA) is disabled for these domains. When hot standby is
11 used for redundancy, a time-aware system operates two PTP Instances simultaneously, each in its own
12 domain. One of the domains is considered the primary domain, and the other domain is considered the
13 secondary domain. A time-aware system that has a primary domain and a secondary domain available uses
14 the primary domain via the associated PTP instance. If the primary domain becomes unavailable (e.g., due to
15 temporary or permanent failure of a physical link) and the secondary domain is still available to the
16 time-aware system, the time-aware system begins using the secondary domain.

17 Examples of hot standby are shown in 7.3.3 of this standard.

18 18.2 Overview

19 Figure 18-1 provides a model of hot standby for time synchronization.

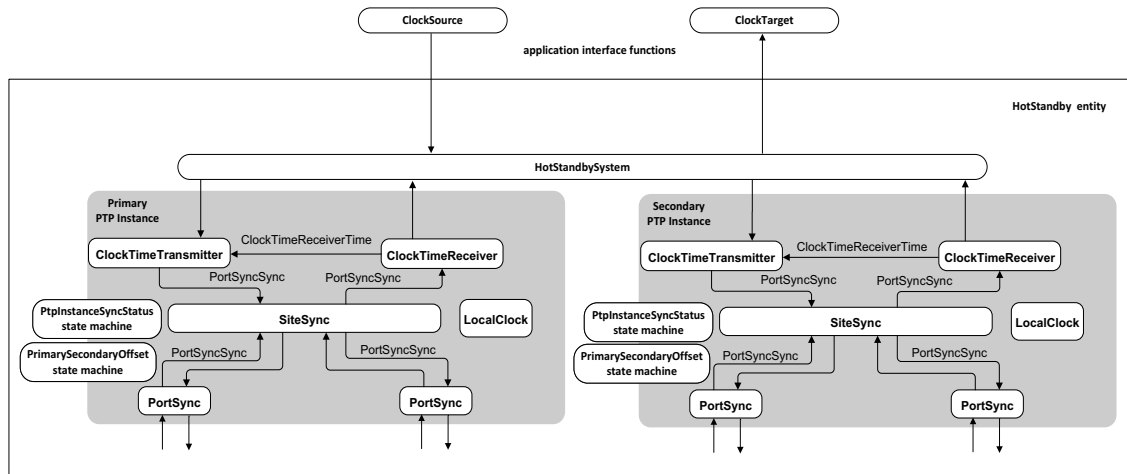


Figure 18-1—Model for hot standby entity in a time-aware system, and its interfaces to higher-layer applications

20 Each PTP Instance has a corresponding PtpInstanceSyncStatus state machine (see 18.4). The
21 PtpInstanceSyncStatus state machine monitors its PTP Instance to determine whether it is synchronized
22 according to the requirements of the respective application or profile standard.

1 There is one HotStandbySystem entity, which interacts with the primary and secondary PTP Instances via the HotStandbySystem state machine (see 18.5) in order to provide a single value of time to the ClockTarget (see Clause 9). This single value of time is based on the redundant values of time provided by either the primary PTP Instance, or the secondary PTP Instance, or both. There is one HotStandbySystem for both PTP Instances that use hot standby. The primary and secondary PTP Instances use the same timescale (see 8.2).

The ClockSource entity is used to transfer the source of time when at least one of the PTP Instances is grandmaster-capable. The ClockSource and its application interfaces are analogous to the ClockSource specified in Clause 9. The ClockSource might be present if the PTP Instance is capable of becoming a Grandmaster PTP Instance.

There is one LocalClock entity for each PTP Instance. Therefore, Figure 18-1 shows two LocalClock entities.

NOTE—The LocalClock entities can be traceable to the same oscillator or to different oscillators.

18.3 PTP Instance configuration

PTP Instances that support hot standby are configured as follows:

- a) The two PTP Instances are enabled using domainNumbers as specified by the respective application or profile standard;
- b) For both PTP Instances, externalPortConfigurationEnabled is set to TRUE;
- c) If a PTP Instance (primary or secondary) is grandmaster, externalPortConfigurationPortDS.desiredState is configured to TimeTransmitterPort or PassivePort for all PTP Ports (portNumber 1 and higher); otherwise, externalPortConfiguration.desiredState is configured to TimeReceiverPort for one PTP Port, and is configured to TimeTransmitterPort or PassivePort for other PTP Ports; and
- d) Each PTP Instance shall have a corresponding PtpInstanceSyncStatus state machine (see 18.4).

18.4 PtpInstanceSyncStatus state machine

This state machine operates within the PTP Instance, to determine whether it is synchronized according to the requirements of the respective application or profile standard (see 3.23).

18.4.1 PtpInstanceSyncStatus state machine global variables

The following variables are used in the state diagram in Figure 18-2 (in 18.4.4):

18.4.1.1 isSynced: The synchronization status of the PTP Instance. The variable is a Boolean, and is TRUE if synchronization is sufficient according to the operation of the PtpInstanceSyncStatus state machine and FALSE otherwise. An application or profile standard can set the offsetFromTimeTransmitterMax (see 18.4.1.3), rxSyncCountTimeReceiverPThresh (see 18.4.1.5), offsetMaxExceededCountThresh (see 18.4.1.7), and offsetMaxMetCountThresh (see 18.4.1.9) according to its requirements.

18.4.1.2 offsetFromTimeTransmitter: The value of the managed object currentDS.offsetFromTimeTransmitter (see 14.3.3).

18.4.1.3 offsetFromTimeTransmitterMax: The value of the managed object ptpInstanceSyncDS.offsetFromTimeTransmitterMax (see 14.8.3)

1 **18.4.1.4 rxSyncCountTimeReceiverP:** The value of the managed object
2 portStatisticsDS.rxSyncCountTimeReceiverP (see 14.12.20).

3 **18.4.1.5 rxSyncCountTimeReceiverPThresh:** The value of the managed object
4 ptpInstanceSyncDS.rxSyncCountTimeReceiverPThresh (see 14.8.4)

5 **18.4.1.6 offsetMaxExceededCount:** The current number of consecutive exceedances of
6 offsetFromTimeTransmitterMax (see 18.4.1.3) by offsetFromTimeTransmitter (see 18.4.1.2). The data type
7 of offsetMaxExceededCount is UInteger32.

8 **18.4.1.7 offsetMaxExceededCountThresh:** The threshold (see 14.8.5) for the number of consecutive
9 exceedances of offsetFromTimeTransmitterMax (see 18.4.1.3) by offsetFromTimeTransmitter (see
10 18.4.1.2), at which isSynced for the PTP Instance is no longer TRUE (see 18.4.3.3).

11 **18.4.1.8 offsetMaxMetCount:** The current number of consecutive occurrences of
12 offsetFromTimeTransmitter (see 18.4.1.2) being within offsetFromTimeTransmitterMax (see 18.4.1.3). The
13 data type of offsetMaxMetCount is UInteger32.

14 **18.4.1.9 offsetMaxMetCountThresh:** The threshold (see 14.8.6) for the number of consecutive
15 occurrences of offsetFromTimeTransmitter (see 18.4.1.2) being within offsetFromTimeTransmitterMax
16 (see 18.4.1.3), at which isSynced for the PTP Instance is changed from FALSE to TRUE (see 18.4.3.3).

17 **18.4.2 PtpInstanceSyncStatus state machine local variables**

18 **18.4.2.1 lastRxSyncCountTimeReceiverP:** Holds the last value of rxSyncCountTimeReceiverP
19 (see 18.4.1.4) that the state machine read for the TimeReceiverPort, as part of monitoring for a sync event
20 message timeout.

21 **18.4.3 PtpInstanceSyncStatus state machine functions**

22 **18.4.3.1 isGtpCapable():** This function returns a Boolean value that is TRUE when ptpPortEnabled
23 (see 10.2.5.13) and asCapable (see 10.2.5.1) are TRUE, as determined by the function portIsCapable, for at
24 least one PTP Port (i.e., port for which portNumber is not zero).

```
25 isGtpCapable()
26 {
27     if (!instanceEnable)
28         return (FALSE);
29     for (int i = 1; i <= numberPorts; i++)
30         // see 8.6.2.8 for numberPorts
31         {
32             if (portIsCapable (i))
33                 return (TRUE);
34         }
35     return (FALSE);
36 }
```

37 **18.4.3.2 isGm():** This function determines if the PTP Instance is a Grandmaster PTP Instance (TRUE) or
38 not (FALSE) by searching the selectedState array (see 10.2.4.20) for absence or presence of at least one
39 value equal to TimeReceiverPort (see Table 10-2).

```
40 isGm()
41 {
42     for (int i = 1; i <= numberPorts; i++)
43         // see 8.6.2.8 for numberPorts
```

```

1      1      {
2      2      if (selectedState[i] == TimeReceiverPort)
3      3      return (FALSE);
4      4      }
5      5      if (priority1 == 255)
6      6      return (FALSE);
7      7
8      8      return (TRUE);
9      9      }

```

10 **18.4.3.3 portIsCapable(j):** This function returns a Boolean value that is TRUE if the PTP Port whose
11 portList index is j is enabled and asCapable is TRUE, and FALSE otherwise.

```

12 portIsCapable (j)
13 {
14     if (ptpPortEnabled && asCapable)
15         return (TRUE);
16     else
17         return (FALSE);
18 }

```

19 **18.4.4 State diagram**

20 The PtpInstanceSyncStatus state machine shall implement the function specified by the state diagram in
21 Figure 18-2, the variables specified in 18.4.1 and 18.4.2, and the functions specified in 18.4.3.

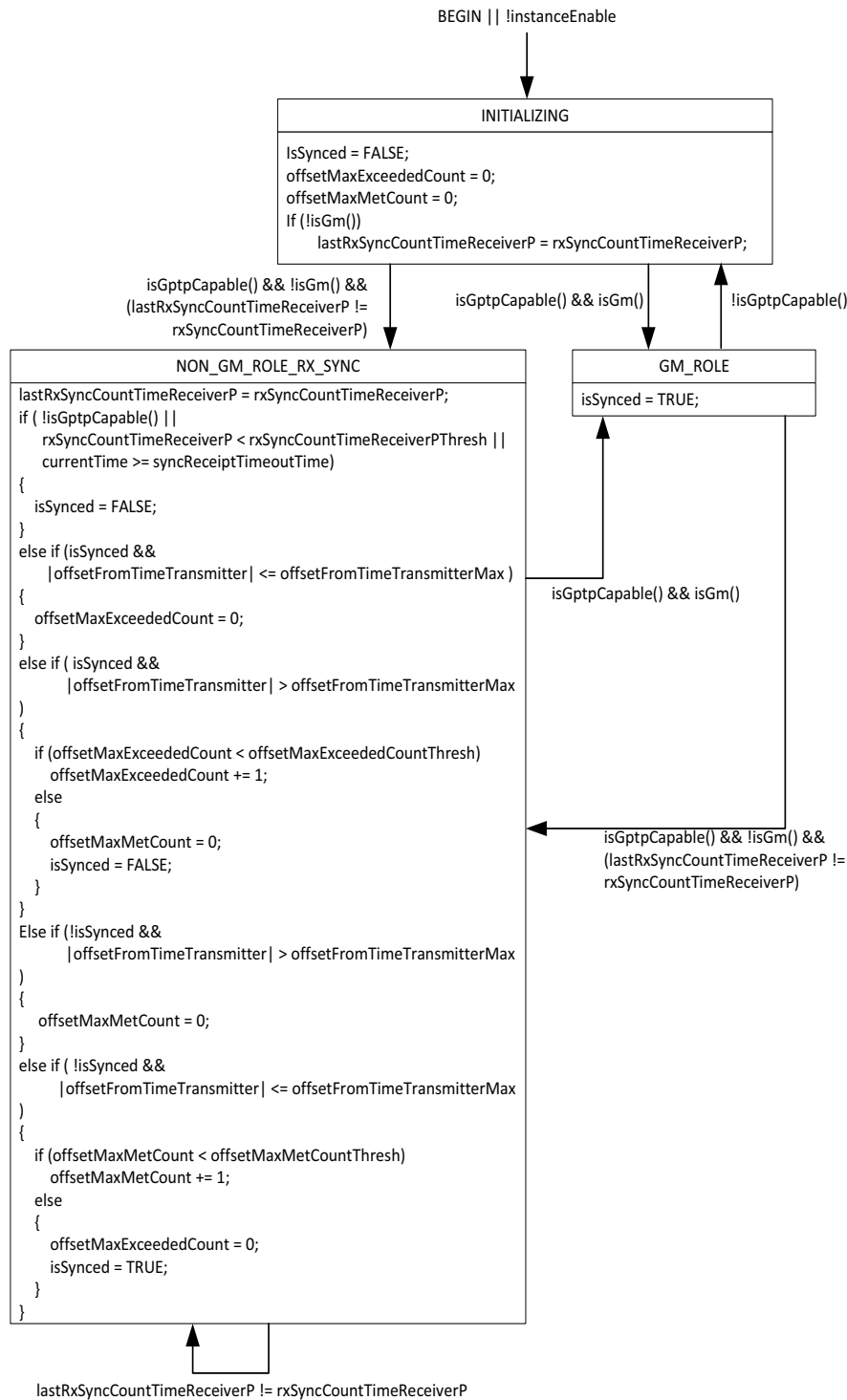


Figure 18-2—PtpInstanceSyncStatus state machine

18.5 HotStandbySystem state machine

This state machine interacts with the primary and secondary PTP Instances in order to provide a single redundant time to the application.

18.5.1 HotStandbySystem state machine global variables

18.5.1.1 hotStandbySystemState: State of the HotStandbySystem. The variable is an enumeration that takes one of the following values:

- a) **INIT:** Initialization after the HotStandbySystem powers on and is enabled. In this state, the system is waiting for both PTP Instances to synchronize.
- b) **REDUNDANT:** Both PTP Instances are synchronized according to the requirements of the respective application or profile standard (see 3.23). Time synchronization is redundant.

NOTE—A PTP Instance is implicitly considered to be synchronized if it is a Grandmaster PTP Instance (see 18.4.3).

- c) **NOT_REDUNDANT:** One PTP Instance is synchronized, and the other PTP Instance is faulted (not synchronized). Time synchronization continues to meet the requirements of the respective application or profile standard (see 3.23). Time synchronization is not redundant.
- d) **OUT_OF_SYNC:** The HotStandbySystem can adjust phase/frequency of its local time using the data stored while the system was in the REDUNDANT or NOT_REDUNDANT state, but the local time will eventually drift relative to other time-aware systems. During OUT_OF_SYNC state, time synchronization might not meet the requirements of the respective application or profile standard (see 3.23).

18.5.1.2 hotStandbySystemEnable: A Boolean variable used to enable/disable the HotStandbySystem.

18.5.1.3 hotStandbySystemLogSyncTimeThresh: The value of the managed object hotStandbySystemLogSyncTimeThreshold (see 14.21.9) of the hotStandbySystemDS. The data type for hotStandbySystemLogSyncTimeThreshold is Integer8.

18.5.1.4 primarySecondaryOffset: The value of the managed object hotStandbySystemDS.primarySecondaryOffset (see 14.21.7), which is the absolute value of the difference between the clockTimeReceiverTime variables (see 10.2.4.3) of the primary and secondary PTP Instances.

18.5.2 HotStandbySystem state machine local variables

18.5.2.1 primary: Reference to the data sets of the PTP Instance configured to use the primary domainNumber as specified by the respective application or profile standard (see 3.23). This references an element of the instanceList specified in 14.1.

18.5.2.2 secondary: Reference to the data sets of the PTP Instance configured to use the secondary domainNumber as specified by the respective application or profile standard (see 3.23). This references an element of the instanceList specified in 14.1.

18.5.2.3 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 18-3). The data type for TEMP is Integer16.

18.5.2.4 transitionTime: The value of currentTime (see 10.2.4.12) after which the hotStandbySystem transitions from the OUT_OF_SYNC state to either the NOT_REDUNDANT or REDUNDANT state, or from the NOT_REDUNDANT to the REDUNDANT state, if all other conditions for the respective transition are met. The data type for transitionTime is UScaledNs.

18.5.2.5 primarySecondaryOffsetThresh: The value of the managed object hotStandbySystemDS.primarySecondaryOffsetThresh, which is the threshold for primarySecondaryOffset (see 18.5.1.4), above which the hotStandbySystemState transitions from REDUNDANT to NOT_REDUNDANT, or stays in NOT_REDUNDANT or OUT_OF_SYNC even if other conditions for transitioning to REDUNDANT are satisfied.

18.5.3 State diagram

The HotStandbySystem state machine shall implement the function specified by the state diagram in Figure 18-3, the variables specified in 18.5.1 and 18.5.2, and the state requirements specified in 18.6.

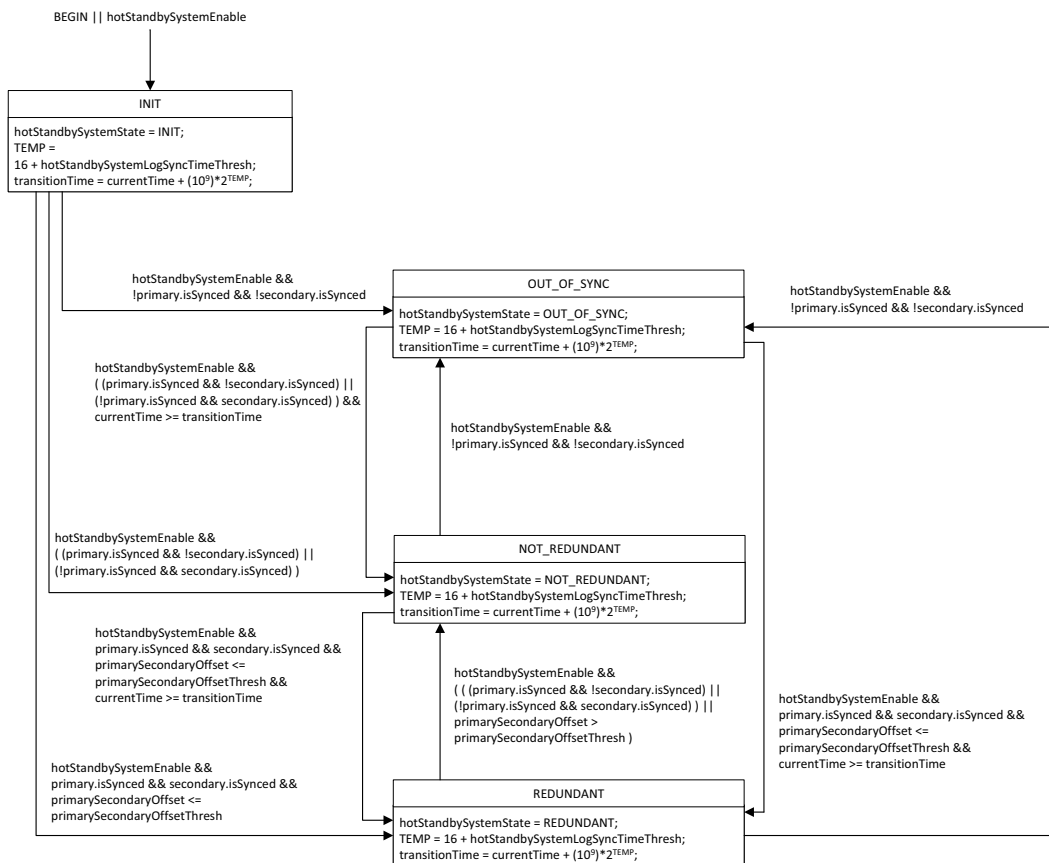


Figure 18-3—HotStandbySystem state machine

18.6 Information transfers of the HotStandbySystem entity associated with the HotStandbySystem state machine

18.6.1 Primary grandmaster

When the primary PTP Instance is a Grandmaster PTP Instance (i.e., no PTP Port in TimeReceiverPort state), the HotStandbySystem entity shall transfer phase, frequency, clockSourceTimeBaseIndicator (see 10.2.4.8), clockSourceLastGmPhaseChange (see 10.2.4.10), and clockSourceLastGmFreqChange (see 10.2.4.11) from the ClockSource to the ClockTimeTransmitter of the primary PTP Instance (see Figure 18-

1 11). If no external source of time is implemented, the ClockSource is equivalent to the LocalClock of this
2 PTP Instance.

3
4 3 When the primary PTP Instance is grandmaster, there is no requirement for the HotStandbySystem entity to
5 4 receive time from the ClockTimeReceiver of the secondary PTP Instance.

6 7 **5 18.6.2 Secondary grandmaster**

8 9 **6 18.6.2.1 HotStandbySystem in REDUNDANT state**

10
11 7 When the secondary PTP Instance is a Grandmaster PTP Instance (i.e., no PTP Port in TimeReceiverPort
12 8 state), and the hotStandbySystemState is REDUNDANT, the HotStandbySystem entity shall transfer phase,
13 9 frequency, clockSourceTimeBaseIndicator (see 10.2.4.8), clockSourceLastGmPhaseChange (see 10.2.4.10),
14 10 and clockSourceLastGmFreqChange (see 10.2.4.11) from the ClockTimeReceiver of the primary PTP
15 11 Instance to the ClockTimeTransmitter of the secondary PTP Instance (see Figure 18-1). By using phase from
16 12 the primary PTP Instance, the secondary grandmaster can maintain continuity in the event of a fault in the
17 13 primary domain.

18 19 **14 18.6.2.2 HotStandbySystem in NOT_REDUNDANT state**

20
21 15 When the secondary PTP Instance is a Grandmaster PTP Instance (i.e., no PTP Port in TimeReceiverPort
22 16 state), and the hotStandbySystemState is NOT_REDUNDANT, which means that isSynced for the primary
23 17 PTP Instance is FALSE and isSynced for the secondary PTP Instance is TRUE, the HotStandbySystem
24 18 entity shall transfer phase, frequency, clockSourceTimeBaseIndicator (see 10.2.4.8),
25 19 clockSourceLastGmPhaseChange (see 10.2.4.10), and clockSourceLastGmFreqChange (see 10.2.4.11) from
26 20 the ClockSource to the ClockTimeTransmitter of the secondary PTP Instance (see Figure 18-1). If no
27 21 external source of time is implemented, the ClockSource is equivalent to the LocalClock of this PTP
28 22 Instance.

29
30 23 The secondary PTP Instance is considered synchronized if it is a Grandmaster PTP Instance (see 18.4.4 and
31 24 Figure 18-2). Since the hotStandbySystemState is NOT_REDUNDANT, this means that either the primary
32 25 PTP Instance is not synchronized or primarySecondaryOffset exceeds primarySecondaryOffsetThresh.
33 26 However, in the former case primarySecondaryOffset can exceed primarySecondaryOffsetThresh because in
34 27 this case the primary and secondary PTP Instances receive timing from different sources. The result is that
35 28 the HotStandbySystemState remains in the NOT_REDUNDANT state, even if the primary PTP Instance
36 29 subsequently becomes synchronized again by the primary Grandmaster PTP Instance, because
37 30 primarySecondaryOffset continues to exceed primarySecondaryOffsetThresh. This situation does not occur
38 31 if it is the primary PTP Instance that is synchronized and the secondary PTP Instance that is not
39 32 synchronized due to some fault condition. When the fault condition is cleared, the secondary Grandmaster
40 33 PTP Instance ClockTimeTransmitter would receive timing from the primary PTP Instance
41 34 ClockTimeReceiver (see 18.6.2.1) and, as a result, it would become synchronized again and the
42 35 hotStandbySystemState would change to REDUNDANT.

43
44 36 The situation described in the previous paragraph is avoided if the optional split functionality (see 18.6.4) is
45 37 used. In this case, timing is transferred directly from the secondary SiteSync entity to the primary SiteSync
46 38 entity. This will synchronize the primary PTP Instance to the secondary PTP Instance; hence, the
47 39 primarySecondaryOffset changes to within primarySecondaryOffsetThresh. The HotStandbySystemState
48 40 changes to REDUNDANT and the secondary Grandmaster PTP Instance begins receiving timing from the
49 41 primary PTP Instance. However, even if the optional split functionality is not used, an application can
50 42 choose to re-initialize the secondary PTP Instance after the primary PTP Instance becomes synchronized. If
51 43 this is done, the secondary Grandmaster PTP Instance receives timing from the primary PTP Instance
52 44 ClockTimeReceiver after initialization and hotStandbySystemState is REDUNDANT.

1 NOTE—If the secondary PTP Instance is re-initialized after the primary PTP Instance becomes synchronized, this can
2 cause a jump in time when the secondary PTP Instance synchronizes to the primary PTP Instance.

3 On initialization, if the primary PTP Instance is synchronized, the secondary PTP Instance shall receive
4 timing from the primary PTP Instance. When the conditions as specified the HotStandbySystem state
5 machine (see Figure 18-3) are satisfied, the hotStandbySystemState machine transitions to the
6 REDUNDANT state.

7 18.6.3 TimeReceiver HotStandbySystem

8 18.6.3.1 General

9 A HotStandbySystem entity is said to be a TimeReceiver HotStandbySystem if and only if neither PTP
10 Instance is a Grandmaster PTP Instance.

11 18.6.3.2 HotStandbySystem in REDUNDANT state

12 When the HotStandbySystem entity is a timeReceiver (i.e., neither PTP Instance is a grandmaster), and
13 hotStandbySystemState is REDUNDANT, the HotStandbySystem entity shall transfer phase, frequency,
14 gmTimeBaseIndicator (see 10.2.4.15), lastGmPhaseChange (see 10.2.4.16), and lastGmFrequencyChange
15 (see 10.2.4.17) from the ClockTimeReceiver of the primary PTP Instance to the ClockTarget (application) if
16 a ClockTarget is present.

17 18.6.3.3 HotStandbySystem in NOT_REDUNDANT state

18 18.6.3.3.1 Transfer of synchronized time to the ClockTarget

19 When the HotStandbySystem entity is a timeReceiver, and the HotStandbySystemState is in the
20 NOT_REDUNDANT state, the HotStandbySystem entity shall transfer phase, frequency,
21 gmTimeBaseIndicator (see 10.2.4.15), lastGmPhaseChange (see 10.2.4.16), and lastGmFrequencyChange
22 (see 10.2.4.17) from the ClockTimeReceiver of the synchronized PTP Instance (i.e., isSynced == TRUE) to
23 the ClockTarget if a ClockTarget is present.

24 18.6.4 Split functionality

25 The HotStandbySystem entity may provide an optional split functionality. If the managed object
26 hotStandbySystemDS.hotStandbySystemSplitFunctionality is set to TRUE, the split functionality is
27 invoked. If the managed object hotStandbySystemDS.hotStandbySystemSplitFunctionality is set to FALSE,
28 the split functionality is not invoked. The split functionality shall provide an interworking function (IWF)
29 that transfers time synchronization information from the primary PTP Instance to the secondary PTP
30 Instance when isSynced is FALSE (e.g., the timeReceiver port of the PTP Instance is in failure condition,
31 see 18.4.1.1) for the secondary PTP Instance and isSynced is TRUE (see 18.4.1.1) for the primary PTP
32 Instance, or from the secondary PTP Instance to the primary PTP Instance when isSynced is FALSE for the
33 primary PTP Instance and isSynced is TRUE for the secondary PTP Instance. The IWF provides the most
34 recently received PortSyncSync structure at the timeReceiver port of the synchronized PTP Instance
35 (isSynced == TRUE, see 18.4.1.1) to the unsynchronized PTP Instance (isSynced == FALSE) SiteSyncSync
36 entity, in order to generate a PortSyncSync structure that is used by the timeTransmitter ports of the
37 unsynchronized PTP Instance, as follows:

- 38 a) The domainNumber of the received PortSyncSync structure is changed by the IWF from the
39 synchronized PTP Instance domainNumber to the unsynchronized PTP Instance domainNumber;
- 40 b) The localPortNumber of the received PortSyncSync structure is changed by the IWF to the
41 portNumber of the unsynchronized PTP Instance timeReceiver port; and

- c) All other members (i.e., syncReceiptTimeoutTime, followUpCorrectionField, sourcePortIdentity, logMessageInterval, preciseOriginTimestamp, upstreamTxTime, rateRatio, gmTimeBaseIndicator, lastGmPhaseChange, and lastGmFreqChange) of the received PortSyncSync structure of the synchronized PTP Instance are provided to the unsynchronized PTP Instance SiteSync entity unchanged.
 - d) The flags leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, and frequencyTraceable, and the fields currentUtcOffset and timeSource of received Announce messages on the timeReceiver port of the unsynchronized PTP Instance are ignored. The IWF provides the values of these corresponding members of the timePropertiesDS of the synchronized PTP Instance to the unsynchronized PTP Instance. These values are then copied to:
 - i) The respective members of the timePropertiesDS of the unsynchronized PTP Instance;
 - ii) The unsynchronized PTP Instance global variables leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, and frequencyTraceable, and the fields currentUtcOffset and timeSource, respectively; and
 - iii) The unsynchronized PTP Instance global variables annLeap61, annLeap59, annCurrentUtcOffsetValid, annPtpTimescale, annTimeTraceable, and annFrequencyTraceable, and the fields annCurrentUtcOffset and annTimeSource, respectively.
 - e) The messagePriorityVector (see 10.3.4 and 10.3.5) information of received Announce messages on the timeReceiver port of the unsynchronized PTP Instance is ignored. The IWF provides the values of the corresponding members of the synchronized PTP Instance parentDS, i.e., grandmasterIdentity, grandmasterClockQuality.clockClass, grandmasterClockQuality.clockAccuracy, grandmasterClockQuality.offsetScaledLogVariance, grandmasterPriority1, grandmasterPriority2, which are copied to:
 - i) The corresponding members of the parentDS of the unsynchronized PTP Instance; and
 - ii) The gmPriorityVector of the unsynchronized PTP Instance.
- The IWF also provides the value of the stepsRemoved member of the synchronized PTP Instance currentDS, which is copied to:
- iii) The stepsRemoved member of the currentDS of the unsynchronized PTP Instance; and
 - iv) The gmPriorityVector of the unsynchronized PTP Instance.

NOTE 1—With the above, the secondary/primary PTP Instance state machines operate as though the time synchronization information had been received from the secondary/primary PTP Instance timeReceiver port. The SiteSync entity of the secondary/primary PTP Instance transfers the timing information to the PortSync entity of each timeTransmitter port of the secondary/primary PTP Instance. Each PortSyncSync state machine computes rateRatio, which now is relative to the primary/secondary PTP Instance GM. Each MDSyncSend state machine computes the fields of transmitted Sync and, in the two-step case, Follow_Up messages. The copied syncReceiptTimeout time is less than currentTime because sync receipt timeout has not occurred at the primary/secondary PTP Instance.

NOTE 2—The split functionality is used to transfer time synchronization information from the PTP Instance for which isSynced is TRUE to the PTP Instance for which isSynced is FALSE. It is not meant to cover the case where isSynced is FALSE for both the primary and secondary PTP Instances.

18.6.5 Both PTP Instances have isSynced FALSE

When isSynced is FALSE for both the primary and secondary PTP Instances (i.e., both are not synchronized), the HotStandbySystemState is OUT_OF_SYNC and the HotStandbySystem entity shall transfer phase, frequency, gmTimeBaseIndicator (see 10.2.4.15), lastGmPhaseChange (see 10.2.4.16), and lastGmFrequencyChange (see 10.2.4.17) from the ClockTimeReceiver of the PTP Instance for which isSynced was TRUE, before the HotStandbySystemState transitioned to the OUT_OF_SYNC state, to the ClockTarget if a ClockTarget is present.

1 When the HotStandbySystemState is OUT_OF_SYNC, time synchronization performance is not required to
2 meet the respective application or profile standard (see 3.23) requirements. Nevertheless, in order to
3 mitigate LocalClock frequency drift, the PTP Instance slowly adjusts the frequency of its LocalClock using
4 the data stored while isSynced for that PTP Instance was TRUE and the HotStandbySystemState was in the
5 REDUNDANT or NOT_REDUNDANT state.

6 18.7 PrimarySecondaryOffset state machine

7 This state machine updates the value of the global variable primarySecondaryOffset (see 18.5.1.4) whenever
8 a new value of clockTimeReceiverTime (see 10.2.4.3) is computed for either the primary or secondary PTP
9 Instance. A new value of clockTimeReceiverTime is computed when the global variable rcvdPSSyncCSS
10 (see 10.2.4.29) or the global variable rcvdLocalClockTickCSS (see 10.2.4.30) is set to TRUE.

11 18.7.1 State diagram

12 The PrimarySecondaryOffset state machine shall implement the function specified by the state diagram in
13 Figure 18-4.

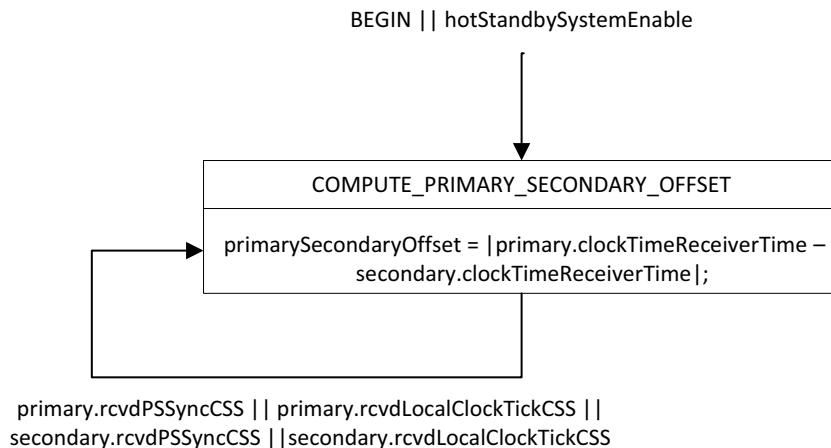


Figure 18-4—PrimarySecondaryOffset state machine

14 18.8 Transition of hotStandbySystemState from REDUNDANT to 15 NOT_REDUNDANT

16 The time synchronization provided to the application can fulfill the time synchronization requirements of the
17 respective application or profile standard (see 3.23) during a HotStandbySystemState change from
18 REDUNDANT to NOT_REDUNDANT (or NOT_REDUNDANT to REDUNDANT)

19 NOTE 1—The secondary grandmaster is responsible for maintaining continuity (no “jump” in time) when a fault occurs
20 in the primary grandmaster.

1 NOTE 2—The primary domain global variables `lastGmPhaseChange` and `lastGmFrequencyChange` are propagated
2 throughout the secondary domain, and can be used to compensate for any phase or frequency jump that occurs when the
3 `HotStandbySystemState` becomes `NOT_REDUNDANT`.

4

1 Annex A

2 (normative)

3 Protocol Implementation Conformance Statement (PICS)

4 proforma²³

5 A.1 Introduction

6 The supplier of a protocol implementation that is claimed to conform to this standard shall complete the
7 following PICS proforma.

8 A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of
9 which capabilities and options of the protocol have been implemented. The PICS can have a number of uses,
10 including the following:

- 11 a) By the protocol implementer, as a checklist to reduce the risk of failure to conform to the standard
12 through oversight;
- 13 b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication
14 of the capabilities of the implementation, stated relative to the common basis for understanding
15 provided by the standard PICS proforma;
- 16 c) By the user—or potential user—of the implementation, as a basis for initially checking the
17 possibility of interworking with another implementation (note that, while interworking can never be
18 guaranteed, failure to interwork can often be predicted from incompatible PICS);
- 19 d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for
20 conformance of the implementation.

21 A.2 Abbreviations and special symbols

22 A.2.1 Status symbols

23	M	mandatory
24	O	optional
25	<i>O.n</i>	optional, but support of at least one of the group of options labeled by the same numeral n
26		is required
27	X	prohibited
28	pred:	conditional-item symbol, including predicate identification (see A.3.4)
29	¬	logical negation, applied to a conditional item's predicate

30 A.2.2 General abbreviations

31	N/A	not applicable
32	PICS	Protocol Implementation Conformance Statement

²³ Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

1 A.3 Instructions for completing the PICS proforma

2 A.3.1 General structure of the PICS proforma

3 The first part of the PICS proforma, implementation identification and protocol summary, is to be completed
4 as indicated with the information necessary to identify fully both the supplier and the implementation.

5 The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each
6 containing a number of individual items. Answers to the questionnaire items are to be provided in the
7 rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No), or
8 by entering a value or a set or range of values. (Note that there are some items where two or more choices
9 from a set of possible answers can apply; all relevant choices are to be marked.)

10 Each item is identified by an item reference in the first column. The second column contains the question to
11 be answered; the third column records the status of the item—whether support is mandatory, optional, or
12 conditional (see also A.3.4). The fourth column contains the reference or references to the material that
13 specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

14 A supplier might also provide (or be required to provide) further information, categorized as either
15 Additional Information or Exception Information. When present, each kind of further information is to be
16 provided in a further subclause of items labeled *A_i* or *X_i*, respectively, for cross-referencing purposes, where
17 *i* is any unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on
18 its format and presentation.

19 A completed PICS proforma, including any Additional Information and Exception Information, is the
20 Protocol Implementation Conformation Statement for the implementation in question.

21 NOTE—Where an implementation is capable of being configured in more than one way, a single PICS might be able to
22 describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering
23 some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of
24 the information.

25 A.3.2 Additional Information

26 Items of Additional Information allow a supplier to provide further information intended to assist the
27 interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS
28 can be considered complete without any such information. Examples might be an outline of the ways in
29 which a (single) implementation can be set up to operate in a variety of environments and configurations, or
30 information about aspects of the implementation that are outside the scope of this standard but that have a
31 bearing upon the answers to some items.

32 References to items of Additional Information may be entered next to any answer in the questionnaire and
33 may be included in items of Exception Information.

34 A.3.3 Exception Information

35 It occasionally happens that a supplier will wish to answer an item with mandatory status (after any
36 conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer
37 will be found in the Support column for this; instead, the supplier shall write the missing answer into the
38 Support column, together with an *X_i* reference to an item of Exception Information, and shall provide the
39 appropriate rationale in the Exception item itself.

1 An implementation for which an Exception item is required in this way does not conform to this standard.

2 NOTE—A possible reason for the situation described above is that a defect in this standard has been reported, a
3 correction for which is expected to change the requirement not met by the implementation.

4 **A.3.4 Conditional status**

5 **A.3.4.1 Conditional items**

6 The PICS proforma contains a number of conditional items. These are items for which both the applicability
7 of the item itself, and its status if it does apply—mandatory or optional—are dependent upon whether or not
8 certain other items are supported.

9 Where a group of items is subject to the same condition for applicability, a separate preliminary question
10 about the condition appears at the head of the group, with an instruction to skip to a later point in the
11 questionnaire if the Not Applicable (N/A) answer is selected. Otherwise, individual conditional items are
12 indicated by a conditional symbol in the Status column.

13 A conditional symbol is of the form “**pred:** S” where **pred** is a predicate as described in A.3.4.2, and S is a
14 status symbol, M or O.

15 If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated
16 by the status symbol following the predicate; the answer column is to be marked in the usual way. If the
17 value of the predicate is false, the N/A answer is to be marked.

18 **A.3.4.2 Predicates**

19 A predicate is one of the following:

- 20 a) An item-reference for an item in the PICS proforma; the value of the predicate is true if the item is
21 marked as supported, and is false otherwise;
- 22 b) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item-
23 references using the Boolean operator OR; the value of the predicate is true if one or more of the
24 items is marked as supported;
- 25 c) The logical negation symbol “¬” prefixed to an item-reference or predicate-name; the value of the
26 predicate is true if the value of the predicate formed by omitting the “¬” symbol is false, and vice
27 versa.

28 Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for
29 grouped conditional items, is indicated by an asterisk in the Item column.

1**A.4 PICS proforma for IEEE Std 802.1AS-2020**

A.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) of machines and/or operating system names	

2 NOTE 1—Only the first three items are required for all implementations; other information may be completed as
3 appropriate in meeting the requirement for full identification.

4 NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier’s terminology
5 (e.g., Type, Series, Model).

6

A.4.2 Protocol summary, IEEE Std 802.1AS

Identification of protocol specification	IEEE Std 802.1AS-2020, IEEE Standard for Local and metropolitan area networks—Timing and Synchronization for Time-Sensitive Applications			
Identification of amendments and corrigenda to the PICS proforma which have been completed as part of the PICS	Amd.	:	Corr.	:
	Amd.	:	Corr.	:
Have any Exception items been required? (See A.3.3: the answer Yes means that the implementation does not conform to IEEE Std 802.1AS-2020)	No []		Yes []	

Date of Statement	
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7

A.5 Major capabilities

Item	Feature	Status	References	Support
DOMADD	Does the time-aware system support one or more PTP Instances with domainNumber in the range 1 to 127?	O	item e) of 5.4.2, 8.1	Yes [] No []
DOM0	Does the time-aware system support one or more PTP Instances with domainNumber 0	O	item e) of 5.4.2, 8.1	Yes [] No []
MINTA	Does the PTP Instance support at least one PTP Port with minimal requirements?	M	10.2.13, item c) of 5.4, A.7	Yes []
BTCX	Does any PTP Instance with domainNumber in the range of 1 to 127 implement the best timeTransmitter clock algorithm?	DOMADD:O	10.2.13, item g) of 5.4.3, 10.3, A.9	Yes [] No []
SIG	Does the PTP Instance transmit Signaling messages?	M	item h) of 5.4.1, 10.4, 10.6.4, A.8	Yes []
GMCAP	Is the PTP Instance capable of acting as a Grandmaster PTP Instance?	O	10.2.13, item c) of 5.4.2, 10.1.3, A.10	Yes [] No []
BRDG	Does the PTP Instance act as a PTP Relay Instance on two or more PTP Ports?	O	item d) of 5.4.2, 5.4.4	Yes [] No []
MITT	Does the PTP Instance support media-independent timeTransmitter functionality on at least one PTP Port?	GMCAP or BRDG:M	item b) of 5.4.2, A.11	Yes [] N/A []
MIPERF	Does the PTP Instance support the performance requirements?	M	10.2.13, item j) of 5.4, A.12	Yes []
EXT	Does the PTP Instance support external port configuration?	O	item f) of 5.4.2, A.21	Yes [] No []
MDFDPP	Does the PTP Instance support media-dependent full-duplex point-to-point functionality on one or more PTP Ports?	O.1	5.5, Clause 11, A.6, A.13	Yes [] No []
MDDOT11	Does the PTP Instance support media-dependent IEEE 802.11 link functionality on one or more PTP Ports?	O.1	5.6, Clause 12, A.6, A.14	Yes [] No []
MDEPON	Does the PTP Instance support IEEE 802.3 Passive Optical Networking (EPON)?	O.1	5.7, Clause 13, A.6, A.15	Yes [] No []
MDGHN	Does the PTP Instance support media-dependent ITU-T G.hn functionality on one or more PTP Ports?	O.1	item b) of 5.8, 16.6.3, A.18	Yes [] No []
MDMOCA	Does the PTP Instance support media-dependent MoCA functionality on one or more PTP Ports?	O.1	item b) of 5.8, 16.6.2, A.17	Yes [] No []
MDCSN	Does the PTP Instance support media-dependent CSN functionality on one or more PTP Ports?	MDGHN or MDMOCA:M	5.8, Clause 16, A.6, A.16	Yes [] No []

A.5 Major capabilities *(continued)*

Item	Feature	Status	References	Support
MGT	Is management of the PTP Instance supported?	O	item j) of 5.4.2, Clause 14	Yes [] No []
RMGT	Is a remote management protocol supported?	MGT: O	item k) of 5.4.2, A.19	Yes [] No []
APPL	Does the PTP Instance support one or more of the application interfaces?	O	item i) of 5.4.2, Clause 9, A.20	Yes [] No []
HOTSTDB Y	Does the time-aware system support hot standby as specified in Clause 18?	O	Clause 18, 14.8, item m) of 5.4.2, 9.3.3.4, 9.4.3.4, 9.5.3.4, 9.6.2.6	Yes [] No []
DRFTRK	Does the PTP Instance support the Drift_Tracking TLV as specified in 10.2.4.26, 10.2.4.27, 10.2.4.28, 11.2.14, 11.2.15, and 11.4.4?	O	item n) of 5.4.2, 10.2.4.26, 10.2.4.27, 10.2.4.28, 11.2.14, 11.2.15, and 11.4.4	Yes [] No []

1

A.6 Media access control methods

Item	Feature	Status	References	Support
MAC-IEEE-802.3	Which MAC methods are implemented in conformance with the relevant MAC standards?	O:2	11.1	Yes [] No []
MAC-IEEE-802.11		O:2	12.1	Yes [] No []
MAC-1	Has a PICS been completed for each of the MAC methods implemented as required by the relevant MAC Standards?	M		Yes []
MAC-2	Do all the MAC methods implemented support the MAC Timing aware Service as specified?	M	Clause 11 Clause 12 Clause 13	Yes []

1

A.7 Minimal time-aware system

Item	Feature	Status	References	Support
MINTA-1	Do all PTP Instances of the device implement the functionality specified by the SiteSyncSync state machine in Figure 10-3 in compliance with the requirements of 10.2.7?	M	item f) of 5.4, 10.2.7	Yes []
MINTA-2	Do all PTP Instances of the device implement the functionality specified by the PortSyncSyncReceive state machine in Figure 10-4 on each PTP Port in compliance with the requirements of 10.2.8?	M	item d) of 5.4	Yes []
MINTA-3	Do all PTP Instances of the device implement the functionality specified by the ClockTimeReceiverSync state machine in Figure 10-9 in compliance with the requirements of 10.2.13?	M	10.2.13, item e) of 5.4	Yes []
MINTA-4	For all PTP Instances of the device, does the PTP Port sending a Signaling message that contains a message interval request TLV adjust its syncReceiptTimeoutTimeInterval of this PTP Instance in compliance with the requirements of 10.6.4.3.7 and Table 10-16?	SIG:M	10.6.4.3.7	Yes [] N/A []
MINTA-5	Is the clockIdentity constructed in compliance with the requirements of 8.5.2.2?	M	8.5.2.2	Yes []
MINTA-6	Is the domainNumber for all transmitted messages in the range 0 through 127, in compliance with the requirements of 8.1?	M	8.1	Yes []
MINTA-7	Is the majorSdoId 0x1 and the minorSdoId 0x0 for all transmitted gPTP domain messages?	M	8.1	Yes []
MINTA-8	Is the domainNumber for at least one of the gPTP domains supported by the time-aware system, in compliance with the requirements of 8.1?	M	8.1	Yes []
MINTA-9	Is the IEEE 802.1AS time of domain 0 measured relative to the PTP epoch in compliance with the requirements of 8.2.2?	M	8.2.2	Yes []
MINTA-10	If path delay asymmetry is modeled by this PTP Instance does it comply with the requirements of 8.3?	O	8.3	Yes [] No [] N/A []
MINTA-11	Do all derived data types that are transmitted in IEEE 802.1AS messages and headers comply with 6.4.4?	M	6.4.4	Yes []
MINTA-12	Is the granularity of the local clock 40 ns or better in compliance with the requirements of B.1.2?	M	B.1.2	Yes []

A.7 Minimal time-aware system (continued)

Item	Feature	Status	References	Support
MINTA-13	Is the frequency of the local clock relative to TAI ± 100 ppm in compliance with the requirements of B.1.1?	M	B.1.1	Yes []
MINTA-14	Does the PTP Instance ignore non-propagating TLVs of Announce and Signaling messages that it cannot parse, and attempt to parse the next TLV, in compliance with the requirements of 10.6.1?	M	10.6.1	Yes []
MINTA-15	Does the PTP Instance support the state machines related to signaling gPTP capability?	M	item g) of 5.4.1, 10.4.3	Yes []
MINTA-16	For receive of all messages and for transmit of all messages except Announce, does the PTP Instance support the message requirements?	M	item h) of 5.4.1, 10.5, 10.6, 10.7	Yes []
MINTA-17	Does the PTP Instance support the gPTP requirements specified in Clause 8, including the PTP Instance attributes?	M	item a) of 5.4, Clause 8, 8.6.2	Yes []
MINTA-18	Does the PTP Instance support the requirements for time-synchronization state machines?	M	item b) of 5.4	Yes []
MINTA-19	Does the PTP Instance implement the path trace TLV (i.e., process this TLV when received in an Announce message, and attach this TLV to a transmitted Announce message unless the TLV would cause the maximum frame size to be exceeded)?	M	10.3.11, 10.3.13, 10.3.14, 10.3.16	Yes []
MINTA-20	Does the PTP Instance forward TLVs as required?	M	10.6.1	Yes []
MINTA-21	Do the PTP Ports of this PTP Instance implement the functionality of the GptpCapableIntervalSetting state machine in compliance with the requirements of 10.4.4 and Figure 10-23?	MINTA-15:O	10.4.4, item h) of 5.4.2, item h) of 5.4.1	Yes [] No []
MINTA-22	Does the PTP Instance implement the PtpInstanceSyncStatus state machine of 18.4 and the PtpInstanceStatusDS of 14.8?	¬HOTSTDBY: O	18.4, 14.8	Yes [] No []

1

A.8 Signaling

Item	Feature	Status	References	Support
SIG-1	Do the sequence numbers of Signaling messages comply with the requirements of 10.5.7?	SIG:M	10.5.7	Yes []
SIG-2	Does the Signaling message body comply with the requirements of 10.6.4.1 and Table 10-13?	SIG:M	10.6.4.1	Yes []
SIG-3	Does the Signaling message header comply with the requirements of 10.6.2?	SIG:M	10.6.2	Yes []
SIG-4	Are all Signaling message reserved fields equal to 0 in compliance with the requirements of 10.6.1?	SIG:M	10.6.1	Yes []
SIG-5	Is the destination MAC address for all Signaling messages equal to 01:80:C2:00:00:0E in compliance with the requirements of 10.5.3?	SIG:M	10.5.3	Yes []
SIG-6	Is the EtherType for all Signaling messages equal to 88-F7 in compliance with the requirements of 10.5.4?	SIG:M	10.5.4	Yes []
SIG-7	Does the message interval request TLV for Signaling messages comply with the requirements in 10.6.4.3?	BTC-23:M, MITT-16:M, MDFDPP-5:M	10.6.4.3	Yes []

2

A.9 Best timeTransmitter clock

Item	Feature	Status	References	Support
BTC-1	Does the PTP Instance implement the functionality specified by the PortAnnounceReceive state machine in Figure 10-13 on each PTP Port in compliance with the requirements of 10.3.11?	DOM0 or BTCX	10.3.11	Yes []
BTC-2	Does the PTP Instance implement the functionality specified by the PortAnnounceInformation state machine in Figure 10-14 on each PTP Port in compliance with the requirements of 10.3.12?	DOM0 or BTCX	10.3.12	Yes []
BTC-3	Does the PTP Instance implement the functionality specified by the PortStateSelection state machine in Figure 10-15 on each PTP Port in compliance with the requirements of 10.3.13? NOTE—There is one instance of the PortStateSelection state machine for the PTP Instance, for each gPTP domain. Some of the PortStateSelection state machine computations are performed for each PTP Port, and some of the computations are performed for the PTP Instance as a whole (and all the computations are performed for each gPTP domain).	DOM0 or BTCX	10.3.13	Yes []
BTC-4	If the value of clockA's SystemIdentity is less than that of clockB, is clockA selected as Grandmaster PTP Instance in compliance with the requirements of 10.3.2?	DOM0 or BTCX	10.3.2	Yes []
BTC-5	Does the value of priority1 comply with the requirements of 8.6.2.1?	DOM0 or BTCX	8.6.2.1	Yes []
BTC-6	Does the value of clockClass comply with the requirements of 8.6.2.2?	DOM0 or BTCX	8.6.2.2	Yes []
BTC-7	Does the value of priority2 comply with the requirements of 8.6.2.5?	DOM0 or BTCX	8.6.2.5	Yes []
BTC-8	Does the value of clockAccuracy comply with requirements of 8.6.2.3?	DOM0 or BTCX	8.6.2.3	Yes []
BTC-9	Does the value of offsetScaledVariance comply with the requirements of 8.6.2.4?	DOM0 or BTCX	8.6.2.4	Yes []
BTC-10	Does the value of timeSource comply with requirements of 8.6.2.7 and Table 8-2?	DOM0 or BTCX	8.6.2.7	Yes []
BTC-11	Is the PTP Port number equal to 1 in compliance with the requirements of 8.5.2.3?	¬BRDG:M	8.5.2.3	Yes [] N/A []

A.9 Best timeTransmitter clock (continued)

Item	Feature	Status	References	Support
BTC-12	Are the PTP Ports numbered 1 through N for each of N PTP Ports in compliance with the requirements of 8.5.2.3?	DOM0 or BTCX	8.5.2.3	Yes []
BTC-13	Does the clockIdentity field comply with the requirements of 8.5.2.2?	DOM0 or BTCX	8.5.2.2	Yes []
BTC-14	When no grandmaster-capable PTP Instance is available does the behavior of the PTP Instance comply with the requirements of 10.2.13.2, i.e., the clockTimeReceiverTime should be provided by the local clock?	DOM0 or BTCX	10.2.13.2	Yes []
BTC-15	Does the value of announceReceiptTimeout comply with the requirements of 10.7.3.2?	DOM0 or BTCX	10.7.3.2	Yes []
BTC-16	Does the TimeReceiverPort remove the PTP Port from the BTC selection after announceReceiptTimeout expires in compliance with the requirements of 10.7.3.2?	DOM0 or BTCX	10.7.3.2	Yes []
BTC-17	Does the value of syncReceiptTimeout comply with the requirements of 10.7.3.1?	DOM0 or BTCX	10.7.3.1	Yes []
BTC-18	Does the TimeReceiverPort remove the PTP Port from the BTC selection after syncReceiptTimeout expires in compliance with 10.7.3.1?	DOM0 or BTCX	10.7.3.1	Yes []
BTC-19	Does the PTP Port sending a message interval request Signaling message adjust its announceReceiptTimeoutTimeInterval in compliance with the requirements of 10.6.4.3.8 and Table 10-17?	SIG:M	10.6.4.3.8	Yes []
BTC-20	If the PTP Instance implements the ClockSourceTime interface, does the value of lastGmPhaseChange comply with the requirements of 9.2.2 and 6.4.3.3?	O	9.2.2	Yes [] No []
BTC-21	Does the transmitted timing information comply with the requirements of 10.3.1, including specifications for externalPortConfigurationEnabled value of false?	GMCAP:M	10.3.1	Yes [] N/A []
BTC-22	Does the PTP Instance implement BTCA requirements that are not listed in the preceding BTC rows?	DOM0 or BTCX	10.3.2, 10.3.3, 10.3.4, 10.3.5, 10.3.6, 10.3.8, 10.3.10	Yes []
BTC-23	Do the TimeTransmitterPorts of this PTP Instance implement the functionality of the AnnounceIntervalSetting state machine in compliance with the requirements of 10.3.17 and Figure 10-19?	BTCX:O	10.3.17, item g) of 5.4.2, item h) of 5.4.1	Yes [] No []

A.10 Grandmaster-capable PTP Instance

Item	Feature	Status	References	Support
	If GMCAP not supported, mark N/A.			N/A []
GMCAP-1	Does the PTP Instance implement the functionality specified by the ClockTimeTransmitterSyncSend state machine in compliance with the requirements of 10.2.9 and Figure 10-5?	GMCAP:M	10.2.9	Yes []
GMCAP-2	Does the PTP Instance implement the functionality specified by the ClockTimeTransmitterSyncOffset state machine in compliance with the requirements of 10.2.10 and Figure 10-6?	GMCAP:M	10.2.10	Yes []
GMCAP-3	Does the device implement the functionality specified by the ClockTimeTransmitterSyncReceive state machine in compliance with the requirements of 10.2.11 and Figure 10-7?	GMCAP:M	10.2.11	Yes []

1

A.11 Media-independent timeTransmitter

Item	Feature	Status	References	Support
	If MITT not supported, mark N/A.			N/A []
MITT-1	Does the PTP Instance implement the functionality of the AnnounceIntervalSetting state machine in compliance with the requirements of 10.3.17 and Figure 10-19 on each PTP Port?	MITT:M	10.3.17	Yes []
MITT-2	Does the PTP Instance implement the functionality of the PortSyncSyncSend state machine in compliance with the requirements of 10.2.9 and Figure 10-8 on each PTP Port?	MITT:M	10.2.9	Yes []
MITT-3	Does the PTP Instance implement the functionality of the PortAnnounceTransmit state machine in compliance with the requirements of 10.3.16 and Figure 10-18 on each PTP Port?	MITT:M	10.3.16	Yes []
MITT-4	Does the destination MAC address of all Announce messages equal 01:80:C2:00:00:0E?	MITT:M	10.5.3	Yes []
MITT-5	Does the EtherType of all Announce messages equal 88-F7?	MITT:M	10.5.4	Yes []
MITT-6	Do the sequence numbers of Announce messages comply with the requirements of 10.5.7?	MITT:M	10.5.7	Yes []

A.11 Media-independent timeTransmitter (continued)

Item	Feature	Status	References	Support
MITT-7	Does the Announce message header comply with 10.6.2?	MITT:M	10.6.2	Yes []
MITT-8	Does the Announce message body comply with the requirements in 10.6.3.1 and Table 10-11?	MITT:M	10.6.3.1	Yes []
MITT-9	Are all Announce message reserved fields equal to 0?	MITT:M	10.6.1	Yes []
MITT-10	If it is not otherwise specified, is the logAnnounceInterval equal to zero or within the allowed range?	MITT:M	10.7.2.1	Yes []
MITT-11	Does the value of currentUtcOffset comply with the requirements of 8.2.3?	MITT:M	8.2.3	Yes []
MITT-12	Do the values of the leap59, leap61, and currentUtcOffsetValid flags comply with the requirements of 10.3.8?	MITT:M	10.3.8	Yes []
MITT-13	Does this PTP Instance ensure that messages that traverse it or originate from it are not transmitted with VLAN tags in compliance with the requirements of 11.3.3?	MITT:M	11.3.3	Yes []
MITT-14	Is the computation of cumulative rateRatio in accordance with 10.2.8.3?	MITT:M	10.2.8.3	Yes [] N/A []
MITT-15	For transmit of the Announce message, does the PTP Instance support the message requirements?	MITT:M	10.5, 10.6, 10.7	Yes []
MITT-16	Do the TimeTransmitterPorts of this PTP Instance implement the functionality of the SyncIntervalSetting state machine in compliance with the requirements of 10.3.18 and Figure 10-20?	MITT:O	10.3.18, item b) 3) of 5.4.2, item h) of 5.4.1	Yes [] No []

1

A.12 Media-independent performance requirements

Item	Feature	Status	References	Support
MIPERF-1	Does the PTP Instance comply with the performance requirements of B.1?	M	B.1	Yes []
MIPERF-2	Does the PTP Instance comply with the performance requirements of B.2.4?	M	B.2.4	Yes []
MIPERF-3	Does the PTP Instance comply with the performance recommendations of B.2.2?	O	B.2.2	Yes [] No []
MIPERF-4	Does the PTP Instance comply with the performance recommendations of B.2.3?	O	B.2.3	Yes [] No []

2

A.13 Media-dependent, full-duplex point-to-point link

Item	Feature	Status	References	Support
MDFDPP-1	Does this PTP Port implement the functionality of the MDSyncReceiveSM state machine in compliance with the requirements of 11.2.14 and Figure 11-6?	MDFDPP:M	11.2.14	Yes []
MDFDPP-2	Does this PTP Port implement the functionality of the MDSyncSendSM state machine in compliance with the requirements of 11.2.15 and Figure 11-7?	MITT and MDFDPP:M	11.2.15	Yes []
MDFDPP-3	Does this port implement the functionality of the MDPdelayRequest state machine in compliance with the requirements of 11.2.19 and Figure 11-9?	MDFDPP:M	11.2.19	Yes []
MDFDPP-4	Does this port implement the functionality of the MDPdelayResponse state machine in compliance with the requirements of 11.2.20 and Figure 11-10?	MDFDPP:M	11.2.20	Yes []
MDFDPP-5	Does this port implement the functionality of the LinkDelayIntervalSetting state machine in compliance with the requirements of 11.2.21 and Figure 11-11?	MDFDPP:O	11.2.21, item l) of 5.5, item h) of 5.4.1	Yes [] No []
MDFDPP-6	Does this PTP Port timestamp Sync messages on ingress with respect to the LocalClock in compliance with 11.3.2.1 and 11.3.9?	MDFDPP:M	11.3.2.1	Yes []
MDFDPP-7	Does this PTP Port timestamp Sync messages on egress with respect to the LocalClock in compliance with the requirements of 11.3.2.1 and 11.3.9?	MITT and MDFDPP:M	11.3.2.1	Yes []
MDFDPP-8	Does this port timestamp Pdelay_Req messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 11.3.2.1 and 11.3.9?	MDFDPP:M	11.3.2.1	Yes []
MDFDPP-9	Does this port timestamp Pdelay_Resp messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 11.3.2.1 and 11.3.9?	MDFDPP:M	11.3.2.1	Yes []
MDFDPP-10	Are all IEEE 802.1AS messages on this port sent without a Q-tag in compliance with the requirements of 11.3.3?	MDFDPP:M	11.3.3	Yes []
MDFDPP-11	Do all media-dependent messages transmitted on this port use a destination MAC address taken from Table 11-3 in compliance with the requirements of 11.3.4 [01-80-C2-00-00-0E]?	MDFDPP:M	11.3.4	Yes []
MDFDPP-12	Do all media-dependent messages transmitted on this port use a source MAC address that is assigned to that port in compliance with the requirements of 11.3.4?	MDFDPP:M	11.3.4	Yes []

A.13 Media-dependent, full-duplex point-to-point link *(continued)*

Item	Feature	Status	References	Support
MDFDPP-13	Do all media-dependent message transmitted on this port us an EtherType specified in Table 11-4 [88-F7]?	MDFDPP:M	11.3.5	Yes []
MDFDPP-14	Does the header of all the media-dependent messages on this port comply with the requirements of 11.4.2 and Table 10-7?	MDFDPP:M	11.4.2	Yes [] N/A []
MDFDPP-15	Does the body of Sync messages sent on this PTP Port comply with the requirements of 11.4.3, Table 11-8, and Table 11-9?	MDFDPP:M	11.4.3	Yes []
MDFDPP-16	Does the body of Follow_Up messages sent on this PTP Port comply with the requirements of 11.4.4, (lastGmPhaseChange), and Table 11-10?	MDFDPP:M	11.4.4	Yes []
MDFDPP-17	Does the body of Pdelay_Req messages sent on this port comply with the requirements of 11.4.5 and Table 11-13?	MDFDPP:M	11.4.5	Yes []
MDFDPP-18	Does the body of Pdelay_Resp messages sent on this port comply with the requirements of 11.4.6 and Table 11-14?	MDFDPP:M	11.4.6	Yes []
MDFDPP-19	Does the body of Pdelay_Resp_Follow_Up messages sent on this port comply with the requirements of 11.4.7 and Table 11-15?	MDFDPP:M	11.4.7	Yes []
MDFDPP-20	Are all reserved fields in media-dependent messages sent on this port set to 0 in compliance with the requirements of 11.4.1?	MDFDPP:M	11.4.1	Yes []
MDFDPP-21	Do the Sync message sequence numbers comply with the requirements of 11.3.8?	MITT and MDFDPP:M	11.3.8	Yes [] N/A []
MDFDPP-22	Do the Pdelay_Req message sequence numbers comply with the requirements of 11.3.8?	MDFDPP:M	11.3.8	Yes []
MDFDPP-23	Does the Pdelay mean request transmission interval comply with the requirements of 11.5.2.2?	MDFDPP:M	11.5.2.2	Yes []
MDFDPP-24	Does the Sync mean transmission interval comply with the requirements of 11.5.2.3?	MDFDPP:M	11.5.2.3	Yes []
MDFDPP-25	Does the full-duplex point-to-point media-dependent layer set the asCapable global variable in the media-independent PortSync entity in compliance with the requirements of 11.2.2?	MDFDPP:M	11.2.2	Yes []
MDFDPP-26	Does the device's use of flow control comply with the requirements of 11.2.3 and 11.2.4?	MDFDPP:M	11.2.3, 11.2.4	Yes []

A.13 Media-dependent, full-duplex point-to-point link *(continued)*

Item	Feature	Status	References	Support
MDFDPP-27	Does the PTP Instance or CMLDS consider the PTP Port or Link Port, respectively, to not be exchanging Pdelay messages when a valid response is not received in compliance with the requirements of 11.5.3?	MDFDPP:M	11.5.3	Yes []
MDFDPP-28	Does the PTP Instance ignore TLVs, of PTP messages, that it cannot parse and attempt to parse the next TLV, in compliance with the requirements of 11.4.1?	MDFDPP:M	11.4.1	Yes []
MDFDPP-29	Does the time-aware system initialize meanLinkDelayThresh as specified in 11.2.2?	MDFDPP:M	11.2.2	Yes []
MDFDPP-30	Does this port of the time-aware system support asymmetry measurement mode (see Annex G for informative description)?	MDFDPP:O	14.15, 14.20, 10.2.5, 10.2.8, 10.3.12, 10.3.13, 10.3.15, 10.3.16, 11.2.14, 11.2.15, 11.2.20	Yes [] No []
MDFDPP-31	Does this PTP Port support one-step receive?	MDFDPP:O	11.2.14	Yes [] No []
MDFDPP-32	Does this PTP Port support one-step transmit?	MDFDPP:O	11.2.15	Yes [] No []
MDFDPP-33	Does this PTP Port implement the functionality of the OneStepTxOperSetting state machine in compliance with the requirements of 11.4, 11.2.16, and Figure 11-8?	MDFDPP:O	11.4, 11.2.16	Yes [] No []
MDFDPP-34	Does this port support propagation delay averaging?	MDFDPP:O	11.2.19.3.4	Yes [] No []
MDFDPP-35	If the time-aware system implements more than one domain, does the time-aware system provide CMLDS?	MDFDPP:M	11.2.17.1	Yes []
MDFDPP-36	If the time-aware system implements only one domain, does the time-aware system provide CMLDS?	MDFDPP:O	11.2.17.2	Yes [] No []
MDFDPP-37	Does this port support two-step capability on receive?	MDFDPP:M	11.2.14, item c) of 5.5	Yes []
MDFDPP-38	Does this port support two-step capability on transmit?	MDFDPP:M	11.2.15, item d) of 5.5	Yes []

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A.14 Media-dependent IEEE 802.11 link

Item	Feature	Status	References	Support
MDDOT11-1	Does the IEEE 802.11 MAC implement the timeTransmitter port functionality in compliance with the requirements of 12.5.1?	MDDOT11 and MITT:M	item d) of 5.6, 12.5.1	Yes []
MDDOT11-2	Does the IEEE 802.11 MAC implement the timeReceiver port functionality in compliance with the requirements of 12.5.2?	MDDOT11:M	item a), item b), and item d) of 5.6, 12.5.2	Yes []
MDDOT11-3	Does the IEEE 802.11 MAC determine the value of asCapable in compliance with the requirements of 12.4?	MDDOT11:M	12.4	Yes []
MDDOT11-4	Does the IEEE 802.11 MAC determine the value of mean time interval between synchronization messages in compliance with the requirements of 12.8?	MDDOT11 and MITT:M	12.8	Yes []
MDDOT11-5	Does the IEEE 802.11 MAC support the use of the VendorSpecific information element of 12.7 to carry end-to-end link-independent timing information?	MDDOT11:M	12.7	Yes []
MDDOT11-6	Does the IEEE 802.11 MAC implement Fine Timing Measurement as a timeTransmitter port?	MDDOT11-1:O	item c) and item e) of 5.6, 12.5.1	Yes [] No []
MDDOT11-7	Does the IEEE 802.11 MAC implement Fine Timing Measurement as a timeReceiver port?	MDDOT11-2:O	item c) and item e) of 5.6, 12.5.2	Yes [] No []

2

A.15 Media-dependent IEEE 802.3 EPON link

Item	Feature	Status	References	Support
MDEPON-1	Does the TIMESYNC message format comply with the requirements of 13.3 and Table 13-1?	MDEPON:M	13.3	Yes []
MDEPON-2	Does the PTP Instance implement the functionality specified by the requester state machine in compliance with the requirements of 13.8.1 and Figure 13-3?	MDEPON and MITT:M	13.8.1.4	Yes []
MDEPON-3	Does the PTP Instance implement the functionality specified by the responder state machine in compliance with the requirements of 13.8.2 and Figure 13-4?	MDEPON:M	13.8.2.4	Yes []
MDEPON-4	Does the TIMESYNC message transmission interval comply with the requirements of 13.9.1 and 13.9.2?	MDEPON:M	13.9.1, 13.9.2	Yes []

A.15 Media-dependent IEEE 802.3 EPON link *(continued)*

Item	Feature	Status	References	Support
MDEPON-5	Does the implementation of best timeTransmitter selection comply with the requirements of 13.1.3?	MDEPON:M	13.1.3	Yes []
MDEPON-6	Does the determination of the value of asCapable comply with the requirements of 13.4?	MDEPON:M	13.4	Yes []

1

A.16 Media-dependent CSN link

Item	Feature	Status	References	Support
MDCSN-1	Does the PTP Instance implement the functionality of the MDSyncSendSM state machine in compliance with 11.2.15?	MDCSN and MITT:M	11.2.15	Yes []
MDCSN-2	Does the PTP Instance implement the functionality of the MDSyncReceiveSM state machine in compliance with 11.2.14?	MDCSN:M	11.2.14	Yes []
MDCSN-3	Does the PTP Instance calculate path delay in compliance with the requirement of 16.4?	MDCSN:M	16.4.1, 16.4.2, 16.4.3	Yes []
MDCSN-4	Does the PTP Instance propagate synchronized time in compliance with the requirements of 16.5?	MDCSN:M	16.5.2, 16.5.3	Yes []
MDCSN-5	Does the PTP Instance act as Grandmaster PTP Instance in compliance with the requirements of 16.7?	GMCAP and MDCSN:M	16.7	Yes []
MDCSN-6	Does the PTP Instance comply with the performance requirements of 16.8?	GMCAP and MDCSN:M	16.8	Yes []

2

A.17 Media-dependent MoCA link

Item	Feature	Status	References	Support
MDMOCA-1	Does the MoCA MD entity propagate Sync messages in compliance with the requirements of 16.6.2?	MDMOCA:M	16.6.2	Yes []

3

A.18 Media-dependent ITU-T G.hn link

Item	Feature	Status	References	Support
MDGHN-1	Does the GHN MD entity propagate Sync messages in compliance with the requirements of 16.6.3?	MDGHN:M	16.6.3	Yes []

1

A.19 Remote management

Item	Feature	Status	References	Support
	If item RMGT is not supported, mark N/A.			N/A []
RMGT-1	What management protocol standard(s) or specification(s) are supported?	RMGT:M	item k) 1) of 5.4.2	
RMGT-2	What standard(s) or specification(s) for managed object definitions and encodings are supported for use by the remote management protocol?	RMGT:M	item k) 2) of 5.4.2	
RMGT-3	If the Simple Network Management Protocol (SNMP) is listed in RMGT-2, is the IEEE 8021-AS-MIB module fully supported (per its MODULE-COMPLIANCE)?	RMGT:O	item k) 3) of 5.4.2, Clause 15	Yes [] No [] N/A []
RMGT-4	If a remote management protocol that supports YANG is listed in RMGT-2, is the YANG data model ieee802-dot1as-gtp of Clause 17 supported?	RMGT:O	item k) 4) of 5.4.2, Clause 17	Yes [] No []
RMGT-5	If a remote management protocol that supports YANG is listed in RMGT-2, and if hot standby is supported, is the YANG data model ieee802-dot1as-hs of Clause 17 supported?	RMGT:O	item k) 5) of 5.4.2, Clause 17, Clause 18	Yes [] No [] N/A []

2

A.20 Application interfaces

Item	Feature	Status	References	Support
	If item APPL is not supported, mark N/A.			N/A []
APPL-1	What application interfaces(s) are supported?	APPL:M	item i) of 5.4.2	

3

A.21 External port configuration

Item	Feature	Status	References	Support
	If item EXT is not supported, mark N/A.			NA []

A.21 External port configuration

Item	Feature	Status	References	Support
EXT-1	Does the PTP Instance support the specifications for externalPortConfigurationEnabled value of true?	EXT:M	10.3.1	Yes []
EXT-2	Does the PTP Instance support the PortAnnounceInformationExt state machine?	EXT:M	10.3.14	Yes []
EXT-3	Does the PTP Instance support the PortStateSettingExt state machine?	EXT:M	10.3.15	Yes []

1 **Annex B**

2 (normative)

3 **Performance requirements**

4 **B.1 LocalClock requirements**

5 **B.1.1 Frequency accuracy**

6 The fractional frequency offset of the LocalClock relative to the TAI frequency (see ISO 80000-3:2006 and
7 Annex C) shall be within ± 100 ppm.

8 **B.1.2 Time measurement granularity**

9 The granularity with which the LocalClock measures time shall be less than or equal to $40/(1-0.0001)$ ns.

10 **B.1.3 Noise generation**

11 **B.1.3.1 Jitter generation**

12 The jitter generation of the free-running LocalClock shall not exceed 2 ns peak-to-peak, when measured
13 over a 60 s measurement interval using a band-pass filter that consists of the following low-pass and high-
14 pass filters:

- 15 a) High-pass filter: first-order characteristic (i.e., 0 dB gain peaking), 20 dB/decade roll-off, and 3 dB
16 bandwidth (i.e., corner frequency) of 10 Hz
- 17 b) Low-pass filter: maximally-flat (i.e., Butterworth) characteristic, 60 dB/decade roll-off, and 3 dB
18 bandwidth equal to the Nyquist rate of the LocalClock entity (i.e., one-half the nominal frequency of
19 the LocalClock entity)

20 **B.1.3.2 Wander generation**

21 Wander generation is specified using the Time Deviation (TDEV) parameter. The corresponding values of
22 the Allan Deviation (ADEV) and PTP Deviation (PTPDEV) are given for information; the former is also
23 useful in describing the wander generation of clocks and oscillators, and the latter is related to the
24 offsetScaledLogVariance attribute (see 8.6.2.4). Information on ADEV and TDEV is contained in
25 ITU-T G.810 [B21] and IEEE Std 1139™-1999 [B9]. Information on Allan Deviation and PTP Variance
26 (PTP Deviation is the square root of PTP Variance) is contained in 7.6.3 of IEEE Std 1588-2019.

1 TDEV, denoted $\sigma_x(\tau)$, is estimated from a set of measurements, as shown in Equation (B-1).

$$2 \quad \sigma_x(\tau) = \sqrt{\frac{1}{6n^2(N-3n+1)} \sum_{j=1}^{N-3n+1} \left[\sum_{i=j}^{n+j-1} (x_{i+2n} - 2x_{i+n} + x_i) \right]^2}, n = 1, 2, \dots, \left\lfloor \frac{N}{3} \right\rfloor \quad (\text{B-1})$$

3 where

- 4 τ is $n\tau_0$ = observation interval
- 5 τ_0 is the sampling interval
- 6 N is the total number of samples $[(N-1)\tau_0 = \text{measurement interval}]$
- 7 $\lfloor y \rfloor$ denotes the floor function, i.e., the greatest integer less than or equal to y
- 8 x_i is the measured phase (time) error at the i^{th} sampling time [the units of x_i and $\sigma_x(\tau)$ are the same]

9 ADEV, denoted $\sigma_y(\tau)$, is estimated from a set of measurements, as shown in Equation (B-2).

$$10 \quad \sigma_y(\tau) = \sqrt{\frac{1}{2n^2\tau_0^2(N-2n)} \sum_{i=1}^{N-2n} (x_{i+2n} - 2x_{i+n} + x_i)^2}, n = 1, 2, \dots, \left\lfloor \frac{N-1}{2} \right\rfloor \quad (\text{B-2})$$

11 where the notation is the same as defined above for TDEV.

12 PTPDEV, denoted $\sigma_{PTP}(\tau)$, is estimated from a set of measurements, as shown in Equation (B-3).

$$13 \quad \sigma_{PTP}(\tau) = \sqrt{\frac{1}{6(N-2n)} \sum_{i=1}^{N-2n} (x_{i+2n} - 2x_{i+n} + x_i)^2}, n = 1, 2, \dots, \left\lfloor \frac{N-1}{2} \right\rfloor \quad (\text{B-3})$$

14 where the notation is the same as defined above for TDEV.

15 TDEV, ADEV, and PTPDEV are second-order statistics on the phase error. All three statistics are functions
16 of second differences of the phase error. As a result, these statistics are not affected by a constant frequency
17 offset. This behavior is desired, because these statistics are used here to constrain noise generation.

1 TDEV for the LocalClock entity shall not exceed the mask of Table B-1 and Figure B-1, when measured
2 using:

- 3 a) A measurement interval that is at least 120 s (i.e., at least 12 times the longest observation interval),
- 4 b) A low-pass filter with 3 dB bandwidth of 10 Hz, first-order characteristic, and 20 dB/decade roll-off,
- 5 and
- 6 c) A sampling interval τ_0 that does not exceed 1/30 s.

Table B-1—Wander generation TDEV requirement for LocalClock entity

TDEV limit	Observation interval τ
No requirement	$\tau < 0.05$ s
5.0 τ ns	$0.05 \leq \tau \leq 10$ s
No requirement	$\tau > 10$ s

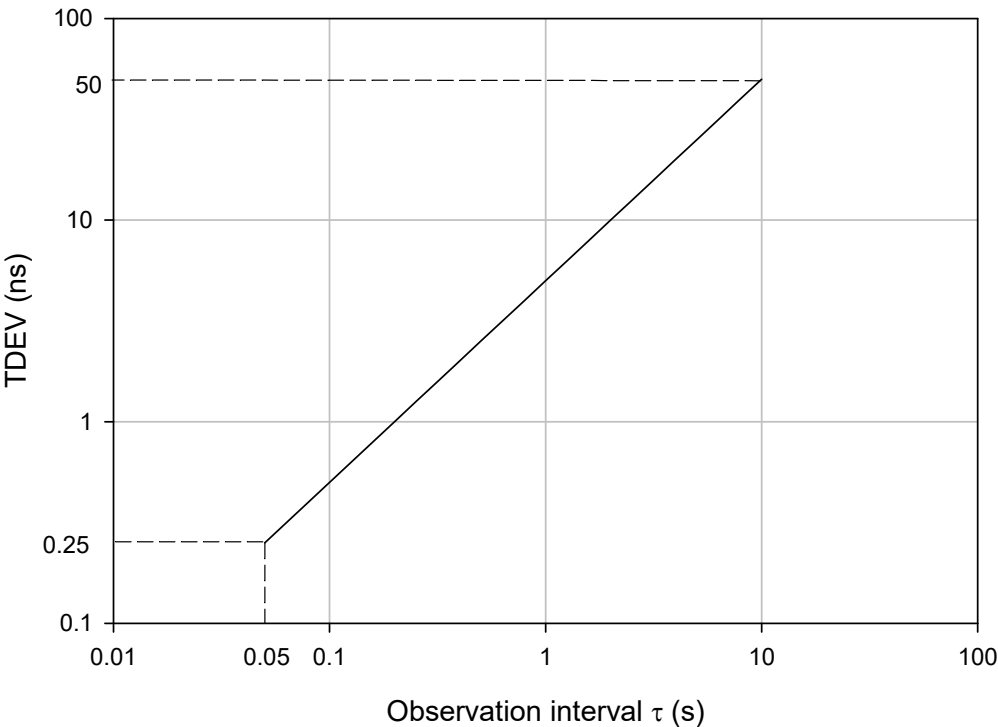


Figure B-1—Wander generation (TDEV) requirement for LocalClock entity

1 The ADEV limit that corresponds to the TDEV requirement of Table B-1 and Figure B-1 is shown in
2 Table B-2 and Figure B-2, respectively.

Table B-2—ADEV limit corresponding to wander generation requirement of Table B-1

ADEV limit	Observation interval τ
No requirement	$\tau < 0.05$ s
1.054×10^{-8}	$0.05 \leq \tau \leq 10$ s
No requirement	$\tau > 10$ s

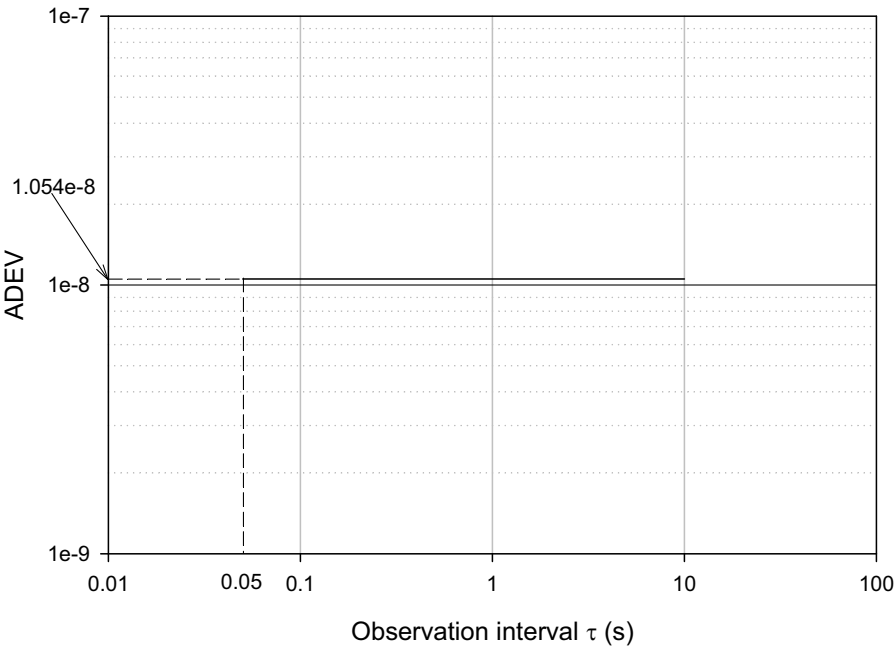


Figure B-2—ADEV limit corresponding to wander generation requirement of Figure B-1

1 The PTPDEV limit that corresponds to the TDEV requirement of Table B-1 and Figure B-1 is shown in
2 Table B-3 and Figure B-3, respectively.

Table B-3—PTPDEV limit corresponding to wander generation requirement of Table B-1

PTPDEV limit	Observation interval τ
No requirement	$\tau < 0.05$ s
6.08τ ns	$0.05 \leq \tau \leq 10$ s
No requirement	$\tau > 10$ s

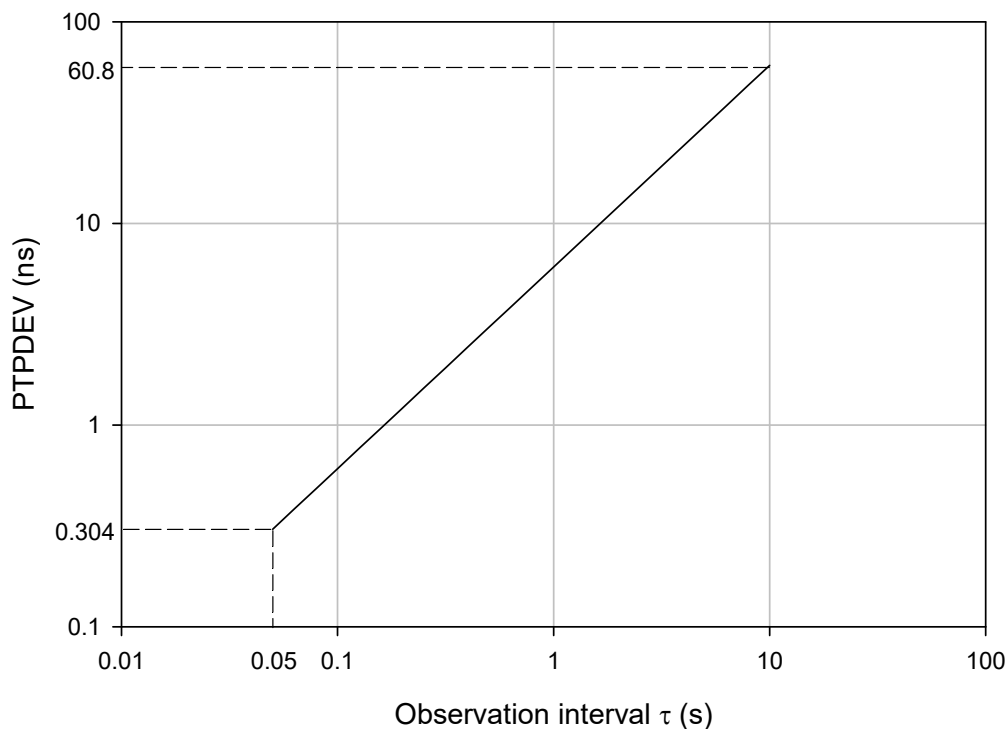


Figure B-3—PTPDEV limit corresponding to wander generation requirement of Figure B-1

1 B.2 PTP Instance requirements

2 B.2.1 General

3 In order to achieve the accuracy goals, certain constraints are placed on the responsiveness and accuracy of
4 PTP Instances.

5 B.2.2 Residence time

6 The residence time (see 3.30) of a PTP Instance, measured relative to the TAI second (see 8.2), should be
7 less than or equal to 10 ms.

8 Any error in the measured frequency offset relative to the Grandmaster Clock (i.e., error in accumulated
9 rateRatio) results in an error in the transported synchronized time that is equal to the frequency offset error
10 multiplied by the residence time (see 7.4.3 and 11.1.3).

11 B.2.3 Pdelay turnaround time

12 The pdelay turnaround time is the duration of the interval between the receipt of a Pdelay_Req message by a
13 port of a time-aware system, and the sending of the corresponding Pdelay_Resp message.

14 The pdelay turnaround time of a time-aware system, measured relative to the TAI second (see 8.2), should
15 be less than or equal to 10 ms.

16 A nonzero pdelay turnaround time and any error in the measured frequency offset between the peer delay
17 initiator and peer delay responder (i.e., error in nrrPdelay) results in an error in the measured mean
18 propagation delay. This in turn results in an error in the transported synchronized time (see 11.1.2 for more
19 details).

20 NOTE—While a larger value of pdelay turnaround time can result in worse time-synchronization performance, the peer
21 delay protocol will still operate as long as the peer delay initiator receives Pdelay_Resp and Pdelay_Resp_Follow_Up
22 within a time interval since sending Pdelay_Req that is less than the current Pdelay_Req message transmission interval
23 (see 11.2.19.4 and 11.5.2.2).

24 B.2.4 Measurement of rate ratio

25 This standard requires the measurement of rate ratio or, equivalently, frequency offset, in several subclauses
26 (see 10.2.10, 10.2.11, 11.2.19, 12.5.2, 16.4.2, and 16.4.3.2). The error inherent in any scheme used to
27 measure rate ratio shall not exceed ± 0.1 ppm.

28 NOTE—This requirement is consistent with a rate ratio measurement made by measuring the default Pdelay_Req
29 message transmission interval (the nominal interval duration is 1 s; see 11.5.2.2) relative to the clocks whose rate ratio is
30 desired, assuming the clocks meet the time measurement granularity requirement of B.1.2 (i.e., no worse than 40 ns).

1 B.3 End-to-end time-synchronization performance

2 Assuming that the requirements of this standard and of standards referenced for each medium are met, any
3 two PTP Instances separated by six or fewer PTP Instances (i.e., seven or fewer hops) will be synchronized
4 to within 1 microsecond peak-to-peak of each other during steady-state operation (i.e., each PTP Instance
5 receives time-synchronization information every sync interval).

6 B.4 End-to-end jitter and wander performance

7 The requirements of this standard and standards referenced by this standard ensure that the synchronized
8 time at a PTP Instance that is separated from the Grandmaster PTP Instance by six or fewer PTP Instances
9 (i.e., seven or fewer hops) will, when filtered by a reference endpoint filter with rolloff of 20 dB/decade,
10 gain peaking that does not exceed 0.1 dB, and bandwidth that does not exceed the value given in each entry
11 of Table B-4, have maximum time interval error (MTIE) (see ITU-T G.810 [B21]) that does not exceed the
12 MTIE for that entry of Table B-4, and jitter that does not exceed the peak-to-peak jitter of Table B-4 when
13 measured through the corresponding high-pass jitter measurement filter given in Table B-4.

14 NOTE—For example, the endpoint filter can be of the following form:

$$15 \quad y_k = a_1 y_{k-1} + a_2 y_{k-2} + \dots + a_n y_{k-n} + b_0 x_k + b_1 x_{k-1} + \dots + b_n x_{k-n}$$

16 where the x_k are the unfiltered synchronized time values, the y_k are the filtered synchronized time values, and the a_k and
17 b_k are filter coefficients. The a_k and b_k are chosen such that the filter has desired bandwidth and gain peaking that does
18 not exceed 0.1 dB. The preceding equation is a general infinite impulse response (IIR) digital filter. Simplified forms,
19 e.g., a second order IIR filter obtained by setting $n = 2$, or a finite impulse response (FIR) filter obtained by setting the a_k
20 to zero are possible.

Table B-4—Maximum endpoint filter bandwidths needed to meet respective MTIE masks and peak-to-peak jitter limits

Endpoint filter maximum bandwidth (Hz)	Corresponding MTIE mask of Figure B-4 that is not exceeded	Corresponding jitter high-pass measurement filter (Hz)	Corresponding peak-to-peak jitter that is not exceeded (ns)
10	Mask 2 (Figure B-4, Table B-6)	8000	10.2
1	Mask 1 (Figure B-4, Table B-5)	200	11.1

21 Mask 1 of Table B-4 and Figure B-4 corresponds to consumer digital audio applications. Mask 2 of
22 Table B-4 and Figure B-4 corresponds to professional digital audio applications. Mask 1 is derived from the
23 requirements given in IEC 60958-3 [B6]. Mask 2 is derived from the requirements given in
24 IEC 60958-4 [B7], AES3-2009 [B1], and AES11-2009 [B2]. Garner describes the methodology for
25 deriving MTIE from the various jitter and synchronization requirements and presents the detailed
26 derivations for masks 1 and 2.

27

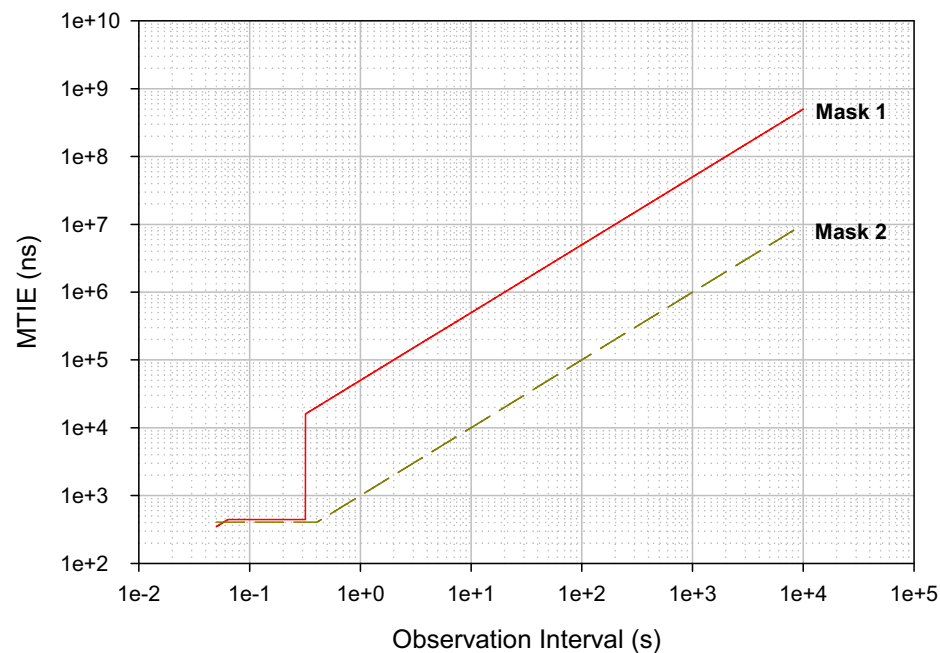


Figure B-4—MTIE masks met for maximum endpoint filter bandwidths of Table B-4

Table B-5—Breakpoints for Mask 1

Observation interval S (s)	MTIE (ns)
$0.05 \leq S < 0.0637$	$6954.8S$
$0.0637 \leq S < 0.3183$	443
$0.3183 \leq S \leq 10000$	$50000S$

Table B-6—Breakpoints for Mask 2

Observation interval S (s)	MTIE (ns)
$0.05 \leq S < 0.4069$	407
$0.4069 \leq S < 10000$	$1000S$

1 Annex C

2 (informative)

3 Timescales and epochs

4 (informative)

5

6 C.1 Overview

7 A more detailed discussion of many of the topics in this annex can be found in Allan et al. [B3].

8 For historical reasons, time is specified in a variety of ways as listed in Table C-1. GPS, PTP, and TAI times
9 are based on values yielded by atomic clocks and advance on each second. NTP and UTC times are similar,
10 but are occasionally adjusted by one leap second, to account for differences between the atomic clocks and
11 the rotation time of the earth.

Table C-1—Timescale parameters

Parameter	Timescale				
	GPS	PTP	TAI	NTP	UTC
Approximate epoch ^a	1980-01-06 1999-08-22 2019-04-07	1970-01-01	No epoch defined	1900-01-01	No epoch defined
Representation	weeks.seconds	seconds	YYYY-MM-DD hh:mm:ss	seconds	YYYY-MM-DD hh:mm:ss
Rollover (years)	19.7	≈8 900 000	10 000	≈136	10 000
NOTE 1—After 1972-01-01 00:00:00 TAI, TAI and UTC differ by only integer seconds.					
NOTE 2—The following represent the same instant in time: 1958-01-01 00:00:00 TAI and 1958-01-01 00:00:00 UTC					

^a Each approximate epoch occurs at 00:00:00 on the respective date.

GPS global positioning satellite
NTP network time protocol
PTP IEEE 1588 precision time protocol
TAI International Atomic Time (from the French term *Temps Atomique International*)
UTC Coordinated Universal Time [The abbreviation is a compromise between the English phrase *coordinated universal time* (CUT) and the French phrase *temps universel coordonné* (TUC).]

12 C.2 TAI and UTC

13 TAI and UTC are international standards for time based on the SI second (see Allan et al. [B3], IAU [B27],
14 and Petit and Luzum [B26]). The SI second is the duration of 9 192 631 770 periods of the
15 radiation corresponding to the transition between the two hyperfine levels of the ground state of the

1 cesium 133 atom.²⁴ TAI is implemented by a suite of atomic clocks and forms the timekeeping basis for
 2 other timescales in common use. The rate at which UTC time advances is normally identical to the rate of
 3 TAI. An exception is an occasion when UTC is modified by adding or subtracting exactly one whole leap
 4 second. The TAI frequency is the frequency of a signal whose period is the TAI second.

5 The TAI and UTC timescales were introduced as of 1958-01-01, and the times 1958-01-01 00:00:00 TAI
 6 and 1958-01-01 00:00:00 UTC represent the same instant in time. However, this instant in time is not an
 7 epoch for TAI and UTC. An epoch, in the sense of the origin of a timescale (see 8.2.2), is not defined for
 8 either TAI or UTC. TAI and UTC are expressed in the form YYYY-MM-DD hh:mm:ss, rather than as
 9 elapsed time since an epoch; here, YYYY-MM-DD denotes the date, and hh:mm:ss denotes the time in
 10 each day.

11 Prior to 1972-01-01, corrections to the offset between UTC and TAI were made by applying fractional-
 12 second corrections to UTC and corrections to the rate at which UTC advanced relative to the rate at which
 13 TAI advanced. After 1972-01-01, leap-second corrections are applied to UTC by inserting or deleting
 14 second(s) at the end of the last minute of preferably the last day of June or December. Also after 1972-01-01,
 15 UTC and TAI advance at the same rate. As of 2006-01-01, TAI and UTC times differed by +33 s (i.e.,
 16 TAI time minus UTC time equals +33 s for 2006-01-01).

17 In computer networks, the common POSIX-based time conversion algorithms are typically used to produce
 18 the correct ISO 8601:2004 [B15] printed representations for both TAI and UTC.

19 The PTP epoch is set such that a direct application of the POSIX algorithm to a PTP timescale timestamp
 20 converts the PTP timestamp to the ISO 8601:2004 [B15] printed representation of TAI. PTP also distributes
 21 the current offset between TAI and UTC in the currentUtcOffset field of Announce messages. Except during
 22 leap seconds, subtracting currentUtcOffset from a PTP timestamp and then applying the POSIX algorithm
 23 result in the ISO 8601:2004 printed representation of UTC. Conversely, except during leap seconds,
 24 applying the inverse POSIX algorithm and adding currentUtcOffset convert from the ISO 8601:2004 printed
 25 form of UTC to the form required to generate a PTP timestamp.

26 For example, at 0h 2 January 1972 TAI, the value of PTP Instance Time was 63 158 400. At this time
 27 currentUtcOffset was 10. The POSIX algorithm applied to the value (63 158 400 – 10) gives a value of
 28 1972-01-01 23:59:50 (10 s before 0h 2 January 1972 UTC). The value of PTP Instance Time on
 29 0h 2 January 1972 TAI is computed by observing that PTP Instance Time = 0 on 0h 1 January 1970 TAI, i.e.,
 30 Modified Julian Day (MJD) 40587 (see Petit and Luzum [B26]). On 0h 2 January 1972 TAI, MJD = 41 318.
 31 Thus, PTP Instance Time on 0h 2 January 1972 TAI is $0 + 86\,400 \times (41\,318 - 40\,587)$. Note that if this
 32 calculation were done for a day in which a leap second occurred, a more complex algorithm would be
 33 required to ensure that, for the duration of the leap second, the ISO 8601:2004 [B15] print form seconds
 34 value was 60 for a positive leap second.

35 International standards specify that if a correction to UTC relative to TAI is required, the leap second occurs
 36 at the last second of the UTC day, preferably at the end of June 30 or December 31. For a negative leap
 37 second, the last minute of the designated day has only 59 seconds. Negative leap seconds have never
 38 occurred and are unlikely to occur in the future. For a positive leap second, the last minute of the designated
 39 day has 61 seconds.

40 Although a negative leap second is unlikely to occur, if such a correction becomes necessary, the
 41 ISO 8601:2004 [B15] printed representation would appear as follows for a hypothetical negative leap
 42 second on 30 June 1972 UTC:

43 1972-06-30 23:59:57, 1972-06-30 23:59:58, 1972-07-01 00:00:00

²⁴ See ITU-R TF.460-6 [B20], Jekeli [B22], Service de la Rotation Terrestre [B28], SI [B14], IAU [B27], and Petit and Luzum [B26] for more details on UTC, TAI, and the SI second.

1 For the positive leap second that actually occurred on 30 June 1972 UTC, the ISO 8601:2004 [B15] printed
2 representation appeared as follows:

3 1972-06-30 23:59:59, 1972-06-30 23:59:60, 1972-07-01 00:00:00

4 Note the 23:59:60 notation to indicate the added second.

5 The update semantics for leap second updates in PTP are discussed in B.2.2 of IEEE Std 1588-2019.

6 C.3 NTP and GPS

7 Two standard time sources of particular interest in implementing PTP Instances: NTP and GPS. Both NTP
8 and GPS systems are expected to provide time references for calibration of the grandmaster-supplied PTP
9 time.

10 NTP represents seconds as a 32-bit unsigned integer that rolls-over every 2^{32} s $\approx$ 136 year, with the first such
11 rollover occurring in the year 2036. The precision of NTP systems is usually in the millisecond range.

12 NTP is a widely used protocol for synchronizing computer systems. NTP is based on sets of servers, to
13 which NTP clients synchronize. These servers themselves are synchronized to time servers that are traceable
14 to international standards.

15 NTP version 4 provides the current UTC time and warning flags indicating that a leap second will be
16 inserted at the end of the current UTC day. The NTP clock effectively stops for one second when the leap
17 second is inserted.

18 GPS time comes from a global positioning satellite system, GPS, maintained by the U.S. Department of
19 Defense. The precision of GPS system is usually in the 10 ns to 100 ns range. GPS system transmissions
20 represent the time as {*weeks, secondsInWeek*}, the number of weeks since the GPS epoch and the number of
21 seconds since the beginning of the current week.

22 GPS provides a leap seconds offset and warning flags marking the introduction of a leap second correction
23 (see IS-GPS-200J [B19]). UTC and TAI times can be computed solely based the information contained in
24 the GPS transmissions.

25 GPS timing receivers generally manage the epoch transitions (1024-week rollovers), providing the correct
26 time (YYYY-MM-DD hh:mm:ss) in TAI and/or UTC timescales, and often also local time; in addition to
27 providing the raw GPS week, second of week, and leap-second information.

28 C.4 Timescale conversions

29 Previously discussed representations of time can be readily converted to/from PTP *time* based on a constant
30 offset and the distributed *utcOffset* value, as specified in Table C-2. Within Table C-2, all variables represent
31 integers; “/” and “%” represent a integer divide and remainder operation, respectively.

32 NOTE—Some of the conversions in Table C-2 will not give the correct result when the current second is a leap second.

Table C-2—Timescale conversions

ta		PTP value tb
Name	Format	
GPS	weeks:seconds	tb = ta.seconds + 315 964 819 + (gpsRollovers * 1024 + ta.weeks) * (7 * DAYSECS);
		ta.weeks = (tb – 315 964 819) / (7 * DAYSECS) – gpsRollovers*1024; ta.seconds = (tb – 315 964 819) % (7 * DAYSECS);
TAI	date{YYYY,MM,DD} :time{hh,mm,ss}	tb = DateToDays(“1970-01-01”, ta.date) * DAYSECS + ((ta.time.hh * 24) + ta.time.mm) *60) + ta.time.ss;
		secs = tb % DAYSECS; ta.date = DaysToDate(“1970-01-01”, tb / DAYSECS); ta.time.hh = secs / 3600; ta.time.mm = (secs % 3600)/60; ta.time.ss = (secs % 60);
NTP	seconds	tb = (ta + utcOffset)–2208988800;
		ta = (tb–utcOffset) +2208988800;
UTC	date{YYYY,MM,DD} :time{hh,mm,ss}	tb = DateToDays(“1970-01-01”, ta.date) * DAYSECS + ((ta.time.hh * 24) + ta.time.mm) *60) + ta.time.ss + utcOffset;
		tc = tb – utcOffset; secs = tc % DAYSECS; ta.date = DaysToDate(“1970-01-01”, tc/DAYSECS); ta.time.hh = secs / 3600; ta.time.mm = (secs % 3600)/60; ta.time.ss = (secs % 60);

gpsRollovers Currently equals 1; changed from 0 to 1 between 1999-08-15 and 1999-08-22

DAYSECS The number of seconds within a day: (60 × 60 × 24)

utcOffset The difference TAI – UTC, in seconds [i.e., currentUtcOffset (see 8.2.3)]

DateToDays For arguments DateToDays(*past*, *present*), returns days between *past* and *present* dates

DaysToDate For arguments DaysToDate(*past*, *days*), returns the current date, *days* after the *past* date

1 C.5 Time zones and GMT

2 The term *Greenwich Mean Time* (GMT) once referred to mean solar time at the Royal Observatory in

3 Greenwich, England. GMT now commonly refers to the timescale UTC; or the UK winter time zone

4 (Western European Time, WET). Such GMT references are, strictly speaking, incorrect but nevertheless

5 quite common. The following representations correspond to the same instant of time:

6 18:07:00 (GMT), commonplace usage13:07:00 (Eastern Standard Time, EST)

7 18:07:00 (UTC)01:07 PM (Eastern Standard Time, EST)

8 18:07:00 (Western European Time, WET)10:07:00 (Pacific Standard Time, PST)

9 06:07 PM (Western European Time, WET)10:07 AM (Pacific Standard Time, PST)

1 Annex D

2 (informative)

3 Reserved for future use

1 Annex E

2 (informative)

3 Reserved for future used

4 (For information on media-dependent layer specification for CSN, see Clause 16.)

1 Annex F

2 (informative)

3 PTP profile included in this standard

4 F.1 General

5 The specification in this standard of synchronized time transport over a full-duplex point-to-point link
6 includes a PTP profile. The information contained in a PTP profile is described in 20.3 of IEEE Std 1588-
7 2019. This annex summarizes the PTP profile for transport of timing over full-duplex point-to-point links.
8 This PTP profile is also used in the transport of timing over CSN when a CSN clock reference is not present
9 (see Clause 16). This PTP profile is not used in the transport of timing over IEEE 802.11 links and IEEE
10 802.3 EPON links; both these transports use native timing mechanisms to assist in the synchronized time
11 transport.

12 F.2 Identification

13 The identification values for this PTP profile (see 20.3.3 of IEEE Std 1588-2019) are as follows:

14 PTP Profile:
15 IEEE 802.1AS PTP profile for transport of timing
16 Profile Name: IEEE 802.1AS PTP profile
17 profileNumber: 0
18 primaryVersion: 2
19 revisionNumber: 0
20 profileIdentifier: 00-80-C2-00-02-00

21 NOTE—In the above profileIdentifier (see 20.3.3 of IEEE Std 1588-2019):

- 22 a) 00-80-C2 (first three octets) is the OUI owned by the IEEE 802.1 Working Group (it identifies the organization
23 that specifies the PTP profile);
24 b) 00 (fourth octet) is a number that identifies the PTP profile, among all profiles specified by the organization that
25 owns the OUI (or, in general, OUI or CID) of item a);
26 c) 02 (fifth octet) is the primary version of this PTP profile; and
27 d) 00 (sixth octet) is the revisionNumber of this PTP profile.

28 This profile is specified by the IEEE 802.1 Working Group of the IEEE 802 LAN/MAN Standards
29 Committee.

30 A copy can be obtained by ordering IEEE Std 802.1AS-2020 from the IEEE Standards Organization.²⁵

31 This PTP profile is a revision of the PTP profile included in IEEE Std 802.1AS-2011, i.e., major changes
32 have been made to the profile relative to Version 1.0. Therefore, the primaryVersion is changed to 2, with
33 revisionNumber 0.

²⁵ IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org>).

1 F.3 PTP attribute values

2 The ranges and default values for time-aware system and PTP Instance attributes covered by this profile are
3 as follows:

- 4 a) A domain whose domainNumber is 0 is present. A domain whose domainNumber is in the range 1
5 through 127 can be present (see 8.1).
- 6 b) The default logAnnounceInterval (see 10.7.2.2) is 0. The value 127 is supported.
- 7 c) The default logSyncInterval (see 11.5.2.3) is -3. The value 127 is supported.
- 8 d) The default logPdelayReqInterval (see 11.5.2.2) is 0. The value 127 is supported.
- 9 e) The default announceReceiptTimeout (see 10.7.3.2) is 3.
- 10 f) The default values of priority1, for different media, are specified in 8.6.2.1, Table 8-1. The value of
11 priority1 for a PTP Instance that is not grandmaster-capable is 255.
- 12 g) The default value of priority2 is 248 (see 8.6.2.5).

13 The default observation interval for offsetScaledLogVariance is equal to the default sync interval, i.e., 0.125
14 s (see 8.6.2.4).

15 F.4 PTP options

- 16 a) The BTCA of this standard is the default BTCA according to the specifications of 9.3 of
17 IEEE Std 1588-2019.
- 18 b) The following options of 17.7 of IEEE Std 1588-2019 are invoked:
 - 19 1) The FAULTY state is not used.
 - 20 2) The UNCALIBRATED state is not used.
 - 21 3) The LISTENING state is not used.
 - 22 4) The PRE_TIME_TRANSMITTER state, and PRE_TIME_TRANSMITTER qualification are
23 not used.
 - 24 5) The foreign timeTransmitter feature is not used.
- 25 c) The management mechanism is the mechanism specified in Clause 14, Clause 15, and Clause 17.
- 26 d) The path delay mechanism is the peer-to-peer delay mechanism (see Clause 11).
- 27 e) The transport mechanism is full-duplex and point-to-point, and it uses attribute values described in
28 Annex E of IEEE Std 1588-2019 for IEEE 802.3 Ethernet. Specifically, the address, EtherType, and
29 subtype are specified in 11.3.4, 11.3.5, and 11.3.6.
- 30 f) A PTP Instance that contains one PortSync and one MD entity is an Ordinary Clock. A PTP Instance
31 that contains more than one PortSync and more than one MD entity is a Boundary Clock.
- 32 g) Each port of a time-aware system measures the frequency offset of its neighbor, at the other end of
33 the attached link, relative to itself (see Clause 11). The frequency offset, relative to the Grandmaster
34 Clock, is accumulated in a standard organization TLV that is attached to the Follow_Up message if
35 the PTP Port is two-step and the Sync message if the PTP Port is one-step (see 11.4.4.3). The
36 standard organization TLV also carries information on Grandmaster Clock traceability and phase
37 and frequency change due to the most recent Grandmaster PTP Instance change. The physical
38 adjustment of the frequency of the LocalClock entity (i.e., physical syntonization) is allowed but not
39 required.

40 NOTE 1—This feature is similar to the cumulative frequency transfer method specified in 16.10 of IEEE Std 1588-2019;
41 however, it is not the same feature and uses a different TLV from the one used in the IEEE 1588 feature. This feature
42 existed in the 2011 edition of the present standard and is retained for backward compatibility and also because the TLV
43 of this standard carries additional information that is not carried in the IEEE 1588 feature [see item g) in this subclause].

- 1 h) A standard organization TLV is defined to allow a PTP Port to signal to its neighbor PTP Port that it
2 is capable of invoking gPTP (see 10.4 and 10.6.4.4).
- 3 i) The path trace feature of 16.2 of IEEE Std 1588-2019 is used (see 10.6.3.2.8).
- 4 j) A standard organization TLV is defined that allows a port of a time-aware system to request that its
5 neighbor slow down or speed up the rate at which it sends Sync/Follow_Up, peer delay, and/or
6 Announce messages (see 10.6.4.3).
- 7 k) The acceptable timeTransmitter table feature of IEEE Std 1588-2019 is used with IEEE 802.3 EPON
8 links to ensure that the OLT is timeTransmitter and ONUs are timeReceivers.

9 NOTE 2—This feature is used with EPON links and therefore could be considered to be outside the PTP profile
10 (because EPON links are not part of the PTP profile). It is included here because it is one of the optional features
11 described in IEEE Std 1588-2019.

- 12 l) The profile isolation feature of IEEE Std 1588-2019 is not explicitly used; however, the PTP profile
13 specified in this standard is isolated from other PTP profiles because it uses sdoId 0x100 (see 8.1).
- 14 m) A time-aware system can have more than one gPTP domain (see 8.1).
- 15 n) The Common Mean Link Delay Service specified in 16.6 of IEEE Std 1588-2019 is required if more
16 than one domain is implemented and is optional if one domain (with domainNumber 0) is
17 implemented (see 11.2.17).
- 18 o) The security mechanism of 16.14 of IEEE Std 1588-2019 and security annex (Annex P) of
19 IEEE Std 1588-2019 are not used.
- 20 p) The external port configuration feature of 17.6 of IEEE Std 1588-2019 is optional in the present
21 standard.
- 22 q) Except for items g) through p) in this subclause, the optional features of Clause 16 and Clause 17
23 and the optional annexes of IEEE Std 1588-2019 are not used.

24 **F.5 LocalClock and PTP Instance performance requirements**

25 The LocalClock performance requirements are as specified in B.1. The PTP Instance performance
26 requirements are as specified in B.2.

27

1 Annex G

2 (informative)

3 The asymmetry compensation measurement procedure based on 4 line-swapping

5 G.1 Introduction

6 This annex describes the asymmetry compensation measurement procedure based on the line-swapping
7 method. The entire procedure is controlled by the Network Management System (NMS), except that the
8 line-swapping is manually operated. This annex is intended to address the delay asymmetry due to length
9 differences between transmit and receive fibers for long fiber runs.

10 NOTE—When a port is put into asymmetry measurement mode, it is put into this mode on all domains. The per-port
11 global variable `asymmetryMeasurement mode` is common to, and accessible by, all domains (see 10.2.5.2).

12 G.2 Pre-conditions for measurement

13 The following pre-conditions should be met to both improve the accuracy of the measurement and make the
14 measurement procedure more convenient:

- 15 a) The measurement environment, including the testing nodes (i.e., the time-aware systems at the
16 endpoint of the link whose asymmetry is being compensated) and related nodes (i.e., nodes in the
17 paths between the testing nodes and the Grandmaster PTP Instance), should enable gPTP and the
18 BTCA so that the test can be made for each link without changing the configuration.
- 19 b) The testing nodes should have redundant paths for synchronization so that they remain synchronized
20 to the Grandmaster Clock during the asymmetry compensation measurement.

21 G.3 Measurement procedure

22 The assumed measurement environment is shown in Figure G-1. Before the measurement starts, every node
23 has enabled syntonization [i.e., by measuring `nrrPdelay` (see 11.2.13.14) for each port of each time-aware
24 system and accumulating the respective values over the synchronization tree paths to obtain `rateRatio` (see
25 10.2.8.1.4) relative to the Grandmaster Clock²⁶] and time synchronization. The testing link is between Port2
26 on node ACC1 and Port1 on node ACC2.

27

28

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²⁶ This standard neither requires nor prohibits syntonization (see 3.35) at the physical layer, e.g., using Synchronous Ethernet as the physical layer. If frequency is syntonized at the physical layer, the respective `nrrPdelay` and `rateRatio` values are expected to be close to 1.

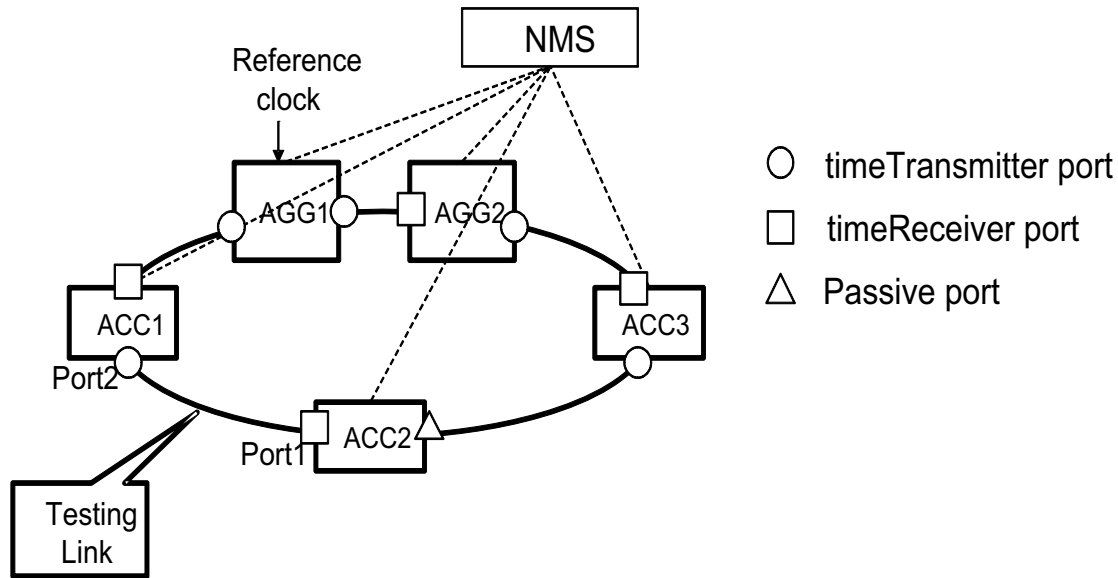


Figure G-1—Asymmetry compensation measurement procedure

1 The criteria of 11.2.17 for determining whether the peer-to-peer delay mechanism is the transport-specific
2 peer-to-peer delay mechanism or CMLDS apply here.

3 The measurement procedure is as follows:

- 4 a) The NMS puts Port1 of ACC2 and Port2 of ACC1 into asymmetry measurement mode through the
5 MIB (i.e., by setting the managed object asymmetryMeasurementMode for each port to TRUE).
6 These two ports will not affect the PTP calculations of either node when the ports are in asymmetry
7 measurement mode. If synchronization flowed over the link connecting these ports prior to their
8 being put into asymmetry measurement mode, the BTCA will result in a reconfiguration of the
9 synchronization tree so that both ACC1 and ACC2 remain synchronized.
- 10 b) Port1 of ACC2 will send Pdelay_Req messages periodically using the peer delay measurement
11 mechanism and receive Pdelay_Resp and Pdelay_Resp_Follow_Up messages. For each set of
12 messages, ACC2 should save t_3 [the pdelayRespEventEgressTimestamp (see 11.3.2.1), carried in
13 the Pdelay_Resp_Follow_Up message] and t_4 (the pdelayRespEventIngressTimestamp, taken when
14 the Pdelay_Resp message timestamp point crosses the reference plane at Port1 of ACC2 on
15 reception).
- 16 c) The NMS reads and saves multiple sets of (t_3, t_4) from ACC2 through the MIB. It is necessary that
17 each set of (t_3, t_4) be from the same measurement, i.e., from the same peer delay message exchange.
18 The number of (t_3, t_4) sets can be decided as required and is outside the scope of gPTP.
- 19 d) The tester manually exchanges the transmit and receive fibers of Port2 of ACC1 and Port1 of ACC2.
20 Then the tester waits until the port status and protocol status become stable again.
- 21 e) Port1 of ACC2 will again make periodic peer delay measurements and save each set of
22 measurement values (t_3', t_4') (the primes are used to denote measurements that have occurred after
23 the transmit and receive fibers have been exchanged).
- 24 f) The NMS reads and saves the multiple sets of (t_3', t_4') from ACC2 through the MIB. Then the NMS
25 can compute the delay asymmetry, in units of time, as $(t_4' - t_4) \times \text{nrrPdelay} - (t_3' - t_3)$. The NMS
26 can use multiple sets of (t_3, t_4, t_3', t_4') to compute average values to get a more accurate result. The
27 averaging method can be decided as required and is outside the scope of gPTP. If ACC1 and ACC2
28 are frequency synchronized, then nrrPdelay is 1, and the delay asymmetry is $(t_4' - t_3') - (t_4 - t_3)$.

- 1 g) Based on the above result, the NMS sets the asymmetry value for Port1 of ACC2 and Port2 of ACC1
 2 through the MIB.
- 3 h) After the NMS sets Port 1 of ACC2 and Port2 of ACC1 into normal mode (i.e., by setting the
 4 managed object `asymmetryMeasurementMode` for each port to `FALSE`), the delay asymmetry
 5 measurement of the testing link is completed. ACC1 and ACC2 will use the computed delay
 6 asymmetry as compensation in the PTP calculations.

7 It is also possible to compute the asymmetry ratio, i.e., the ratio of the delay on the receive fiber at ACC2
 8 (`Delay_rx_fiber`, the delay of the `Pdelay_Resp`) to the delay on the transmit fiber at ACC2 (`Delay_tx_fiber`,
 9 the delay of the `Pdelay_Req`), both after line swapping. In this case, the NMS should collect multiple sets of
 10 (t_1, t_2, t_3, t_4) and (t_1', t_2', t_3', t_4') before and after line-swapping, respectively, where t_1 is the
 11 `pdelayReqEventEgressTimestamp` (taken when the `Pdelay_Req` message timestamp point crosses the
 12 reference plane at Port1 of ACC2 on transmission), t_2 is the `pdelayReqEventIngressTimestamp` (carried in
 13 the `requestReceiptTimestamp` field of the `Pdelay_Resp` message), and t_3 and t_4 are as given above. The
 14 NMS can compute the delay on the receive fiber as $[(t_4' - t_1) \times \text{nrpDelay} - (t_3' - t_2)] / 2$ and the delay on the
 15 transmit fiber as $[(t_4 - t_1') \times \text{nrpDelay} - (t_3 - t_2)] / 2$. Then the asymmetry ratio is as follows:

$$16 \quad \text{Delay_rx_fiber} / \text{Delay_tx_fiber} = [(t_4' - t_1) \times \text{nrpDelay} - (t_3' - t_2)] / [(t_4 - t_1') \times \text{nrpDelay} - (t_3 - t_2)]$$

1 Annex H

2 (informative)

3 Bibliography

4 Bibliographical references are resources that provide additional or helpful material but do not need to be
5 understood or used to implement this standard. Reference to these resources is made for informational use
6 only.

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8 transmission format for two-channel linearly represented digital audio data, Audio Engineering Society.²⁷

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10 Synchronization of digital audio equipment in studio operations, Audio Engineering Society.

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12 Application 1289, 1997.²⁸

13 [B4] Garner, Geoffrey M., Derivation of FTM Parameters in 12.6 of 802.1AS-Rev, presentation to IEEE
14 802.1 TSN TG, 28 Oct. 2018.²⁹

15 Garner, Geoffrey M., End-to-End Jitter and Wander Requirements for ResE Applications, presentation to
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